First Data®

BUYPASS® ATL105

Message Format Specifications

October 09, 2026

Version 2026-4

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Current Revision History

This is a new document. It replaces the following document:

- *BUYPASS® Platform ATL105 Message Format Specifications*
  Version No. 2026-3
  Released on August 07, 2026

The following table lists added, updated, and deleted pages:

Note: Red text and change bars in the page margin indicate updated/added text. Deleted text is indicated only in the following table. Blue text indicates a link to another location in the text.

Changes Made to Ver. No. 2026-3 And Reflected in Ver. No. 2026-4 Released October 09, 2026			
Topic	**2026-3 P. No.**	**2026-4 P. No.**	**Summary of Change**
Incremental Auth Data Updates	NA	I-88	Added table ID '82' (Incremental Auth Data) in Appendix I
	NA	I-67	Added verbiage on the usage of field Terminal Transaction Number in table ID '063' (Visa®/MC/AMEX Incremental Auth Lodging Data) under Appendix I
	NA	13-85	Added value '082' (Incremental Auth Data) to data element '111' under Data Elements in Data Element Number Order
	NA	13-91	Added processing rule for data element '111' value '082' in data element '113' under Data Elements in Data Element Number Order
	NA	AF-1	Added Appendix AF. Estimated and Incremental Authorization Transactions
Online Refund/Refund Authorization Updates	NA	I-27	Removed content from and marked sub-table ID '11' as 'Reserved' under table ID '032' (POS Additional Information Indicator) in Appendix I
	NA	X-1	Removed entire content from Appendix X and marked it as 'Reserved'
Originating Device Type Information Update	NA	I-16	Added additional Originating Device Type Indicators under table ID '022' in Appendix I
Payment Tokenization Updates	NA	10-62	Added additional networks that support payment token usage under Payment Tokenization

Changes Made to Ver. No. 2026-3 And Reflected in Ver. No. 2026-4 Released October 09, 2026			
Topic	**2026-3 P. No.**	**2026-4 P. No.**	**Summary of Change**
	NA	10-77	Updated grid of supported payment tokens with additional networks under Supported Payment Token Usage
	NA	I-55	Updated note with verbiage on Discover cards issued in India in sub-table ID ‘01’ under table ID ‘056’ in Appendix I
	NA	I-56	Updated note with verbiage on Discover cards issued in India in sub-table ID ‘02’ under table ID ‘056’ in Appendix I
	NA	I-57	Added note on Discover cards issued in India in sub-table ID ‘03’ under table ID ‘056’ in Appendix I
	NA	I-57	Added note on Discover cards issued in India in sub-table ID ‘04’ under table ID ‘056’ in Appendix I
	NA	I-58	Updated note with verbiage on Discover cards issued in India in sub-table ID ‘05’ under table ID ‘056’ in Appendix I
	NA	I-59	Updated note with verbiage on Discover cards issued in India in sub-table ID ‘06’ under table ID ‘056’ in Appendix I
	NA	I-60	Added note on Discover cards issued in India in sub-table ID ‘07’ under table ID ‘056’ in Appendix I
Network Token Data Updates	NA	12-4	Updated grid of ‘Data Segments and the Transactions in which They Can Appear’ with Network Token Data Response under Data Segment Formats
	NA	12-93	Updated existing Network Token Data Layouts tables and added additional ones under Network Token Data Request Segment
	NA	12-121	Added Network Token Data Response Segment and associated layouts

Prior Revision History

Changes Made to Ver. No. 2026-2 And Reflected in Ver. No. 2026-3 Released August 07, 2026			
Topic	**2026-2 P. No.**	**2026-3 P. No.**	**Summary of Change**
Valid Authorizer Codes Update	NA	C-1	Updated multiple Valid Authorizer Code Descriptions and Authorizer Types in Appendix C

Changes Made to Ver. No. 2026-1 And Reflected in Ver. No. 2026-2 Released June 05, 2026			
Topic	**2026-1 P. No.**	**2026-2 P. No.**	**Summary of Change**
Visa Category Code Updates	NA	I-51	Added sub-table ID '30' under table ID '049' in Appendix I
	NA	K-2	Added additional information 'Visa Category Code' in Appendix K
	NA	K-33	Added table ID '047' (Visa Category Code) in Appendix K
	NA	12-30	Added data segment '112' information 'Visa Category Code' in Additional Information Data Segment
	NA	13-95	Added value '047' (Visa Category Code) in data element '116' under Data Elements in Data Element Number Order
	NA	13-98	Added processing rule for data element '116' value '047' in data element 118 under Data Elements in Data Element Number Order
Transaction Link Action Indicator Updates	NA	I-50	Added sub-table ID '26' under table ID '049' in Appendix I
	NA	K-2	Added additional information 'Transaction Link Action Indicator' in Appendix K
	NA	K-30	Added table ID '039' (Transaction Link Action Identifier) in Appendix K

Changes Made to Ver. No. 2026-1 And Reflected in Ver. No. 2026-2 Released June 05, 2026			
Topic	**2026-1 P. No.**	**2026-2 P. No.**	**Summary of Change**
	NA	12-30	Added data segment '112' information 'Transaction Link Action Indicator' in Additional Information Data Segment
	NA	13-95	Added value '039' (Transaction Link Action Indicator) in data element '116' under Data Elements in Data Element Number Order
	NA	13-98	Added processing rule for data element '116' value '039' in data element 118 under Data Elements in Data Element Number Order
Digital Commerce Data Update	NA	I-75	Added support for Maestro card type to Digital Commerce Data throughout table ID '066' under Appendix I
General Update	NA	K-34	Corrected reference to sub-table ID '28' of Merchant Supplementary Data in table ID '046' under Appendix K

Changes Made to Ver. No. 2025-6 And Reflected in Ver. No. 2026-1 Released May 04, 2026			
Topic	**2025-6 P. No.**	**2026-1 P. No.**	**Summary of Change**
General Updates	NA	I-73	Updated verbiage on Visa Enabler Verification Value and renamed the table in table ID '065' in Appendix I
	NA	I-78	Updated example of Enabler Verification value in table ID '071' in Appendix I
Visa and MasterCard Enhanced Fleet Updates	NA	I-80	Added table ID '078' (Line Item) in Appendix I
	NA	I-85	Added table ID '079' (Visa Commercial Card Data) in Appendix I
	NA	12-72	Added note on MasterCard Enhanced Fleet EMV transactions in Enhanced Fleet Request Segment
	NA	12-79	Added list of Conexxus Prompt Codes in Enhanced Fleet Request Segment

Changes Made to Ver. No. 2025-6 And Reflected in Ver. No. 2026-1 Released May 04, 2026			
Topic	**2025-6 P. No.**	**2026-1 P. No.**	**Summary of Change**
	NA	I-47	Added notes on MasterCard Enhanced Fleet EMV transactions in sub-table ID '18' under table ID '049' in Appendix I
	NA	K-26	Added support for MasterCard Enhanced Fleet EMV transactions in table ID 034 under Appendix K
	NA	13-70	Added value 'H' in data element '87' under Data Elements in Data Element Number Order
	NA	13-80	Added values 'H' and 'I' in data element '106' under Data Elements in Data Element Number Order
	NA	13-158	Added support for MasterCard Enhanced Fleet EMV in data element '239' under Data Elements in Data Element Number Order
	NA	13-159	Added data element '243' (Extended Unit of Measure) under Data Elements in Data Element Number Order
	NA	I-47	Added value '0' to sub-table ID '18' under table ID '049' in Appendix I
Visa Gateway ID and Staged Wallet ID Updates	NA	I-87	Added table IDs '080' (Staged Wallet ID) and '081' (Visa Gateway ID) in Appendix I
Visa Card Verification Data Update	NA	M-8	Updated descriptions of CVV response values 'M', 'N', 'P', 'S', and 'U' for Visa in table ID '004' under Appendix K

Changes Made to Ver. No. 2025-5 And Reflected in Ver. No. 2025-6 Released November 07, 2025			
Topic	**2025-5 P. No.**	**2025-6 P. No.**	**Summary of Change**
Digital Commerce Data Updates	NA	I-74	Added value '0' and updated associated note in sub-table ID '01' under table ID '066' in Appendix I

Changes Made to Ver. No. 2025-5 And Reflected in Ver. No. 2025-6 Released November 07, 2025			
Topic	**2025-5 P. No.**	**2025-6 P. No.**	**Summary of Change**
	NA	I-74	Added additional values and updated associated note in sub-table ID ‘02’ under table ID ‘066’ in Appendix I
	NA	I-74	Renamed sub-table ID ‘03’ under table ID ‘066’ in Appendix I
	NA	I-75	Renamed sub-table ID ‘04’ under table ID ‘066’ in Appendix I
Transaction Link Identifier Updates	NA	I-77	Added table ID ‘072’ (Transaction Link Identifier) in Appendix I
	NA	K-29	Added verbiage on Moneris to table ID ‘038’ in Appendix K
Visa Discount Program Updates	NA	I-78	Added table ID ‘075’ (Discount Program Data) in Appendix I
	NA	I-78	Added table ID ‘076’ (Discount Line-Item Data) in Appendix I
	NA	I-79	Added table ID ‘077’ (Discount Amount) in Appendix I
Merchant Transaction ID Updates	NA	I-52	Added note on Moneris to field Merchant Tran ID in Appendix I
	NA	I-50	Added sub-table ID ‘25’ under table ID ‘049’ in Appendix I
	NA	K-31	Added table ID ‘044’ (Merchant Tran ID) in Appendix K
Applied Special Service Data Updates	NA	I-50	Added sub-table ID ‘27’ under table ID ‘049’ in Appendix I
	NA	K-31	Added table ID ‘045’ (Applied Special Service) in Appendix K
Voyager Restriction Code Updates	NA	I-50	Added sub-table ID ‘28’ under table ID ‘049’ in Appendix I
	NA	K-31	Added table ID ‘046’ (Voyager Restriction Code) in Appendix K

Changes Made to Ver. No. 2025-4 And Reflected in Ver. No. 2025-5 Released October 10, 2025

Topic	2025-4 P. No.	2025-5 P. No.	Summary of Change
General Update	NA	K-29	Corrected attribute to 'ans' and added associated note in table ID '038' (MasterCard Transaction Link Identifier) under Appendix K
Currency Update	NA	L-4	Updated currency name and currency code for Bulgaria and added associated note in Appendix L

Changes Made to Ver. No. 2025-3 And Reflected in Ver. No. 2025-4 Released August 14, 2025			
Topic	**2025-3 P. No.**	**2025-4 P. No.**	**Summary of Change**
Expiration Date Update	NA	13-32	Added verbiage on Expiration Date for Accel and STAR in data element '12' under Data Elements in Data Element Number Order
General Update	NA	I-44	Corrected sub-table ID '09' length value under table ID '049' in Appendix I

Changes Made to Ver. No. 2025-2 And Reflected in Ver. No. 2025-3 Released June 12, 2025			
Topic	**2025-2 P. No.**	**2025-3 P. No.**	**Summary of Change**
Enabler Verification Value Updates	NA	13-84	Added value '071' (Enabler Verification Value) in data element '111' under Data Elements in Data Element Number Order
	NA	13-90	Added processing rule for data element '111' value '071' under data element '113'
	NA	I-2	Added 'Enabler Verification Value' to the list of variable information in Appendix I
	NA	I-76	Added table ID '071' (Enabler Verification Value) in Appendix I
Comdata Updates	NA	12-12	Added notes on the usage of Product Code Data Segment at the beginning of the segment

Changes Made to Ver. No. 2025-2 And Reflected in Ver. No. 2025-3 Released June 12, 2025			
Topic	**2025-2 P. No.**	**2025-3 P. No.**	**Summary of Change**
	NA	12-105	Added Adjusted Product Code Data Segment under Data segment Formats
	NA	13-12	Added Adjusted Product Segment (No. 157) to applicable data elements under Data Elements in Alphabetical Order
	NA	13-55	Updated data element ‘63’ with references to input data segments in chapter 12
	NA	13-59	Updated data elements 76,77, 81,84,85, 87,106, and 107 with references to data segment 157

Changes Made to Ver. No. 2025-1 And Reflected in Ver. No. 2025-2 Released April 10, 2025			
Topic	**2025-1 P. No.**	**2025-2 P. No.**	**Summary of Change**
Card Expiration Date Format Update	NA	K-14	Added note on card expiration date format in table ID ‘013’ under Appendix K
Visa Authentication Data Quality Program Updates	NA	12-30	Added Authentication Data Quality Indicator in data elements list under Additional Information Data Segment
	NA	13-94	Added value ‘042’ (Authentication Data Quality Indicator) in data element ‘116’ under Data Elements in Data Element Number Order
	NA	13-97	Added processing rule for data element ‘116’ value ‘042’ in data element ‘118’
	NA	I-48	Added sub-table ID ‘23’ under Merchant Supplementary Data in Appendix I
	NA	K-1	Added Authentication Data Quality Indicator in additional information list under Appendix K
	NA	K-28	Added table ID ‘042’ (Authentication Data Quality Indicator) in Appendix K

Changes Made to Ver. No. 2025-1 And Reflected in Ver. No. 2025-2 Released April 10, 2025			
Topic	**2025-1 P. No.**	**2025-2 P. No.**	**Summary of Change**
Visa Token Update First Use Updates	NA	12-30	Added Token Update First Use Indicator in data elements list under Additional Information Data Segment
	NA	13-94	Added value ‘043’ (Token Update First Use Indicator) in data element ‘116’ under Data Elements in Data Element Number Order
	NA	13-97	Added processing rule for data element ‘116’ value ‘043’ in data element ‘118’
	NA	I-49	Added sub-table ID ‘24’ under Merchant Supplementary Data in Appendix I
	NA	K-2	Added Token Update First Use Indicator in additional information list under Appendix K
		K-29	Added table ID ‘043’ (Token Update First Use Indicator) in Appendix K
Visa DAF Indicator Update	NA	K-24	Added additional DAF Indicator values in table ID ‘031’ under Appendix K
Originating Device Type Indicator Updates	NA	I-16	Added additional Originating Device Type Indicator values in table ID ‘022’ in Appendix I
MasterPass Removal Updates	NA	I-23	Removed sub-table ID ‘03’ (MasterPass Transaction Wallet ID) from table ID ‘032’ (POS Additional Information Indicator) in Appendix I
	NA	I-33	Removed value ‘22 = MasterPass transaction’ from table ID ‘036’ (MasterCard UCAF Data) in Appendix I
	NA	Q-3	Removed National Point-of-Service Condition Code ‘22’ from Appendix Q
	NA	Q-4	Removed the line containing ‘101 22 Unattended off premise customer-operated MasterPass Transaction Identifier’ from Appendix Q
MasterCard Future Transaction Amount	NA	I-76	Added table ID ‘073’ (Future Transaction Amount) in Appendix I

Changes Made to Ver. No. 2025-1 And Reflected in Ver. No. 2025-2 Released April 10, 2025			
Topic	**2025-1 P. No.**	**2025-2 P. No.**	**Summary of Change**
Updates	NA	13-84	Added value '073' in data element '111' under Data Elements in Data Element Number Order
	NA	13-90	Added processing rule for data element '111' value '073' in data element '113' under Data Elements in Data Element Number Order
	NA	K-29	Added table IDs '040' (Fraud Score) and '041' (Fraud Score Reason Code) in Appendix K
	NA	K-1	Added Fraud Score and Fraud Score Reason Code in additional information list under Appendix K
	NA	12-30	Added Fraud Score and Fraud Score Reason Code in data elements list under Additional Information Data Segment
	NA	13-94	Added values '040' and '041' in data element '116' and associated processing rules in data element 118 under Data Elements in Data Element Number Order
MasterCard and Maestro Transaction Link Identifier Update	NA	I-48	Added sub-table ID '22' under table ID '049' (Merchant Supplementary Data) in Appendix I
	NA	K-29	Added table ID '038' (MasterCard Transaction Link Identifier) in Appendix K
	NA	K-1	Added Transaction Link Identifier in additional information list under Appendix K
	NA	12-30	Added Transaction Link Identifier in data elements list under Additional Information Data Segment
	NA	13-94	Added value '038' in data element '116' and associated processing rule in data element 118 under Data Elements in Data Element Number Order

Changes Made to Ver. No. 2025-1 And Reflected in Ver. No. 2025-2 Released April 10, 2025			
Topic	**2025-1 P. No.**	**2025-2 P. No.**	**Summary of Change**
Currency Update	NA	L-6	Removed 'Netherlands Antilles' and updated the currency for 'Curacao' and 'Sint Maarten (Dutch Part)' to Caribbean Guilder in Appendix L

Changes Made to Ver. No. 2024-5 And Reflected in Ver. No. 2025-1 Released February 14, 2025			
Topic	**2024-5 P. No.**	**2025-1 P. No.**	**Summary of Change**
Visa Electric Vehicle (EV) Transaction Updates	NA	12-12	Added verbiage with information on EV charging transactions in Product Code Data Segment
	NA	12-100	Added data segment '156' (EV Charging Data Segment) under Data Segment Formats
	NA	12-5	Added data segment '156' in the data segments table under Data Segment Formats
	NA	13-156	Added data element '242' (EV Charging Data) under Data Elements in Data Element Number Order
	NA	13-7	Added data element '242' (EV Charging Data) under Data Elements in Alphabetical Order
	NA	13-84	Added values '68' (Additional Transaction Fee 1) and '69' (Additional Transaction Fee 2) in data element '111' under Data Elements in Data Element Number Order
	NA	13-90	Added processing rules for data element '111' values '68' and '69' in data element '113' under Data Elements in Data Element Number Order
	NA	I-73	Added table IDs '68' and '69' for EV-Additional Transaction Fee 1 and EV-Additional Transaction Fee 2 in Appendix I

Changes Made to Ver. No. 2024-5 And Reflected in Ver. No. 2025-1 Released February 14, 2025			
Topic	**2024-5 P. No.**	**2025-1 P. No.**	**Summary of Change**
	NA	13-80	Added values 'M' and 'W' in data element '106' (Unit of Measure) under Data Elements in Data Element Number Order
	NA	I-64	Updated field Terminal Transaction Number with support for EV and added EV examples for Visa Incremental Auth transactions in Appendix I
	NA	AE-1	Added verbiage containing MCC applicable to Estimated and Incremental authorizations for EV transactions in Appendix AE
Anticipated Completion Amount Updates	NA	13-84	Added value '70' (Anticipated Completion Amount) in data element '111' (Variable Information Indicator) under Data Elements in Data Element Number Order
	NA	13-90	Added processing rule for data element '111' value '70' in data element '113' under Data Elements in Data Element Number Order
	NA	I-74	Added table ID '070' (Anticipated Completion Amount) in Appendix I

Changes Made to Ver. No. 2024-4 And Reflected in Ver. No. 2024-5 Released October 04, 2024			
Topic	**2024-4 P. No.**	**2024-5 P. No.**	**Summary of Change**
MasterCard Digital Commerce Data Updates	NA	I-2	Added 'Digital Commerce Data' in the list of variable information under Appendix I. Variable Information Data Layouts
	NA	I-70	Added table ID '066' (Digital Commerce Data) in Appendix I. Variable Information Data Layouts

Changes Made to Ver. No. 2024-4 And Reflected in Ver. No. 2024-5 Released October 04, 2024			
Topic	**2024-4 P. No.**	**2024-5 P. No.**	**Summary of Change**
	NA	13-84	Added value '066' to data element '111' (Variable Information Indicator) under Data Elements in Data Element Number Order
	NA	13-90	Added processing rule for Digital Commerce Data in data element 113 under Data Elements in Data Element Number Order
MasterCard Consumer Multi-Use Virtual Card Number Update	NA	K-22	Added Merchant Advice Code value '43' in table ID '030' (Merchant Advice Code) under Appendix K. Additional Information Data Layouts
MasterCard/Maestro Merchant Payment Gateway Update	NA	I-70	Updated Merchant Payment Gateway description in table ID '065' under Appendix I. Variable Information Data Layouts
Currency Update	NA	L-9	Updated currency name and currency code for Zimbabwe in Appendix L. Valid Currency Codes
Acquiring Institution Country Update	NA	AC-3	Removed 'Netherlands Antilles' and associated code from Country Codes table in Appendix AC
MasterCard Identity Check Update	NA	I-33	Removed text 'UCAF data collection is not supported.' from UCAF Security Level position '3' value '4' description in table ID '036' under Appendix I. Variable Information Data Layouts
Recurring or Installment Transaction (For Card Types Other than Credit) Update	NA	I-48	Added sub-table ID '21' under table ID '049' (Merchant Supplementary Data) in Appendix I. Variable Information Data Layouts
General Updates	NA	12-54	Corrected Source to 'Device' for EMV Additional Information elements '191', '192', and '118' in EMV Request Data Segment

Changes Made to Ver. No. 2024-4 And Reflected in Ver. No. 2024-5 Released October 04, 2024			
Topic	**2024-4 P. No.**	**2024-5 P. No.**	**Summary of Change**
	NA	1-2	Replaced First Data Spec Portal URL with Direct Platform Specifications Portal URL throughout the document

Changes Made to Ver. No. 2024-3 And Reflected in Ver. No. 2024-4 Released August 09, 2024			
Topic	**2024-3 P. No.**	**2024-4 P. No.**	**Summary of Change**
WEX OTR Updates	NA	12-74	Added additional Product Code Categories for electric vehicle charge levels for WEX OTR in Enhanced Fleet Request Segment
	NA	12-77	Added additional Prompt Tokens for WEX OTR in Enhanced Fleet Request Segment

Changes Made to Ver. No. 2024-2 And Reflected in Ver. No. 2024-3 Released June 07, 2024			
Topic	**2024-2 P. No.**	**2024-3 P. No.**	**Summary of Change**
Enhanced Fleet Data Updates	NA	12-72	Renamed table ID ‘001’ as ‘Enhanced Fleet Request Flags’ and updated it with associated information under Enhanced Fleet Request Segment
	NA	12-73	Renamed table ID ‘002’ as ‘Non-Fuel Product Data (Request)’ and added associated information under Enhanced Fleet Request Segment
	NA	12-77	Added list of valid prompt tokens for WEX OTR transactions to table ID ‘004’ (Prompt Data (request)) under Enhanced Fleet Request Segment
	NA	12-77	Added table IDs 006 (Money Code Payee Name), 007 (Money Code Check Number), and 008 (Cash Advance Limit) under Enhanced Fleet Request Segment

Changes Made to Ver. No. 2024-2 And Reflected in Ver. No. 2024-3 Released June 07, 2024			
Topic	**2024-2 P. No.**	**2024-3 P. No.**	**Summary of Change**
	NA	13-26	Added verbiage on Money Code/Check transactions under Valid Codes/Values in data element 2 (Account Number)
	NA	13-32	Added verbiage on Money Code/Check request under Valid Codes/Values in data element 12 (Card Discretionary Block Data)
	NA	C-3	Modified authorizer name to 'WEX OTR Fleet' and authorizer type to 'Fleet' for authorizer code '81' in Appendix C
	NA	E-2	Added valid card type codes '091', '092', and '093' and associated information in Appendix E
Multiple Fuel Type Codes Update	NA	12-12	Added note on multiple fuel types in a single transaction in Product Code Data Segment

Changes Made to Ver. No. 2024-1 And Reflected in Ver. No. 2024-2 Released April 05, 2024			
Topic	**2024-1 P. No.**	**2024-2 P. No.**	**Summary of Change**
Valid Card Type Codes Update	NA	E-3	Added Card Type Code 150 to the table of Valid Card Type Codes Used in the Table Load Response under Appendix E
PULSE Update	NA	I-25	Removed Discover® exclusivity for the value "07" of Wallet Identifier in sub-table ID '07' under table ID '032' (POS Additional Information Indicator) in Appendix I
Maestro Merchant Payment Gateway ID Data Update	NA	I-69	Updated note to add support for Maestro in table ID '065' (Merchant Payment Gateway ID Data) under Appendix I

Changes Made to Ver. No. 2024-1 And Reflected in Ver. No. 2024-2 Released April 05, 2024			
Topic	**2024-1 P. No.**	**2024-2 P. No.**	**Summary of Change**
Maestro Mobile POS Support Updates	NA	I-21	Updated example and note to add support for Maestro in sub-table ID '01' under table ID '032' (POS Additional Information Indicator) in Appendix I
	NA	I-43	Added support for Maestro to sub-table ID '06' under table ID '049' (Merchant Supplementary Data) in Appendix I
Visa Enabler Verification Value Updates	NA	I-69	Renamed table ID '065' as 'Merchant Payment Gateway ID Data/Enabler Verification Value' and added support to identify Enablers for Visa in Appendix I
	NA	13-84	Renamed value '065' as 'Merchant Payment Gateway ID Data/Enabler Verification Value' in data element '111' under Data Elements in Data Element Number Order
	NA	13-89	Updated Processing Rule for data element '111' value '065' in data element '113' under Data Elements in Data Element Number Order to add support for Visa Enabler Verification Value
	NA	I-2	Renamed variable information 'Merchant Payment Gateway ID Data' as 'Merchant Payment Gateway ID Data/Enabler Verification Value' in Appendix I
Visa Anticipated Amount, Account Type, and Account Funding Source Information Updates	NA	I-70	Added table ID '067' (Anticipated Amount) in Appendix I
	NA	I-2	Added variable information 'Anticipated Amount' in Appendix I
	NA	13-84	Added value '067' in data element '111' under Data Elements in Data Element Number Order
	NA	13-89	Added Processing Rule for data element '111' value '067' in data element '113' under Data Elements in Data Element Number Order

Changes Made to Ver. No. 2024-1 And Reflected in Ver. No. 2024-2 Released April 05, 2024			
Topic	**2024-1 P. No.**	**2024-2 P. No.**	**Summary of Change**
	NA	I-48	Added sub-table ID ‘20’ under table ID ‘049’ (Merchant Supplementary Data) in Appendix I
	NA	K-1	Added ‘Account Type’ and ‘Account Funding Source’ to the list of additional information used in conjunction with data element 116 and data element 118 in Appendix K
	NA	13-92	Added codes ‘036’ and ‘037’ in data element ‘116’ under Data Elements in Data Element Number Order
	NA	13-95	Added Processing Rules for data element ‘116’ values ‘036’ and ‘037’ in data element ‘118’ under Data Elements in Data Element Number Order
	NA	K-26	Added table ID ‘036’ (Account Type) and table ID ‘037’ (Account Funding Source) in Appendix K

Changes Made to Ver. No. 2023-6 And Reflected in Ver. No. 2024-1 Released February 09, 2024			
Topic	**2023-6 P. No.**	**2024-1 P. No.**	**Summary of Change**
Real Time Account Updater Updates	NA	12-7	Added reference to data element 2 in field no. 6 of Standard Message Data Segment
	NA	12-7	Added reference to data element 12 in field no. 7 of Standard Message Data Segment
	NA	13-25	Added section and removed defunct verbiage on Real Time Account Updater in data element 2 under Data Elements in Data Element Number Order
	NA	13-31	Added section and removed defunct verbiage on Real Time Account Updater in data element 12 under Data Elements in Data Element Number Order

Changes Made to Ver. No. 2023-6 And Reflected in Ver. No. 2024-1 Released February 09, 2024			
Topic	**2023-6 P. No.**	**2024-1 P. No.**	**Summary of Change**
	NA	AD-1	Added reference to data element 2 and data element 12 in Appendix AD. Real Time Account Updater
General Update	NA	13-32	Updated note on expired cards throughout the document

Changes Made to Ver. No. 2023-5 And Reflected in Ver. No. 2023-6 Released November 15, 2023			
Topic	**2023-5 P. No.**	**2023-6 P. No.**	**Summary of Change**
Fuel Price Data Updates	NA	12-83	Added Fuel Price Update Request Segment under Data Segment Formats
	NA	13-155	Added Data Element 241 (Price Data) under Data Elements in Data Element Number Order
	NA	13-12	Added Data Element 241 (Price Data) under Data Elements in Alphabetical Order
	NA	12-84	Added Fuel Price Update Response Segment under Data Segment Formats
Comdata Updates	NA	12-71	Added support for Comdata to Enhanced Fleet Request Segment
	NA	12-76	Added verbiage to exclude table Non-fuel Product Limits (Response) from Comdata response messages
	NA	E-2	Added additional card type codes for Comdata to the list of Valid Card Type Codes in Appendix E
	NA	C-1	Updated Valid Authorizer Code ‘33’ as COMDATA Fleet in Appendix C
	NA	13-32	Added information Comdata Express Code or Check requests to data element 12 under Data Elements in Data Element Number Order

Changes Made to Ver. No. 2023-4 And Reflected in Ver. No. 2023-5 Released October 06, 2023			
Topic	**2023-4 P. No.**	**2023-5 P. No.**	**Summary of Change**
MasterCard Consumer Non-Reloadable Prepaid Products and Single-Use Virtual Card Numbers Update	NA	K-22	Added Merchant Advice Codes '40' and '41' to table ID 30 in Appendix K
MasterCard Merchant Payment Gateway ID Updates	NA	I-2	Added 'Merchant Payment Gateway ID Data' to the list of variable information in Appendix I
	NA	I-69	Added table ID 065 (Merchant Payment Gateway ID Data) in Appendix I
MasterCard and Maestro Service Location Information Update	NA	I-67	Updated Service Location Information description and its applicability in table ID 064 under Appendix I
Valid Authorizer Codes Update	NA	C-1	Updated Authorizer Names for Authorizer Codes '25' and '26' in Appendix C
General Update	NA	13-31	Updated note on expired cards for Visa and MasterCard throughout the document

Changes Made to Ver. No. 2023-3 And Reflected in Ver. No. 2023-4 Released August 18, 2023			
Topic	**2023-3 P. No.**	**2023-4 P. No.**	**Summary of Change**
Incomm OTC Updates	NA	1-44	Added sub-table ID 09 under table ID 049 in Appendix I
	NA	12-82	Added Incomm OTC Market Basket Data Request and Response Segments under Data Segment Formats
Network Token Update	NA	12-84	Added Network Token Data Request Segment under Data Segment Formats
eWIC Data Update	NA	10-42	Added verbiage on POS Condition Code under eWIC Data Elements
Data Section Updates	NA	11-3	Updated specifics of Data Section No. 3 under Financial Transactions

Changes Made to Ver. No. 2023-3 And Reflected in Ver. No. 2023-4 Released August 18, 2023			
Topic	**2023-3 P. No.**	**2023-4 P. No.**	**Summary of Change**
	NA	11-41	Updated specifics of Data Section No. 3 under EMV Financial Transactions
	NA	12-2	Updated information in Data Segment table under Data Segment Formats
General Updates	NA	11-17	Updated Segment No. value to '119' for field 3 under Data Section 3 of Totals with Proprietary Data Load Request
	NA	13-31	Added note on expired cards for Visa and MasterCard to Data Element 12 under Data Elements in Data Element Number Order
Speedpass Feature Removal Update	NA	NA	Removed instances of the defunct Speedpass feature throughout the document

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1. Introduction

This document provides information about the BUYPASS® platform (BUYPASS) ATL105 message formats (software version ATL105). This format is generally used when a BUYPASS customer device accesses BUYPASS directly.

The scope of the information contained in this document is described in the chapter descriptions listed below.

The audiences of this document include decision makers evaluating BUYPASS services and the programmers and analysts who implement and support these services.

Note: It is important to note that sending of Payment Application Data Security Standard (PA-DSS) sensitive information, for example PAN, in any field, other than those designated to contain such information, is strictly prohibited and can lead to adverse consequences.

This document contains the following chapters and appendices:

1. Introduction. This chapter lists and describes each chapter in this document.

2. Message Flow Overview. This chapter describes the flow of messages from a BUYPASS customer device to BUYPASS.

3. Specifications Requirements. This chapter provides information about the hardware and software requirements necessary to support these specifications.

4. Supported Industries. This chapter provides information about the industries supported by these specifications.

5. Supported Payment Network Types. This chapter provides information about the payment network types supported by these specifications.

6. PIN Encryption Processing. This chapter provides diagrams and textual information about PIN encryption key management relative to using these specifications.

7. Network Management Processing. This chapter provides information about messages involved in the network management process relative to using these specifications.

8. Store-and-Forward Processing. This chapter discusses store-and-forward processing for automated fuel pumps and the restaurant and lodging industry.

9. End-of-Day Processing. This chapter provides sample information about end-of-day processing using a customer-designed electronic journal feature.

10. Processing Requirements. This chapter provides information about credit card, debit card, electronic benefits transfer card, fleet card, stored value card, check, loyalty, software update, electronic mail, TransArmor, Payment Tokenization, and EMV processing requirements.

11. Message Formats. This chapter provides information about message formats.

12. Data Segment Formats. This chapter provides information about data segment formats.

13. Data Element Descriptions. This chapter provides detailed information about the data elements used in the messages described in chapter 11, “Message Formats.”

14. Support, Testing, and Certification. This chapter provides information about development support provided to customers by BUYPASS before, during, and after project development.

Appendix A. TCP/IP Message Header. This appendix contains TCP/IP message header information.

Appendix B. POS Purchase Example. This appendix provides an example of the data layout for a POS purchase.

Appendix C. Valid Authorizer Codes. This appendix lists valid entries for Element No. 7, Authorizer Code.

Appendix D. Valid State Codes. This appendix lists valid State Code entries (alphabetical and ANSI®) used to construct Element No. 4 (Address Line 2), Element No. 12 (Card Discretionary Block Data), Element No. 102 (Terminal Identifier), and Element No. 124 (State Code).

Appendix E. Valid Card Type Codes. This appendix lists valid entries for Element No. 14, Card Type. Card Type codes are also used to construct Element No. 78, Prompt Code.

Appendix F. Valid Payment Systems Product Codes. This appendix lists valid entries for Element No. 77, Product Code.

Appendix G. Valid Transaction Type Codes. This appendix lists valid Transaction Type codes used to construct Element No. 78, Prompt Code.

Appendix H. Decline Codes Permitting Clerk/Customer Intervention. This appendix lists valid s (Element No. 26) that can be used to prompt a clerk/customer to call for an authorization, to reenter indicated data, to resubmit a timed out transaction, or to indicate that a TransArmor Token Only transaction has been declined due to nonretrieval of Token.

Appendix I. Variable Information Data Layouts. This appendix contains data layout information for the variable information used in conjunction with Element No. 111, Variable Information Indicator.

Appendix J. Valid Point-of-Service Entry Mode Codes. This appendix contains valid point-of-service entry mode information for the variable information used in conjunction with Element No. 113, Variable Information.

Appendix K. Additional Information Data Layouts. This appendix contains data layout information for the additional information used in conjunction with Element No. 116, Additional Information Indicator.

Appendix L. Valid Currency Codes. This appendix contains a list of currencies and their corresponding valid Currency Codes for Element No. 20 (Currency Code). These codes identify the currency used in a transaction and whether they are used for Visa and/or MasterCard.

Appendix M. Element Format Examples for EBT Program Data. This appendix contains two format examples for Element No. 164 (EBT Program Data) which is used with the Healthy Incentives Program (HIP).

Appendix N. First Data™ Premium Gift Card Magnetic Stripe Data Layout. This appendix contains a table that describes the layout for magnetic stripe data in Track II for the First Data™ Premium Gift Card.

Appendix O. Payment Tokenization Terminologies. This appendix lists payment tokenization terminologies and its respective descriptions.

Appendix P. TransArmor VeriFone Edition: Responses to Administrative Commands. For detailed information on TransArmor, refer the BUYPASS®_Platform_ATL105_Specification_Updates_for_TransArmor_Processing document, available on the Direct Platform Specifications Portal.

Appendix Q. Valid National Point-of-Service Condition Codes. This appendix lists valid entries for Variable Information (Element No. 113), Table ID 030.

Appendix R. EMV Chip Data Example. This appendix contains an EMV chip data packet example.

Appendix S. Valid CA Public Key File Record Layout. This appendix contains the record layout information of the CA Public Key File used in conjunction with element CA Public Key File Data.

Appendix T. EMV Additional Information Data Layouts. This appendix contains data layout information for EMV Table Data (Code 001) and Card Authentication Results Code (CARC).

Appendix U. Visa Digital Wallet. This appendix contains detailed information on Visa Pass-through and Staged Digital Wallet Transactions.

Appendix V. Moneris Data Layout. This appendix contains detailed information on Moneris data layout for request and response messages.

Appendix W. Download Data Layout. This appendix contains detailed information on Download data layout for Data Segment No.DL7, Supplemental Terminal Data Segment.

2. Message Flow Overview

This chapter discusses the following message flow information:

- Message flow diagram
- Successful financial transaction description
- Unsuccessful EFT transactions
  - ◊ Time-out by customer's host/device system (no response)
  - ◊ Time-out by customer's host/device system (late approved response)
  - ◊ BUYPASS response not recognized by customer's host/device system

Notes: For information about specific message formats, refer to chapter 11, "Message Formats."

For information about specific data segment formats, refer to chapter 12, "Data Segment Formats."

For information about individual data elements within the messages, refer to chapter 13, "Data Element Descriptions."

2.1 Message Flow Diagram

The following diagram describes the flow of messages from a device to BUYPASS.

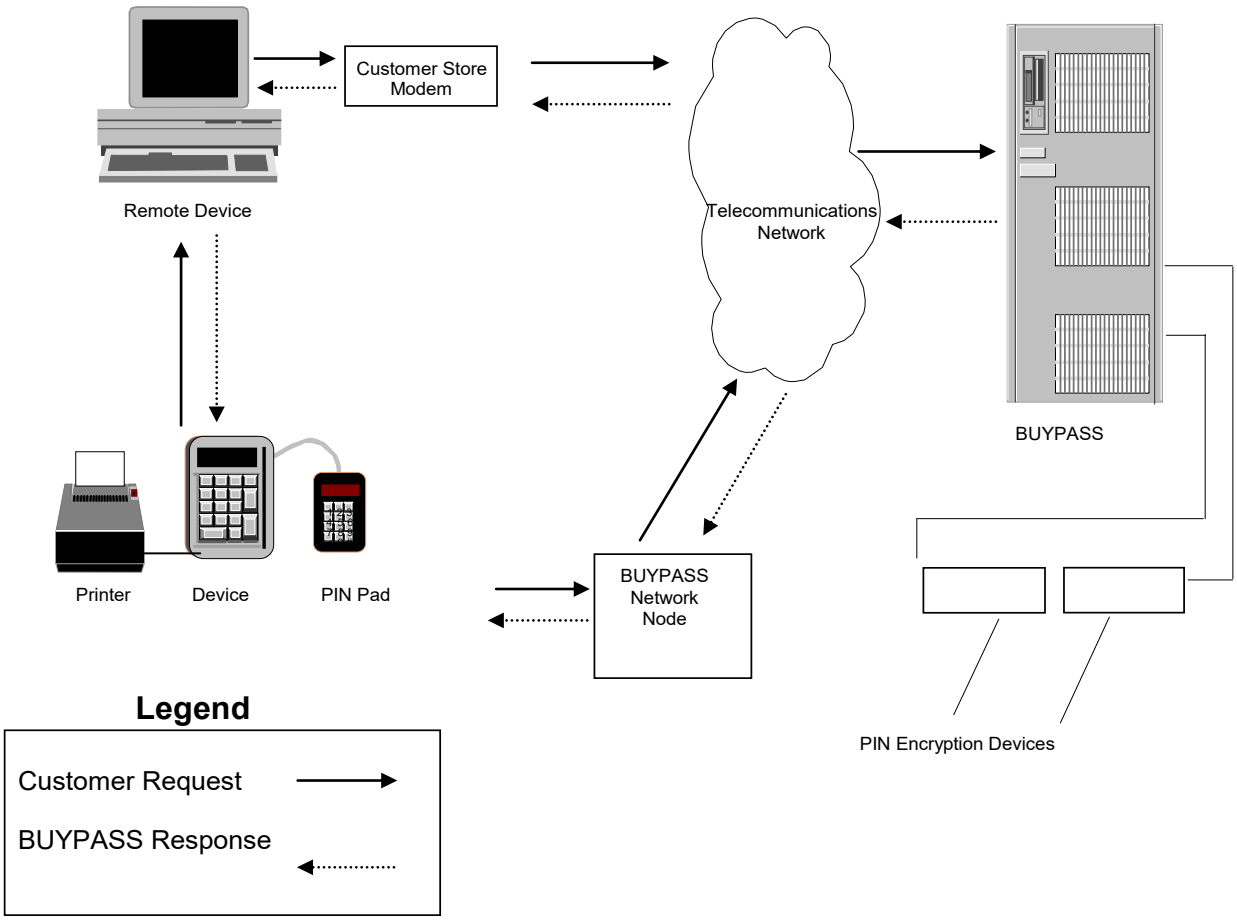


Fig. 2-1. Message flow overview.

2.2 Successful Financial Transaction

The following information illustrates a successful financial transaction (request and response).

Customer Device	Via Communications Protocol Connection	BUYPASS
The customer's device verifies the data and formats the request. Upon verification, the customer's device forwards the request to the BUYPASS network node, and starts a 30-second timer.	The BUYPASS network node forwards the request to BUYPASS. ---------->	BUYPASS determines the card type, and routes the transaction to the appropriate authorization center. After receiving an approval or decline from the authorization center, BUYPASS routes the response back to the customer's device, via the network node.
If an approved or declined response is received by the customer's device during the 30-second response interval, the customer's device forwards the response data to the originating device.	The BUYPASS network node forwards the response to the customer's device. <----------	

2.3 Unsuccessful Financial Transactions

This section contains the following examples of unsuccessful financial transactions:

- Time-out by customer's host/controller/POS system
- Time-out by customer's host/device system (late approved response)
- BUYPASS response not recognized by customer's host/controller/POS system

A time-out reversal (TOR) is created for an unsuccessful financial transaction. A TOR never initiates a new financial transaction request.

Notes: A TOR is generated when any of the following transactions is unsuccessful:

- POS Purchase/Capture transaction
- CAT Purchase/Capture transaction
- Mail/Phone Purchase transaction

A TOR is not supported for either of the following transactions:

- Debit POS Capture transaction
- Debit CAT Capture transaction

2.3.1 Time-out by Customer's Host/Controller/POS System

This section describes the transaction flow for an unsuccessful financial transaction (request and response) resulting from a time-out by a customer's host/controller/POS system.

This transaction flow describes single threading (one transaction at a time); however, any combination of requests and responses can be in process at any given time.

A time-out occurs at the customer's host/controller/POS system when either of the following situations occurs:

- The customer's 30-second response interval expires without a response from BUYPASS.
- A response from BUYPASS is received after the customer's 30-second response interval expires.

If the customer's host/controller/POS system receives a late response to the original transaction request from BUYPASS, the late response is ignored. In the event the customer's host/controller/POS system receives more than three consecutive time-outs in an IP environment, the line to BUYPASS is marked "down," and a communications test/health message is sent every 30 seconds until a valid response is received from BUYPASS.

Once a time-out occurs for a device, the customer's host/controller/POS system formats a TOR, which is queued in the customer's host/controller/POS system until the next financial transaction is initiated for that device. A TOR must be sent from the same device as the original financial transaction request that resulted in the TOR. When the next financial transaction is initiated from that device, the queued TOR is transmitted to BUYPASS ahead of the new financial transaction request. If a new financial transaction is not submitted between the occurrence of the TOR and the end-of-day settlement, the TOR should be sent to the host immediately prior to the settlement request. Some customer host/controller/POS systems allow transactions to be approved and processed in an off-line mode. When the host/controller/POS system detects that the line to BUYPASS is back "up," the off-line transactions are forwarded to BUYPASS only after a valid response is received for all queued TOR's.

The Sequence Number, Account Number, and dollar amount—assigned by the customer's host/controller/POS system—are the elements used to match a request with its respective response. The Sequence Number of a TOR must be the same as the Sequence Number of the financial transaction being reversed. Each time any financial transaction request is forwarded to BUYPASS, the financial transaction should be assigned a new Sequence Number by the customer's host/controller/POS system.

Any response from BUYPASS to a TOR, whether it is an approval or a decline, is considered a valid response and should clear the TOR from the customer's host/controller/POS queue. The customer's host/controller/POS system should continue to send the TOR—subject to health message rules—to BUYPASS until a valid response is received. If no valid response is received after three attempts on each available connection route, an error should be logged in the electronic journal, and the TOR should be cleared from the queue.

BUYPASS declines a TOR request as an unmatched reversal when BUYPASS has not received the original financial transaction request related to the TOR; or when the original financial transaction has already been reversed.

The following message flow illustrates an unsuccessful financial transaction (request and response) resulting from a time-out by a customer's host/controller/POS system:

Customer Host/Controller/POS System	**Via Communications Protocol Connection**	**BUYPASS**
The customer's host/ controller/POS system verifies the financial transaction data and formats the transaction request.		
Upon verification, the customer's host/controller /POS system forwards the financial transaction request to BUYPASS network node, and starts a 30-second timer.	BUYPASS network node forwards the financial transaction request to BUYPASS. ---------->	BUYPASS does not respond within the customer's host/controller POS system 30-second response interval; or there is a response within the response interval, but BUYPASS network node is not able to forward the response to the customer's host/controller/POS system.
The customer's host/controller POS system 30-second response interval expires.		
A time-out now occurs at the customer's host/controller /POS system.		
The customer's host/controller POS system formats and queues a TOR that remains with the customer's host /controller/POS system until a new financial transaction is initiated at the device associated with the TOR.		
The device for which the TOR is queued prepares to transmit a new financial transaction with a new Sequence Number.		
The customer's host/controller /POS system verifies the new financial transaction data and		

Customer Host/Controller/POS System	Via Communications Protocol Connection	BUYPASS
formats the new transaction request. Upon verification, the customer's host/controller /POS system transmits the TOR and the new financial transaction request to BUYPASS network node, and starts a 30-second timer. The TOR precedes the new financial transaction request in the transmission.	BUYPASS network node forwards the TOR and the new financial transaction request to BUYPASS. ---------->	BUYPASS processes the TOR. An approved or declined response to the TOR signifies completion of a TOR. BUYPASS processes the new financial transaction request. This activity is also subject to a TOR if BUYPASS does not respond within the customer's host/controller/POS system 30-second response interval; or there is a response within the response interval, but BUYPASS network node is not able to forward the response to the customer's host/controller/POS system.

2.3.2 Time-out by Customer's Host/Device System (Late Approved Response)

The following information illustrates an unsuccessful EFT transaction (request and response) resulting from a time-out by a customer's host/device system (late approved response). This transaction flow describes single threading (one transaction at a time); however, any combination of requests and responses can be in process at any given time. The sequence number, assigned by the customer's host/device system, is the element used to match a request with its respective response.

Customer Host/Device System	**Via Leased Line Connection**	**BUYPASS**
The customer's host/device system verifies the data and formats the request. Upon verification, the customer's host/device system forwards the request to the BUYPASS network node, and starts a 30-second timer.	The BUYPASS network node forwards the request to BUYPASS. ---------->	BUYPASS does not respond within the 30-second response interval.
The customer's host/device system 30-second response interval expires. The customer's host/device system formats a time-out reversal and forwards it to BUYPASS.	---------->	BUYPASS sends a late approval before receiving the time-out reversal.
	<----------	
The customer's host/device system receives the late approval. The customer's host/device system generates a time-out reversal to the late approval.	---------->	
The customer's host/device system forwards the off-line transactions to BUYPASS only after the customer's host/device system has received responses (approvals or declines) for all associated TOR's. The customer's host/device system continues to send the		

Customer Host/Device System	Via Leased Line Connection	BUYPASS
TOR (subject to health message rules) until the response is received. The customer's host/device should have a maximum number of retries for sending a TOR. It is recommended that the TOR be forwarded at least 30 seconds after the original transaction has timed out. Refer to chapter 9, "End-of-Day Processing," for additional information.		

2.3.3 BUYPASS Response Not Recognized by Customer's Host/Controller/POS System

This section discusses an unsuccessful financial transaction (request and response) resulting from the BUYPASS response not being recognized by the customer's host/controller/POS.

This transaction flow describes single threading (one transaction at a time); however, any combination of requests and responses can be in process at any given time. This flow applies to financial transactions including TOR's.

If a customer's host/controller/POS system receives a response from BUYPASS that is not correctly formatted or for which the CRC value is incorrect, it should recognize this as a system error. Once a system error is detected, the customer's host/controller/POS system formats a TOR, which is queued in the customer's host/controller/POS system until the next financial transaction is initiated for that device. A TOR must be sent from the same device as the original financial transaction request that resulted in the TOR. When the next financial transaction is initiated from that device, the queued TOR is transmitted to BUYPASS ahead of the new financial transaction request. If a new financial transaction is not submitted between the occurrence of the TOR and the end-of-day settlement, the TOR should be sent to the host immediately prior to the settlement request.

The Sequence Number, Account Number, and dollar amount—assigned by the customer's host/controller/POS system—are the elements used to match a request with its respective response. The Sequence Number of a TOR must be the same as the Sequence Number of the financial transaction being reversed. Each time any financial transaction request is forwarded to BUYPASS, the financial transaction should be assigned a new Sequence Number by the customer's host/controller/POS system.

Any response from BUYPASS to a TOR, whether it is an approval or a decline, is considered a valid response and should clear the TOR from the queue.

BUYPASS declines a TOR request as an unmatched reversal when BUYPASS has not received the original financial transaction request related to the TOR; or when the original financial transaction has already been reversed.

The following information illustrates an unsuccessful financial transaction (request and response) resulting from a time-out by a customer's host/controller/POS system:

Customer Host/Controller/POS System	**Via Communications Protocol Connection**	**BUYPASS**
The customer's host/controller /POS system verifies the financial transaction data and formats the request.		
Upon verification, the customer's host/controller /POS system forwards the financial transaction request to BUYPASS network node, and starts a 30-second timer.	BUYPASS network node forwards the financial transaction request to BUYPASS. ---------->	BUYPASS determines the card type, and routes the financial transaction to the

Customer Host/Controller/POS System	**Via Communications Protocol Connection**	**BUYPASS**
		appropriate authorization center. After receiving an approval or decline, BUYPASS routes the response back to the customer's host /controller/POS system, via the network node within 30 seconds.
The customer's host/controller /POS system cannot parse the formatted BUYPASS response. The customer's operator console displays an edit error message. The customer's host/controller POS system formats and queues a TOR that remains with the customer's host /controller/POS system until a new financial transaction is initiated at the device associated with the TOR. The device for which the TOR is queued prepares to transmit a new financial transaction with a new Sequence Number. The customer's host/controller /POS system verifies the new financial transaction data and formats the new transaction request. Upon verification, the customer's host/controller /POS system transmits the TOR and the new financial transaction request to the BUYPASS network node, and starts a 30-second timer. The TOR precedes the new financial transaction request in the transmission.	<----------	
	The BUYPASS network node forwards the TOR and the new financial transaction request to BUYPASS. ---------->	BUYPASS processes the

Customer Host/Controller/POS System	Via Communications Protocol Connection	BUYPASS
		TOR. An approved or declined response to the TOR signifies completion of a TOR. BUYPASS processes the new financial transaction request. This activity is also subject to a TOR if BUYPASS does not respond within the customer's host/controller/POS system 30-second response interval; or there is a response within the response interval, but the BUYPASS network node is not able to forward the response to the customer's host/controller/POS system.

2.3.4 Summary of TOR Processing Rules

Refer to the below list for a summary of TOR processing rules that are described in detail in the above sections:

- A TOR should never be attempted more than 3 times.
- A TOR is considered complete with any response from the host, except a timeout or no response from the host. An approval and decline response are both considered to be a successful termination of the TOR and the TOR should be removed from the transaction queue.
- A TOR can only be sent for financial transactions. To remove the hold on an authorization only transaction, the device must send a cancellation transaction instead of a TOR.
- A TOR is not supported for a debit completion.
- Multiple transactions, and multiple TORs, can be in-flight at any given time. The terminal or device is responsible for tracking the status and maintaining the queue of the outstanding transactions.
- When attempting to submit a TOR for a completion transaction, the device must finalize the TOR processing prior to resubmitting the completion. If the completion is resent before the TOR processing has been completed, the device risks voiding the resubmitted completion.
- A TOR must never be sent until the device's 30-second timeout has expired.
- In the event a device has multiple queued up TOR transactions, the device must throttle the submission of these transactions, submitting no more than one every 30 seconds.
- A TOR should only be sent during the same business day as the original transaction.

3. Specifications Requirements

This chapter provides information about the hardware, software, and telecommunications requirements necessary to support these specifications.

A customer's host hardware and software must be capable of interfacing with BUYPASS hardware and software using a communications protocol that BUYPASS supports.

3.1 Hardware Requirements

Industry standard EFT processing hardware is supported by these specifications.

Note: If a BUYPASS customer device supports any payment type that requires secure entry of a cardholder's PIN, that device must be DUKPT-certified.

3.2 Software Requirements

Industry standard EFT processing software is supported by these specifications.

If a vendor is a third-party processor, BUYPASS assigns a TPP (third party processor) ID or VAR (value added reseller) ID to the vendor. Once this ID has been assigned, the vendor must send it to BUYPASS in the Variable Information (Element No. 113) of the Authorization Request. Also, please see valid entries in Appendix I. Variable Information Data Layouts, Table ID 10 (Visa® Trusted Agent Program Data) for additional information.

A terminal ID must be assigned to each Point-of Sale (POS) entry point. The terminal ID must be unique and the vendor must send it to BUYPASS in the Variable Information (Element No. 113) of the Authorization Request. Please see Appendix I. Variable Information Data Layouts, Table ID 18 (Card Acceptor Terminal Identifier Information) for additional information.

Note: Due to higher instances of fraud, it is highly recommended that the off-line/forced authorization key on the software/terminals be coded with a flag that will disable the key. The flag should be defaulted to off, so that the software/terminal does not have off-line/forced authorization key capability. The flag should be coded so that key can be enabled/disabled through setup within the software terminal.

It is also recommended that a standalone password be added to the off-line/forced authorization key, so that when the key is enabled, the owner/manager has the capability to allow only certain personnel to perform the key function. The software/terminal should have the capability of changing the password independent of downloads and should not be linked to the default software/terminal passwords.

3.3 Telecommunications Requirements

This interface supports several telecommunications infrastructures and protocols.

In addition to this section's information, the latest versions of the following documents may be useful as indicated:

- For additional dial support information, please refer to *BUYPASS® Platform Multithreaded Asynchronous Protocol Specifications*
- For additional VSAT, frame relay, or other nondial access methods, please refer to Appendix A. TCP/IP Message Header, in this document
- For additional Internet support information, please refer to the Secure Transport section on the First Data® Web site (https://www.firstdata.com/en_us/customer-center/merchants/support/secure-transport-api)

3.3.1 Infrastructures

The following leased line infrastructures are supported by these specifications:

- Standard frame relay connection to multiple First Data data centers with dial backup
- Any facility that emulates private leased lines, such as packet networks or satellite networks

3.3.2 Dial Backup and Diverse Facilities

BUYPASS supports and encourages merchant locations to implement dial backup facilities. Any location that generates more than 1,500 transactions per month must subscribe to dial backup. Any location that generates more than 500,000 transactions per month must have at least two diverse connections that support automatic rerouting of traffic.

The First Data standard requires that diverse connections use different carriers and are directed to two separate First Data data centers. Transactions should be load-balanced across all connections. BUYPASS directs responses over the same circuit as the initiating customer request.

4. Supported Industries

This chapter lists the industries supported by these specifications. It discusses the following industry types:

- General retail
- Hospitality
- Supermarket
- Mail order
- Utilities
- Petroleum

4.1 General Retail

The following general retail business types are supported by these specifications:

- Discount chains
- Pharmacies
- Hardware stores
- Department stores

4.2 Hospitality

The following hospitality business types are supported by these specifications:

- Restaurants
- Casinos
- Hotels
- Motels
- Arenas
- Resorts

4.3 Supermarket

The following supermarket business types are supported by these specifications:

- Supermarkets
- Hypermarkets
- Grocery wholesalers

4.4 Mail Order

Direct marketing with address verification service (AVS) is supported by these specifications.

4.5 Utilities

Utility payment is supported by these specifications.

4.6 Petroleum

The following petroleum payment processing types are supported by these specifications:

- Convenience store
- Service station
- Unattended fuel sites

5. Supported Payment Network Types

The following payment network types are supported by these specifications:

- Automated clearing house (ACH) preauthorized debit cards
- Automated teller machine (ATM) debit cards
  - ◊ PIN debit
  - ◊ PINless debit
- Check guarantee services
- Check verification services provided by BUYPASS
- Check verification services provided by others
- Credit cards
  - ◊ Nonproprietary
  - ◊ Proprietary
- Electronic benefits transfer (EBT) programs
  - ◊ Food stamp benefits
  - ◊ Cash benefits
  - ◊ eWIC
- Proprietary cards
- Petroleum proprietary cards
- Fleet cards
- Stored value cards
- Loyalty cards

Note: Check with your BUYPASS representative for information about specific programs.

6. PIN Encryption

Key management involves the creation, distribution, maintenance, storage, and destruction of cryptographic keys. This process is required between a customer's store device and the customer's host/device system, and between the customer's host/device system and BUYPASS.

These specifications support the following PIN encryption methods:

- DES
- 3DES DUKPT

This chapter discusses the following PIN encryption topics:

- Requirements
- 3DES DUKPT PIN Encryption Processing
- DUKPT Derivation Key (DKey)
- Key Sequence Number (KSN)
- Test Key
- PIN Block Construction
- PIN Block Message Placement
- Declines

6.1 Requirements

DUKPT PIN encryption (DUKPT) is the required PIN encryption method for BUYPASS processing.

In addition, for effective key management, the following criteria must be met:

- All personnel are excluded from handling keys in the clear except when performing an initial installation or when adding new or modifying existing customer store devices.
- Personal identification numbers (PIN's) and cryptographic keys never appear in the clear.
- PIN's and cryptographic keys are never retrievable by application programs.
- For DUKPT PIN encryption, the customer devices and BUYPASS share the same derivation key (DKey).

6.2 Triple DES (3DES) DUKPT PIN Encryption Processing

This section contains the following information about Triple DES (3DES) DUKPT PIN encryption processing:

- Customer PIN encryption
- BUYPASS PIN encryption

3DES DUKPT PIN encryption processing does not require the periodic exchange of PIN encryption keys as does master session PIN encryption processing. This is accomplished by storing a 3DES DUKPT key at BUYPASS and using the 3DES DUKPT key to encrypt a serial number at each PIN pad.

One 3DES DUKPT key exists between BUYPASS and a customer. This means that potentially several hundred to several thousand PIN pads contain the same 3DES DUKPT key.

A unique serial number is assigned to each PIN pad.

The 3DES DUKPT key encrypts a PIN pad's serial number; and the serial number and its encrypted form (seed key) are inserted in the assigned PIN pad.

The seed key and its corresponding serial number can generate approximately one million unique PIN encryption keys.

6.2.1 Customer PIN Encryption

Hardware security modules must be used to encrypt all PIN's, following ANSI® PIN management standards.

Transmission of a PIN among interface participants must use DES-encrypted ANSI® PIN blocks.

The following data are required to encrypt a PIN block for transmission to BUYPASS:

- Card's PAN
- Cardholder's PIN
- Generated KPE

6.2.1.1 Point-of-Entry PIN Encryption

The following events occur to accomplish PIN encryption of a PIN block for transmission to the customer's host system from the customer's card reading device:

1. The PAN is read from the magnetic stripe card placed in the point-of-entry device by the cardholder. Other information may be formatted on the card and can be read by the system.
2. The point-of-entry device prompts the cardholder to enter his/her 4- to 12-digit PIN using a secure PIN pad.
3. The PIN is encrypted with the generated KPE.
4. The encrypted PIN block and the KSN are sent to the customer's host system for forwarding to BUYPASS.

6.2.2 BUYPASS PIN Encryption

The following events occur to accomplish PIN encryption of the encrypted PIN block received from the customer's host system by BUYPASS:

1. The customer's host system receives the encrypted PIN block and KSN from the point-of-entry device.
2. BUYPASS retrieves from its database the 3DES DUKPT key of the customer. This key is encrypted under the BUYPASS master file key contained in the BUYPASS HSM.
3. The encrypted PIN block, the PAN, the KSN, and the 3DES DUKPT key are passed to the BUYPASS HSM and translated into a new translated encrypted PIN block using the BUYPASS master key.
4. The encrypted PIN block is translated in the BUYPASS HSM to a new translated PIN block using the network's interchange working key.
5. The encrypted PIN is transmitted to the network host system.
6. The pathway terminal configuration for the PIN pad value must be defined for 3DES DUKPT.
7. At the Bank FIID level the key representation for 3DES DUKPT must be defined.

6.3 DUKPT Derivation Key (DKey)

DUKPT uses a derivation key (DKey) and a key sequence number (KSN) to generate a unique PIN encryption key for every transaction that takes place at a PIN pad.

The DKey is comprised of two 16-hexadecimal (HEX) character halves. These halves are loaded onto the processor's host and are associated with a given customer base. When concatenated, these halves form a key part. BUYPASS supports 2- and 3-part keys.

6.4 Key Sequence Number (KSN)

Another ingredient in the DUKPT algorithm is the key sequence number (KSN), which is a 20-byte HEX field. The format is discussed below.

During the initialization process, the DKey and an initial key sequence number (IKSN) are used to create the base PIN encryption key. Future PIN encryption keys for any PIN entry device are derived from the base PIN encryption key.

The IKSN is 20 bytes but the right-most (low order) 4 bytes—the Transaction Counter—must be initialized to 0's. Each PIN-based transaction causes the KSN to increment.

The key is changed at the PIN pad each time the KSN increments. Because the same DKey is stored on the host and used in the injection of the PIN pad, the host is able to derive the PIN block in each transaction.

The initial value of a 20-byte IKSN must follow the format below:

PPPPXXXXXXXXXXEZ0000

Where:

PPPP represents HEX characters (*F*'s);

XXXXXXXXXX represents any 10-byte HEX character string;

E represents any even HEX character (*0, 2, C, E*, etc.);

Z represents zero; and

0000 represents the Transaction Counter value.

Note: The Transaction Counter can be manually included as part of the IKSN or programmatically concatenated by vendor-supplied software such as MKIXOR.

6.5 Test Key

A test key is used for certifications. The clear key components are given to the customer/vendor for PIN pad injection along with the KSN. Overall left and right halves are given.

Contact your BUYPASS certification representative for additional test key information.

6.6 PIN Block Construction

The encrypted PIN block is constructed from the combination of the entered PIN and PAN fields.

The block must be at least 4 digits and no greater than 12 digits in length. A digit is a 4-bit binary number (value $0–15_{10}$ or $0–F_{16}$). Numeric values are stored as individual digits (value 0–9).

The PIN field is constructed as follows:

C 1	L 2	3	4	5	6	N F 7	N F 8	N F 9	N F 10	N F 11	N F 12	N F 13	N F 14	F 15	R 16

Where: C = control field—always zero

L = length of entered PIN as a 4-bit binary number (4–12)

N = entered PIN, left aligned, including any leading zeros

F = (14–L) pad digits of *1111* (F_{16})

Calculate the PAN field from the Track 2 data on a swiped card as follows:

If the Track 2 data is *;1234567890123456789=0312*, then drop the check digit (9) prior to the field separator (=), and use the remaining right-most 12 digits (789012345678).

C 0 1	C 0 2	C 0 3	C 0 4	P 7 5	P 8 6	P 9 7	P 0 8	P 1 9	P 2 10	P 3 11	P 4 12	P 5 13	P 6 14	P 7 15	P 8 16

Where:

C = control field—always zero

P = right-most 12 digits of the Track 2 data, excluding the check digit. (The last digit of the PAN is considered a check digit whether it is or not.)

6.7 PIN Block Message Placement

The 16-byte PIN block in field no. 8 of the Standard Message Segment is followed by the KSN.

The authorization message sent to BUYPASS contains a 16-byte encrypted PIN block and a 20-byte KSN. The host derives the PIN encryption based on the DKey stored at the host for the device associated with the transaction.

6.8 Declines

If a device receives either a *4R* or *4<* Decline Code while sending encrypted transactions to the host, one of the following situations may be the cause:

- The encrypted PIN data is either not included in the format or not sent by the PIN encryption device.
- The KSN is either missing or not sent by the PIN encryption device.
- The KSN is not formatted correctly.
- The Terminal Profile PIN Encryption type is not set to *UKP*.
- The PIN pad is encrypted incorrectly.
- The KSN counter is not incrementing for each PIN transaction.
- The KSN is not the correct length.
- The BANK FIID is incorrect.

7. Network Management Processing

BUYPASS provides a facility to verify the communications link between a BUYPASS customer device and BUYPASS.

This chapter contains information about the following communications test messages:

- Communications Test Request
- Communications Test Response

7.1 Communications Test Request

The Communications Test Request is used to verify a communications link between a customer device and BUYPASS.

It has the following format:

STX	Message Format Version Identifier	Number of Segments	FS	COMMTEST	ETX

where *COMMTEST* is the Network Management Message.

Note: Refer to section 11.6.1, "Request," for additional format information.

7.2 Communications Test Response

The Communications Test Response is used to verify a communications link between a customer device and BUYPASS. It is the response to the Communications Test Request.

It has the following format:

STX	Response Code	Download Indicator	COMM_TEST_PASSED	ETX

where *COMM_TEST_PASSED* is the Terminal Display Communications Message.

Note: Refer to section 11.6.2, "Response," for additional format information.

8. Store-and-Forward Processing

This chapter discusses store-and-forward processing for the following types of business:

- Automated fuel pump (AFP)
- Restaurant and lodging

Note: Please contact your BUYPASS representative about additional specifications that discuss the above business types.

8.1 Automated Fuel Pump Processing Overview

The BUYPASS automated fuel pump (AFP) interface provides a controller-based AFP system with the ability to communicate with BUYPASS in order to request preauthorizations for and draft capture of credit card, debit card, fleet card, and oil company card fuel purchases.

The following information is determined at the island device portion of the AFP system:

- Card type
- Intended purchase amount or preauthorization amount
- Indication whether payment request is from a customer-activated device
- Any other required input from the purchaser or station attendant

8.1.1 Configuration Options

The following diagram illustrates a likely controller-based AFP configuration.

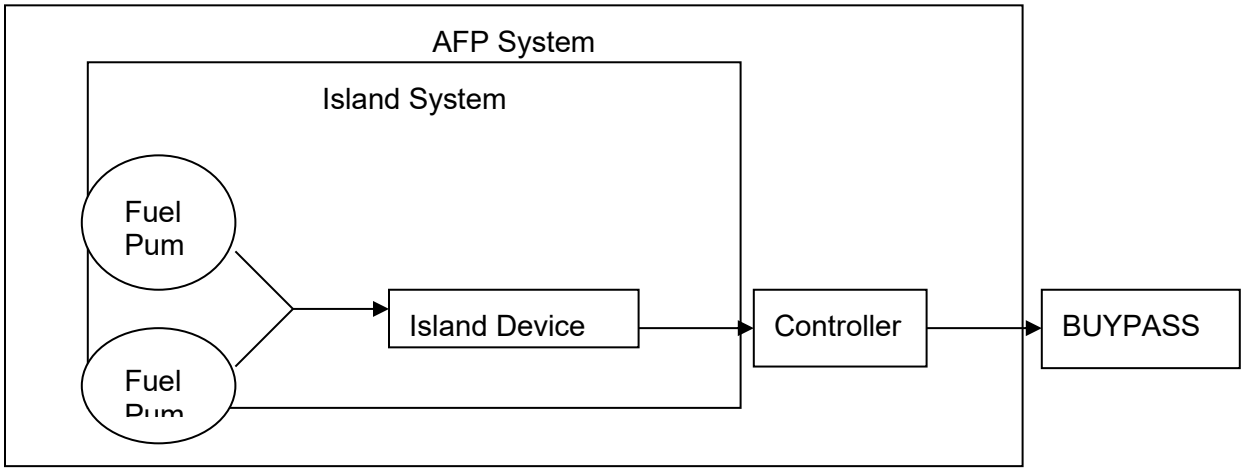


Fig. 8-1. AFP System.

The following conditions can exist in a controller-based AFP configuration:

- There may be multiple occurrences of island systems within an AFP system.
- BUYPASS assigns one merchant number to the overall AFP system.
- BUYPASS assigns one device number to the overall AFP system.
- The controller and/or island device should have the following capabilities:
  - ◊ DUKPT
  - ◊ Journal/Receipt printer for logging transactions and printing receipts
  - ◊ Nonvolatile media for electronic journal storage and transaction queuing
  - ◊ System console
  - ◊ Ability to request, receive, and store download parameters from BUYPASS
  - ◊ Ability to retrieve a download after the transaction queue is empty if a download is pending
  - ◊ Ability to configure preauthorization amounts
  - ◊ Ability to allow an automated end-of-day close based on the BUYPASS-designated merchant cut time

8.1.2 Configuration Requirements

The following off-line transaction parameters should be configurable:

- Dollar amount
- Number of transactions
- Total off-line dollar amount

If transactions go into off-line mode, the numbers for controller-generated approvals should start with *LA0001* and be incremented by *1* for each transaction.

If a transaction has been sent to BUYPASS for any reason other than a host decline, that transaction may be resubmitted up to three more times; or reformatted and resubmitted one more time. If, after these attempts, the transaction is not captured at BUYPASS, log that transaction in the Daily Exception Log. The Daily Exception Log attempts to retransmit its information one more time before the end-of-day settlement. That log should print at the end of the day, along with the Totals Report.

8.1.3 Automated Fuel Pump Controller System Activities

The following table describes, in sequential order, the activities performed by the customer's AFP controller system during transaction processing:

Notes: If the controller does not receive a Transaction Response from the host, it should generate a Local Approval Response.

If the DLL flag is set in a host response message, the controller connects to BUYPASS and requests a PIN encryption load once the queue is empty.

AFP Controller System Activities	
Step No.	**Description**
1.	Before pumping fuel, the customer completes the following tasks: • Selects card type • Swipes/inserts card • Enters PIN (required for debit and some fleet cards) • Enters additional fields (determined by card type) • Select product (optional)
2.	The island device/controller must verify the following card data when applicable: • ISO number • Expiration data • Check digit calculation • Card range
3.	The island device/controller should also validate the following card data: • Account number • Purchase amount • Card issuer
4.	The controller formats the data into a Preauthorization Request. The format must indicate that the transaction was initiated by a customer-activated device.
5.	The controller supplies the Preauthorization amount, the anticipated fuel type, and the price per gallon. This is overridden when the actual fuel is dispensed but is required for fuel limits and fuel-only card validation.
6.	The controller connects to BUYPASS if not already connected.
7.	The controller forwards the Authorization Only Request.
8.	The controller parses the Authorization Only Response, storing the transaction data and approval code if the transaction is approved.
9.	The controller signals the island device of the approval or decline.
10.	If an approval is received, the island device signals the AFP system to dispense the fuel up to the preapproved Preauthorization amount.
11.	The controller services any Authorization Requests subsequently initiated if the controller is still connected to BUYPASS and the line disconnect timer has not expired.
12.	If the controller is still connected to BUYPASS, the controller forwards any queued transactions (preauthorization completions), parses the responses, and updates associated electronic journal data.

AFP Controller System Activities	
Step No.	**Description**
13.	If the controller is still connected to BUYPASS, no transactions are pending, and the load request flag is set to *yes*, request the load before hanging up the line.
14.	After the fuel is pumped, the island device stores the following information: • Actual purchase amount • Fuel price per gallon • Fuel type • Gallons dispensed to the controller A receipt is printed reflecting the actual purchase amount. For debit and stored value cards, the transaction can only be stored for a maximum of fifteen minutes, and then the controller must connect to BUYPASS for processing the transaction. For credit cards, the transaction can only be stored for a maximum of 55 minutes, and then the controller must connect to BUYPASS for processing the transaction. For all other card types, this transaction should be stored until the next Authorization or other request that initiates a connection to BUYPASS. A stored preauthorized completion transaction should never initiate a connection to BUYPASS.
15.	Prior to requesting end-of-day totals, all queued transactions, including the Daily Exception Log transactions, should be sent to BUYPASS. The device should connect to BUYPASS 15 minutes prior to the cut time if the location has not processed its totals.

8.2 Restaurant and Lodging Processing Overview

The BUYPASS restaurant and lodging interfaces provide controller-based systems with the ability to communicate with BUYPASS in order to request authorizations for restaurant and lodging purchases and draft capture of restaurant and lodging purchases.

Restaurant Credit card sales are terminal-capture and initiated as POS Authorization Only transactions. Transactions are stored in the terminal batch as pending transactions until they have (or have not) been adjusted for tips. Tip adjustments are completed offline at the terminal. When the batch is closed, the pending sales are marked to send and uploaded as POS Purchase/Capture transactions, including the approval number from the original transaction request. A Totals Request (999999 – Shift Read and Reset) follows the successful completion of a batch upload. If a communication error occurs during the batch upload, the terminal reconnects to the host and follows normal TOR logic for any sent transactions that have not received an ACK from the host. The terminal then sends only the remaining transactions in the batch and does not resend transactions that have already received an ACK. An exception report prints after a successful batch upload. It includes any transactions declined during the upload process.

Restaurant Debit, EBT, and gift sales are host-capture and initiated as POS Purchase/Capture transactions. Transactions prompt for tips at the time of the sale. Tip adjustment is not allowed with these card types.

At the completion of the business day, the terminal sends a Totals Request (222222 – Read and Cut Settlement) after sending the last batch.

9. End-of-Day Processing

This chapter contains information about the following end-of-day processing activities:

- Electronic journal
- Local totals
- BUYPASS totals

9.1 Electronic Journal

Note: An electronic journal feature is not provided by BUYPASS; however, it can be designed, created, and implemented by a BUYPASS customer. The information in this section provides an example of how an electronic journal can be designed and used.

In this example an electronic journal needs to contain all approved transactions, excluding authorization only transactions. Also, local totals need to be an accumulation of the transactions in the electronic journal, including authorization only transactions. The electronic journal and local totals features are separate functions for this example, and the local totals are calculated from the electronic journal.

In this example an electronic journal provides a listing of each approved transaction for local review. When the host response to an approved transaction is received, the customer device—in this example a device—immediately records the transaction in the electronic journal. The electronic journal is structured as a circular queue. It can be accessed by a designated key for each device type.

In this example, the following data for each transaction should be stored in an electronic journal:

- Account Number
- Card Type
- Manual slide indicator
- Transaction day (1–31)
- Transaction time (HH:MM)
- Total transaction amount
- Transaction type (approval, reversal, or merchandise return)
- Approval Number
- Sequence Number

9.1.1 Displaying the Electronic Journal

In this example, the electronic journal can be searched for approved purchase transactions by pressing a designated key for the particular device type. If no entries are present in the electronic journal, a message displays indicating there are no entries, and the device returns to the idle prompt.

In this example, if entries are present in the electronic journal, a message displays indicating there are entries. A designated key enables the user to proceed to subsequent prompts for displaying the electronic journal. Some sample prompts, together with brief descriptions, are listed below:

- ENTER ACCOUNT NUMBER—Enter a complete account number to display the account having this number in the electronic journal. Enter a partial number to display a list of accounts beginning with this number in the electronic journal. Press a designated key without entering an account number to display all account numbers in the electronic journal.
- ENTER DOLLAR AMOUNT—Enter a complete dollar amount to display this amount in the electronic journal. Enter a partial dollar amount to display a list of amounts beginning with this number. Press a designated key without entering an amount to display all amounts in the electronic journal.

In this example, if the electronic journal is searched and no matches are found, a message displays indicating there are no matches, and the device returns to the idle prompt.

In this example, if the electronic journal is searched and a match is found, the transaction(s) display(s) on the device screen. Press a designated key to advance the display one transaction at a time and scroll through all the electronic journal entries. The most recent entry displays first.

Fig. 9-1 below illustrates the device display of a single line of the electronic journal.

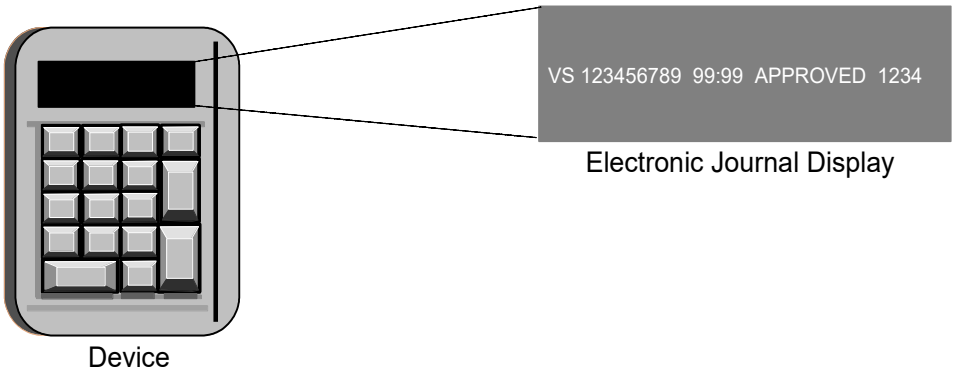


Fig. 9-1. End-of-day balancing—Displaying the electronic journal.

The following table lists the data fields contained in the electronic journal for this example, together with their respective field data descriptions:

Field No.	Field Format	Field Description
1	Sixteen digits	a minimum of the first 16 positions of the account number
2	xxhh:mmzzz	*xx* = The day the transaction was performed. *hh:mm* = The time the transaction was performed. *zzz* = The sequence number of the transaction, with *1* representing the oldest transaction.
3	TOTAL	*$999999.99*
4	APPROVED XXXXXX REVERSAL XXXXXX MDSE RETN XXXXXX	*XXXXXX* = The host approval code.

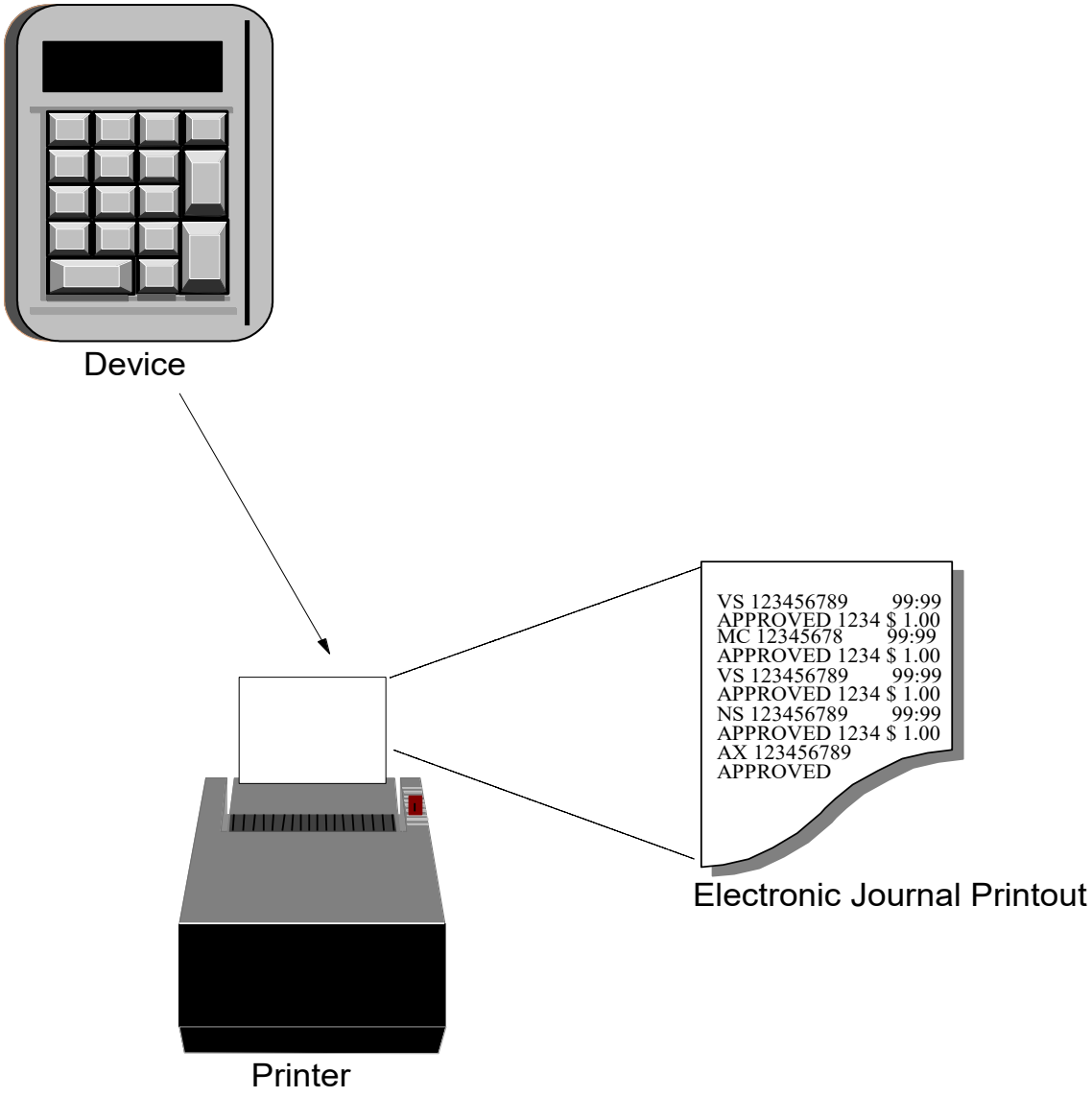


Fig. 9-2. End-of-day balancing—Printing the electronic journal.

9.1.2 Printing the Electronic Journal

Fig. 9-2, “End-of-day balancing—Printing the electronic journal,” illustrates the printing of the electronic journal in this example.

In this example, the approved purchase transactions are printed from the electronic journal by pressing a designated key for the particular device type. If no entries are present in the electronic journal, a message displays indicating there are no entries, and the device returns to the idle prompt.

In this example, the electronic journal is searched for a specific account number or dollar amount and, if a match is found, the transaction detail for that specific match is printed. If the entire electronic journal is printed, the listing begins with the most recent transaction.

In this example, the electronic journal is printed in the following format:

Line No.	Line Data and Description
1	a code indicating the account number was keyed; or a space indicating the account number was swiped
	a code identifying the card type (See section 10.1.7.3, “Card Type ID.”)
	a maximum number of characters identifying the account number
	a transaction time (HH:MM)
2	*APPROVED* message identifying an approved transaction
	REVERSAL message identifying a reversal transaction
	MDSE RTN message identifying a merchandise return transaction
	an Approval Number identifying the approval
	a date identifying the date the transaction was performed

Note: Some device applications print the date the transaction was performed; however, this feature is not supported by all device applications.

9.2 Local Totals

In this example, local totals are maintained by the device. They are calculated from the electronic journal entries and are grouped in the same way as host totals. The Card Label is displayed at the device, along with the number of transactions and total dollar amount per category.

The Card Label categories are listed below—in bucket number order—for this example:

Card Label Categories	
Card Label	**Categories**
CC	MasterCard®, Visa®
TE	American Express
DS	Discover® Network, Diners Club®, JCB®
AO	Authorization only
DB	Debit
FL	Fleet
CS	Cash
PR	Proprietary
CK	Check
EF	Electronic benefits transfer—Food stamps
EC	Electronic benefits transfer—Cash benefit
SV1	Stored value card—Activate/Deactivate a stored value card
SV2	Stored value card (POS/CAT)—Purchase/Capture; Merchandise Return
SV3	Stored value card—Replace a stored value card
SV4	Stored value card—Recharge a stored value card
ECA	ECA/ TeleCheck® service
EWIC	EBT version of WIC
EK	EBT child care
TOTAL	Grand total
HD	Host Discounts— Total amount of host discounts applied to the financial transactions in shift or business day totals.

The grand total is calculated by totaling the transactions from the electronic journal entries.

Note: The grand total includes all transactions except the non-financial transactions grouped in the following categories:

- AO—Authorization only
  Note: Cancellations of non-gift cards must subtract from the AO bucket.
- SV1—Stored value—Activate/deactivate
- SV3—Stored value—Replace card
- SV4—Stored value—Recharge
- HD—Host Discounts used for informational purposes only

Note: Transactions from these excluded categories must be stored separately from the electronic journal.

A Purchase/Capture increases the count and dollar amount of the corresponding category as well as the grand total.

A Reversal decreases the count and dollar amount of the corresponding category as well as the grand total.

A Merchandise Return increases the count and decreases the dollar amount for the corresponding category as well as the grand total.

An Account Verify increases the count of the corresponding category as well as the count of the grand total. It does not increase the dollar amount of the corresponding category or the grand total.

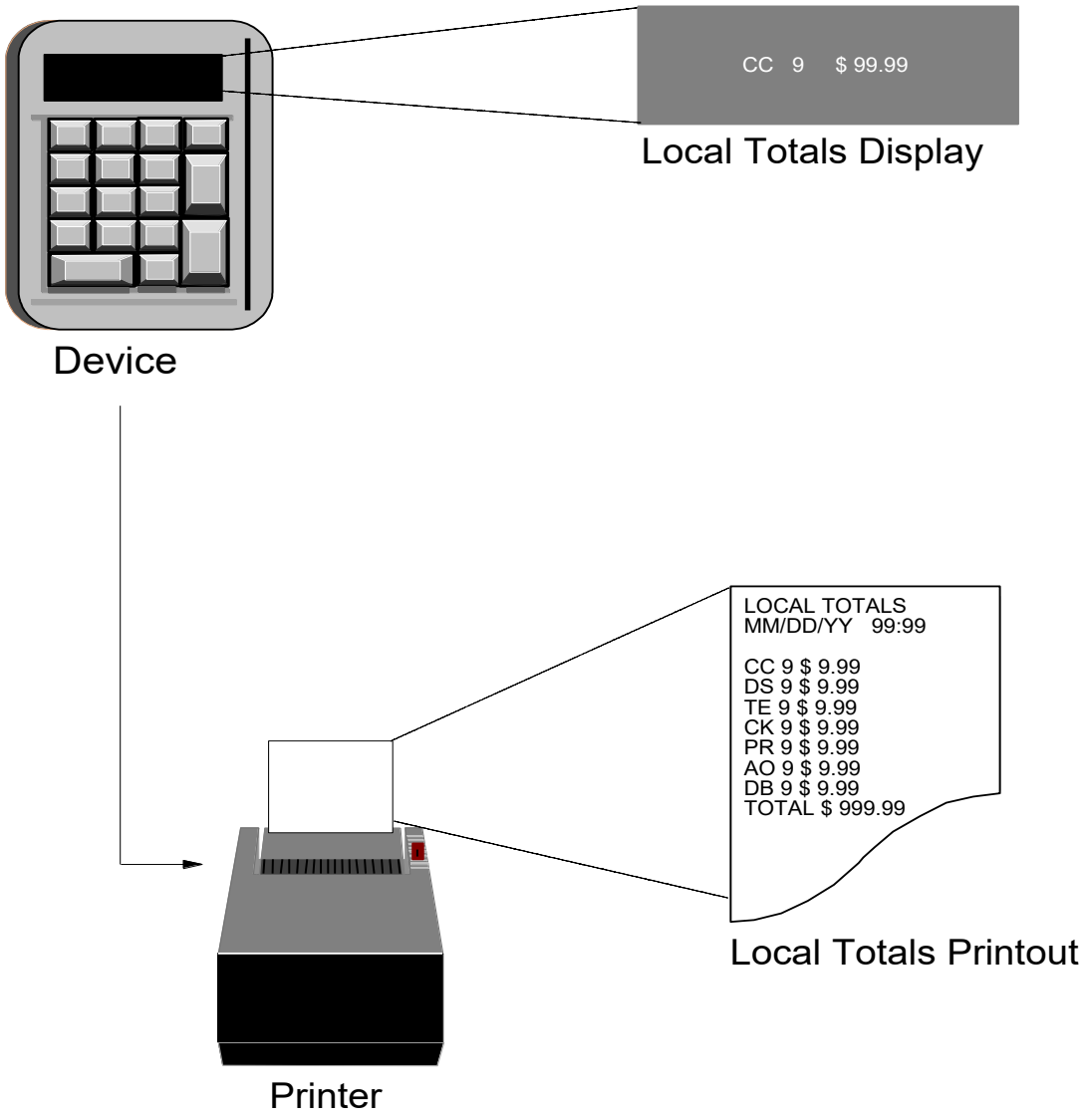


Fig. 9-3. End-of-day balancing—Displaying/printing local totals.

9.2.1 Displaying Local Totals

Fig. 9-3, “End-of-Day balancing—Displaying/printing local totals,” illustrates the process of printing local totals in this example.

In this example, the accumulated totals for each card label category can be displayed by pressing a designated key for the particular device type. If no entries are present in the electronic journal, a message indicating no entries are present displays, and the device returns to the idle prompt. If entries are present, the device displays a message indicating that processing is taking place to calculate the totals from the electronic journal.

In this example, each category displays as follows:

Message	Description
XX XXX $9999.99	where: *XX* represents the card label; *XXX* represents the total number of transactions per group; and *$9999.99* represents the total dollar amount per group.

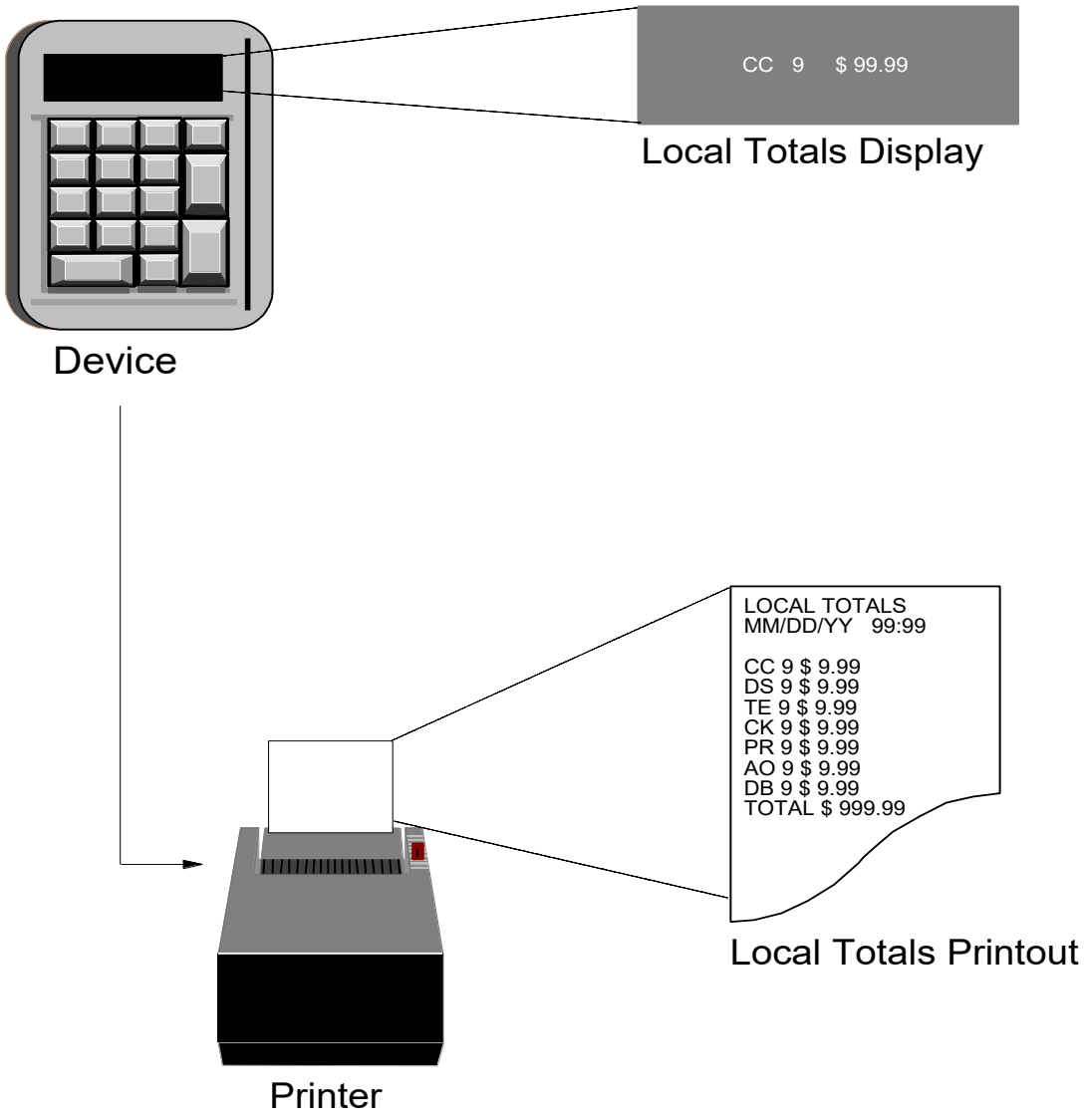


Fig. 9-4. End-of-day balancing—Displaying/printing local totals.

9.2.2 Printing Local Totals

Fig. 9-4, “End-of-Day balancing—Displaying/printing local totals,” illustrates the process of printing local totals in this example.

In this example, the accumulated totals for each card label category can be printed by pressing a designated key for the particular device type. If no entries are present in the electronic journal, a message indicating no entries are present displays, and the device returns to the idle prompt. If entries are present, the device prints a local totals report.

In this example, each category displays as follows:

Message	Description
XX XXX $9999.99	where: *XX* represents the card label; *XXX* represents the total number of transactions per group; and *$9999.99* represents the total dollar amount per group.

In this example, the local totals from the device are compared to the host totals obtained from BUYPASS. Local totals correspond to the requested shift or settlement totals request. The device prints the count and dollar totals for each card label category, including the grand total. The printout displays the local totals summary and the host totals summary indicating the date, time, and type of totals requested from the host. The result of the totals comparison is indicated at the bottom of the printout as *OUT OF BALANCE* when the grand totals are not equal.

9.3 Host Totals

This section discusses the activities involved in requesting host totals from BUYPASS.

The device must be able to support all requests for merchant shift totals and settlement totals.

When requesting host totals, the device displays a prompt for entering a password and date. The password must match the data in the BUYPASS merchant profile that is obtained via a Table Load. The host totals are accessed by entering a 6-digit Totals Date code (Please see chapter 13, "Data Element Descriptions," for a list of valid codes for Element No. 105, Totals Date.)

BUYPASS maintains two sets of daily totals for each merchant:

- Shift totals
- Settlement totals

9.3.1 Merchant Shift Totals

This section discusses the activities involved in requesting merchant shift totals from BUYPASS.

Shift totals accumulate for each merchant from work shift to work shift. These totals are scratch totals and do not affect settlement. They can accumulate beyond a merchant's cut time. These totals can be requested as *read only* or *read and reset*. Descriptions of both of these total requests follow, together with their respective 6-digit Totals Date codes.

9.3.1.1 Read Only Total Request

To request shift totals as *read only*, enter the following 6-digit Totals Date code in the date field displayed on the device: 111111.

Entering this code enables a merchant to view shift totals without resetting the BUYPASS shift total amounts to zero. BUYPASS returns the host shift totals for transactions done since the last shift cut [Read and Reset Total Request (Totals Date code: 999999)].

9.3.1.2 Read and Reset Total Request

To request shift totals as *read and reset*, enter the following 6-digit Totals Date code in the date field displayed on the device: 999999.

Entering this code enables a merchant to view shift totals and reset the BUYPASS shift total amounts to zero. BUYPASS returns the host shift totals for transactions done since the last shift cut [Read and Reset Total Request (Totals Date code: 999999)], ends the current shift, and resets the merchant shift total amounts to zero.

Use these totals to balance against the local totals stored in the merchant's device. If the merchant fails to request shift totals at the end of a shift, the totals continue to accumulate at BUYPASS until a Read and Reset Total Request (Totals Date code: 999999) is made. The host maintains the shift totals and they are available until the first transaction in a new shift has been completed. This allows retrieval of the shift totals from the most recently completed shift.

9.3.2 Settlement Totals

Settlement totals accumulate over a 24-hour period and are totaled once a day. The settlement time, also known as the cut time, is determined by the merchant.

The merchant chooses a default settlement time and has the option of cutting totals before that time. A merchant exercising this option is referred to as a variable-cut merchant.

Settlement totals can be requested as *read only* or *read and cut*. Descriptions of both of these total requests follow, together with their respective 6-digit Totals Date codes.

Business day totals can be requested as *read only* for the current business day, the previous business day, or the business day before the previous business day. Descriptions of these total requests follow, together with their respective 6-digit Totals Date codes.

9.3.2.1 Read Only Settlement Totals Request

To request settlement totals for a specific business day as *read only*, enter the desired 6-character settlement date in the date field displayed on the device in the following format: MMDDYY.

Entering this specific date enables a merchant to view settlement totals for a specific date without resetting the BUYPASS settlement total amounts to zero. BUYPASS returns the host settlement totals for transactions that settle on the specified date.

Using the Totals Date code of *MMDDYY* returns totals for transactions that settled on the specified date.

Note: *Settlement Date* refers to the date when transactions actually settle. Unless the merchant chooses to end the business day at 2:00 p.m., settlement time does not coincide with the BUYPASS end-of-business day. In most cases, to retrieve totals for a current business day, request totals using a settlement date equivalent to the business day plus one day.

Based on the merchant-selected end-of-business daytime, BUYPASS groups all transactions done in that 24-hour period and totals them; however, BUYPASS does not run a settlement until 2:00 p.m. Therefore, if the merchant's end-of-day is 6:00 p.m., transactions for that day settle at 2:00 p.m. on the following day.

9.3.2.2 Read and Cut Settlement Totals Request

This option can be used only by variable-cut merchants.

To request totals for the current settlement date as *read and cut*, enter the following 6-digit Totals Date code in the date field displayed on the device: 222222.

Entering this code enables a variable-cut merchant to close its business day before the predetermined cut time as designated on the BUYPASS merchant profile. If the variable-cut

merchant does not request *read and cut* totals, BUYPASS automatically closes the merchant's business day at the designated cut time. Transactions performed after close of business are included in the next business day's totals.

Notes: In a controller-based automated fuel pump environment, the controller should be designed to connect to BUYPASS 30 minutes before the designated cut time and forward all queued trailer transactions to the host. This ensures that all authorization only/preauthorization transaction pairs are captured on the same business day. After all trailer transactions are transmitted to BUYPASS, the controller completes a *read and cut* total request (Totals Date code: 222222); however, if a *read and cut* total request is not made, BUYPASS automatically closes the merchant's business day at the designated cut time.

If a device supports table loads from BUYPASS, receiving the Card Type (Element No. 14) value of "164" enables Auto Close in a device.

9.3.2.3 Current Business Day Read Only Total Request

To request settlement totals for the current business day as *read only*, enter the following 6-digit Totals Date code in the date field displayed on the device: 333333.

Entering this code enables a merchant to view up-to-date current business day totals.

Note: Using this code produces the same result as following the instructions for using the Read Only Settlement Totals Request described in section 9.3.2.1, "Read Only Settlement Totals Request."

9.3.2.4 Previous Business Day Read Only Total Request

To request settlement totals for the previous business day as *read only*, enter the following 6-digit Totals Date code in the date field displayed on the device: 444444.

Entering this code enables a merchant to view the previous business day's totals.

Note: It is possible for calendar days to elapse between business days. For example, if a Previous Business Day Read Only Total Request is made on a Monday, and there were no transactions done the preceding Saturday and Sunday, totals are returned for the previous Friday.

9.3.2.5 Business Day Before Previous Business Day Cut Total Request

To request settlement totals for the business day before the previous business day (the third most recent business day), enter the following 6-digit Totals Date code in the date field displayed on the device: 555555.

Entering this code enables a merchant to view the totals for the business day before the previous business day.

Note: It is possible for calendar days to elapse between business days. For example, if a Business Day Before Previous Business Day Cut Total Request is made on a Monday, and there were no transactions done the preceding Saturday and Sunday, totals are returned for the previous Thursday.

10. Processing Requirements

This chapter contains information about the following processing types:

- Credit card processing requirements
- Debit card processing requirements
- Electronic benefits transfer card processing requirements
- Fleet card processing requirements
- Stored value card processing requirements
- Check processing requirements
- Loyalty card processing requirements
- Software update processing requirements
- Electronic mail processing requirements
- TransArmor processing requirements
- Payment Tokenization processing requirements
- EMV processing requirements

10.1 Credit Card Processing Requirements

This section discusses the following credit card processing requirements:

- Card requirements
- Supported transaction types
- Partial Approval transaction
- Authorization Only Reversal transaction
- Address Verification Service
- Bill payment/Recurring and Installment payment transactions
- Radio frequency identification (RFID) service
- Healthcare auto-substantiation service
- Receipt requirements
- Regulatory requirements

10.1.1 Card Requirements

This section discusses the following card requirements for credit card processing:

- Expiration date
- Track support
- Manual entry

10.1.1.1 Expiration Date

The device performs expiration date validation in a credit card transaction.

Note: Dates up to and including December, 2049 (Format: *1249*) are assumed to be in the twenty-first century.

10.1.1.2 Track Support

The default track that is sent to the host should be Track 2 if the device supports Track 2. If the device cannot read Track 2, then the device should send unaltered Track 1 before attempting manual entry. If a Pre-Auth transaction is approved offline, the full track data must be sent. Also if a Sale transaction is approved offline, the full track data must be sent.

Note: Track 1 data is mandatory for all (both MSR and MSD) Amex AFD transactions and preferred for all other Amex transactions.

If the AFD transaction being submitted contains encrypted data in the TransArmor Encryption Block, Track 1 data must be present inside the Encryption Block with Encryption Target as Track 1.
For all other card types, the default track should be Track 2. If the device cannot read Track 2, then the device should send unaltered Track 1 on the initial transaction, before attempting manual entry.
All follow-up transactions should use truncated Track 2 or truncated Track 2 facsimile.

10.1.1.3 Manual Entry

Manual entry is allowed on a credit card transaction. If the device is unable to read the card's magnetic strip, it should prompt the user to swipe the card again—up to three times. If unable to read the magnetic strip after three attempts, the device should prompt for manual entry of the Account Number and Expiration Date (Format: MMYY).

Note: If manual entry is used, this fact is indicated on the receipt and journal by an asterisk (*) before the Account Number.

10.1.2 Supported Transaction Types

The following point-of-sale transactions are supported for credit card processing:

- Purchase/Capture
- Purchase Reversal
- Authorization Only
- Merchandise Return
- Time-out Reversal
- Mail/Phone Purchase
- Mail/Phone Authorization Only
- Mail/Phone Reversal
- Void of Merchandise Return
- Balance Inquiry
- Partial Approval
- Authorization Only Reversal

Note: The following customer-activated transactions are supported for credit card processing:

- Purchase/Capture
- Authorization Only
- Time-out Reversal
- Partial Approval
- Authorization Only Reversal

10.1.2.1 Partial Approval Transaction

A Partial Approval transaction enables a merchant to receive approval for the amount available on the presented credit card when the account balance is more or less than the amount of the Transaction Request.

Credit networks supporting the Partial Approval transaction return an approval containing the following information:

Note: This information is returned when a Partial Approval Indicator value of 1 is sent in Partial Approval Indicator (Element No. 121).

- A Response Code (Element No. 83) value of *F* (Indicates “approval, partial amount approved.”)
- Approved Amount (Element No. 6) contains the approved partial amount, and the POS device must prompt for payment of the remaining balance due on the sale.
- The card balance—if returned by the issuer—is in the Additional Information Data Segment (Segment No. 112).

Partial Approval transactions are supported for:

- POS Purchase
- POS Authorization Only
- CAT Authorization Only

The amount in a Reversal (Void) request must be the amount approved in the original transaction response when a transaction has been partially approved. BUYPASS supports reversal of a partially approved original authorization and NOT a partial reversal of an original transaction.

The amount in a TOR request must be the amount requested in the original transaction response when a transaction has been partially approved.

The Sequence Number must be the Sequence Number contained in the original transaction request.

10.1.2.2 Authorization Only Reversal Transaction

In support of the following requirements, BUYPASS supports the Authorization Only Reversal transaction:

- All approved and partially approved authorization transactions must be either cleared or reversed.
- Acquirers must ensure that their merchants process authorization reversals for authorizations submitted in error and/or cardholder cancelled card-present transactions within 24 hours. Card-absent merchants are allowed up to 72 hours to process the required partial or full authorization reversals.
- Issuers must immediately process matched reversal transactions, subject to the issuer's risk or fraud control policies, and release any applicable holds on cardholder available funds.
- It is recommended for merchants who use velocity processing to send the SIC Information Code (Data Segment 111, Table ID 007) of "5542" for CAT (outside) transactions and "5541" for inside transactions.

This transaction enables a merchant to reverse the following credit card transactions within required timeframes:

- POS Authorization Only (Transaction Type code "3")
- CAT Authorization Only (Transaction Type code "5")

Successful reversal matching requires the following:

- Element No. 5 (Approval Number)—present in the original transaction—must be present and identical in the Authorization Only Reversal transaction.
- Element No.86 (Sequence Number)—present in the original transaction—must be present and identical in the Authorization Only Reversal transaction.
- Element No. 78 (Prompt Code) must contain a Transaction Type code "S" to indicate cancellation.

10.1.3 Address Verification Service

Address Verification Service (AVS) is a credit card authorization tool used for transactions performed primarily in a non-face-to-face environment, such as mail order, phone order, or customer-activated terminal (CAT).

The device sends a transaction request to BUYPASS containing the AVS information in the Variable Information Data Segment (Please refer to sections 11.1, "Financial Transactions," and 12.10, "Variable Information Data Segment," for additional information.).

The authorizer returns a response containing both the transaction authorization Response Code and the AVS information in the Additional Information Data Segment (Please refer to sections 11.1, "Financial Transactions," and 12.11, "Additional Information Data Segment," for additional information.).

BUYPASS processes all Purchase/Capture transactions when the Response Code is an approval, regardless of the AVS information. If the merchant decides not to accept the purchase due to the value of the AVS information, the merchant must perform a Purchase Reversal.

10.1.4 Bill Payment/Recurring and Installment Payment Transactions

Support for bill payment/recurring and installment payment transactions is an optional service. These transactions must be identified by a Variable Information Indicator located in the Variable Information section of the Variable Information Data Segment (Segment No. 111). (See section 12.10, "Variable Information Data Segment.")

10.1.5 Radio Frequency Identification Service

Radio frequency identification (RFID) service is an optional service. It is a contactless data transmission and capture technology. When applied to credit card processing, it refers to the capability of a point-of-sale device to capture and transmit full, unaltered Track 2 data from a card without requiring the card to have contact with the point-of-sale device.

An RFID card has a chip embedded in the card that transmits Track 2 data wirelessly when the card is in close proximity to an RFID receiver. Most RFID cards also have a magnetic stripe on the back of the card that can be used with a card reader if an RFID receiver is not present.

RFID processing requires that the appropriate values be sent in Point-of-Sale Entry Mode (Contained in Element No. 113, Variable Information.) on all transactions when an RFID receiver is connected to the point-of-sale device or controller. Even if the transaction is swiped, manually keyed, or uses a card type for which RFID is not applicable, Point-of-Sale Entry Mode must be populated correctly in the transaction request from the point-of-sale when an RFID receiver is present.

10.1.6 Healthcare Auto-Substantiation Service

The Healthcare Auto-Substantiation Service provided by credit card issuers (Amex®, Discover® Network, MasterCard®, Visa®) and PIN debit card issuers, is supported by these specifications.

This service identifies and substantiates medical purchases, including over-the-counter (OTC) purchases, for flexible spending accounts (FSA) and health reimbursement arrangement (HRA) accounts.

This section discusses the following FSA/HRA card processing requirements:

- Card requirements
- Transaction Request requirements
- Transaction Response requirements
- Supported transaction types
- Receipt requirements

10.1.6.1 Card Requirements

This section discusses the following card requirements for FSA/HRA card processing:

- Eligible benefit cards
- Partial Approval
- Split tender processing
- Qualified healthcare products

10.1.6.1.1 Eligible Benefit Cards

Eligible benefit cards are identified by a 9-digit BIN. The merchant's POS system must be able to store BIN information for use at the time the card is swiped at the device.

10.1.6.1.2 Partial Approval

If Partial Approvals are not supported, the transaction request will either be approved for the full amount, or the transaction will be declined.

10.1.6.1.3 Split Tender Processing

A Split Tender transaction results when a Partial Approval is returned to the POS in the transaction response. The POS device must then prompt for payment of the remaining balance due on the sale. The cardholder uses a different form of payment for nonqualified items, such as another payment card, cash, or a check.

10.1.6.1.4 Qualified Healthcare Products

Qualified medical expenses are generally those items that qualify for IRS medical and dental expense deductions. Eligible items are categorized by OTC grouping and by brand name.

The following table lists supported product categories, whether or not they are required in messages, their Product Codes, and brief descriptions:

Qualified Healthcare Products (QHP)		
Product	**Message Requirement**	**Product Code and Description**
Clinical Services (includes tax and shipping, less discounts)	Optional	Code: 890 Identifies the total amount of clinic or other medical services. This amount is included in Total QHP Amount.
Dental Services (includes tax and shipping, less discounts)	Optional	Code: 891 Identifies the total amount of dental services. This amount is included in Total QHP Amount.
Prescription/Rx Only (includes tax and shipping, less discounts)	Optional	Code: 892 Identifies the total amount of prescriptions used to gain access to Rx-only purses. This amount is included in the Total QHP Amount.
Vision/Optical (includes tax and shipping, less discounts)	Optional	Code: 893 Identifies the total amount of vision care items. This amount is included in Total QHP Amount.
Total QHP Amount (includes tax and shipping, less discounts)	Required	Code: 894 Identifies the total QHP amount. This includes all Clinical Services, Dental Services, Prescription/Rx, and Vision/Optical amounts.
Co-Payment	Optional	Code: 895 Identifies the total amount for co-payments
Transit	Optional	Code: 896 Identifies the total amount for transit purchases

10.1.6.2 Transaction Request Requirements

FSA/HRA transaction requests include the following data fields:

- Market-Specific Data Indicator (MSDI) (found in Variable Information Data Segment [No. 111])
- Partial Approval Indicator (found in Standard Message Data Segment [No. 100])
- Product Code (found in Product Code Data Segment [No. 102])

10.1.6.3 Transaction Response Requirements

FSA/HRA transaction responses contain standard response data. There are no data elements specific to FSA/HRA transactions.

10.1.6.4 Supported Transaction Types

This section discusses the following supported transaction types for FSA/HRA card processing:

- Purchase/Capture
- Purchase Reversal
- Merchandise Return
- Time-out Reversal

10.1.6.4.1 Purchase/Capture

The Partial Approval Indicator and Product Code Data Segment fields are required in a FSA/HRA Purchase/Capture transaction.

All approved Purchase/Capture transactions are added to the count and amount of the card type totals bucket.

10.1.6.4.2 Purchase Reversal

The Partial Approval Indicator is required in a FSA/HRA Purchase Reversal transaction.

All approved Purchase Reversal transactions are subtracted from the count and amount of the card type totals bucket.

The amount in a Reversal (Void) request must be the amount approved in the original transaction response when a transaction has been partially approved.

The Sequence Number must be the Sequence Number contained in the original transaction

10.1.6.4.3 Merchandise Return

All approved Merchandise Return transactions are added to the count and subtracted from the amount of the card type totals bucket.

10.1.6.4.4 Time-out Reversal

The Partial Approval Indicator is required in a FSA/HRA TOR transaction.

All approved TOR transactions are subtracted from the count and amount of the card type totals bucket.

The amount in a TOR request must be the amount approved in the original transaction response when a transaction has been partially approved.

The Sequence Number must be the Sequence Number contained in the original transaction

10.1.6.5 Receipt Requirements

The credit card receipt requirements listed in section 10.1.7, "Receipt Requirements," apply to FSA/HRA card transactions.

In addition, the following information prints on all approved FSA/HRA receipts:

- Indicator that identifies the QHP items on the customer receipt
- Subtotal of the QHP amount, including taxes and discounts

10.1.7 Receipt Requirements

This section discusses the following information that must print on credit card receipts:

- Merchant information
- Transaction Type
- Card Type ID
- Account Number
- Expiration Date
- Transaction Date
- Transaction Time
- Sequence Number
- Product information
- Balance information
- Approved/Declined message
- Signature line

10.1.7.1 Merchant Information

The merchant's name and address prints on all credit card receipts.

10.1.7.2 Transaction Type

The Transaction Type prints on all receipts.

10.1.7.3 Card Type ID

The type of card prints on all credit card receipts. Card Type ID's are:

Amex®:	AX
Debit:	DB
Diners Club®	DCI-DISC
Discover® Network:	DS
EBT:	EB
JCB®	JCB-DISC
MasterCard®:	MC
Visa®:	VISA
STAR® Signature Debit:	STAR

10.1.7.4 Account Number

The Account Number (Element No. 2) must be truncated—suppressed to its last four digits—on all transaction receipts.

A series of *X*'s equal to the number of suppressed digits precedes these last four digits for a field length equal to the original Account Number length.

Example. The Account Number *1234567890123456* prints as: *XXXXXXXXXXXX3456*.

The full Account Number can be retrievable from the terminal journal or transaction log for retrieval requests.

Note: If manual entry is used, this fact is indicated on the receipt and journal by an asterisk (*) before the Account Number.

10.1.7.5 Expiration Date

The Expiration Date is the expiration date of the card presented in a transaction.

The Expiration Date must be suppressed on all transaction receipts.

10.1.7.6 Transaction Date

The date of the transaction prints on all credit card receipts.

10.1.7.7 Transaction Time

The time of the transaction prints on all credit card receipts.

10.1.7.8 Sequence Number

The transaction's Sequence Number appears on all credit card receipts.

10.1.7.9 Product Information

The following product information must appear on a receipt for an "approved" transaction:

- Fuel product information
  - ◊ Fuel type
  - ◊ Quantity in gallons
  - ◊ Price per gallon
  - ◊ Total fuel purchase amount
- Nonfuel product information
  - ◊ Number of units of product
  - ◊ Unit price of product
  - ◊ Total product purchase amount

10.1.7.10 Balance Information

Balance information prints on a credit card receipt for a Partial Approval transaction and a Balance Return with Purchase transaction. In these cases, the available card balance prints on an approved or declined transaction receipt when that amount is returned in the transaction response. This pertains to transactions in which an available card balance is returned by the issuer to the POS device.

10.1.7.11 Approved/Declined Message

The approval phrase, including the approval code, must be on a receipt for an "approved" transaction that involves a point-of-sale device.

The decline message must be on a receipt for a "declined" transaction that involves a point-of-sale device if the device uses the "preprint" feature.

The decline message is displayed on the device's screen for a customer-activated device.

10.1.7.12 Signature Line

A signature line prints only on the merchant’s copy of the credit card transaction receipt.

10.1.8 Regulatory Requirements

This section discusses the following regulatory requirements that must be met when processing all electronic credit (American Express®, Diners Club®, Discover® Network, JCB®, MasterCard®, MasterCard® Fleet, Visa®, and Visa® Fleet) transactions:

- Data storage requirements
- Track support requirements

10.1.8.1 Data Storage Requirements

The following data storage requirements must be met when processing all electronic credit transactions:

- A device at the point of interaction must not display/store any card-read data except the card's account number, card's expiration date, and cardholder's name, if present.
- The merchant device must not store full magnetic stripe data.
- The merchant—or any agent representative thereof—may record only the card's account number, card's expiration date, and cardholder's name on paper, microfiche, or an online authorization file, for research purposes at its site. This data must be kept and maintained in a secure manner.
- Full card-read data (including discretionary data) must not be stored. The merchant—or any agent or representative thereof—must not record, store, replicate, or use full card-read data for any other purpose. By way of example—but not limited to—card data provided in an authorization or clearing transaction must not be the result of previously captured full card-read data.

10.1.8.2 Track Support Requirements

The following track support requirements must be met when processing all electronic credit transactions:

- The default track that is sent to the host should be Track 2 if the device supports Track 2. If the device cannot read Track 2, then the device should send unaltered Track 1 before attempting manual entry.
  Note: For American Express, the default track should be Track 1. If the device cannot read Track 1, then the device should send unaltered Track 2 on the initial transaction, before attempting manual entry.
- Regulatory requirements regarding data storage dictate that full card-read data (including discretionary data) must not be stored.
- Card data provided in a clearing transaction must not be the result of previously captured full card-read data.
  Note: This does not apply to the TransArmor-Verifone Edition processing.
  The host message should only contain the following card data information in the Account Number and Card Discretionary Block Data fields:
  - ◊ Account Number
  - ◊ Expiration Date (Field format: YYMM)

  Note: These two items must be separated by an "=" sign, rather than a Field Separator. (Please see Processing Rules for Card Discretionary Block Data (Element No. 12) in chapter 13, "Data Element Descriptions," for additional information.)
- The host message should only contain the Account Number, followed by an account delimiter (=), followed by the 4 digit Expiration Date (Storage format – YYMM). No other

Card Discretionary Data field follows the 4-digit Expiration Date. This scenario applies only to the following credit transactions:

- ◊ Credit Completions
- ◊ Credit POS Purchase Reversals
- ◊ Credit Time Out Reversals
- ◊ Credit POS Cancellations
- ◊ Credit Cancellations

There is one exception to these track support requirements:

- In Flight Table
- Track 2 data is kept in this table while the transaction is in process. It is removed from the table when the transaction is completed.
  Note: All offline transactions must send full unaltered track data. This includes both Track 1 and Track 2 formats

10.2 Debit Card Processing Requirements

This section discusses the following debit card processing requirements:

- Card requirements
- Supported transaction types
- Healthcare auto-substantiation service
- Receipt requirements
- Regulatory requirements

10.2.1 Card Requirements

This section discusses the following card requirements for debit card processing:

- Expiration date
- Duplicate Completion transactions
- PIN entry
- Manual entry

10.2.1.1 Expiration Date

The device does not perform expiration date validation on a debit card when the card is processed as a debit transaction. On-line authorization must be requested for the transaction regardless of the expiration date on the card.

10.2.1.2 Duplicate Completion Transactions

When processing a Debit Card Completion transaction, the transaction can only be processed one time by BUYPASS and debit networks. This transaction cannot be reversed or processed again by the device.

In the event the device sends a Debit Card Completion transaction to BUYPASS and does not receive a response, the device should resend the request on the next dial attempt or 30 minutes later—whichever occurs first. If the device receives a Response Code value of *1* (indicating a "decline") and a Decline Code value of *24* in the transaction response message, then the device should treat the response as already captured on BUYPASS and update the device totals as "approved."

Any response from BUYPASS to a Debit Card Completion transaction should end attempts to retry a Debit Card Completion whether "approved" or "declined."

10.2.1.3 PIN Entry

When processing a debit card transaction, the entry of the customer PIN is required on a DES-compliant entry device, and it must be encrypted using the DUKPT PIN encryption method.

All transactions—except the Purchase Reversal transaction—require PIN entry.

10.2.1.4 Manual Entry

Manual entry is not allowed on a debit card transaction. If the device is unable to read the card's magnetic strip, it should prompt the user to swipe the card again—up to three times. If unable to read the magnetic strip after three attempts, a message should be displayed to the clerk and the transaction ended.

10.2.2 Supported Transaction Types

The following point-of-sale transactions are supported for debit card processing:

- Purchase/Capture
- Purchase with Cashback
- Purchase Reversal
- Authorization Only
- Merchandise Return
- Time-out Reversal

Note: Time-out Reversal for debit Completion transaction is not supported.

- Void of a Merchandise Return
- Balance Inquiry
- Partial Approval
- Authorization Only Reversal

The following customer-activated transactions are supported for debit card processing:

- Purchase/Capture
- Authorization Only
- Time-out Reversal
- Partial Approval
- Authorization Only Reversal

10.2.2.1 Partial Approval Transaction

A Partial Approval transaction enables a merchant to receive approval for a PIN debit card's balance if that balance is more or less than the transaction request amount.

Debit networks supporting the Partial Approval transaction return an approval containing the following information:

Note: This information is returned when a Partial Approval Indicator value of 1 is sent in Partial Approval Indicator (Element No. 121).

- A Response Code (Element No. 83) value of *F* (Indicates "approval, partial amount approved.")
- Approved Amount (Element No. 6) contains the approved partial amount, and the POS device must prompt for payment of the remaining balance due on the sale.
- The card balance—if returned by the issuer—is in the Additional Information Data Segment (Segment No. 112).

Partial Approval transactions are supported for:

- POS Purchase
- POS Purchase with Cashback
- POS Authorization Only
- CAT Authorization Only

Note: As of April, 2013, Interlink requires that merchants supporting cashback must also support Partial Approval transactions. Most merchants support Partial Approval transactions as part of a previous Maestro mandate. This note is to clarify that merchants supporting cashback must also support Partial Approval transactions. Merchants must ensure that their systems allow for this mandate. If any debit transaction requesting authorization for purchase with

cashback is partially approved, then the partially approved amount will be for the purchase only and not cashback.

The amount in a Reversal (Void) request must be the amount approved in the original transaction response when a transaction has been partially approved.

The amount in a TOR request must be the amount approved in the original transaction response when a transaction has been partially approved.

The Sequence Number must be the Sequence Number contained in the original transaction request.

10.2.2.2 Authorization Only Reversal Transaction

In support of the following requirements, BUYPASS supports the Authorization Only Reversal transaction:

- All approved and partially approved authorization transactions must be either cleared or reversed.
- Acquirers must ensure that their merchants process authorization reversals for authorizations submitted in error and/or cardholder cancelled card-present transactions within 24 hours. Card-absent merchants are allowed up to 72 hours to process the required partial or full authorization reversals.
- Issuers must immediately process matched reversal transactions, subject to the issuer's risk or fraud control policies, and release any applicable holds on cardholder available funds.

This transaction enables a merchant to reverse the following debit card transactions within required timeframes:

- POS Authorization Only (Transaction Type code "3")
- CAT Authorization Only (Transaction Type code "5")

Successful reversal matching requires the following:

- Element No. 5 (Approval Number)—present in the original transaction—must be present and identical in the Authorization Only Reversal transaction.
- Element No.86 (Sequence Number)—present in the original transaction—must be present and identical in the Authorization Only Reversal transaction.
- Element No. 78 (Prompt Code) must contain a Transaction Type code "S" to indicate cancellation.

10.2.3 Healthcare Auto-Substantiation Service

The Healthcare Auto-Substantiation Service provided by credit card issuers (Amex®, Discover® Network, MasterCard®, Visa®) and PIN debit card issuers, is supported by these specifications.

This service identifies and substantiates medical purchases, including over-the-counter (OTC) purchases, for flexible spending accounts (FSA) and health reimbursement arrangement (HRA) accounts.

This section discusses the following FSA/HRA card processing requirements:

- Card requirements
- Transaction Request requirements
- Transaction Response requirements
- Supported transaction types
- Receipt requirements

10.2.3.1 Card Requirements

This section discusses the following card requirements for FSA/HRA card processing:

- Eligible benefit cards
- Partial Approval
- Split tender processing
- Qualified healthcare products

10.2.3.1.1 Eligible Benefit Cards

Eligible benefit cards are identified by a 9-digit BIN. The merchant's POS system must be able to store BIN information for use at the time the card is swiped at the device.

10.2.3.1.2 Partial Approval

If Partial Approvals are not supported, the transaction request will either be approved for the full amount, or the transaction will be declined.

10.2.3.1.3 Split Tender Processing

A Split Tender transaction results when a Partial Approval is returned to the POS in the transaction response. The POS device must then prompt for payment of the remaining balance due on the sale. The cardholder uses a different form of payment for nonqualified items, such as another payment card, cash, or a check.

10.2.3.1.4 Qualified Healthcare Products

Qualified medical expenses are generally those items that qualify for IRS medical and dental expense deductions. Eligible items are categorized by OTC grouping and by brand name.

The following table lists supported product categories, whether or not they are required in messages, their Product Codes, and brief descriptions:

Qualified Healthcare Products (QHP)		
Product	**Message Requirement**	**Product Code and Description**
Clinical Services (includes tax and shipping, less discounts)	Optional	Code: 890 Identifies the total amount of clinic or other medical services. This amount is included in Total QHP Amount.
Dental Services (includes tax and shipping, less discounts)	Optional	Code: 891 Identifies the total amount of dental services. This amount is included in Total QHP Amount.
Prescription/Rx Only (includes tax and shipping, less discounts)	Optional	Code: 892 Identifies the total amount of prescriptions used to gain access to Rx-only purses. This amount is included in the Total QHP Amount.
Vision/Optical (includes tax and shipping, less discounts)	Optional	Code: 893 Identifies the total amount of vision care items. This amount is included in Total QHP Amount.
Total QHP Amount (includes tax and shipping, less discounts)	Required	Code: 894 Identifies the total QHP amount. This includes all Clinical Services, Dental Services, Prescription/Rx, and Vision/Optical amounts.
Co-Payment	Optional	Code: 895 Identifies the total amount for co-payments
Transit	Optional	Code: 896 Identifies the total amount for transit purchases

10.2.3.2 Transaction Request Requirements

FSA/HRA transaction requests include the following data fields:

- Market-Specific Data Indicator (MSDI) (found in Variable Information Data Segment [No. 111])
- Partial Approval Indicator (found in Standard Message Data Segment [No. 100])
- Product Code (found in Product Code Data Segment [No. 102])

10.2.3.3 Transaction Response Requirements

FSA/HRA transaction responses contain standard response data. There are no data elements specific to FSA/HRA transactions.

10.2.3.4 Supported Transaction Types

This section discusses the following supported transaction types for FSA/HRA card processing:

- Purchase/Capture
- Purchase Reversal
- Merchandise Return
- Time-out Reversal

10.2.3.4.1 Purchase/Capture

The Partial Approval Indicator and Product Code Data Segment fields are required in a FSA/HRA Purchase/Capture transaction.

All approved Purchase/Capture transactions are added to the count and amount of the card type totals bucket.

10.2.3.4.2 Purchase Reversal

The Partial Approval Indicator is required in a FSA/HRA Purchase Reversal transaction.

All approved Purchase Reversal transactions are subtracted from the count and amount of the card type totals bucket.

The amount in a Reversal (Void) request must be the amount approved in the original transaction response when a transaction has been partially approved.

The Sequence Number must be the Sequence Number contained in the original transaction

10.2.3.4.3 Merchandise Return

All approved Merchandise Return transactions are added to the count and subtracted from the amount of the card type totals bucket.

10.2.3.4.4 Time-out Reversal

The Partial Approval Indicator is required in a FSA/HRA TOR transaction.

All approved TOR transactions are subtracted from the count and amount of the card type totals bucket.

The amount in a TOR request must be the amount approved in the original transaction response when a transaction has been partially approved.

The Sequence Number must be the Sequence Number contained in the original transaction

10.2.3.5 Receipt Requirements

The debit card receipt requirements listed in section 10.2.4, "Receipt Requirements," apply to FSA/HRA card transactions.

In addition, the following information prints on all approved FSA/HRA receipts:

- Indicator that identifies the QHP items on the customer receipt
- Subtotal of the QHP amount, including taxes and discounts

10.2.4 Receipt Requirements

The credit card receipt requirements listed in section 10.1.7, "Receipt Requirements," also apply to debit card receipts. Optionally, a PIN USED message can be printed on transaction receipts when a PIN is entered.

No signature line is required when printing a debit receipt. Optionally, a SIGNATURE NOT REQUIRED message can be printed.

10.2.5 Regulatory Requirements

This section discusses the following regulatory requirements that must be met when processing all electronic debit transactions:

- Data storage requirements
- Track support requirements

10.2.5.1 Data Storage Requirements

The following data storage requirements must be met when processing all electronic debit transactions:

- A device at the point of interaction must not display/store any card-read data except the card's account number, card's expiration date, and cardholder's name, if present.
- The merchant device must not store full magnetic stripe data.
- The merchant—or any agent representative thereof—may record only the card's account number, card's expiration date, and cardholder's name on paper, microfiche, or an online authorization file, for research purposes at its site. This data must be kept and maintained in a secure manner.
- Full card-read data (including discretionary data) must not be stored. The merchant—or any agent or representative thereof—must not record, store, replicate, or use full card-read data for any other purpose. By way of example—but not limited to—card data provided in an authorization or clearing transaction must not be the result of previously captured full card-read data.
- PIN block data is not included in follow-on transactions (Completions/Reversals).

10.2.5.2 Track Support Requirements

The following track support requirements must be met when processing all electronic debit transactions:

- The default track that is sent to BUYPASS should be Track 2.
- Regulatory requirements regarding data storage dictate that full card-read data (including discretionary data) must not be stored.
- Card data provided in a clearing transaction must not be the result of previously captured full card-read data.
  Note: This does not apply to the TransArmor-Verifone Edition processing.
  The host message should only contain the following card data information in the Account Number and Card Discretionary Block Data fields:
  - ◊ Account Number
  - ◊ Expiration Date (Field format: YYMM)

  Note: These two items must be separated by an "=" sign, rather than a Field Separator. (Please see Processing Rules for Card Discretionary Block Data (Element No. 12) in chapter 13, "Data Element Descriptions," for additional information.)
- The host message should only contain the Account Number, followed by an account delimiter (=), followed by the 4 digit Expiration Date (Storage format – YYMM). No other Card Discretionary Data field follows the 4-digit Expiration Date. This scenario applies only to the following debit transactions:
  - ◊ Debit Completions
  - ◊ Debit POS Purchase Reversals
  - ◊ Debit Time Out Reversals
  - ◊ Debit POS Cancellations
  - ◊ Debit Cancellations

There is one exception to these track support requirements:

- In Flight Table
  Track 2 data is kept in this table while the transaction is in process. It is removed from the table when the transaction is completed.

10.3 PINless POS Debit Card Processing Requirements

Note: Please check with your First Data Relationship Manager for the availability of this functionality.

This section discusses the following PINless POS Debit card processing requirements:

- Card requirements
- Supported transaction types

10.3.1 Card Requirements

PINless POS Debit cards allow certain POS Debit transactions to be processed without a PIN. Card requirements for PINless POS Debit card processing include the following:

- Card must be issued by a supported Debit network
- For MSR, Card must be swiped at a POS device
- For EMV, the card may be dipped or tapped for contact or contactless transactions

Note: For EMV, POS PINless transactions are ONLY eligible with the US Common Debit AID as 'No CVM' and 'No PIN Block'.

- Card Track Data must be Track 2

10.3.2 Supported Transaction Types

The following point-of-sale transactions are supported for PINless POS Debit card processing:

- Purchase
- Merchandise Return\Refund
- Time-out Reversal (TOR)

Note: A Time-out Reversal (TOR) transaction is used to reverse a transaction on one of the following occasions:

- If the POS device fails.
- If the transaction is aborted after initial communication with the host.

Supported transactions do not require a PIN when less than or equal to $50.00. The PINless POS feature has been expanded to support over $50.00 for Non-PIN debit transactions (PINless POS Plus product) for specific BINs of debit networks.

Note: The amount may be updated as required by debit networks.

The following exclusions are applicable for PINless POS Debit transactions:

- Transactions with cashback
- Healthcare (FSA) transactions. HSA is not excluded, since they are treated as prepaid cards
- Automated Fuel Dispenser (MCC 5542) transactions
- Transactions where the final amount is not known (restaurant with tip, hotels, car rental)
- Store and Forward transactions

10.4 Signature Debit Card Processing Requirements

Note: Please check with your First Data Relationship Manager for the availability of this functionality.

This section discusses the following Signature Debit card processing requirements:

- Card requirements
- Supported transaction types
- Restrictions for Processing STAR® Single and Dual Message
- Additional Restrictions for Processing STAR® Single Message
- Follow-up transaction requirements
- Address Verification Service
- Bill Payment/Recurring payment transactions
- Healthcare auto-substantiation service
- Receipt requirements
- Regulatory requirements

10.4.1 Card Requirements

Signature Debit cards allow certain debit transactions to be processed without a PIN. Card requirements for Signature Debit card processing include the following:

- Card must be issued by a supported debit network
- Card must support all entry modes, such as, manual, swiped, contact chip, and contactless chip

Note: STAR® is a debit network that is expanding its transaction set to support Signature Debit in addition to its current support of PIN and PINless debit. This expansion may provide additional routing choices for merchants. This new offering will allow merchants in most processing environments the ability to take advantage of the various cardholder verification methods, i.e. PIN, signature, etc., available for the transaction based on a merchant's chosen point-of-sale device. Please contact your Account Manager for additional information.

10.4.2 Supported Transaction Types

The following point-of-sale transactions are supported for Signature Debit card processing:

- Purchase/Capture
- Purchase Reversal
- Authorization Only
- Time-out Reversal
- Authorization Only Cancellation
- Partial Approval

Note: A Time-out Reversal (TOR) transaction is used to reverse a transaction on one of the following occasions:

- If the POS device fails
- If the transaction is aborted after initial communication with the host

10.4.3 Restrictions for Processing STAR® Single and Dual Message

The following are the BUYPASS Platform restrictions to process STAR Single and Dual message:

- Transactions processed as STAR Signature Debit must occur within US and US territories (Guam, Puerto Rico, Virgin Islands etc.)
- The transaction must be settled only in US Dollar (840)
- Cash back is not supported
- Recharge, Activate, or Inquiry transactions are not supported
- eCommerce transactions, including those using 3-D Secure Protocol, such as Verified by Visa, MC Secure code transactions, Amex SafeKey and Discover ProtectBuy are not supported

10.4.4 Additional Restrictions for Processing STAR® Single Message

Besides the restrictions listed in the above section, STAR Single message processing has the following additional restrictions:

- The transactions must be card present, full mag-stripe data, or full EMV chip data.

- The transactions cannot be keyed
- Transactions must have the final amount same as the original authorization amount
- Mail Order/Telephone Order (MOTO) is not supported
- The following MCCs are excluded:

MCC	Descriptions
4813	Key entry Telecom merchant providing single local and long distance phone calls using a central access number in a non-face-to-face environment using key entry
4829	Money transfer
5542	Fuel dispenser automated
5960	Direct marketing insurance
5962	Direct marketing travel related arrangement service
5964	Direct marketing catalog merchants
5966	Direct marketing outbound telemarketing merchants
5967	Direct marketing inbound telemarketing merchants
5968	Direct marketing continuity/subscription merchants
5969	Direct marketing other direct marketers not elsewhere classified
6010	Member financial institution – manual cash disbursement
6011	Member financial institution – automated cash disbursements
6012	Financial institution – merchandise and services
6050	Quasi cash – member financial institution
6051	Quasi cash – merchant
6531	Payment service provider – money transfer for a purchase
6532	Payment transaction – member financial institution
6533	Payment transaction – merchant
6534	Money transfer – member financial institution
7995	Gambling transactions
9405	Intra-Government purchases – Government only
9700	International automated referral service
9702	GCAS emergency services
9754	Gambling – horse racing, dog racing, state lotteries
9950	Intra-company purchases

Note: If the transaction does not meet the requirements noted in section 10.4.3 (Restrictions for Processing STAR® Single and Dual Message) and section 10.4.4 (Additional Restrictions for Processing STAR® Single Message), then it will be routed to the card associations such as Visa, MC, as applicable.

10.4.5 Follow-Up Transaction Requirements

When an original transaction response comes back with a Transaction Attributes Data Segment (Data Segment No.134) and the Settlement Type (Data Element 198) has a value of 'X' (where X= Non-traditional Signature Debit), then any associated transactions (Reversals, Completions, Incremental Authorizations) must contain the Settlement Data Acceptance Flag, Table ID 21 in

the Variable Information Data Segment (Data Segment No. 111), with a PINless Debit Acceptance field value of 'X'.

10.4.6 Address Verification Service

Address Verification Service (AVS) listed in section 10.1.3, Address Verification Service for credit card also apply to Signature Debit card.

10.4.7 Bill Payment/Recurring Payment Transactions

Bill Payment/Recurring Payment Transactions listed in section 10.1.4, Bill Payment/Recurring Payment Transaction for credit card also apply to Signature Debit card.

10.4.8 Healthcare Auto-Substantiation Service

The Healthcare Auto-Substantiation Service provided by credit card issuers (Amex®, Discover® Network, MasterCard®, Visa®), PIN debit card issuers, and Signature Debit card issuers is supported by these specifications.

This service identifies and substantiates medical purchases, including over-the-counter (OTC) purchases, for flexible spending accounts (FSA) and health reimbursement arrangement (HRA) accounts.

This section discusses the following FSA/HRA card processing requirements:

- Card requirements
- Transaction Request requirements
- Transaction Response requirements
- Supported transaction types
- Receipt requirements

10.4.8.1 Card Requirements

This section discusses the following card requirements for FSA/HRA card processing:

- Eligible benefit cards
- Partial Approval
- Split tender processing
- Qualified healthcare products

10.4.8.1.1 Eligible Benefit Cards

The eligible benefit card requirements listed in section 10.1.6.1.1, Eligible Benefit Cards for credit cards also apply to Signature Debit card.

10.4.8.1.2 Partial Approval

The partial approval requirements listed in section 10.1.6.1.2, Partial Approval for credit cards also apply to Signature Debit card.

10.4.8.1.3 Split Tender Processing

The split tender processing requirements listed in section 10.1.6.1.3, Split Tender Processing for credit cards also apply to Signature Debit card.

10.4.8.1.4 Qualified Healthcare Products

The qualified healthcare products listed in section 10.1.6.1.4, Qualified Healthcare Products for credit cards also apply to Signature Debit card.

10.4.8.2 Transaction Request Requirements

FSA/HRA transaction request requirements listed in section 10.1.6.2, Transaction Request Requirements for credit cards also apply to Signature Debit card.

10.4.8.3 Transaction Response Requirements

FSA/HRA transaction response requirements listed in section 10.1.6.3, Transaction Response Requirements for credit cards also apply to Signature Debit card.

10.4.8.4 Supported Transaction Types

This section discusses the following supported transaction types for FSA/HRA card processing:

- Purchase/Capture
- Purchase Reversal
- Merchandise Return
- Time-out Reversal

10.4.8.4.1 Purchase/Capture

The Purchase/Capture transaction requirements listed in section 10.1.6.4.1, Purchase/Capture for credit card also apply to Signature Debit card.

10.4.8.4.2 Purchase Reversal

The Purchase Reversal transaction requirements listed in section 10.1.6.4.2, Purchase Reversal for credit card also apply to Signature Debit card.

10.4.8.4.3 Merchandise Return

The Merchandise Return transaction requirements listed in section 10.1.6.4.3, Merchandise Return for credit card also apply to Signature Debit card.

10.4.8.4.4 Time-out Reversal

The Time-out Reversal transaction requirements listed in section 10.1.6.4.4, Time-out Reversal for credit card also apply to Signature Debit card.

10.4.8.5 Receipt Requirements

The FSA/HRA card transaction receipt requirements as listed in section 10.1.6.5, Receipt Requirements for credit card also apply to Signature Debit card.

10.4.9 Receipt Requirements

The credit card receipt requirements listed in section 10.1.7, Receipt Requirements for credit card also apply to Signature Debit card.

10.4.10 Regulatory Requirements

This section discusses the following regulatory requirements that must be met when processing all Signature Debit transactions:

- Data storage requirements
- Track support requirements

10.4.10.1 Data Storage Requirements

The data storage requirements for processing all electronic credit transactions as described in section 10.1.8.1, Data Storage Requirements also applies to Signature Debit cards.

10.4.10.2 Track Support Requirements

The track support requirements for processing all electronic credit transactions as described in section 10.1.8.2, Track Support Requirements also applies to Signature Debit cards.

10.5 Electronic Benefits Transfer Card Processing Requirements

This section discusses the following electronic benefits transfer (EBT) card processing requirements:

- Card requirements
- Food stamp electronic vouchers
- Supported transaction types
- Receipt requirements
- eWIC transaction processing
- Healthy Incentives Program (HIP) Transaction Processing

10.5.1 Card Requirements

This section discusses the following card requirements for EBT card processing:

- Expiration date
- PIN entry
- Manual entry
- Stand-in mode
- Tender types
- Balance inquiry

10.5.1.1 Expiration Date

The device does not perform expiration date validation in an EBT card transaction.

10.5.1.2 PIN Entry

All transactions—except the Food Stamp Electronic Voucher transaction, Food Stamp Reversal (Void) transaction, and Cash Benefit Reversal (Void) transaction—require PIN entry.

10.5.1.3 Manual Entry

Manual entry of EBT must be supported; however, the store's device must prompt for the PIN regardless of Account Number entry mode.

For manually entered Account Numbers, the device must also prompt for the card's expiration date. If no expiration date is provided, use 1249.

10.5.1.4 Stand-in Mode

Currently, there is no "stand-in" mode for EBT transactions. The USDA guidelines do not specifically allow for the off-line processing of EBT transactions. In addition, a defined transaction format that supports stored-and-forwarded (SAF) EBT transactions does not exist. If the retailer chooses to support SAF EBT transactions, the retailer assumes all risk for transactions submitted once the connection is reestablished. The PIN must be stored in an encrypted format. The balance field must not print on the receipt (Do not print zeros.).

10.5.1.5 Tender Types

EBT processing supports the following tender types:

- Food stamps
- Cash benefits

10.5.1.6 Balance Inquiry

Devices must support inquiries by tender type.

10.5.2 Food Stamp Electronic Vouchers

Electronic vouchers electronically submit food stamp approvals received over the phone during a device, BUYPASS, or authorizing host outage. Electronic vouchers must prompt for the following items:

- Account Number prompt
- Approval Number prompt
- Voucher Number prompt

10.5.2.1 Account Number Prompt

If the store system can retain the account information from an original transaction attempt, the remainder of the information can be entered at the controller level or directly into the device in the lane. For stand-alone equipment, the electronic voucher transaction must allow for manual entry of the Account Number and the Expiration Date.

In cases where the PAN is printed on transaction receipts, it must be truncated to its last four digits. A series of *X*'s equal to the number of truncated digits precede these four digits for a field length equal to the original Account Number's length.

Example. The Account Number *1234567890123456* prints as *XXXXXXXXXXXX3456*.

10.5.2.2 Approval Number Prompt

The device must allow for entry of the Approval Number received over the phone.

10.5.2.3 Voucher Number Prompt

The device must allow for entry of the Voucher Number from the manual voucher ticket.

10.5.3 Supported Transaction Types

The following transaction types are supported for EBT processing:

- Food Stamp Purchase
- Food Stamp Reversal (Void)
- Food Stamp Return
- Cash Benefit Purchase
- Cash Benefit Reversal (Void)
- Food Stamp Balance Inquiry
- Food Stamp Electronic Voucher
- Cash Benefit Purchase with Cash Back
- Cash Benefit Balance Inquiry
- Time-out Reversal
- Food Stamp Void of a Merchandise Return

10.5.4 Receipt Requirements

The credit card receipt requirements listed in section 10.1.7, "Receipt Requirements," also apply to EBT card receipts with the exception of Approved/Declined Message.

In addition to those requirements, the following information must appear on EBT card receipts:

- Tender Type and Transaction Type
- Clerk ID
- Voucher Number
- Transaction amounts
- Balances
- Approved messages
- Declined messages
- Approved HIP transaction data

No signature line is required when printing an EBT receipt. Optionally, a SIGNATURE NOT REQUIRED message can be printed.

10.5.4.1 Tender Type and Transaction Type

The receipt must print whether the transaction is a Food Stamp transaction or a Cash Benefit transaction. It must also indicate the type of transaction attempted.

Examples. *Food Stamp Purchase*

Cash Benefit Purchase with Cash Back

10.5.4.2 Clerk ID

Note: The Clerk ID is an optional receipt requirement.

As an option, it may be retrieved from the transaction prompting sequence or a log on/log off function and printed on the receipt.

Also, as an option, the device may prompt for a Clerk ID, whether it is on a transaction basis or within a log on/log off function. The Clerk ID may be sent within the transaction's message request and may be printed on the receipt.

10.5.4.3 Voucher Number

For Food Stamp Electronic Voucher transactions only, the manually entered Voucher Number must appear on the receipt.

10.5.4.4 Transaction Amounts

The food stamp dollar amount must be printed on a Food Stamp transaction receipt.

The cash benefit dollar amount(s) must be printed on a Cash Benefit transaction receipt. It/They must be broken down for purchases versus cash back and include a total for both—if both amounts were entered.

10.5.4.5 Balances

For both Food Stamp transactions and Cash Benefit transactions, the ending balance must be printed on the receipt.

As an option, the beginning EBT balance may be printed on the receipt by calculating the balance based on the ending balance and the amount of the transaction.

EBT authorizers may return either a food stamp balance only, a cash benefit balance only, or both balances. If a particular balance is not returned, do not print a balance of zero; omit the

particular balance line altogether. Balances of zero dollars should only be printed when zero is actually returned as the balance from the authorizer.

10.5.4.6 Approved Transactions

The Approval Number and message from the transaction response must be printed on the receipts for all "approved" EBT transactions.

In the case when a PIN is entered, a PIN USED message prints on the transaction receipt.

10.5.4.7 Declined Transactions

All "declined" transactions must print a receipt. The reason for the decline must be indicated on the receipt. Transactions can be declined for the following reasons:

- Insufficient funds
- Bad PIN

In the case of a decline due to "insufficient funds," the ending balance is returned and must be printed on the declined transaction receipt.

In the case of a decline due to a "bad PIN," a declined transaction receipt must be printed. The device may not prompt for reentry of the PIN without printing a receipt and retrying the transaction.

10.5.4.8 Approved HIP transaction data

The following fields must be printed on the receipt for the EBT FOOD approved HIP transactions:

EBT HIP Transactions:

- Incentive Eligible Amount
- Incentive Earned or Incentive Returned
- Month to date Incentive Earned

Note: Incentive Earned is applicable to Sale transactions. Incentive Returned is applicable to Return transactions.

10.5.5 eWIC Transaction Processing

This section discusses the following processing requirements for implementing the EBT version of Women, Infants and Children Nutrition Program solution (eWIC) for transaction processing involving BUYPASS:

- eWIC Transaction Types
- eWIC Prompt Codes
- eWIC Payment Type
- eWIC Authorizer
- eWIC Totals
- eWIC Product Data
- eWIC Receipt Requirements

10.5.5.1 eWIC Transaction Types

Note: These specifications do not support Return transactions.

The following nonfinancial transaction types are supported for eWIC:

- eWIC Authorization (a.k.a. Benefits Inquiry)

  This transaction enables an eWIC client to query the eWIC host for the following information:

  ◊ eWIC client list of eWIC-approved products

  ◊ eWIC client account balance information

  This transaction is initiated by swiping an existing eWIC card at a POS device (device). When prompted, the eWIC client keys the necessary PIN into the device, and the device sends the eWIC Authorization Request to BUYPASS. BUYPASS processes the Request, and forwards the eWIC Authorization Response to the device.
- eWIC Authorization (a.k.a. Benefits Inquiry) Cancellation

  This transaction cancels a previously submitted Authorization.

  Note: If there is a time-out or the eWIC client cancels the order, an eWIC Authorization Cancellation transaction is required.

  This transaction is initiated by swiping an existing eWIC card at a POS device (device). When prompted, the eWIC client keys the necessary PIN into the device, and the device sends the eWIC Authorization Cancellation Request to BUYPASS. BUYPASS processes the Request, and forwards the eWIC Authorization Cancellation Response to the device. The device displays/prints the eWIC account balance information from the Response.
- eWIC Balance Inquiry

  This transaction enables an eWIC client to query the eWIC host for remaining eWIC account balance information.

  This transaction is initiated by swiping an existing eWIC card at a POS device (device). When prompted, the eWIC client keys the necessary PIN into the device, and the device sends the eWIC Balance Inquiry Request to BUYPASS. BUYPASS processes the Request, and forwards the eWIC Balance Inquiry Response to the device. The device displays/prints the eWIC account balance information from the Response.

The following nonfinancial transaction types are supported for eWIC:

- eWIC Purchase Completion

  This transaction performs an eWIC client's purchase.

  Note: If there is a time-out or the eWIC client cancels the order, an eWIC Authorization Cancellation transaction is required.

  This transaction is initiated by swiping an existing eWIC card at a POS device (device). When prompted, the eWIC client keys the necessary PIN into the device, and the device sends the eWIC Authorization Request to BUYPASS. BUYPASS processes the Request, and forwards the eWIC Authorization Response to the device. The device displays/prints the eWIC account balance information from the Response. The clerk scans the items and enters any applicable manufacturer/cents-off discount coupon amount(s). The device sends the eWIC Purchase Completion Request to BUYPASS. BUYPASS processes the Request, and sends the eWIC Purchase Completion Response to the device. The device receives the Response.

- eWIC Purchase Reversal/Void

  This transaction guarantees reversal/void of the effects of an eWIC Purchase Request at the eWIC host when an eWIC Purchase Response is not received or is received "late" (a time-out or a late response) at the POS device (device).

  Notes: This transaction must be completed (The eWIC Purchase Reversal/Void Response must be received by the device.) before any additional eWIC transactions can be sent from the device that originated the eWIC Purchase Reversal/Void Request.

  If the eWIC host responds to the Purchase Reversal/Void Request, the Purchase Reversal/Void transaction is considered "processed" regardless of the Response Code.

  If no Purchase Reversal/Void Response is received, the Merchant must repeat the process at its earliest opportunity until a Purchase Reversal/Void Response to the eWIC Purchase Reversal/Void Request is received.

  This transaction is initiated by the clerk determining that one of the following situations has occurred:

  ◊ An eWIC Purchase Response has been received late; or

  ◊ A time-out situation exists.

  The clerk sends the eWIC Purchase Reversal/Void Request from the device to BUYPASS. BUYPASS processes the Request, and sends the eWIC Purchase Reversal/Void Response to the device. The device displays/prints the eWIC account balance information from the Response.

- eWIC Voucher Clear

  Note: This transaction is an optional transaction.

  This transaction allows the settlement of a previously authorized eWIC Voucher by electronically submitting—within established Merchant time limits—eWIC approvals received over the phone during a device/BUYPASS/eWIC host outage.

  This transaction is initiated by the clerk determining that the device/BUYPASS/eWIC host is offline. The eWIC client's purchase (including manufacturer discount(s)/cents-off coupons) is called in to the authorization center (center). The center notifies the clerk whether the purchase is approved/declined.

  If the center declines the transaction, the eWIC client leaves.

  If the center approves the transaction, the center gives the clerk an eWIC Approval Number, and the clerk issues a paper eWIC Voucher to the eWIC client. The eWIC Voucher contains the following information:

  ◊ eWIC Voucher Number
  ◊ eWIC Account Number
  ◊ eWIC Approval Number
  ◊ eWIC Transaction Amount

  Note: This information must be electronically submitted to the eWIC host within the Merchant's specified time limit in order for the Merchant to receive payment.

  Using the information on the Merchant's copy of the eWIC Voucher, the clerk keys the above information into the device together with any applicable manufacturer/cents-off discount coupon amount(s). The device sends the eWIC Voucher Clear Request to BUYPASS. BUYPASS processes the Request, and sends an eWIC Voucher Clear Response to the device. The device displays/prints the eWIC information from the eWIC Voucher Clear Response.

10.5.5.2 eWIC Prompt Codes

The following Processing Codes and Message Codes are supported by these specifications:

Note: Please refer to the following appendixes for additional information:

Appendix E. Valid Card Type Codes

Appendix G. Valid Transaction Type Codes

eWIC Prompt Codes	
Prompt Code	**Description**
3086	eWIC Authorization
S086	eWIC Authorization Cancellation
E086	eWIC Balance Inquiry
0086	eWIC Purchase Completion
8086	eWIC Purchase Reversal/Void
0086	eWIC Voucher Clear **Note**: This is an optional transaction.

10.5.5.3 eWIC Payment Type

These specifications support the following eWIC Payment Type: EBT WIC (50).

10.5.5.4 eWIC Authorizers

These specifications support the following eWIC Authorizers: eWIC Nevada (74), eWIC Kentucky (75), and ACS® WIC (83).

10.5.5.5 eWIC Totals

The following eWIC totals are maintained:

- eWIC Dollar Amount
- eWIC Count

The following fields have been added to Totals Response for eWIC processing:

- eWIC Dollar Amount
- eWIC Count
- Card Label Code: EWIC

Totals information is obtained from section 12.6, "Totals Data Segment." (Please refer to section 10.5.5.6, "eWIC Data Elements," for additional information.

10.5.5.6 eWIC Data Elements

These specifications support the following new eWIC data elements:

- EBT Data Segment (Segment No. 103)
  - WIC Discount Amount
  - WIC Product Data

The following existing data elements now support eWIC data processing:

- Prompt Code (Element No. 78; constructed from Transaction Type Code and Card Type [No. 14])
- Authorizer Code (Element No. 7)
- EBT Data Segment (Segment No. 105)
- POS Condition Code (Segment No. 111): Attended self-checkout lanes are considered unattended by the eWIC processors. This should be reflected in the value of POS Condition Code.

Note: Please refer to chapter 13, "Data Element Descriptions," for additional information about the above data elements.

10.5.5.7 eWIC Receipt Requirements

The credit card receipt requirements listed in section 10.1.7, "Receipt Requirements," also apply to eWIC receipts with the exception of Approved/Declined message.

In addition to those requirements, the following information must appear on eWIC card receipts:

- Payment Tender Type
- Transaction Type
- Clerk ID
- Voucher ID
- Transaction amounts
- Balances
- Approved messages
- Declined messages
- Benefit expiration date (Subelement "Earliest WIC Benefit Expiration Date" contained in Element No. 154 [WIC Product Data])

Note: Please refer to chapter 13, "Data Element Descriptions," for additional information about the above data elements.

10.5.6 Healthy Incentives Program (HIP) Transaction Processing

This section provides information about support for the Healthy Incentives Program (HIP) to evaluate health and nutrition promotion in the Supplemental Nutrition Assistance Program (SNAP).

Electronic benefit card programs from some US states are offering HIP in order to award bonuses for healthy food purchases. The goal is to provide an incentive to purchase a target group of healthy foods.

The incentive for each purchase is calculated by the authorizer. Running totals are maintained by the authorizer.

Additional balances are returned providing the incentive earned for the transaction and month-to-date incentive earned. The balances are reset at the beginning of the fiscal month.

A new data element (No. 164 [EBT Program Data]) supports this feature. For a complete description of this element, please see section 13.2, “Data Elements in Data Element Number Order.” Data layout examples for this data element are contained in appendix M, “Element Format Examples for EBT Program Data.”

This new data element is used in the EBT Data Segment (Data Segment No. 103). For a complete description of this addition, please see section 12.4, “EBT Data Segment.”

The receipt requirements listed in section 10.1.7, "Receipt Requirements," also apply to HIP receipts.

Please contact your Buypass representative for a list of states that support HIP.

10.6 Fleet Card Processing Requirements

Please refer to the most recent version of the following documents for fleet card processing requirements:

- *BUYPASS® Platform Petroleum Industry Processing Specifications*
- Any other documents published since the publication date of this document

10.7 Stored Value Card Processing Requirements

This section discusses the following stored value card (SVC) processing requirements:

- Process overview
- Card requirements
- Supported transaction types
- Receipt Requirements
- Postal Code Capture
- Track Support Requirements

10.7.1 Process Overview

A stored value card cannot be used for transactions until it has been activated. There are three activation methods:

- Activation without using a device
- Activation using a device (predefined monetary limit)
- Activation using a device (customer-defined monetary limit; issuer-activated)

Under the first method, a stored value card is activated before a customer purchases it. The card is ready to use immediately. This method requires receipt by the vendor of a fulfillment letter confirming the cards' shipment to the client.

Under the second method, a customer purchases a sealed SVC package that has a predefined monetary limit displayed on the cover. The card is activated by swiping the package through the card reader of an indoor device. If the card's magnetic stripe is invalid, the entire package is discarded, and the process is repeated with a new package. If an "approved" Response is received, the card is active and ready to use—up to the predefined monetary limit.

Under the third method, a customer purchases a stored value card that has no predefined value. The card is activated by swiping it through a card reader and entering the activation amount at an indoor device. If an “approved” Response is received, the card is active and ready to use—up to the defined monetary limit.

The card can be used until its monetary limit is exhausted. Clients can determine whether purchases can be made under a specific amount (i.e. $1.00).

If an activated card is not used within the time frame established by its authorizer, it can be deactivated at any of the participating client’s devices.

10.7.2 Card Requirements

This section discusses the following card requirements for SVC processing:

- Expiration date
- Manual entry
- Amount
- Partial Approval
- Embossing standards—First Data™ Premium Gift Card

10.7.2.1 Expiration Date

The device does not perform expiration date validation in a stored value card transaction.

10.7.2.2 Manual Entry/Bar Code

Manual entry may be allowed on a stored value card transaction. It is dependent on the authorizer.

If the POS device is unable to read the card's magnetic stripe, it should prompt the user to swipe the card again—up to three times.

When manual entry is allowed, and the POS device is unable to read the magnetic stripe after three attempts, the POS device should prompt for manual entry of the Account Number and Expiration Date (Manual entry format: MMYY). The value to be used as the expiration date appears as an embossed 4-digit number on the First Data™ Standard Gift Card and the Stored Value Systems card. An expiration date is not embossed on the front of the First Data™ Premium Gift Card; therefore, a default value of *1249* (Manual entry format: MMYY) is sent by the POS device. (**Note**: The First Data™ Premium Gift Card has a 16-digit Account Number.)

When manual entry is not allowed, and the POS device is unable to read the magnetic stripe after three attempts, a message should be displayed to the clerk, and the transaction ended.

If manual entry is used, this fact is indicated on the receipt and journal by an asterisk (*).

Bar Code Scanned Gift card should be treated as a manually entered transaction. It should be send like a manual transaction with Account Number and Expiration Date in the designated fields. The entry mode should be “01”.

Blackhawk Gift card transactions should be treated as swiped transactions. The entry mode should be received as “90” for Blackhawk Scanned transactions.

10.7.2.3 Amount

When an activation (Please see section 10.7.3.6, “Activation Transaction.”) transaction, deactivation (Please see section 10.7.3.7, “Deactivation (Cash Out) Transaction.”) transaction, or a recharged/issued (Please see section 10.7.3.8, “Recharge/Issue Card Transaction.”) transaction is performed, a prompt for the activated/recharged/issued amount is required.

10.7.2.4 Partial Approval

Stored value card issuers have the ability to approve a transaction request for an amount that is different from the request amount. When this occurs, the device will process the transaction for the approved amount returned in the transaction response and handle the remaining balance of the sale as a split tender transaction.

When a transaction is partially approved by the issuer, transaction responses contain the following information:

- A Response Code (Element No. 83) value of *F* (Indicates “approval, partial amount approved.”)

- Approved Amount (Element No. 6) contains the approved partial amount, and the POS device must prompt for payment of the remaining balance due on the sale.
- Approval Number (Element No. 5) contains the card balance—if returned by the issuer.

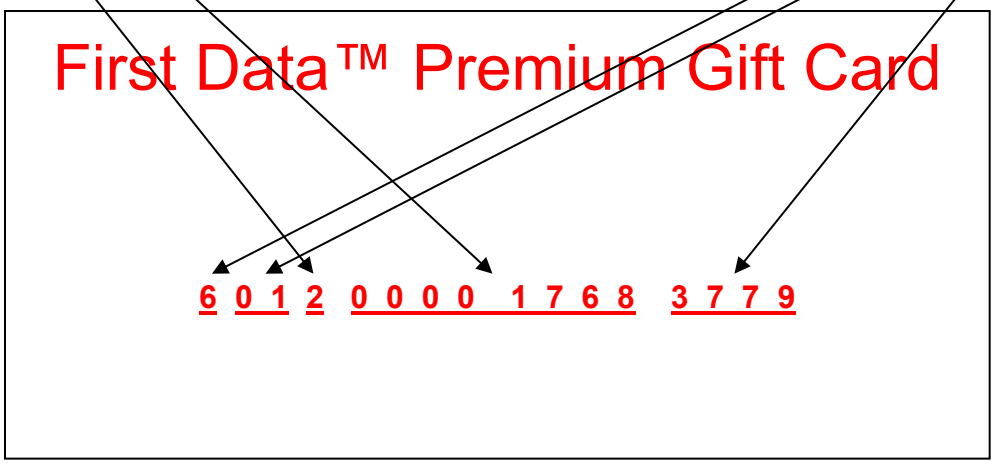
When the transaction is processed as a split tender transaction the device will print a transaction receipt indicating the amount due on the transaction, display the amount due and offer options to complete the remaining balance of the sale. Other options for payment include credit, debit, EBT, cash, or cancellation of the transaction.

10.7.2.5 Embossing Standards—First Data™ Premium Gift Card

The following table contains information about embossing standards for the First Data™ Premium Gift Card:

First Data™ Premium Gift Card
Card Number Translation
Track 2 Data to Embossed Account Number

SS	Account Number			FS	Exp Date	Service Code	PVKI	PVV	Discretionary Data		ES
	BIN	Partial Acct #	CD						Leading Digits	Check Sum	
;	601056	2 00001768	5	=	0812	000	4	0000	6 0 01	3779	?



Track Data and Alpha Character Translation:

Track 2 (alpha): ;601056XYYYYYYYY5=000100040000V0WWZZZZ?
Track 2: ;6010562000017685=000100040000600013779?

Card number (alpha): VWWX YYYY YYYY ZZZZ
Card number: 6012 0000 1768 3779

Note: The First Data™ Premium Gift Card ISO is not part of the embossed card number.

For a detailed layout of Track II magnetic stripe data, please refer to Appendix N, "First Data™ Premium Gift Card Magnetic Stripe Data Layout."

10.7.3 Supported Transaction Types

The following point-of-sale transactions are supported for SVC processing:

- Purchase/Capture
- Purchase Reversal
- Authorization Only
- Authorization Only Reversal
- Merchandise Return
- Activation
- Deactivation
- Recharge/Issue Card
- Balance Merge
- Replace Card
- Balance Inquiry
- Cancellation
- Cancellation (Time-out Reversal)

The following customer-activated transactions are supported for SVC processing:

- Purchase/Capture
- Authorization Only
- Authorization Only Reversal
- Cancellation (Time-out Reversal)

The following additional transaction types are supported for InComm:

- Account Verification (Preauthorization Activation)

10.7.3.1 Purchase/Capture Transaction

If an approved response is received, the card issuer deducts the purchase amount from the card's balance; otherwise, a declined response is sent to the device.

10.7.3.2 Purchase Reversal Transaction

This transaction adds the transaction amount to the card's balance. This amount must be the exact amount of a previous SVC transaction. (The previous transaction must have been done in the same business day.) The POS device must subtract this amount from the SVC bucket and subtract this transaction from the SVC quantity.

10.7.3.3 Authorization Only Transaction

This transaction places a lock on the card. The client decides the duration of the lock. The Authorization Only Response contains the card's balance and the authorization number.

10.7.3.4 Authorization Only Reversal Transaction

This transaction is used if the device fails, or a communication problem occurs after the device receives an approved Authorization Only Response. This transaction has an amount previously set by the client. When this transaction is received, the card issuer unlocks the SVC account.

10.7.3.5 Merchandise Return Transaction

This transaction adds the given amount to the card's balance. The POS device must subtract this amount from the SVC bucket and add this transaction to the SVC quantity.

10.7.3.6 Activation Transaction

This transaction activates the SVC's Account Number for a specific dollar amount on the card issuer's system. After a successful Activation Transaction, charges can be made against this SVC account.

10.7.3.7 Deactivation (Cash Out) Transaction

For non-First Data™ Standard Gift Card transactions, this transaction is called a Deactivation transaction. The deactivation amount in the transaction request must match the account balance. Depending upon the client's configuration, after a successful Deactivation transaction, the stored value card's Account Number can be closed on the card issuer's system, and charges can no longer be made against this Account Number.

For First Data™ Premium Gift Card transactions, this transaction is called a Cash Out transaction. The cash out amount in the transaction request must be greater than $0.00 and equal to or less than the account balance. After the request is received, the account's balance is taken to $0.00, and the amount used to bring the balance to $0.00 is returned to the POS device in the transaction response's Approved Amount field. Depending upon the client's configuration, after a successful Cash Out transaction, the stored value card's Account Number can remain open on the card issuer's system with an account balance of $0.00. (**Note**: The First Data™ Premium Gift Card has a 16-digit Account Number.)

The POS device must add this count to the SV1 count total.

The POS device must subtract this amount from the SV1 amount total.

10.7.3.8 Recharge/Issue Card Transaction

This transaction adds the amount entered at the device to the card's current balance. The added amount cannot cause the balance to be greater than the original value of the card.

10.7.3.9 Balance Merge Transaction

This transaction enables a customer to combine the balances of two stored value cards in his/her possession, and place that merged balance on one of those cards. The account balance for the remaining card is reduced to $0.00.

Note: The maximum number of cards that can be involved in this transaction is two. If a customer has more than two stored value cards in his/her possession for which balances need to be merged, the transaction can be repeated until all balances have been merged to a single card, and the remaining card balances are reduced to $0.00.

10.7.3.10 Replace Card Transaction

This transaction enables a customer to transfer the balance of any lost, stolen, or damaged card to a new card with a completely new account number. Receipts may be used to obtain the account number for lost or stolen cards. This transaction replaces the lost, stolen, or damaged card. The balance on the account for the lost, stolen, or damaged card is reduced to $0.00.

10.7.3.11 Balance Inquiry Transaction

This transaction is an information only transaction. It provides the SVC's remaining balance. This transaction should not change the SVC quantity or amount totals.

10.7.3.12 Cancellation Transaction

This transaction is used to reverse any previous stored value gift card transaction. The transaction amount must be the same amount as the transaction that is being reversed. When this transaction is received, the card issuer unlocks the stored value card account.

The POS device must subtract this count from the corresponding count total impacted by the original transaction.

The POS device must subtract this amount from the corresponding amount total impacted by the original transaction.

10.7.3.13 Cancellation (Time-out Reversal) Transaction

This transaction is used if the device fails, or a communication problem occurs during a transaction. The transaction has an amount previously set by the client. When this transaction is received, the card issuer unlocks the SVC account.

10.7.3.14 Account Verification (Preauthorization Activation) Transaction

This transaction is used to verify that the POS, about to load/preload, is valid and ready to accept funds prior to accepting tender for the card.

After performing the account verification, the POS may proceed with accepting tender and the activation. The subsequent activation is not associated with the original account verification.

Note: This transaction type is only supported for InComm cards.

10.7.4 Receipt Requirements

The credit card receipt requirements listed in section 10.1.7, "Receipt Requirements," also apply to stored value card receipts.

In addition to those requirements, the recharge amount prints on all approved stored value card receipts.

In addition to those requirements, the remaining balance (positive/negative) prints on all approved and declined stored value card receipts. The remaining balance (positive/negative) is returned as the Approval Number in the Authorization Response.

No signature line is required when printing a stored value card receipt. Optionally, a SIGNATURE NOT REQUIRED message can be printed.

In cases where the PAN is printed on transaction receipts, it must be truncated to its last four digits. A series of X's equal to the number of truncated digits precedes these four digits for a field length equal to the original Account Number's length. Truncation of the Account Number is required on all customer-activated transactions. The full Account Number should be retrievable from the POS device journal or transaction log for retrieval requests.

Note: The last four digits of the 16-digit Account Number are truncated on First Data™ Premium Gift Card receipts only. In this case, these last four digits are replaced by a series of four X's.

In the case of either a Balance Merge transaction or a Replacement/Balance Transfer transaction, it is preferred that the receipt contain the Account Number for the card with the final balance.

10.7.5 Postal Code Capture

Support has been added to allow a device to capture and send a customer's ZIP Code™ in a closed loop prepaid card transaction.

The device sends a transaction request to BUYPASS containing the ZIP Code™ in Table ID 003 of Element No. 111 (Variable Information Indicator). (Please refer to chapter 13, "Data Element Descriptions," for additional information about this data element.)

The value of the ZIP Code™ is not validated by the issuer and does not have any bearing on the approval or decline of a transaction. The ZIP Code™ is used for reporting purposes only.

10.7.6 Track Support Requirements

The following track support requirements must be met when processing all electronic stored value card transactions:

- The default track that is sent to BUYPASS on an original transaction should be Track 2.
- Card data provided in a Stored Value System (SVS) follow-on transaction can be either full or partial Track 2 Data.
- Card data provided in a First Data™ Premium Gift Card (BIN 601056) follow-on transaction using either the token from the response or 601056* card number may be sent to BUYPASS as key entered transaction.
- Card data provided in all other First Data™ Premium Gift Card follow-on transactions must be full Track 2 Data.

Note: Partial Track 2 Data consists of the Primary Account Number, followed by "=" sign—rather than a field separator—followed by the 4-digit expiration date (Storage format: YYMM). No other discretionary data follows the 4-digit expiration date.

10.8 Check Processing Requirements

This section discusses the following check processing requirements:

- Check requirements
- Supported transaction types
- Receipt requirements—Non- ECA/ TeleCheck® service processing
- Receipt requirements—ECA/ TeleCheck® service processing
- Receipt requirements— Denial Record Receipt Processing

10.8.1 Check Requirements

This section discusses the following check requirements for check processing:

- Swiped entry requirements
- Manual entry requirements

10.8.1.1 Swiped Entry Requirements

The following check processors require MICR data for check processing when swiped data entry is used:

- SCAN (ETC)
- ECA/ TeleCheck® service

10.8.1.2 Manual Entry Requirements

The following check processors require driver's license data and the alphabetical State Code (See Appendix D, Valid State Codes), for check processing when manual data entry is used:

- SCAN (ETC)
- ECA/ TeleCheck® service

The following check processor requires driver's license data, the alphabetical State Code (See Appendix D, Valid State Codes), and date of birth for check processing when manual data entry is used:

- Certegy (Formerly Telecredit)

10.8.2 Supported Transaction Types

The following transaction is supported for check processing:

- Purchase

10.8.3 Receipt Requirements—Non- ECA/ TeleCheck® Service Processing

No receipt is required for non- ECA/ TeleCheck® service processing.

The Approval/Decline response from the authorizer is displayed on the device only.

10.8.4 Receipt Requirements— ECA/ TeleCheck® Service Processing

A merchant authorization and a copy of the authorization for the check writer must be printed for each approved ECA/ TeleCheck® service transaction. The check writer's copy may be printed on the back of the paper check or on a separate document and provided to the check writer. The check writer must sign the merchant's copy before the transaction is considered complete.

Once the check writer signs the authorization, the paper check must be voided and handed back to the check writer at the POS. Merchants must have a process in place to support voiding the check.

10.8.4.1 Required Fields

The following information must appear on each copy of the receipt:

- Merchant Name
- City, State and ZIP Code™/ZIP+4®
- Merchant Telephone number
- ECA/ TeleCheck® Service Telephone Number (in trailer text)
- Terminal ID
- ECA/ TeleCheck® Merchant ID
- Date and Time
- Transaction Number
- ECA/ TeleCheck® Service Trace ID
- Approval Number
- Total Check Amount
- Response Check Number
- Return Fee Amount
- ECA/ TeleCheck® Acceptance Statement allowing the electronic deposit of the customer's check including the appropriate check return fee.
- Authorization for single point-of-sale, warranty or nonwarranty, ECA/ TeleCheck® notice for consumer or company accounts:

  "When you provide a check as payment, you authorize us to use information from your check to process a one-time Electronic Funds Transaction (EFT) or draft from your account, or process the payment as a check transaction. You also authorize us to process credit adjustment, if applicable. If your payment is returned unpaid, you authorize us to collect your payment and the Return Fee amount below by an EFT(s) or drafts(s) from your account. If you are presenting a corporate check, you make these representations as an authorized corporate representative."

Note: Line for customer's signature must be printed below the ECA/ TeleCheck® Authorization notice.

Additional recommended fields include:

- Customer Telephone Number, if prompted
- Custom Field prompt and data
- Merchant Trace ID

The Merchant copy of the receipt—at the discretion of ECA/ TeleCheck® service—must also print the following:

- Line for customer's name
- Line for customer's street address
- Line for customer's state and ZIP Code™/ZIP+4®

10.8.5 Receipt Requirements— Denial Record Receipt Processing

This section describes the print requirements for Denial Record service transaction receipts.

The Denial Record Receipt is used to print the details and reference number (Denial Record Number) for the consumer in a transaction decline scenario. The alternative would be to provide TeleCheck pre-printed Denial Record receipt (TML-1791-R.).

The Denial Record number is used with all TeleCheck products.

10.8.5.1 Required Fields

The following information must be printed on each copy of Version 2 Denial Receipt. This receipt is to be used if cashiers are NOT utilizing courtesy cards.

- Denial Record Number
- Check Amount
- NACHA Language
- TeleCheck Phone Number

Note: For sample of the receipt, please contact your BUYPASS TeleCheck representative.

10.9 Loyalty Card Processing Requirements

Loyalty card programs provide merchants with a system that allows consumers to accumulate real-time rewards, discounts, and messaging based on POS transactions.

This section discusses the following loyalty card processing requirements:

- Card requirements
- Supported transaction types
- Supported loyalty advice functions
- Receipt requirements

10.9.1 Card Requirements

This section discusses the following card requirements for loyalty card processing:

- Expiration date
- Manual entry

10.9.1.1 Expiration Date

The device does not perform expiration date validation in a loyalty card transaction.

10.9.1.2 Manual Entry

Manual entry is allowed on a loyalty card transaction. If the device is unable to read the card's magnetic strip, it should prompt the user to swipe the card again—up to three times. If unable to read the magnetic strip after three attempts, the device should prompt for manual entry of the Account Number and Expiration Date (MMYY). The user is allowed to skip through the Expiration Date prompt when manually keyed data is entered and an expiration date is not present on the card. When no expiration data is present, the device sends a default value of 1249 (MMYY).

Note: When a loyalty card is not present, the Street Address and Loyalty Phone Number associated with the consumer's existing loyalty account are manually keyed at the device in lieu of the Loyalty Account Number.

10.9.2 Supported Transaction Types

The following point-of-sale transactions can include loyalty card information:

- Purchase/Capture
- Purchase Reversal
- Time-out Reversal (TOR)

10.9.3 Supported Loyalty Advice Functions

The following loyalty advice functions are supported for loyalty card processing:

- Add account

  This transaction is used to add a new loyalty account to the loyalty host from a POS device. The clerk swipes a new loyalty card and enters the customer's street number and telephone number.

- Coupon redemption

  This transaction is used to redeem coupons for payment from a POS device. The clerk swipes the loyalty card or manually keys the loyalty account number and enters the Coupon ID and Coupon Amount to redeem for payment. If the loyalty card is not available, in lieu of the Loyalty Account Number, the clerk enters both the consumer's street number and telephone number.

- Expiration date update

  This transaction is used to update the expiration date of an account on the loyalty host from a POS device. The clerk swipes the loyalty card or manually keys the account number. If the loyalty card is not available, in lieu of the Loyalty Account Number, the clerk enters both the consumer's street number and telephone number. The host will then extend the loyalty account for a predefined amount of time.

- Account inquiry

  This transaction is used to inquire about the status of a loyalty account from a POS device. The clerk swipes the loyalty card or manually keys the account number. If the loyalty card is not available, in lieu of the Loyalty Account Number, the clerk enters both the consumer's street number and telephone number. The host returns the Loyalty Print Data in the response. The data returned to the device is a pre-programmed message maintained by the merchant at the loyalty host.

- Points redemption

  This transaction is used to redeem points for payment from a POS device. The clerk swipes the loyalty card or manually keys the Loyalty Account Number and enters the number of points to redeem for payment. If the loyalty card is not available, in lieu of the Loyalty Account Number, the clerk enters both the consumer's street number and telephone number.

- Reversal of coupon redeem

  This transaction is used to reverse coupons redeemed on a previously approved coupon redemption transaction from a POS device. The clerk swipes the loyalty card or manually keys the account number, and enters the Coupon ID, Coupon Amount, and approval number of the previous transaction receipt. If the loyalty card is not available, in lieu of the Loyalty Account Number, the clerk enters both the consumer's street number and telephone number.

- Reversal of points redeemed

  This transaction is used to reverse points redeemed on a previously approved point redemption transaction from a POS device. The clerk swipes the loyalty card or manually keys the account number, and enters the point amount and approval number of the previous transaction receipt. If the loyalty card is not available, in lieu of the Loyalty Account Number, the clerk enters both the consumer's street number and telephone number.

- Sale update

This transaction is used to update the loyalty host with noncredit card sale data for a loyalty account from a POS device. The clerk swipes the loyalty card or manually keys the Loyalty Account Number, and enters the amount of the noncredit card or cash sale. If the loyalty card is not available, in lieu of the Loyalty Account Number, the clerk enters both the consumer's street number and telephone number.

- Account update

  This transaction is used to update street address and phone number information on a loyalty account from a POS device. The clerk swipes the loyalty card and must enter both the new street number and new telephone number the customer wants to associate with the existing loyalty account.

- Totals Report, Loyalty

  This transaction is used to request a summary of all loyalty transactions and subtotal by card number. The clerk selects the totals menu option from the loyalty menu at the device, which initiates a request to the loyalty host. The host returns the loyalty totals report to the device to print.

10.9.4 Receipt Requirements

The credit card receipt requirements listed in section 10.1.7, "Receipt Requirements," also apply to loyalty cards.

Additional loyalty card information that can be included on loyalty card receipts is described in Appendix K. Additional Information Data Layouts, in Table 008, "Loyalty Information – Version 1," and Table 010, "Loyalty Information – Version 2." Any of this information that is received by the device is printed.

10.10 Software Update Processing Requirements

Note: Vendors who support their own applications do not use software update processing.

Software update processing involves an automated request by an active device for a full application download to the device from the device management system supported by BUYPASS.

The following steps must be included in this process:

1. A bit is set on the merchant profile at BUYPASS indicating that the device must request a DLL.
2. BUYPASS sends a Download Indicator of *1* in the response to the next transaction request from the device.
3. The device sends a Table Load request to BUYPASS.
4. BUYPASS responds to the Table Load request by sending the Software Dial Load Data Segment and Software IP Load Data Segment.
5. The device stores the data received in the Software Load response:
    a. New Software Version
    b. Software Terminal Record ID
    c. Software Load Phone Number or Software Load IP/URL Address
    d. Software Load Request Date
    e. Software Load Request Time
6. At the designated date and time, the device dials the Software Load Phone Number /Software Load IP/URL Address and initiates a request for a full application load.
7. Upon completion of the full application load, the device dials BUYPASS and requests a Table Load.
8. BUYPASS responds by sending the Table Load without requiring the bit to be set on the merchant profile.

Notes: The device can attempt to request the automated full application load a maximum of three times.

If a load request to Software Load Phone Number/Software Load IP/URL Address is unsuccessful, the device prints the “decline” message.

10.11 Electronic Mail Processing

This chapter contains information about electronic mail processing involving a client, BUYPASS, and a client's store point-of-sale (POS) device.

10.11.1 Electronic Mail Retrieval

Data originating from a client can be sent to the client's store POS device via BUYPASS. The data is sent to BUYPASS where it is stored until transmission to the client's store POS device in one of the following ways:

- A device operator, using the client's store POS device, sends an Electronic Mail Request for the file to BUYPASS; or
- BUYPASS responds to an automated Electronic Mail Request from the client's store POS device.

Either of these methods can be used prior to the end-of-business day, up to the time when a device ends its day.

The Electronic Mail Request in either case uses a Prompt Code of *981*.

Note: For retrieval of proprietary card data, the Electronic Mail Request uses a Prompt Code of *996*.

There can be up to 750 bytes of print data for each Electronic Mail Request.

10.11.2 Electronic Mail Submission

In addition, a client's store POS device can send data to the client via BUYPASS in one of the following ways:

- A device operator, using the client's store POS device, sends data to BUYPASS using an Electronic Mail Request; or
- The client's store POS device sends data to BUYPASS using an automated Electronic Mail Request.

The Electronic Mail Request uses a Prompt Code of *995*.

The Electronic Mail Request contains up to 217 bytes of data.

BUYPASS sends an Electronic Mail Response to the client's store POS device to confirm receipt of the Electronic Mail Request.

The data is stored at BUYPASS.

10.12 TransArmor Processing Requirements

Note: For detailed information on TransArmor, refer the BUYPASS®_Platform_ATL105_Specification_Updates_for_TransArmor_Processing document, available on the Direct Platform Specifications Portal.

10.13 Payment Tokenization

Note: Please check with your First Data Certification Representative for the availability of this new feature.

This chapter provides an overview and describes processing requirements for implementing the Payment Token processing feature involving the BUYPASS platform ATL105 message formats (software version ATL105).

With the introduction of new payments, such as, mobile phones, and tablets, etc., the need for advanced protection against counterfeit, account misuse, and other forms of fraud, has increased.

Visa, MasterCard, Discover, American Express, Interlink, Maestro STAR, NYCE, Pulse, Accel, and Shazam have introduced payment token which is the replacement of the Primary Account Number (PAN) with a token number that represents the PAN by device and/or e-commerce payment transactions.

Payment token issuance is performed by a token vault that generates and stores the payment token and the associated cardholder PAN. The operator of the token vault is referred to as the token service provider. The token requestor is typically an acquirer, merchant or digital wallet provider that will use the payment token to replace the PAN in its database and in transaction processing. Payment processes that need to support token processing may include authorization, capture, clearing, and exception processing.

Note: For additional information on payment tokenization terminologies, please refer to Appendix O (Payment Tokenization Terminologies) in this document.

10.13.1 Salient Features of Payment Tokenization

Following are the general features of Tokenization:

- Payment Token ranges will be in the existing ranges of the relevant card brand
- Payment Tokenization is the replacement of the Primary Account Number (PAN) with a token number that represents the PAN by device and/or eCommerce payment transactions
- Payment Tokens are mapped to the PAN, leveraging the existing payments infrastructure and messaging formats for authorization and processing
- The Payment Token will be used for the entire life-cycle of the transaction for all card brands
- EMVCo manages the Payment Token Standard

10.13.2 Payment Tokenization Usages

A Payment token transaction, where the actual card primary account number is replaced by an association payment token, could take place in any of the following scenarios:

- NFC transactions, when customer's NFC device is sending Payment Token instead of actual card number
- eCommerce transactions with security data that contains token cryptogram, like VPAS, UCAF
- Card-on-file transactions, where the initial transaction contains the token with valid cryptogram. The token and token related data could be saved by merchants, and used for subsequent transactions, like recurring payment

The following sections describe the Payment Token transactions which are currently supported by BUYPASS:

10.13.2.1 Near Frequency Communication (NFC) - In Person Purchase

Consumer has a device such as tablet, smartphone, etc. and the device has been provisioned with payment capability. A token is provisioned in the consumer’s device. The consumer walks to a store, for example, Coffee Is Us and buys a double espresso. He uses his “digital wallet” instead of the Coffee Is Us Merchant App on his smartphone/tablet to tap and pay. The tokenized card information is read and the appropriate contactless data is captured by the POS. Merchant submits the authorization as a contactless transaction exactly as they do today.

10.13.2.2 Mobile App (In App) – eCommerce

Consumer has a device such as tablet, smartphone, etc. and the device has been provisioned with payment capability. A token is provisioned in the consumer’s device. Consumer activates the “digital wallet” on the device when using a Merchant App for purchasing. For example: the consumer downloads the “TV Store” app (an example of Merchant App) to his/her device. The consumer then decides to buy a TV and uses the TV Store app instead of going to TVStore.com website or to the physical store location to initiate the transaction. The TV Store App asks if the consumer wants to pay with the “digital wallet”. The Merchant App interacts with the wallet to get the token and merchant sends authorization request with appropriate data including the cryptogram.

10.13.2.3 Mobile App (In App) – eCommerce Subsequent Transactions (such as Recurring, etc)

Consumer purchases 4 pairs of shoes on Fancyshoe.com via the Merchant App on their tablet/smartphone and decides to pay with the “digital wallet” on the tablet/smartphone. The merchant realizes that they have only 3 pairs in stock and 1 pair is on back order. Depending on the timeframe, the card brand allows the authorization to remain valid, while a new authorization may be required once the 4th pair of shoe is shipped. If a new authorization is required, there are specific field requirements that must be included in the authorization request for a token transaction to ensure that the subsequent authorization is not declined. Specific field requirements vary by card brand.

10.13.2.4 Card On File (COF)

Consumer has established a relationship with an eCommerce merchant, where they select the option of storing their cardholder data (card number, expiration date) with the merchant for future transactions. In this scenario the payment token provided by the card brand replaces the PAN that is stored at the merchant’s end.

10.13.2.4.1 Stored Credentials

A stored credential is information, including, but not limited to, an account number or a payment token. It is stored by a merchant, its agent, a payment facilitator or a staged digital wallet operator to process future transactions for a cardholder.

A credential is not considered a stored credential when details of a card are stored to:

- Complete a single transaction
- Complete multiple authorizations related to the initial transaction, such as Incremental transactions

Types of Stored Credential transactions are as follows:

- **Cardholder-Initiated Transaction (CIT) –** A cardholder-initiated transaction is any transaction where the cardholder is actively participating in the transaction. This can be either at a terminal in-store or through a checkout experience online, or a subsequent stored credential transaction.
- **Merchant-Initiated Transaction (MIT) –** A merchant-initiated transaction is a subsequent transaction with already stored credentials. The cardholder has given prior consent to the merchant to store payment credentials for future use without active engagement from the cardholder.

Subsequent Transaction with respect to Stored Credentials:

Once the card credentials are shared for stored credential at a merchant location or website, the following transactions are considered to be subsequent transactions:

Standing instructions MITs:

- Separate transaction to buy a different item, once again, using the same card at the same merchant location or website
- Recurring transaction associated with the original transaction
- Installment transaction associated with the original transaction

10.13.2.4.2 Use Cases

Note: The use cases provided below are not exhaustive and all encompassing. The list of use cases only describes some of the common stored credential scenarios.

References: The use cases refer to the following sections/fields of the ATL105 specification. The sections/fields have been provided for ease of reference:

- Variable Information Indicator (Data Element 111), Table ID 32 and Sub-table ID 08 (Initial/Credential-on-file) indicator
- Variable Information Indicator (Data Element 111), Table ID 05 (POS Entry Mode Information)
- Variable Information Indicator (Data Element 111), Table ID 49 (Merchant Supplementary Data), Sub Table ID 02 (Transaction Initiation Indicator)
- Variable Information Indicator (Data Element 111), Table ID 44 (Original Transaction Identifier)
- Variable Information Indicator (Data Element 111), Table ID 15 (Discover® Network Retrieval Reference Number)
- Appendix K (Additional Information Data Layouts), Table ID 028 (Transaction Identifier)
- Appendix K (Additional Information Data Layouts), Table ID 012 (Discover® Network Retrieval Reference Number)
- Appendix J (Valid Point-of-Service Entry Mode Codes)
- Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 01-08 (Total Payment Count, Payment Type, MIT Amount, Unique ID, Frequency of MIT, Validation Flag, Validation Reference, Sequence Indicator).
- Variable Information Indicator (Data Element 111), Table ID 49 (Merchant Supplementary Data), Sub Table ID 13 Identifies Installment Payment Type
- Soft Descriptor – Merchant Name (Data Element 111) Table ID 047
  When this field is sent from a Third Party Payment Providor, the underlying merchant name field must also be present in the transaction.
  Note: The Merchant Name must be submitted in the format as follows:
  Third party installment provider name followed by an asterisk (*) followed by the underlying retailer name, e.g. “XYZ*A SMALL CO”.
- SIC Information (Data Segment 111, Table ID 007) with MCC/SIC of underlying retailer of a third-party payment provider for Installments must be sent for Discover.
- Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 (MIT/CIT category and subcategory data) used to indicate category and subcategory data for Cardholder initiated transaction (CIT) and Merchant initiated transactions (MIT) for MasterCard card type.

10.13.2.4.2.1 Example # 1 – CIT (Cardholder Initiated)

Transaction Sequence	**Transaction Attributes**
Storing credentials for the first time	• Cardholder walks in a pharmacy to buy prescribed drugs • At the check-out, the merchant offers or the POS prompts to store the credential for automatic prescription drugs pickup and refills • Cardholder chooses to do so • The transaction will be a Cardholder initiated transaction with the following fields: ○ **Data Element 111, Table ID 05 - Point-of-Service Entry Mode** - original POS Entry Mode (for example: 02, 90 etc. respectively as defined in Appendix J) ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'I' (Initial) ○ For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to CIT transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.
Subsequent transaction initiated by cardholder, using stored credentials (Ad hoc or Unscheduled)	• After 2 months the cardholder walks in the same pharmacy with a different prescription for pickup or refill the previous prescription • The transaction will be a Cardholder initiated transaction with the following fields : ○ **Data Element 111, Table ID 05 - Point-of-Service Entry Mode** - POS Entry Mode value of '10' ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08**

Transaction Sequence	Transaction Attributes
	- Value 'C' (Credential on File) ○ **Data Element 111, Table ID 49 - Merchant Supplementary Data, Sub Table ID 02** - Value 'C' (Cardholder Initiated) ○ For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to CIT ad hoc/unscheduled transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.
Recurring - Standing order (Variable amount and Fixed frequency) for Cardholder initiated transactions (CIT)	• Cardholder agrees to store the credential-on-file and initiates a first transaction in a series intended to be for a variable amount and a fixed frequency. Example: - The initial transaction to store the credential-onfile for a monthly utility payment. • The transaction will be a Cardholder initiated transaction with the following fields: ○ **Data Element 111, Table ID 05 - Point-of-Service Entry Mode** - original POS Entry Mode (for example: 02, 90 etc. respectively as defined in Appendix J) ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'I' (Initial) ○ **Data Element 111, Table ID 13 – Special Payments Indicator**

Transaction Sequence	Transaction Attributes
	Information – Value 'R' (Recurring). ○ **Data Element 111, Table ID 49 - Merchant Supplementary Data, Sub Table ID 02** - Value 'C' (Customer Initiated) ○ For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to CIT transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.
Recurring - Subscription (Fixed amount and Fixed frequency) for Cardholder initiated transactions (CIT).	• Cardholder agrees to store the credential-on-file and initiates a first transaction in a series intended to be for a fixed amount and a fixed frequency. Example: - The initial transaction to store the credential-onfile for a monthly newspaper subscription. • The transaction will be a Cardholder initiated transaction with the following fields: ○ **Data Element 111, Table ID 05 - Point-of-Service Entry Mode** - original POS Entry Mode (for example: 02, 90 etc. respectively as defined in Appendix J) ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'I' (Initial). ○ **Data Element 111, Table ID 13 – Special Payments Indicator Information** – Value 'R' (Recurring). ○ **Data Element 111, Table ID 49 - Merchant Supplementary Data, Sub Table ID 02** - Value 'C' (Customer Initiated)

Transaction Sequence	Transaction Attributes
	○ For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to CIT transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.
Cardholder initiated Installment transaction	• Cardholder agrees to store the credential-on-file to establish an installment plan and initiates a first transaction in a series. The installment plan should reflect the single purchase of goods/ services with a known amount and set frequency over a specified duration. Example: - The initial transaction to store the credential-onfile for a purchase to be paid through installment payments. • The transaction will be a Cardholder initiated transaction with the following fields: ○ **Data Element 111, Table ID 05 - Point-of-Service Entry Mode** - original POS Entry Mode (for example: 02, 90 etc. respectively as defined in Appendix J) ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'I' (Initial) ○ **Data Element 111, Table ID 13 – Special Payments Indicator Information** – Value 'I' (Installment). ○ **Data Element 111, Table ID 49 - Merchant Supplementary Data, Sub Table ID 02** - Value 'C' (Customer Initiated)

Transaction Sequence	Transaction Attributes
	○ For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to CIT transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.

10.13.2.4.2.2 Example # 2 - MIT (Merchant Initiated) Recurring

Transaction Sequence	**Transaction Attributes**
Recurring – first transaction with no supplies	• Cardholder registers at B.com on the 1st of the month and stores credentials to receive monthly supplies of paper towels and cleaning liquid on the 30th of each month • The first transaction will be a cardholder initiated $0.00 transaction with the following fields : ○ **Data Element 111, Table ID 13 – Special Payments Indicator Information** – Value 'R' (Recurring) ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'I' (Initial) • The merchant receives the Visa/AmexTransaction ID in Segment 112, Table ID 028 to be used later in subsequent recurring transactions • The merchant receives the Discover Network Reference ID in Segment 112, Table ID 012 to be used later in subsequent recurring transactions
Recurring – first transaction with supplies	• Cardholder registers at B.com and stores credentials on the 30th of the month to order the monthly supplies of paper towels and cleaning liquid the same day • The first transaction will be a cardholder initiated transaction with actual authorization amount and the following fields : ○ **Data Element 111, Table ID 13 – Special Payments Indicator Information** – Value 'R' (Recurring) ○ **Data Element 111, Table ID 32 – POS Additional Data Information (Table ID 33) Sub table ID 08** - Value 'I' (Initial) • The merchant receives the Visa/Amex Transaction ID in Segment 112, Table ID

Transaction Sequence	Transaction Attributes
	028 to be used later in subsequent recurring transactions • The merchant receives the Discover Network Reference ID in Segment 112, Table ID 012 to be used later in subsequent recurring transactions
Recurring – subsequent transaction initiated by merchant with stored credentials	• All subsequent recurring transaction on 30th of every month will be MIT with the following fields: ○ **Data Element 111, Table ID 05 - Point-of-Service Entry Mode** - POS Entry Mode value of '10' ○ **Data Element 111, Table ID 13 – Special Payments Indicator Information** - Value 'R' (Recurring) ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'C' (Credential on File) ○ **Data Element 111, Table ID 49 - Merchant Supplementary Data, Sub Table ID 02** - Value 'M' (Merchant Initiated) ○ For Visa, Discover, and Amex recurring transactions, merchants should include the original Visa Tran ID/Amex Tran ID (Data Element 111, Table ID 44) and the Discover Network Reference ID (Data element 111, Table ID 15) in request ○ For Discover subsequent recurring transactions, merchants should also send the First Authorization Amount in Element No. 111 Variable Information Indicator, table ID 045. ○ For Discover subsequent recurring transactions, merchants should send the Stored Credential Additional Data (Data Element 111, Table ID 56, Sub Table ID 01 to 08) if cards are issued in India and the amount is above $52

Transaction Sequence	Transaction Attributes
	○ For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to MIT transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.

10.13.2.4.2.3 Example # 3 - MIT (Merchant Initiated) Installment

Transaction Sequence	Transaction Attributes
Installment – first transaction	• Cardholder walks in to Store B, and buys a television for $1000. • Cardholder registers and stores credentials for a monthly installment of $100. • The transaction will be a Cardholder initiated transaction with the following fields : ○ **Data Element 111, Table ID 13 – Special Payments Indicator Information** – Value 'I' (Installment) ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'I' (Initial) • The merchant receives the Transaction ID (Visa and Amex) in Segment 112, Table ID 028 to be used later in subsequent installment transactions • The merchant receives the Discover Network Reference ID in Segment 112, Table ID 012 to be used later in subsequent installment transactions
Installment – subsequent transaction initiated	• All subsequent installment payments will be MIT with the following fields: ○ **Data Element 111, Table ID 05 -**

Transaction Sequence	Transaction Attributes
by merchant with stored credential	**Point-of-Service Entry Mode** - POS Entry Mode value of '10' ○ **Data Element 111, Table ID 49 - Merchant Supplementary Data, Sub Table ID 02** - Value 'M' (Merchant Initiated) ○ **Data Element 111, Table ID 13 – Special Payments Indicator Information** - Value 'I' (Installment) ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'C' (Credential on File) ○ For Visa, Discover, and Amex installment transactions, merchants should include the original Visa Tran ID/Amex Tran ID (Data Element 111, Table ID 44) and the Discover Network Reference ID (Data element 111, Table ID 15) in request ○ For Discover subsequent installment transactions, merchants should also send the First Authorization Amount in Element No. 111 Variable Information Indicator, Table ID 045. ○ **For Discover subsequent installment transactions:** ○ Merchants should send the Stored Credential Additional Data (Data Element 111, Table ID 56, Sub Table ID 01 to 08) ○ Merchants should include Installment Payment Type (Data Element 111, Table ID 49, Sub table ID 13) to identify the type of installment payment ○ Third-party payment provider should send name in the format 'Third Party* Merchant Name' in Soft Descriptor – Merchant name (Data Element 111, Table ID 47) e.g. "XYZ*A SMALL CO"

Transaction Sequence	Transaction Attributes
	○ SIC Information should be MCC/SIC code of third-party payment provider underlying retailer (Data Element 111, Table ID 007) ○ For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to MIT transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.

10.13.2.4.2.4 Example # 4 - Unscheduled Credential on File

Unscheduled Credential on File is a transaction, with fixed or variable amount using stored credential, which doesn't occur on a scheduled date. The cardholder has provided consent for the merchant to initiate one or more future transactions. An example of such transaction is an account auto-top up transaction. In BUYPASS, any merchant initiated COF transaction where the recurring or installment indicators are not received, are treated as unscheduled.

Transaction Sequence	Transaction Attributes
First transaction	• Cardholder registers at TOLLROAD.com for toll payment. The credentials are stored for future unscheduled toll payment transactions. • The transaction will be a Cardholder initiated transaction with the following fields : ○ **Data Element 111, Table ID 05 - Point-of-Service Entry Mode** - POS Entry Mode value of '01' ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'I' (Initial) • The merchant receives the Visa/Amex Transaction ID in Segment 112, Table ID 028 to be used later in subsequent merchant initiated transactions

Transaction Sequence	Transaction Attributes
	• The merchant receives the Discover Network Reference ID in Segment 112, Table ID 012 to be used later in subsequent merchant initiated transactions • For Discover card type Delayed card sale transactions, Variable information Data Element 111, Table ID 16 Delayed card sale Indicator must be 'D'. • For Discover card type No Show Charge transactions, Variable information Data Element 111, Table ID 17 No Show Charge Indicator must be 'N'.
Unscheduled – subsequent transaction initiated by merchant with stored credentials	• Merchant initiated unscheduled subsequent transaction (as and when paying for toll) with the following fields: ○ **Data Element 111, Table ID 05 - Point-of-Service Entry Mode** - POS Entry Mode value of '10' ○ **Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08** - Value 'C' (Credential on File) ○ **Data Element 111, Table ID 49 - Merchant Supplementary Data, Sub Table ID 02** - Value 'M' (Merchant Initiated) ○ Visa Tran ID, Amex Tran ID, and Discover Network Reference number received in the response of the original authorization request must be sent in all subsequent merchant initiated transactions ○ For Discover subsequent merchant initiated transactions, merchants should also send the First Authorization Amount in Element No. 111 Variable Information Indicator, Table ID 045. ○ For Discover subsequent recurring transactions, merchants should send the Stored Credential

Transaction Sequence	Transaction Attributes
	Additional Data (Elelment 111, Table ID 56, Sub Table ID 01 to 08) if cards are issued in India and amount is above $52. ○ For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to MIT transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details. ○ For Discover card type Delayed card sale transactions, Variable information Data Element 111, Table ID 16 Delayed card sale Indicator must be 'D'. ○ For Discover card type No Show Charge transactions, Variable information Data Element 111, Table ID 17 No Show Charge Indicator must be 'N'.

Note: For further details, refer to Appendix Z (Stored Credential).

10.13.3 Supported Payment Token Usage

The following grid displays the supported payment token usage by card types:

Card Type/Network	In App	NFC	COF
Visa	√	√	√
MasterCard	√	√	√
AMEX	√	√	√
Discover	√	√	√
STAR		√	√

Card Type/Network	In App	NFC	COF
NYCE		√	√
PULSE		√	√
ACCEL		√	√
Shazam		√	√
Interlink		√	
Maestro		√	

10.13.4 Payment Tokenization Processing Rules

This section provides information on the new values and new tables added to support Payment Tokenization processing through the ATL105 message formats (software version ATL105).

The following data elements/data segments have been impacted with the addition of new value to support Payment Tokenization processing:

- Data Segment No. 123 (NFC Payment Tokenization Data Segment): This data segment applies to NFC Payment Tokenization only. It is required on all initial and recurring transactions involving tokenized data.
- Data Element No.195 (CAVV Revised Format): This data element applies to Visa only. It identifies Verified By Visa Format. Authentication Tracking Number (ATN) replaces XID to support payment tokenization.
- Data Element No.196 (Token Requestor ID): This data element identifies merchant's Token Requestor ID
- Data Element No.197 (Token PAN Suffix): This data element identifies the 4 digits of the cardholder PAN to be printed on the transaction receipt.
- Data Element No. 202 (Cryptogram Token Data): This data element is specific to In App transactions and represents merchant's cryptogram token data block A (or block A and B).

The following data elements/data segments have been impacted with the addition of new tables/sub tables to support Payment Tokenization processing:

- Data Element No.111 (Variable Information Indicator): Sub-Table ID "06". This new sub table has been added under POS Additional Data Information. This sub-table is specific to NFC payment transaction and indicates whether NFC payment tokenization is supported.

10.14 EMV Processing Requirements

Note: For EMV processing -

- Refer to First Data EMV Implementation Guide, available on the Direct Platform Specifications Portal.
- The EMV Implementation Guide must be used in conjunction with the ATL105 Message Format Specifications.

This section discusses the following EMV card processing requirements:

- Process Overview
- Card requirements
- Supported transaction types
- EMV PIN (Personal Identification Number) Processing

10.14.1 Process Overview

EMV stands for Europay, MasterCard and Visa. It represents a global standard for inter-operation of integrated circuit cards (IC cards or "chip cards") through IC card capable point-of-sale (POS) terminals and automated teller machines (ATMs). EMV is used for authenticating credit and debit card transactions.

The most widely known chip card implementations of EMV standard are:

- VSDC - VISA
- M/Chip - MasterCard
- AEIPS - American Express
- J Smart - JCB
- D-PAS - Discover/Diners Club International

There are two types of chip card technology – contact and contactless:

- Contact - When a chip card is inserted into a card acceptance device, the contact allows the chip to connect to a reader. This connection enables the chip to get power from and exchange data with the terminal.
- Contactless – When a contactless chip card is held within a couple of inches of a contactless-capable reader, the reader energizes the chip embedded in the card and allows exchange of data via radio frequency without the card ever leaving the cardholder's possession.

The benefits of EMV contact and contactless payment cards include:

- Improved security against credit card fraud through use of a unique online cryptogram
- Enhanced cardholder verification methods
- Chip stores more information than magnetic stripe cards
- Possibility for finer control of "offline" credit card transaction approvals

In order to perform offline data authentication, a terminal must be configured with the necessary card scheme public keys.

The card scheme (MasterCard, Visa, and Amex) Certification Authority distributes the card scheme Certificate of Authority (CA) Public Key to acquirers (i.e. First Data). Acquirers are responsible for the distribution of the card scheme CA Public Key to all terminals that support offline data authentication.

Common AID is another application that incorporates the EMV standard. This application can exist on the issued card and the terminal that can be selected for performing a financial

transaction. There are four categories of Common AIDs. Any of these AIDs may be used in processing a Common AID transaction.

- Visa Common AID
- US Maestro AID
- DNA Shared Debit AID
- Discover Common AID

The Common AID applications are actually implementations of VSDC (for Visa Common AID), M/Chip (for US Maestro AID) and D-PAS (for DNA Shared Debit AID and Discover Common AID).Those applications are licensed to other debit networks (like Star, NYCE, Pulse, etc.) and provide compliance with the Durbin amendment.
DNA Shared Debit AID is used for 'debit only' cards that do not support MasterCard, Visa or Discover Signature Debit.
Merchants should submit Visa and US Maestro Common AID transactions as 'credit / 0100 / Dual message' requests. This will provide merchants with optimal routing options. If First Data receives a credit transaction request with cashback with one of these Common AIDs, First Data will route this transaction, except Discover, to a 'single message' debit network. Discover already supports cashback on credit requests. First Data does not support cashback for Visa or MasterCard credit / Signature Debit transactions.
DNA Shared Debit AID is limited to the 'debit / 0200 / single message' networks. As a result, merchants should submit DNA Shared Debit AID transactions as 'debit/0200/single message' requests. Cards supporting the DNA Shared Debit AID do not support Visa or MasterCard Signature Debit. If First Data receives a debit transaction request with cashback with this Common AID, First Data will route this transaction to a 'single message' debit network.
Note: Please check with your First Data Certification Representative for the availability of this new feature.

10.14.2 Card Requirements

This section discusses the following card requirements for EMV card processing:

- Expiration Date
- Manual Entry
- Supported Cards
- Supported Transaction Types
- Receipt Requirements

10.14.2.1 Expiration Date

The device does not perform expiration date validation in an EMV card transaction.

10.14.2.2 Manual Entry

Manual entry of the EMV account number is allowed on all EMV card transactions. If the account number is keyed, the device prompts for Expiration Date (Format: MMYY) and any other optional or conditional data, such as card present, AVS, or CVV code.

Note: If manual entry is used, this fact is indicated on the receipt and journal by an asterisk (*) before the account number.

10.14.2.3 Supported Cards

The following credit cards are supported for EMV processing:

- American Express

- Discover®
- MasterCard®
- Visa®
- STAR® Signature Debit

The following debit cards are supported for EMV processing:

- Interlink®
- Maestro®
- STAR® West
- STAR® Southeast
- STAR® Northeast
- NYCE®
- PULSE®
- Accel®
- Shazam

The following proprietary and fleet cards are supported for EMV processing:

- Generic Proprietary (Some issuers participating)
- Voyager®
- Wright Express®

10.14.2.4 Supported Transaction Types

The following point-of-sale transactions are supported for EMV card processing:

- Purchase
- Purchase/Capture
- Authorization Only
- Merchandise Return
- Purchase Reversal
- Balance Inquiry
- Cancellation
- Time-out Reversal (TOR)

Note: A Time-out Reversal (TOR) transaction is used to reverse a transaction, if the device fails, or the transaction is aborted after communication has been initiated to the host.

EMV data is not required on Reversal transactions.

10.14.2.5 Receipt Requirements

The credit card receipt requirements listed in section 10.1.7, “Receipt Requirements,” also apply to EMV cards.

Two elements of the receipt (whether to print a signature line and what to display as a card type) are provided in the authorization response message for a Common AID transaction. The disposition of the transaction is also provided indicating how the transaction was authorized – as a single message (debit) transaction or as a Dual message (credit) transaction.

10.14.3 EMV PIN (Personal Identification Number) Processing

Note: Please check with your First Data Certification Representative for the availability of these new features.

MasterCard has mandated that Point-of-Sale devices that support EMV PIN must support both online PIN and offline PIN processing. Visa also supports this functionality but does not mandate online PIN support. This means that POS Devices processing transactions that are initiated from chip cards for credit and signature debit must have the capability to submit a PIN in the authorization request message submitted.

Following are the two modes of PIN validation during EMV payment card processing:

- Online PIN: This is applicable for credit or debit transactions, in which the PIN is electronically sent to and validated by the card issuer. The potential PIN requirement for credit transactions where a PIN pad is present is new with EMV.
- Offline PIN: Instead of sending the PIN to the issuer, the PIN entered by the consumer is matched to that on an application housed on the EMV chip card. This functionality is exclusive to EMV card transactions. The authorization request may still be sent to the issuer. **Note:** In a traditional PIN Debit transaction, the PIN field is mandatory. Either a PIN or all F's, for offline PIN, must be sent in this mandatory field or the transaction will decline as "invalid transaction". In a credit/signature debit transaction, a PIN of all F's is NOT required since PINs are an optional field for a credit/signature debit transaction.

10.14.4 EMV Fallback Transaction Processing

For the purposes of this document, the term "Fallback" is defined as the acceptance of chip cards via magnetic-stripe processing at a chip-capable device. EMV POS Solutions must accept both chip and magnetic-stripe cards and thus have both chip and magnetic-stripe readers.

If a chip card is presented at a chip capable terminal and a chip transaction cannot be performed, then the merchant may process the chip card as a magnetic stripe read (swipe). The resultant transaction is referred to as a fallback transaction and must be properly identified as such in the authorization message. Fallback transactions are less secure than EMV but continue to protect the merchant from the liability shift when performed correctly.

Regardless, a Fallback will occur whenever an EMV Certified solution cannot read or obtain a viable matching AID. MSR/Fallback Entry Mode sample usage for contact transactions are as following:

IF	THEN	STATUS
• The device is EMV enabled and • The card is chip and • The card is swiped	Always send the Fallback Entry Mode of '80'or spec equivalent	EMV Fallback transaction
• The card is NOT a chip card or • The device is not chip capable and • The card is swiped	Send as a MSR Entry Mode of '90'or spec equivalent	Regular MSR swipe transaction

11. Message Formats

This chapter discusses the BUYPASS variable length message formats.

Note: For detailed information about individual data elements within the messages, refer to chapter 13, "Data Element Descriptions." The element descriptions in this chapter focus on the purpose of the data, rather than a detailed description of the content of the data.

The request and response messages supported by these specifications are listed below:

- Financial transactions
- Loyalty card transactions
- ECA/ TeleCheck® service transactions
- Totals
  - ◊ Sending totals to BUYPASS
  - ◊ Sending totals with proprietary data to BUYPASS
- Electronic mail
  - ◊ Retrieving electronic mail from BUYPASS
  - ◊ Sending electronic mail to BUYPASS
- Communications test
- Downloads
  - ◊ Table Load (Load Type: P)
  - ◊ Phone Load (Load Type: P)
  - ◊ Date and Time Load (Load Type: D)
  - ◊ Software Load (Load Type: P)

A request is a message from a customer's device.

A response is a message from BUYPASS.

The following message format characteristics are used—where applicable—for each message:

- **Data Section Number**

  Unique number (1–3) that identifies one of the three sections within a Financial Transaction/Totals Request message record.
- **Field Number**

  Relative number that identifies the relative position, beginning with *1*, of each data element within a message record.
- **Segment Number**

  Unique number (100–105, 108–114, and DL1–DL5) that identifies one of the data segments within a message record.

  Note: Data Segment Nos. 106 and 107 are reserved for proprietary use.
- **Segment Name**

  Unique name associated with a specific data segment within a message record. Each segment contains a specific type of data.
- **Element Number**

  Unique number associated with a data element. Use this number to locate an element and its description in chapter 13, "Data Element Descriptions."
- **Element Name**

  Identifies a specific unit of information, regardless of where it appears in a message record.
- **Position**

  Identifies the starting position for a data element within a message record, relative to *1*.
- **Length**

  Identifies the maximum or fixed number of characters a data segment or data element can contain.
- **Required/Optional/Conditional Entry**

  R indicates a data element is required in the message; *O* indicates a data element is optional in the message; and *C* indicates a data element is conditional in the message.
- **Description**

  Describes the purpose of the data element. For additional information, refer to chapter 13, "Data Element Descriptions."

11.1 Financial Transactions

This section contains message formats for all financial transaction requests and responses.

Please refer to section 11.2, "Loyalty Card Transactions," for Loyalty Card message formats.

Please refer to section 11.3, "ECA/ TeleCheck® Service Transactions," for ECA/ TeleCheck® service message formats.

Please refer to section 11.4, "Totals," for Totals message formats.

Please refer to section 11.5, "Electronic Mail," for Electronic Mail message formats.

Please refer to section 11.6, "Communications Test," for Communications Test message formats.

Please refer to section 11.7, "Downloads," for Download message formats.

11.1.1 Request

Each Financial Transaction Request contains three data sections:

- Data Section No. 1

  Data Section No. 1 contains the following two data elements that are the same for each Financial Transaction Request message:

  ◊ Message Format Version Identifier (Element No. 55)
  ◊ Number of Segments (Element No. 63)

  Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 2

  Data Section No. 2 contains the following data segment that is the same for each Financial Transaction Request message:

  ◊ Standard Message Data Segment (Data Segment No. 100)

  Please refer to chapter 12, " Data Segment Formats" for additional information about this data segment.

- Data Section No. 3

  Data Section No. 3 contains none, one, or more of the data segments in Chapter 12. See the Data Section No. 3 table below for specifics on what segments may be included.

  ◊ Fleet Data Segment (Data Segment No. 101)
  ◊ Product Code Data Segment (Data Segment No. 102)
  ◊ EBT Data Segment (Data Segment No. 103)
  ◊ Purchase Card Data Segment (Data Segment No. 104)
  ◊ Variable Information Data Segment (Data Segment No. 111)

Please refer to chapter 12, "Data Segment Formats," for additional information about these data segments.

For a layout example of a POS purchase, please refer to Appendix B. POS Purchase Example.

The following table illustrates the layout of a Financial Transaction Request:

Message Format
Financial Transaction Request

Data Section No. 1

All Financial Transaction Requests contain Element Nos. 55 and 63.

Note: A Field Separator separates Field Nos. 1 and 2; and a Field Separator follows Field No. 2.

Field No.	Ele. No.	Element Name	Len.	Entry: R/O/C[1]	Description
1	55	Message Format Version Identifier	6	R	Identifies the message format type and release version.
2	63	Number of Segments	2	R	Identifies the number of BUYPASS-defined segments included in the message.

Data Section No. 2

All Financial Transaction Requests contain Segment No. 100.

Field No.	Seg. No.	Segment Name	Max. Len.	Entry: R/O/C[1]	Description
3	100	Standard Message Data Segment	218	R	Contains the standard data required for all financial transactions. This segment is the only segment required for all financial transactions.

Message Format
Financial Transaction Request

Data Section No. 3

All Financial Transaction Requests contain none, one, or more of the following data segments in Field Nos. 4–9.

Field No.	Seg. No.	Segment Name	Max. Len.	Entry: R/O/C[1]	Description
4–9	101	Fleet Data Segment	308	C	Contains fleet-specific data. Sent only on transactions requiring fleet data.
	102	Product Code Data Segment	381	C	Contains product-specific data. Sent only on transactions requiring product data.
	103	EBT Data Segment	3,334	C	Contains EBT-specific data. Sent only on transactions requiring EBT data.
	104	Purchase Card Data Segment	86	C	Contains purchase card-specific data. Sent only on transactions requiring purchase card data.
	111	Variable Information Data Segment	999	C	Contains data elements unique to transactions requiring variable information.
	123	NFC Payment Tokenization Data Segment	186	C	Contains data needed to process NFC tokenized transactions. It is required on all initial and recurring transactions involving tokenized data.
	135	Moneris Data (Request) Segment	110	C	Contains data required by the Moneris authorizer. It is required only for transactions destined for the Moneris authorizer.
	143	Tax by Product Data Segment	399	C	Contains tax data for each product included in segment 102. This segment has limited applicability and should only be included if instructed by your account manager.
	145	Enhanced Fleet Request Segment	999	C	Contains data that applies to enhanced fleet offerings. See the section 12.32 Enhanced Fleet Request Segment to determine the applicability of this segment.

	146	Enhanced Fleet Response Segment	999	C	Contains data that applies to enhanced fleet offerings. This is the data returned by the authorization network. See section 12.32 Enhanced Fleet Response Segment to determine the applicability of this segment.
	151	Incomm Market Basket Data (Request) Segment	2309	C	Contains data required by Incomm transactions that contain Market Basket Data. See section 12.35 to determine the applicability of this segment.
	152	Incomm Market Basket Data (Response) Segment	2309	C	Contains data returned from Incomm when Incomm Market Basket Data (Request) segment is included in the request. See section 12.36 to determine the applicability of this segment.
	153	Network Token Data Request Segment	999	C	Contains Network Token Data associated with this transaction.

[1]Required/Optional/Conditional

11.1.2 Response

Each Financial Transaction Response contains two data sections:

- Data Section No. 1

  Data Section No. 1 contains the following data elements:

  ◊ Response Code
  ◊ Download Indicator
  ◊ Initiation Date
  ◊ Initiation Time
  ◊ Sequence Number
  ◊ Approved Amount
  ◊ Approval Number
  ◊ Account Number
  ◊ Decline Code
  ◊ Authorizer Code
  ◊ Authorizer Response Code
  ◊ Settlement Date
  ◊ Additional Information Data Segment Flag
  ◊ Number of Print Lines
  ◊ Terminal Display/Printer Message

  Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 2

  Data Section No. 2 contains the following data segments:

  ◊ Additional Information Data Segment (Data Segment No. 112)
  ◊
  ◊ Print Data Segment (Data Segment No. 115)

  Please refer to chapter 12, "Data Segment Formats" for additional information about these data segments.

The following table illustrates the layout of a Financial Transaction Response:

Note: This response is a variable length (Element No. 101, Terminal Display /Printer Message) can be repeated a maximum of three times.) message; it has no Field Separators.

Message Format
Financial Transaction Response

Data Section No. 1
All Financial Transaction Responses contain Data Section No. 1.

Field No.	Element No.	Element Name	Pos.	Len.	Entry is: R/O/C[1]	Description
1	83	Response Code	1	1	R	Identifies the type of response. Source: Host
2	30	Download Indicator	2	1	R	Indicates whether a download is required; and, if so, whether it is a partial or full download. Source: Host
3	45	Initiation Date	3	6	R	Identifies the date the transaction was initiated. It is calculated by BUYPASS. Source: Host
4	46	Initiation Time	9	4	R	Identifies the time the transaction was initiated. It is calculated by BUYPASS. Source: Host
5	86	Sequence Number	13	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
6	6	Approved Amount	19	9	C	Identifies the amount for which the transaction was approved. Source: Host
7	5	Approval Number	28	6	C	Identifies the approval number for an approved transaction. Source: Host
8	2	Account Number	34	25	R	Identifies a portion of the cardholder's account number or identification number. Source: Device
9	26	Decline Code	59	2	C	Identifies the reason for the decline. Source: Host
10	7	Authorizer Code	61	2	R	Identifies the authorizer. Source: Host

Message Format
Financial Transaction Response

Data Section No. 1
All Financial Transaction Responses contain Data Section No. 1.

Field No.	Element No.	Element Name	Pos.	Len.	Entry is: R/O/C[1]	Description
11	8	Authorizer Response Code	63	2	R	Identifies the response type. Source: Authorizer via Host
12	119	Settlement Date	65	4	R	Identifies the date of actual financial settlement for the transaction. Source: Host
13	115	Additional Information Data Segment Flag	69	1	R	Indicates whether an Additional Information Data Segment (Segment No. 112) follows in the message (Data Section No. 2). Source: Host
14	61	Number of Print Lines	70	1	R	Indicates the number of times Terminal Display/Printer Message is in the response. Source: Host
15 or 15–16 or 15–17	101	Terminal Display/Printer Message	71	31	R	Identifies a device display or printer message for the merchant from BUYPASS. This data element can be repeated up to a maximum of three times. Source: Host

Message Format
Financial Transaction Response

Data Section No. 2

If Element No. 115 (Additional Information Data Segment Flag) above indicates that an Additional Information Data Segment (Segment No. 112) follows in the message, the following segment follows in Field No. 16/17/18.

Field No.	Segment No.	Segment Name	Max. Len.	Entry is: R/O/C[1]	Description
16 /17 /18	112	Additional Information Data Segment	349	C	Contains additional information unique to transactions requiring additional information. Source: Host

If Element No. 115 (Additional Information Data Segment Flag) above indicates that a Print Data Segment (Segment No. 115) follows in the message, and the loyalty Version Number value in the request is "2," the following segment follows in Field No. 17/18/19.

17 /18 /19	115	Print Data Segment	910	C	Contains a large block of print data in a segment that uses a 4-byte length field. Source: Host

[1]Required/Optional/Conditional

11.2 Loyalty Card Transactions

This section contains message formats for the loyalty card transaction request and response.

Please refer to section 11.1, "Financial Transactions," for Financial Transaction message formats.

Please refer to section 11.3, "ECA/ TeleCheck® Service Transactions," for ECA/ TeleCheck® service message formats.

Please refer to section 11.4, "Totals," for Totals message formats.

Please refer to section 11.5, "Electronic Mail," for Electronic Mail message formats.

Please refer to section 11.6, "Communications Test," for Communications Test message formats.

Please refer to section 11.7, "Downloads," for Download message formats.

11.2.1 Request

The Loyalty Card Transaction Request contains three data sections:

- Data Section No. 1

  Data Section No. 1 contains the following data elements:

  ◊ Message Format Version Identifier (Element No. 55)

  ◊ Number of Segments (Element No. 63)

  Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 2

  Data Section No. 2 contains the following data segment:

  ◊ Standard Message Data Segment (Data Segment No. 100)

  Please refer to chapter 12, "Data Segment Formats," for additional information about this data segment.

- Data Section No. 3

  Data Section No. 3 contains none, one, or more of the following data segments:

  ◊ Data Segment No. 108, Loyalty Card Data Segment

  ◊ Data Segment No. 114, SKU Data Segment

  Please refer to chapter 12, "Data Segment Formats," for additional information about these data segments.

The following table illustrates the layout of a Loyalty Card Transaction Request:

Message Format
Loyalty Card Transaction Request

Data Section No. 1
The Loyalty Card Transaction Request contains Element Nos. 55 and 63.
Note: A Field Separator separates Field Nos. 1 and 2; and a Field Separator follows Field No. 2.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	55	Message Format Version Identifier	6	R	Identifies the message format type and release version.
2	63	Number of Segments	2	R	Identifies the number of BUYPASS-defined segments included in the message.

Data Section No. 2
The Loyalty Card Transaction Request contains Segment No. 100.

Field No.	Segment No.	Segment Name	Max. Len.	Entry is: R/O/C[1]	Description
3	100	Standard Message Data Segment	218	R	Contains the standard data required for all loyalty card transactions. This segment is the only segment required for all loyalty card transactions.

Data Section No. 3
The Loyalty Card Transaction Request contains Segment Nos. 108 and 114.

Field No.	Segment No.	Segment Name	Max. Len.	Entry is: R/O/C[1]	Description
4	108	Loyalty Card Data Segment	84	R	Contains the loyalty card data required for all loyalty card transactions.
5	114	SKU Data Segment	1009	O	Contains bar code SKU data.

[1]Required/Optional/Conditional

11.2.2 Response

Each Loyalty Card Transaction Response message contains the same information in the same layout as the Financial Transaction Response message (See section 11.1.2, “Response.”).

11.3 ECA/ TeleCheck® Service Transactions

This section contains message formats for the ECA/ TeleCheck® service transaction request and response.

Please refer to section 11.1, "Financial Transactions," for Financial Transaction message formats.

Please refer to section 11.2, "Loyalty Card Transactions," for Loyalty Card message formats.

Please refer to section 11.4, "Totals," for Totals message formats.

Please refer to section 11.5, "Electronic Mail," for Electronic Mail message formats.

Please refer to section 11.6, "Communications Test," for Communications Test message formats.

Please refer to section 11.7, "Downloads," for Download message formats.

11.3.1 Request

The ECA/ TeleCheck® Service Transaction Request contains three data sections:

- Data Section No. 1

  Data Section No. 1 contains the following data elements:

  ◊ Message Format Version Identifier (Element No. 55)
  ◊ Number of Segments (Element No. 63)

  Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 2

  Data Section No. 2 contains the following data segment:

  ◊ Standard Message Data Segment (Data Segment No. 100)

  Please refer to chapter 12, "Data Segment Formats," for additional information about this data segment.

- Data Section No. 3

  Data Section No. 3 contains none, one, or more of the following data segments:

  ◊ Data Segment No. 110, Check Data Segment
  ◊ Data Segment No. 111, Variable Information Data Segment
  ◊ Data Segment No. 113, ECA/ TeleCheck® Data Segment

  Please refer to chapter 12, "Data Segment Formats," for additional information about these data segments.

The following table illustrates the layout of a ECA/ TeleCheck® Service Transaction Request:

Message Format
ECA/ TeleCheck® Service Transaction Request

Data Section No. 1
The ECA/ TeleCheck® Service Transaction Request contains Element Nos. 55 and 63.
Note: A Field Separator separates Field Nos. 1 and 2; and a Field Separator follows Field No. 2.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	55	Message Format Version Identifier	6	R	Identifies the message format type and release version.
2	63	Number of Segments	2	R	Identifies the number of BUYPASS-defined segments included in the message.

Data Section No. 2
The ECA/ TeleCheck® Service Transaction Request contains Segment No. 100.

Field No.	Segment No.	Segment Name	Max. Len.	Entry is: R/O/C[1]	Description
3	100	Standard Message Data Segment	218	R	Contains the standard data required for all ECA/ TeleCheck® transactions. This segment is the only segment required for all ECA/ TeleCheck® transactions.

Data Section No. 3
The ECA/ TeleCheck® Service Transaction Request contains the following segments in Field Nos. 4–6.

Field No.	Segment No.	Segment Name	Max. Len.	Entry is: R/O/C[1]	Description
4	110	Check Data Segment	168	R	Contains the ECA/ TeleCheck® service data required for all ECA/ TeleCheck® transactions.
5	111	Variable Information Data Segment	20	R	Contains the ECA/ TeleCheck® service data required for all ECA/ TeleCheck® transactions.
6	113	ECA/ TeleCheck® Data Segment	156	R	Contains the ECA/ TeleCheck® service data required for all ECA/ TeleCheck® transactions.

[1]Required/Optional/Conditional

11.3.2 Response

The ECA/ TeleCheck® Service Financial Transaction Response message contains the same information in the same layout as the Financial Transaction Response message (See section 11.1.2, “Response.”).

11.4 Totals

This section contains message formats for the Totals request and response.

Please refer to section 11.1, "Financial Transactions," for Financial Transaction message formats.

Please refer to section 11.2, "Loyalty Card Transactions," for Loyalty Card message formats.

Please refer to section 11.3, "ECA/ TeleCheck® Service Transactions," for ECA/ TeleCheck® service message formats.

Please refer to section 11.5, "Electronic Mail," for Electronic Mail message formats.

Please refer to section 11.6, "Communications Test," for Communications Test message formats.

Please refer to section 11.7, "Downloads," for Download message formats.

11.4.1 Request

This section contains request messages for Totals Request and Totals with Proprietary Data Load Request.

11.4.1.1 Totals Request

Each Totals Request message contains the following two data sections:

- Data Section No. 1

  Data Section No. 1 contains the following two data elements:

  ◊ Message Format Version Identifier (Element No. 55)
  ◊ Number of Segments (Element No. 63)

  Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 3

  Data Section No. 3 contains the following data segment:

  ◊ Totals Data Segment (Data Segment No. 105)

  Please refer to chapter 12, "Data Segment Formats," for additional information about this data segment.

Note: Totals Request messages do not contain Data Section No. 2 [Standard Message Data Segment (Data Segment No. 100)].

The following table illustrates the layout of a Totals Request message:

Message Format
Totals Request

Data Section No. 1
All Totals Request messages contain Element Nos. 55 and 63.
Note: A Field Separator separates Field Nos. 1 and 2; and a Field Separator follows Field No. 2.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	55	Message Format Version Identifier	6	R	Identifies the message format version.
2	63	Number of Segments	2	R	Identifies the number of BUYPASS-defined segments included in the message. Fixed value: 01

Data Section No. 3
All Totals Request messages contain the following segment in Field No. 3.

Field No.	Segment No.	Segment Name	Len.	Entry is: R/O/C[1]	Description
3	105	Totals Data Segment	409	C	Contains totals-specific data. Sent only on transactions requiring totals data.

[1]Required/Optional/Conditional

11.4.1.2 Totals with Proprietary Data Load Request

Each Totals with Proprietary Data Load Request message contains the following two data sections:

- Data Section No. 1

  Data Section No. 1 contains the following two data elements:

  ◊ Message Format Version Identifier (Element No. 55)
  ◊ Number of Segments (Element No. 63)

  Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 3

  Data Section No. 3 contains the following data segment:

  ◊ Totals with Proprietary Data Load Request (Data Segment No. 116)

  Please refer to chapter 12, "Data Segment Formats," for additional information about this data segment.

Note: Totals with Proprietary Data Load Request messages do not contain Data Section No. 2 [Standard Message Data Segment (Data Segment No. 100)].

The following table illustrates the layout of a Totals Request message:

Message Format
Totals with Proprietary Data Load Request

Data Section No. 1

All Totals Request messages contain Element Nos. 55 and 63.

Note: A Field Separator separates Field Nos. 1 and 2; and a Field Separator follows Field No. 2.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	55	Message Format Version Identifier	6	R	Identifies the message format version.
2	63	Number of Segments	2	R	Identifies the number of BUYPASS-defined segments included in the message. Fixed value: 01

Data Section No. 3

All Totals Request messages contain the following segment in Field No. 3.

Field No.	Segment No.	Segment Name	Len.	Entry is: R/O/C[1]	Description
3	119	Totals with Proprietary Data Load Data Segment	389	R	Contains totals-specific data. Sent only on transactions requiring totals data.

[1]Required/Optional/Conditional

11.4.2 Response

This section contains formats for the following messages:

- Approved Totals Response
- Declined Totals Response

11.4.2.1 Approved Totals Response

Each Approved Totals Response message contains the following data elements and sections:

- Response Code
- Totals Date
- Sequence Number
- Grand Total
- Fee Amount
- Net Amount
- Category Total Data Section
  - ◊ Card Label
  - ◊ Card Type Total Count
  - ◊ Card Type Total Amount

The following table illustrates the layout of an Approved Totals Response message:

Note: This response is a single-block, fixed length message. It has no Field Separators. It contains no print data, and is intended for use by complex systems that manipulate the host totals data after it is received.

Message Format Approved Totals Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	83	Response Code	1	1	R	Identifies the type of response. Source: Host
2	105	Totals Date	2	6	R	Identifies the requested type of totals. Source: Host
3	86	Sequence Number	8	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device

Message Format Approved Totals Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
4	42	Grand Total	14	8	R	Identifies the final total amount. Source: Host
5	38	Fee Amount	22	8	R	Identifies the fee amount. Source: Host
6	56	Net Amount	30	8	R	Identifies the difference between Grand Total and Fee Amount. Source: Host

Category Total Data Section

Field Nos. 7, 8, and 9 are sent 18 times—once for each of the following card types—in the order shown whether or not there are totals accumulated for a card type with the exception of Nos. 16, 17 and 18. These three card types are sent only if they have totals.

Use Card Labels (in parentheses) instead of field numbers to identify total buckets.

1. MasterCard®/Visa® (CC)[1]
2. Amex® (TE)
3. Discover® Network, Diners Club®, JCB ® (DS)
4. Preauthorization (AO)
5. Debit (DB)
6. Fleet (FL)
7. Cash (CS)
8. Proprietary card (PR)
9. Check (CK)
10. EBT Food Stamp (EF)
11. EBT Cash Benefit (EC)
12. Stored Value Card Activation/Deactivation (SV1)
13. Stored Value Card Purchase/Capture and Merchandise Return (SV2)
14. Stored Value Card Card Replacement (SV3)
15. Stored Value Card Recharge (SV4)
16. ECA/ TeleCheck® service(ECA)
17. EBT version of eWIC (EWIC)
18. EBT child care (EK)

Note: [1]STAR Signature Debit Totals transaction will be a part of the MasterCard®/Visa® (CC) bucket

7	13	Card Label	38	4	R	Identifies the card type. Source: Host
8	16	Card Type Total Count	42	5	R	Identifies the transaction count for a particular card type. Source: Host
9	15	Card Type Total Amount	47	8	R	Identifies the total amount for a particular card type. Source: Host

[1]Required/Optional/Conditional

11.4.2.2 Declined Totals Response

Each Declined Totals Response message contains the following data elements:

- Response Code
- Totals Date
- Sequence Number
- Terminal Display/Printer Message Grand Total

The following table illustrates the layout of a Declined Response message:

Note: This response is a single-block, fixed length message. It has no Field Separators. It contains no print data, and is intended for use by complex systems that manipulate the host totals data after it is received.

Message Format Declined Totals Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	83	Response Code	1	1	R	Identifies the type of response. Source: Host
2	105	Totals Date	2	6	R	Identifies the requested type of totals. Source: Host
3	86	Sequence Number	8	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
4	101	Terminal Display /Printer Message	14	31	R	Identifies the decline message for a Declined Totals Response. Source: Host

[1]Required/Optional/Conditional

11.5 Electronic Mail

This section contains message formats for the Electronic Mail request and response.

Please refer to section 11.1, “Financial Transactions,” for Financial Transaction message formats.

Please refer to section 11.2, “Loyalty Card Transactions,” for Loyalty Card message formats.

Please refer to section 11.3, “ECA/ TeleCheck® Service Transactions,” for ECA/ TeleCheck® service message formats.

Please refer to section 11.4, “Totals,” for Totals message formats.

Please refer to section 11.6, “Communications Test,” for Communications Test message formats.

Please refer to section 11.7, “Downloads,” for Download message formats.

Please refer to chapter 11, “Electronic Mail Processing,” for additional electronic mail information.

11.5.1 Request

An Electronic Mail Request message contains the following two data sections:

- Data Section No. 1

  Data Section No. 1 contains the following two data elements:

  ◊ Message Format Version Identifier (Element No. 55)
  ◊ Number of Segments (Element No. 63)

  Please refer to chapter 13, “Data Element Descriptions,” for additional information about these data elements.

- Data Section No. 2

  Data Section No. 2 contains the following data segment:

  ◊ Electronic Mail Data Segment (Data Segment No. 109)

  Please refer to chapter 12, “Data Segment Formats,” for additional information about this data segment.

The following table illustrates the layout of an Electronic Mail Request message:

Message Format
Electronic Mail Request

Data Section No. 1

All Electronic Mail Request messages contain Element Nos. 55 and 63.

Note: A Field Separator separates Field Nos. 1 and 2; and a Field Separator follows Field No. 2.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	55	Message Format Version Identifier	6	R	Identifies the message format version.
2	63	Number of Segments	2	R	Identifies the number of BUYPASS-defined segments included in the message.

Data Section No. 2

All Electronic Mail Request messages contain the following segment in Field No. 3.

Field No.	Segment No.	Segment Name	Len.	Entry is: R/O/C[1]	Description
3	109	Electronic Mail Data Segment	232	C	Contains electronic mail-specific data. Sent only on transactions requiring electronic mail.

11.5.2 Response

The following table illustrates the layout of the Electronic Mail Response:

Note: This response is a variable length (Element No. 51, Mail Text Data, has a variable length.) message; it has no Field Separators.

Message Format Electronic Mail Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	83	Response Code	1	1	R	Identifies the type of response. Source: Host
2	30	Download Indicator	2	1	R	Indicates whether a download is required; and, if so, whether it is a partial or full download. Source: Host
3	45	Initiation Date	3	6	R	Identifies the date the transaction was initiated. Source: Host
4	46	Initiation Time	9	4	R	Identifies the time the transaction was initiated. Source: Host
5	86	Sequence Number	13	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
6	11	Block Number	19	3	R	Identifies the specific electronic mail data block requested by a device or sent by the host. Source: Host
7	52	Mail Text Data Length	22	3	C	Identifies the Mail Text Data length. Source: Host
8	51	Mail Text Data	25	Var.	C	Identifies the data in the response. Source: Host

[1]Required/Optional/Conditional

11.6 Communications Test

This section contains message formats for the Communications Test (Transaction Type 900) request and response.

Please refer to section 11.1, "Financial Transactions," for Financial Transaction message formats.

Please refer to section 11.2, "Loyalty Card Transactions," for Loyalty Card message formats.

Please refer to section 11.3, "ECA/ TeleCheck® Service Transactions," for ECA/ TeleCheck® service message formats.

Please refer to section 11.4, "Totals," for Totals message formats.

Please refer to section 11.5, "Electronic Mail," for Electronic Mail message formats.

Please refer to section 11.7, "Downloads," for Download message formats.

11.6.1 Request

The following table contains format information about the Communications Test request.

Message Format Communications Test Request						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	55	Message Format Version Identifier	1	6	R	Indicates the type and release version of message format. Source: Device
	63	Number of Segments	7	2	R	Identifies the number of BUYPASS-defined data segments included in the Financial Transaction Request, Totals Request, Loyalty Card Transaction Request, and Electronic Mail Request. Source: Device
2	120	Network Management Message	9	8	R	Identifies the communications test message from the device: COMMTEST Source: Device

[1]Required/Optional/Conditional

11.6.2 Response

The following table contains format information about the Communications Test Response.

Note: This response is a fixed length message; it has no Field Separators.

Message Format Communications Test Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	83	Response Code	1	1	R	Indicates a single-message response. Fixed value: 1 Source: Host
2	30	Download Indicator	2	1	R	Indicates whether a download is required; and, if so, whether it is a partial or full download. Fixed value: 0 Source: Host
3	100	Terminal Display Communications Message	3	16	C	Identifies the following message for display at the transmitting device: COMM_TEST_PASSED. Source: Host

[1]Required/Optional/Conditional

11.7 Downloads

This section contains message formats for the following downloads:

- Table Load
- Phone Load
- Date and Time Load
- Software Load

  Note: The Software Dial Load Data Segment and Software IP Load Data Segment are used only for devices loaded through the BUYPASS device management system.
- TransArmor Key and Key ID Load
- Proprietary Data Load

The following items must be downloadable:

- Dialing strings
- BUYPASS Merchant Numbers
- Card types
- Cut times
- Dates
- Times
- Passwords

Note: The device application/system software should not be activated until at least one successful Table Load has been performed. When the software is not activated, all BUYPASS transactions except downline loads should be prohibited at the device.

Downloads must be capable of being requested by manual user intervention, such as selecting a function. The device must also be capable of automatically requesting a download based on the host's response to a device's transaction request.

In the case of dial communications, when indicated by the host's response that a download is requested, the device receives an ENQ after acknowledging the host's response, rather than an EOT. The device should stay on-line and send up a Partial Load Request.

Note: If the controller is capable of queuing multiple transactions, the load request should not be sent until the controller has cleared its queue of all transactions.

The response is a Table Load or a subset of the complete table load containing only dial string information. The information in the download is used in the very next transaction performed by the device.

Please refer to section 11.1, "Financial Transactions," for Financial Transaction message formats.

Please refer to section 11.2, "Loyalty Card Transactions," for Loyalty Card message formats.

Please refer to section 11.3, "ECA/ TeleCheck® Service Transactions," for ECA/ TeleCheck® service message formats.

Please refer to section 11.4, "Totals," for Totals message formats.

Please refer to section 11.5, "Electronic Mail," for Electronic Mail message formats.

Please refer to section 11.6, "Communications Test," for Communications Test message formats.

11.7.1 Table Load

This section contains request and response message formats for a Table Load.

The load flag in the merchant's profile at BUYPASS must be set to *TABL* for this load to take place. A BUYPASS representative can change this load flag. If the load flag is not set, the Downline Load Response will consist only of an error message block and a terminating block.

11.7.1.1 Request

The following table contains format information about the Table Load Request:

Note: There is no Field Separator between fields in this message.

Message Format Table Load Request						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	44	Information Byte	1	1	R	Indicates a Download Request. Fixed value: ? Source: Device
2	102	Terminal Identifier	2	13	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
3	48	Load Type	15	1	R	Identifies the load type. Fixed value: P (Partial Load) Source: Device
4	43	Hardware Version	16	4	R	Identifies the version number of the device's hardware level. Source: Device
5	96	Software Version	20	8	R	Identifies the version number of the device's software application. Source: Device
6	39	Firmware Version	28	8	R	Identifies the version number of the device's EPROM level. Source: Device

[1]Required/Optional/Conditional

11.7.1.2 Response

The Table Load Response contains four data blocks.

The receiving device is expected to parse the download message based on the Data Type Indicator within each data block in the download message.

The following table illustrates the layout for a Table Load Response:

Notes: Please refer to the data segment descriptions in chapter 12, "Data Segment Formats," and the data element descriptions in chapter 13, "Data Element Descriptions," for additional information.

For TCP/IP, Start-of-Data Block Indicator is NOT present before Data Block Nos. 2 and 3. It occurs only before Data Block No. 1.

Message Format Table Load Response					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	97	Start-of-Data Block Indicator	1	R	Indicates the start of a data block. Fixed value:) Source: Host
Data Block No. 1					
Field No.	**Segment No.**	**Segment Name**	**Max. Len.**	**Entry is: R/O/C[1]**	**Description**
2	DL1	Merchant Data Segment	399	R	Contains merchant-specific data. Sent only on a Table Load Response.
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
3	97	Start-of-Data Block Indicator	1	R	Indicates the start of a data block. Fixed value:) For TCP/IP, not present. Source: Host
Data Block No. 2					
Field No.	**Segment No.**	**Segment Name**	**Max. Len.**	**Entry is: R/O/C[1]**	**Description**
4	DL2	Dial String Data Segment	69	C	Contains Dial String-specific data. Sent only on a Table Load Response and a Phone Load Response.

Message Format Table Load Response					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
5	97	Start-of-Data Block Indicator	1	R	Indicates the start of a data block. Fixed value:) For TCP/IP, not present. Source: Host
Data Block No. 3					
Field No.	**Segment No.**	**Segment Name**	**Max. Len.**	**Entry is: R/O/C[1]**	**Description**
6	DL3	Date and Time Data Segment	23	C	Contains Date and Time-specific data. Sent only on a Table Load Response or a Date and Time Load Response.
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
7	35	End-of-Load Indicator	1	R	Indicates the end of a table load. Fixed value: * Source: Host
Data Block No. 4 If Segment No. DL1 (Merchant Data Segment) includes a Card Type value of '173' in Card Type (Element No. 14), the following segment follows in response					
Field No.	**Segment No.**	**Segment Name**	**Max. Len.**	**Entry is: R/O/C[1]**	**Description**
8	DL6	Store and Forward Data segment	9	C	Contains time-specific data to block Store and Forward processing. Sent only on a Table Load Response.
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
9	35	End-of-Load Indicator	1	R	Indicates the end of a table load Fixed value: * Source: Host

[1]Required/Optional/Conditional

11.7.2 Phone Load

This section contains request and response message formats for a phone load.

This load provides the device with the dialing telephone number, redial count, access code, and pause indicator information needed to operate properly.

The load flag in the merchant's profile on BUYPASS must be set to *PHON* for this load to take place. A BUYPASS representative can change this load flag, or it can be changed automatically following a full load. If the load flag is not set, the Downline Load Response will consist only of an error message block and a terminating block.

11.7.2.1 Request

The following table contains format information about the Phone Load Request:

Note: There is no Field Separator between fields in this message.

Message Format Phone Load Request						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	44	Information Byte	1	1	R	Indicates a Download Request. Fixed value: ? Source: Device
2	102	Terminal Identifier	2	13	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
3	48	Load Type	15	1	R	Identifies the load type. Fixed value: P (Partial Load) Source: Device

[1]Required/Optional/Conditional

11.7.2.2 Response

The following table contains format information about the Phone Load Response:

Message Format Phone Load Response					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	97	Start-of-Data Block Indicator	1	R	Indicates that dial string data follows. Fixed value: ! Source: Host
A Phone Load Response contains the following segment:					
Field No.	**Segment No.**	**Segment Name**	**Max. Len.**	**Entry is: R/O/C[1]**	**Description**
2	DL2	Dial String Data Segment	69	R	Contains Dial String-specific data. Sent only on a Table Load Response and a Phone Load Response.

[1]Required/Optional/Conditional

11.7.3 Date and Time Load

This load is initiated by the device. The merchant's load flag does not have to be set at BUYPASS for this load.

This load includes adjustments for the time zone of the initiating device and daylight saving time.

11.7.3.1 Request

The following table contains format information about the Date and Time Load Request:

Note: There is no Field Separator between fields in this message.

Message Format Date and Time Load Request						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	44	Information Byte	1	1	R	Identifies a Download Request. Fixed value: ? Source: Device
2	102	Terminal Identifier	2	13	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
3	48	Load Type	15	1	R	Identifies the load type. Fixed value: *D* (Date and Time Load) Source: Device

[1]Required/Optional/Conditional

11.7.3.2 Response

The following table contains format information about the Date and Time Load Response:

Note: There is no Field Separator between fields in this message.

Message Format Date and Time Load Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	1	R	Indicates that the following data contains the current BUYPASS day of the week, date, and time. Fixed value: : Source: Host
2	25	Day of the Week	2	1	R	Identifies the current day of the week. Source: Host
3	21	Current Date	3	6	R	Identifies the current date. Source: Host
4	22	Current Time	9	4	R	Identifies the current time. Source: Host
5	23	Cut Time	13	4	R	Identifies the merchant's end-of-day settlement time. Source: Host
6	65	Password	17	6	R	Identifies the device password used for requesting host totals, etc. Source: Host

[1]Required/Optional/Conditional

11.7.4 Software Load

This section contains request and response message formats for a Software Load.

The load flag in the merchant's profile on BUYPASS must be set to *SOFT* for this load to take place. A BUYPASS representative can change this load flag. If the load flag is not set, the Download Response will consist only of an error message block and a terminating block.

11.7.4.1 Request

The following table contains format information about the Software Load Request:

Note: There is no Field Separator between fields in this message.

Message Format Software Load Request						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	44	Information Byte	1	1	R	Indicates a Download Request. Fixed value: ? Source: Device
2	102	Terminal Identifier	2	13	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
3	48	Load Type	15	1	R	Identifies the load type. Fixed value: P (Partial Load) Source: Device
4	43	Hardware Version	16	4	R	Identifies the version number of the device's hardware level. Source: Device
5	96	Software Version	20	8	R	Identifies the version number of the device's software application. Source: Device
6	39	Firmware Version	28	8	R	Identifies the version number of the device's EPROM level. Source: Device

[1]Required/Optional/Conditional

11.7.4.2 Response

The Software Load Response contains one data block.

The receiving device is expected to parse the download message based on the Data Type Indicator within each data block in the download message.

The following table illustrates the layout for a Software Load Response:

Note: Please refer to the data element descriptions in chapter 13, "Data Element Descriptions," for additional information.

Message Format Software Load Response					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	97	Start-of-Data Block Indicator	1	R	Indicates the start of a data block. Fixed value:) Source: Host

Data Block No. 1					
Field No.	**Segment No.**	**Segment Name**	**Max. Len.**	**Entry is: R/O/C[1]**	**Description**
2	DL4	Software Dial Load Data Segment	52	R	Contains merchant-specific dial load data. Sent only on a Software Load Response.
3	DL5	Software IP Load Data Segment	66	R	Contains merchant-specific IP/URL address load data. Sent only on a Software Load Response.

[1]Required/Optional/Conditional

11.7.5 TransArmor Key and Key ID Load

Note: For detailed information on TransArmor, refer the BUYPASS®_Platform_ATL105_Specification_Updates_for_TransArmor_Processing document, available on the Direct Platform Specifications Portal.

11.7.6 Proprietary Data Load

This section contains request and response message formats for a Proprietary Data Load.

Proprietary Data Load is used to send Custom Receipt Text, support the Site Configuration, Fuel Volume, and Dynamic Card Table Data load, and allow the Host Discount processing. Data Element 83 (Response Code) which is returned on an End of Day Totals Response, indicates that the Proprietary Data is available for loading. When indicated that this data is available, the POS device will format a Proprietary Data Load Request.

Note: The Proprietary Data Load functions occur at end of day only as part of the Day Close processing. These functions are prompted as part of End of Day Totals Response and cannot be supported at any time of the day.

Data Element 78 (Prompt Code) indicates the request/response and format of the Proprietary Data. For example the 901 Custom Receipt Text format will differ from the 902 Dynamic Card Table Load.

11.7.6.1 Request

Each Proprietary Data Load Request contains the following two data sections:

- Data Section No. 1

  Data Section No. 1 contains the following two data elements:

  ◊ Message Format Version Identifier (Element No. 55)
  ◊ Number of Segments (Element No. 63)

Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 3

  Data Section No. 3 contains the following data segment:

  ◊ Proprietary Data Load Segment (Data Segment No. 118)

  Please refer to chapter 12, "Data Segment Formats," for additional information about this data segment.

Note: Proprietary Data Load Request messages do not contain the Standard Data Message Data Segment (Data Segment 100).

The following table illustrates the layout of a Proprietary Data Load Request message:

Message Format
Proprietary Data Load Request

Data Section No. 1
All Custom Receipt Text Load Request messages contain. Element Nos. 55 and 63.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	55	Message Format Version Identifier	6	R	Identifies the message format version.
2	63	Number of Segments	2	R	Identifies the number of BUYPASS-defined segments included in the message. Fixed value of 1

Data Section No. 3
Note: This request does not contain the Standard Message Data Segment (Data Segment No.100)

Field No.	Segment No.	Segment Name	Len.	Entry is: R/O/C[1]	Description
3	118	Proprietary Data Load Segment	3800	R	Indicates that this is a Proprietary Data Load Request.

[1]Required/Optional/Conditional

11.7.7 Response

The following table illustrates the layout of a Proprietary Data Load Response:

Message Format **Proprietary Data Load Response**						
Data Section No.1 All Proprietary Data Load Responses contain Data Section No. 1						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	83	Response Code	1	1	R	Identifies the type of response. Source: Host
2	30	Download Indicator	2	1	R	Indicates whether a download is required; and, if so, whether it is a partial or full download. Source: Host
3	45	Initiation Date	3	6	R	Identifies the date the transaction was initiated. It is calculated by BUYPASS Source: Host
4	46	Initiation Time	9	4	R	Identifies the time the transaction was initiated. It is calculated by BUYPASS Source: Host
5	86	Sequence Number	13	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
Data Section No.3 All Proprietary Data Load Responses contain Data Section No. 3						
Field No.	**Segment No.**	**Segment Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**	
6	118	Proprietary Data Load Segment	3,800	R	Contains the proprietary load-specific data required for all proprietary load transactions.	

11.7.8 Moneris Key Load

This section contains request and response message formats for a Moneris Key Load.

Note: There is no need to set any load flag on the terminal record. This load can be performed at any time.

11.7.8.1 Request

The following table contains format information about the Moneris Key Load Request:

Note: There is no Field Separator between fields in this message.

Message Format Moneris Key Load Request						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	44	Information Byte	1	1	R	Indicates a Download Request. Fixed value: ? Source: Device
2	102	Terminal Identifier	2	13	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
3	48	Load Type	15	1	R	Identifies the load type. Fixed value: K (Key Load) Source: Device
4	205	Load Subtype	16	1	R	Identifies the load subtype as Moneris. Fixed value: M (Moneris Key Load) Source: Device
5	206	SPDH Header	17	48	R	Contains merchant, terminal, and transaction data required for MAC Encryption. Source: Device
6	207	Moneris Terminal Identifier	65	8	R	Identifies the Moneris Terminal ID provided by Moneris during terminal initialization. Source: Device
7	208	Moneris Merchant ID	73	13	R	Identifies the Moneris Merchant ID provided by Moneris during terminal initialization. Source: Device

[1]Required/Optional/Conditional

11.7.8.2 Response

The following table contains format information about the Moneris Key Load Response.

Note: There is no Field Separator between fields in this message.

Message Format Moneris Key Load Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	1	R	Indicates that the following data contains the Moneris Key Information. Fixed value: K Source: Host
2	206	SPDH Header	2	48	R	Contains merchant, terminal, and transaction data required for MAC Encryption. Source: Moneris
3	207	Moneris Terminal Identifier	50	8	R	Identifies the Moneris Terminal ID provided by Moneris during terminal initialization. Source: Host
3	208	Moneris Merchant ID	58	13	R	Identifies the Moneris Merchant ID provided by Moneris during terminal initialization. Source: Host
4	209	Moneris Response Code	71	2	R	Moneris Response Code from for MAC encryption. Source: Moneris
5	210	MAC	73	16	R	MAC value to be validated at terminal. Source: Moneris
6	211	Key Indicators	89	3	R	Indicates which keys are included followed by, up to three 16 byte keys. Source: Moneris
Key Data This section is repeated for a maximum of 3 keys for a total variable length of up to 48 bytes. The three key types are MAC Encryption Key, Data Encryption Key, and PIN Encryption Key. Presence is based on the Key Indicators field.						
7	212	Moneris Key Data	92	16	R	Key Data returned from Moneris. Source: Moneris

11.8 EMV Financial Transaction

This section contains message formats for the EMV card transaction request and response.

Please refer to section 11.1, "Financial Transactions," for Financial Transaction message formats.

11.8.1 Request

Each Financial Transaction Request contains three data sections:

- Data Section No. 1

Data Section No. 1 contains the following two data elements that are the same for each Financial Transaction Request message:

- ◊ Message Format Version Identifier (Element No. 55)
- ◊ Number of Segments (Element No. 63)

  Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 2

Data Section No. 2 contains the following data segment that is the same for each Financial Transaction Request message:

- ◊ Standard Message Data Segment (Data Segment No. 100)

  Please refer to chapter 12, "Data Segment Formats," for additional information about this data segment.

- Data Section No. 3

Data Section No. 3 contains one, or more of the data segments in Chapter 12. See the Data Section No. 3 table below for specifics on what segments may be included.

- ◊ Fleet Data Segment (Data Segment No. 101)
- ◊ Product Code Data Segment (Data Segment No. 102)
- ◊ Purchase Card Data Segment (Data Segment No. 104)
- ◊ Variable Information Data Segment (Data Segment No. 111)
- ◊ EMV Request Data Segment (Data Segment No. 130)

Please refer to chapter 12, "Data Segment Formats," for additional information about these data segments.

Note: The maximum length of a Financial Transaction Request Message is 4,000 bytes.

The following table illustrates the layout of an EMV Financial Transaction Request:

Message Format
EMV Financial Transaction Request

Data Section No. 1
The EMV Financial Transaction Request contains Elements Nos. 55 and 63.
Note: A Field Separator separates Field Nos. 1 and 2; and a Field Separator follows Field No. 2.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	55	Message Format Version Identifier	6	R	Identifies the message format type and release version.
2	63	Number of Segments	2	R	Identifies the number of BUYPASS-defined segments included in the message.

Data Section No. 2
The EMV Financial Transaction Request contains Segment No. 100.

Field No.	Segment No.	Segment Name	Max. Len.	Entry is: R/O/C[1]	Description
3	100	Standard Message Data Segment	619	R	Contains the standard data required for all EMV card transactions. This segment is required for all EMV card transactions.

Data Section No. 3 All EMV Financial Transaction Requests contain one, or more of the following data segments in Field Nos. 4–8.					
Field No.	**Segment No.**	**Segment Name**	**Max. Len.**	**Entry is: R/O/C[1]**	**Description**
4–8	101	Fleet Data Segment	308	C	Contains fleet-specific data. Sent only on transactions requiring fleet data.
	102	Product Code Data Segment	381	C	Contains product-specific data. Sent only on transactions requiring product data.
	103	EBT Data Segment	3,334	C	Contains EBT-specific data. Sent only on transactions requiring EBT data.
	104	Purchase Card Data Segment	86	C	Contains purchase card-specific data. Sent only on transactions requiring purchase card data.
	111	Variable Information Data Segment	999	C	Contains data elements unique to transactions requiring variable information.
	123	NFC Payment Tokenization Data Segment	186	C	Contains data needed to process NFC tokenized transactions. It is required on all initial and recurring transactions involving tokenized data.
	130	EMV Request Data Segment	9999	R	Contains the EMV data required for all EMV card transactions. This segment is the only segment required for all EMV financial transactions.
	135	Moneris Data (Request) Segment	110	C	Contains data required by the Moneris authorizer. It is required only for transactions destined for the Moneris authorizer.
	143	Tax by Product Data Segment	399	C	Contains tax data for each product included in the segment 102. This segment has limited applicability and should only be included if instructed by your account manager.

Data Section No. 3 All EMV Financial Transaction Requests contain one, or more of the following data segments in Field Nos. 4–8.					
Field No.	**Segment No.**	**Segment Name**	**Max. Len.**	**Entry is: R/O/C[1]**	**Description**
	145	Enhanced Fleet Request Segment	999	C	Contains data that applies to enhanced fleet offerings. See section 12.32 Enhanced Fleet Request Segment to determine the applicability of this segment.
	146	Enhanced Fleet Response Segment	999	C	Contains data that applies to enhanced fleet offerings. This is the data returned by the authorization network. See section 12.32 Enhanced Fleet Response Segment to determine the applicability of this segment.
	151	Incomm Market Basket Data (Request) Segment	2309	C	Contains data required by Incomm transactions that contain Market Basket Data. See section 12.35 to determine the applicability of this segment.
	152	Incomm Market Basket Data (Response) Segment	2309	C	Contains data returned from Incomm when Incomm Market Basket Data (Request) segment is included in the request. See section 12.36 to determine the applicability of this segment.
	153	Network Token Data Request Segment	999	C	Contains Network Token Data associated with this transaction.

[1]Required/Optional/Conditional

11.8.2 Response

The EMV Financial Transaction Response Message contains two data sections:

- Data Section No. 1

  Data Section No. 1 contains the following data elements:

  ◊ Response Code (Element No. 83)
  ◊ Download Indicator (Element No. 30)
  ◊ Initiation Date (Element No. 45)
  ◊ Initiation Time (Element No. 46)
  ◊ Sequence Number (Element No. 86)
  ◊ Approved Amount (Element No. 6)
  ◊ Approval Number (Element No. 5)
  ◊ Account Number (Element No. 2)
  ◊ Decline Code (Element No. 26)
  ◊ Authorizer Code (Element No. 7)
  ◊ Authorizer Response Code (Element No. 8)
  ◊ Settlement Date (Element No. 119)
  ◊ Additional Information Data Segment Flag (Element No. 115)
  ◊ Number of Print Lines (Element No. 61)
  ◊ Terminal Display/Printer Message (Element No. 101)

Please refer to chapter 13, “Data Element Descriptions,” for additional information about these data elements.

- Data Section No. 2

Data Section No. 2 contains the following data segments:

◊ EMV Response Data Segment (No. 131)
◊ Transaction Attributes Data Segment (No. 134)

Please refer to chapter 12, “Data Segments,” for additional information about these data segments.

Note: The maximum length of a Financial Transaction Response Message is 4,000 bytes.

The following table illustrates the layout of an EMV Financial Transaction Response:

Note: This response is a fixed length message; it has no Field Separators.

Message Format EMV Financial Transaction Response					
Data Section No. 1 All EMV Financial Transaction Responses contain Data Section No.1					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	83	Response Code	1	R	Identifies the type of response. Source: Host
2	30	Download Indicator	1	R	Indicates whether a download is required; and, if so, whether it is a partial or full download. Source: Host
3	45	Initiation Date	6	R	Identifies the date the transaction was initiated. It is calculated by the BUYPASS host system. Source: Host
4	46	Initiation Time	4	R	Identifies the time the transaction was initiated. It is calculated by the BUYPASS host system. Source: Host
5	86	Sequence Number	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
6	6	Approved Amount	9	C	Identifies the amount for which the transaction was approved. Source: Host
7	5	Approval Number	6	C	Identifies the approval number for an approved transaction. Source: Host
8	2	Account Number	25	R	Identifies a portion of the cardholder's account number or identification number. Source: Device

Message Format
EMV Financial Transaction Response

Data Section No. 1
All EMV Financial Transaction Responses contain Data Section No.1

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
9	26	Decline Code	2	C	Identifies the reason for the decline. Source: Host
10	7	Authorizer Code	2	R	Identifies the authorizer. Source: Host
11	8	Authorizer Response Code	2	R	Identifies the response type. Source: Host
12	119	Settlement Date	4	R	Identifies the date of actual financial settlement for the transaction. Source: Host
13	115	Additional Information Data Segment Flag	1	R	Indicates whether an Additional Information Data Segment (Segment No. 112) follows in the message (Data Section No. 2). Source: Host
14	61	Number of Print Lines	1	R	Indicates the number of times Terminal Display/Printer Message is in the response. Source: Host
15 or 15-16 or 15-17	101	Terminal Display/Printer Message	31	R	Identifies a device display or printer message for the merchant from the BUYPASS host. This data element can be repeated up to a maximum of three times. Source: Host

Message Format
EMV Financial Transaction Response

Data Section No. 2
If Element No. 115 (Additional Information Data Segment Flag) above indicates that an Additional Information Data Segment (Segment No. 112) follows in the message, the following segment follows in Field No. 16/17/18.

Field No.	Segment No.	Segment Name	Max .Len.	Entry is: R/O/C[1]	Description
16 /17 /18	112	Additional Information Data Segment	349	C	Contains additional information unique to transactions requiring additional information. Source: Host
If Element No. 115 (Additional Information Data Segment Flag) above indicates that a Print Data Segment (Segment No. 115) and Print Data 2 Segment (Segment No. 120) follows in the message, and the loyalty Version Number value in the request is "2," the following segment follows in Field No. 17/18/19.					
17 /18 /19	115	Print Data Segment	999	C	Contains a large block of print data in a segment that uses a 4-byte length field. Source: Host
	120	Print Data 2 Segment	999	C	Contains a large block of print data in a segment that uses a 4-byte length field. Source: Host
The following EMV Data Segment follows in Field No. 17/18/19/20.					
/17 /18 /19 /20	131	EMV Response Data Segment	3850	C	Contains EMV information unique to transactions requiring EMV data. Source: Host
The following Transaction Attributes Data Segment follows in Field No. 17/18/19/20					
/17 /18 /19 /20	134	Transaction Attributes Data Segment	12	C	Contains information about how the transaction was processed. Source: Host

[1]Required/Optional/Conditional

11.9 CA Public Key File Load

This section contains request and response message formats for a CA Public Key File Load.

11.9.1 Request

Each Financial Transaction Request contains three data sections:

- Data Section No. 1

Data Section No. 1 contains the following two data elements that are the same for each Financial Transaction Request message:

- ◊ Message Format Version Identifier (Element No. 55)
- ◊ Number of Segments (Element No. 63)

Please refer to chapter 13, "Data Element Descriptions," for additional information about these data elements.

- Data Section No. 2

Data Section No. 2 contains the following data segment that is the same for each Financial Transaction Request message:

- ◊ Standard Message Data Segment (Data Segment No. 100)

- Data Section No. 3

Data Section No. 3 contains none, one, or more of the following data segment(s) that is/are unique to the Financial Transaction Request message type.

- ◊ Fleet Data Segment (Data Segment No. 101)
- ◊ Product Code Data Segment (Data Segment No. 102)
- ◊ Purchase Card Data Segment (Data Segment No. 104)
- ◊ Variable Information Data Segment (Data Segment No. 111)
- ◊ CA Public Key File Load Segment (Data Segment No. 132)

Please refer to chapter 12, "Data Segment Formats," for additional information about these data segments.

The following table contains format information about the CA Public Key File Load Request:

Note: Field Separators are present between Field No. 1 and Field No. 2; Field No. 2 and Data Section No. 2; Field Nos. 3 and 4; 4 and 5; and 5 and 6. There are no Field Separators between Field Nos. 6 and 7; 7 and 8, 8 and 9; 9 and 10, 10 and 11, and 11 and 12. A Field Separator follows Field No. 12.

Message Format **CA Public Key File Load Request**						
Data Section No. 1						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	55	Message Format Version Identifier	1	6	R	Indicates the message format version. Fixed value: ATL105 Source: Device
<FS>						

Message Format **CA Public Key File Load Request**						
2	63	Number of Segments	7	2	R	Indicates the number of segments in the message. Fixed value: 01 Source: Device
<FS>						

Data Section No. 2 **Note:** This request does not contain the Standard Message Data Segment (Data Segment No. 100).						
Data Segment No. 132 (CA Public Key File Load Segment) **Maximum Segment Length:** 79 bytes						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
3	85	Segment Type	10	3	R	Identifies the segment. Fixed value: 132 Source: Device
<FS>						
4	84	Segment Length	14	3	R	Describes the segment's length. Source: Device
<FS>						
5	86	Sequence Number	18	6	R	Identifies the merchant-assigned Sequence Number for the transaction. **Note:** Limited sequence number range (100000-199999) is ONLY applicable when using the Multithreaded Dial Protocol Communications header for a CAPK File Load for the ATL105 message format. Source: Device
<FS>						
6	102	Terminal Identifier	25	13	R	Identifies the transaction device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. For EMV, the first two characters (Device Type) of this element's value must be "+*" regardless of the actual device type. Source: Device

Data Section No. 2 **Note:** This request does not contain the Standard Message Data Segment (Data Segment No. 100).						
7	48	Load Type	38	1	R	Describes the load type. Fixed value: K (Indicates that Public Key information is requested.) Source: Device
8	43	Hardware Version	39	4	R	Identifies the version number of the device's hardware level. Source: Device
Data Segment No. 132 (CA Public Key File Load Segment) **Maximum Segment Length:** 79 bytes						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
9	96	Software Version	43	8	R	Identifies the version number of the device's software application. Source: Device
10	39	Firmware Version	51	8	R	Identifies the version number of the device's EPROM level. Source: Device
11	187	CA Public Key File Checksum	59	25	R	Identifies the checksum received in the last CA public key download and currently in use at the host and device. Source: Device
12	11	Block Number	84	3	R	Identifies the specific data block requested by a device or sent by the host. Source: Device
<FS>						

1Required/Optional/Conditional

11.9.2 Response

The following table contains format information about the CA Public Key File Load Response:

Note: There are no Field Separators in this message.

Message Format CA Public Key File Load Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	1	R	Describes the returned data type. Valid Value: & For EMV, indicates that a new CA Public Key file is being returned. Source: Host
2	86	Sequence Number	2	6	R	Must be echoed from the CA Public Key File Load Request. Source: Request
3	83	Response Code	8	1	R	Indicates whether the key load transaction has been approved or rejected. Valid Values: M (Indicates an approved key load transaction and more load data pending.) L (Indicates an approved key load transaction, last block.) X (Indicates that either the key update is not required or that a key load request is rejected. For details on the error codes returned in the terminal when response code is 'X', refer Data Element No. 194 (CA Public Key File Block) below.) Source: Host
4	187	CA Public Key File Checksum	9	25	R	Identifies the checksum associated with CA Public Key File. Must be used with all subsequent EMV transactions from this device. Source: Host

Message Format CA Public Key File Load Response						
Field No.	**Element No.**	**Element Name**	**Pos.**	**Len.**	**Entry is: R/O/C[1]**	**Description**
5	193	CA Public Key File Block Length	49	4	R	Identifies the length of the following field (CA Public Key File). Valid Values: 0000—9999 Source: Host
6	194	CA Public Key File Block	53	9,999	R	Identifies the new Public Key file or the Error Message. For an approved key load transaction, the maximum length is 9,999 bytes. For a declined key load transaction, a message is returned in this field and displays at the device. RQST/CURRENT HASH MATCH code returned in the terminal indicates that the key update is not required. The following error codes returned in the terminal indicate that the key load transaction is not successful. BLOCK NBR NOT NUMERIC BLOCK NBR NOT 000-xxx Valid Value: Any valid key Source: Host
7	11	Block Number	Varies	3	R	Identifies the specific data block requested by a device or sent by the host. Source: Host

1Required/Optional/Conditional

12. Data Segment Formats

This chapter describes each data segment in the messages described in chapter 12, “Data Segment Formats.”

The following table lists all data segments and the transactions in which they can be included. Please refer to chapter 11, "Message Formats." for additional information about specific message contents.

Data Segments and the Transactions in which They Can Appear

Seg. No.	Segment Name	Financial Transaction Request	Financial Transaction Response	Loyalty Card Transaction	ECA/ TeleCheck®	EMV Financial Transaction Request	EMV Financial Transaction Response	Totals Request	Electronic Mail Request	Table Load Response	Phone Load Response	Date & Time Load Response	Software Load Response	TransArmor Key Load Request	Proprietary Data Load	Proprietary Data Load Response
100	Standard Message	X		X	X	X										
101	Fleet	X				X										
102	Product Code	X				X										
103	EBT	X				X										
104	Purchase Card	X				X										
105	Totals							X								
106	Reserved															
107	Reserved															
108	Loyalty	X		X		X										
109	Electronic Mail								X							
110	Check	X			X											
111	Variable Information	X			X	X										
112	Additional Information		X				X									
113	ECA/ TeleCheck®	X			X											
114	SKU	X		X		X										

Data Segments and the Transactions in which They Can Appear

Seg. No.	Segment Name	Financial Transaction Request	Financial Transaction Response	Loyalty Card Transaction	ECA/ TeleCheck®	EMV Financial Transaction Request	EMV Financial Transaction Response	Totals Request	Electronic Mail Request	Table Load Response	Phone Load Response	Date & Time Load Response	Software Load Response	TransArmor Key Load Request	Proprietary Data Load	Proprietary Data Load Response
115	Print Data Segment		X				X									
116	TransArmor Load Data Segment													X		
118	Proprietary Data Load														X	X
119	Totals with Proprietary Data Load							X								
120	Print Data 2 Segment		X				X									
123	NFC Payment Tokenization Data Segment	X														
130	EMV Request Data Segment					X										
131	EMV Response Data Segment						X									
134	Transaction Attributes Data Segment		X				X									
135	Moneris Data (Request) Segment	X				X										

Data Segments and the Transactions in which They Can Appear

Seg. No.	Segment Name	Financial Transaction Request	Financial Transaction Response	Loyalty Card Transaction	ECA/ TeleCheck®	EMV Financial Transaction Request	EMV Financial Transaction Response	Totals Request	Electronic Mail Request	Table Load Response	Phone Load Response	Date & Time Load Response	Software Load Response	TransArmor Key Load Request	Proprietary Data Load	Proprietary Data Load Response
136	Moneris Data (Response) Segment		X				X									
143	Tax by Product Data Segment	X				X										
145	Enhanced Fleet Request Segment	X				X										
146	Enhanced Fleet Response Segment		X				X									
148	WEX Available Product Fleet Information Data Segment		X				X									
151	Incomm Market Basket Data (Request)	X				X										
152	Incomm Market Basket Data (Response)		X				X									
153	Network Token Data Request	X				X										
154	Network Token Data Response		X				X									

Data Segments and the Transactions in which They Can Appear

Seg. No.	Segment Name	Financial Transaction Request	Financial Transaction Response	Loyalty Card Transaction	ECA/ TeleCheck®	EMV Financial Transaction Request	EMV Financial Transaction Response	Totals Request	Electronic Mail Request	Table Load Response	Phone Load Response	Date & Time Load Response	Software Load Response	TransArmor Key Load Request	Proprietary Data Load	Proprietary Data Load Response
155	Real Time Account Updater Response Data Segment		X				X									
156	EV Charging Data	X				X										
DL1	Merchant									X						
DL2	Dial String									X	X					
DL3	Date and Time									X		X				
DL4	Software Dial Load												X			
DL5	Software IP Load												X			
DL6	Store and Forward Data									X						

12.1 Standard Message Data Segment

This section describes Data Segment No. 100, Standard Message Data Segment.

All Financial Transaction Request messages contain this data segment in Data Section No. 2, Field No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 2.)

It is the only required data segment on all Financial Transaction Request messages.

The data elements in this data segment are the same for each Financial Transaction Request message.

It originates at the device.

Any trailing fields within the data segment that are not needed should not be transmitted. For example, if a data segment has five fields defined, but the last two fields are optional, only three fields should be transmitted with the third field followed by a Field Separator.

The following table illustrates the layout of the Standard Message Data Segment:

Note: All fields are separated by a Field Separator. When a field is not populated, still send the Field Separator.

Standard Message Data Segment Format Data Segment No. 100					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 100 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	44	Information Byte	1	R	Indicates whether the request is a single-message request or a multimessage request. Source: Device
4	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
5	78	Prompt Code	4	R	Identifies the Transaction Type and Card Type. Source: Device

Standard Message Data Segment Format Data Segment No. 100					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
6	2	Account Number	Var.	R	Identifies a portion of the cardholder's account number or identification number. Source: Device **Note:** Merchants using Real Time Account Updater should refer to Account Number - Data Element 2 for additional information.
7	12	Card Discretionary Block Data	51	C	Identifies the card's expiration date and discretionary data. Source: Device **Note:** Merchants using Real Time Account Updater should refer to Card Discretionary Block Data - Data Element 12 for additional information.
8	33	Encrypted PIN Block Data	36	C	For DUKPT PIN encryption, this is a 16- to 20-byte KSN plus a 16-byte PIN block. Source: Cardholder
9	79	Pump/Lane Number	2	C	Identifies the pump or lane number at the point-of-service location. Source: Device
10	41	Fuel Purchase Amount	8	C	Identifies the net total dollar amount of all fuel purchase amounts less any applicable discounts and/or coupons. Source: Device
11	58	Nonfuel Amount	8	C	Identifies the total net dollar amount of all nonfuel, noncash, and nontax amounts less any applicable discounts and/or coupons. Source: Device
12	99	Tax Amount	8	C	Identifies the total dollar amount of all tax amounts. Source: Device
13	17	Cash Amount	8	C	Identifies the total dollar amount of all cash amounts. Source: Device
14	86	Sequence Number	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
15	5	Approval Number	6	C	Identifies the assigned Approval Number. Source: Device

Standard Message Data Segment Format Data Segment No. 100					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
16	49	Local Date and Local Time	10	C	Identifies the local date and local time of the preauthorized transaction. Source: Device
17	121	Partial Approval Indicator	1	R	Indicates whether or not a merchant supports credit/debit Partial Approvals. Source: Device

[1]Required/Optional/Conditional

12.2 Fleet Data Segment

This section describes Data Segment No. 101, Fleet Data Segment.

This data segment can appear in any of the fields in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.) It has a maximum length of 61 alphanumeric characters (01–61/a–z/A–Z). It originates at the device.

The Fleet Data Segment is included in all fleet card transaction requests.

Note: In spite of the above, Merchants should not send the Fleet segment (Data Segment No. 101) and the Enhanced fleet segment together (Data Segment No. 145).

The following table illustrates the layout of the Fleet Data Segment:

Note: All fields are separated by Field Separators. When a field is not populated, still send the Field Separator.

Fleet Data Segment Format
Data Segment No. 101

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 101 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	64	Odometer	8	C	Identifies the vehicle's odometer reading keyed in by the customer or clerk. Source: Device
4	108	Vehicle Number	10	C	Identifies the Vehicle Number. Source: Device
5	47	Job Number	10	C	Identifies the Job Number. Source: Device
6	31	Driver /Identification Number	10	C	Identifies the unencrypted Driver Number or Identification Number. Source: Device
7	40	Fleet Employee Number	10	C	Identifies the Fleet Employee Number. Source: Device
8	158	License #	10	C	Identifies the License # of the fleet card user. Source: Device
9	159	Job ID	12	C	Identifies the Job ID. Source: Device
10	160	Department #	12	C	Identifies the Department #. Source: Device

Fleet Data Segment Format Data Segment No. 101					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
11	161	Customer Data	12	C	Identifies the Customer Data. Source: Device
12	162	User ID	12	C	Identifies the User ID. Source: Device
13	163	Vehicle ID#	8	C	Identifies the Vehicle ID#. Source: Device
14		Fleet Tag 1	34	C	Sent only in Auth Completion message in the following data format: Tag – 3 bytes fixed length Data – Up to 31 bytes
15		Fleet Tag 2	34	C	Sent only in Auth Completion message in the following data format: Tag – 3 bytes fixed length Data – Up to 31 bytes
16		Fleet Tag 3	34	C	Sent only in Auth Completion message in the following data format: Tag – 3 bytes fixed length Data – Up to 31 bytes
17		Fleet Tag 4	34	C	Sent only in Auth Completion message in the following data format: Tag – 3 bytes fixed length Data – Up to 31 bytes
18		Fleet Tag 5	34	C	Sent only in Auth Completion message in the following data format: Tag – 3 bytes fixed length Data – Up to 31 bytes

[1]Required/Optional/Conditional

Refer the table below for the valid values of the 3 byte tag that can be sent on a Auth Completion (0220) message only when Host Prompts are supported:

Valid Values of 3 Byte Fleet Tag Used in the Completion Message		
Code	**Max Length**	**Description**
DLS	an3	Driver License State/Province Abbreviation
DLN	an22	Driver License nameation
PON	an31	Work Order/P.O. Number
INV	an31	Invoice Number
TRP	an15	Trip Number
UNT	an31	Unit Number

Valid Values of 3 Byte Fleet Tag Used in the Completion Message		
Code	**Max Length**	**Description**
TLH	n6	Trailer Hours/Refer Hours
DOB	n8	Date of Birth
ZIP	an9	ZIP/Postal Code
END	an31	Entered Data (Alphanumeric)
MID	an31	Maintenance ID
VIN	an17	VIN
TRA	an15	Tractor Number
HUB	n9	Hubometer
TLR	an15	Trailer Number
CBA	n6	Cashback Amount
VHT	an10	Vehicle Tag

12.3 Product Code Data Segment

This section describes Data Segment No. 102, Product Code Data Segment.

Note: This segment can be used for all card types except for Comdata cards.

Note: This segment is mutually exclusive with segment 157 Adjusted Product Code Data. Merchants should send one or the other but never both. If both the segments are sent, the transaction will be declined.

When a Financial Transaction request includes the Product Code Data Segment, the products within this segment must meet the following requirements:

- Fuel products must always be the first products in the segment.
- For EV charging transactions, the EV fuel product code should be the first product code in the segment.
- A maximum of ten products is allowed in the segment.
- The total of product amounts in the segment must equal the total of the amounts [Element No. 41 (Fuel Purchase Amount), Element No. 58 (Nonfuel Amount), Element No. 99 (Tax Amount) and Element No. 17 (Cash Amount)] in the Standard Message Data Segment (Data Segment No. 100).
- When tax product codes are included in the segment, the total dollar amount of tax products is also reported in Element No. 99 (Tax Amount) in the Standard Message Data Segment (Data Segment No. 100).
- All fuel merchants must send fuel and nonfuel product data.
  Note: A unique Product Code must be sent for each type of fuel purchase.

Note: With the growth of fleet industry and to fulfil the needs of the Over the Road Trucking (OTR) industry, BUYPASS can accept transactions with multiple fuels (e.g., purchase of Diesel, DEF, REEFER), all on 1 transaction. It is estimated that up to 30% of fuel transactions in OTR require fill-ups involving multiple Fuel Type Codes. Now, fuel merchants can allow multiple fuel fill-ups in 1 transaction. With the support of multiple fuels, a merchant must send the Product segment to BUYPASS with all the fuel codes and any non-fuel codes with all the relevant information like quality, unit of measure, unit price, and product amount. Also, the merchant must ensure to send the primary fuel (fuel used for the primary fuel need of the truck/vehicle) as the very first product code within the Product segment. Refer to the Product Code Data segment for the details on the product information layout and rules.

This data segment can appear in any of the fields in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 381 alphanumeric characters (001–381/a–z/A–Z).

It originates at the device.

The following table illustrates the layout of the Product Data Segment:

Notes: A Field Separator always follows Segment Type and Segment Length. A Field Separator follows Product Amount when Product Amount is the very last element in the segment. When a field is not populated, still send the Field Separator.

A Product Data Field Delimiter always follows Quantity and Unit Price. A Product Data Field Delimiter follows Product Amount when Product Amount is not the last element in the segment. When a field is not populated, still send the Product Data Field Delimiter.

For an example of this segment, please refer to Appendix B. POS Purchase Example.

In the following table, the ▲ symbol represents a Field Separator, and the \ character is a Product Data Field Delimiter.

Product Code Data Segment Format Data Segment No. 102					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 102 Source: Device
< ▲ >					
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
< ▲ >					
3	87	Service Level	1	R	Identifies the type of sale. Source: Device
4	62	Number of Products	2	R	Identifies the count of products reported for this transaction. Source: Device
Product Data This section is repeated—by product—for a maximum of 10 products for a total variable length of up to 370 bytes. Fuel products are always the first products in the segment. **Note:** A unique Product Code must be sent for each type of fuel purchase.					
5	77	Product Code	3	R	Identifies the type of product sold or applicable discounts and/or coupons (negative Product Codes). Source: Device
6	106	Unit of Measure	1	R	Identifies the type of measurement for the product sold. Source: Device
7	81	Quantity	9	R	Identifies the number of units of the product sold. Quantity value must contain a digit for every position of assumed decimal places. Example: If the quantity is 0.05 and the number of assumed decimal places is 2, then this should be represented as '205'. '25' will be considered invalid. Source: Device
< \ >					

Product Code Data Segment Format Data Segment No. 102					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
8	107	Unit Price	9	R	Identifies the price per unit of measure for the product sold. Unit Price value must contain a digit for every position of assumed decimal places. Example: If the unit price is $0.059 and the number of assumed decimal places is 3, then this should be represented as '3059'. '359' will be considered invalid. Source: Device
< \ >					
9	76	Product Amount	12	R	Identifies the monetary value of the product sold or applicable discounts and/or coupons. Source: Device
A Field Separator follows Product Amount if Product Amount is the very last element in the segment. OR A Product Data Field Delimiter follows Product Amount if Product Amount is not the last element in the segment.					

[1]Required/Optional/Conditional

12.4 EBT Data Segment

This section describes Data Segment No. 103, EBT Data Segment.

This data segment can appear in any of the fields in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 3,334 alphanumeric characters (01–3,334/a–z/A–Z).

It originates at the device.

The following table illustrates the layout of the EBT Data Segment:

Note: When a Financial Transaction request includes the EBT Data Segment, all fields are separated by Field Separators. When a field is not populated, still send the Field Separator.

When a Financial Transaction response includes the EBT Data Segment, there is no Field Separator between fields in this message.

EBT Data Segment Format
Data Segment No. 103

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 103 Source: Device
2	84	Segment Length	3 or 4	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device **Note:** Length should be "4" for EBT with eWIC data transactions.
3	18	Clerk ID	10	C	Identifies the clerk's Identification Number. Source: Device
4	109	Voucher ID	10	C	Identifies the preprinted EBT voucher used in a local Approval transaction. Source: Device
5	153	WIC Discount Amount	40	C	Identifies discounts/cents-off coupon amounts unique to eWIC transactions. Source: Device/Host
6	154	WIC Product Data	3001	C	Identifies data unique to eWIC transactions. Source: Device/Host
7	164	EBT Program Data	267	C	Identifies data unique to EBT transactions. Source: Device/Host

[1]Required/Optional/Conditional

12.5 Purchase Card Data Segment

This section describes Data Segment No. 104, Purchase Card Data Segment.

This data segment can appear in any of the fields in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 86 alphanumeric characters (01–86/a–z/A–Z).

It originates at the device.

The following table illustrates the layout of the Purchase Card Data Segment:

Note: All fields are separated by Field Separators. When a field is not populated, still send the Field Separator.

Purchase Card Data Segment Format Data Segment No. 104					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 104 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	80	Purchase Code	16	C	Identifies the Purchase Code associated with the transaction. Source: Device
4	74	PC Tax Amount	7	C	Identifies the amount of tax in the transaction. Source: Device
5	73	PC Freight Amount	7	C	Identifies the freight charges in the transaction. Source: Device
6	72	PC Duty Amount	7	C	Identifies the duty charges in the transaction. Source: Device
7	89	Ship-to Country Code	3	C	Identifies the shipping code for the country to which the merchandise is being shipped. Source: Device
8	90	Ship-to Postal Code	10	C	Identifies the postal code for the location to which the merchandise is being shipped. Source: Device

Purchase Card Data Segment Format Data Segment No. 104					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
9	88	Ship-from Postal Code	10	C	Identifies the postal code for the location from which the merchandise is being shipped. Source: Device
10	29	Direct Marketing Invoice Number	10	C	Identifies the invoice number in a direct marketing transaction. Source: Device

[1]Required/Optional/Conditional

12.6 Totals Data Segment

This section describes Data Segment No. 105, Totals Data Segment.

It always appears in Field No. 3 in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 409 alphanumeric characters (001–409/a–z/A–Z).

It originates at the device.

The following table illustrates the layout of the Totals Data Segment:

Note: Field Nos. 1–14 are separated by Field Separators, with a Field Separator following Field No. 14. When a field is not populated, still send the Field Separator. Field Nos. 15–17 are not separated by Field Separators.

Totals Data Segment Format
Data Segment No. 105

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 105 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	44	Information Byte	1	R	Indicates whether the request is a single-message request or a multimessage request. Source: Device
4	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
5	78	Prompt Code	3	R	Identifies the format type for the requested totals. Fixed value: 990 Source: Device
6	32	Employee Number	4	C	Identifies a BUYPASS customer's employee. Source: Device
7	65	Password	6	C	Identifies the end-of-day function's password. Source: Device
8	105	Totals Date	6	R	Identifies the requested type of totals. Source: Device

Totals Data Segment Format Data Segment No. 105					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
9	43	Hardware Version	4 or 8	R	Identifies the device's hardware level version number. Source: Device
10	96	Software Version	8	R	Identifies the device's software version number. Source: Device
11	39	Firmware Version	8	R	Identifies the device's EPROM level. Source: Device
12	86	Sequence Number	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
13	20	Currency Code	3	O	Identifies the transaction's currency type according to country. Source: Device
14	42	Grand Total	8	R	Identifies the final total amount. Source: Device

Totals Data Segment Format Data Segment No. 105					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
Category Total Data Section Field Nos. 15–17 are sent up to a maximum of 18 times—once for each of the following card types—in the order shown and only when data occurs for a card type. Use Card Labels (in parentheses) instead of field numbers to identify total buckets. 1. MasterCard®/Visa® (CC) 2. Amex® (TE) 3. Discover® Network, Diners Club®, JCB® (DS) 4. Preauthorization (AO) 5. Debit (DB) 6. Fleet (FL) 7. Cash (CS) 8. Proprietary Card (PR) 9. Check (CK) 10. EBT Food Stamp (EF) 11. EBT Cash Benefit (EC) 12. Stored Value Card Activation/Deactivation (SV1) 13. Stored Value Card Purchase/Capture /Merchandise Return (SV2) 14. Stored Value Card Replacement (SV3) 15. Stored Value Card Recharge (SV4) 16. ECA/ TeleCheck® service (ECA) 17. EBT version of WIC (EWIC) 18. EBT child care (EK)					
15	13	Card Label	4	R	Identifies the card type. Source: Device
16	16	Card Type Total Count	5	R	Identifies the transaction count for a particular card type. Source: Device
17	15	Card Type Total Amount	8	R	Identifies the total amount for a particular card type. Source: Device

[1]Required/Optional/Conditional

12.7 Loyalty Card Data Segment

This section describes Data Segment No. 108, Loyalty Card Data Segment.

It always appears in Field No. 4 in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 142 alphanumeric characters (01–142/a–z/A–Z).

It originates at the device.

The following table illustrates the layout of the Loyalty Card Data Segment:

Note: All fields are separated by Field Separators, with a Field Separator following Field No. 13. When a field is not populated, still send the Field Separator.

Loyalty Card Data Segment Format Data Segment No. 108					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 108 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	138	Loyalty Program ID	6	R	Identifies the loyalty program ID. Source: Device
4	139	Loyalty Account Number	24	C	Identifies the loyalty cardholder's account number. Source: Device
5	140	Points to Redeem	6	O	Identifies the loyalty points to redeem for a purchase accepted by the transmitting device. Source: Device
6	141	Coupon ID	19	O	Identifies the ID of the loyalty coupon accepted by the transmitting device. Source: Device
7	142	Coupon Amount	8	O	Identifies the amount of the loyalty coupon accepted by the transmitting device. Source: Device
8	143	Update Code	1	C	Identifies the type of loyalty functions being performed by the transmitting device. Source: Device
9	144	Street Address	5	C	Identifies the loyalty cardholder's street number. Source: Device

Loyalty Card Data Segment Format Data Segment No. 108					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
10	145	Phone Number, Loyalty	10	C	Identifies the loyalty cardholder's phone number. Source: Device
11	146	Expiration Date	4	C	**Note:** This data element is reserved for future use.
12	148	Payment Tender Type	2	R	Used to determine whether the transaction is a multiple-card transaction involving a loyalty card. Source: Device
13	147	Loyalty Track 2 Data	38	O	Identifies the Track 2 data on the loyalty card used in the loyalty transaction. Source: Device
14	150	Loyalty Information Version	1	O	Identifies the version of the loyalty information supported by the device. Source: Device
15	151	Unit of Work	19	O	Identifies a unique number used by the loyalty host to identify a transaction. Source: Device

[1]Required/Optional/Conditional

12.8 Electronic Mail Data Segment

This section describes Data Segment No. 109, Electronic Mail Data Segment.

It always appears in Field No. 3 in Data Section No. 2. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 232 alphanumeric characters (001–232/a–z/A–Z).

It originates at the device.

The following table illustrates the layout of the Electronic Mail Segment:

Note: All fields are separated by Field Separators. When a field is not populated, still send the Field Separator.

Electronic Mail Data Segment Format Data Segment No. 109					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 109 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	44	Information Byte	1	R	Indicates whether the request is a single-message request or a multimessage request. Source: Device
4	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
5	78	Prompt Code	3	R	Identifies the request's format type. Source: Device
6	11	Block Number	3	R	Identifies the specific electronic mail data block requested by a device or sent by the host. Source: Host
7	32	Employee Number	4	C	Identifies a BUYPASS customer's employee. Source: Device
8	65	Password	6	C	Identifies the end-of-day function's password. Source: Device

Electronic Mail Data Segment Format Data Segment No. 109					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
9	86	Sequence Number	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
10	50	Local Time	4	C	Identifies the time of the request. Source: Device
11	36	Extract Date	6	C	Identifies the date the information was extracted. Source: Device
12	37	Extract Time	4	C	Identifies the time the information was extracted. Source: Device
13	104	Text Data Length	3	C	Identifies Text Data's length. Source: Device
14	103	Text Data	150	C	Identifies the request's data. Source: Device

[1]Required/Optional/Conditional

12.9 Check Data Segment

This section describes Data Segment No. 110, Check Data Segment.

This data segment can appear in any of the fields in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 168 alphanumeric characters (001–168/a–z/A–Z).

It originates at the device.

The following table illustrates the layout of the Check Data Segment:

Note: All fields are separated by Field Separators. When a field is not populated, still send the Field Separator.

MICR readers provide the flexibility to format the MICR fields transmitted to the host for authorization. There are two types of MICR formats used in check services:

1) Full MICR Line – TAC Format - Example **T**999999999**A**99999999999999999**C**999999
2) Full MICR Line - Format RAW TOAD

TOAD is a symbol substitution of the MICR characters appearing on the check.

Here,

T replaces 

This symbol is used to indicate the beginning and ending of the bank's 9-digit routing transit number.

O replaces 

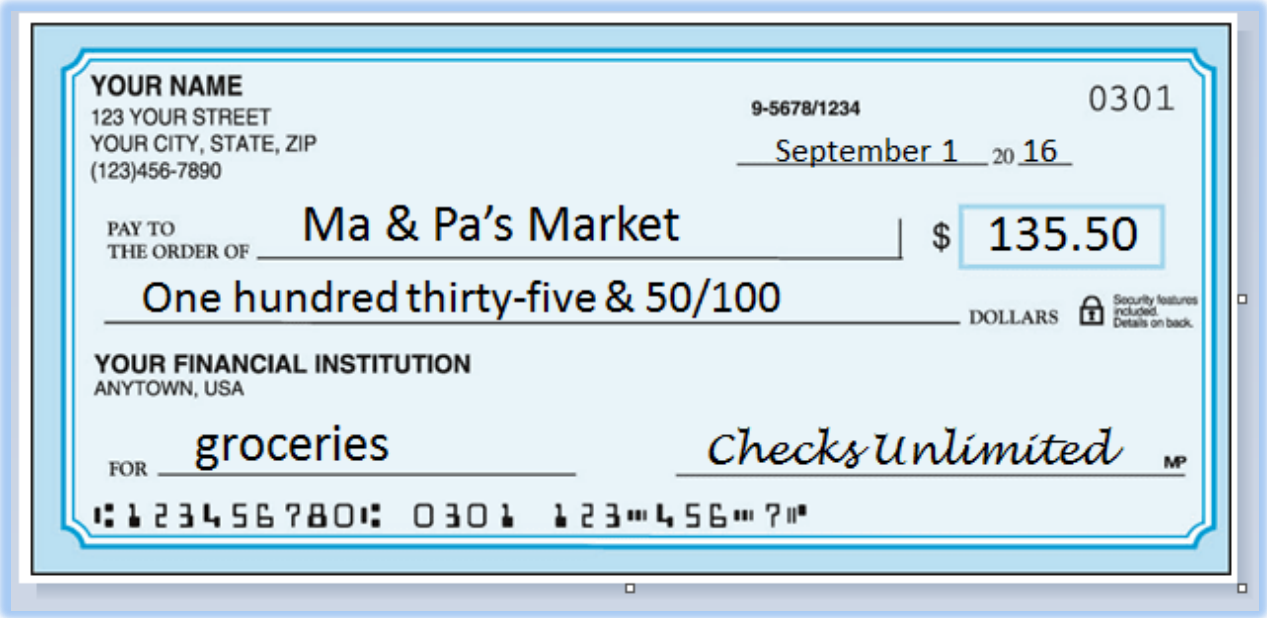
This symbol is used to indicate the 'On-Us' field usually grouped around the account number or check serial number.

A replaces 

This symbol is used to indicate the beginning and ending of the encoded dollar amount field.

D replaces

This symbol is a dash usually appearing in the account number field in the MICR line.



TOAD Example:

⑆123456780⑆0301 123⑈456⑈7⑈

MICR reader output: **T**123456780**T**0301 123**D**456D7**O**

'BUY2' and 'ALB1' are the two types of alphanumeric headers assigned by Certegy (Tele credit) Check Services to uniquely identify each customer. Buypass has to support the check data segment for both the 'BUY2' and 'ALB1' formats. Therefore, a new field, Alternate MICR Indicator, is introduced to identify the 'ALB1' format so that Buypass can support both the RAW TOAD MICR and TAC MICR data.

A merchant, who wants to support the Alternate MICR format, should send 'Y' in the new field Alternate MICR Indicator introduced in the 110-check data segment and typically use this field in a check transaction along with the MICR/Driver's License number.

The element '131' is currently in use. Hence, the element number '239' is used for Alternate MICR Indicator.

Note: Merchants who want support for the RAW MICR format should send Alternate MICR Indicator.

Check Data Segment Format Data Segment No. 110					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**

Check Data Segment Format Data Segment No. 110					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 110 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	122	MICR Data	50	R	Identifies the check's MICR data. Source: Device
4	123	Driver's License	40	C	Identifies the driver's license number. Source: Device
5	124	State Code	2	C	Identifies the State Code for the driver's license number or another form of consumer identification. Source: Device
6	125	Date of Birth	8	C	Identifies the consumer's date of birth. Source: Device
7	126	Check Type	1	R	Identifies the type of check. Source: Device
8	127	Check Number	8	C	Identifies the check number. Required for manually keyed check data. Source: Device
9	128	Customer Phone Number	10	O	Identifies the consumer's phone number. Source: Device

Check Data Segment Format Data Segment No. 110					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
10	129	Customer Last Name	24	O	Identifies the consumer's last name. Source: Device
11	130	Check Issue Date	8	O	Identifies the issue date of the check. Source: Device
12	239	Alternate MICR IND	1	O	Denotes that the merchant supports the Alternate MICR format. This field is optional. Hence, this field is not required for merchants who do not support the Alternate MICR format. Value 'Y' - Indicates that the Alternate MICR format is being sent from the terminal. Fixed: 1 Byte Source: Device

[1]Required/Optional/Conditional

12.10 Variable Information Data Segment

This section describes Data Segment No. 111, Variable Information Data Segment.

This data segment can appear in any of the fields in data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 999 alphanumeric characters (01–999/a-z/A–Z).

It originates at the device.

The following table illustrates the layout of the Variable Information Data Segment:

Note: There is a Field Separator between Field Nos. 1 and 2 and between Field No. 2 and Variable Information Section. Field Separators are not between the elements in Variable Information Section. Field Separators are not between repetitions of Variable Information Section. A Field Separator follows the final Variable Information Section.

Variable Information Data Segment Format Data Segment No. 111					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 111 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
Variable Information Section This section is repeated—by Variable Information Indicator—for a total variable length of up to 991 bytes. The Field Separator that follows Field No. 5 is included with the last repetition.					
3	111	Variable Information Indicator	3	R	Identifies the type of variable information. Source: Device
4	112	Variable Information Length	3	R	Identifies the length of Variable Information (Element No. 113). Source: Device
5	113	Variable Information	Var.	R	Identifies the variable information. Source: Device

[1]Required/Optional/Conditional

12.11 Additional Information Data Segment

This section describes Data Segment No. 112, Additional Information Data Segment.

Segment information can include:

- Balances—Gift, EBT, Credit, Phone
- AVS or NVS Result Code or both
- CVV
- ECA/ TeleCheck® Trace ID
- ECA/ TeleCheck® Denial Record Number
- ECA/ TeleCheck® Return Check Data
- Loyalty—Version 1
- Visa Products
- Loyalty—Version 2
- User data
- Discover® Network
- Expiration Date (MMYY) for TransArmor - VeriFone edition processing
- PIN-on-Receipt Information
- Visa Spend Qualified Indicator Information
- Universal Unique Identifier
- PAR (Payment Account Reference) Data Information
- Merchant Advice Code
- DAF Indicator
- Agreement ID
- Host-based Purchase Restriction
- Cardholder Additional Information Code
- Account Type
- Account Funding Source
- Transaction Link Identifier
- Transaction Link Action Indicator
- Fraud Score
- Fraud Score Reason Code
- Authentication Data Quality Indicator
- Token Update First Use Indicator
- Merchant Tran ID Indicator
- Applied Special Service Indicator
- Voyager Restriction Code Indicator
- Visa Category Code

Note: This data segment may be used in the future for additional information.

This data segment appears at the end of a Financial Transaction response.

It has a maximum length of 999 alphanumeric characters (001–999/a-z/A–Z).

BUYPASS originates the transmission of the above-listed data.

The following table illustrates the layout of the Additional Information Data Segment:

Additional Information Data Segment Format
Data Segment No. 112

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 112 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Host
Additional Information Section This section is repeated—by Additional Information Indicator—for a maximum length of 990 bytes.					
3	116	Additional Information Indicator	3	R	Identifies the type of additional information being transmitted. Source: Host
4	117	Additional Information Length	3	R	Identifies the length of Additional Information (Element No. 118). Source: Host
Additional Information Section This section is repeated—by Additional Information Indicator—for a maximum length of 990 bytes.					
5	118	Additional Information	Var.	R	Identifies the additional information. Source: Host

[1]Required/Optional/Conditional

12.12 ECA/ TeleCheck® Data Segment

This section describes Data Segment No. 113, ECA/ TeleCheck® Data Segment.

This data segment can appear in any of the fields in data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 156 alphanumeric characters (001–156/a-z/A–Z).

It originates at the device.

The following table illustrates the layout of the ECA/ TeleCheck® Data Segment:

Note: All fields are separated by Field Separators. When a field is not populated, still send the Field Separator.

ECA/ TeleCheck® Data Segment Format Data Segment No. 113					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 113 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	131	ECA/ TeleCheck® Clerk ID	6	R	Identifies the clerk's ID. Source: Device
4	132	ECA/ TeleCheck® Product Code	6	C	Identifies the ECA/ TeleCheck® product code for the type of merchandise sold by the merchant and used with ECA/ TeleCheck® Service Risk Control. Source: Device
5	133	ECA/ TeleCheck® Phone Number	10	O	Identifies the phone number of the check writer. Source: Device
6	134	ECA/ TeleCheck® Trace ID	22	C	Identifies the trace ID of the original ECA/ TeleCheck® transaction. This data element must be present in all Void transaction requests. Source: Device
7	135	Merchant Trace ID	25	O	Identifies a unique identifier assigned by the merchant. Source: Device

ECA/ TeleCheck® Data Segment Format Data Segment No. 113					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
8	136	Denial Record Number	7	C	Identifies the record number of the declined ECA/ TeleCheck® transaction. Source: Device
9	137	Extended MICR Data	65	C	Identifies the check MICR data when the MICR length is greater than 50-bytes. Source: Device

[1]Required/Optional/Conditional

12.13 SKU Data Segment

This section describes Data Segment No. 114, SKU Data Segment.

It always appears in Field No. 5 in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 1010 alphanumeric characters (001–1010/a-z/A–Z).

It originates at the device.

The following table illustrates the layout of the SKU Data Segment:

Note: All fields are separated by Field Separators. A Field Separator follows Field No. 3. When a field is not populated, still send the Field Separator.

SKU Data Segment Format Data Segment No. 114					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Source: Device
2	84	Segment Length	4	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	149	SKU Data	1000	R	Identifies the bar code SKU data. Source: Device

[1]Required/Optional/Conditional

12.14 Print Data Segment

This section describes Data Segment No. 115, Print Data Segment

It is only sent on Financial Transaction Response messages requiring large amounts of print data. It contains data elements that are unique to transactions requiring print data.

It always appears at the end of a Financial Transaction response.

It has a maximum length of 1,009 alphanumeric characters (01–1,009/a-z/A–Z).

The BUYPASS host originates the transmission of loyalty information.

No other data segments are contained in the Financial Transaction Response.

The following table illustrates the layout of the Print Data Segment:

Note: There is a Field Separator between Field Nos. 1 and 2 and between Field Nos. 2 and 3.

Print Data Segment Format Data Segment No. 115					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 115 Source: Host
2	84	Segment Length	4	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Host
3	152	Print Data	999	R	Identifies the print data being transmitted to the device. **Note:** This field contains the terms & conditions for Blackhawk phone activation and recharge receipts. Source: Host

[1]Required/Optional/Conditional

12.15 TransArmor Load Data Segment

Note: For detailed information on TransArmor, refer the BUYPASS®_Platform_ATL105_Specification_Updates_for_TransArmor_Processing document, available on the Direct Platform Specifications Portal.

12.16 Proprietary Data Load Segment

This section describes Data Segment No. 118, Proprietary Data Load Segment.

It is used to request Customer Specific Proprietary data for loading or Customer Specific Proprietary data for capturing at the BUYPASS FEP. This function will be similar to Electronic Mail and be used to support the requirements for Dynamic Card Table, Custom Receipt Text, Host Discounts, and Site Configuration, and Fuel Volume. Dynamic Card Table, and Custom Receipt Text, and Host Discounts are driven by the host in a data push to the POS, while Site Configuration data and Fuel Volume data are driven by the POS in data push to the host.

It has a maximum length of 3,800 alphanumeric characters (01–3,800/a–z/A–Z) in a Request.

It has a maximum length of 3,800 alphanumeric characters (01-3,800/a–z/A–Z in a Response.

It originates at the device.

The following table illustrates the layout of the Proprietary Data Load Segment.

Note: For transaction requests, all fields are separated by Field Separators. When a field is not populated, still send the Field Separator. A Field Separator follows Field No. 13.

For transaction responses, there are no Field Separators between fields in this message.

Proprietary Data Load Segment Format
Data Segment No. 118

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C1	Description
1	85	Segment Type	3	R	Indicates the type of data segment being formatted. Fixed Value: 118 Source: Device
2	84	Segment Length	4	R	Indicates the data segment's length, including the Segment Type's length and Field Separators. Source: Device
3	86	Sequence Number	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source:
4	44	Information Byte	1	R	Indicates whether the request is a single message request or a multimessage request. Source: Device
5	102	Terminal Identifier	13	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number Source: Device

Proprietary Data Load Segment Format Data Segment No. 118					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C1**	**Description**
6	78	Prompt Code	3	R	Identifies the prompt code for requested special transaction data. Valid values: 901, 902, 903, 904, 905 Appendix E, "Valid Card Type Codes for Special Transactions" Source: Device
7	182	Prompt Code, Pending	4	R	Identifies the next prompt code for requested special transaction data. Valid values: 0901, 0902, 0904, 0981 Appendix E, "Valid Card Type Codes for Special Transactions" Source: Device
8	176	Device Card Table Version	35	R	This is the device's current version numbers for the Card Table Master ID, BIN table version, RULES table version, RESTRICTIONS table version, SAF table version, PROMPT table version, PRODUCT table version all strung together. Each of these version values is 5 characters individually. Source: Device
9	177	Card Table Load Version	35	R	This represents the versions of the various Card Table components that are being currently loaded to the device. Source: This value comes from the Host and is echoed back in each subsequent block request.
10	174	Card Table Type	4	R	Identifies the specific Table type this segment addresses. Source: This value comes from the host and is echoed back in each subsequent block reques.t
11	178	Load Control Key	60	R	Identifies the load control key. Source: This value comes from the host and is echoed back in each subsequent block request.
12	179	Host Discount Timestamp	12	R	Identifies the timestamp used to support host discounts. Format: CCYYMMDDHHMM Source: Host

Proprietary Data Load Segment Format Data Segment No. 118					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C1**	**Description**
13	11	Block Number	3	C	Identifies the specific block number being sent by a device or by the Host. The Host will use this value to determine the next block to be sent. Source: This value comes from the host and is echoed back in the subsequent block request

Proprietary Load Request Data

In addition to Field Nos. 1-13, this section is included in a Proprietary Data Load Request. The data contained in this section corresponds to Element No. 78 (Prompt Code). The maximum variable length of this data is not to exceed 3,800 bytes.

Note: Please refer to the data segment description in the following section:

- ◊ 12.17.3, Proprietary Load Request Data – Prompt Code 903, Site Configuration Data
- ◊ 12.17.5, Proprietary Load Request Data – Prompt Code 905, Fuel Volume Data

Proprietary Load Response Data

In addition to Field Nos. 1-13, this section is included in a Proprietary Data Load Response. The data contained in this section corresponds to Element No. 78 (Prompt Code). The maximum variable length of this data is not to exceed 3,800 bytes.

Note: Please refer to the data segment descriptions in the following sections:

- ◊ 12.17.1, Proprietary Load Response Data – Prompt Code 901, Custom Receipt Text Data
- ◊ 12.17.2, Proprietary Load Response Data – Prompt Code 902, Dynamic Card Table Data
- ◊ 12.17.4, Proprietary Load Response Data – Prompt Code 904, Host Discount Data

12.16.1 Proprietary Load Response Data, Prompt Code 901

The following table illustrates the format of Field Nos. 14–20 when the value of Data Element No. 78 (Prompt Code) is 901, Custom Receipt Text Data:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Proprietary Load Response Data Prompt Code 901, Custom Receipt Text Data					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C1**	**Description**
14	165	Start Date	8	R	Indicates the date to begin using the Receipt Text Data. Format: CCYYMMDD Source: Host
15	166	Start Time	4	R	Indicates the time to begin using the receipt text data. Format: HHMM Source: Host
16	165	End Date	8	R	Indicates the date to stop using the receipt text data. Format: CCYYMMDD
17	166	End Time	4	R	Indicates the time to stop using the receipt text data. Format: HHMM
18	168	Number of Receipt Text Lines	2	R	Indicates the number of times Receipt Text Data is in the response.
Receipt Text Line Data This section is repeated – by Number of Receipt Text Lines – for a total variable length of up to 220 bytes.					
19	169	Receipt Text Data Length	2	R	Identifies the length of the following field (Receipt Text Data). Source: Host
20	170	Receipt Text Data	20	R	Identifies the receipt text data. Source: Host

[1]Required/Optional/Conditional

12.16.2 Proprietary Load Response Data, Prompt Code 902

A value of 902 in the Prompt Code is used for Dynamic Card Table loads. This segment type is used to return one or more of the Card Table types as part of a Load Request. In most cases this will be a multiple message exchange where the indicated tables are provided through a series of exchanges between the Site System and the BUYPASS FEP.
The following table illustrates the format of Field No. 14 when the value of Data Element No. 78 (Prompt Code) is 902, Dynamic Card Table Data:
Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Proprietary Load Response Data Prompt Code 902, Dynamic Card Table Data					
Field No.	**Element No.**	**Element Name**	**Max. Len.**	**Entry is: R/O/C1**	**Description**
14	175	Card Table Data	3,600	R	The data for the indicated table type. Field length determined by segment length in field 2

[1]Required/Optional/Conditional

12.16.3 Proprietary Load Request Data, Prompt Code 903

A Prompt Code value of 903 is used for Site Configuration Data loads. This load type is used to send the site configuration data tables in a Proprietary Data Load request. In most cases, this will be a multiple message exchange between the Site System and the BUYPASS FEP.
The Block Number starts at 1 and is incremented with each segment of site configuration data as it is sent. A value of 0 in the Block Number field indicates that all data was sent in the previous blocks. A value of 0 in the Block Number field is sent as a final message and does not contain any site configuration data on it.
Site configuration data is sent to the host when a configuration change occurs at the site. If multiple configuration changes occur at a site during a business day, only the final site configuration data is sent to the host. If no configuration changes occur during a business day, then no data is sent from the site to the host.
The Response Codes for the Site Configuration Data load are:

- T – Approved—Proprietary load data, no more data pending
- U – Declined—Proprietary load data, no more data pending

The following table illustrates the format of Field No. 14 when the value of Data Element No. 78 (Prompt Code) is 903, Site Configuration Data:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows. A Field Separator follows Field No. 14.

Proprietary Load Request Data Prompt Code 903, Site Configuration Data					
Field No.	**Element No.**	**Element Name**	**Max Len.**	**Entry is: R/O/C1**	**Description**
14	180	Site Configuration Data	3,600	R	The data for the indicated table type. Field length determined by segment length in field 2. Source: Vendor

1Required/Optional/Conditional

12.16.4 Proprietary Load Response Data, Prompt Code 904

A Prompt Code value of 904 is used when the Response Code in a Totals Response indicates that proprietary load data is pending. The device then sends a Proprietary Data Load Request, using a Prompt Code value of 904. Once the device receives the proprietary load response data, it then applies the host discount data to applicable transactions during the designated date and time period.
The amount of the discount in a Financial Transaction Request is included in the Product Code Data Segment, Segment No. 102 using Product Code 941 or 991.
Note: Product Code 941 indicates a negative amount and is used in a post-pay environment when the discount is applied to the transaction amount.
Product Code 991 indicates an administrative amount and is used in a pre-pay environment when the price is adjusted to include the discount amount.

The following table illustrates the format of Field Nos. 14-27 when the value of Data Element No. 78 (Prompt Code) is 904, Host Discount Data:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Proprietary Load Response Data Prompt Code 904, Host Discount Data					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C1**	**Description**
14	179	Host Discount Timestamp	12	R	Identifies the new host discount timestamp to use after a successful load. Format: CCYYMMDDHHMM Source: Host
15	171	Number of Discounts	2	R	Indicates the number of times Host Discount Data is in the response. Source: Host
Host Discount Data This section is repeated – by Number of Discounts – for a total variable length not to exceed 3,600 bytes.					
16	165	Start Date	8	R	Indicates the date to begin using the host discount data. Source: Host
17	166	Start Time	4	R	Indicates the time to begin using the host discount data. Source: Host
18	165	End Date	8	R	Indicates the date to stop using the host discount data. Source: Host

Proprietary Load Response Data Prompt Code 904, Host Discount Data					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C1**	**Description**
19	166	End Time	4	R	Indicates the time to stop using the host discount data. Source: Host
20	173	BUYPASS Card Type	4	R	The BUYPASS Card Type to which a Discount applies – this value is matched to the Card Type value of the BIN table to determine if a Host Discount is applicable Source: Host
21	183	Card BIN Range, Beginning	12	R	Indicates the beginning card BIN range to apply the host discount data. Source: Host
22	184	Card BIN Range, Ending	12	R	Indicates the ending card BIN range to apply the host discount data. Source: Host
23	77	Product Code	3	R	The Product Code to which a Discount applies. Source: Host
24	172	Product Discount Amount	5	R	The amount of discount to apply to the product indicated in the preceding field Source: Host
25	77	Product Code	3	R	Indicates the Product Code for the device to send in the Product Code Data Segment to identify the type of discount. Source: Host
26	185	Discount Quantity Limit	3	R	Indicates the quantity limit for host discount data. Source: Host
27	186	Discount Program Description	15	R	Indicates the description of the host discount data. Source: Host

1Required/Optional/Conditional

12.16.5 Proprietary Load Request Data, Prompt Code 905

A Prompt Code value of 905 is used for Fuel Volume Data loads. This load type is used to send the fuel volume data tables in a Proprietary Data Load request.

The following table illustrates the format of Field No. 14 when the value of Data Element No. 78 (Prompt Code) is 905, Fuel Volume Data:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows. A Field Separator follows Field No. 14.

Proprietary Load Request Data Prompt Code 905, Fuel Volume Data					
Field No.	**Element No.**	**Element Name**	**Max Len.**	**Entry is: R/O/C1**	**Description**
14	201	Fuel Volume Data	3,600	R	The data for the indicated table type. Field length determined by segment length in field 2. Source: Vendor

[1]Required/Optional/Conditional

12.17 Totals with Proprietary Data Load Data Segment

This section describes Data Segment No. 119, Totals with Proprietary Data Load Data Segment.

It always appears in Field No. 3 in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 493 alphanumeric characters (001–493/a–z/A–Z).

It originates at the device with the purpose of requesting Totals. This data segment also provides the current Card Table Version and Host Discount data from the device. This function is identical to the existing Totals processing and is needed to support the functionality related to Dynamic Card Table.

The following table illustrates the layout of the Totals with Proprietary Data Load Data Segment:

Note: Field Nos. 1–17 are separated by Field Separators, with a Field Separator following Field No. 17. When a field is not populated, still send the Field Separator. Field Nos. 18–20 are not separated by Field Separators. A Field Separator follows the last occurrence of Field No. 20.

Totals with Proprietary Data Load Data Segment Format
Data Segment No. 119

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 119 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	44	Information Byte	1	R	Indicates whether the request is a single-message request or a multimessage request. Source: Device
4	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
5	78	Prompt Code	3	R	Identifies the format type for the requested totals. Fixed value: 990 Source: Device
6	32	Employee Number	4	C	Identifies a BUYPASS customer's employee. Source: Device

Totals with Proprietary Data Load Data Segment Format Data Segment No. 119					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
7	65	Password	6	C	Identifies the end-of-day function's password. Source: Device
8	105	Totals Date	6	R	Identifies the requested type of totals. Source: Device
9	43	Hardware Version	4	R	Identifies the device's hardware level version number. Source: Device
10	96	Software Version	8	R	Identifies the device's software version number. Source: Device
11	39	Firmware Version	8	R	Identifies the device's EPROM level. Source: Device
12	86	Sequence Number	6	R	Identifies the unique transaction sequence number assigned to and associated with a transaction throughout its life. Source: Device
13	176	Device Card Table Version	35	R	Identifies the current version numbers of the card table loaded at the device. Source: Device
14	179	Host Discount Timestamp	12	R	Identifies the timestamp used to support host discounts. Format: CCYYMMDDHHMM Source: Host
15	20	Currency Code	3	O	Identifies the transaction's currency type according to country. Source: Device
16	42	Grand Total	8	R	Identifies the final total amount. Source: Device

Category Total Data Section

Field Nos. 17–19 are sent up to a maximum of 20 times—once for each of the following card types—in the order shown. Card types 1 through 15 will be included in all approved responses. Card types 16 through 20 will be included only when data occurs for a card type.

Use Card Labels (in parentheses) instead of field numbers to identify total buckets.

Totals with Proprietary Data Load Data Segment Format Data Segment No. 119					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
19. MasterCard®/Visa® (CC) 20. Amex® (TE) 21. Discover® Network, Diners Club®, JCB® (DS) 22. Preauthorization (AO)* 23. Debit (DB) 24. Fleet (FL) 25. Cash (CS) 26. Proprietary Card (PR) 27. Check (CK) 28. EBT Food Stamp (EF) 29. EBT Cash Benefit (EC) 30. Stored Value Card Activation/Deactivation (SV1)* 31. Stored Value Card Purchase/Capture /Merchandise Return (SV2) 32. Stored Value Card Replacement (SV3)* 33. Stored Value Card Recharge (SV4)* 34. TeleCheck® ECA® service (ECA) 35. EBT version of WIC (EWIC) 36. EBT child care (EK) 37. Host Discounts (HD)* **Note**: Card categories indicated by * are considered non-financial. The dollar amounts are not included in the Grand Total.					
17	13	Card Label	4	R	Identifies the card type. Source: Device
18	16	Card Type Total Count	5	R	Identifies the transaction count for a particular card type. Source: Device
19	15	Card Type Total Amount	8	R	Identifies the total amount for a particular card type. Source: Device

[1]Required/Optional/Conditional

12.18 Print Data 2 Segment

This section describes Data Segment No. 120, Print Data 2 Segment

It is only sent on Financial Transaction Response messages requiring large amounts of print data. It contains data elements that are unique to transactions requiring print data.

It always appears at the end of a Financial Transaction response.

It has a maximum length of 1,009 alphanumeric characters (01–1,009/a-z/A–Z).

The following table illustrates the layout of the Print Data 2 Segment.

Note: There is a Field Separator between Field Nos. 1 and 2 and between Field Nos. 2 and 3.

Print Data 2 Segment Data Segment No. 120					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 120 Source: Host
2	84	Segment Length	4	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Host
3	152	Print Data	999	R	Identifies the print data being transmitted to the device. **Note:** This field contains the receipt text for Blackhawk phone activation and recharge. A field delimiter "\" will follow each line of text sent to the device. **Ex.:** Line of text\line of text\line of text **Source:** Host

[1]Required/Optional/Conditional

12.19 NFC Payment Tokenization Data Segment

This section describes Data Segment No. 123, NFC Payment Tokenization Data Segment
This data segment originates from the device and provides information needed for processing NFC tokenized transactions. It is required on all initial and recurring transactions involving tokenized data and for transactions that include MasterCard Token or Digital Secure Remote Payment (DSRP) or Visa TAVV data.
The Token PAN suffix is returned in the response, if supplied by the authorizer.
It has a maximum length of 186 alphanumeric characters (001–186/a-z/A–Z).

The following table illustrates the layout of the NFC Payment Tokenization Data Segment.

Note: All fields are separated by Field Separator. When a field is not populated, still send the Field Separator. A Field Separator follows Field No. 4. In the following table, the ▲ symbol represents a Field Separator.

NFC Payment Tokenization Data Segment Format Data Segment No. 123					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 123 Source: Device
< ▲ >					
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
< ▲ >					
3	195	CAVV, Revised Format	20	O	Identifies the Verified By Visa® Data – Revised Format. This data consists of: **Pos.** **Description** 1-20 Cardholder authentication verification value data (ATN (Authentication Tracking Number) field replaces the need for the XID. Source: Device
< ▲ >					
4	196	Token Requestor ID	11	O	Identifies the Merchant's Token Requestor ID. Source: Device
5	197	Token PAN Suffix	4	C	Identifies the last 4 digits of the cardholder PAN to be printed on the transaction receipt. Source: Issuer/Authorizer
< ▲ >					

NFC Payment Tokenization Data Segment Format Data Segment No. 123					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
6	202	Cryptogram Token Data	28 or 56	O	Identifies the Merchant's Cryptogram Token data Block A (or block A and B). Source: Device
< ▲ >					
7	203	SafeKey Data	58	O	Identifies Safekey cryptogram indicator "SK" – 2 bytes AEVV (American Express Verification Value) – 28 bytes: This is a cryptographic value derived by the Issuer during the AESK (American Express SafeKey) payment authentication that can provide evidence of the results of payment authentication during an online purchase. AESK (American Express SafeKey) – 28 bytes: AESK Transaction Identifier is determined by the Merchant during AESK payment authentication. When the first 28 bytes of value is not applicable, it should be space filled. If AESK is optional for merchants then the last 28 bytes should be space filled. Source: Device
8	204	SafeKey Response	1	O	Identifies SafeKey Cryptogram AEVV response value Source: Authorizer
< ▲ >					

NFC Payment Tokenization Data Segment Format Data Segment No. 123					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
9	237	TAVV Cryptogram	28	O	Identifies the Merchant's Token Authentication Verification Value (TAVV) Cryptogram data base64 format for payment token-based, card on file, digital secure remote payment (DSRP), e-commerce, and application based e-commerce transactions. Merchant can only populate the TAVV Cryptogram – 28 bytes in this segment during transaction request. Please refer to Appendix-Y for using this field in a MasterCard DSRP transaction or for Visa 3D Secure. Source: Device Attribute type : b64 (Base 64) Valid values: a-z, A-Z, 0-9,+,=, /
10	238	TAVV Result Code	1	O	Identifies whether the TAVV Cryptogram result received from Visa is the same response value sent back to the merchant's terminal. This field will be populated as appropriate based on the TAVV cryptogram. Source: Authorizer
< ▲ >					

[1]Required/Optional/Conditional

12.20 EMV Request Data Segment

This section describes Data Segment No. 130, EMV Request Data Segment.

It always appears in Data Section No. 3 of an EMV Financial Transaction Request.

It has a maximum length of 3043 alphanumeric characters (001–3043/a-z/A–Z).

It originates at the device.

The following table illustrates the layout of the EMV Request Data Segment:

Note: There is a Field Separator between Field Nos.1 and 2, Field Nos. 2 and 3, Field Nos. 3 and 4, Field Nos. 4 and 5, Field Nos. 5 and 6 and between Field No. 6 and EMV Additional Information Section. Even when a field is not populated, you still need to send the Field Separator. Field Separators are not present between the elements in EMV Additional Information Section. Field Separators are not present between repetitions of EMV Additional Information Section. A Field Separator follows the final EMV Additional Information Section.

EMV Request Data Segment Data Segment No. 130					
Field No.	**Element No.**	**Element Name**	**Max .Len .**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 130 Source: Device
2	84	Segment Length	4	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	187	CA Public Key File Checksum	25	O	Identifies the checksum received in the last CA Public Key download and currently in use at the host and device. Source: Device/Host
4	188	EMV Card Sequence Number	3	C	Identifies the sequence number of the EMV card. Source: Device
5	189	EMV Chip Data Length	3	R	Identifies the length of the following field (EMV Chip Data). Valid values: 000—999 Source: Device

EMV Request Data Segment Data Segment No. 130					
Field No.	**Element No.**	**Element Name**	**Max .Len .**	**Entry is: R/O/C[1]**	**Description**
6	190	EMV Chip Data	999	R	Identifies the EMV chip data. Source: Device

EMV Request Data Segment Data Segment No. 130					
EMV Additional Information Section This section is repeated – by EMV Additional Information Indicator – for a maximum length of 2,000 bytes. The Field Separator that follows Field No. 9 is included with the last repetition.					
Field No.	**Element No.**	**Element Name**	**Max .Len .**	**Entry is: R/O/C[1]**	**Description**
7	191	EMV Additional Information Indicator	3	R	Identifies the type of EMV additional information being transmitted. Source: Device
8	192	EMV Additional Information Length	3	R	Identifies the length of the EMV Additional Information. Source: Device
9	118	EMV Additional Information	Var.	R	Identifies the EMV additional information. Source: Device

[1]Required/Optional/Conditional

12.21 EMV Response Data Segment

This section describes Data Segment No. 131, EMV Response Data Segment.

It always appears in Field No. 4 in Data Section No. 3.

It has a maximum length of 3834 alphanumeric characters (001–3834/a-z/A–Z).

It originates at BUYPASS.

The following table illustrates the layout of the EMV Response Data Segment:

Note: Fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

EMV Response Data Segment Data Segment No. 131					
Field No.	**Element No.**	**Element Name**	**Max .Len .**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 131 Source: Host
2	84	Segment Length	4	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Host
3	187	CA Public Key File Checksum	25	R	Identifies the checksum received in the last CA Public Key download and currently in use at the host and device. This is echoed from the Request. Source: Device/Host
4	189	EMV Chip Data Length	3	R	Identifies the length of the following field (EMV Chip Data). Source: Device
5	190	EMV Chip Data	999	R	Identifies the EMV chip data. Source: Device

EMV Additional Information Section This section is repeated – by EMV Additional Information Indicator – for a maximum length of 2,800 bytes.					
6	191	EMV Additional Information Indicator	3	R	Identifies the type of EMV additional information being transmitted. Source: Host
7	192	EMV Additional Information Length	3	R	Identifies the length of the EMV Additional Information. Source: Host
8	118	EMV Additional Information	Var.	R	Identifies the EMV additional information. Source Host

[1]Required/Optional/Conditional

12.22 CA Public Key File Segment

This section describes Data Segment No. 132, CA Public Key File Segment.

It has a maximum length of 77 alphanumeric characters (01–77/a-z/A–Z).

The following table illustrates the layout of the CA Public Key File Data Segment:

CA Public Key File Data Segment Data Segment No. 132					
Field No.	**Element No.**	**Element Name**	**Max .Len .**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 132 Source: Device
<FS>					
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
<FS>					
3	86	Sequence Number	6	R	Identifies the merchant-assigned Sequence Number for the transaction. **Note:** Limited sequence number range (100000-199999) is ONLY applicable when using the Multithreaded Dial Protocol Communications header for a CAPK File Load for the ATL105 message format. Source: Device
<FS>					
4	102	Terminal Identifier	13	R	Identifies the transaction device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. For EMV, the first two characters (Device Type) of this element's value must be "+*" regardless of the actual device type. Source: Device

CA Public Key File Data Segment Data Segment No. 132					
Field No.	**Element No.**	**Element Name**	**Max .Len .**	**Entry is: R/O/C[1]**	**Description**
5	48	Load Type	1	R	Describes the load type. Fixed value: K (Indicates that Public Key information is requested.) Source: Device
6	43	Hardware Version	4	R	Identifies the version number of the device's hardware level. Source: Device
7	96	Software Version	8	R	Identifies the version number of the device's software application. Source: Device
8	39	Firmware Version	8	R	Identifies the version number of the device's EPROM level. Source: Device
9	187	CA Public Key File Checksum	25	R	Identifies the checksum received in the last CA public key download and currently in use at the host and device. Source: Device
10	11	Block Number	3	R	Identifies the specific data block requested by a device or sent by the host. Source: Device

[1]Required/Optional/Conditional

12.23 Transaction Attributes Data Segment

This section describes Data Segment No. 134, Transaction Attributes Data Segment.

It originates at BUYPASS.

It has a maximum length of 19 alphanumeric characters (01–19/a-z/A–Z).

The following table illustrates the layout of the Transaction Attributes Data Segment.

Note: The fields are not separated by by Field Separators. When a field is not populated, the next field immediately follows

Transaction Attributes Segment Data Segment No. 134					
Field No.	**Element No.**	**Element Name**	**Max .Len .**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 134 Source: Host
2	84	Segment Length	4	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Host
3	198	Settlement Type	1	R	Identifies the disposition of the transaction. Values: D = Dual message S = Single message X = Non-traditional Signature Debit **Note:** Please check with your First Data Project Manager or Relationship Manager for availability of Signature Debit functionality.
4	199	Signature Required	1	R	Values: T = Signature is required F = Signature is not required Space = Device software logic determines if signature is required
5	200	Receipt Card Description	10	R	Contains description of card type. Left justified and space filled. (i.e. “STAR “)

[1]Required/Optional/Conditional

12.24 Moneris Data (Request) Segment

This section describes Data Segment No. 135, Moneris Data (Request) Segment.

This segment contains data required by the Moneris authorizer.

The following table illustrates the layout of the Moneris Data (Request) Segment:

Note: Field Nos 1 and 2 and 2 and 3 are separated by Field Separators. The segment should end with a field separator. Data fields contained within field 3 are not separated by Field Separators.

Moneris Data Segment (Request) Format Data Segment No. 135					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed Value 135 Source: Device
2	84	Segment Length Indicator	3	R	Identifies the data segment's length, including Segment Type's length. Source: Device
3	213	Moneris Data	100	R	All or some of the Moneris Data. Data is in <tag><len><data> format. For details on Moneris Data Layout, refer to Appendix V (Moneris Data layouts).

[1]Required/Optional/Conditional

12.25 Moneris Data (Response) Segment

This section describes Data Segment No. 136, Moneris Data (Response) Segment.

This segment is used for both financial transactions and Key Load transactions.

The following table illustrates the layout of the Moneris Data (Response) Segment:

Note: A Field Separator will follow the segment.

Moneris Data Segment (Response) Format Data Segment No. 136					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed Value 136 Source: Device
2	84	Segment Length Indicator	3	R	Identifies the data segment's length, including Segment Type's length. Source: Device
3	213	Moneris Data	100	R	All or some of the Moneris Data specified below. Data is in <tag><len><data> format. For details on Moneris Data Layout, refer to Appendix V (Moneris Data layouts).

[1]Required/Optional/Conditional

12.26 Moneris Day End Batch Balance (Request) Segment

This section describes Data Segment No. 139, Moneris Day End Batch Balance (Request) Segment.

The purpose of this segment is to retrieve debit totals for the previous period that are to be used in the day end batch close transaction.

The following table illustrates the layout of the Moneris Day End Batch Balance (Request) Segment:

Moneris Day End Batch Balance Segment Format (Request)
Data Segment No. 139

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 139 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
4	206	SPDH Header	48	R	Contains merchant, terminal, and transaction data required for MAC Encryption. Source: Device
5	207	Moneris Terminal Identifier	8	R	Identifies the Moneris Terminal ID for this batch. Source: Device
6	208	Moneris Merchant ID	13	R	Identifies the Moneris Merchant ID for this batch. Source: Device
7	214	Batch Number	3	R	Identifies the batch number of this batch. Source: Device
8	215	Language Indicator	1	R	Moneris Language Indicator from the Moneris Language Matrix. Source: Device

[1]Required/Optional/Conditional

12.27 Moneris Day End Batch Balance (Response) Segment

This section describes Data Segment No. 140, Moneris Day End Batch Balance (Response) Segment.

This segment consists of a reiteration of the first 6 fields in the request, followed by a text response provided by Moneris, followed by the total of all debit transactions for this batch.

The following table illustrates the layout of the Moneris Day End Batch Balance (Response) Segment.

Moneris Day End Batch Balance Segment Format (Response) Data Segment No. 140					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 140 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
4	206	SPDH Header	48	R	Contains merchant, terminal, and transaction data required for MAC Encryption. Source: Moneris
5	207	Moneris Terminal Identifier	8	R	Identifies the Moneris Terminal ID for this batch. Source: Device
6	208	Moneris Merchant ID	13	R	Identifies the Moneris Merchant ID for this batch. Source: Device
7	214	Batch Number	3	R	Identifies the batch number of this batch. Source: Moneris
8	216	Response Display	16	R	Text of response to batch close request from Moneris. Source: Moneris
9	217	Number of Debits	4	R	Number of debit transactions in this batch. Source: Moneris

Moneris Day End Batch Balance Segment Format (Response) Data Segment No. 140					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
10	218	Debit Dollar Value	19	R	Dollar value of all debit transactions in this batch. Format is +/-9(16)v99. Source: Moneris
11	219	Number of Credits	4	R	Number of credit transactions in this batch. Source: Moneris
12	220	Credit Dollar Value	19	R	Dollar value of all credit transactions in this batch. Format is +/-9(16)v99. Source: Moneris
13	221	Number of Corrections	4	R	Number of correction transactions in this batch. Source: Moneris
14	222	Corrections Dollar Value	19	R	Dollar value of all correction transactions in this batch. Format is +/-9(16) v99. Source: Moneris

[1]Required/Optional/Conditional

12.28 Moneris Day End Batch Close (Request) Segment

This section describes Data Segment No.141, Moneris Day End Batch Close (Request) Segment.

This segment is used to close out each pay point to Moneris. It should be sent up once for each pay point at day end.

Note: All fields are separated by field separator characters.

Moneris Day End Bath Close Segment Format (Request) Data Segment No. 141					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 141 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
3	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
4	206	SPDH Header	48	R	Contains merchant, terminal, and transaction data required for MAC Encryption. Source: Device
5	207	Moneris Terminal Identifier	8	R	Identifies the Moneris Terminal ID for this batch. Source: Device
6	208	Moneris Merchant ID	13	R	Identifies the Moneris Merchant ID for this batch. Source: Device
7	214	Batch Number	3	R	Identifies the batch number of this batch. Source: Device
8	215	Language Indicator	1	R	Moneris Language Indicator, from the Moneris Language Matrix. Source: Device
9	217	Number of Debits	4	R	Number of debit transactions in this batch. Source: Device
10	218	Debit Dollar Value	19	R	Dollar value of all debit transactions in this batch. Format is +/-9(16)v99. Source: Device

Moneris Day End Bath Close Segment Format (Request) Data Segment No. 141					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
11	219	Number of Credits	4	R	Number of credit transactions in this batch. Source: Device
12	220	Credit Dollar Value	19	R	Dollar value of all credit transactions in this batch. Format is +/-9(16)v99. Source: Device
13	221	Number of Corrections	4	R	Number of correction transactions in this batch. Source: Device
14	222	Corrections Dollar Value	19	R	Dollar value of all correction transactions in this batch. Format is +/-9(16)v99. Source: Device

[1]Required/Optional/Conditional

12.29 Moneris Day End Batch Close (Response) Segment

This section describes Data Segment No.142, Moneris Day End Batch Close (Response) Segment.

The response to a Moneris Day End Batch Close (Request) will have a special format that will consist of a reiteration of the first 6 fields in the request followed by a text response provided by Moneris.

Moneris Batch Close Segment Format (Response) Data Segment No. 142					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 142 Source: Host
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Host
3	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number. Source: Device
4	206	SPDH Header	48	R	Contains merchant, terminal, and transaction data required for MAC Encryption. Source: Moneris
5	207	Moneris Terminal Identifier	8	R	Identifies the Moneris Terminal ID for this batch. Source: Device
6	208	Moneris Merchant ID	13	R	Identifies the Moneris Merchant ID for this batch. Source: Device
7	214	Batch Number	3	R	Identifies the batch number of this batch. Source: Moneris
8	216	Response Display	16	R	Text of response to batch close request from Moneris. Source: Moneris
9	210	MAC	16	R	MAC value to be validated at terminal. Source: Moneris

[1]Required/Optional/Conditional

12.30 Tax by Product Data Segment

This section describes Data Segment No.143, Tax by Product Data Segment.

When a Financial Transaction request includes the Tax by Product Segment, the following rules should be followed:

- A Product Code Data Segment (Data Segment No. 102) must be present in the transaction.
- Each entry in the Product Code Data Segment (Data Segment No. 102) should have an entry in this segment too and the entries should be in the same order in both the segments.
- Up to three taxes can be reported for a single product. If less than three taxes are to be used for a particular product, merchants can end the entry for that product early by using a field separator after the tax entry instead of a "\".
- If there is no tax associated with a particular product, the inclusive/exclusive flag should be set to "N". In this case, the tax type and amount should be omitted. So, the entry for that product would only be the product number, followed by "N", followed by field separator.
- The only validation that will be done for this segment is whether the format is correct. No judgement will be made of the values contained.
- For Canadian transactions, the valid values for Tax Type are GST (Goods and Services Tax), HST (Harmonized Sales Tax), and PST (Provincial State Tax). Any given product can have between 0 and 3 of these taxes specified in this segment.

The following table illustrates the layout of the Tax by Product Segment:
Notes: A Field Separator always follows Segment Type and Segment Length. A Field Separator follows the last Tax Amount for each product.
A Tax by Product Field Delimiter follows each tax amount when there is another tax amount for this product.
Several examples of the segment accompanied by the Product Code Data Segment follow the table.
In the following table, the ▲ symbol represents a Field Separator, and the \ character is a Tax by Product Field Delimiter.

Tax by Product Data Segment Format Data Segment No. 143					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 143 Source: Device
< ▲ >					
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
< ▲ >					

Tax by Product Data Segment Format Data Segment No. 143					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
3	62	Number of Products	2	R	Identifies the count of products reported for this transaction. Should match the number of products from the Product Code Data Segment. Source: Device
Tax by Product Data This section is repeated—by product—for a maximum of 10 products for a total variable length of up to 360 bytes.					
4	77	Product Code	3	R	Identifies the product code from the same entry in the Product Code Data Segment. Source: Device
5	223	Inclusive/ Exclusive for Tax 1	1	R	Identifies whether this tax is inclusive (“I”) or exclusive (“E”) to the product amount in the Product Code Data Segment. If this tax amount is not applicable to this product, this value should be “N” and the tax type and amount should be skipped. Source: Device
6	224	Tax Type 1	3	C	Identifies the tax type for this tax item. If the Inclusive/Exclusive flag is “N”, this field is omitted. Source: Device
7	225	Tax Amount 1	Var. Max 9	C	Identifies the first tax amount for this product. Decimal places are assumed by the currency of the transaction. Should be omitted if its Inclusive/Exclusive flag is “N”. Source: Device
< \ > if there is another tax amount for this product, otherwise < ▲ > and continue to the next product.					
8	226	Inclusive/ Exclusive for Tax 2	1	C	Identifies whether this tax is inclusive (“I”) or exclusive (“E”) to the product amount in the Product Code Data Segment. If this tax amount is not applicable to this product, this value should be “N” and the tax type and amount should be skipped. Source: Device
9	227	Tax Type 2	3	C	Identifies the second tax type for this tax item. If the Inclusive/Exclusive flag is “N”, this field is omitted. Source: Device

Tax by Product Data Segment Format Data Segment No. 143					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
10	228	Tax Amount 2	Var. Max 9	C	Identifies the second tax amount for this product. Decimal places are assumed by the currency of the transaction. Should be omitted if its Inclusive/Exclusive flag is "N". Source: Device
< \ > if there is another tax amount for this product, otherwise < ▲ > and continue to the next product.					
11	229	Inclusive/ Exclusive for Tax 3	1	C	Identifies whether this tax is inclusive ("I") or exclusive ("E") to the product amount in the Product Code Data Segment. If this tax amount is not applicable to this product, this value should be "N" and the tax type and amount should be skipped. Source: Device
12	230	Tax Type 3	3	C	Identifies the tax type for this tax item. If the Inclusive/Exclusive flag is "N", this field is omitted. Source: Device
13	231	Tax Amount 3	Var. Max 9	C	Identifies the third tax amount for this product. Decimal places are assumed by the currency of the transaction. Should be omitted if its Inclusive/Exclusive flag is "N". Source: Device
< ▲ > and continue to the next product. If this is the last product, the < ▲ > signifies the end of segment.					

[1]Required/Optional/Conditional

Example:
102.033.S01020L318198\31099\2000.
The product segment contains a single product: 020 for $20.00
143.023.01020IGST200.
The product "020" had an inclusive GST of 2.00 and no other tax. Therefore the tax amount is followed by a FS.

102.041.S02963U01\225\25\400U01\2500\500.
This product segment contains 2 products: 963 and 400.
143.032.02963N.400EGST25\IPST50.
In this scenario, product "963" had no taxes associated with it (since it's already a tax itself), so it includes the product. But with an "N" as inclusive/exclusive followed immediately by FS indicates end of the process for that product. This is followed by product "400" with an exclusive GST of .25 and an inclusive PST of .50. The tax entries are separated by "\" and the whole process ends with a FS.

102.033.S01001L320000\31000\2000.

143.023.01001IHST200.
A single product “001” with an inclusive HST of 2.00 followed by a FS to indicate no more taxes.

102.161.S09963U01\2535\535\102U01\21300\1300\400U01\2500\500\101U01\2500\500\100U01\25000\5000\440U01\2200\200\115U01\22500\2500\460U01\2400\400\401U01\2300\300.

143.097.09963N.102EGST65.400EGST25.101EGST25.100EGST250.440EGST10.115EGST125.460EGST20.401EGST15.
In this scenario, there are 9 products. Product “963” is already a tax, so no tax amount is associated with this product. Each other entry has an exclusive GST of 5%. Interestingly, if you add all the exclusive taxes, they add up to the total of product 963 in the product segment.

12.31 Enhanced Fleet Request Segment

This section describes Data Segment No. 145, Enhanced Fleet Request Segment.

The Enhanced Fleet Request Segment contains fleet data that applies to the enhanced fleet offerings such as Wex OTR, Comdata, Voyager EMV, and Visa Fleet 2.0. This segment will potentially hold multiple sub-segments, each of which contains different fleet transaction-related data. The sub-segments will be in the standard TLV format (sub-segment type, sub-segment length, sub-segment value).

The following table illustrates the layout of the Enhanced Fleet Request Segment:

Note: Only the Prompt Table sub-segment is valid for the Voyager EMV transactions.

Note: Only the Prompt Table sub-segment is valid for Visa Fleet 2.0 transactions.

Note: Only the Prompt Table sub-segment is valid for Comdata transactions.

Note: Only the Prompt Table sub-segment is valid for MasterCard Enhanced Fleet EMV transactions.

Availability Notice: This functionality for MasterCard Enhanced transactions is expected to be enabled in First Data production no earlier than June 2026. Until then, this field is intended for development and certification preparation only and should not be used for live production transactions.

Note: Merchants should not send Fleet segment (Data Segment No. 101) and Enhanced fleet segment together (Data Segment No. 145).

Enhanced Fleet Request Format Data Segment No. 145					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Indicates the type of data segment being formatted Fixed Value 145 Source: Device
2	84	Segment Length Indicator	3	R	Indicates the data segment length including the Segment Type length Source: Device
3	239	Enhanced Fleet Data	999	R	All or some of the Enhanced Fleet sub-segment data is specified below. Data is in <tag><len><data> format.

[1]Required/Optional/Conditional

Enhanced Fleet Data Layouts		
Enhanced Fleet Request Flags		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 001
Table Length	n3	Variable Value: Max 020.

Table Data	an10	A series of flags to cover the current and future simple data that needs to be communicated between the POS and the EFS authorizer. Flags should be separated by ",". If a flag is not present, it can be skipped by including just the "," separator. If no flags are present, this field should not be included. Flag 1: Commercial/Retail Flag Values: "C": Terminal is a Commercial Terminal. Any other value: Terminal is not a Commercial Terminal.

Example: "001001C"

Enhanced Fleet Data Layouts

Non-Fuel Product Data (Request)

Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 002
Table Length	n3	Variable Value: Max 350. Overall length of product data including separator characters.
Table Data	an***	Product Data formatted as below.
Element Name	**Len.**	**Description**
< | > Separator between products		
Product Category	4	Identifies a product category contained in the transaction. This value should be the product category for the product that is found in the PRODUCTS table from the Dynamic Card Table. If the product category for a product is not found in the PRODUCTS table, use the default product category MERC for that product. If multiple products in a transaction map to the same category, they should be summed and included in a single category. This is done to accommodate the fact that multiple products may map to a single product category.
Colon	1	":" to separate product code from the data pertaining to that code.
Quantity	Var. Max 3	Quantity of the product
Comma	1	"," separator.
Amount	Var. Max 9	Amount of product purchased. Decimal point should be included, e.g. – 10.00.
< | > Separator between products		

Repeat Product Data for as many products as needed, separated by "|" and ending with "|".

For WEX OTR transactions, the following are the product categories that will be used for this purpose:

Product Code Categories	Description	Product Code Categories	Description
ADD	Additives	PADV	Personal funds cash advance
ANFR	Anti-freeze	PART	Parts
BRAK	Brakes and wheels	PHON	Phone time
CADV	Company funds cash advance	PNT	Paint
CLTH	Clothing	RECP	Tire Recap
DEF	DEF Container	REPR	Repair service
DELI	Deli items	REST	Restaurant purchases
ELEC	Electronics	SCAN	Imaging
ETAX	Exempt tax amount	SCLE	Scale
EVC1	Electric Vehicle Charge Level 1	SHWR	Shower
EVC2	Electric Vehicle Charge Level 2	TCHN	Tire Chains
EVC3	Electric Vehicle Charge Level 3	TIRE	Tire
FAX	Fax	TOLL	Toll Charges
FLAT	Flat Repair	TRAL	Trailer
GROC	Groceries	TRPP	Trip permit
HARD	Hardware	UREA	UREA Container
IDLE	Idleaire	WASH	Car/Truck wash
LMPR	Lumper Fee	WWFL	Windshield Washer Fluid
LUBE	Lube		
MERC	Default category for merchandise		
OIL	Oil		
OILC	Oil change		

Example: To include a non-fuel Scale product: "002014|SCLE:1,10.00|"

Enhanced Fleet Data Layouts

Prompt Data (request)

Field	Attributes	Description
Table ID	n3	Fixed value: 004
Table Length	n3	Variable Value: Max 200. Overall length of all prompt data including separator characters
Table Data	An200	Prompt Data is formatted as below

Element Name	Length	Description
< | > Separator between prompts		
Prompt	Var. Max 4	Identifies a prompt to which the terminal is responding
Colon	1	":" to separate prompt from the response
Prompt Response	Var. Max 20	Value
< | > Separator between prompts		
Includes as many occurrences of the "Prompt Token and Prompt Response" entries as needed, separating each by a "|" and including a leading and ending "|" around the Prompt Data as a whole		

Since different authorizers support different prompts, Prompt Tokens are listed based on authorizers:

Valid Prompt Token for Voyager EMV transactions:

EMV Tag	Prompt Tokens	Description	Maximum Length
DF10	AUID	Voyager ID	6
DF11	VHID	Vehicle ID	6
DF12	DRID	Driver ID	6
DF13	ODRD	Odometer	6
DF20	TRLR	Trailer ID	10
DF21	HRRD	Reefer Hours	7
DF22	TRIP	Trip Number	10
DF23	EXID	Extended Driver ID	20
DF24	UNIT	Unit ID	10
DF25	PONB	Contract/PO	10
DF26	DLIC	Driver License	20

EMV Tag	Prompt Tokens	Description	Maximum Length
DF27	DLST	Driver License State	2
DF28	LCSN	Vehicle License	12
DF29	LCST	Vehicle License State	2
DF30	TRLN	Trailer License	12
DF31	TRLS	Trailer License State	2
DF32 – DF6F	*	Voyager Data	20

* DF32 – DF6F represents multiple tags that are reserved for future use. In order to ensure the fastest possible time to make these tags ready for use, they are to be handled differently so that neither the terminal nor the ATL105 spec needs to be updated to support them. Therefore, for these EMV Tag values, the Prompt Token will match the EMV Tag. The terminal should support up to 20 characters of input data for each of these 'future use' EMV Tags.

Example: Customer responded to prompts for Unit ID and Driver ID:
"004020|UNIT:4321|DRID:14B|"

Example: Customer responded to prompts for Unit ID, Driver ID, and DF32:
"004032|UNIT:4321|DRID:14B|DF32:333333|"

For Visa Fleet EMV 2.0 transactions, the following are the available prompts:

Prompt Tokens	Description	Maximum Length
AD1	Additional Prompted Data 1	20
AD2	Additional Prompted Data 2	20
DRI	Driver ID*	17
EMP	Employee Number	12
GEN	Generic ID*	17
ODM	Odometer	7
TRL	Trailer Number	16
VHI	Vehicle ID*	17
WON	Work Order Number	25

Note: * The issuers can only prompt for one of the three prompts—Generic ID, Vehicle ID, and Driver ID—which maps to a single field in the authorization and clearing messages.

For Comdata transactions, the following are the available prompts:

Prompt Tokens	Description	Maximum Length

UNIT	Unit ID	10
DRID	Driver Number	16
TRLR	Trailer ID	10
HBRD	(Tractor) Hub reading	6
TRHB	Trailer Hub	6
HRRD	Trailer (Reefer) Hours	6
TRIP	Trip Number	10
DLIC	Driver License	20
DLST	Driver License State	2
PONB	PO Number	10
INVC	Invoice	8
XPRS	Express Check Number	25
PAYE	Payee's Name (Last, First)	20
XNID	Transaction ID	10

Example: Customer responded to prompts for Unit Number and Driver ID:

"004020|UNIT:4321|DRID:14B|"

Valid prompt tokens for WEX OTR transactions:

Prompt Tokens	Description	Prompt Tokens	Description
BDAY	Birthday information in MMDDYY format	LSNB	Lessee number
BLID	Billing ID	LSTN	Last Name
CARR	Carrier ID	NAME	Customer name
CLCD	Class Code	ODRD	Odometer reading
CNTN	Control number	PDLN	Product line
CUNB	Customer number	PLDS	Product line description
CVNM	Customer vehicle name	PONB	Purchase order number
DLIC	Driver's license number	PPIN	Non-encrypted PIN
DLST	Driver's license state	RTMP	Reefer temperature
DMLC	Domicile Location	SLDS	Subproduct line desc
DRID	Driver ID	SPLN	Subproduct line code
DSCD	District Code	SSEC	Social Security Number (should not be used)
DTKT	Delivery ticket number	SSUB	Sub-fleet number
FSTI	First name initial	TRIP	Trip number

HBRD	Hubometer reading	TRLR	Trailer number
HRRD	Reefer hour meter reading; 1 implied decimal position	TRNB	Transaction number (check authorizations)
ISNB	Issuer Number (check authorizations)	UNIT	Unit number
LCCD	Location Code	VEHN	Vehicle number
LCST	Vehicle license plate state	VHNB	VIN
LICN	Vehicle license plate number	VHTP	Vehicle Type

Example: Customer responded to prompts for Unit Number and Driver ID:
"004020|UNIT:4321|DRID:14B|"

Conexxus Prompt Codes:

List of universal prompt codes. Currently being used by MasterCard Enhanced Fleet EMV transactions.

Conexxus Prompt Codes			
Prompt Code	Prompt	Prompt Code	Prompt
001	User ID	038	Reefer Temperature
002	Vehicle Number	040	Driver or Vehicle Card
003	Vehicle Tag	041	Customer Number
004	Driver ID	042	Additional Card Data
005	Odometer	043	Additional Vehicle Data
009	Work Order / P.O. Number	044	Engine Hours
010	Invoice Number	045	Tank Level Start
011	Trip Number	046	Fuel Gauge Level
012	Unit Number	047	Battery Voltage
013	Reefer Hours	048	Coolant Temperature

Conexxus Prompt Codes			
Prompt Code	Prompt	Prompt Code	Prompt
015	ZIP Code / Postal Code	049	Check Engine Status
016	Replacement Car	050	Fuel Economy
017	Entered Data Alpha	051	Engine RPM
018	Web Portal Data	052	Engine Load
019	Entered Data Num	053	Engine Oil Temperature
021	Job Number	054	Engine Time Total
022	Maintenance ID	055	Hard Breaking
023	Department Number	056	Hard Acceleration
024	Trailer Number	058	Idle Time
025	Delivery Ticket Number	065	Total Idle Time
026	Hubometer	067	Engine Oil Pressure
033	Sub Fleet Number	068	Engine Oil Life
035	Transaction Number	069	Billing ID
036	Control Number	999	No prompts issued

Example: “004029|001:123456|002:123456|”

Enhanced Fleet Data Layouts

Money Code Payee Name

Field	Attributes	Description
Table ID	n3	Fixed value: 006

Table Length	n3	Variable Value: Max 017.
Table Data	an***	Payee Name in the format: “Last Name,First Initial”. Optional for Money Code transactions

Example: “006007SMITH,B“

Enhanced Fleet Data Layouts		
Money Code Check Number		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 007
Table Length	n3	Variable Value: Max 015.
Table Data	an***	Money Code check number. Required for all Money Code transactions.

Example: “007009249567891“

Enhanced Fleet Data Layouts		
Cash Advance Limit		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 008
Table Length	n3	Variable Value: Max 009
Cash Advance Limit	Var. Max 9	Maximum dollar amount the host should return for the cash advance limit. Decimal point should be included.

Example: “008006500.00“

12.32 Enhanced Fleet Response Segment

This section describes Data Segment No. 146, Enhanced Fleet Response Segment.

Enhanced Fleet Data Segment (Response) Format Data Segment No. 146					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Indicates the type of data segment being formatted. Fixed Value 146 Source: Device
2	84	Segment Length Indicator	3	R	Indicates the data segments length, including the Segment Types length Source: Device
3	239	Enhanced Fleet Data	999	R	All or some of the Enhanced Fleet Data specified below. Data is in <tag><len><data> format.

[1]Required/Optional/Conditional

Enhanced Fleet Data Layouts		
Enhanced Fleet Response Flags		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 001
Table Length	n3	Variable value: Max 020.
Table Data	an10	A series of flags to cover current and future simple data that needs to be communicated between the EFS authorizer and the POS. Flags should be separated by ",". If a flag is not present, it can be skipped by including just the "," separator. If no flags are present, this field will not be included. Flag 1: Settlement Indicator Values: CP – Cost Plus DB – Direct Bill DF – Direct Bill Funded (Reserved) RC – Recourse (Reserved) RF – Recourse Funded (Reserved) FN – Funded

Enhanced Fleet Data Layouts

Non-fuel Product Limits (Response)

Field	Attributes	Description
Table ID	n3	Fixed value: 002
Table Length	n3	Variable Value: Max 350. Overall length of product limits data including separator characters.
Table Data	an***	Product Limits Data formatted as below.

Element Name	Len.	Description
< | > Separator between products		
Product Category	4	Identifies a product category that is allowed for this customer. The product category will correspond to the product category column in the Dynamic Card Table PRODUCTS table.
Colon	1	":" to separate product category from the data pertaining to that code.
Amount	Var. Max 9	Amount of product category allowed. The sum of all products in this category should not exceed this amount. Decimal point will be included.
< | > Separator between products		
Repeat Product Data for as many products as needed, separated by "|" and ending with "|".		

Example: Customer is allowed to include Oil, Additives, Windshield Washer Fluid, and Cash Advance: "002047|OIL:900.00|ADD:900.00|WWFL:900.00|CADV:200.00|"

See the previous section for the list of product categories.

This table will not be present in Comdata response messages.

Enhanced Fleet Data Layouts

Fuel Product Limits (Response)

Field	Attributes	Description
Table ID	n3	Fixed value: 003
Table Length	n3	Variable Value: Max 350. Overall length of product limits including separator characters.
Table Data	an***	Product Limits formatted as below.

Element Name	Len.	Description
< | > Separator between products		

Element Name	Len.	Description
Product Code	3	Identifies a product contained in the transaction. Uses standard product codes as found in ATL105 Appendix F. Valid Payment Systems Product Codes. Note that unlike non-fuel products, fuel products use the product codes in the appendix.
Colon	1	":" separator.
Total Price	Var. Max 8	Total currency allowed for this product. Decimal point will be included.
< | > Separator between products		
Repeat Product Data for as many products as needed, separated by "|" and ending with "|".		

Example: Authorizer returns limit of $3509.10 for has ultra-low sulfur diesel (use type 1), $100.00 for ultra-low sulfur diesel reefer (use type 2), and $2519.10 for Diesel Exhaust Fluid (DEF) (use type 1):
"003046|051:3509.10|032:100.00|062:2519.10|"

Enhanced Fleet Data Layouts

Prompt Formats (Response)

Field	Attributes	Description
Table ID	n3	Fixed value: 004
Table Length	n3	Variable Value: Max 200. Overall length of all prompt formats including separator characters.
Table Data	an***	Prompt Formats, formatted as below.

Element Name	Len.	Description
< | > Separator between prompts		
Prompt	Var. Max 4	Identifies a prompt to which the terminal should respond.
Colon	1	":" to separate prompt from the format.
Prompt Format	Var. Max 20	Prompt Format using the edit masks below with a ";" between each mask.
< | > Separator between prompts		
Repeat Prompt Data for as many prompts as needed, separated by "|" and ending with "|".		

See the previous section for the list of available prompts. Here are the formatting details that will be included for each requested prompt.

Edit Mask	Description
?	Re-prompt using previously received formatting

<table>
<tr><td>A,B</td><td>Alpha numeric; as of 01/02/2015, types ‘A’ and ‘B’ are treated the same</td></tr>
<tr><td>N</td><td>Numeric only</td></tr>
<tr><td>I</td><td>Data is free format</td></tr>
<tr><td>O</td><td>Prompt, but response is optional.</td></tr>
<tr><td>Z</td><td>Capture, but do not send the data on subsequent transactions. Captured data must meet any other formatting requirement.</td></tr>
<tr><td>T</td><td>Identifies the data type of the value:
N = Number
S = String</td></tr>
<tr><td>Mn</td><td>Minimum of n, which is dependent upon the ‘T’ edit mask type:
TN = Minimum value of the number; an edit mask of M999 indicates 999 as the minimum value
TS = Minimum length of the string; an edit mask of M4 indicates 4 as the minimum string length</td></tr>
<tr><td>Xn</td><td>Maximum of n, which is dependent upon the ‘T’ edit mask type:
TN = Maximum value of the number; an edit mask of X2500 indicates 2500 as the maximum value
TS = Maximum length of the string; an edit mask of X7 indicates 7 as the maximum string length</td></tr>
<tr><td>Vn</td><td>Validates exact match, where n is the data value to match; an edit mask of V1234 indicates that the data value must match 1234</td></tr>
<tr><td>Pn</td><td>Validates match against provided pattern, where n is the pattern to match.
<table>
<tr><th>Pattern Mask</th><th>Description</th></tr>
<tr><td>@</td><td>Position must contain an Alpha character</td></tr>
<tr><td>#</td><td>Position must contain a Numeric character</td></tr>
<tr><td>*</td><td>Position must contain an Alphanumeric character</td></tr>
<tr><td>Letter or Number</td><td>This letter or number is required in this position.</td></tr>
</table>
For example, an edit mask of PUN#### indicates that the data must start with “UN” and be followed by four numeric characters.</td></tr>
</table>

Example 1: Authorizer returns that the customer must enter Driver ID “1100”, an Odometer Reading that is a numeric string between 1 and 25 characters long, and a Unit number that is “UN” followed by four digits:

“004029|DRID:V1100|ODRD:N;TS;M1;X25|UNIT:O;PUN####|”

Example 2: Authorizer returns that the customer must enter Driver ID “1100”, an Hub Reading that must have a value greater than x and less than y:

"004029|DRID:V1100|HBRD:N;TN;M123;X456|"
See the previous section for the list of available prompts.

Enhanced Fleet Data Layouts

Customer Information

Field	Attributes	Description
Table ID	n3	Fixed value: 005
Table Length	n3	Variable Value: Max 070.
Table Data	an***	Customer information in the format: Name for 25, city for 15, state for 2, account code for the rest.

Example: "005049ABC TRUCKING DENVER CO234W987"

Enhanced Fleet Data Layouts

Additional Response Data

Field	Attributes	Description
Table ID	n3	Fixed value: 010
Table Length	n3	Variable Value: Max 200.
Table Data	an***	Additional Response data formatted as below:

Element Name	Len.	Description
< | > Separator between data elements		
Data Label	4	Identifies type of data being returned. See table below
Colon	1	":" separator.
Data Value	Var. by Label	Data being returned.
< | > Separator between Data elements		
Repeat data elements for as many elements as needed, separated by "|" and ending with "|".		

Data Label	Max Len	Description
FNAM	15	Cardholder's First Name
LNAM	20	Cardholder's Last Name
ATHN	4	Transaction's Auth Number (differs from Approval Code)
ACCT	5	Account Code (on Express Check Encashment Response)
DMSG	80	Driver Message

Example: “010032|FNAM:JOHN|LNAM:SMITH|ATHN:1234|“

12.33 WEX Available Product Fleet Information Data Segment

This section describes Data Segment No. 148, WEX Available Product Fleet Information Data Segment.

This segment is used to send available product and fleet information from Wex to the terminal. This data segment appears for Financial Transaction response for WEX. It has a maximum length of 23 alphanumeric characters (001–999/a-z/A–Z) and special characters.

WEX Available Product Fleet Information Data Segment No. 148					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Indicates the type of data segment being formatted Fixed Value 148 Source: Host
2	84	Segment Length Indicator	3	R	Indicates the data segment length including the Segment Type length Source: Host

WEX Available Product Fleet Information
Data Segment No. 148

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
3	240	Available Product information	17	R	All of the available product information from WEX. Source: Host

Data Element Name	LEN	Format	Notes
Restriction Code	2-3	AN	A 2 digit code here is the fuel product group which limits which products can be added to this transaction. Allowed values: 00 – No fuel product restriction 01 – Any gasoline 06 - Biodiesels 07 – Any Diesel, including Reefer 08 - Any off-road diesel, including Reefer 10 – Alternative fuels 11 – Hybrid (gas and electric) 12 – Aviation fuels 13 – Marine fuels
Filler	1		"="
Restriction Code amount	1-5	AN	The Fuel Amount limit to be applied to the Restriction Code if the restriction code has a length of 2.
Filler	1		","
Restriction Code Quantity	1-5	AN	The Fuel Quantity Limit to be applied to the Restriction Code if the restriction code has a length of 2.
Filler	1		Space
Restriction Code Unit of Measure	1	AN	UOM for the Quantity.
Example : 00 = 100,50 _ G _			

12.34 Fuel Price Update Request Segment

This section describes Data Segment No.149, Fuel Price Update Segment.

This segment contains data required by the Comdata authorizer.

The following table illustrates the layout of the Fuel Price Update Segment:

Note: Field No.s 1 and 2, 2 and 3, and 3 and 4 are separated by Field Separators. The segment should end with a field separator. Data fields contained within field 4 are not separated by Field Separators.

Fuel Price Update Request Segment Format Data Segment No. 149					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted Fixed value: 149 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators Source: Device
3	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number Source: Device
4	241	Price Data	Var.	R	Fuel Price Data

[1]Required/Optional/Conditional

Example Fuel Price Update request message:

"ATL105<FS>01<FS>149<FS>045<FS>1D13500101001<FS>|CASS:03.00|CRSS:3.05|<FS>"

12.35 Fuel Price Update Response Segment

This section describes Data Segment No.150, Fuel Price Update Response Segment.

This segment contains an indication of success or failure of the Fuel Price Update Request.

The following table illustrates the layout of the Fuel Price Update Response Segment:

Note: A Field Separator will follow the segment.

Fuel Price Update Response Segment Format Data Segment No. 150					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted Fixed value: 150 Source: Device
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators Source: Device
3	102	Terminal Identifier	Var.	R	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number Source: Device
4	26	Decline Code	2	R	Indicates success or failure of the request

[1]Required/Optional/Conditional

12.36 Incomm OTC Market Basket Data (Request) Segment

This section describes Data Segment No. 151, Incomm OTC Market Basket Data (Input) Segment.
This data segment can appear in any of the fields in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)
It has a maximum length of 2,309 alphanumeric (01–3,334/a–z/A–Z) and field separator characters.
It originates at the device.
The following table illustrates the layout of the Incomm OTC Market Basket Data Segment:

Request InComm OTC Market Basket Data
Data Segment No. 151

Note:
Please include a Field Separator between each of the 3 fields.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C	Description
1	85	Segment Type	3	R	Indicates the type of data segment being formatted Fixed Value 151 Source: Host
2	84	Segment Length Indicator	4	R	Indicates the data segment length including the Segment Type length Source: Host
3		Market Basket Data	Var. Max 2300	R	Describes market basket data for the current OTC transaction. Consists of one (1) DV dataset followed by up to 10 PI datasets. See Buypass Incomm Market Basket Data format document for format details of the Market Basket Data.

12.37 Incomm OTC Market Basket Data (Response) Segment

This section describes Data Segment No. 152, Incomm OTC Market Basket Data Segment (Response).
The following table illustrates the layout of the Incomm OTC Market Basket Data Segment (Response):

Response InComm OTC Market Basket Data
Data Segment No. 152

Note:
A Field Separator will follow the segment.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C	Description
1	85	Segment Type	3	R	Indicates the type of data segment being formatted Fixed Value 152 Source: Host
2	84	Segment Length Indicator	4	R	Indicates the data segment length including the Segment Type length Source: Host
3		Market Basket Data	Var. Max 2300	R	Describes market basket data for the current OTC transaction. Consists of one (1) DV dataset, up to 10 PI datasets, and up to 10 PU datasets. See Buypass Incomm Market Basket Data format document for format details of the Market Basket Data.

12.38 Network Token Data Request Segment

This segment is used to send Network Token transaction information from Ucom to BUYPASS.
This section describes Data Segment No. 153, Network Token Data Request Segment.
The Network Token Data Request Segment contains Network Token Data associated with this transaction. This segment will potentially hold multiple sub-segments, each of which contains different network token-related data. The sub-segments will be in the standard TLV format (sub-segment type, sub-segment length, sub-segment value). There is a Field Separator between Field Nos. 1 and 2. Field Separators are not there between the elements in Network Token Data Layouts.
When a merchant is entitled to use both First Data provisioned Network Tokens and Auth Optimization service, the merchant is required to send the same and exact information from the original transaction request in all the follow-up transactions (Reversals, Completions, Cancellations, Void of Merchandise Return, Timeout Reversals). This rule applies even when the merchant receives a new Card Number or a new Expiration Date in their original transaction response.
The following table illustrates the layout of the Network Token Data Request Segment:

Network Token Data Request Format
Data Segment No. 153

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Indicates the type of data segment being formatted Fixed Value 153 Source: Device
2	84	Segment Length Indicator	3	R	Indicates the data segment length including the Segment Type length Source: Device
3	239	Network Token Data	999	R	All or some of the Network Token sub-segment data is specified below. Data is in <tag><len><data> format.

[1]Required/Optional/Conditional

Network Token Data Layouts

Network Token

Field	Attributes	Description
Table ID	n3	Fixed value: 001

Table Length	n3	Variable
Table Data	n	Max. Len.: 19 Identifies the Network Token Data.

Network Token Data Layouts

Network Token Expiration Date

Field	Attributes	Description
Table ID	n3	Fixed value: 002
Table Length	n3	Fixed Value: 004
Table Data	n	Fixed Len.: 4 Identifies the Network Token Expiration Date. Format: YYMM

Network Token Data Layouts

Network Token Provisional Fee Indicator

Field	Attributes	Description
Table ID	n3	Fixed value: 003
Table Length	n3	Fixed Value: 001
Table Data	an	Fixed Len.: 1 Identifies the per provision billing indicator of the Tokenized transaction. It is applicable only to Original (example: Auth, Sale, Return) transactions.

Network Token Data Layouts

Network Token Input Indicator

Field	Attributes	Description
Table ID	n3	Fixed value: 004
Table Length	n3	Fixed Value: 001
Table Data	an	Network Token Input Indicator

		Identifies that the Token or PAN is sent by the merchant Valid values are: Y – Network Token N – Clear PAN

Network Token Data Layouts		
Network Token Eligible Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 005
Table Length	n3	Fixed Value: 001
Table Data	an	Fixed Len.: 1 Identifies token eligible information.

Network Token Data Layouts		
Network Token PAN Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 006
Table Length	n3	Fixed Value: 001
Table Data	an	Network Token PAN Indicator Identifies that the original transaction is authorized by either PAN or Token. This data is applicable only to subsequent transactions (Example: Reversal, Completion, Cancellation). Ucom should echo back this data in the response from original transaction.

Network Token Data Layouts		
Network Token TransArmor Billing Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 007

Table Length	n3	Fixed
Table Data	an	Fixed Len.: 1 This field contains the TransArmor Billing Indicator received from the uCom platform. Presence of this field indicates that the uCom platform has performed the TransArmor tokenization and detokenization of the card number. **Note:** This field should only be sent in the transaction request for the First Data Provisioned Network Token based transactions that is also using the TransArmor tokenization and detokenization service. This field should be sent for the Original and Subsequent transactions. Valid value: Y – TransArmor is invoked at uCom

12.39 Real Time Account Updater Response Data Segment

The following table illustrates the layout of the Real Time Account Updater Response Data Segment.

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Real Time Account Updater Response Segment Format
Data Segment No. 155

Note:
This segment can only be sent when the transaction qualifies for First Data Auth Optimizer service and when the First Data Auth Optimizer service is used.

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 155
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators.

Account Updater Card Number		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 001
Table Length	n3	Variable length of up to 019 bytes
Table Data	an19	Max Len.: 19 This field contains the new account number information when provided by First Data Auth Optimizer service. Note: This field is only included when the merchant has sent the Account Updater Request Indicator field (Variable Information segment 111 Table ID 060 Sub table ID 01) with the value of "Y" in the request. Note: This field is applicable only to Visa and MasterCard. This field will not be sent in the response for TransArmor token-based transactions.

Account Updater TransArmor Token		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 002
Table Length	n3	Variable length of up to 019 bytes

Table Data	an19	Max Len.: 19 This field only included when the merchant sent the Account Updater Request Indicator field (Variable Information segment 111 Table ID 060 Sub table ID 01) with the value of "Y" in the request. Note: This field is applicable only for: 'Visa', 'MasterCard'. This field will not be sent in the response for Card number or PAN based transactions.
Account Updater Expiration Date		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 003
Table Length	n3	Fixed value: 004
Table Data	an4	Fixed Len.: 4 This field contains the new expiration date information when provided by First Data Auth Optimizer service. Format: YYMM Note: This field is only included when the merchant has sent the Account Updater Request Indicator field (Variable Information segment 111 Table ID 060 Sub table ID 01) with the value of "Y" in the request. Note: This field is applicable only to Visa and MasterCard.
Account Updater Card Status		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 004
Table Length	n3	Fixed value: 001
Table Data	An1	Fixed Len.: 1 This field contains the account status received from the First Data Auth Optimizer. Valid Values: A = New Acct # & Exp Date E = New Exp Date Q = Contact Cardholder C = Closed U = Unknown Card Note: This field is included in the response only when the merchant has sent the Account Updater Request Indicator field (Variable Information segment 111 Table ID 060 Sub table ID 01) with the value of "Y" or "I" in the transaction request. Note: This field is applicable only to Visa and MasterCard.

Original Association Response Code		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 005
Table Length	n3	Variable length of 3 bytes
Table Data	an4	Max Len.: 4 This field contains the original decline code from the Authorizing Network that prompted First Data to perform Auth Optimization. This field is included in the response segment only when the merchant has sent the Account Updater Request Indicator field (Variable Information segment 111 Table ID 060 Sub table ID 01) with the value of "Y" or "I" in the transaction request. Note: This field is applicable only to Visa and MasterCard. This field will be returned in the response only when the authorization has been declined by the Card network and then reattempted with the Card network, based on the direction from First Data Auth Optimizer.
Account Updater Result Code		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 006
Table Length	n3	Fixed value: 006

Table Data	an6	Fixed Len.: 6 This field contains the result code received from the Network or Card Issuer Auth Optimization request. Valid Values: VAU001 – Transaction did not qualify for Real Time VAU because the transaction contains token VAU002 – Real Time VAU is supported only for Visa branded PAN VAU003 – Real Time VAU is not supported for this network VAU004 – Transaction is not original purchase, return, bill payment, account funding, original credit VAU005 – Transaction contains CVV2 VAU006 – Transaction is not a qualifying transaction type VAU007 – Real Time VAU is not supported for this MCC VAU008 – Acquirer or processor is not activated for Real Time VAU VAU009 – Issuer does not support Real Time VAU VAU010 – Issuer or Visa blocked the merchant VAU011 – Preauthorized Payment Cancellation Service (PPCS) stop-payment order for this transaction VAU012 – Credentials in the authorization request is the latest VAU data VAU013 – Expiry date in the authorization request is later than VAU data VAU014 – PAN has been opted out of VAU VAU016 – PAN has stop advice set up Note: This field is included in the response only when the merchant has sent the Account Updater Request Indicator field (Variable Information segment 111 Table ID 060 Sub table ID 01) with the value of “Y” or “I” in the transaction request. Note: This field is applicable only to the card type: Visa.

12.40 EV Charging Data Segment

This section describes Data Segment No.156, EV Charging Data Segment.
The following table illustrates the layout of the EV Charging Data Segment.
This segment is applicable to the Visa card type only.
Note: The fields are not separated by field separators. When a field is not populated, the next field immediately follows.

EV Charging Data Segment
Data Segment No. 156

Note:
This segment is sent for an EV charging transaction only (EV fuel code in Product Code Segment 102).

Field No.	Element No.	Element Name	Len.	Entry is: R/ O/ C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 156
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and field separators.
3	242	EV Charging Data	..200	R	All or some of the sub-segment data is specified below. The data is in the <tag><len><data> format.

[1]Required/Optional/Conditional

EV Charging Data Layouts

EV Transaction Indicator

Field	Attributes	Description
Table ID	n2	Fixed value: 01
Table Length	n3	Fixed value: 001
Table Data	a1	Y' – Indicates that the EV charging data is present **Note:** This is a mandatory field for an EV transaction. Without this field, the transaction will not be considered as an EV transaction. If the EV segment is sent by a merchant, this indicator should also be sent as 'Y', without which the transaction will be declined by BUYPASS.

EV Charging Data Layouts

Total Time Plugged In

Field	Attributes	Description
Table ID	n2	Fixed value: 02
Table Length	n3	Fixed Value: 006
Table Data	n6	Contains the total time the vehicle was plugged in. The format of this field is hhmmss, where: • hh = Hours (00–99) • mm = Minutes (00–59) • ss = Seconds (00–59)

EV Charging Data Layouts

Total Charging Time

Field	Attributes	Description
Table ID	n2	Fixed value: 03
Table Length	n3	Fixed value: 006
Table Data	n6	Contains the actual time taken to charge the vehicle. The format of this field is hhmmss, where: • hh = Hours (00–99) • mm = Minutes (00–59) • ss = Seconds (00–59)

EV Charging Data Layouts

Start Time of Charge

Field	Attributes	Description
Table ID	n2	Fixed value: 04
Table Length	n3	Fixed value: 006
Table Data	n6	Contains the start time of charge expressed in the local time of the card acceptor location. The format of this field is hhmmss, where: • hh = Hours (00–99) • mm = Minutes (00–59) • ss = Seconds (00–59)

EV Charging Data Layouts

Finish Time of Charge

Field	Attributes	Description
Table ID	n2	Fixed value: 05
Table Length	n3	Fixed value: 006
Table Data	n6	Contains the finish time of charge expressed in the local time of the card acceptor location. The format of this field is hhmmss, where: • hh = Hours (00–99) • mm = Minutes (00–59) • ss = Seconds (00–59)

EV Charging Data Layouts

Charging Power Output Capacity

Field	Attributes	Description
Table ID	n2	Fixed value: 06
Table Length	n3	Variable 001-006
Table Data	..n6	Maximum length: 6 Contains the charging station power output capacity represented in kW (kilowatts).

EV Charging Data Layouts

Charging Reason Code

Field	Attributes	Description
Table ID	n2	Fixed value: 07
Table Length	n3	Variable 001-003
Table Data	..n3	Maximum length: 3 Contains the charging reason code for the transaction. For Visa, below are the valid values: **Value** **Description** 010 Other Error (Abnormal completion of charging)

		011	Connector Lock Failure (Failure to lock or unlock connector)
		012	EV Communication Error (Communication failure with vehicle)
		013	Ground Failure (Ground fault circuit interrupter has been activated)
		014	High Temperature (Temperature inside charge point is too high)
		015	Internal Error (Error in internal hardware or software component)
		016	Over Current Failure (Over current protection device has tripped)
		017	Over Voltage (Voltage has risen above an acceptable level)
		018	Power Meter Failure (Failure to read power meter)
		019	Power Switch Failure (Failure to control power switch)
		020	Reader Failure (Failure with idTag reader)
		021	Reset Failure (Unable to perform a reset)
		022	Under Voltage (Voltage has dropped below an acceptable level)
		023	Weak Signal (Wireless communication device reports a weak signal)
		100	No Error (No error to report)
		200	Payment Related Error (Abnormal completion of charging related to payment)

EV Charging Data Layouts

Estimated KM/Miles Added

Field	Attributes	Description
Table ID	n2	Fixed value: 08
Table Length	n3	Variable 001-006
Table Data	..n6	Maximum Length: 6 Contains an estimate of the total distance added, in km/miles, based on the kWh added and the type of vehicle engine charged.

EV Charging Data Layouts

Carbon Footprint		
Field	**Attributes**	**Description**
Table ID	n2	Fixed value: 09
Table Length	n3	Variable 001-012
Table Data	..n12	Maximum length :12 Contains the carbon footprint avoidance measurement (how much was saved from this charge) for the purchase on the transaction, measured in grams of carbon dioxide equivalent (CO2e).

Estimated Vehicles KM/Miles Available		
Field	**Attributes**	**Description**
Table ID	n2	Fixed value: 10
Table Length	n3	Variable 001-006
Table Data	..n6	Maximum length: 6 Contains the estimated mileage the car will have once it leaves the electric vehicle charging station after completion of the charge.

Maximum Power Dispensed		
Field	**Attributes**	**Description**
Table ID	n2	Fixed value: 11
Table Length	n3	Variable 001-006
Table Data	..n6	Maximum length: 6 Contains the maximum power dispensed during the specific charge session from that charging station. This can be different from the power output capacity of the station, based on the power management by the site operator.

Connector Type		
Field	**Attributes**	**Description**
Table ID	n2	Fixed value: 12
Table Length	n3	Variable 001-003

Table Data	..an3	Maximum length: 3 Contains a Visa-defined code for a specific connector type to identify the connection for the charge session. For Visa, below are the valid values: **Value** **Description** 001 AC - J1772 Type 1 002 AC - Mennekes - Type 2 003 AC - GB/T 100 DC - CCS1 101 DC - CHAdeMO 102 DC - CCS2 103 DC - GB/T 200 Tesla

12.41 Adjusted Product Code Data Segment

This section describes Data Segment No. 157, Adjusted Product Code Data Segment.

Note: This segment is used exclusively and allowed for Comdata cards only. Any transaction containing this segment that is not for a Comdata card will be declined.

Note: This segment is mutually exclusive with segment 102 Product Code Data. Merchants should send one or the other but never both. If both the segments are sent, the transaction will be declined.

Note: This segment does not allow any product codes above '899' except for '955' Cash Back. If any of these product codes is present, the transaction will be declined.

When a Financial Transaction request includes the Adjusted Product Code Data Segment, the products within this segment must meet the following requirements:

Fuel products must always be the first products in the segment.

A maximum of ten products is allowed in the segment.

The total of adjusted product amounts in the segment must equal the total of the amounts [Element No. 41 (Fuel Purchase Amount), Element No. 58 (Nonfuel Amount), Element No. 99 (Tax Amount), and Element No. 17 (Cash Amount)] in the Standard Message Data Segment (Data Segment No. 100).

Since tax amounts are being added to the individual product amounts, there should not be separate tax product codes included in the segment. Instead, tax amount should be reported in Element No. 99 (Tax Amount) in the Standard Message Data Segment (Data Segment No. 100).

Since Discount, Tax, and Coupon amounts are being accounted for in the individual product amounts, there should not be separate discount, tax, or coupon product codes included in the segment.

All fuel merchants must send fuel and nonfuel product data if applicable.

Note: A unique Product Code must be sent for each type of fuel purchased.

Note: With the growth of fleet industry and to fulfil the needs of the Over the Road Trucking (OTR) industry, BUYPASS can accept transactions with multiple fuels (e.g., purchase of Diesel, DEF, REEFER), all in 1 transaction. It is estimated that up to 30% of fuel transactions in OTR require fill-ups involving multiple Fuel Type Codes. Now, fuel merchants can allow multiple fuel fill-ups in 1 transaction. With the support of multiple fuels, a merchant must send the Product segment to BUYPASS with all the fuel codes and any non-fuel codes with all the relevant information like quantity, unit of measure, unit price, and product amount. Also, the merchant must send the primary fuel (fuel used for the primary fuel need of the truck/vehicle) as the very first product code within the Adjusted Product Code Segment.

This data segment can appear in any of the fields in Data Section No. 3. (Please refer to section 11.1, "Financial Transactions," for additional information about Data Section No. 3.)

It has a maximum length of 381 alphanumeric characters (001–381/a–z/A–Z).

It originates at the device.

The following table illustrates the layout of the Adjusted Product Code Data Segment:

Notes: A Field Separator always follows Segment Type and Segment Length. A Field Separator follows Adjusted Product Amount when Adjusted Product Amount is the very last element in the segment. When a field is not populated, still send the Field Separator.

A Product Data Field Delimiter always follows Quantity and Unit Price. A Product Data Field Delimiter follows Adjusted Product Amount when Adjusted Product Amount is not the last

element in the segment. When a field is not populated, still send the Adjusted Product Data Field Delimiter.

In the following table, the ▲ symbol represents a Field Separator, and the \ character is a Product Data Field Delimiter.

Adjusted Product Code Data Segment Format **Data Segment No. 157**					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 157 Source: Device
< ▲ >					
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators. Source: Device
< ▲ >					
3	87	Service Level	1	R	Identifies the type of sale. Source: Device
4	62	Number of Products	2	R	Identifies the count of products reported for this transaction. Source: Device
Product Data This section is repeated—by product—for a maximum of 10 products for a total variable length of up to 370 bytes. Fuel products are always the first products in the segment. **Note:** A unique Product Code must be sent for each type of fuel purchase.					
5	77	Product Code	3	R	Identifies the type of product sold including any cash back. Source: Device
6	106	Unit of Measure	1	R	Identifies the type of measurement for the product sold. Source: Device

Adjusted Product Code Data Segment Format **Data Segment No. 157**					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
7	81	Quantity	9	R	Identifies the number of units of the product sold. Quantity value must contain a digit for every position of assumed decimal places. Example: If the quantity is 0.05 and the number of assumed decimal places is 2, then this should be represented as '205'. '25' will be considered invalid. Source: Device
< \ >					
8	107	Unit Price	9	R	Identifies the price per unit of measure for the product sold. Unit Price value must contain a digit for every position of assumed decimal places. Example: If the unit price is $0.059 and the number of assumed decimal places is 3, then this should be represented as '3059'. '359' will be considered invalid. Source: Device
< \ >					
9	76	Product Amount	12	R	Identifies the monetary value of the product sold AFTER tax and applicable discounts or coupons. Source: Device
A Field Separator follows Product Amount if Product Amount is the very last element in the segment. OR A Product Data Field Delimiter follows Product Amount if Product Amount is not the last element in the segment.					

[1]Required/Optional/Conditional

For example: if a purchase included a $30 fuel product A, a $10 non-fuel product B, a $20 non-fuel product C, and $3 of tax, then

the Adjusted Product Amount for product A would be adjusted to be $30, (fuel product amounts don't get adjusted because they already include tax and discount amounts in their unit prices)

the Adjusted Product Amount for product B would be adjusted to be $11,

the Adjusted Product Amount for product C would be adjusted to be $22,

there would be no Tax product included

below is an example of the what the segment should look like.

“157+127+103020U3000010000\3000003000\000000030000\400U3000001000\3000010000\000000001100\401U3000002000\3000010000\000000002200+”

12.42 Merchant Data Segment

This section describes Data Segment No. DL1, Merchant Data Segment.

It has a maximum length of 399 alphanumeric characters (001–399/a–z/A–Z).

It originates at BUYPASS.

The following table illustrates the layout of the Merchant Data Segment:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Merchant Data Segment Format Data Segment No. DL1					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	R	Indicates that merchant data follows. Fixed value: # Source: Host
2	53	Merchant Name	24	R	Identifies the merchant associated with the transmitting device. Source: Host
3	98	Store Number	16	R	Identifies the store associated with the transmitting device. Source: Host
4	3	Address Line 1	24	R	Identifies the street address associated with the transmitting device. Source: Host
5	4	Address Line 2	21	R	Identifies the city, state, and ZIP Code™/ZIP+4® associated with the transmitting device. Source: Host
6	54	Merchant Phone Number	13	R	Identifies the telephone number associated with the transmitting device. Source: Host
7	59	Number of Card Types	2	R	Identifies the number of times that Field No. 8 is repeated. It indicates the number of card types processed by the transmitting device. Source: Host

Merchant Data Segment Format Data Segment No. DL1					
Field No. 8 can be present from 01–99 time(s).					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
8	14	Card Type	3	R	Identifies the card type accepted by the transmitting device. Source: Host
9	34	End-of-Data Indicator	1	R	Indicates the end of the data segment. Fixed value: ~ Source: Host

[1]Required/Optional/Conditional

12.43 Dial String Data Segment

This section describes Data Segment No. DL2, Dial String Data Segment.

It has a maximum length of 69 alphanumeric characters (01–69/a–z/A–Z).

It originates at BUYPASS.

The following table illustrates the layout of the Dial String Data Segment:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Dial String Data Segment Format Data Segment No. DL2					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	R	Indicates that dial string data follows. Fixed value: ! Source: Host
2	28	Dial String Type	1	R	Indicates transaction dial strings. Fixed value: 1 Source: Host
Primary Phone Number (Field Nos. 3–6)					
3	82	Redial Count	1	R	Indicates the number of times to redial the subsequent phone number. Source: Host
4	1	Access Code	12	C	Identifies data that must be dialed prior to dialing the actual phone number. Source: Host
5	66	Pause Indicator	1	C	Indicates that the device should pause approximately one second before proceeding. Fixed value: B Source: Host
6	75	Phone Number	18	R	Identifies the first phone number used in transaction dialing. Source: Host
7	27	Dial String Terminator	1	R	Indicates the end of the first dial string. Fixed value: A Source: Host

Secondary Phone Number
(Field Nos. 8–11)

This information is identical to the Primary Phone Number information with the exception of Phone Number. A secondary phone number is used. The secondary phone number is used after all attempts using the primary phone number are exhausted in accordance with the error recovery logic as set forth in the *Asynchronous Communications Protocol Specifications*.

12	27	Dial String Terminator	1	R	Indicates the end of the second dial string. Fixed value: F Source: Host
13	34	End-of-Data Indicator	1	R	Indicates the end of the data segment. Fixed value: ~ Source: Host

[1]Required/Optional/Conditional

12.44 Date and Time Data Segment

This section describes Data Segment No. DL3, Date and Time Data Segment.

It has a maximum length of 23 alphanumeric characters (01–23/a–z/A–Z).

It originates at BUYPASS.

The following table illustrates the layout of the Date and Time Data Segment:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Date and Time Data Segment Format
Data Segment No. DL3

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	24	Data Type Indicator	1	R	Indicates that the following data contains the current BUYPASS day of the week, date, and time. Fixed value: : Source: Host
2	25	Day of the Week	1	R	Identifies the current day of the week. Source: Host
3	21	Current Date	6	R	Identifies the current date. Source: Host
4	22	Current Time	4	R	Indicates the current time. Source: Host
5	23	Cut Time	4	R	Identifies the merchant's end-of-day settlement time. Source: Host
6	65	Password	6	R	Identifies the end-of-day function's password. Source: Device
7	34	End-of-Data Indicator	1	R	Indicates the end of the data segment. Fixed value: ~ Source: Host

[1]Required/Optional/Conditional

12.45 Software Dial Load Data Segment

This section describes Data Segment No. DL4, Software Dial Load Data Segment.

It has a maximum length of 52 alphanumeric characters (01–52/a–z/A–Z).

It originates at BUYPASS.

Note: The Software Dial Load Data Segment is used to download an application from a device management system supported by BUYPASS. Vendors who support their own applications do not use this segment.

The following table illustrates the layout of the Software Dial Load Data Segment:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Software Dial Load Data Segment Format Data Segment No. DL4					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	R	Indicates that software dial load data follows. Fixed value: @ Source: Host
2	57	New Software Version	8	R	Identifies the new software version number. Source: Host
3	95	Software Terminal Record ID	13	R	Identifies the Software Terminal Record ID associated with the device management system. Source: Host
4	91	Software Load Phone Number	18	R	Identifies the telephone number associated with the device management system. Source: Host
5	92	Software Load Request Date	6	R	Identifies the date the device should request a full load of the new software. Source: Host
6	93	Software Load Request Time	4	R	Identifies the time the device should request a full load of the new software. Source: Host
7	94	Software Load Type	1	R	Identifies the type of requested software load. Source: Host
8	34	End-of-Data Indicator	1	R	Indicates the end of the data segment. Fixed value: ~ Source: Host

12.46 Software IP Load Data Segment

This section describes Data Segment No. DL5, Software IP Load Data Segment.

It has a maximum length of 64 alphanumeric characters (01–64/a–z/A–Z).

It originates at BUYPASS.

Note: The Software IP Load Data Segment is used to download an application from a device management system supported by BUYPASS. Vendors who support their own applications do not use this segment.

The following table illustrates the layout of the Software IP Load Data Segment:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Software IP Load Data Segment Format
Data Segment No. DL5

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	24	Data Type Indicator	1	R	Indicates that software IP load data follows. Fixed value: $ Source: Host
2	57	New Software Version	8	R	Identifies the new software application name in the device management system. Source: Host
3	95	Software Terminal Record ID	13	R	Identifies the Software Terminal Record ID associated with the device management system. Source: Host
4	114	Software Load IP/URL Address	30	R	Identifies the IP/URL address used to access the device management system. Source: Host
5	92	Software Load Request Date	6	R	Identifies the date the device should request a full load of the new software. Source: Host
6	93	Software Load Request Time	4	R	Identifies the time the device should request a full load of the new software. Source: Host
7	94	Software Load Type	1	R	Identifies the type of requested software load. Source: Host
8	34	End-of-Data Indicator	1	R	Indicates the end of the data segment. Fixed value: ~ Source: Host

[1]Required/Optional/Conditional

12.47 Store and Forward Data Segment

This section describes Data Segment No.DL6, Store and Forward Data Segment.

It has a maximum length of 9 alphanumeric characters (01–9/a–z/A–Z).

It originates at BUYPASS.

Note: The Store and Forward Data Segment is only sent in a Table Load Response when Segment No. DL1 (Merchant Data Segment) includes a Card Type value of 173 in Data Element No. 14 (Card Type).

The following table illustrates the layout of the Store and Forward Data Segment:

Note: The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

Store and Forward Data Segment Format Data Segment No. DL6					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	R	Indicates that software IP load data follows. Fixed value: \ Source: Host
2	166	Start Time	4	R	Indicates the time to begin blocking store and forward processing. Format: HHMM Source: Host
3	166	End Time	4	R	Indicates the time to stop blocking store and forward processing. Format: HHMM
4	34	End-of-Data Indicator	1	R	Indicates the end of the data segment. Fixed value: ~ Source: Host

[1]Required/Optional/Conditional

12.48 Supplemental Terminal Data Segment

This section describes Data Segment No.DL7, Supplemental Terminal Data Segment.

This data segment will be identified by the special character "^".

The following table illustrates the layout of the Supplemental Terminal Data Segment.

Supplemental Terminal Data Segment Format Data Segment No. DL7					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	R	Indicates Supplementary Terminal data follows. Fixed Value "^".
2	84	Segment Length Indicator	3	R	Indicates the data segments length, exclusive of the Data Type Indicator Source: Device
3	232	Download Data	100	R	All or some of the Download Data. Data is in <tag><len><data> format. Refer to Appendix W for Download Data Layout information.

[1]Required/Optional/Conditional

12.49 EMV Terminal Floor Limits Data Segment

This section describes Data Segment No.DL8, EMV Terminal Floor Limits Data Segment.

This new segment will contain floor limits for each card type (as RID) to be used by this terminal. Inclusion of this segment in a table load will be dependent on a Special set at the terminal level. This data will be maintained at the Buypass Host and when this data is changed, the table download flag will be set for all terminals that have this special set.

The following table illustrates the layout of the EMV Terminal Floor Limits Data Segment.

EMV Terminal Floor Limits Data Segment Format Data Segment No. DL8					
Field No.	**Element No.**	**Element Name**	**Len.**	**Entry is: R/O/C[1]**	**Description**
1	24	Data Type Indicator	1	R	Indicates EMV Floor Limits data. Fixed Value "%".
2	84	Segment Length Indicator	3	R	Indicates the data segment's length, exclusive of the Data Type Indicator Source: Device
Floor Limit Data This section is repeated—by RID—for a maximum of 24 RIDs for a total variable length of up to 624 bytes.					
3	233	RID	10	R	EMV Application Identifier Source: Host
4	234	Stand-in Indicator	1	R	Stand-in Indicator. Source: Host
5	235	Floor Limit	12	R	Maximum allowable stand-in value. Source: Host
6	236	BUYPASS RID Card Type	3	R	BUYPASS card type associated with this RID. Source: Host

[1]Required/Optional/Conditional

12.50 Network Token Data Response Segment

This section describes Data Segment No. 154, Network Token Data Response Segment.
The Network Token Data Response Segment contains Network Token Data associated with this transaction. This segment will potentially hold multiple sub-segments, each of which contains different network token-related data. The sub-segments will be in the standard TLV format (sub-segment type, sub-segment length, sub-segment value).

The fields are not separated by Field Separators. When a field is not populated, the next field immediately follows.

The following table illustrates the layout of the Network Token Data Response Segment:

Network Token Data Response Format
Data Segment No. 154

Field No.	Element No.	Element Name	Len.	Entry is: R/O/C[1]	Description
1	85	Segment Type	3	R	Identifies the type of data segment being formatted. Fixed value: 154
2	84	Segment Length	3	R	Identifies the data segment's length, including Segment Type's length and Field Separators.
3	239	Network Token Data	999	R	All or some of the Network Token sub-segment data is specified below. Data is in <tag><len><data> format.

[1]Required/Optional/Conditional

Network Token Data Layouts

Network Token

Field	Attributes	Description
Table ID	N3	Fixed value: 001
Table Length	N3	Variable
Table Data	an	Max. Len.: 19 Identifies the Network Token Data.

Network Token Data Layouts

Network Token/PAN Indicator

Field	Attributes	Description
Table ID	N3	Fixed value: 002
Table Length	N3	Fixed
Table Data	an	Fixed Len.: 1 Identifies that the original transaction is authorized by either PAN or Token. Valid values are T – Network Token P – Clear PAN

Network Token Data Layouts

Auth Indicator 1st Try

Field	Attributes	Description
Table ID	N3	Fixed value: 003
Table Length	N3	Fixed
Table Data	an	Fixed Len.: 1 Identifies the value sent from First Data to network in the first auth request as PAN or NT. Valid values are T – Network Token P – Clear PAN

Network Token Data Layouts

Auth Indicator 2nd Try

Field	Attributes	Description
Table ID	N3	Fixed value: 004
Table Length	N3	Fixed
Table Data	an	Fixed Len.: 1 Identifies the value sent from First Data to network in auth request for the 2nd try as PAN or NT, if applicable Valid values are T – Network Token P – Clear PAN

Network Token Data Layouts

Auth Indicator 3rd Try		
Field	**Attributes**	**Description**
Table ID	N3	Fixed value: 005
Table Length	N3	Fixed
Table Data	an	Fixed Len.: 1 Identifies the value sent from First Data to network in auth request for the 3rd try as PAN or NT, if applicable Valid values are T – Network Token P – Clear PAN

13. Data Element Descriptions

This chapter describes each data element in the messages described in chapter 11, "Message Formats."

Section No. 13.1, “Data Elements in Alphabetical Order,” lists the data elements in alphabetical order, together with their corresponding data element number and location in chapter 11, "Message Formats"and chapter 12, “Data Segment Formats.”

Section No. 13.2, “Data Elements in Data Element Number Order,” lists the data elements in data element number order.

13.1 Data Elements in Alphabetical Order

The following table contains a list of all data elements used in chapter 11, "Message Formats." and chapter 12, Data Segment Formats." The list is arranged in alphabetical order and consists of the following information:

- Data element name
- Data element number
- Message(s) and/or data segment(s) in which the data element is found
- Page number for the data element's message format/data segment description

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Access Code	1	Dial String Data Segment (No. DL2)	12-113
Account Number	2	• Financial Transaction Response • Standard Message Data Segment (No. 100)	• 11-7 • 12-6
Additional Information	118	Additional Information Data Segment (No. 112)	12-30
Additional Information Data Segment Flag	115	Financial Transaction Response	11-7
Additional Information Indicator	116	Additional Information Data Segment (No. 112)	12-30
Additional Information Length	117	Additional Information Data Segment (No. 112)	12-30
Address Line 1	3	Merchant Data Segment (No. DL1)	12-50
Address Line 2	4	Merchant Data Segment (No. DL1)	12-50
Approval Number	5	• Financial Transaction Response • Standard Message Data Segment (No. 100)	• 11-7 • 12-6
Approved Amount	6	Financial Transaction Response	11-7
Authorizer Code	7	Financial Transaction Response	11-7
Authorizer Response Code	8	Financial Transaction Response	11-7
Available Product Information	240	WEX Available Product Fleet Information Data Segment (No. 148)	12-73
Merchant Trace ID	135	ECA/ TeleCheck® Data Segment (No. 113)	12-32

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Batch Number	214	• Moneris Day End Batch Balance (Request) Segment (No. 139) • Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141) • Moneris Day End Batch Close (Response) Segment (No. 142)	• 12-62 • 12-63 • 12-65 • 12-67
Block Number	11	• Electronic Mail Response • Electronic Mail Data Segment (No. 109)	• 11-23 • 12-23
BUYPASS Card Type	173	• Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 904 is returned.	• 12-37
BUYPASS RID Card Type	236	• EMV Terminal Floor Limits Data Segment (No.DL8)	• 12-120
CA Public Key File Block	194	• CA Public Key File Load Response	• 11-53
CA Public Key File Block Length	193	• CA Public Key File Load Response	• 11-53
CA Public Key File Checksum	187	• EMV Request Data Segment (No. 130) • EMV Response Data Segment (No. 131) • CA Public Key File Data Segment (No. 132)	• 12-53 • 12-55 • 12-57
Card BIN Range, Beginning	183	• Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 904 is returned.	• 12-37
Card BIN Range, Ending	184	• Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 904 is returned.	• 12-37
Card Discretionary Block Data	12	Standard Message Data Segment (No. 100)	12-6
Card Label	13	• Approved Totals Response • Totals Data Segment (No. 105)	• 11-18 • 12-18

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Card Table Data	175	• Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 902 is returned.	• 12-37
Card Table Load Version	177	• Proprietary Data Load Segment (No.118)	• 12-37
Card Table Type	174	• Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 902 is returned.	• 12-37
Card Type	14	Merchant Data Segment (No. DL1) **Note:** Sometimes used to construct a Prompt Code. See also Prompt Code.	12-50
Card Type Total Amount	15	• Approved Totals Response • Totals Data Segment (No. 105)	• 11-18 • 12-18
Card Type Total Count	16	• Approved Totals Response • Totals Data Segment (No. 105)	• 11-18 • 12-18
Cash Amount	17	Standard Message Data Segment (No. 100)	12-6
CAVV Revised Format	195	NFC Payment Tokenization Data Segment (No.123)	12-50
Check Issue Date	130	Check Data Segment (No. 110)	12-25
Check Number	127	Check Data Segment (No. 110)	12-25
Check Type	126	Check Data Segment (No. 110)	12-25
Clerk ID	18	EBT Data Segment (No. 103)	12-15
Corrections Dollar Value	222	• Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141)	• 12-63 • 12-65
Coupon Amount	142	Loyalty Card Data Segment (No. 108)	12-21
Coupon ID	141	Loyalty Card Data Segment (No. 108)	12-21
Credit Dollar Value	220	• Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141)	• 12-63 • 12-65
Cryptogram Token Data	202	NFC Payment Tokenization Data Segment (No.123)	12-50
Currency Code	20	Totals Data Segment (No. 105)	12-18

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Current Date	21	• Date and Time Load Response • Date and Time Data Segment (No. DL3)	• 11-33 • 12-115
Current Time	22	• Date and Time Load Response • Date and Time Data Segment (No. DL3)	• 11-33 • 12-115
Customer Data	161	Fleet Data Segment (No. 101)	12-9
Customer Last Name	129	Check Data Segment (No. 110)	12-25
Customer Phone Number	128	Check Data Segment (No. 110)	12-25
Cut Time	23	• Date and Time Load Response • Date and Time Data Segment (No. DL3)	• 11-33 • 12-115
Data Type Indicator	24	• Date and Time Load Response • TransArmor Key and Key ID Load Response • Merchant Data Segment (No. DL1) • Dial String Data Segment (No. DL2) • Date and Time Data Segment (No. DL3) • Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)	• 11-33 • 12-50 • 12-113 • 12-115 • 12-116 • 12-117
Date of Birth	125	Check Data Segment (No. 110)	12-25
Day of the Week	25	• Date and Time Load Response • Date and Time Data Segment (No. DL3)	• 11-33 • 12-115
Decline Code	26	• Financial Transaction Response • Fuel Price Update Response Segment Format	• 11-7 • 12-84
Debit Dollar Value	218	• Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141)	• 12-63 • 12-65
Denial Record Number	136	ECA/ TeleCheck® Data Segment (No. 113)	12-32
Department #	160	Fleet Data Segment (No. 101)	12-9
Device Card Table Version	176	Proprietary Data Load Segment (No. 118)	12-37

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Dial String Terminator	27	Dial String Data Segment (No. DL2)	12-113
Dial String Type	28	Dial String Data Segment (No. DL2)	12-113
Direct Marketing Invoice Number	29	Purchase Card Data Segment (No. 104)	12-16
Discount Program Description	186	Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 904 is returned.	12-37
Discount Quantity Limit	185	Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 904 is returned.	12-37
Download Indicator	30	• Financial Transaction Response • Electronic Mail Response • Communications Test Response	• 11-7 • 11-23 • 11-25
Download Data	232	• Supplemental Terminal Data Segment (No. DL7)	• 12-119
Driver/Identification Number	31	Fleet Data Segment (No. 101)	12-9
Driver's License	123	Check Data Segment (No. 110)	12-25
EBT Program Data	164	EBT Data Segment	12-15
ECA/ TeleCheck® Clerk ID	131	ECA/ TeleCheck® Data Segment (No. 113)	12-32
ECA/ TeleCheck® Phone Number	133	ECA/ TeleCheck® Data Segment (No. 113)	12-32
ECA/ TeleCheck® Product Code	132	ECA/ TeleCheck® Data Segment (No. 113)	12-32
ECA/ TeleCheck® Trace ID	134	ECA/ TeleCheck® Data Segment (No. 113)	12-32
Employee Number	32	• Totals Data Segment (No. 105) • Electronic Mail Data Segment (No. 109)	• 12-18 • 12-23
EMV Additional Information Indicator	191	• EMV Request Data Segment (No.131) • EMV Response Data Segment (No.132)	• 12-53 • 12-55
EMV Additional Information Length	192	• EMV Request Data Segment (No.131) • EMV Response Data Segment (No.132)	• 12-53 • 12-55
EMV Card Sequence Number	188	• EMV Request Data Segment (No.131)	• 12-53

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
EMV Chip Data	190	• EMV Request Data Segment (No.131) • EMV Response Data Segment (No.132)	• 12-53 • 12-55
EMV Chip Data Length	189	• EMV Request Data Segment (No.131) • EMV Response Data Segment (No.132)	• 12-53 • 12-55
Encrypted PIN Block Data	33	Standard Message Data Segment (No. 100)	12-6
End-of-Data Indicator	34	• Merchant Data Segment (No. DL1) • Dial String Data Segment (No. DL2) • Date and Time Data Segment (No. DL3) • Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)	• 12-50 • 12-113 • 12-115 • 12-116 • 12-117
End-of-Load Indicator	35	Table Load Response	11-28
Enhanced Fleet Data	239	Enhanced Fleet Request Segment (No. 145)	12-72
EV Charging Data	242	EV Charging Data Segment (No. 156)	12-100
Expiration Date	146	Loyalty Card Data Segment (No. 108)	12-21
Extended MICR Data	137	ECA/ TeleCheck® Data Segment (No. 113)	12-32
Extended Unit of Measure	243	• Product Code Data Segment (No. 102) • Variable Information Data Segment (No. 111)	12-12 12-29
Extract Date	36	Electronic Mail Data Segment (No. 109)	12-23
Extract Time	37	Electronic Mail Data Segment (No. 109)	12-23
Fee Amount	38	Approved Totals Response	11-18
Firmware Version	39	• Table Load Request • Software Load Request • TransArmor Key and Key ID Load Request • Totals Data Segment (No. 105) • TransArmor Load Data Segment (No. 116)	• 11-27 • 11-34 • 12-18 • 12-36
Fleet Employee Number	40	Fleet Data Segment (No. 101)	12-9

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Floor Limit	235	• EMV Terminal Floor Limits Data Segment (No.DL8)	• 12-120
Fuel Purchase Amount	41	Standard Message Data Segment (No. 100)	12-6
Fuel Volume Data	201	Proprietary Data Load Segment (No.118)	12-37
Grand Total	42	• Approved Totals Response • Totals Data Segment (No. 105)	• 11-18 • 12-18
Hardware Version	43	• Table Load Request • Software Load Request • TransArmor Key and Key ID Load Request • Totals Data Segment (No. 105) • TransArmor Load Data Segment (No. 116)	• 11-27 • 11-34 • 12-18 • 12-36
Host Discount Timestamp	179	• Proprietary Data Load Segment (No.118)	• 12-37
Identification Number	See Driver/Identification Number.		
Information Byte	44	• Table Load Request • Phone Load Request • Date and Time Load Request • Software Load Request • Standard Message Data Segment (No. 100) • Totals Data Segment (No. 105) • Electronic Mail Data Segment (No. 109)	• 11-27 • 11-30 • 11-32 • 11-34 • 12-6 • 12-18 • 12-23
Inclusive/Exclusive for Tax 1	223	Tax by Product Data Segment (No. 143)	12-68
Inclusive/Exclusive for Tax 2	226	Tax by Product Data Segment (No. 143)	12-68
Inclusive/Exclusive for Tax 3	229	Tax by Product Data Segment (No. 143)	12-68
Initiation Date	45	• Financial Transaction Response • Electronic Mail Response	• 11-7 • 11-23
Initiation Time	46	• Financial Transaction Response • Electronic Mail Response	• 11-7 • 11-23
Job ID	159	Fleet Data Segment (No. 101)	12-9
Job Number	47	Fleet Data Segment (No. 101)	12-9

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Key Data	157	TransArmor Key and Key ID Load Response	
Key Data Length	156	TransArmor Key and Key ID Load Response	
Key ID	155	TransArmor Key and Key ID Load Response	
Lane Number	See Pump/Lane Number.		
Language Indicator	215	• Moneris Day End Batch Balance (Request) Segment (No. 139) • Moneris Day End Batch Close (Request) Segment (No. 141)	• 12-62 • 12-65
License #	158	Fleet Data Segment (No. 101)	12-9
Load Control Key	178	Proprietary Data Load Segment (No.118)	12-37
Load Subtype	205	Moneris key Load Request	11-39
Load Type	48	• Table Load Request • Phone Load Request • Date and Time Load Request • Software Load Request • TransArmor Key and Key ID Load Request • TransArmor Load Data Segment (No. 116)	• 11-27 • 11-30 • 11-32 • 11-34 • 12-36
Local Date and Local Time	49	Standard Message Data Segment (No. 100)	12-6
Local Time	50	Electronic Mail Data Segment (No. 109)	12-23
Loyalty Account Number	139	Loyalty Card Data Segment (No. 108)	12-21
Loyalty Information Version	150	Loyalty Card Data Segment (No. 108)	12-21
Loyalty Phone Number	See Phone Number, Loyalty.		
Loyalty Program ID	138	Loyalty Card Data Segment (No. 108)	12-21
Loyalty Track 2 Data	147	Loyalty Card Data Segment (No. 108)	12-21
MAC	210	• Moneris Key Load Response • Moneris Data (Response) Segment (No.136) • Moneris Day End Batch Close (Response) Segment (No. 142)	• 11-40 • 12-61 • 12-67

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Mail Text Data	51	Electronic Mail Response	11-23
Mail Text Data Length	52	Electronic Mail Response	11-23
Merchant Name	53	Merchant Data Segment (No. DL1)	12-50
Merchant Phone Number	54	Merchant Data Segment (No. DL1)	12-50
Message Format Version Identifier	55	• Data Section No. 1 – Financial Req. • Data Section No. 1 – Loyalty Req. • Data Section No. 1 – ECA/ TeleCheck® Req. • Data Section No. 1 – Totals Req. • Data Section No. 1 – Email Req. • Data Section No. 1 – Comm. Req. • Data Section No. 1 – TransArmor Key and Key ID Load Request	• 11-3 • 11-11 • 11-13 • 11-15 • 11-21 • 11-24
MICR Data	122	Check Data Segment (No. 110) **Note:** The maximum length is 50 bytes	12-25
Moneris Data	213	• Moneris Data (Request) Segment (No.135) • Moneris Data (Response) Segment (No.136)	• 12-60 • 12-61
Moneris Key Data	212	• Moneris Key Load Response	• 11-40
Moneris Key Indicators	211	• Moneris Key Load Response	• 11-40
Moneris Merchant ID	208	• Moneris Key Load Request • Moneris Key Load Response • Moneris Data (Request) Segment (No.135) • Moneris Data (Response) Segment (No.136) • Moneris Day End Batch Balance (Request) Segment (No. 139) • Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141) • Moneris Day End Batch Close (Response) Segment (No. 142)	• 11-39 • 11-40 • 12-60 • 12-61 • 12-62 • 12-63 • 12-65 • 12-67

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Moneris Response Code	209	• Moneris Key Load Response	• 11-40
Moneris Terminal Identifier	207	• Moneris Key Load Request • Moneris Key Load Response • Moneris Data (Request) Segment (No.135) • Moneris Data (Response) Segment (No.136) • Moneris Day End Batch Balance (Request) Segment (No. 139) • Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141) • Moneris Day End Batch Close (Response) Segment (No. 142)	• 11-39 • 11-40 • 12-60 • 12-61 • 12-62 • 12-63 • 12-65 • 12-67
Net Amount	56	Approved Totals Response	11-18
Network Management Message	120	Communications Test Request	11-24
New Software Version	57	• Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)	• 12-116 • 12-117
Nonfuel Amount	58	Standard Message Data Segment (No. 100)	12-6
Number of Card Types	59	Merchant Data Segment (No. DL1)	12-50
Number of Corrections	221	• Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141)	• 12-63 • 12-65
Number of Credits	219	• Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141)	• 12-63 • 12-65

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Number of Debits	217	• Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141)	• 12-63 • 12-65
Number of Discounts	171	Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 904 is returned.	12-37
Number of Receipt Text Lines	168	Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 901 is returned.	12-37
Number of Print Lines	61	Financial Transaction Response	11-7
Number of Products	62	Product Code Data Segment (No. 102) and Adjusted Product Segment (No. 157)	• 12-12 • 12-105
Number of Segments	63	• Data Section No. 1 – Financial Req. • Data Section No. 1 – Loyalty Req. • Data Section No. 1 – ECA/ TeleCheck® Req. • Data Section No. 1 – Totals Req. • Data Section No. 1 – Email Req. • Data Section No. 1 – Comm. Req. • Data Section No. 1 – TransArmor Key and Key ID Load Request	• 11-3 • 11-11 • 11-13 • 11-15 • 11-21 • 11-24
Odometer	64	Fleet Data Segment (No. 101)	12-9
Token PAN Suffix	197	NFC Payment Tokenization Data Segment (No.123)	12-50
Partial Approval Indicator	121	Standard Message Data Segment (No. 100)	12-6
Password	65	• Date and Time Load Response • Totals Data Segment (No. 105) • Electronic Mail Data Segment (No. 109) • Date and Time Data Segment (No. DL3)	• 11-33 • 12-18 • 12-23 • 12-115
Pause Indicator	66	Dial String Data Segment (No. DL2)	12-113
Payment Tender Type	148	Loyalty Card Data Segment (No. 108)	12-21
PC Duty Amount	72	Purchase Card Data Segment (No. 104)	12-16
PC Freight Amount	73	Purchase Card Data Segment (No. 104)	12-16
PC Tax Amount	74	Purchase Card Data Segment (No. 104)	12-16
Phone Number	75	Dial String Data Segment (No. DL2)	12-113

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Phone Number, Loyalty	145	Loyalty Card Data Segment (No. 108)	12-21
Points to Redeem	140	Loyalty Card Data Segment (No. 108)	12-21
Price Data	241	Fuel Price Update Request Segment Format (No. 149)	12-83
Print Data	152	Print Data Segment (No. 115)	12-35
	See also Terminal Display /Printer Message and Number of Print Lines.		
Printer Message	See Terminal Display /Printer Message and Number of Print Lines.		
Product Amount	76	Product Code Data Segment (No. 102) and Adjusted Product Segment (No. 157)	• 12-12 • 12-105
Product Code	77	Product Code Data Segment (No. 102) and Adjusted Product Segment (No. 157)	• 12-12 • 12-105
Product Discount Amount	172	Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 904 is returned.	12-37
Prompt Code	78	• Standard Message Data Segment (No. 100) • Totals Data Segment (No. 105) • Electronic Mail Data Segment (No. 109) **Note:** Constructed using Card Type. See also Card Type.	• 12-6 • 12-18 • 12-23
Prompt Code, Pending	182	• Proprietary Data Load Segment (No.118)	• 12-37
Pump/Lane Number	79	Standard Message Data Segment (No. 100)	12-6
Purchase Code	80	Purchase Card Data Segment (No. 104)	12-16
Quantity	81	Product Code Data Segment (No. 102) and Adjusted Product Segment (No. 157)	• 12-12 • 12-105
Redial Count	82	Dial String Data Segment (No. DL2)	12-113
Response Code	83	• Financial Transaction Response • Approved Totals Response • Declined Totals Response • Electronic Mail Response • Communications Test Response • TransArmor Key and Key ID Load Response	• 11-7 • 11-18 • 11-20 • 11-23 • 11-25

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Response Display	216	• Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Response) Segment (No. 142)	• 12-63 • 12-67
Receipt Card Description	200	• Transaction Attributes Data Segment (No. 134)	• 12-53
Receipt Text Data	170	• Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 901 is returned.	• 12-37
Receipt Text Data Length	169	• Proprietary Data Load Segment (No.118) **Note:** Appears when Prompt Code 901 is returned.	• 12-37
RID	233	• EMV Terminal Floor Limits Data Segment (No.DL8)	• 12-120
SafeKey Data	203	• NFC Payment Tokenization Data Segment (No.123)	• 12-50
SafeKey Response	204	• NFC Payment Tokenization Data Segment (No.123)	• 12-50
Segment Length	84	• TransArmor Key and Key ID Load Request • Standard Message Data Segment (No. 100) • Fleet Data Segment (No. 101) • Product Code Data Segment (No. 102) • EBT Data Segment (No. 103) • Purchase Card Data Segment (No. 104)	• 10-61 • 12-6 • 12-9 • 12-12 • 12-15 • 12-16

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
		• Totals Data Segment (No. 105) • Loyalty Card Data Segment (No. 108) • Electronic Mail Data Segment (No. 109) • Check Data Segment (No. 110) • Variable Information Data Segment (No. 111) • Additional Information Data Segment (No. 112) • ECA/ TeleCheck® Data Segment (No. 113) • SKU Data Segment (No. 114) • Print Data Segment (No. 115) • TransArmor Load Data Segment (No. 116) • Proprietary Data Load Segment (No. 118) • Totals with Proprietary Data Load Data Segment (No.119) • Print Data 2 Segment (No.120) • NFC Payment Tokenization Data Segment (No. 123) • EMV Request Data Segment (No.130) • EMV Response Data Segment (No. 131) • CA Public Key File Data Segment (No. 132) • Transaction Attributes Data Segment (No. 134) • Adjusted Product Segment (No 157)	• 12-18 • 12-21 • 12-23 • 12-25 • 12-29 • 12-30 • 12-32 • 12-34 • 12-35 • 12-36 • 12-37 • 12-46 • 12-49 • 12-50 • 12-53 • 12-55 • 12-57 • 12-59 • 12-105

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Segment Type	85	• TransArmor Load Data Segment (No. 116) • Standard Message Data Segment (No. 100) • Fleet Data Segment (No. 101) • Product Code Data Segment (No. 102) • EBT Data Segment (No. 103) • Purchase Card Data Segment (No. 104) • Totals Data Segment (No. 105) • Loyalty Card Data Segment (No. 108) • Electronic Mail Data Segment (No. 109) • Check Data Segment (No. 110) • Variable Information Data Segment (No. 111) • Additional Information Data Segment (No. 112) • ECA/ TeleCheck® Data Segment (No. 113) • SKU Data Segment (No. 114) • Print Data Segment (No. 115) • Print Data 2 Segment (No. 120) • TransArmor Load Data Segment (No. 116)	• 12-36 • 12-6 • 12-9 • 12-12 • 12-15 • 12-16 • 12-18 • 12-21 • 12-23 • 12-25 • 12-29 • 12-30 • 12-32 • 12-34 • 12-35 • 12-37 • 12-36
Sequence Number	86	• Financial Transaction Response • Approved Totals Response • Declined Totals Response • Electronic Mail Response • TransArmor Key and Key ID Load Request • TransArmor Key and Key ID Load Response • Standard Message Data Segment (No. 100) • Totals Data Segment (No. 105) • Electronic Mail Data Segment (No. 109) • TransArmor Load Data Segment (No. 116)	• 11-7 • 11-18 • 11-20 • 11-23 • 12-6 • 12-18 • 12-23 • 12-36

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Service Level	87	Product Code Data Segment (No. 102)	12-12
Settlement Date	119	Financial Transaction Response	11-7
Settlement Type	198	Transaction Attributes Data Segment (No. 134)	12-53
Ship-from Postal Code	88	Purchase Card Data Segment (No. 104)	12-16
Ship-to Country Code	89	Purchase Card Data Segment (No. 104)	12-16
Ship-to Postal Code	90	Purchase Card Data Segment (No. 104)	12-16
Signature Required	199	Transaction Attributes Data Segment (No. 134)	12-53
Site Configuration Data	180	Proprietary Data Load Segment (No. 118)	12-37
SKU Data	149	SKU Data Segment (No. 114)	12-34
Software Load IP/URL Address	114	Software IP Load Data Segment (No. DL5)	12-117
Software Load Phone Number	91	Software Dial Load Data Segment (No. DL4)	12-116
Software Load Request Date	92	• Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)	• 12-116 • 12-117
Software Load Request Time	93	• Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)	• 12-116 • 12-117
Software Load Type	94	• Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)	• 12-116 • 12-117
Software Terminal Record ID	95	• Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)	• 12-116 • 12-117
Software Version	96	• Table Load Request • Software Load Request • TransArmor Key and Key ID Load Request • Totals Data Segment (No. 105) • TransArmor Load Data Segment (No. 116)	• 11-27 • 11-34 • 12-18 • 12-36

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
SPDH Header	206	• Moneris Key Load Request • Moneris Key Load Response • Moneris Data (Request) Segment (No.135) • Moneris Data (Response) Segment (No.136) • Moneris Day End Batch Balance (Request) Segment (No. 139) • Moneris Day End Batch Balance (Response) Segment (No. 140) • Moneris Day End Batch Close (Request) Segment (No. 141) • Moneris Day End Batch Close (Response) Segment (No. 142)	• 11-39 • 11-40 • 12-60 • 12-61 • 12-62 • 12-63 • 12-65 • 12-67
Stand-in Indicator	234	• EMV Terminal Floor Limits Data Segment (No.DL8)	• 12-120
Start-of-Data Block Indicator	97	• Table Load Response • Phone Load Response • Software Load Response	• 11-28 • 11-31 • 11-35
State Code	124	Check Data Segment (No. 110)	12-25
Start Date or End Date	165	Proprietary Data Load Segment (No.118)	12-37
Start Time or End Time	166	Proprietary Data Load Segment (No.118)	12-37
Stock Keeping Unit Data	See SKU Data.		
Store Number	98	Merchant Data Segment (No. DL1)	12-50
Street Address	144	Loyalty Card Data Segment (No. 108)	12-21
TAVV Cryptogram	237	NFC Payment Tokenization Data Segment (No.123)	12-50
TAVV Result Code	238	NFC Payment Tokenization Data Segment (No.123)	12-50
Tax Amount	99	Standard Message Data Segment (No. 100)	12-6
Tax Amount 1	225	Tax by Product Data Segment (No. 143)	12-68
Tax Amount 2	228	Tax by Product Data Segment (No. 143)	12-68
Tax Amount 3	231	Tax by Product Data Segment (No. 143)	12-68
Tax Type 1	224	Tax by Product Data Segment (No. 143)	12-68
Tax Type 2	227	Tax by Product Data Segment (No. 143)	12-68

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Tax Type 3	230	Tax by Product Data Segment (No. 143)	12-68
Terminal Display Communications Message	100	Communications Test Response	11-25
Terminal Display /Printer Message	101	• Financial Transaction Response • Declined Totals Response	• 11-7 • 11-20
Terminal Identifier	102	• Table Load Request • Phone Load Request • Date and Time Load Request • Software Load Request • TransArmor Key and Key ID Load Request • Standard Message Data Segment (No. 100) • Totals Data Segment (No. 105) • Electronic Mail Data Segment (No. 109) • TransArmor Load Data Segment (No. 116) • Fuel Price Update Request Segment Format (No. 149) • Fuel Price Update Response Segment Format (No. 150)	• 11-27 • 11-30 • 11-32 • 11-34 • 12-6 • 12-18 • 12-23 • 12-36 • 12-83 • 12-84
Text Data	103	Electronic Mail Data Segment (No. 109)	12-23
Text Data Length	104	Electronic Mail Data Segment (No. 109)	12-23
Token Requestor ID	196	NFC Payment Tokenization Data Segment (No.123)	12-50
Totals Date	105	• Approved Totals Response • Declined Totals Response • Totals Data Segment (No. 105)	• 11-18 • 11-20 • 12-18
Unit of Measure	106	Product Code Data Segment (No. 102) and Adjusted Product Segment (No. 157)	• 12-12 • 12-105
Unit of Work	151	Loyalty Card Data Segment (No. 108)	12-21
Unit Price	107	Product Code Data Segment (No. 102) and Adjusted Product Segment (No. 157)	• 12-12 • 12-105
Update Code	143	Loyalty Card Data Segment (No. 108)	12-21
User ID	162	Fleet Data Segment (No. 101)	12-9
Variable Information	113	Variable Information Data Segment (No. 111)	12-29

Alphabetical List of Data Element Names			
This Data Element:	**Has this Number:**	**And Is found in:**	**On Page No.**
Variable Information Indicator	111	Variable Information Data Segment (No. 111)	12-29
Variable Information Length	112	Variable Information Data Segment (No. 111)	12-29
Vehicle ID#	163	Fleet Data Segment (No. 101)	12-9
Vehicle Number	108	Fleet Data Segment (No. 101)	12-9
Voucher ID	109	EBT Data Segment (No. 103)	12-15
WIC Discount Amount	153	EBT Data Segment (No. 103)	12-15
WIC Product Data	154	EBT Data Segment (No. 103)	12-15

13.2 Data Elements in Data Element Number Order

This section lists the data elements in data element number order. The following table lists information that may appear in each data element description:

Element Information	Description	Codes
Number	Unique identifier for the element	
Name	General name for the element	
Character type	Alpha Numeric (Right justified, zero filled padding to the left) Alphanumeric (Left justified; space filled padding to the right) Signed numeric Special character	A N AN SN S
Maximum Length	Maximum data length in bytes	
Representation	Fixed number of characters Maximum variable number of characters Character type Number of decimal places Sign Right/left alignment Zero-/space-filled Upper/lower case	
Purpose	Processing intent Information conveyed	
Processing rules	Required attributes for data entry or use	
Valid codes/values	Supported entry options and their definitions	

Number:	1	**Name:**	Access Code
Character Type:	AN	**Maximum Length:**	12 bytes

Representation:	Variable length of up to 12 alphanumeric characters
Purpose:	Identifies data that must be dialed prior to dialing the phone number in the Telephone Table Load Response. **Examples.** PBX access codes Codes for disabling custom calling features
Processing Rules:	If Access Code information is not necessary, the dial string contains neither this element nor the Pause Indicator that immediately follows it; however, this element may be used to indicate a much longer than standard pause for use with phone systems that are slow to provide a dial tone. In these cases, one or more *B* characters may be inserted in this element. Each one should be interpreted as a 1-second pause.
Valid Codes/Values:	Any number B

Number: 2 **Name:** Account Number

Character Type: ANS **Maximum Length:**

Non-TransArmor (all requests): 24 bytes

Non-TransArmor (all responses): 25 bytes

TransArmor

PKI Encryption and Tokenization:

Initial Request	425 bytes
Initial Response	25 bytes
Follow-on Request	50 bytes
Follow-on Response	25 bytes

Token Only:

Initial Request	425 bytes
Initial Response	25 bytes
Follow-on Request	425 bytes
Follow-on Response	25 bytes

TransArmor - VeriFone Edition

Encryption and Tokenization:

All Requests	425 bytes
All Responses	25 bytes

Representation:

Non-TransArmor—Initial Transaction—Request:

Format: 123456789012nn…nn

Where 123456789012nn…nn = a variable length (up to 24 alphanumeric bytes) Account Number.

Non-TransArmor—Initial Transaction—Response:

Format: 123456789012nnnnnnnnnnnnn

Where 123456789012nnnnnnnnnnnnn = a fixed length (25 alphanumeric bytes) Account Number.

Non-TransArmor— Subsequent Transaction —Request:

Format: 123456789012nn…nn

Where 123456789012nn…nn = a variable length (up to 24 alphanumeric bytes) Account Number.

Non-TransArmor— Subsequent Transaction —Response:

Format: 123456789012nnnnnnnnnnnnn

Where 123456789012nnnnnnnnnnnnn = a fixed length (25 alphanumeric bytes) Account Number.

TransArmor – All Messages:

For all TransArmor formatting of the Account Number field, see Appendix AA. TransArmor Processing Considerations.

Purpose:

Non-TransArmor: Identifies the card or account.

TransArmor (Either Encryption Method): Identifies the 25-byte TransArmor header plus the 400-byte EDATA block that contains the encrypted/unencrypted information that identifies the account for TransArmor.

Processing Rules:

For **non-TransArmor** and **TransArmor Token Only**, the POS device maintains the card's Account Number until it receives a response from the host.

For **TransArmor PKI Encryption and Tokenization**, the POS device must not maintain the card's Account Number after the initial card swipe.

For **Non-TransArmor and TransArmor**, if the Account Number is obtained from a Track 2 card reader, instead of being manually keyed, the Account Number and Card Discretionary Block Data should consist of all data read from the card between, but not including, the Start Sentinel and End Sentinel. Account Number is considered the data before the = character, and Card Discretionary Block Data is considered the data after the = character. The = character is used instead of a Field Separator.

For **Non-TransArmor and TransArmor**, if the Account Number is obtained from a Track 1 card reader (credit cards only), instead of

being manually keyed, the Account Number and Card Discretionary Block Data should consist of all unaltered data read from the card between, but not including, the Start Sentinel and End Sentinel. Account Number is considered the data before the first ^ character, and Card Discretionary Block Data is considered the data after the first ^ character. The ^ character is used instead of a Field Separator. For **Non-TransArmor** – Account Number and Card Discretionary Block Data combined do not exceed 76 bytes for Track 1 or 37 bytes for Track 2.

For **TransArmor (Either Encryption Method)** – Account Number and Card Discretionary Block Data combined do not exceed 476 bytes for Track 1 or Track 2.

Note: Track data placed in the combined data elements of Account Number and Card Discretionary Block Data cannot exceed the total length of those data elements

Real Time Account Updater:

When a merchant receives Account Updater Card Number (New PAN) in the original transaction response in the Real Time Account Updater Response Segment – Segment 155, the merchant can send either the new PAN or the original PAN in the 'Follow-on' transactions in this Account Number field.

When a merchant receives Account Updater TransArmor Token (New TransArmor Token) in the original transaction response in the Real Time Account Updater Response Segment – Segment 155, the merchant can send either the new TransArmor token or the original TransArmor token in the 'Follow-on' transactions in this Account Number field as TAP request.

In the event of a 'Retry' of a 'Follow-on' transaction (Reversals, Completions, Cancellations, Void of Merchandise Return, and Timeout reversals) by the merchant, the merchant should use the same credentials for the retry as were used in the initial attempt.

Refer to Appendix AD – Real Time Account Updater for more details.

Valid Codes/Values:

Credit, debit, and EBT cards: Please refer to the most recent version of *BUYPASS® Platform Issuer Identification Number (IIN) Reference Manual*.

Fleet cards: Please refer to the most recent version of *BUYPASS® Platform Petroleum Industry Processing Specifications*.

Stored Value cards: The length of the Account Number should be 16 or 19 bytes.

Note: For a First Data™ Premium Gift Card Transaction Response, the calculated card number is returned in the Account Number field.

Check Services: All MICR data (scanned or manually entered) up to 23 bytes are included in this field. This data must also be included in MICR Data (Element No. 122) of the Check Data Segment (Segment No. 110) .The Account Number should contain the following keyed or swiped information:

Service Provider	Pos. Nos.	Data
SCAN (ETC)	1–23	MICR data or driver's license
ECA/ TeleCheck® service	1–23	MICR data or driver's license
Certegy (Formerly Telecredit)	1–24	Driver's license

Loyalty cards: For a Loyalty Totals Request, Update Code value of "T", the words "TOTALS REQUEST" must appear in the Account Number field. In any other loyalty advice transaction, the value is the actual Loyalty Account Number.

Express Code transactions: For an Express Code request (card type code 095), the 25 digit Express Code Value should appear in the Account Number area.

ComCheck transactions: For a ComCheck request (card type code 096), the 10 digit Check Value should appear in the Account Number area.

Money Code/Check transactions: For a Money Code/Check request (card type code 093), the 10-21 digit Money Code Value should appear in the Account Number area.

Number:	3	**Name:**	Address Line 1
Character Type:	AN	**Maximum Length:**	24 bytes

Representation:	Fixed length of up to 24 alphanumeric characters
Purpose:	Identifies the street address of the transmitting device location.
Processing Rules:	
Valid Codes/Values:	

Number:	4	**Name:**	Address Line 2
Character Type:	AN	**Maximum Length:**	21 bytes

Representation:	Fixed length of 21 alphanumeric characters	
Purpose:	Identifies the city, state, and ZIP Code™/ZIP+4® of the transmitting device location.	
Processing Rules:	Used in Data Segment No. DL1, Merchant Data Segment.	
Valid Codes/Values:	**Pos. Nos.**	**Valid Entry**
	1–12	Any valid city name assigned by BUYPASS personnel.
	13	Space
	14–15	Any valid alphabetical State Code. Please refer to Appendix D. Valid State Codes for a list of valid alphabetical State Codes.
	16	Space
	17–21	Any valid ZIP Code™/ZIP+4® assigned by BUYPASS personnel.

Number:	5	**Name:**	Approval Number
Character Type:	AN	**Maximum Length:**	6 bytes

Representation:	Fixed length of six alphanumeric characters; space-filled
Purpose:	Identifies the assigned Approval Number.
Processing Rules:	Required on reversals for matching information to the original purchase. If it is included in a Purchase Request, then the Purchase Request is considered preauthorized, and BUYPASS does not go out for authorization. In the case of an Authorization Only Reversal, this element—present in the original transaction—must be present and identical in the Authorization Only Reversal transaction. (**Note**: The Authorization Only Reversal transaction enables a merchant to reverse the following credit card transactions within required timeframes: POS Authorization Only [Transaction Type code "3"]); POS Mail/Phone Authorization Only [Transaction Type code "B"]); and CAT Authorization Only [Transaction Type code "5"]).
Valid Codes/Values:	Any alphanumeric representation, including spaces

Number:	6	**Name:**	Approved Amount
Character Type:	N	**Maximum Length:**	9 bytes

Representation:	Fixed length of nine digits with two assumed decimal places Zero-filled and right aligned
Purpose:	Identifies the amount approved by the issuer.
Processing Rules:	The Approved Amount may not be the same as the requested amount. If the transaction is run from an AFP device, the pump should be set to pump the Approved Amount—not the requested amount. If the transaction is a Partial Approval transaction, the amount approved is contained in this field, and the POS device must prompt for payment of the remaining balance due in the sale.
Valid Codes/Values:	000000001–999999999

Number:	7	**Name:**	Authorizer Code
Character Type:	N	**Maximum Length:**	2 bytes

Representation:	Fixed length of two digits
Purpose:	Identifies the authorizer.
Processing Rules:	BUYPASS defines this code.
Valid Codes/Values:	Please refer to Appendix C. Valid Authorizer Codes, for a list of valid entries.

Number:	8	**Name:**	Authorizer Response Code
Character Type:	AN	**Maximum Length:**	2 bytes

Representation:	Fixed length of two alphanumeric characters; space-filled
Purpose:	Identifies the response type.
Processing Rules:	This code comes from BUYPASS, which has obtained it from the authorizer.
Valid Codes/Values:	Any valid Authorizer Response Code, including spaces.

Number:	11	**Name:**	Block Number
Character Type:	N	**Maximum Length:**	3 bytes

Representation: Fixed length of three digits

Purpose: Identifies the specific proprietary or public key data block requested by a device or sent by the host.

Processing Rules: The host always uses a Block Number to track the data block it sends in an Electronic Mail response and a CA Public Key File Load response. If response processing is interrupted, the host knows the Block Number for the last complete data block sent in the response.

A device is not required to track the Block Number it receives in an Electronic Mail response. If it is required to do so, and processing between the host and device is interrupted, the device can reinstate transmission of the interrupted data block using that Block Number in the next Electronic Mail request it sends to the host.

A device always uses a Block Number to track the data block it sends in a Proprietary Data Load request. If request processing is interrupted, the host knows the Block Number for the last complete data block sent in the request.

A device is required to track the Block Number it receives in a CA Public Key File Load response. If CA Public Key File Load processing between the host and device is interrupted, the device reinstates transmission of the interrupted data block using that Block Number in the next request it sends to the host.

Valid Codes/Values:

Codes	Description
000	The host sends the first data block for CA Public Key File Load.
000	The device sends this value indicating the last block of site / fuel volume data is being sent in the Proprietary Data Load request.
001	The host restarts data block processing, and sends the block that was Block No. 1 in the Electronic Mail response as the next block.
<specific Block No.>	The device can request a particular block by sending that Block Number in the next Electronic Mail request or CA Public Key File Load request.

Number:	12	**Name:**	Card Discretionary Block Data
Character Type:	AN	**Maximum Length:**	51 bytes

Representation:	Variable length of up to 51 alphanumeric characters
Purpose:	Identifies the card's expiration date and/or discretionary data.
Processing Rules:	The format varies, depending on the application. If the Account Number is obtained from a Track 2 card reader, instead of being manually keyed, the Account Number and Card Discretionary Block Data should consist of all data read from the card between, but not including, the Start Sentinel and End Sentinel. Account Number is considered the data before the = character, and Card Discretionary Block Data is considered the data after the = character. The = character is used instead of a Field Separator. If the Account Number is obtained from a Track 1 card reader (credit cards only), instead of being manually keyed, the Account Number and Card Discretionary Block Data should consist of all unaltered data read from the card between, but not including, the Start Sentinel and End Sentinel. Account Number is considered the data before the first ^ character, and Card Discretionary Block Data is considered the data after the first ^ character. The ^ character is used instead of a Field Separator. Account Number and Card Discretionary Block Data combined do not exceed 76 characters for Track 1 or 37 characters for Track 2. **Note:** Track data placed in the combined data elements of Account Number and Card Discretionary Block Data cannot exceed the total length of those data elements For either TransArmor processing method, the expiration date must be included here. The expiration date is stored for follow-on transactions associated with a Token. **Real Time Account Updater:** When a merchant receives Account Updater Expiration Date (New Expiration Date) in the original transaction response in the Real Time Account Updater Response Segment – Segment 155, the merchant can send either the new expiration date or the original expiration date in the 'Follow-on' transactions in this Card Discretionary Block Data field. A merchant should send the Expiration Date corresponding to the original PAN or the new PAN / the original or the new TransArmor Token that is being sent in the cardholder Account Number field – Data Element 2 in the Standard Message Data Segment 100.

In the event of a 'Retry' of a 'Follow-on' transaction (Reversals, Completions, Cancellations, Void of Merchandise Return, and Timeout reversals) by the merchant, the merchant should use the same credentials for the retry as were used in the initial attempt.

Refer to Appendix AD – Real Time Account Updater for more details.

Valid Codes/Values:

For a card whose number has been manually entered, the expiration date is in MMYY format, and the other required data depends upon the application.

Note: For all authorizations, merchants must send a value even for expired cards.

For a Certegy (Formerly Telecredit) transaction, the information consists of the 2-character alphabetical State Code (See Appendix D. Valid State Codes.) and the 6-character birth date (MMDDYY).

If a driver's license number is used as identification for a SCAN (ETC) transaction, the information consists of the 2-character alphabetical State Code (See Appendix D. Valid State Codes.).

For either TransArmor processing method, the expiration date must be included here. The expiration date is stored for follow-on transactions associated with a Token.

For a Comdata Express Code or Check request (card type code 095/096), this field should be skipped. Field separators should be included as usual.

For a Money Code/Check request (card type code 093), this field should be skipped. Field separators should be included as usual.

For Accel and STAR: Expiration date is mandatory for e-commerce and one-time bill payment transactions. Effective May 1, 2025, expiration date is optional for Accel and STAR transactions, where the merchant retrieves the payment details from the merchant's internal customer database to submit the transaction. These credentials on file (COF) transactions include bill payment, recurring, and installment payment transactions.

Number: 13 Name: Card Label

Character Type: AN Maximum Length: 4 bytes

Representation: Fixed length of four alphanumeric characters; left-aligned and space-filled

Purpose: Identifies the card type in a Totals Response.

Processing Rules:

Valid Codes/Values:

Card Type	Code
MasterCard®/ Visa®	CC
Amex®	TE

Discover® Network, Diners Club®, JCB®	DS
Authorization only	AO
Debit	DB
Fleet	FL
Cash	CS
Proprietary	PR
Check	CK
EBT food stamp	EF
EBT cash	EC
Stored value activation/deactivation	SV1
Stored value purchase/capture /merchandise return	SV2
Stored value card replacement	SV3
Stored value recharge	SV4
ECA/ TeleCheck® service	ECA
EBT version of WIC	EWIC
EBT child care	EK

Number:	14	**Name:**	Card Type
Character Type:	AN	**Maximum Length:**	3 bytes

Note: See also the description for Element No. 78, Prompt Code.

Representation:	Fixed length of three alphanumeric characters
Purpose:	In the case of a Table Load response, this code identifies a card type the device accepts or a specific feature the device performs.. In the case of a financial transaction, this code identifies the card used in the transaction. In the case of a special transaction, this code identifies the type of requested data.
Processing Rules:	Must be a valid Card Type code. In the case of a Table Load response, the host sends this code multiple times to the device instructing the device about which cards to accept and which features to perform. In the case of a financial transaction, the device sends this code to the host as part of the 4-character Prompt Code. In the case of a special transaction, the device sends this code to the host as a 3-character Prompt Code requesting specific data.
Valid Codes/Values:	For a list of valid entries, please see Appendix E. Valid Card Type Codes.

Number:	15	**Name:**	Card Type Total Amount
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Fixed length of eight digits with two assumed decimal places
Purpose:	Identifies the total amount for a particular card type.
Processing Rules:	
Valid Codes/Values:	00000000–99999999

Number:	16	**Name:**	Card Type Total Count
Character Type:	N	**Maximum Length:**	5 bytes

Representation:	Fixed length of five digits
Purpose:	Identifies the transaction count for a particular card type in a Totals Response.
Processing Rules:	
Valid Codes/Values:	00001–99999

Number:	17	**Name:**	Cash Amount
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Variable length of up to eight digits with two assumed decimal places
Purpose:	Identifies the cash back amount of a transaction.
Processing Rules:	If a merchant chooses to support Interlink's Optional Cash Back Fee, this field includes the total of the cash back amount and the cash back fee.
Valid Codes/Values:	1–99999999

Number:	18	**Name:**	Clerk ID
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable amount of up to 10 digits
Purpose:	Identifies the clerk's identification number.
Processing Rules:	When identifying a Clerk ID for ECA/ TeleCheck® processing, please refer to Data Element No. 131, ECA/ TeleCheck® Clerk ID.
Valid Codes/Values:	1–9999999999

Number:	20	**Name:**	Currency Code
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of three digits
Purpose:	Identifies the currency type according to country.
Processing Rules:	If not sent in Transaction Request, the value defaults to *840* (United States currency). This field will be ignored during transaction processing. Due to totaling restrictions, all transactions for a given merchant will use the currency associated with that merchant.
Valid Codes/Values:	Please see Appendix L. Valid Currency Codes.

Number:	21	**Name:**	Current Date
Character Type:	N	**Maximum Length:**	6 bytes

Representation:	Fixed length of six digits (MMDDYY)	
Purpose:	Identifies the current date.	
Processing Rules:	Originates at BUYPASS.	
Valid Codes/Values:	**Code**	**Description**
	01–12	Month of the year
	01–31	Day of the month
	00–99	Last two digits of the year

Number:	22	**Name:**	Current Time
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of four digits (HHMM)	
Purpose:	Identifies the current time.	
Processing Rules:	Originates at BUYPASS. Includes adjustments for the time zone and daylight saving time.	
Valid Codes/Values:	**Code**	**Description**
	01–24	Hour of the day
	01–60	Minute of the hour

Number:	23	**Name:**	Cut Time
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length 4-digit number (HHMM)	
Purpose:	Identifies the merchant's end-of-day settlement time.	
Processing Rules:	For devices that use an automatic cut time, the device should start the settlement procedure 30 minutes prior to the merchant's cut time if the settlement has not been completed by the clerk for that business day.	
Valid Codes/Values:	**Code**	**Description**
	01–24	Hour of the day
	01–60	Minute of the hour

Number:	24	**Name:**	Data Type Indicator
Character Type:	AN	**Maximum Length:**	1 byte

Representation: Fixed length of one alphanumeric character

Purpose: Identifies specific characteristics of the following data in a host response.

Processing Rules: Required for TransArmor PKI Encryption and Tokenization.
Required for EMV CA Public Key File Load

Valid Codes/Values:

Code	Description
#	Identifies the Merchant Data Segment (No. DL1).
!	Identifies the Dial String Data Segment (No. DL2).
:	Identifies the Date and Time Data Segment (No. DL3).
@	Identifies the Software Dial Load Data Segment (No. DL4).
$	Identifies the Software IP Load Data Segment (No. DL5).
\	Identifies the Store and Forward Data Segment (No. DL6).
&	Indicates that a new Key and Key ID are being returned to a device using TransArmor PKI Encryption and Tokenization.
&	For EMV, indicates that a block of a new CA Public Key file is being returned to the device.
K	Indicates that a new Key and Key ID are being returned to a device using TransArmor PKI Encryption and Tokenization
^	Identifies the Supplemental Terminal Data Segment (No. DL7)
%	Identifies EMV Floor Limits Segment (No. DL8)

Number:	25	**Name:**	Day of the Week
Character Type:	N	**Maximum Length:**	1 byte

Representation: Fixed length of one digit

Purpose: Identifies the current day of the week.

Processing Rules: Originates at BUYPASS.

Valid Codes/Values:

Code	Description

0	Sunday
1	Monday
2	Tuesday
3	Wednesday
4	Thursday
5	Friday
6	Saturday

Number: 26 **Name:** Decline Code
Character Type: AN **Maximum Length:** 2 bytes

Representation: Fixed length of two alphanumeric characters

Purpose: Identifies the reason for the decline.

Processing Rules: BUYPASS defines this code.

For some Decline Codes, a device can be programmed to display a decline message prompting intervention by a clerk/customer.

For the other Decline Codes, a device displays a decline message and completes the transaction.

For a TransArmor Token Only transaction that has been declined because BUYPASS cannot retrieve the necessary token for the associated PAN, please refer to the appendix indicated below for additional information.

Valid Codes/Values: Any valid Decline Code. For a list of valid Decline Codes permitting clerk/customer intervention following a decline, please refer to Appendix H. Decline Codes Permitting Clerk/Customer Intervention.

Number: 27 **Name:** Dial String Terminator
Character Type: AN **Maximum Length:** 1 byte

Representation: Fixed length of one alphanumeric character

Purpose: Indicates the end of a dial string.

Processing Rules:

Valid Codes/Values:

Value	Description
A	Indicates the end of the first dial string in the Dial String Data Segment (Data Segment No. DL2).
F	Indicates the end of the second dial string in the Dial String Data Segment (Data Segment No. DL2).

Number:	28	**Name:**	Dial String Type
Character Type:	N	**Maximum Length:**	1 byte

Representation:	Fixed length of one digit
Purpose:	Indicates transaction dial strings.
Processing Rules:	Fixed value: *1*
Valid Codes/Values:	**Value** — **Description** 1 — Transaction dial strings

Number:	29	**Name:**	Direct Marketing Invoice Number
Character Type:	AN	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 alphanumeric characters
Purpose:	Identifies the invoice number in a Direct Marketing/Address Verification System Request.
Processing Rules:	
Valid Codes/Values:	

Number:	30	**Name:**	Download Indicator
Character Type:	N	**Maximum Length:**	1 byte

Representation:	Fixed length of one digit
Purpose:	Indicates whether the device should request a load; and, if so, what type.
Processing Rules:	BUYPASS leaves communications established so that a Load Request can be initiated. Required for TransArmor PKI Encryption and Tokenization. This value indicates the device must update its Key ID and Key because the current Key ID is due to expire or has expired. Required for EMV Public Key loads. This value indicates that the device must update its CA Public Key File because the checksum in the EMV Financial Transaction Request does not match the checksum value at the host.
Valid Codes/Values:	**Code** — **Description** 0 — No Download Request.

1	Indicates device should request a partial load.
8	Indicates the device must update its Key ID and Key
9	Indicates that the device can update its Public Key

Number:	31	**Name:**	Driver/Identification Number
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 digits
Purpose:	Used for fleet cards that require an unencrypted Driver/Identification Number.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by the issuer, this number must be an unencrypted Driver/Identification Number. If required by the issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any numeric representation

Number:	32	**Name:**	Employee Number
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of four digits
Purpose:	Identifies the employee who performs a utility transaction. **Note:** Used for reporting purposes only.
Processing Rules:	
Valid Codes/Values:	Valid employee numbers. **Note:** Use *1111* if no employee numbers are assigned.

Number:	33	**Name:**	Encrypted PIN Block Data
Character Type:	AN	**Maximum Length:**	36 bytes

Representation:	For DUKPT PIN encryption, this is a 16- to 20-byte Key Serial Number (KSN) plus a 16-byte PIN block.
Purpose:	Required for debit and EBT. It is not required for a completion. **Note:** Refer to chapter 6, "PIN Encryption," for composition of PIN block.
Processing Rules:	Customer-keyed, must be encrypted using DUKPT. If not included in a request, insert a Field Separator. For EMV offline PIN validated debit transactions, the 16-byte PIN block portion of this data element contains either a PIN or all F's, otherwise the transaction will decline as "invalid transaction".
Valid Codes/Values:	PIN block composition is validated by hardware PIN encryption devices. If composition is invalid or the device is out-of-sync with BUYPASS, the transaction is declined.

Number:	34	**Name:**	End-of-Data Indicator
Character Type:	A	**Maximum Length:**	1 byte

Representation:	Fixed length of one alpha character
Purpose:	Indicates the end of a data segment.
Processing Rules:	It is used in the following data segments: • Merchant Data Segment (No. DL1) • Dial String Data Segment (No. DL2) • Date and Time Data Segment (No. DL3) • Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)
Valid Codes/Values:	~

Number:	35	**Name:**	End-of-Load Indicator
Character Type:	A	**Maximum Length:**	1 byte

Representation:	Fixed length of one alpha character
Purpose:	Indicates the end of a Table Load.
Processing Rules:	It is used in the Table Load Response.
Valid Codes/Values:	*

Number:	36	**Name:**	Extract Date
Character Type:	N	**Maximum Length:**	6 bytes

Representation:	Fixed length of six digits (MMDDYY)
Purpose:	Identifies the date the information was extracted for an Electronic Mail Data Segment.
Processing Rules:	
Valid Codes/Values:	Any valid date

Number:	37	**Name:**	Extract Time
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of four digits (HHMM)
Purpose:	Identifies the time the information was extracted for an Electronic Mail Data Segment.
Processing Rules:	
Valid Codes/Values:	Any valid time

Number:	38	**Name:**	Fee Amount
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Fixed length of eight digits with two assumed decimal places
Purpose:	Identifies the fee amount calculated by BUYPASS.
Processing Rules:	
Valid Codes/Values:	00000000–99999999

Number:	39	**Name:**	Firmware Version
Character Type:	AN	**Maximum Length:**	8 bytes

Representation:	Fixed length of eight alphanumeric characters
Purpose:	Identifies the device's firmware version at the customer location.
Processing Rules:	Required for Totals Requests, Table Load Request, and TransArmor PKI Encryption and Tokenization Load Request.
Valid Codes/Values:	

Number:	40	**Name:**	Fleet Employee Number
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 digits
Purpose:	Used for fleet cards that require a Fleet Employee Number.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by the issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any valid Fleet Employee Number

Number:	41	**Name:**	Fuel Purchase Amount
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Variable length of up to eight digits with two assumed decimal places **Note:** A maximum length of 8 bytes is allowed for American Express cards only. A maximum length of 7 bytes is allowed for all other card types.
Purpose:	Identifies the total dollar amount of all fuel sold for this transaction.
Processing Rules:	Included on any transactions with fuel sales. For a transaction involving only a fuel purchase and discounts and/or coupons, this amount is the net amount derived by subtracting the discount and/or coupon amount from the sum of fuel products in Product Amount (No. 76). For a transaction involving fuel and nonfuel amounts together with discounts and/or coupons, the discounts and/or coupons are first applied to the nonfuel amount.
Valid Codes/Values:	1–99999999

Number:	42	**Name:**	Grand Total
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Fixed length of eight digits with two assumed decimal places
Purpose:	Identifies the final total amount in a Totals Response.
Processing Rules:	
Valid Codes/Values:	00000000–99999999

Number:	43	**Name:**	Hardware Version
Character Type:	AN	**Maximum Length:**	4/8 bytes

Representation:	Fixed length of four or eight alphanumeric characters
Purpose:	Identifies the device's hardware version.
Processing Rules:	Required for Totals Requests, Table Load Request, and TransArmor PKI Encryption and Tokenization Load Request. For Total Requests, some devices send 8 bytes of length.
Valid Codes/Values:	Any valid hardware version number

Number:	44	**Name:**	Information Byte
Character Type:	AN	**Maximum Length:**	1 byte

Representation:	Fixed length of one alphanumeric character	
Purpose:	Identifies the type of request	
Processing Rules:	Download Request signifies type of special load. Single-message Request signifies one transaction is performed, and then communication is discontinued. Multimessage Request signifies that after a transaction has been processed, the communication line remains established for further transactions.	
Valid Codes/Values:	**Character**	**Description**
	?	Download Request
	0	Single-message transmission
	1	Multimessage transmission

Number:	45	**Name:**	Initiation Date
Character Type:	N	**Maximum Length:**	6 bytes

Representation:	Fixed length of six digits (MMDDYY)
Purpose:	Identifies the local date that the transaction was performed.
Processing Rules:	Calculated by BUYPASS using the BUYPASS system clock, the merchant profile time zone, and daylight saving time flag.
Valid Codes/Values:	

Number:	46	**Name:**	Initiation Time
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of four digits (HHMM)
Purpose:	Identifies the local time that the transaction was performed.
Processing Rules:	Calculated by BUYPASS using the BUYPASS system clock, the merchant profile time zone, and daylight saving time flag.
Valid Codes/Values:	

Number:	47	**Name:**	Job Number
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 digits
Purpose:	Identifies the Job Number.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by the issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any numeric representation

Number:	48	**Name:**	Load Type
Character Type:	A	**Maximum Length:**	1 byte

Representation: Fixed length of one alpha character

Purpose: Identifies the load type.

For TransArmor PKI Encryption and Tokenization, it indicates that Key information is requested in the Key and Key ID request.

For EMV, it indicates that CA Public Key File information is requested in the CA Public Key File Load Request.

Processing Rules: Required for TransArmor PKI Encryption and Tokenization.

Required for EMV Public Key loads.

Valid Codes/Values:

Codes	Description
P	Partial Load (Phone, Table)
D	Date and Time Load
K	TransArmor PKI Encryption and Tokenization Load Unsigned Key or Signed Key
K	CA Public Key File Load **Note:** BUYPASS Device Type must be "+*" for the CA Public Key File Load Request.
S	Signing Key ID

Number:	49	**Name:**	Local Date and Local Time
Character Type:	N	**Maximum Length:**	10 bytes

Representation: Fixed length of up to 10 digits

Purpose: Identifies the local date (MMDDYY) and local time (HHMM) of a preauthorized transaction.

Processing Rules: The local date and local time is only sent in a POS and CAT Capture transaction. Merchant should also send terminal local date and time for Debit Cancellation and Time out Reversal transactions. Debit Cancellation are handled as TORs so unless Merchant send local date and time to BUYPASS, BUYPASS cannot send the local date and time to associations.

Valid Codes/Values:

Number:	50	**Name:**	Local Time
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of four digits (HHMM)
Purpose:	Identifies the time of an Electronic Mail Data Segment.
Processing Rules:	
Valid Codes/Values:	Any valid time

Number:	51	**Name:**	Mail Text Data
Character Type:	AN	**Maximum Length:**	750 bytes

Representation:	For an IP response: A variable length of up to 750 alphanumeric characters For a dial up response: A variable length of up to 150 alphanumeric characters
Purpose:	Identifies the data in the Electronic Mail Response.
Processing Rules:	
Valid Codes/Values:	

Number:	52	**Name:**	Mail Text Data Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of three digits; zero-filled and right aligned
Purpose:	Identifies the length of Mail Text Data.
Processing Rules:	
Valid Codes/Values:	For an IP response: 001–750 For a dial up response: 001–150

Number:	53	**Name:**	Merchant Name
Character Type:	AN	**Maximum Length:**	24 bytes

Representation:	Fixed length of up to 24 alphanumeric characters
Purpose:	Identifies the merchant associated with the transmitting device location.
Processing Rules:	
Valid Codes/Values:	

Number:	54	**Name:**	Merchant Phone Number
Character Type:	AN	**Maximum Length:**	13 bytes

Representation:	Fixed length of 13 alphanumeric characters
Purpose:	Identifies the telephone number [(nnn)nnn-nnnn] of the transmitting device location.
Processing Rules:	Used in Data Segment No. DL1, Merchant Data Segment.
Valid Codes/Values:	Any valid telephone number

Number:	55	**Name:**	Message Format Version Identifier
Character Type:	AN	**Maximum Length:**	6 bytes

Representation: Fixed length of six alphanumeric characters

Purpose: Identifies the type and release version of message format.

Processing Rules: All Financial Transaction Request messages, Totals Request messages, Loyalty Card Request messages, and Electronic Mail Request messages contain this data element in Field No. 1. It originates at the device.

For Financial Transaction Requests, the following data element and data segment always follow this data element:

- Element No. 63, Number of Segments
- Data Segment No. 100, Standard Message Data Segment

plus none, one, or more of the following data segments:

- Data Segment No. 101, Fleet Data Segment
- Data Segment No. 102, Product Code Data Segment
- Data Segment No. 103, EBT Data Segment
- Data Segment No. 104, Purchase Card Data Segment
- Data Segment No. 111, Variable Information Data Segment

For Financial Transaction Responses, the following data segment can follow the last data element in the message:

- Additional Information Data Segment (Data Segment No. 112)

For Totals Request messages, the following data element and data segment always follow this data element;

- Element No. 63, Number of Segments
- Data Segment No. 105, Totals Data Segment

For Loyalty Card Request messages, the following data element and data segment always follow this data element;

- Element No. 63, Number of Segments
- Data Segment No. 108, Loyalty Card Data Segment

For Electronic Mail Request messages, the following data element and data segment always follow this data element;

- Element No. 63, Number of Segments
- Data Segment No. 109, Electronic Mail Data Segment

For TransArmor PKI Encryption and Tokenization Load Request messages, the following data element and data segment always follow this data element

- Element No. 63, Number of Segments
- Data Segment No. 116, TransArmor Load Data Segment

Valid Codes/Values:

Value	Description
ATL105	Device/Controller (default)

Number: 56 **Name:** Net Amount
Character Type: N **Maximum Length:** 8 bytes

Representation: Fixed length of eight digits with two assumed decimal places
Purpose: Identifies the difference between Grand Total and Fee Amount in a Totals Response.
Processing Rules:
Valid Codes/Values: 00000000–99999999

Number: 57 **Name:** New Software Version
Character Type: AN **Maximum Length:** 8 bytes

Representation: Fixed length of eight alphanumeric characters
Purpose: Identifies the new software application version number.
Processing Rules: Used in the following data segments:

- Software Dial Load Data Segment (No. DL4)
- Software IP Load Data Segment (No. DL5)

Valid Codes/Values:

Number:	58	**Name:**	Nonfuel Amount
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Variable length of up to eight digits with two assumed decimal places **Note:** A maximum length of 8 bytes is allowed for American Express cards only. A maximum length of 7 bytes is allowed for all other card types.
Purpose:	Identifies the transaction's dollar amount that is not a fuel, tax, or cash amount.
Processing Rules:	For a stored value card activation transaction, this data contains the activation amount. For a stored value card deactivation transaction, this data contains the deactivation amount**.** For a stored value card recharge transaction, this data contains the recharge amount**.** For a transaction involving only a nonfuel purchase and discounts and/or coupons, this amount is the net amount derived by subtracting the discount and/or coupon amount from the sum of nonfuel products in Product Amount (No. 76). For a transaction involving fuel and nonfuel amounts together with discounts and/or coupons, the discounts and/or coupons are first applied to the nonfuel amount.
Valid Codes/Values:	1–99999999

Number:	59	**Name:**	Number of Card Types
Character Type:	N	**Maximum Length:**	2 bytes

Representation:	Fixed length of up to 2 digits
Purpose:	Identifies the number of times that Field No. 8 is repeated in the Merchant Data Segment. It indicates the number of card types processed by the transmitting device.
Processing Rules:	
Valid Codes/Values:	01–99

Number:	61	**Name:**	Number of Print Lines
Character Type:	N	**Maximum Length:**	1 byte

Representation: Fixed length of one digit

Purpose: Indicates the number of times Terminal Display/Printer Message is in the response.

Note: This count does not include Data Element No. 168, Count of Receipt Text Line which is considered additional Custom Receipt Text. The count represented in Data Element No 168 does not reflect the total from Data Element No 61, Number of Print Lines.

Processing Rules: This information comes from BUYPASS.

Valid Codes/Values:

Value	Description
1	Indicates that Terminal Display/Printer Message appears one time in the Financial Transaction Response.
2	Indicates that Terminal Display/Printer Message appears two times in the Financial Transaction Response.
3	Indicates that Terminal Display/Printer Message appears three times in the Financial Transaction Response.

Number:	62	**Name:**	Number of Products
Character Type:	N	**Maximum Length:**	2 bytes

Representation: Fixed length of two digits

Purpose: Identifies the count of products reported for a transaction.

Processing Rules: Always precede single digits (1–9) with a zero (01–09).

Valid Codes/Values: 01–10

Number:	63	**Name:**	Number of Segments
Character Type:	N	**Maximum Length:**	2 bytes

Representation: Variable length of up to two digits

Purpose: Identifies the number of BUYPASS-defined data segments included in the Financial Transaction Request, Totals Request, Loyalty Card Transaction Request, and Electronic Mail Request.

Processing Rules: It originates at the device.

Financial Transaction Requests: This data element is always followed by Data Section No. 2 which contains Data Segment No. 100, Standard Message Data Segment, plus Data Section No. 3 which contains none, one, or more of the of the input data segments listed in Chapter 12.

Loyalty Card Transaction Requests: This data element is always followed by Data Section No. 2 which contains Data Segment No. 100, Standard Message Data Segment, plus Data Section No. 3 which contains Data Segment No. 108, Loyalty Card Data Segment, and Data Segment 114, SKU Data Segment.

ECA/ TeleCheck® Service Transaction Requests: This data element is always followed by Data Section No. 2 which contains Data Segment No. 100, Standard Message Data Segment, plus Data Section No. 3 which contains Data Segment No. 110, Check Data Segment, and Data Segment 111, Variable Information Data Segment.

Totals Requests: This data element is always followed by Data Segment No. 105, Totals Data Segment.

Electronic Mail Requests: This data element is always followed by Data Section No. 2, which contains Data Segment No. 109, Electronic Mail Data Segment.

Communications Test Requests: This data element is always followed by Data Section No. 2 which contains Element No. 120, Network Management Message.

TransArmor PKI Encryption and Tokenization: This data element is used for a Key and Key ID Load. In the requests, it is always followed by Data Section No. 2, which contains Data Segment No. 116, TransArmor Load Data Segment.

Valid Codes/Values: The value of this field, call it N, indicates that the transaction will have the 100 segment + N-1 of the input segments listed in Chapter 12.

Number:	64	**Name:**	Odometer
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Variable length of one to eight digits
Purpose:	Identifies the odometer reading keyed in by the customer or clerk.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by the issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	1–99999999

Number:	65	**Name:**	Password
Character Type:	N	**Maximum Length:**	6 bytes

Representation:	Variable length of up to six digits
Purpose:	Identifies the password entered during the end-of-day function.
Processing Rules:	Password required for totals request or electronic mail request. This is matched against the password defined in the merchant's profile at BUYPASS. If entry is less than six characters long, right-align the entry using spaces.
Valid Codes/Values:	Default password: 123456.

Number:	66	**Name:**	Pause Indicator
Character Type:	AN	**Maximum Length:**	1 byte

Representation:	Fixed length of one alphanumeric character
Purpose:	Used for telephone systems that are slow to provide a dial tone.
Processing Rules:	Indicates that the device should pause approximately one second before proceeding. Differs from loader to loader.

	A device using a modem that interprets the standard Hayes command set would replace the *B* character with a comma in the dial string.
Valid Codes/Values:	B

Number:	72	**Name:**	PC Duty Amount
Character Type:	N	**Maximum Length:**	7 bytes

Representation:	Variable length of up to seven digits with two assumed decimal places
Purpose:	Identifies the duty amount in a Purchase Card Transaction Request.
Processing Rules:	Used for Purchase Card Transaction Request.
Valid Codes/Values:	

Number:	73	**Name:**	PC Freight Amount
Character Type:	N	**Maximum Length:**	7 bytes

Representation:	Variable length of up to seven digits with two assumed decimal places
Purpose:	Identifies the freight charges included in a Direct Marketing/Address Verification System Request.
Processing Rules:	Used in Direct Marketing/Address Verification System Request.
Valid Codes/Values:	

Number:	74	**Name:**	PC Tax Amount
Character Type:	N	**Maximum Length:**	7 bytes

Representation:	Variable length of up to seven digits with two assumed decimal places
Purpose:	Identifies the transaction tax amount in Data Segment No. 104, Purchase Card Data Segment.
Processing Rules:	Note the difference between this data element and Element No. 99, Tax Amount.
Valid Codes/Values:	

Number:	75	**Name:**	Phone Number
Character Type:	N	**Maximum Length:**	18 bytes

Representation:	Variable length of up to 18 digits
Purpose:	Identifies the primary, secondary, or tertiary (assigned, but not used) Phone Number used in transaction dialing.
Processing Rules:	Transaction dialing always begins with the primary Phone Number. The secondary Phone Number is used once attempts using the primary Phone Number are exhausted. A tertiary Phone Number is assigned to but not used by the device. Consists of all data up to the following *C* (Log-on Indicator) or *A* (Delimiter) character.
Valid Codes/Values:	

Number:	76	**Name:**	Product Amount
Character Type:	N	**Maximum Length:**	12 bytes

Representation:	Variable length of up to 12 digits with two assumed decimal places
Purpose:	Identifies the monetary value of product purchased in Data Segment No. 102, Product Code Data Segment or Data Segment No. 157, Adjusted Product Code Data Segment.
Processing Rules:	The decimal point is implied by the optional Currency Code. The default value has two assumed decimal places. This data element, along with the following data elements, is repeated for up to a maximum of 10 products in Data Segment No. 102 or Data Segment No. 157: • Product Code (No. 77) • Unit of Measure (No. 106) • Quantity (No. 81) • Unit Price (No. 107)
Valid Codes/Values:	1–999999999999

Number:	77	**Name:**	Product Code
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of three digits
Purpose:	Identifies the type of product in Data Segment No. 102, Product Code Data Segment or Data Segment No. 157, Adjusted Product Code Data Segment. In the case of a Proprietary Load Response (Prompt Code 904), the first instance identifies the type of product which is eligible for discount, and the second instance identifies the corresponding discount product code to send in Data Segment No. 102, Product Code Data Segment.
Processing Rules:	This data element, along with the following data elements, is repeated for up to a maximum of 10 products in Data Segment No. 102 or Data Segment No. 157: • Unit of Measure (No. 106) • Quantity (No. 81) • Unit Price (No. 107) • Product Amount (No. 76) **Note:** A unique Product Code must be sent for each type of fuel purchase. When a device uses a Dynamic Card Table, only the product codes defined in the table are valid for transaction processing. In the case of a Proprietary Load Response (Host Discount Data, Prompt Code 904), the value in the second or third positions of the Product Code may contain a wild card character of *. If that occurs, any digit is allowed in that position. For example, if the value is 01*, product codes 010, 011, 012, 013, 014, 015, 016, 017, 018 and 019 are all available for discount.
Valid Codes/Values:	Please refer to Appendix F. Valid Payment Systems Product Codes, for lists of valid codes.

Number:	78	**Name:**	Prompt Code
Character Type:	AN	**Maximum Length:**	4 bytes

Note: See also the description for Element No. 14, Card Type.

Representation:	Variable length of three or four alphanumeric characters
Purpose:	In the case of a financial transaction, this is a 4-character Prompt Code that identifies the transaction's Transaction Type (one character) and Card Type (three characters).

	In the case of a special transaction, this is a 3-character Prompt Code that identifies requested special data (3-character Card Type).
Processing Rules:	Must be a valid Transaction Type code and/or Card Type code. In the case of a financial transaction, a device transmits a 4-character Prompt Code when the transaction is initiated. In the case of a special transaction, a device transmits a 3-character Prompt Code (3-character Card Type) for a special transaction when the transaction is initiated. In the case of an Authorization Only Reversal, this element—present in the original transaction—must be present and identical in the Authorization Only Reversal transaction. (**Note**: The Authorization Only Reversal transaction enables a merchant to reverse the following credit card transactions within required timeframes: POS Authorization Only [Transaction Type code "3"]); POS Mail/Phone Authorization Only [Transaction Type code "B"]); and CAT Authorization Only [Transaction Type code "5"]). Signature Debit eligible transactions must be sent with a Credit Prompt Code, and NOT a Debit Prompt Code. In EMV transactions, if 'D' is returned in response, then subsequent transactions i.e. completion or void or reversal must be sent as credit. This will be reflected in the Prompt Code (Data Segment 100: Data Element 78) of the subsequent transaction. In EMV transactions, if 'S' is returned in response, then subsequent transactions i.e. completion or void or reversal must be sent as debit. This will be reflected in the Prompt Code (Data Segment 100: Data Element 78) of the subsequent transaction.
Valid Codes/Values:	In the case of a financial transaction, position no. 1 is the Transaction Type (For a list of valid entries, please see Appendix G. Valid Transaction Type Codes.); and position nos. 2–4 are the Card Type (For a list of valid entries, please see Appendix E. Valid Card Type Codes.). In the case of a special transaction, position nos. 1–3 are the Card Type (For a list of valid entries, please see Appendix E. Valid Card Type Codes.) that identifies requested special data.

Number: 79 Name: Pump/Lane Number

Character Type: N Maximum Length: 2 bytes

Representation:	Variable length of up to two digits
Purpose:	Identifies the pump or lane number at the point-of-sale location.

Processing Rules:	Required on all fuel transactions. If Data Element 2 (Account Number) EDATA Identifier is of value 7, then this data element will be used as pump/lane number for decryption purposes only.
Valid Codes/Values:	1–99

Number:	80	**Name:**	Purchase Code
Character Type:	AN	**Maximum Length:**	16 bytes

Representation:	Variable length of up to 16 alphanumeric characters
Purpose:	Identifies the Purchase Code associated with the transaction.
Processing Rules:	
Valid Codes/Values:	

Number:	81	**Name:**	Quantity
Character Type:	N	**Maximum Length:**	9 bytes

Representation:	Variable length of up to nine digits with a maximum of three assumed decimal places
Purpose:	Identifies the number of product units sold in Data Segment No. 102, Product Code Data Segment or Data Segment No. 157, Adjusted Product Code Data Segment.
Processing Rules:	The leading number indicates the number of assumed decimal places. There is a maximum number of three assumed decimal places. This data element, along with the following data elements, is repeated for up to 10 products in Data Segment No. 102, Product Code Data Segment or Data Segment No. 157, Adjusted Product Code Data Segment: • Product Code (No. 77) • Unit of Measure (No. 106) • Unit Price (No. 107) • Product Amount (No. 76)
Valid Codes/Values:	00000000. 01–399999999 **Examples:** 010 = 10, where *0* is the number of assumed decimal places and *10* represents the quantity of 10. 34170 = 4.170, where *3* is the number of assumed decimal places and *4170* represents the quantity of 4.170. For an example of this data element, please see Appendix B. POS Purchase Example.

Number:	82	**Name:**	Redial Count
Character Type:	N	**Maximum Length:**	1 byte

Representation:	Fixed length of one digit
Purpose:	Identifies the number of times the subsequent phone number in Data Segment No. DL2, Dial String Data Segment, can be redialed.
Processing Rules:	
Valid Codes/Values:	1–3

Number:	83	**Name:**	Response Code
Character Type:	AN	**Maximum Length:**	1 byte

Representation:	Fixed length of one alphanumeric character
Purpose:	For financial transactions, it identifies an approval with purchase/capture, an approval without purchase/capture, or a decline. For totals, it identifies an approval, an approval with electronic mail retrieval pending, decline with electronic mail retrieval pending, approved with electronic mail retrieval pending and proprietary data pending, declined with electronic mail retrieval pending and proprietary data pending, approved with proprietary data pending, or declined with proprietary data pending. For electronic mail, it identifies an approval or an approval with more mail pending. For communications test, it identifies an approval. For a TransArmor PKI Encryption and Tokenization transaction, it identifies one of the following: • The transaction is an approved TransArmor PKI Encryption and Tokenization transaction. • The transaction is a rejected TransArmor PKI Encryption and Tokenization transaction. • The merchant is not a TransArmor merchant. For an EMV transaction, it identifies one of the following: • The transaction is an approved EMV transaction. • The transaction is a rejected EMV transaction.
Processing Rules:	

Valid Codes/Values:	**Transaction Code**	**Description**
	0	Approved—Purchase/Capture
	1	Approved—Communications Test
		Declined—All other transactions
	2	Approved—Authorization only
	3	Approved—Authorization only with AVS
	4	Approved—Purchase/Capture with AVS
	5	Approved—Totals
	6	Approved—Totals with electronic mail retrieval pending
	7	Declined—Totals with electronic mail retrieval pending
	8	Approved—Electronic mail
	9	Approved—Electronic mail and more mail pending
	B	Approved—Totals with electronic mail retrieval pending and proprietary data pending
	C	Declined—Totals with electronic mail retrieval pending and proprietary data pending
	D	Approved—Totals with proprietary data pending
	E	Declined—Totals with proprietary data pending
	F	Approved—Partial approval
	G	Approved—Electronic mail and end of proprietary data block
	H	Approved—Proprietary data retrieval, more pending
	J	Approved—Electronic Mail, reset
	K	Approved—TransArmor Load
	L	Rejected for one of the following reasons— 1) Merchant is not TransArmor enabled 2) Signing Key is invalid or outdated
	L	Approved—EMV Key Load, last block
	M	Approved—EMV Key Load, more pending
	M	Declined—Totals with Proprietary Host Discount data pending
	N	Approved—Totals with Proprietary Host Discount data pending

O	Approved—Proprietary data load, more pending
P	Successful Signing/Signed Key Load Request
S	Declined—Retry the transaction.
T	Approved—Proprietary data load, no more data pending
U	Declined—Proprietary data load, no more data pending
V	Declined—Totals with Proprietary Custom Receipt Text data pending
W	Approved—Totals with Proprietary Custom Receipt Text pending
X	Declined—Proprietary data load, proceed to next pending Prompt Code
X	Not Required/Rejected—EMV Key Load, or the checksum in the request matches the host
Y	Approved—Proprietary data load, proceed to next pending Prompt Code

Note: In some instances, proprietary data can be downloaded.

If a TransArmor Key Load (Signing/Signed/Unsigned) request is declined, the last element in the message will be Data Element 83 (Response Code).

Number:	84	**Name:**	Segment Length
Character Type:	N	**Maximum Length:**	4 bytes

Representation: Variable length of three or four digits

Note: Segment Length has a length of four digits in the following seven instances only:

- EBT Data Segment (No. 103)
- SKU Data Segment (No. 114)
- Print Data Segment (No. 115)
- Proprietary Data Load Segment (No. 118)
- Print Data 2 Segment (No. 120)
- EMV Request Data Segment (No.130)
- EMV Response Data Segment (No. 131)

Purpose: Identifies the length of the segment in which it appears.

Processing Rules: This figure is calculated using the lengths of all data elements in the segment plus all Field Separators/Product Data Field Delimiters related to those elements. (Please refer to chapter 12, “Data Segment Formats,” for additional information about specific data segments—including their composition, data element lengths, Field Separators, and Product Data Field Delimiters.)

Note: Segment Length can only have a length of 4 digits when transmitting the following Data Segments. It cannot have a length of 4 digits when sending any other segment.

- EBT Data Segment (No. 103)
- SKU Data Segment (No. 114)
- Print Data Segment (No. 115)
- Proprietary Data Load Segment (No. 118)
- Print Data 2 Segment (No. 120)
- EMV Request Data Segment (No.130)
- EMV Response Data Segment (No. 131)

Valid Codes/Values:	**Data Length**	**Data Segment**
	001–218	Standard Message Data Segment (No. 100)
	001–61	Fleet Data Segment (No. 101)
	001–381	Product Code Data Segment (No. 102)
	001–3334	EBT Data Segment (No. 103)
	001–86	Purchase Card Data Segment (No. 104)
	001–409	Totals Data Segment (No. 105)
	001–142	Loyalty Card Data Segment (No. 108)
	001–232	Electronic Mail Data Segment (No. 109)
	001–168	Check Data Segment (No. 110)
	001–999	Variable Information Data Segment (No. 111)
	001–999	Additional Information Data Segment (No. 112)
	001–156	ECA/ TeleCheck® Data Segment (No. 113)
	0001–1010	SKU Data Segment (No. 114)
	001-1009	Print Data Segment (No. 115)
	01–50	TransArmor Load Data Segment (No. 116)
	0001-3800	Proprietary Data Load Segment (No. 118)
	001-493	Totals with Proprietary Data Load Data Segment (No. 119)
	001-1009	Print Data 2 Segment (No. 120)
	001-186	NFC Payment Tokenization Data Segment (No. 123)
	001-3043	EMV Request Data Segment (No.130)
	001-3834	EMV Response Data Segment (No. 131)
	01-77	CA Public Key File Data Segment (No. 132)
	01-19	Transaction Attributes Data Segment (No. 134)
	001–381	Adjusted Product Code Data Segment (No. 157)

Number:	85	**Name:**	Segment Type
Character Type:	N	**Maximum Length:**	3 bytes

Representation: Fixed length of three digits

Purpose: Identifies the type of segment being formatted in a Financial Transaction request, a Totals request, a Loyalty Card transaction request, Electronic Mail request, or a TransArmor PKI Encryption and Tokenization Load Request.

For TransArmor PKI Encryption and Tokenization, this value indicates that the segment contains Key and Key ID information.

Processing Rules: Required for the TransArmor PKI Encryption and Tokenization Load Request.

Valid Codes/Values:

Code	Description
100	Data Segment No. 100, Standard Message Data Segment
101	Data Segment No. 101, Fleet Data Segment
102	Data Segment No. 102, Product Code Data Segment
103	Data Segment No. 103, EBT Data Segment
104	Data Segment No. 104, Purchase Card Data Segment
105	Data Segment No. 105, Totals Data Segment
106	Data Segment No. 106 is reserved for proprietary use.
107	Data Segment No. 107 is reserved for proprietary use.
108	Data Segment No. 108, Loyalty Card Data Segment
109	Data Segment No. 109, Electronic Mail Data Segment
111	Data Segment No. 111, Variable Information Data Segment
112	Data Segment No. 112, Additional Information Data Segment
116	Data Segment No. 116, TransArmor Load Data Segment
118	Data Segment No 118, Proprietary Data Load Segment
119	Data Segment No. 119, Totals with Proprietary Data Load Load Data Segment
120	Data Segment No. 120, Print Data 2 Segment
123	Data Segment No. 123, NFC Payment Tokenization Data Segment
130	Data Segment No. 130, EMV Request Data Segment
131	Data Segment No. 131, EMV Response Data Segment
132	Data Segment No. 132, CA Public Key File Segment
134	Data Segment No. 134, Transaction Attributes Data Segment

157 Data Segment No. 157, Adjusted Product Code Data Segment
DL1 Data Segment No. DL1, Merchant Data Segment.
DL2 Data Segment No. DL2, Dial String Data Segment
DL3 Data Segment No. DL3, Date and Time Data Segment
DL4 Data Segment No. DL4, Software Dial Load Data Segment
DL5 Data Segment No. DL5, Software IP Load Data Segment
DL6 Data Segment No.DL6, Store and Forward Data Segment

Number:	86	**Name:**	Sequence Number
Character Type:	N	**Maximum Length:**	6 bytes

Representation: Fixed length of six digits

Purpose: **Note:** Identifies a transaction for the life of the transaction. Sequence Number must persist in all the subsequent transactions that are associated with the initial authorization request.

In the case of a preauthorized transaction, the Sequence Number in the Purchase/Capture identifies the Sequence Number in the transaction's Authorization Only request.

Processing Rules: Each transaction has a unique Sequence Number. It is included on all transaction types, and originates at the device.

In the case of a preauthorized transaction, the Sequence Number in the Purchase/Capture is the same Sequence Number as in the transaction's Authorization Only request.

In the case of a purchase reversal/void or time-out reversal transaction, the Sequence Number must be the same as in the original transaction request.

In the case of an Authorization Only Reversal, this element—present in the original transaction—must be present and identical in the Authorization Only Reversal transaction. (**Note**: The Authorization Only Reversal transaction enables a merchant to reverse the following credit card transactions within required timeframes: POS Authorization Only [Transaction Type code "3"]); POS Mail/Phone Authorization Only [Transaction Type code "B"]); and CAT Authorization Only [Transaction Type code "5"]).

For TransArmor PKI Encryption and Tokenization, this element is included in a new data segment (No. 116 [TransArmor Load Data Segment]).

Valid Codes/Values: 000001–999999

100000-199999

Note: This limited sequence number range (100000-199999) consists of the only allowed values when using the Multithreaded Dial Protocol Communications header for a CAPK File Load for the ATL105 message format

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Number:	87	**Name:**	Service Level
Character Type:	A	**Maximum Length:**	1 byte

Representation: Fixed length of one alpha character

Purpose: Identifies the sale type for a product in Data Segment No. 102, Product Code Data Segment or Data Segment No. 157, Adjusted Product Code Data Segment.

Processing Rules:

Valid Codes/Values:

Value	Description
F	Full serve
S	Self-serve
N	Mini-serve
X	Maxi-serve
H	High-Speed Dispense
O	Other/Fuel not present.
0–9	Reserved for private use.
Other	Reserved for future use.

Number:	88	**Name:**	Ship-from Postal Code
Character Type:	AN	**Maximum Length:**	10 bytes

Representation: Fixed length of nine digits plus a hyphen (*nnnnn-nnnn*).

Purpose: Identifies the postal code of the shipping location.

Processing Rules:

Valid Codes/Values: Any valid postal code

Number:	89	**Name:**	Ship-to Country Code
Character Type:	N	**Maximum Length:**	3 bytes

Representation: Fixed length of three digits

Purpose: Identifies the postal code of the country to which a shipment is sent.

Processing Rules:

Valid Codes/Values: Any valid postal country code

Number:	90	**Name:**	Ship-to Postal Code
Character Type:	AN	**Maximum Length:**	10 bytes

Representation:	Fixed length of nine digits plus a hyphen (*nnnnn-nnnn*).
Purpose:	Identifies the postal code of the location to which a shipment is sent.
Processing Rules:	
Valid Codes/Values:	Any valid postal code

Number:	91	**Name:**	Software Load Phone Number
Character Type:	AN	**Maximum Length:**	18 bytes

Representation:	Variable length of up to eighteen alphanumeric characters
Purpose:	Identifies the telephone number associated with the device management system.
Processing Rules:	Used in the Software Dial Load Data Segment (No. DL4).
Valid Codes/Values:	

Number:	92	**Name:**	Software Load Request Date
Character Type:	N	**Maximum Length:**	6 bytes

Representation:	Fixed length of six digits (MMDDYY)
Purpose:	Identifies the date the device will request a full load of the new application from the device management system.
Processing Rules:	Used in the following data segments: • Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)
Valid Codes/Values:	

Number:	93	**Name:**	Software Load Request Time
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of four digits (HHMM)
Purpose:	Identifies the time the device will request a full load of the new application from the device management system.
Processing Rules:	Used in the following data segments: • Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)
Valid Codes/Values:	

Number:	94	**Name:**	Software Load Type
Character Type:	A	**Maximum Length:**	1 byte

Representation:	One alphanumeric character	
Purpose:	Identifies the type of software the device will request from the device management system.	
Processing Rules:	Used in the following data segments: • Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)	
Valid Codes/Values:	**Codes**	**Description**
	F	Full application load
	P	Partial application load

Number:	95	**Name:**	Software Terminal Record ID
Character Type:	AN	**Maximum Length:**	13 bytes

Representation:	Fixed length of 13 alphanumeric characters
Purpose:	Identifies the Software Terminal Record ID associated with the device management system.
Processing Rules:	Used in the following data segments: • Software Dial Load Data Segment (No. DL4) • Software IP Load Data Segment (No. DL5)
Valid Codes/Values:	

Number:	96	**Name:**	Software Version
Character Type:	AN	**Maximum Length:**	8 bytes

Representation:	Fixed length of eight alphanumeric characters
Purpose:	Identifies the device's software version at the customer location.
Processing Rules:	Required for Totals Request, Table Load Request, and TransArmor PKI Encryption and Tokenization Load Request.
Valid Codes/Values:	

Number:	97	**Name:**	Start-of-Data Block Indicator
Character Type:	A	**Maximum Length:**	1 byte

Representation:	Fixed length of one alpha character
Purpose:	Indicates the start of a data block in a Table Load Response.
Processing Rules:	
Valid Codes/Values:	Fixed value:)

Number:	98	**Name:**	Store Number
Character Type:	N	**Maximum Length:**	16 bytes

Representation:	Fixed length of up to 16 digits
Purpose:	Identifies the store associated with the transmitting device.
Processing Rules:	If the data is masked by all asterisks, the data is neither displayed nor printed.
Valid Codes/Values:	0000000000000001-9999999999999999

Number:	99	**Name:**	Tax Amount
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Variable length of up to eight digits with two assumed decimal places **Note:** A maximum length of 8 bytes is allowed for American Express cards only. A maximum length of 7 bytes is allowed for all other card types.
Purpose:	Identifies the transaction tax amount in Data Segment No. 100, Standard Message Data Segment.
Processing Rules:	Note the difference between this data element and Element No. 74, PC Tax Amount. **Note:** Sales tax is applied to EBT Cash Benefits only. Sales tax is not applicable to EBT Food Stamps.
Valid Codes/Values:	1–9999999

Number:	100	**Name:**	Terminal Display Communications Message
Character Type:	A	**Maximum Length:**	16 bytes

Representation:	Fixed length of 14 upper case alpha characters and two spaces
Purpose:	Identifies the message returned from BUYPASS that indicates a successful communications test.
Processing Rules:	
Valid Codes/Values:	COMM_TEST_PASSED

Number:	101	**Name:**	Terminal Display /Printer Message
Character Type:	AN	**Maximum Length:**	31 bytes

Representation: Fixed length of 31 alphanumeric characters; space-filled

Purpose: Identifies a device display or a printer message for the merchant from BUYPASS.

Examples.
Fill_out_change_of_address_form
Get_driver's_license_number.___
*Approved_990001_*
*PASSWORD_ERR_5A_*

In the first example, there are 26 characters and 5 spaces (indicated by the _ character) for a total of 31 alphanumeric characters.

In the second example, there are 25 characters and 6 spaces (indicated by the _ character) for a total of 31 alphanumeric characters.

In the third example, there are 14 characters and 2 spaces (indicated by the _ character) for a total of 16 alphanumeric characters.

In the fourth example, there are 13 characters and 3 spaces (indicated by the _ character) for a total of 16 alphanumeric characters.

Processing Rules: Print what is sent.

This data element can be repeated up to a maximum of three times in a Financial Transaction Response. The number of times it is repeated is indicated by the Number of Print Lines value.

This data element cannot be repeated in a Declined Totals Response.

Required for both the TransArmor PKI Encryption and Tokenization and TransArmor Token Only Financial Transaction Response in which a Token has not been retrieved from the host.

Valid Codes/Values: Thirty-one alphanumeric characters; space-filled

For TransArmor, the message indicates that BUYPASS has not been able to retrieve a token for the associated PAN for a TransArmor transaction.

Number:	102	**Name:**	Terminal Identifier
Character Type:	AN	**Maximum Length:**	22 bytes

Representation: Variable length up to 22 alphanumeric characters

Purpose: Identifies the transaction device by Device Type, State Code (ANSI® Code), BUYPASS Merchant Number, and Device Number.

Processing Rules: Required for all transactions and messages.

The Terminal Identifier must be assigned by and set up at BUYPASS.

Note: In the case of a Table Load Request, Phone Load Request, Date and Time Load Request, or Software Load Request, the field length is 13 bytes; otherwise, the field length is a maximum of 22 bytes.

For the TransArmor PKI Encryption and Tokenization Load Request, the first two characters (Device Type) of this element's value must be "++" regardless of the actual device type.

For the EMV Key Load Request, the first two characters (Device Type) of this element's value must be "+*" regardless of the actual device type.

Valid Codes/Values:

Device Type (Pos. Nos. 1–2)

Any valid 2-character Device Type assigned by BUYPASS personnel.

For the TransArmor PKI Encryption and Tokenization Load Request, this element's value must be "++" regardless of the actual device type.

For the EMV Key Load Request, this element's value must be "+*" regardless of the actual device type.

State Code (Pos. Nos. 3–4)

Any valid 2-character State Code (ANSI® Code). Please see Appendix D. Valid State Codes, for a list of valid ANSI® Code entries.

BUYPASS Merchant Number (Starts immediately in Pos. No. 5.)

Any valid 6- to 15-character BUYPASS Merchant Number assigned by BUYPASS personnel. The Device Number immediately follows the BUYPASS Merchant Number.

Note: If using TransArmor, the first 6 digits of the Merchant Number are used in **Error! Reference source not found.**. P lease refer to the description of Element No. 2 (Account Number) for additional information.

Device Number (Starts immediately after the BUYPASS Merchant Number.)

Any valid 3-character Device Number assigned by BUYPASS personnel.

Number:	103	**Name:**	Text Data
Character Type:	AN	**Maximum Length:**	150 bytes

Representation:	Variable length of up to 150 alphanumeric characters
Purpose:	Identifies the data in an Electronic Mail Data Segment.
Processing Rules:	This is free-form text.
Valid Codes/Values:	Free-form text

Number:	104	**Name:**	Text Data Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length three digits Zero-filled and right aligned
Purpose:	Identifies the length of Text Data.
Processing Rules:	
Valid Codes/Values:	001–750

Number:	105	**Name:**	Totals Date
Character Type:	N	**Maximum Length:**	6 bytes

Representation: Fixed length of six digits

Purpose: Identifies the date for which totals are being requested or the date for which totals and an end-of-day are being requested.

Processing Rules: The customer assigns this information. Totals cannot be requested for a date prior to the third most recent date with activity.

Note: The value of this field in a Totals Response can be different than the value sent in the Totals Request.

Valid Codes/Values:

Request	Response	Description
MMDDYY	YYMMDD	Identifies a request for totals for a specified settlement date.
111111	111111	Identifies a request for totals for the period of time since the last *999999* code was used.
222222	YYMMDD	Identifies a request for totals for the current settlement date; ends the current settlement date; and rolls the date to the next settlement date.
333333	YYMMDD	Identifies a request for totals for the most recent date with activity.
444444	YYMMDD	Identifies a request for totals for the second most recent date with activity.
555555	YYMMDD	Identifies a request for totals for the third most recent date with activity.
999999	999999	Identifies a request to clear all totals since the last *999999* code was used. Used with the *111111* code.

Number:	106	**Name:**	Unit of Measure
Character Type:	A	**Maximum Length:**	1 byte

Representation: Fixed length of one alpha character

Purpose: Identifies the measurement type for a product in Data Segment No. 102, Product Code Data Segment.

Processing Rules: This data element, along with the following data elements, is repeated for up to 10 products in Data Segment No. 102, Product Code Data Segment or Data Segment No. 157, Adjusted Product Code Data Segment:

- Product Code (No. 77)
- Quantity (No. 81)
- Unit Price (No. 107)
- Product Amount (No. 76)

Valid Codes/Values:

Value	Description
C	Case/Carton
G	Gallon
H	Hours
I	Imperial Gallons
K	Kilogram
L	Liter
M	Charging Minutes (EV)
P	Pound
Q	Quart
U	Unit
W	Kilowatt per hour (EV)
Z	Ounce
O	Other/Unknown unit
0–9	Reserved for private use.
Other	Reserved for future use.

Number:	107	**Name:**	Unit Price
Character Type:	N	**Maximum Length:**	9 bytes

Representation:	Variable length of up to nine digits with a maximum of three assumed decimal places
Purpose:	Identifies the price per Unit of Measure of a product in Data Segment No. 102, Product Code Data Segment or Data Segment No. 157, Adjusted Product Code Data Segment.
Processing Rules:	The leading number indicates the number of assumed decimal places. There is a maximum number of three assumed decimal places. This data element, along with the following data elements, is repeated for up to 10 products in Data Segment No. 102, Product Code Data Segment or Data Segment No. 157, Adjusted Product Code Data Segment: • Product Code (No. 77) • Unit of Measure (No. 106) • Quantity (No. 81) • Product Amount (No. 76)
Valid Codes/Values:	000000.000–999999.999 For an example of this data element, please see Appendix B. POS Purchase Example.

Number:	108	**Name:**	Vehicle Number
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 digits
Purpose:	Identifies the Vehicle Number.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by the issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any numeric representation

Number:	109	**Name:**	Voucher ID
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 digits
Purpose:	Identifies the preprinted voucher used in a local approval transaction.
Processing Rules:	Required by some states when an EBT transaction is approved locally.
Valid Codes/Values:	

Number:	111	**Name:**	Variable Information Indicator
Character Type:	AN	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 alphanumeric characters
Purpose:	Identifies the type of variable information in a Variable Information Data Segment.
Processing Rules:	Used in the Variable Information Data Segment (No. 111). Balance transfer (value *008*) supports both the Balance Merge (Transaction Type code *M*) and Replace (Transaction Type code *N*) transaction types for stored value gift card. The account number of the account that is being reduced to $0.00 is placed in Variable Information (Element No. 113).

Valid Codes/Values:

Value	Description
001	Universal Product Code (UPC) indicator
002	Program Identifier indicator
003	Address verification service (AVS) indicator
004	Card verification (CVV, CVV2, CVC2, CID, DTVV) indicator
005	Point-of-service entry mode indicator
006	Currency code indicator
007	Standard Industrial Classification (SIC) indicator
008	Balance transfer—stored value gift card account number indicator
009	Partial Approval Indicator
010	Visa® Trusted Agent Program (TAP) indicator
011	Tax Collector indicator
012	Market-Specific Data indicator
013	Special Payments indicator
014	User Data Information
015	Discover® Network Retrieval Reference Number indicator
016	Visa payWave™ card data indicator
017	Suspected Fraud Reversal indicator
018	Card Acceptor Terminal Identifier Information Indicator
019	Recharge Type Information
020	Top Up Phone Number Information
021	Settlement Data Acceptance Flag
022	Originating Device Type Indicator
024	Manual Check MICR Type
025	Gift Card Number
026	Order ID
027	Customer Name
028	Use Unique ID
030	POS Condition Code
031	Point-of-Service Terminal Capability Code
032	POS Additional Data Information
034	Name Verification Service (NVS) Indicator
035	Verified By Visa® Data
036	MasterCard UCAF Data
037	MCX Checkout Token
038	Paydiant Tender ID
039	Fraud Enhanced Data

040	Final Amount Indicator
041	Country Subdivision
042	Cardholder Language
043	Cardholder Postal Code (for AVS)
044	Original Transaction Identifier
045	First Auth Amount
046	Previous Transaction ID
047	Soft Descriptor – Merchant Name
048	MasterCard Authentication Data
049	Merchant Supplementary Data
050	Additional Security Code
051	Additional Card ID
052	Additional TransArmor Data
053	Remote Commerce Acceptor Identifier
054	Merchant Tran ID
055	Experience Provider Value
056	Stored Credentials Transaction Additional Data
057	Soft Descriptor Merchant City
058	Soft Descriptor Merchant State
059	Soft Descriptor Merchant Country
060	Real Time Account Updater Request Data
062	Customer Additional Information
063	Visa® Incremental Auth Lodging Data
064	Service Location Information
065	Merchant Payment Gateway ID Data
066	Digital Commerce Data
067	Anticipated Amount
068	Additional Transaction Fee 1
069	Additional Transaction Fee 2
070	Anticipated Completion Amount
071	Enabler Verification Value
072	Transaction Link Identifier
073	Future Transaction Amount
075	Discount Program Data
076	Discount Line-Item Data
077	Discount Amount
078	Line Item
079	Visa Commercial Card Data
080	Staged Wallet ID
081	Visa Gateway ID

082 Incremental Auth Data

Number:	112	**Name:**	Variable Information Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 bytes
Purpose:	Identifies the length of Variable Information (Element No. 113) in the Variable Information Data Segment.
Processing Rules:	Used in the Variable Information Data Segment (No. 111).
Valid Codes/Values:	001–999

Number:	113	**Name:**	Variable Information
Character Type:	AN	**Maximum Length:**	982 bytes

Representation: Variable length of up to 982 alphanumeric characters

Note: Please refer to section 12.10, "Variable Information Data Segment," for conditions affecting this element's length.

Purpose: Identifies the data in a Variable Information Data Segment.

Processing Rules: Used in the Variable Information Data Segment (No. 111).

When Element No. 111 (Variable Information Indicator) has a value of ***001*** (UPC indicator), the actual UPC is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***002*** (Program Identifier Indicator), the actual 3-byte Program Identifier is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***003*** (AVS indicator), address information (street name, ZIP Code™/ZIP+4®) is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***004*** [card verification value (CVV, CVV2, CVC2, CID) indicator], the card's actual verification code is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***005*** (point-of-service entry mode indicator), the actual point-of-service entry mode (swiped/manual/RFID) is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***006*** (currency code indicator), the actual currency code (always 3 bytes) is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***007*** (SIC indicator), the actual SIC (4 or 5 or 6 numeric bytes) is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***008*** (balance transfer—stored value card account number indicator), the account number (maximum of 19 digits) of the account that is being reduced to $0.00 is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***009*** (Partial Approval Indicator), the Partial Approval Indicator (fixed length of 1 alphanumeric characters) is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***010*** (Visa® Trusted Agent Program (TAP) Indicator), the Visa® Trusted Agent Program (TAP) Indicator is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***011*** (Tax Collector Indicator), the code indicating whether the transaction is taxable and whether sales tax is included is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***012*** (Market-Specific Data Indicator), the Market-Specific Data Indicator is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***013*** (Special Payments Indicator), the Special Payments information is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***014*** (User Data Information Indicator), the User Data information is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***015*** (Discover® Network Retrieval Reference Number Indicator), the Discover® Network Retrieval Reference Number from the associated financial response is included here. It has a fixed alphanumeric length of 15 bytes. It is included in all Discover® Network follow-on transactions.

When Element No. 111 (Variable Information Indicator) has a value of ***016*** (Visa payWave™ card indicator), the Visa payWave ™ card information is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***017*** (Suspected Fraud Reversal Indicator Information Indicator), the Suspected Fraud Reversal Indicator information is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***018*** (Card Acceptor Terminal Identifier Information Indicator), the card acceptor terminal identifier information is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***019*** (Recharge Type Information), the recharge type information is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***020*** (Top Up Phone Number Information), the top up phone number information is included here.

When Element No. 111 (Variable Information Indicator) has a value of ***021*** (Settlement Data Acceptance Flag), the transaction will be processed as PINless.

When Element No. 111 (Variable Information Indicator) has a value of **022** (Originating Device Type Indicator Information), the originating device type information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **024** (Manual Check MICR Type), the Manual Check MICR Type information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **025** (Gift Card Number), the Gift Card Number information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **026** (Order ID), the Order ID information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **027** (Customer Name), the Customer Name information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **028** (Use Unique ID), the Use Unique ID information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **030** (POS Condition Code), the POS Condition Code information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **031** (Point-of-Service Terminal Capability Code), the Point-of-Service Terminal Capability Code information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **032** (POS Additional Data Information), the POS additional data information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **034** (Name Verification Service (NVS) Indicator), the NVS information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **035** (Verified By Visa® Data), the Verified By Visa® Data information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **036** (MasterCard UCAF Data), the MasterCard UCAF Data information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **037** (MCX Checkout Token), the MCX Checkout Token information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **038** (Paydiant Tender ID), the Paydiant Tender ID information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **039** (Fraud Enhanced Data), the Fraud Enhanced Data information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **040** (Final Amount Indicator), the Final Amount Indicator information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **041** (Country Subdivision), the Country Subdivision information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **042** (Cardholder Language), the Cardholder Language information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **043** (Cardholder Postal Code (for AVS)), the Cardholder Postal Code (for AVS) information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **044** (Original Transaction Identifier), the Original Transaction Identifier information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **045** (First Auth Amount), the First Auth Amount information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **046** (Previous Transaction ID), the Previous Transaction ID information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **047** (Soft Descriptor – Merchant Name), the Merchant Name information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **048** (MasterCard Authentication Data), the MasterCard Authentication Data information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **049** (Merchant Supplementary Data), the Merchant Supplementary Data information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **050** (Additional Security Code), the Additional Security Code information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **051** (Additional Card ID), the Additional Card ID information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **052** (Additional TransArmor Data), the Additional TransArmor Data information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **053** (Remote Commerce Acceptor Identifier), the Remote Commerce Acceptor Identifier information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **054** (Merchant Tran ID), the Merchant Tran ID information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **055** (Experience Provider), the Experience Provider Value information is included here.

When Element No. 111 (Variable Information Indicator) has a value of **056** (Stored Credentials Transaction Additional Data), the additional data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **057** (Soft Descriptor Merchant City), the Soft Descriptor Merchant City data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **058** (Soft Descriptor Merchant State), the Soft Descriptor Merchant State data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **059** (Soft Descriptor Merchant Country), the Soft Descriptor Merchant Country data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **060** (Real Time Account Updater Request Data), the Real Time Account Updater Request data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **062** (Customer Additional Information Data), the Customer Additional Information data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **063** (Visa® Incremental Auth Lodging Data), the Visa® Incremental Auth Lodging Data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **064** (Service Location Information), the Service Location Information data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **065** (Merchant Payment Gateway ID Data), the Merchant Payment Gateway ID Data is included here.

When Element No. 111 (cVariable Information Indicator) has a value of **066** (Digital Commerce Data), the Digital Commerce Data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **067** (Anticipated Amount), the Anticipated Amount data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **068** (Additional Transaction Fee 1), the Transaction Fee data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **069** (Additional Transaction Fee 2), the Transaction Fee data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **070** (Anticipated Completion Amount), the Anticipated Completion Amount data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **071** (Enabler Verification Value), the Enabler Verification Value data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **072** (Transaction Link Identifier), the Transaction Link Identifier data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **073** (Future Transaction Amount), the Future Transaction Amount data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **075** (Discount Program Data), the Discount Program Data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **076** (Discount Line-Item Data), the Discount Line-Item Data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **077** (Discount Amount), the Discount Amount data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **078** (Line Item), the Line Item data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **079** (Visa Commercial Card Data), the Visa Commercial Card Data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **080** (Staged Wallet ID), the Staged Wallet ID Data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **081** (Visa Gateway ID), the Visa Gateway ID Data is included here.

When Element No. 111 (Variable Information Indicator) has a value of **082** (Incremental Auth Data), the Incremental Auth Data is included here.

Valid Codes/Values: 001–982

a–z

A–Z

For data layout information, please refer to Appendix I. Variable Information Data Layouts.

Number:	114	**Name:**	Software Load IP/URL Address
Character Type:	AN	**Maximum Length:**	30 bytes

Representation: Fixed length of 30 alphanumeric characters

Purpose: Identifies the terminal number used to access the device management system.

Processing Rules: Used in the Software IP Load Data Segment (No. DL5).

Valid Codes/Values: 01–999

a–z

A–Z

Number:	115	**Name:**	Additional Information Data Segment Flag
Character Type:	N	**Maximum Length:**	1 byte

Representation:	Fixed length of one digit	
Purpose:	Indicates whether an Additional Information Data Segment (Segment No. 112) follows in the message.	
Processing Rules:	Used in the Financial Transaction Response.	
Valid Codes/Values:	**Value**	**Description**
	0	No Additional Information Data Segment follows.
	1	An Additional Information Data Segment follows.

Number:	116	**Name:**	Additional Information Indicator
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 digits
Purpose:	Identifies the type of additional information being transmitted in a Financial Transaction Response.
Processing Rules:	Used in the Additional Information Data Segment (No. 112). BUYPASS initiates transmittal of balance information (*001*) in a Financial Transaction Response. A Financial Transaction Request initiates transmittal of AVS information (*003*) and CVV information (*004*) in a Financial Transaction Response. A Financial Transaction Request initiates transmittal of Visa® product result information (*009*) in a Financial Transaction Response.

Valid Codes/Values:

Code	Description
001	Balance information for gift card, EBT card, credit card, or phone card transactions
002	Reserved
003	Address Verification Service (AVS) information or Name Verification Service (NVS) or both the information
004	Card verification (CVV, CVV2, CVC2, CID, DTVV) information
005	ECA/ TeleCheck® Trace ID
006	ECA/ TeleCheck® Denial Record Number
007	ECA/ TeleCheck® Return Check Data
008	Loyalty Information – Version 1
009	Visa® product result
010	Loyalty Information – Version 2
011	User Data Information
012	Discover® Network Retrieval Reference Number
013	Expiration Date (MMYY) for TransArmor - VeriFone edition processing
016	PIN-on-Receipt Information
017	BUYPASS Host Card Type
018	PINless Debit Information
019	Visa Spend Qualified Indicator Information
020	Host Prompts Information
021	Re-Price Data Response Information
022	CAVV Result
023	MCX Reference Number
024	Carwash Indicator
025	Language Indicator
026	DST Response
027	Universal Unique Identifier
028	Transaction Identifier
029	PAR (Payment Account Reference) Data
030	Merchant Advice Code
031	DAF Indicator
032	Agreement ID
034	Host-based Purchase Restriction
035	Cardholder Additional Information Result Code
036	Account Type
037	Account Funding Source

038	Transaction Link Identifier
039	Transaction Link Action Indicator
040	Fraud Score
041	Fraud Score Reason Code
042	Authentication Data Quality Indicator
043	Token Update First Use Indicator
044	Merchant Tran ID
045	Applied Special Service
046	Voyager Restriction Code
047	Visa Category Code

Number:	117	**Name:**	Additional Information Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 digits
Purpose:	Identifies the length of Additional Information (Element No. 118).
Processing Rules:	Used in the Additional Information Data Segment (No. 112).
Valid Codes/Values:	001–985

Number:	118	**Name:**	Additional Information
Character Type:	AN	**Maximum Length:**	984 bytes

Representation:	Variable length of up to 984 alphanumeric characters **Note:** Please refer to section 12.11, "Additional Information Data Segment," for conditions affecting this element's length.
Purpose:	Identifies the additional data being transmitted in the Financial Transaction response.
Processing Rules:	Used in the Additional Information Data Segment (No. 112). When Element No. 116 (Additional Information Indicator) has a value of *001*, gift card, EBT card, credit card, or phone card balance information is included here. When Element No. 116 (Additional Information Indicator) has a value of *003*, AVS information is included here. When Element No. 116 (Additional Information Indicator) has a value of *004*, card verification value information (CVV, CVV2, CVC2, and CID) is included here. When Element No. 116 (Additional Information Indicator) has a value of *005*, ECA/ TeleCheck® Trace ID information is included here. When Element No. 116 (Additional Information Indicator) has a value of *006*, ECA/ TeleCheck® Denial Record Number information is included here. When Element No. 116 (Additional Information Indicator) has a value of *007*, ECA/ TeleCheck® Return Check Data is included here. When Element No. 116 (Additional Information Indicator) has a value of *008*, loyalty information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *009*, Visa® product result information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *010*, loyalty information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *011*, user data information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *012*, the Discover® Network Retrieval Reference Number is included here. It has a fixed alphanumeric length of 15 bytes. It is included in all Discover® Network follow-on transactions.

When Element No. 116 (Additional Information Indicator) has a value of *013*, the Expiration Date (MMYY) is included here. It has a fixed length of 4 bytes.

When Element No. 116 (Additional Information Indicator) has a value of *016*, the PIN-on-Receipt Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of 017, the BUYPASS Host Card Type is included here. It has a fixed length of three bytes.

When Element No. 116 (Additional Information Indicator) has a value of *018*, the PINless transaction processing information is included here. It has a fixed numeric length of 3.

When Element No. 116 (Additional Information Indicator) has a value of *019*, the Visa Spend Qualified Indicator Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *020*, the Host Prompts Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *021*, the Re-Price Data Response Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *022*, the CAVV Result Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *023*, the MCX Reference Number Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *024*, the Carwash Indicator Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *025*, the Language Indicator Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *026*, the DST Response Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *027*, the Universal Unique Identifier is included here.

When Element No. 116 (Additional Information Indicator) has a value of *028*, the Transaction Identifier Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of *029*, the PAR (Payment Account Reference) Data Information is included here.

When Element No. 116 (Additional Information Indicator) has a value of 030, the Merchant Advice code is included here.

When Element No. 116 (Additional Information Indicator) has a value of *031*, the DAF Indicator is included here.

When Element No. 116 (Additional Information Indicator) has a value of *032*, the Agreement ID is included here.

When Element No. 116 (Additional Information Indicator) has a value of *034*, the Host-based Purchase Restriction is included here.

When Element No. 116 (Additional Information Indicator) has a value of *035*, the Cardholder Additional Information Result Code is included here.

When Element No. 116 (Additional Information Indicator) has a value of *036*, the Account Type is included here.

When Element No. 116 (Additional Information Indicator) has a value of *037*, the Account Funding Source is included here.

When Element No. 116 (Additional Information Indicator) has a value of *038*, the Transaction Link Identifier is included here.

When Element No. 116 (Additional Information Indicator) has a value of *039*, theTransaction Link Action Indicator is included here.

When Element No. 116 (Additional Information Indicator) has a value of *040*, the Fraud Score is included here.

When Element No. 116 (Additional Information Indicator) has a value of *041*, the Fraud Score Reason Code is included here.

When Element No. 116 (Additional Information Indicator) has a value of *042*, the Authentication Data Quality Indicator is included here.

When Element No. 116 (Additional Information Indicator) has a value of *043*, the Token Update First Use Indicator is included here.

When Element No. 116 (Additional Information Indicator) has a value of *044*, the Merchant Tran ID is included here.

When Element No. 116 (Additional Information Indicator) has a value of *045*, the Applied Special Service is included here.

When Element No. 116 (Additional Information Indicator) has a value of *046*, the Voyager Restriction Code is included here.

When Element No. 116 (Additional Information Indicator) has a value of *047*, the Visa Category Code is included here.

Valid Codes/Values: 001–984

a–z

A–Z
Please refer to Appendix K. Additional Information Data Layouts.

Number:	118	**Name:**	EMV Additional Information
Character Type:	ANSB	**Maximum Length:**	984 bytes

Representation:	Variable length of up to 984 alphanumeric characters
Purpose:	Identifies the additional data being transmitted in the Financial Transaction request or response.
Processing Rules:	When EMV Additional Information Indicator has a value of *001*, EMV table data is included here.
Valid Codes/Values:	001–984 a–z, A–Z Please refer to Appendix T. EMV Additional Information Data Layouts.

Number:	119	**Name:**	Settlement Date
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of 4 digits (MMDD)	
Purpose:	Identifies the date of actual financial settlement for the transaction.	
Processing Rules:		
Valid Codes/Values:	**Value**	**Description**
	01–12	January through December
	01–31	Days of the month

Number:	120	**Name:**	Network Management Message
Character Type:	A	**Maximum Length:**	8 bytes

Representation:	Fixed length of 8 alphabetic characters
Purpose:	Identifies the communications test message from the transmitting device.
Processing Rules:	
Valid Codes/Values:	COMMTEST

Number:	121	**Name:**	Partial Approval Indicator
Character Type:	N	**Maximum Length:**	1 byte

Representation: Fixed length of 1 digit

Purpose: Indicates whether or not a merchant supports credit/debit/fleet Partial Approvals.

Processing Rules: Required on all transactions.

Valid Codes/Values:

Value	Description
0	Not specified (default)/Not supported This value is the default value. If the Merchant does not support credit/debit/fleet Partial Approvals, then this value is transmitted.
1	Supported This value indicates that the Merchant supports credit/debit/fleet Partial Approval transactions. If the Merchant transmits this value with a transaction—and the corresponding network supports credit/debit/fleet Partial Approval transactions—the network can approve an amount that is less than the amount in the Authorization request; and BUYPASS will send a Partial Approval response. The network may return a card balance that is printed on the customer's copy of the receipt.
5	This value indicates that the Merchant only supports the receipt of the card balance, not Partial Approval transaction processing. The network may return a card balance that is printed on the customer's copy of the receipt. **Note:** This value is only supported for American Express prepaid cards.

Note: Support for Partial Authorization is mandatory for all card brands. Data Element 121 (Partial Approval Indicator) should be populated correctly for all card present transactions, showing that the terminal is capable of partial authorization responses.

Note: For American Express transactions, Merchants sending transaction request amount as 'zero' and using values '1' or '5' for Prepaid card balances, will not be receiving the card balance back in response.

Number:	122	**Name:**	MICR Data
Character Type:	AN	**Maximum Length:**	50 bytes

Representation:	Variable length of up to 50 alphanumeric characters
Purpose:	Identifies the data encoded along the bottom of a check.
Processing Rules:	Required on all check transactions. If MICR data is greater than 50 bytes, include the first 50 bytes (from left to right) in this field. **Note:** MICR Data is also included in Account Number (Element No. 2) of the Standard Message Data Segment (Segment No. 100). If MICR data > 50 bytes and the prompt code is ECA/ TeleCheck® check service then refer ECA/ TeleCheck® Data Segment (Segment No. 113).
Valid Codes/Values:	

Number:	123	**Name:**	Driver's License
Character Type:	AN	**Maximum Length:**	40 bytes

Representation:	Variable length of up to 40 alphanumeric characters
Purpose:	Identifies the driver's license number used as identification in a check transaction.
Processing Rules:	Required on all manually entered check transactions.
Valid Codes/Values:	

Number:	124	**Name:**	State Code
Character Type:	AN	**Maximum Length:**	2 bytes

Representation:	Fixed length of 2 alphanumeric characters
Purpose:	Identifies the 2-character alphabetical State Code when a driver's license number or other form of consumer identification is used in a check transaction.
Processing Rules:	Required when Driver's License (Element No. 123) is included in a check transaction.
Valid Codes/Values:	See Appendix D. Valid State Codes, for a list of valid alphabetical entries.

Number:	125	**Name:**	Date of Birth
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Fixed length, 8-digit number (MMDDYYYY)
Purpose:	Identifies the consumer's date of birth in a check transaction.
Processing Rules:	
Valid Codes/Values:	

Number:	126	**Name:**	Check Type
Character Type:	AN	**Maximum Length:**	1 byte

Representation:	Fixed length of 1 alphanumeric character	
Purpose:	Identifies the type of check used in a check transaction.	
Processing Rules:	Required on all manually entered check transactions.	
Valid Codes/Values:	**Code**	**Description**
	P	Personal
	C	Company

Number:	127	**Name:**	Check Number
Character Type:	AN	**Maximum Length:**	8 bytes

Representation:	Variable length of up to 8 alphanumeric characters
Purpose:	Identifies the manually entered check number.
Processing Rules:	Required on all manually entered check transactions.
Valid Codes/Values:	

Number:	128	**Name:**	Customer Phone Number
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 digits
Purpose:	Identifies the phone number of the consumer presenting the check.
Processing Rules:	
Valid Codes/Values:	

Number:	129	**Name:**	Customer Last Name
Character Type:	AN	**Maximum Length:**	24 bytes

Representation:	Variable length of up to 24 alphanumeric characters
Purpose:	Identifies the last name of the consumer presenting the check.
Processing Rules:	
Valid Codes/Values:	

Number:	130	**Name:**	Check Issue Date
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Fixed length, 8-digit number (MMDDYYYY)
Purpose:	Identifies the issue date of the check.
Processing Rules:	
Valid Codes/Values:	

Number:	131	**Name:**	ECA/ TeleCheck® Clerk ID
Character Type:	AN	**Maximum Length:**	6 bytes

Representation:	Variable length of up to 6 alphanumeric characters
Purpose:	Identifies the clerk's identification number.
Processing Rules:	
Valid Codes/Values:	1–999999

Number:	132	**Name:**	ECA/ TeleCheck® Product Code
Character Type:	AN	**Maximum Length:**	6 bytes

Representation:	Variable length of up to 6 alphanumeric characters
Purpose:	Identifies the ECA/ TeleCheck® product code for the merchandise type sold by the merchant and used by ECA/ TeleCheck® Service Risk Control.
Processing Rules:	
Valid Codes/Values:	

Number:	133	**Name:**	ECA/ TeleCheck® Phone Number
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 digits
Purpose:	Identifies the phone number of the consumer presenting the check.
Processing Rules:	
Valid Codes/Values:	

Number:	134	**Name:**	ECA/ TeleCheck® Trace ID
Character Type:	AN	**Maximum Length:**	22 bytes

Representation:	Variable length of up to 22 alphanumeric characters
Purpose:	Identifies the unique trace identification number generated by the ECA/ TeleCheck® service host.
Processing Rules:	
Valid Codes/Values:	

Number:	135	**Name:**	Merchant Trace ID
Character Type:	AN	**Maximum Length:**	25 bytes

Representation:	Variable length of up to 25 alphanumeric characters
Purpose:	Identifies an optional unique identifier assigned by the merchant, entered at the time of the transaction and later used by the merchant to identify the customer.
Processing Rules:	
Valid Codes/Values:	

Number:	136	**Name:**	Denial Record Number
Character Type:	AN	**Maximum Length:**	7 bytes

Representation:	Variable length of up to 7 alphanumeric characters
Purpose:	Identifies the record number of a declined ECA/ TeleCheck® transaction.
Processing Rules:	
Valid Codes/Values:	

Number:	137	**Name:**	Extended MICR Data
Character Type:	AN	**Maximum Length:**	65 bytes

Representation:	Variable length of up to 65 alphanumeric characters
Purpose:	Identifies the data encoded along the bottom of a check.
Processing Rules:	This field must be populated in addition to MICR Data (Element No. 122) in Check Data Segment (Segment No. 110) when the raw MICR data exceeds the length.
Valid Codes/Values:	

Number:	138	**Name:**	Loyalty Program ID
Character Type:	N	**Maximum Length:**	6 bytes

Representation:	Variable length of up to 6 digits
Purpose:	Identifies the loyalty card's program ID.
Processing Rules:	Required on all loyalty card transactions. Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	ERN

Number:	139	**Name:**	Loyalty Account Number
Character Type:	N	**Maximum Length:**	24 bytes

Representation:	Variable length of up to 24 digits
Purpose:	Identifies the loyalty card's account number.
Processing Rules:	Required on all loyalty card transactions. Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	

Number:	140	**Name:**	Points to Redeem
Character Type:	N	**Maximum Length:**	6 bytes

Representation:	Variable length of up to 6 digits
Purpose:	Identifies the loyalty card's points to redeem.
Processing Rules:	Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	

Number:	141	**Name:**	Coupon ID
Character Type:	N	**Maximum Length:**	19 bytes

Representation:	Variable length of up to 19 digits
Purpose:	Identifies the coupon ID in a loyalty transaction.
Processing Rules:	Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	

Number:	142	**Name:**	Coupon Amount
Character Type:	N	**Maximum Length:**	8 bytes

Representation:	Variable length of up to 8 digits
Purpose:	Identifies the coupon amount in a loyalty transaction.
Processing Rules:	Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	

Number:	143	**Name:**	Update Code
Character Type:	AN	**Maximum Length:**	1 byte

Representation:	Fixed length of 1 alphanumeric character	
Purpose:	Identifies the update code used in a loyalty transaction.	
Processing Rules:	Appears in Loyalty Card Data Segment (Segment No. 108).	
Valid Codes/Values:	**Value**	**Description**
	A	Add account
	C	Coupon redemption
	E	Expiration date update
	I	Account Inquiry
	P	Points redemption
	S	Sale update
	T	Totals Report, Loyalty
	U	Update account

Number:	144	**Name:**	Street Address
Character Type:	N	**Maximum Length:**	5 bytes

Representation:	Variable length of up to 5 digits
Purpose:	Identifies the consumer's street number in a loyalty transaction.
Processing Rules:	Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	

Number:	145	**Name:**	Phone Number, Loyalty
Character Type:	N	**Maximum Length:**	10 bytes

Representation:	Variable length of up to 10 digits
Purpose:	Identifies the consumer's phone number in a loyalty transaction.
Processing Rules:	Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	

Number:	146	**Name:**	Expiration Date
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Variable length of 4 digits (MMYY)
Purpose:	Identifies the expiration date in a loyalty transaction.
Processing Rules:	Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	

Number:	147	**Name:**	Loyalty Track 2 Data
Character Type:	AN	**Maximum Length:**	38 bytes

Representation:	Variable length of up to 38 alphanumeric characters
Purpose:	Identifies the Track 2 data on the loyalty card used in the loyalty transaction.
Processing Rules:	Appears in Loyalty Card Data Segment (Segment No. 108).
Valid Codes/Values:	

Number:	148	**Name:**	Payment Tender Type
Character Type:	AN	**Maximum Length:**	2 bytes

Representation: Fixed length of 2 alphanumeric characters

Purpose: Used to determine whether the transaction is a multiple-card transaction involving a loyalty card. If the value is not "CS," then two cards must be swiped—a loyalty card and a payment.

Processing Rules: Appears in Loyalty Card Data Segment (Segment No. 108).

Valid Codes/Values:

Value	Description
AX	American Express
CK	Check
CS	Loyalty Only—either a cash payment or an administrative transaction (card activation or account inquiry)
DB	Debit
DN	Diners Club®
DS	Discover® Network
EB	EBT Food Stamps
EC	EBT Cash Benefits
FL	Fleet
GC	Gift Card
JC	JCB®
MC	MasterCard
PC	Phone Card
PR	Proprietary
VS	Visa®

Number:	149	**Name:**	SKU Data
Character Type:	AN	**Maximum Length:**	1000 bytes

Representation: Variable length of up to 1000 alphanumeric characters

Purpose: Identifies the SKU (stock keeping unit) data from a bar code.

Processing Rules: Appears in SKU Data Segment (Segment No. 114).

Valid Codes/Values:

Number:	150	**Name:**	Loyalty Information Version
Character Type:	N	**Maximum Length:**	1 byte

Representation: Fixed length of 1 byte

Purpose: Identifies the version of the loyalty information supported by the device.

Processing Rules: If not sent in a transaction request, the value defaults to *1*. Appears in Loyalty Card Data Segment (Segment No. 108).

Valid Codes/Values:

Value	Description
1	Loyalty information, Table ID 008
2	Loyalty Information, Table ID 010

Number:	151	**Name:**	Unit of Work
Character Type:	N	**Maximum Length:**	19 bytes

Representation: Fixed length of 19 bytes

Purpose: Identifies the unique number used by the Loyalty host to identify a transaction.

Processing Rules: Required on loyalty reversals for matching information to the original purchase.

Appears in Loyalty Card Data Segment (Segment No. 108).

Valid Codes/Values:

Number:	152	**Name:**	Print Data
Character Type:	ANS	**Maximum Length:**	900 bytes

Representation: Variable length of up to 900 alphanumeric and special characters

Purpose: Identifies the data to print at the device.

Processing Rules: Appears in Print Data Segment (Segment No. 115).

Valid Codes/Values:

Number:	153	**Name:**	WIC Discount Amount
Character Type:	N	**Maximum Length:**	40 bytes

Representation: Variable length of up to 40 digits

Purpose: Identifies merchant discount/cents-off coupon amounts unique to eWIC transactions.

Processing Rules: Appears in EBT Data Segment (Segment No. 103).

eWIC merchant discount/cents-off information is transmitted in this field, using the following format:

Position	Data
1–2	Account Type
3–4	Amount Type
5–7	Currency Code
8–20	Amount

Valid Codes/Values:

Data	Value
Account Type	97
Amount Type	52
Currency Code	See Appendix L. Valid Currency Codes
Amount	(x + n12, where x = 0 (credit amount); C (credit amount); or D (debit amount)

Number:	154	**Name:**	WIC Product Data
Character Type:	AN	**Maximum Length:**	3001 bytes

Representation: Variable length of up to 3,001 alphanumeric characters

Purpose: Identifies data unique to eWIC transactions.

Processing Rules: Appears in EBT Data Segment (Segment No. 103).

Composed of two or more of the following subelements:

- Total Length
- One or more of the following:
  - Earliest WIC Benefit Expiration Date
  - WIC Prescription Balance Information
  - WIC UPC Exception/Denial Information
  - WIC UPC Purchase Information

Earliest WIC Benefit Expiration Date is included in the eWIC Balance Inquiry Response, eWIC Authorization Response, and eWIC Purchase Completion Response.

WIC Prescription Balance is included in the eWIC Balance Inquiry Response, eWIC Authorization Response, eWIC Purchase Completion Response.

For WIC purchases involving fruits and vegetables via the Cash Value Benefit (CVB), please set the Quantity equal to the price in the WIC Purchase Information.

WIC UPC Exception/Denial Information is included in the eWIC Purchase Completion Response, and eWIC Voucher Clear Response.

In cases where the WIC Purchase price exceeds the value set by the state (and provided in the Approved Product List [APL] file), an exception record is returned in the WIC Product Data field. The exception record reflects both the approved and original prices for the transaction. The total transaction amount (Settlement Amount) is adjusted to reflect the total price adjustments, and is returned in the Approved Amount field.

Valid Codes/Values: The above five components are described below:

Subelement Name:	Total Length
Data Set Identifier:	N/A
Maximum Length:	4 bytes
Character Type:	n
Maximum Value:	2997
Format:	000n
Description:	Identifies the total count of the bytes in the subelement to follow.

Subelement Name:	Earliest WIC Benefit Expiration Date
Data Set Identifier:	EF
Maximum Length:	8 bytes
Character Type:	n
Maximum Value:	N/A
Format:	CCYYMMDD
Description:	Identifies the earliest date from across all the WIC Prescription Balance Information data returned in WIC Product Data, upon which one or more of the benefits will expire.

Subelement Name: WIC Prescription Balance Information
Data Set Identifier: EA
Maximum Length: 14 bytes
Character Type: an
Maximum Value: N/A
Format: EA01414200200099999

Type	Bit #	Field Description
Bit Map (an 4)	-	'1420' (hexadecimal value) 16-Bit Translation: 0001010000100000 (Bit Nos.: 4,6,11)
n 2	4	Category
n 3	6	Sub-category
n 5	11	Quantity Allowed; 999v99

Description: Identifies WIC Prescription Balance Information.

Subelement Name: WIC UPC Exception/Denial Information
The POS device should be prepared to receive any of the bit combinations shown below.

Data Set Identifier: PS

Maximum Length: 47 bytes

Character Type: an

Maximum Value: N/A

Format:

PS04747E0000000111104010140002390010000002490010012
PS 042 47A0 00000123456789012 000285 00100 26 000300 12
PS 041 4760 00000123456789012 000285 00100 03 00300 12
PS 036 4720 00000123456789012 000285 00100 00 12
PS 034 4620 00000123456789012 000285 00100 12

Type	Bit #	Field Description
Bit Map (an 4)	-	'47E0' (hexadecimal value) (Exception/Denial) 16-Bit Translation: 0100011111100000 (Bit Nos.: 2,6,7,8,9,10,11) '47A0' (hexadecimal value) (Exception/Denial) 16-Bit Translation: 0100011110100000 (Bit Nos.: 2,6,7,8,9,11) '4760' (hexadecimal value) (Exception/Denial) 16-Bit Translation: 0100011101100000 (Bit Nos.: 2,6,7,8,10,11) '4720' (hexadecimal value) (Exception/Denial) 16-Bit Translation: 0100011100100000 (Bit Nos.: 2,6,7,8,11) '4620' (hexadecimal value) (Exception/Denial) 16-Bit Translation: 0100011000100000 (Bit Nos.: 2,6,7,11)
n 17	2	UPC/PLU data; left-padded with zeroes. First position indicates UPC/PLU (0=UPC; 1=PLU). Exact length (from right-most digit) specified in data length Bit No. 11.
n 6	6	Item price—Approved price for item, 9999v99. If Item Action Code (Bit No. 8) is "00", product was approved, but maximum price was exceeded. This field contains the price.
n 5	7	Purchase Quantity—Number of UPC's affected by this exception/denial record;

Type	Bit #	Field Description
		999v99
n 2	8	Item Action Code 00 Approved 01 Category not prescribed 02 Sub-category not prescribed 03 Insufficient units 04 UPC/PLU not prescribed 05 Add record 06 Delete record 26 Approved for a lower price due to max price exceeded 27 Approved for a lower price due to max price exceeded, and approved for less units than originally requested due to insufficient units 28-99 Reserved for later use
n 6	9	Original item price of UPC/PLU; 9999v99
n 5	10	Original number of UPC's/PLU's; 999v99
n 2	11	UPC/PLU data length, starting from the right-most digit.

Description: Identifies WIC UPC Exception/Denial Information.

Subelement Name: WIC UPC Purchase Information
Data Set Identifier: PS
Maximum Length: 34 bytes
Character Type: an
Maximum Value: N/A
Format: PS03446200000001111060905200099900100120 11

Type	Bit #	Field Description
Bit Map (an 4)	-	'4620' (hexadecimal value) (Purchase) 16-Bit Translation: 0100011000100000 (Bit Nos.: 2,6,7,11)
n 17	2	UPC/PLU data; left-padded with zeroes. First position indicates UPC/PLU (0=UPC; 1=PLU). Exact length (from right-most digit) specified in data length Bit No. 11.
n 6	6	Item price—Approved price for item, 9999v99. If Item Action Code (Bit No. 8) is "00", product was approved, but maximum price was exceeded. This field contains the price.
n 5	7	Purchase Quantity—Number of UPC's affected by this exception /denial record; 999v99
n 2	11	UPC/PLU data length, starting from the right-most digit.

Description: Identifies WIC UPC Purchase Information.

Number:	155	**Name:**	Key ID
Character Type:	AN	**Maximum Length:**	11 bytes

Representation:	Fixed length of 11 alphanumeric bytes
Purpose:	Used in TransArmor PKI Encryption and Tokenization to identify the Key ID associated with the encryption Key included in Element No. 157 (Key Data) in the TransArmor Load Response.
Processing Rules:	Required in the TransArmor Load Response. Once requested and approved, this Key ID must be used for all subsequent TransArmor transactions from this device.
Valid Codes/Values:	Any valid Key ID (including CA Key ID that is returned in response to a Signing Key request).

Number:	156	**Name:**	Key Data Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Maximum variable length of 3 bytes
Purpose:	Used in TransArmor PKI Encryption and Tokenization to identify the length of the following field (Element No. 157 [Key Data]) in the TransArmor Load Response.
Processing Rules:	Required in the TransArmor Load Response.
Valid Codes/Values:	000–999

Number:	157	**Name:**	Key Data
Character Type:	AN	**Maximum Length:**	999 bytes

Representation:	Variable maximum length of 999 alphanumeric characters
Purpose:	Identifies either the new Encryption Key and the Key ID (Element No. 155) associated with it in the TransArmor Load Response or the Error Message.
Processing Rules:	Required in the TransArmor Load Response. For an approved TransArmor transaction, the maximum length of Signing and Signed Keys can be 999 bytes. For a declined TransArmor transaction, a message is returned in this field and displays at the device.
Valid Codes/Values:	Any valid key.

Number:	158	**Name:**	**License #**
Character Type:	AN	**Maximum Length:**	10 bytes

Representation:	Variable maximum length of 10 alphanumeric characters
Purpose:	If required by the issuer, it identifies the License # of the fleet card user.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any valid License #.

Number:	159	**Name:**	**Job ID**
Character Type:	AN	**Maximum Length:**	12 bytes

Representation:	Variable maximum length of 12 alphanumeric characters
Purpose:	If required by the issuer, it identifies the Job ID.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any valid Job ID.

Number:	160	**Name:**	Department #
Character Type:	AN	**Maximum Length:**	12 bytes

Representation:	Variable maximum length of 12 alphanumeric characters
Purpose:	If required by the issuer, it identifies the Department #.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any valid Department #.

Number:	161	**Name:**	Customer Data
Character Type:	AN	**Maximum Length:**	12 bytes

Representation:	Variable maximum length of 12 alphanumeric characters
Purpose:	If required by the issuer, it identifies the Customer Data.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any valid Customer Data.

Number:	162	**Name:**	User ID
Character Type:	AN	**Maximum Length:**	12 bytes

Representation:	Variable maximum length of 12 alphanumeric characters
Purpose:	If required by the issuer, it identifies the User ID.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any valid User ID, except zero [0].

Number:	163	**Name:**	Vehicle ID#
Character Type:	AN	**Maximum Length:**	8 bytes

Representation:	Variable maximum length of 8 alphanumeric characters
Purpose:	If required by the issuer, it identifies the Vehicle ID#.
Processing Rules:	Please refer to the latest release of *BUYPASS® Platform Petroleum Industry Processing Specifications* for issuer requirements. If required by issuer, this element is found in the Fleet Data Segment (Segment No. 101).
Valid Codes/Values:	Any valid Vehicle ID#.

Number:	164	**Name:**	EBT Program Data
Character Type:	AN	**Maximum Length:**	267 bytes

Representation:	Variable maximum length of up to 267 alphanumeric characters It is composed of two subelements: • Total Length (Fixed length of 3 digits; appears only once.) • Program Data (Variable maximum length of 264 alphanumeric bytes; 1–6 subelements of a maximum variable length of 44 bytes each.) The total length of the element is based on the number of Program Data subelements sent/received.
Purpose:	Identifies additional transaction data unique to EBT programs.
Processing Rules:	This element is transmitted in Data Segment No. 103, EBT Data Segment. It is composed of two subelements: • Total Length (Fixed length of 3 digits; appears only once.) • Program Data (Variable maximum length of 264 alphanumeric bytes; 1–6 subelements of a maximum variable length of 44 bytes each.) The total length of the element is based on the number of Program Data subelements sent/received.
Valid Codes/Values:	This element contains two subelements: • Total Length (Fixed length of 3 digits; appears only once.) • Program Data (Variable maximum length of 264 alphanumeric bytes; 1–6 subelements of a maximum variable length of 44 bytes each.) The following tables describe the content of these two subelements:

The following table describes the first subelement in this new element:

Description of Subelement 1 (Total Length)	
Descriptor	**Description**
Subelement Name:	Total Length
Data Set Identifier:	N/A
Maximum Length:	3 Bytes
Character Type:	N
Maximum Value:	264
Format:	A fixed length of 3 digits; right-aligned; and padded with zeroes.
Description:	Identifies the total count of all the bytes in all occurrences of the Program Data subelement(s) to follow. Appears only once.

The following table describes the second subelement in this new element:

<table>
<tr><th colspan="3">Description of Subelement 2 (EBT Program Data)</th></tr>
<tr><th>Descriptor</th><th>Pos. Nos.</th><th>Description</th></tr>
<tr><td>Subelement Name:</td><td>N/A</td><td>Program Data</td></tr>
<tr><td>Maximum Length:</td><td>N/A</td><td>264 Bytes
Up to 6 occurrences with each occurrence having a maximum variable length of 44 bytes each.</td></tr>
<tr><td>Character Type</td><td>N/A</td><td>AN</td></tr>
<tr><td rowspan="3">Valid Codes/Values:</td><td>1–2</td><td><u>New Label</u> <u>Description</u>
TAG n2 Program type of data to follow:
Valid values for requests:
<u>Value</u> <u>Description</u>
50 HIP purchase/return amount
IT HIP Internet purchase shipping address/zip code
Valid values for responses:
<u>Value</u> <u>Description</u>
51 HIP incentive earned/returned
52 HIP month-to-date incentive earned</td></tr>
<tr><td>3–4</td><td><u>New Label</u> <u>Description</u>
LEN n2 Length of the detail data to follow.</td></tr>
<tr><td>5–6</td><td><u>New Label</u> <u>Description</u>
ACCOUNT TYPE Fixed value: 98 (HIP Account Type)
Note: This data is not required when TAG = IT.</td></tr>
</table>

<table>
<tr><th colspan="3">Description of Subelement 2 (EBT Program Data)</th></tr>
<tr><th>Descriptor</th><th>Pos. Nos.</th><th>Description</th></tr>
<tr><td rowspan="5">Valid Codes/Values:
(continued)</td><td>7–8</td><td><u>New Label</u> <u>Description</u>
AMOUNT TYPE Valid value for requests:
<u>Value</u> <u>Description</u>
50 HIP purchase/return amount
Valid values for responses:
<u>Value</u> <u>Description</u>
51 HIP incentive earned/returned
52 HIP month-to-date incentive earned
Note: This data is not required when TAG = IT.</td></tr>
<tr><td>9–11</td><td><u>New Label</u> <u>Description</u>
CURRENCY CODE Fixed value: 840 (US Dollars)
Note: This data is not required when TAG = IT.</td></tr>
<tr><td>12</td><td><u>New Label</u> <u>Description</u>
AMOUNT DESCRIPTOR Fixed value: 0/C/D (credit/credit/debit amount to follow)
Note: This data is not required when TAG = IT.</td></tr>
<tr><td>13–24</td><td><u>New Label</u> <u>Description</u>
DETAIL In the case where TAG = 50, 51, or 52, this will be a HIP dollar amount with a fixed length of 12 digits right-aligned; padded with zeroes; and 2 assumed decimal places.</td></tr>
<tr><td>5–32

33–41</td><td>In the case where TAG = IT, this will be a shipping address with a fixed length of 37 alphanumeric characters:
The “Address” portion will have a fixed length of 28 alphanumeric characters, left-aligned, space-filled.
The “Zip Code” portion will have a maximum variable length of 9 digits, left-aligned, space-filled.</td></tr>
</table>

Please refer to Appendix M, Element Format Examples for EBT Program Data, for examples of the element's format in the following messages:

- Request containing a HIP purchase/return amount
- Request containing a HIP purchase/return amount and Internet shipping information

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Number:	165	**Name:**	Start Date or End Date
Character Type:	N	**Length:**	8 bytes

Representation:	Fixed length of eight digits (CCYYMMDD)
Purpose:	Identifies the start date or end date the device uses custom receipt text or host discount data.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118). When it appears as a Start Date and contains 00000000 this indicates the data is used immediately. When it appears as an End Date and contains 99999999 this indicates the data is used until further notice. Used with Element No. 166, Start Time or End Time.

Valid Codes/Values:

Code	Description
00–99	Century
00–99	Year of the century
01–12	Month of the year
01-31	Day of the month
00000000	Start immediately
99999999	No end date

Number:	166	**Name:**	Start Time or End Time
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of four digits (HHMM)
Purpose:	Identifies the start time or end time the device uses custom receipt text or host discount data.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118). Used with Element No. 165, Start Date or End Date.
Valid Codes/Values:	0000–2359

Number:	168	**Name:**	Number of Receipt Text Lines
Character Type:	N	**Maximum Length:**	2 bytes

Representation:	Fixed length of two digits
Purpose:	Provides a count of Receipt Text Line items to follow as part of Customized Receipt Text.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 901 is returned (Customized Receipt Text).
Valid Codes/Values:	1 to a maximum of 10

Number:	169	**Name:**	Receipt Text Data Length
Character Type:	N	**Maximum Length:**	2 bytes

Representation:	Fixed length of two digits
Purpose:	Provides the length of characters to follow representing a line of Receipt Text to be retained by the POS Device for printing on receipts.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 901 is returned (Customized Receipt Text).
Valid Codes/Values:	1 to a maximum of 40

Number:	170	**Name:**	Receipt Text Data
Character Type:	AN	**Maximum Length:**	20 bytes

Representation:	Variable length of up to 20 alphanumeric characters
Purpose:	Provides the text to be printed on a Customers receipt.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 901 is returned (Customized Receipt Text). This data is to be retained by the POS Device and used on subsequent receipt printing.
Valid Codes/Values:	

Number:	171	**Name:**	Number of Discounts
Character Type:	N	**Fixed Length:**	2 bytes

Representation:	Fixed length of two digits
Purpose:	Provides the number of host discount items that will follow.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 904 is returned (Host Discount). This is to be used by the POS Device to parse the following fields that contain Product Codes and associated Discounts that may apply.
Valid Codes/Values:	

Number:	172	**Name:**	Product Discount Amount
Character Type:	N	**Fixed Length:**	5 bytes

Representation:	Fixed length of five digits
Purpose:	Provides the flat rate discount amount to be applied to the previously indicated Product Code.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 904 is returned (Host Discount). This data is to be used by the POS Device for Authorization/Pre-Pay transaction discount calculation. FORMAT: 99v999 where the first 2 digits are dollars and the following 3 digits are cents.
Valid Codes/Values:	

Number:	173	**Name:**	BUYPASS Card Type
Character Type:	N	**Fixed Length:**	4 bytes

Representation:	Fixed length of four digits
Purpose:	Provides the BUYPASS recognized Card Type values. This data is matched to the BUYPASS Card Type in the Dynamic BIN tables. This allows the device to determine any applicable Host Discounts by Card Type.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 904 is returned (Host Discount). This data is used by the POS Device for Authorization/Pre-Pay transaction discount calculation. FORMAT: 9999
Valid Codes/Values:	

Number:	174	**Name:**	Card Table Type
Character Type:	N	**Fixed Length:**	4 bytes

Representation:	Fixed length of four digits
Purpose:	Defines the specific sub-table type within Dynamic Card Table is being provided in the associated data element 175.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 902 is returned (Dynamic Card Table Load)... This data is to be used by the POS Device for creating/replacing its Card Table information FORMAT: 9999
Valid Codes/Values:	0001 BIN Table 0002 RULES Table 0003 RESTRICTIONS Table 0004 SAF Table 0005 PROMPT Table 0006 PRODUCT Table

Number:	175	**Name:**	Card Table Data
Character Type:	AN	**Variable Length:**	3,600 bytes

Representation:	Variable length of up to 3,600 alphanumeric characters
Purpose:	Provides the actual Card Table data for the table identified in Field 174.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 902 is returned (Dynamic Card Table Load). This is the actual table data is to be used by the POS Device for creating/replacing its Card Table information
Valid Codes/Values:	

Number:	176	**Name:**	Device Card Table Version
Character Type:	N	**Fixed Length:**	35 bytes

Representation:	Fixed length of 35 digits
Purpose:	Provides the complete Card Table Version data that the device currently has loaded.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) so that the Host can check for updates to the Card Table data. If the value of the Device Card Table Version begins with *99999*, this indicates there is no card table used by the location.
Valid Codes/Values:	

Number:	177	**Name:**	Card Table Load Version
Character Type:	N	**Fixed Length:**	35 bytes

Representation:	Fixed length of 35 digits
Purpose:	Provides the complete Card Table Version data that is being sent to the device.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) so that the Host can retain the operation in process in a multi-block request and reply data exchange.
Valid Codes/Values:	

Number:	178	**Name:**	Load Control Key
Character Type:	AN	**Fixed Length:**	60 bytes

Representation:	Fixed length of 60 alphanumeric characters
Purpose:	Provides the load control key for tracking block data.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118).
Valid Codes/Values:	

Number:	179	**Name:**	Host Discount Timestamp
Character Type:	N	**Fixed Length:**	12 bytes

Representation:	Fixed length of 12 digits
Purpose:	Provides the timestamp used to support host discounts. In the case of Proprietary Load Request Data, this represents the existing host discount timestamp used by the device. In the case of Proprietary Load Response Data, this represents the new host discount timestamp used by the device after a successful load.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118). Format: CCYYMMDDHHMM
Valid Codes/Values:	

Number:	180	**Name:**	Site Configuration Data
Character Type:	AN	**Variable Length:**	3,600 bytes

Representation:	Variable length of up to 3,600 alphanumeric characters
Purpose:	Provides the Site Configuration data for merchant location.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 903 (Proprietary Load Response Data) is sent in the request. Site configuration data is sent to the host when a configuration change occurs at the site. If multiple configuration changes occur during a business day, only the final site configuration data is sent to the host. If no configuration changes occur during a business day, then no data is sent from the site. This is the site configuration data used by the POS Device.
Valid Codes/Values:	

Number:	182	**Name:**	Prompt Code, Pending
Character Type:	AN	**Max. Length:**	4 bytes

Representation:	Fixed length of 4 alphanumeric characters
Purpose:	In the case of a special transaction, this is a 4-character Prompt Code that identifies the pending requested data (4-character Car Type.)
Processing Rules:	Must be a valid Transaction Type and/or Card Type Code. In the case of a special transaction, a device transmits a 4-character Prompt Code

Valid Codes/Values:

Value	Description
0901	Custom Receipt Test Data
0902	Proprietary Data Load
0904	Host Discount Data
0981	Electronic Mail

Number:	183	**Name:**	Card BIN Range, Beginning
Character Type:	AN	**Max. Length:**	12 bytes

Representation:	12 digits; left-justified, space-filled
Purpose:	Identifies the beginning card BIN range to apply the host discount data.
Processing Rules:	Appears in Proprietary Load Response Data, Prompt Code 904 (Host Discount Data).
Valid Codes/Values:	

Number:	184	**Name:**	Card BIN Range, Ending
Character Type:	AN	**Max. Length:**	12 bytes

Representation:	12 digits; left-justified, space-filled
Purpose:	Identifies the ending card BIN range to apply the host discount data.
Processing Rules:	Appears in Proprietary Load Response Data, Prompt Code 904 (Host Discount Data).
Valid Codes/Values:	

Number:	185	**Name:**	Discount Quantity Limit
Character Type:	N	**Max. Length:**	3 bytes

Representation:	Fixed length of 3 digits
Purpose:	Identifies the quantity limit for host discount data.
Processing Rules:	Appears in Proprietary Load Response Data, Prompt Code 904 (Host Discount Data).
Valid Codes/Values:	

Number:	186	**Name:**	Discount Program Description
Character Type:	AN	**Max. Length:**	15 bytes

Representation:	Fixed length of 15 alphanumeric characters
Purpose:	Identifies the program description of the host discount data.
Processing Rules:	Appears in Proprietary Load Response Data, Prompt Code 904 (Host Discount Data).
Valid Codes/Values:	

Number:	187	**Name:**	CA Public Key File Checksum
Character Type:	N	**Maximum Length:**	25 bytes

Representation:	Fixed length of 25 bytes
Purpose:	Identifies the checksum of the CA Public Key File Block.
Processing Rules:	Required in the EMV Financial Transaction Request.

	Once received, the checksum must be used for all subsequent EMV transaction requests from this device. Required in the CA Public Key File Load Response. The checksum is provided in each CA Public Key File Load Response.
Valid Codes/Values:	

Number:	188	**Name:**	EMV Card Sequence Number
Character Type:	AN	**Maximum Length:**	3 bytes

Representation:	Fixed length of three bytes, right-aligned and padded with zeroes Space-filled if EMV Tag 5F34 is not provided by the EMV chip card
Purpose:	Identifies the card sequence number supplied by the EMV chip card. It is used to differentiate cards that have the same PAN.
Processing Rules:	For transactions containing ICC Data, the Card Sequence Number is required when the card product includes the Card Sequence Number in the EMV data. **Note:** If EMV Tag 5F34 is not provided by the card, send three spaces
Valid Codes/Values:	000—099 EMV Card Sequence Number is supplied by the EMV chip card Spaces Three spaces if EMV Tag 5F34 is not provided by the EMV chip card

Number:	189	**Name:**	EMV Chip Data Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of three digits
Purpose:	Identifies the length of the following field, EMV Chip Data, in an EMV transaction request and response.
Processing Rules:	Required in all EMV transactions.
Valid Codes/Values:	000—999

Number:	190	**Name:**	EMV Chip Data
Character Type:	ANSB	**Maximum Length:**	999 bytes

Representation: Variable length of 999 characters, preceded by three digit length indicator

All fields after the table length are in binary format.

Purpose: Identifies the data sent in the EMV card to terminal interface.

Processing Rules: Required in all EMV transactions.

EMV data elements include a series of TLV (Tag/Length/Value) combinations which are required for processing EMV transactions.

All data elements must contain the exact values that were present on the card to terminal interface.

If the same data element appears several times on the card to terminal interface, this field must contain the last value that was present on the card to terminal interface during the transaction.

Note: For the EMV transaction to be processed properly, the Application ID (AID) TLV (tag 9F06) or Dedicated File Name (tag 84) must be included in Data Element No. 190 (EMV Chip Data).
Note: Tag '5A' (PAN) and Tag '57' (Track 2 equivalent data) must not be included in Data Element 190 — EMV Chip Data.

Valid Codes/Values: The components of EMV data elements are described below:

Component Name: Tag

Each sub element or Tag can be either one or two bytes long. To determine the length, check the last five bits of the first byte, if they are all on (1) the tag of the sub element is two bytes long, including the current byte.

Example 1: Tag x'9F10' is a two byte tag - the last five bits of the first byte (x'9F') are all ones.

Example 2: Tag x'72' is a one byte tag - the last five bits of the first byte (x'72') are not all ones.

Component Name: Length (ANSB …127)

The next one or more bytes are the length indicator of the tag. To determine the number of length bytes used for this inspect the first byte of the length field. If the first bit of the first byte is a 1, the remaining bits (next seven) are a binary representation of the number of following bytes of the length field. If the first bit is 0, the remaining bits indicate the actual length of the data that follows.

Example 1: If the length byte is x'0C', then 12 is the length of the data that follows the sub element.

Example 2: If the tag is x'82', the length is contained in the next byte of the sub element. The value in the length byte indicates the number of bytes of data following.

Component Name: Value (ANSB …255)

The data of each Tag/Length/Value combination is up to 255 bytes long, depending on the length of the field specified by the preceding length portion of the sub element.

For an example of this data element, please see Appendix R. EMV Chip Data.

Number:	191	**Name:**	EMV Additional Information Indicator
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 digits	
Purpose:	Identifies the type of EMV additional information transmitted in the Financial Transaction request or response.	
Processing Rules:		
Valid Codes/Values:	**Code**	**Description**
	001	EMV table data

Number:	192	**Name:**	EMV Additional Information Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 digits
Purpose:	Identifies the length of EMV Additional Information.
Processing Rules:	
Valid Codes/Values:	001–985

Number:	193	**Name:**	CA Public Key File Block Length
Character Type:	N	**Maximum Length:**	3 bytes

Representation:	Fixed length of three digits
Purpose:	Identifies the length of the following field, CA Public Key File Block, in the CA Public Key File Load Response.
Processing Rules:	Required in the CA Public Key File Load Response.
Valid Codes/Values:	000—999

Number:	194	**Name:**	CA Public Key File Block
Character Type:	AN	**Maximum Length:**	9,999 bytes

Representation:	Variable length of 9,999 alphanumeric characters
Purpose:	Identifies either the Public Key file in the CA Public Key File Load Response or the Error Message.
Processing Rules:	Required in the CA Public Key File Load Response. The CA Public Key File contains all active CA Public Keys that apply for processing EMV transactions. The CA Public Key File completely replaces the key file previously stored at the device. For an approved EMV load transaction, the maximum length is 9,999 bytes. **Note**: The device concatenates all of the CA Public Key File Block data to construct the CA Public Key File. For a declined EMV load transaction, a message is returned in this field and displays at the device.
Valid Codes/Values:	Any valid key. Following are the error codes that are returned in the terminal:

Code	**Description**
RQST/CURRENT HASH MATCH	Key update not required
BLOCK NBR NOT NUMERIC	Key load transaction not successful
BLOCK NBR NOT 000-xxx	Key load transaction not successful

Please refer to Appendix S. Valid CA Public Key File Record Layout.

Number:	195	**Name:**	CAVV Revised Format
Character Type:	AN	**Max. Length:**	20 bytes

Representation:	Fixed length of 20 bytes	
Purpose:	Identifies the Verified by Visa Data – Revised format.	
Processing Rules:	This Data Element is used only for Visa and Interlink payment tokenized transactions.	
Valid Codes/Values:	**Position**	**Description**
	1-20	Cardholder authentication verification value data (ATN (Authentication Tracking Number) field replaces the need for the XID).

For CAVV cryptogram usage details, refer to the Verified by Visa section in Appendix Y (3-D Secure).

Number:	196	**Name:**	Token Requestor ID
Character Type:	AN	**Max. Length:**	11 bytes

Representation: Fixed length of 11 bytes

Purpose: Identifies merchant's Token Requestor ID.

Processing Rules: This Data Element is used only for Visa/MasterCard payment tokenized transactions.

Valid Codes/Values:

Position	Description
1-11	Token Requestor ID, Left justified, space-filled

Note: In late 2014, MasterCard introduced the Card on File tokenization service and static card on file token type to support token payments for e-Commerce merchants that traditionally stored credit card numbers. MasterCard required merchants to submit the Token Requestor ID for Card on File tokenized transactions. Effective 8 June 2021, MasterCard renamed the Card on File token type to "Static Token" type. Also, they removed the requirement to submit the Token Requestor ID. However, if submitted, MasterCard will validate and will decline if invalid.

Merchants are advised not to send the Token Requestor ID if submitting a Card on File tokenized transaction/Static Token type. Cryptographic Data is still not required for Static Tokens.

Merchants that submit Secure Element and Cloud-based token types are still required to submit the Token Requestor ID along with Cryptographic data, if applicable.

Number:	197	**Name:**	Token PAN Suffix
Character Type:	AN	**Max. Length:**	4 bytes

Representation:	Fixed length of 4 bytes.
Purpose:	Identifies the last 4 digits of the cardholder PAN to be printed on the transaction receipt.
Processing Rules:	This Data Element is returned only for Visa/MasterCard, and Interlink/Maestro tokenized transactions
Valid Codes/Values:	**Position** — **Description** 1-4 — Token PAN Suffix, 4 bytes.

Number:	198	**Name:**	Settlement Type
Character Type:	AN	**Maximum Length:**	1 byte

Representation:	Fixed length 1 alphanumeric character.
Purpose:	Identifies how the transaction was processed – Single message (debit) or Dual message (credit).
Processing Rules:	Required in the Common AID EMV and non-traditional Signature Debit Financial Transaction Response. **Note:** Please check with your First Data Project Manager or Relationship Manager for availability of Signature Debit functionality.
Valid Codes/Values:	Following are the values that are returned in the terminal: **Value** — **Description** D — Transaction was processed Dual message (credit) S — Transaction was processed Single message (debit) X — Transaction was processed Dual message (non-traditional Signature Debit) **Note:** In EMV transactions, if 'D' is returned in response, then subsequent transactions i.e. completion or void or reversal must be sent as credit. This will be reflected in the Prompt Code (Data segment 100: Data Element 78) of the subsequent transaction. In EMV transactions, if 'S' is returned in response, then subsequent transactions i.e. completion or void or reversal must be sent as debit. This will be reflected in the Prompt Code (Data segment 100: Data Element 78) of the subsequent transaction.

Number:	199	**Name:**	Signature Required
Character Type:	AN	**Maximum Length:**	1 byte

Representation:	Fixed length 1 alphanumeric character.
Purpose:	Indicates if a signature is required for this transaction.
Processing Rules:	Required in the Common AID EMV and non-traditional Signature Debit Financial Transaction Response. **Note:** Please check with your First Data Project Manager or Relationship Manager for availability of Signature Debit functionality.
Valid Codes/Values:	Following are the values that are returned in the terminal:

Value	Description
T	Signature is required
F	Signature is not required
Space	Device software logic determines if signature is required

Number:	200	**Name:**	Receipt Card Description
Character Type:	AN	**Maximum Length:**	10 bytes

Representation:	Fixed length 10 alphanumeric characters.
Purpose:	Contains the text that can be used on the printed receipt to indicate the card type.
Processing Rules:	Required in the Common AID EMV and non-traditional Signature Debit Financial Transaction Response. **Note:** Please check with your First Data Project Manager or Relationship Manager for availability of Signature Debit functionality.
Valid Codes/Values:	A left-justified, space filled, fixed length text value that indicates the card type of the transaction. For example, a Star Northeast authorized transaction would contain the following text “STAR“. This can be used to print in the receipt to indicate the card type.

Number:	201	**Name:**	Fuel Volume Data
Character Type:	AN	**Variable Length:**	3,600 bytes

Representation:	Variable length of up to 3,600 alphanumeric characters.
Purpose:	Provides the fuel volume data for merchant location.
Processing Rules:	Appears in Proprietary Data Load Segment (Segment No. 118) when Prompt Code 905 (Proprietary Load Response Data) is sent in the request.

Valid Codes/Values:

Number: 202 **Name:** Cryptogram Token Data

Character Type: B64 **Variable Length:** 28/56 bytes

Representation: Fixed length of 28 or 56 bytes.

Purpose: Identifies the cryptogram token data A and B from Merchant. Merchant can send only token data block A or can send both Token data block A and B in the request.

Processing Rules: This data element is being used for In App payment tokenization transactions.

Valid Codes/Values:

Position	Description
1-28	Cryptogram token data block A – 28 bytes in base64 format
1-56	Cryptogram token data block A and B - 56 bytes in base64 format

Refer the below table for Base64 data format representation:

Base64 Data Format Representation			
B64	Base64	a-z,A-Z, 0-9,+,/,=	Base64 Encoded data

Number: 203 **Name:** Safekey Data

Character Type: B64 **Maximum Length:** 58 bytes

Representation: Fixed length of 58 bytes.

Purpose: Identifies SafeKey cryptogram indicator

Processing Rules: Used in SafeKey payment tokenization transaction. Appears in Data Segment No. 123 (NFC Payment Tokenization Data Segment).

Valid Codes/Values:

Position	Description
1-2	Fixed value: “SK” Indicates SafeKey cryptogram transaction
3-30	SafeKey AEVV (American Express Verification Value) in Base64 format (Should be space filled when not applicable)
31-58	SafeKey AESK (American Express SafeKey) value in Base64 format (Should be space filled when not applicable)

Refer the below table for sample data format representation:

Sample Data Format				
eCommerce Secure ID (Data Segment 111 Table ID 030 Position 9-10 - POS Condition Code)	**Indicator**	**AEVV**	**AESK**	**Sample Data**
25	SK	Mandatory	Optional	SK<Mandatory 28 bytes of AEVV><Optional 28 bytes of AESK>
26	SK	Mandatory	Optional	SK<Mandatory 28 bytes of AEVV><Optional 28 bytes of AESK>
27	SK	Optional	Optional	SK<Optional 28 bytes of AEVV><Optional 28 bytes of AESK>
Note: If Optional data is received as spaces from merchants, AEVV, AESK values will be in Base 64 format.				

Number: 204 Name: SafeKey Response

Character Type: AN Maximum Length: 1 byte

Representation: Fixed length of 1 byte.

Purpose: Identifies SafeKey Cryptogram AEVV response value

Processing Rules: Indicates the AEVV validation result and appears in Data Segment No. 123 (NFC Payment Tokenization Data Segment)

Valid Codes/Values:

Value	Description
0	Reserved for future use
1	AEVV Failed – Authentication Issuer Key
2	AEVVPassed – Authentication Issuer key
3	AEVVPassed – Attempt, Issuer Key
4	AEVV Failed – Attempt, Issuer Key
5	Reserved for future use
6	Reserved for future use
7	AEVV Failed – Attempt, Issuer not participating, Network Key
8	AEVV Passed – Attempt, Issuer not participating, Network Key

9	AEVV Failed – Attempt, Participating, ACS not available, Network Key
A	AEVV Passed – Attempt, Participating, ACS not available, Network Key
B	Reserved for future use
C	Reserved for future use
D	Reserved for future use
U	AEVV Unchecked

Number: 205 **Name:** Load Subtype

Character Type: AN **Maximum Length:** 1 byte

Representation: Fixed length of 1 byte.

Purpose: Identifies the load subtype as Moneris.

Processing Rules: Appears in Moneris Key Load Request

Valid Codes/Values:

Value	Description
M	Moneris Key Load

Number: 206 **Name:** SPDH Header

Character Type: AN **Maximum Length:** 48 bytes

Representation: Fixed length of 48 bytes.

Purpose: Contains merchant, terminal, and transaction data required for MAC Encryption.

Processing Rules: Appears in both Moneris Key Load request and response.

Valid Codes/Values: Refer to Appendix V (Moneris Data layouts) for further details.

Number: 207 **Name:** Moneris Terminal Identifier

Character Type: AN **Maximum Length:** 8 bytes

Representation: Fixed length of 8 bytes.

Purpose: Identifies Moneris terminal in request and response messages.

Processing Rules: Obtained from Moneris during terminal initialization.

Number:	208	**Name:**	Moneris Merchant ID
Character Type:	AN	**Maximum Length:**	13 bytes

Representation:	Fixed length of 13 bytes.
Purpose:	Identifies Moneris merchant in request and response messages.
Processing Rules:	Obtained from Moneris during terminal initialization.

Number:	209	**Name:**	Moneris Response Code
Character Type:	AN	**Maximum Length:**	2 bytes

Representation:	Fixed length of 2 bytes.
Purpose:	Required for MAC Encryption.
Processing Rules:	Obtained from Moneris in transaction response.

Number:	210	**Name:**	MAC
Character Type:	AN	**Maximum Length:**	16 bytes

Representation:	Fixed length of 16 bytes.
Purpose:	Represents MAC for a transaction via algorithm supplied by Moneris.
Processing Rules:	Sent in the transaction request and received in the transaction response for Moneris debit transactions.
Valid Codes/Values:	Any alpha-numeric characters

Number:	211	**Name:**	Moneris Key Indicators
Character Type:	AN	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 bytes followed by up to three 16 byte keys.	
Purpose:	Indicates which keys are included in Moneris Key Load Response.	
Processing Rules:	Set based on presence of keys in transaction response	
Valid Codes/Values:	**Value**	**Description**
	Character 1	“Y” if MAC Encryption key is present

Character 2	"Y" if Data Encryption Key is present
Character 3	"Y" if PIN Encryption Key is present

Number:	212	**Name:**	Moneris Key Data
Character Type:	AN	**Maximum Length:**	16 bytes

Representation:	Fixed length of 16 bytes.
Purpose:	Contains the keys to be used by the terminal for transaction PIN processing.
Processing Rules:	One iteration will be included for each "Y" in Element 211, Moneris Key Indicators.
Valid Codes/Values:	Any alpha-numeric characters

Number:	213	**Name:**	Moneris Data
Character Type:	AN	**Maximum Length:**	100 bytes

Representation:	Variable Length up to 100 bytes.
Purpose:	Identifies all or some of the Moneris data.
Processing Rules:	This data should be in <tag><len><data> format.
Valid Codes/Values:	Any alpha-numeric characters, formatted in <tag><len><data> format.

Number:	214	**Name:**	Batch Number
Character Type:	AN	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 bytes.
Purpose:	Identifies the batch number of a Moneris batch.
Processing Rules:	Supplied by the terminal to identify the batch to Moneris.
Valid Codes/Values:	Number from 000 to 999

Number:	215	**Name:**	Language Indicator
Character Type:	AN	**Maximum Length:**	1 byte

Representation:	Fixed length of 1 byte.
Purpose:	Indicates language from Moneris matrix.
Processing Rules:	Required to identify language in Moneris's format.
Valid Codes/Values:	Refer to Appendix V for the Moneris Language Indicator Matrix.

Number:	216	**Name:**	Response Display
Character Type:	AN	**Maximum Length:**	16 bytes

Representation:	Fixed length of 16 bytes.
Purpose:	Indicates text of response to batch close request from Moneris.
Processing Rules:	Supplied by Moneris in batch close response.
Valid Codes/Values:	Any alpha-numeric characters

Number:	217	**Name:**	Number of Debits
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of 4 bytes.
Purpose:	Indicates number of debit transactions in a batch.
Processing Rules:	
Valid Codes/Values:	0001-9999

Number:	218	**Name:**	Debit Dollar Value
Character Type:	AN	**Maximum Length:**	19 bytes

Representation:	Fixed length of 19 bytes with leading +/- and two implied decimal places.
Purpose:	Indicates dollar value of all debit transactions in a batch.
Processing Rules:	
Valid Codes/Values:	(+/-)000000000000000000-(+/-)999999999999999999

Number:	219	**Name:**	Number of Credits
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of 4 bytes.
Purpose:	Indicates number of credit transactions in a batch.
Processing Rules:	
Valid Codes/Values:	0001-9999

Number:	220	**Name:**	Credit Dollar Value
Character Type:	AN	**Maximum Length:**	19 bytes

Representation:	Fixed length of 19 bytes with leading +/- and two implied decimal places.
Purpose:	Indicates dollar value of all credit transactions in a batch.
Processing Rules:	
Valid Codes/Values:	(+/-)000000000000000000-(+/-)999999999999999999

Number:	221	**Name:**	Number of Corrections
Character Type:	N	**Maximum Length:**	4 bytes

Representation:	Fixed length of 4 bytes.
Purpose:	Indicates number of correction transactions in a batch.
Processing Rules:	
Valid Codes/Values:	0001-9999

Number:	222	**Name:**	Corrections Dollar Value
Character Type:	AN	**Maximum Length:**	19 bytes

Representation:	Fixed length of 19 bytes with leading +/- and two implied decimal places.
Purpose:	Indicates dollar value of all correction transactions in a batch.
Processing Rules:	

Valid Codes/Values: (+/-)000000000000000000-(+/-)999999999999999999

Number: 223 **Name:** Inclusive/Exclusive for Tax 1

Character Type: AN **Maximum Length:** 1 byte

Representation: Fixed length of 1 byte.

Purpose: Identifies whether this tax is inclusive or exclusive or not applicable to the product.

Processing Rules: If this tax amount value is "N", both tax type and amount should be skipped.

Valid Codes/Values:

Value	Description
I	Tax is inclusive
E	Tax is exclusive
N	Tax amount not applicable to product

Number: 224 **Name:** Tax Type 1

Character Type: AN **Maximum Length:** 3 bytes

Representation: Fixed length of 3 bytes.

Purpose: Identifies the tax type for this tax item.

Processing Rules: If the Inclusive/Exclusive flag is "N", this field is omitted.

Valid Codes/Values: Valid values are country dependent. For Canada values are:

Value	Description
GST	Goods and Services Tax
HST	Harmonized Sales Tax
PST	Provincial Sales Tax
QST	Quebec Province Sales Tax

Number: 225 **Name:** Tax Amount 1

Character Type: N **Variable Length:** 9 bytes

Representation: Variable length of 9 bytes.

Purpose: Identifies the first tax amount for this product. Decimal places are assumed by the currency of the transaction.

Processing Rules: If the Inclusive/Exclusive flag is "N", this field should be omitted.

Valid Codes/Values: 000000000-999999999

Number: 226 **Name:** Inclusive/Exclusive for Tax 2

Character Type: AN **Maximum Length:** 3 bytes

Representation: Fixed length of 3 bytes.

Purpose: Identifies whether this tax is inclusive or exclusive or not applicable to the product amount in the Product Code Data Segment.

Processing Rules: If this tax amount value is “N”, both tax type and amount should be skipped.

Valid Codes/Values:

Value	Description
I	Tax is inclusive
E	Tax is exclusive
N	Tax amount not applicable to product

Number: 227 **Name:** Tax Type 2

Character Type: AN **Maximum Length:** 3 bytes

Representation: Fixed length of 3 bytes.

Purpose: Identifies the tax type for this tax item.

Processing Rules: If the Inclusive/Exclusive flag is “N”, this field is omitted.

Valid Codes/Values: Valid values are country dependent. For Canada values are:

Value	Description
GST	Goods and Services Tax
HST	Harmonized Sales Tax
PST	Provincial Sales Tax
QST	Quebec Province Sales Tax

Number: 228 **Name:** Tax Amount 2

Character Type: N **Variable Length:** 9 bytes

Representation: Variable length of 9 bytes.

Purpose: Identifies the second tax amount for this product. Decimal places are assumed by the currency of the transaction.

Processing Rules: If the Inclusive/Exclusive flag is “N”, this field should be omitted.
Valid Codes/Values: 000000000-999999999

Number: 229 **Name:** Inclusive/Exclusive for Tax 3
Character Type: AN **Maximum Length:** 1 byte

Representation: Fixed length of 1 byte.
Purpose: Identifies whether this tax is inclusive or exclusive or not applicable to the product amount in the Product Code Data Segment.
Processing Rules: If this tax amount value is “N”, both tax type and amount should be skipped.
Valid Codes/Values:

Value	Description
I	Tax is inclusive
E	Tax is exclusive
N	Tax amount not applicable to product

Number: 230 **Name:** Tax Type 3
Character Type: AN **Maximum Length:** 3 bytes

Representation: Fixed length of 3 bytes.
Purpose: Identifies the tax type for this tax item.
Processing Rules: If the Inclusive/Exclusive flag is “N”, this field is omitted.
Valid Codes/Values: Valid values are country dependent. For Canada values are:

Value	Description
GST	Goods and Services Tax
HST	Harmonized Sales Tax
PST	Provincial Sales Tax
QST	Quebec Province Sales Tax

Number: 231 **Name:** Tax Amount 3
Character Type: N **Variable Length:** 9 bytes

Representation: Variable length of 9 bytes.

Purpose:	Identifies the third tax amount for this product. Decimal places are assumed by the currency of the transaction.
Processing Rules:	If the Inclusive/Exclusive flag is "N", this field should be omitted.
Valid Codes/Values:	000000000-999999999

Number:	232	**Name:**	Download Data
Character Type:	AN	**Variable Length:**	100 bytes

Representation:	Variable length up to 100 bytes.
Purpose:	Identifies all or some of the Download Data
Processing Rules:	
Valid Codes/Values:	Any alpha-numeric data

Number:	233	**Name:**	RID
Character Type:	AN	**Maximum Length:**	10 bytes

Representation:	Fixed length of 10 bytes.
Purpose:	Identifies EMV application.
Processing Rules:	
Valid Codes/Values:	A valid EMV Registered Application Provider identifier

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Number:	234	**Name:**	Stand-in Indicator
Character Type:	N	**Maximum Length:**	1 byte

Representation:	Fixed length of 1 byte.
Purpose:	Indicates stand in indicator.
Processing Rules:	

Valid Codes/Values:

Value	Description
1	No Stand-in
2	Domestic only Stand-in
3	Domestic & Foreign Stand-in

Number:	235	**Name:**	Floor Limit
Character Type:	N	**Maximum Length:**	12 bytes

Representation:	Fixed length of 12 bytes.
Purpose:	Indicates maximum allowable stand-in value.
Processing Rules:	
Valid Codes/Values:	000000000000-999999999999

Number:	236	**Name:**	BUYPASS RID Card Type
Character Type:	AN	**Maximum Length:**	3 bytes

Representation:	Fixed length of 3 bytes.
Purpose:	Indicates BUYPASS card type associated with RID.
Processing Rules:	Appears in Data Segment No.DL8, EMV Terminal Floor Limits Data Segment
Valid Codes/Values:	

Number:	237	**Name:**	TAVV Cryptogram
Character Type:	AN	**Maximum Length:**	28 bytes

Representation:	Fixed length of 28 bytes.
Purpose:	Identifies the Merchant's TAVV Cryptogram data for payment token-based card on file, Digital Secure Payment Cryptogram, e-commerce and application based e-commerce transactions.
Processing Rules:	Appears in Data Segment No.123, NFC Payment Tokenization Data Segment. For TAVV cryptogram usage in a Visa transaction, refer to the Verified by Visa section in Appendix Y (3-D Secure). For TAVV cryptogram usage in a MasterCard transaction, refer to the MasterCard Digital Secure Remote Payment section in Appendix Y (3-D Secure).

Number:	238	**Name:**	TAVV Result Code
Character Type:	AN	**Maximum Length:**	1 byte

Representation: Fixed length of 1 byte.

Purpose: This field will be populated as appropriate based on the TAVV cryptogram.

Processing Rules: Appears in Data Segment No.123, NFC Payment Tokenization Data Segment.

Valid Codes/Values: TAVV response values:

Value	Description
1	TAVV Cryptogram failed validation
2	TAVV Cryptogram passed validation
3	DTVV or Visa defined format Cryptogram failed validation
4	DTVV or Visa defined format Cryptogram passed validation

Number:	239	**Name:**	Enhanced Fleet Data
Character Type:	AN	**Maximum Length:**	999 bytes

Representation: Variable length of up to 999 alphanumeric characters

Purpose: Contains fleet data that applies to enhanced fleet offerings such as Wex OTR, Visa Fleet 2.0, MasterCard Enhanced Fleet EMV, and Voyager EMV

Processing Rules: Used in the Enhanced Fleet Data Segment (No 145). When Element No. 239 (Enhanced Fleet Data) has a value of '004' (Prompt Data Request), the list of enhanced prompts and the associated information is included here.

Valid Codes/Values: Refer to Data Segment 145— Enhanced Fleet Request Format

Note: Visa Fleet 2.0 is a new Fleet EMV Standard that harnesses the additional capabilities of EMV at the fuel pump in a solution that is straightforward for fuel merchants to implement. In addition to more driver prompt options, this functionality enables commercial clients to restrict

card use at fuel Merchants to certain fuel and non-fuel product categories. For example, cards that can be used to purchase diesel fuel or used for EV charging.

Availability Notice: This functionality for MasterCard Enhanced transactions is expected to be enabled in First Data production no earlier than June 2026. Until then, this field is intended for development and certification preparation only and should not be used for live production transactions.

Number:	240	**Name:**	Available Product Information
Character Type:	AN	**Variable Length:**	17 bytes

Representation:	Variable length up to 17 bytes
Purpose:	Appears in Data Segment 148 - WEX Available Product Fleet Information. Identifies available product/fleet information from WEX.
Processing Rules:	If WEX Available Product Fleet Indicator is set to '4' during request, this segment is returned to the terminal during response.
Valid Codes/Values:	Any alphanumeric data and special characters are allowed.

Number:	241	**Name:**	Price Data
Character Type:	AN	**Variable Length:**	999 bytes

Representation:	Variable length of up to 999 alphanumeric characters
Purpose:	Appears in Data Segment 149 - Fuel Price Update Request Segment. Identifies Fuel Price Data.
Processing Rules:	Unique to the Fuel Price Update Request Segment (No 149). Contains the Cash and Credit unit prices for the location's primary Diesel #2 fuel.
Valid Codes/Values:	Refer to Data Segment 149— Fuel Price Update Request Format

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Number:	242	**Name:**	EV Charging Data
Character Type:	AN	**Variable Length:**	200 bytes

Representation:	Variable length of up to 200 alphanumeric characters
Purpose:	Appears in Data Segment 156 - EV Charging Data Segment. Identifies EV charging transactions.
Processing Rules:	Unique to the EV Charging Data Segment (No. 156). Contains the EV charging data.
Valid Codes/Values:	Refer to Data Segment 156 - EV Charging Data Segment.

Number:	243	**Name:**	Extended Unit of Measure
Character Type:	A	**Variable Length:**	4 bytes

Representation:	Fixed length of 4 alpha characters
Purpose:	Identifies the measurement code for a Non-Fuel product in Data Segment No. 102, Product Code Data Segment especially for MasterCard Enhanced Fleet EMV transactions.
Processing Rules:	This data element, along with the following data elements, is repeated for up to 10 Non-Fuel products in Data Segment No. 102, Product Code Data Segment. Used in the Variable Information Data Segment (No. 111) in Table ID 78 with sub-table ID 14. **Availability Notice:** This functionality for MasterCard Enhanced transactions is expected to be enabled in FirstData production no earlier than June 2026. Until then, this field is intended for development and certification preparation only and should not be used for live production transactions.

Valid Codes/Values:

Codes	Description
ACRE	Acre
ARES	Are
CELI	Centiliter
CMET	Centimeter
EACH	Each
FOOT	Foot

GBGA	GBGallon
GBOU	GBOunce
GBPI	GBPint
GBQA	GBQuart
GRAM	Gram
HECT	Hectare
INCH	Inch
KILO	Kilogram
KMET	Kilometer
LITR	Liter
METR	Meter
MILE	Mile
MILI	Milliliter
MMET	Millimeter
PIEC	Piece
PUND	Pound
SCMT	Square Centimeter
SMET	Square Meter
SMIL	Square Millimeter
SQFO	Square Foot
SQIN	Square Inch
SQKI	Square Kilometer
SQMI	Square Mile
SQYA	Square Yard
TONS	Ton
USGA	USGallon
USOU	USOunce
USPI	USPint
USQA	USQuart
YARD	Yard

14. Support, Testing, and Certification

This chapter discusses the following topics:

- Project management
- Development support
- General testing
- Certification

14.1 Project Management

BUYPASS uses a project management process. One of the main objectives of the process is to provide the best definition possible of the customer's requirements. A customer project is submitted by a sponsor for approval. When a project has been approved, a project leader is assigned. The project sponsor identifies a customer representative who is included in meetings and receives project updates. The customer representative works with the project sponsor to construct the requirement document and identify critical benchmarks. The customer representative also works with the functional areas involved in the project itself. The actual management of the project takes place at BUYPASS.

14.2 Development Support

Support is available for BUYPASS customers before, during, and after implementation of the interface between a customer's device and BUYPASS.

14.3 General Testing

BUYPASS provides a QA test system for the customer, vendor, or third-party processor to test new software code, modifications/enhancements to existing software code, communications, host interface functions, and other general testing prior to scheduling certification time.

The client is assigned a test BUYPASS Merchant Number and access number to interface to the test system. Support by BUYPASS QA analysts is available.

General testing includes the following areas:

- Telecommunications
  - ◊ Connections
  - ◊ Line down
- Message formats
  - ◊ Requests
    - Transaction types
    - Card types
  - ◊ Responses
    - Authorizer-generated for approved transactions
    - Authorizer-generated for denied transactions
    - Transactions denied by BUYPASS

14.4 Certification

A customer, vendor, or third-party processor is required to submit a Request for Certification form once general testing is completed. This form can be obtained by contacting the BUYPASS QA department. It provides the BUYPASS QA department with the pertinent information needed to coordinate and schedule certification time.

The certification process includes test plans and scripts, designed for the specific application requirements of the customer, vendor, or third-party processor. A BUYPASS QA analyst is assigned to the certification and provides support throughout the process.

The scope of the certification may also require regression testing. Regression testing is a variation of certification testing.

Appendix A. TCP/IP Message Header

TCP/IP is a streaming protocol, meaning that data received may consist of part of a message or parts of more than one message.

Example. Message 1: “Hello, I am message one.”
Message 2: “Message 2 is my name.”
Message 3: “A third message I am.”

Each time a receive is performed, one or more parts of these messages may be received:

Example. Receive results 1: “Hello, I”
Receive results 2: “ am message one.Mes”
Receive results 3: “sage 2 is my name.A thi”
Receive results 4: “rd message I am.”

With no header or delimiter between each message, it is not possible to determine where one message ends and another begins. A simple delimiter such as a carriage return character would work well for messages that consist entirely of printable text:

Example. Message 1: “Hello, I am message one.\n”
Message 2: “Message 2 is my name.\n”
Message 3: “A third message I am.\n”

BUYPASS has chosen the standard header method as described below.

BUYPASS Header		
Field Name	**Size**	**Description**
Message Length	2 bytes (16 bits)	Length of the following data. High order byte is first (network or “big-endian order”).
TPDU Protocol ID	1 byte	0x60 (Always use this value.)
TPDU Destination Address	2 bytes (16 bits)	Reserved for internal use.
TPDU Source Address	2 bytes (16 bits)	Reserved for internal use.

The message length includes the length of all subsequent data in the message including the TPDU area, but does not include the two bytes of the message length itself. The message length is binary data, represented in network (a.k.a. little-endian) order. Be aware that some processor architecture, such as the Intel® processors used on PC's store 16-bit integers in "little-endian" order, so the bytes should be reversed before sending or after receiving.

Example. A 4660-byte message would have a length of *0x1234* in hexadecimal. The first byte in the header should contain *0x12* and the second byte should contain *0x34*. Be aware that a 16-bit integer with a value of *4660* may be represented internally on your architecture as *0x3412*, so moving this integer directly into or out of the message length field would not work.

The sockets library on your system may have built-in library functions to convert to and from network order and the local host order on your machine.

Note: The TCP/IP socket aborts without notification.

Also be aware that the streaming nature of TCP may cause your application to receive the beginning of a message that contains only one byte of the length. The actual length of the next message will not be known until more data, containing the second byte of the length, is received.

The TPDU area is reserved for internal use. Set all four bytes to *0x00* in messages to BUYPASS.

Testing may be performed against BUYPASS certification system using the private BUYPASS IP network.

Note: Check with BUYPASS Communications Support at the time of installation for the IP address to be used.

Appendix B. POS Purchase Example

This appendix contains an example of a POS purchase that includes the following items:

- Credit Partial Approval support
- 4.170 gallons of unleaded fuel at $1.199 per gallon (self-serve)
  Note: A unique Product Code must be sent for each type of fuel purchase.
- 10 packs of cigarettes at $1.79 per pack (6% tax)
- Card verification information request

The parsed message format data is:

ATL105 ▲ 03 ▲ 100 ▲ 078 ▲ 0 ▲ 1234567890123 ▲ 020 ▲ 12341234567890123456 ▲ 1200 ▲ ▲ 01 ▲ 500 ▲ 1790 ▲ 97 ▲ ▲ 000001 ▲ ▲ ▲ 1 ▲ 102 ▲ 049 ▲ S 02 001 G 34170 \ 31199 \ 500 \ 411 U 010 \ 2179 \ 1790 ▲ 111 ▲ 054 ▲ 004 003 969 010 030 00000000000000000000VAR001 ▲

where:

- spaces have been added for clarification;
- ▲ represents a Field Separator; and
- \ is a Product Data Field Delimiter

The following table describes the data layout for the above information:

POS Purchase Data Layout

Data Section No. 1				
Field Data in Example	**Len. in Ex.**	**Max. Len.**	**Element Name**	**Description**
ATL105	6	6	Message Format Version Identifier	Identifies the message format type [ATL = Device/Controller (default)] and release version (105 = 1.05).
<Field Separator>				
03	2	2	Number of Segments	Identifies the number (3) of segments in the message.
<Field Separator>				

POS Purchase Data Layout (*continued*)

Data Section No. 2 Standard Message Data Segment (Data Segment No. 100)				
Field Data in Example	**Len. in Ex.**	**Max. Len.**	**Element Name**	**Description**
100	3	3	Segment Type	This value (100) identifies the Standard Message Data Segment.
<Field Separator>				
078	3	3	Segment Length	Identifies the length (78 bytes) of the Standard Message Data Segment, including field separators.
<Field Separator>				
0	1	1	Information Byte	Identifies the request as a single-message (0) request.
<Field Separator>				
1234567890123	13	22	Terminal Identifier	Identifies the device by Device Type, State Code, BUYPASS Merchant Number, and Device Number.
<Field Separator>				
020	3	4	Prompt Code	Identifies the Transaction Type as a point-of-sale purchase/capture (0) and the Card Type as a credit card type (20).
<Field Separator>				
1234123456789 0123456	20	24	Account Number	Identifies the portion of the Account Number used for manual entry.
<Field Separator>				
1200	4	51	Card Discretionary Block Data	Identifies the card's expiration date (12/00) used in manual entry.
<Field Separator>				
	0	36	Encrypted PIN Block Data	
<Field Separator>				
01	2	2	Pump/Lane Number	Identifies the pump or lane number (No.1) at the point-of-sale location.
<Field Separator>				
500	3	8	Fuel Purchase Amount	Identifies the total dollar amount ($5.00) of all fuel purchases.
<Field Separator>				

POS Purchase Data Layout (*continued*)

Data Section No. 2 (*continued*) **Standard Message Data Segment (Data Segment No. 100)** (*continued*)				
Field Data in Example	**Len. in Ex.**	**Max. Len.**	**Element Name**	**Description**
1790	4	8	Nonfuel Amount	Identifies the total dollar amount ($17.90) of all nonfuel, noncash, and nontax purchases.
<Field Separator>				
97	2	8	Tax Amount	Identifies the total dollar amount ($.97) of all tax amounts of the transaction.
<Field Separator>				
	0	8	Cash Amount	
<Field Separator>				
000001	6	6	Sequence Number	Identifies the unique transaction sequence number (000001) assigned to and associated with the transaction throughout its life.
<Field Separator>				
	0	6	Approval Number	Identifies the assigned Approval Number.
<Field Separator>				
	0	10	Local Date and Local Time	Identifies the local date and local time of the preauthorized transaction.
<Field Separator>				
1	0	1	Partial Approval Indicator	Indicates whether or not a merchant supports credit/debit Partial Approvals.
<Field Separator>				

POS Purchase Data Layout (*continued*)

Data Section No. 3 Product Code Data Segment (Data Segment No. 102)				
Field Data in Example	**Len. in Ex.**	**Max. Len.**	**Element Name**	**Description**
102	3	3	Segment Type	This value (102) identifies the Product Code Data Segment.
<Field Separator>				
049	3	3	Segment Length	Identifies the length (49 bytes) of the Product Code Data Segment, including Field Separators and Product Data Field Delimiters.
<Field Separator>				
S	1	1	Service Level	Identifies the type of sale as a self-serve (S) sale.
02	2	2	Number of Products	Identifies the count (2) of products reported for this transaction.
001	3	3	Product Code	Identifies unleaded fuel (001) as the first product sold.
G	1	1	Unit of Measure	Identifies gallons (G) as the type of measurement used for selling unleaded fuel.
34170	5	9	Quantity	The leading number indicates the number of assumed decimal places. Identifies 4.170 gallons as the number of units of unleaded fuel sold.
<Product Data Field Delimiter>				
31199	5	9	Unit Price	The leading number indicates the number of assumed decimal places. Identifies $1.199/gallon as the price per unit of measure for selling unleaded fuel.
<Product Data Field Delimiter>				
500	3	12	Product Amount	Identifies the monetary value ($5.00) of the product sold.
<Product Data Field Delimiter>				
411	3	3	Product Code	Identifies the type of product sold as cigarettes (411).
U	1	1	Unit of Measure	Identifies the type of measurement used for selling cigarettes as unit (U).

POS Purchase Data Layout (*continued*)

Data Section No. 3 (*continued*)
Product Code Data Segment (Data Segment No. 102) (*continued*)

Field Data in Example	Len. in Ex.	Max. Len.	Element Name	Description
010	3	9	Quantity	The leading number indicates the number of assumed decimal places. Identifies the number of units (10 packs) of cigarettes sold.
<Product Data Field Delimiter>				
2179	4	9	Unit Price	The leading number indicates the number of assumed decimal places. Identifies the price per unit of measure ($1.79/pack) for the product sold.
<Product Data Field Delimiter>				
1790	4	12	Product Amount	Identifies the monetary value ($17.90) of the cigarettes sold.
<Field Separator>				

POS Purchase Data Layout (*continued*)

Data Section No. 3 (*continued*)
Variable Information Data Segment (Data Segment No. 111)

Field Data in Example	Len. in Ex.	Max. Len.	Element Name	Description
111	3	3	Segment Type	This value (111) identifies the Variable Information Data Segment.
<Field Separator>				
054	3	3	Segment Length	Identifies the length (017 bytes) of the Variable Information Data Segment, including Field Separators.
<Field Separator>				
004	3	3	Variable Information Indicator	Identifies the content of Variable Information as card verification value (CVV, CVV2, CVC2, CID, DTVV) information.
003	3	3	Variable Information Length	Identifies the length of the content of Variable Information.
969	3	999	Variable Information	Identifies the card's actual verification (CVV, CVV2, CVC2, CID, DTVV) information.
010	3	3	Variable Information Indicator	Identifies the content of Variable Information as Visa® Trusted Agent Program (TAP) Indicator information.
030	3	3	Variable Information Length	Identifies the length of the content of Variable Information.
0000000000000000000000VAR001	30	999	Variable Information	Identifies the assigned VAR (value added reseller) ID.
<Field Separator>				

Appendix C. Valid Authorizer Codes

The following table contains a list of valid entries for Element No. 7, Authorizer Code:

Note: This list includes some codes that are not in Production as of the date of this publication. Please contact a BUYPASS representative for additional information.

Valid Authorizer Codes		
Authorizer Code	**Authorizer Name**	**Authorizer Type**
00	Other	Other
01	American Express	Credit
02	Visa®	Credit
03	MasterCard®	Credit
04	Discover® Network	Credit
05	Valero	Proprietary
06	PULSE®	Debit
07	STAR® Network Southeast	Debit
08	Certegy℠	Check
09	ECA/ TeleCheck® Service	Check
11	Moneris (Canada)	Credit
12	GE Consumer Finance	Other
13	GECC Wright Express	Other
14	BUYPASS	Internal
15	Fleetcor2	Fleet
16	InComm	Stored value/Phone/Gift
17	STAR® Network Northeast	Debit
18	Safeway	Stored value/Phone/Gift
19	Accel®	Debit
20	Texaco®	Proprietary
21	FleetCor	Fleet
22	NetSpend	Stored Value
23	STAR® Network West	Debit
24	ESI Loyalty (Canada)	Stored Value
25	Chase Acquiring (ChaseNet)	Credit
26	Chase Acquiring (VisaNet)	Credit
27	Everywhere Gift Card	Stored value/Gift
28	NYCE®	Debit
29	Chase Payment Tech	Credit
30	Cardlock (BP PayPoint)	Fleet
31	Acormida (BP PayPoint)	Fleet
32	Citi	Proprietary
33	COMDATA Fleet	Fleet
34	Sun Oil	Proprietary

Valid Authorizer Codes		
Authorizer Code	**Authorizer Name**	**Authorizer Type**
35	First Data℠ Loyalty solutions	Loyalty
36	ECA/TeleCheck® Debit	Proprietary
37	HSBC Direct	Proprietary
38	A Toda Hora Banco Popular®	EBT
39	Car Care	Proprietary
40	Maestro® Direct	Debit
41	Bank of America® PayPoint	Debit
42	Wells Fargo PayPoint	Debit
43	Bank of America® Direct	+Debit
44	ConnectPay℠	Debit
45	Wells Fargo Direct	+Debit
46	PayPoint	Stored value/Gift
47	PayPoint	Check
48	Interlink®	Debit
49	HRS	Stored value/Gift
50	Walmart MDEX	WIC
52	Bank of America®/Deluxe	EBT
53	Transactive EBT (Texas)	EBT
54	The Home Depot® Gift Card	Stored value/Gift
55	First Data™ Standard Gift Card	Stored value/Gift
56	HSBC PayPoint	Proprietary
57	WEX™ ISO	Fleet
58	Shazam	Debit
60	Deluxe	EBT
62	Transactive EBT (Illinois)	EBT
63	A Toda Hora Global	EBT
64	Northeast Coalition of States	EBT
65	Western States EBT Association	EBT
66	Southern Alliance of States	EBT
67	Lockheed Martin℠ INS	EBT
68	Credit Union 24®—FID	Debit
69	Accel®	Debit
70	A Toda Hora Banco Popular®	Debit
71	Stored Value Systems	Stored value/Gift
72	Wright Express	Fleet
73	Gascard®	Fleet
74	Nevada	eWIC
75	Kentucky	eWIC

Valid Authorizer Codes		
Authorizer Code	**Authorizer Name**	**Authorizer Type**
77	Sinclair®	Proprietary
78	A Toda Hora Global	Debit
79	ACS® Direct	EBT
80	BUYCheck	Check
81	WEX OTR Fleet	Fleet
82	EDS	EBT
83	ACS® WIC	WIC
84	STAR Single Message	Signature Debit
85	STAR Dual Message	Signature Debit
86	Armed Forces Financial Network® Direct	Other
87	Comdata Corporation	Proprietary
88	Optis (Omaha) Cycle	Debit
89	LML™ Check Service	Check
91	PreSolutions	Phone
92	Alliance Data Systems®	Proprietary
94	Voyager℠	Fleet
95	Unocal Corporation	Proprietary
96	Fleet One	Fleet
97	Jeanie® Direct	Debit
98	Optum	Proprietary

+AUTHFILE in Production has been updated with this information; however, feature is not available as of the release date of this document. Check with your First Data Corporation representative for additional information.

Appendix D. Valid State Codes

This appendix contains the following tables:

- State Codes for the states of the United States of America
- State Codes for the territories of the United States of America
- State Codes for military organizations
- State Codes for the provinces of Canada

Each table contains the following information:

- State Code Description
- Valid alphabetical State Codes
- Valid ANSI® State Codes

The alphabetical State Codes are used in the following data elements:

- Address Line 2 (No. 4)
- Card Discretionary Block Data (No. 12)
- State Code (No. 124)

The ANSI® State Codes are used in the following data element:

- Terminal Identifier (No. 102)

The following table contains a list of valid State Codes for the states of the United States of America:

Valid State Codes United States of America		
State	**Alphabetical Code**	**ANSI® Code**
Alabama	AL	01
Alaska	AK	02
Arizona	AZ	04
Arkansas	AR	05
California	CA	06
Colorado	CO	08
Connecticut	CT	09
Delaware	DE	10
District of Columbia	DC	11
Florida	FL	12
Georgia	GA	13
Hawaii	HI	15
Idaho	ID	16

Valid State Codes United States of America		
State	**Alphabetical Code**	**ANSI® Code**
Illinois	IL	17
Indiana	IN	18
Iowa	IA	19
Kansas	KS	20
Kentucky	KY	21
Louisiana	LA	22
Maine	ME	23
Maryland	MD	24
Massachusetts	MA	25
Michigan	MI	26
Minnesota	MN	27
Mississippi	MS	28
Missouri	MO	29
Montana	MT	30
Nebraska	NE	31
Nevada	NV	32
New Hampshire	NH	33
New Jersey	NJ	34
New Mexico	NM	35
New York	NY	36
North Carolina	NC	37
North Dakota	ND	38
Ohio	OH	39
Oklahoma	OK	40
Oregon	OR	41
Pennsylvania	PA	42
Rhode Island	RI	44
South Carolina	SC	45

Valid State Codes United States of America		
State	**Alphabetical Code**	**ANSI® Code**
South Dakota	SD	46
Tennessee	TN	47
Texas	TX	48
Utah	UT	49
Vermont	VT	50
Virginia	VA	51
Washington	WA	53
West Virginia	WV	54
Wisconsin	WI	55
Wyoming	WY	56

The following table contains a list of valid State Codes for the territories of the United States of America:

Valid State Codes Territories of the United States of America		
Territory	**Alphabetical Code**	**ANSI® Code**
Guam	GU	43
Puerto Rico	PR	14
Virgin Islands	VI	52

The following table contains the valid State Code for military organizations:

Valid State Codes Military ID's		
Military ID	**Alphabetical Code**	**ANSI® Code**
Armed Forces Americas	AA	57
Armed Forces Europe	AE	58
Armed Forces Pacific	AP	59
Military State--Reserved	XX	60

The following table contains a list of valid State codes for the provinces of Canada:

Valid State Codes Provinces of Canada		
Province of Canada	**Alphabetical Code**	**ANSI® Code**
Alberta	AB	61
British Columbia	BC	62
Manitoba	MB	63
New Brunswick	NB	64
Newfoundland and Labrador	NL	65
Nova Scotia	NS	66
Northwest Territories	NT	67
Nunavut	NU	76
Ontario	ON	68
Prince Edward Island	PE	69
Quebec	QC	70
Saskatchewan	SK	71
Yukon	YT	72

The following table contains valid state code that should only be used for locations outside of US and Canada.

Valid State Codes Non-US and Non- Canada		
Province of Canada	**Alphabetical Code**	**ANSI® Code**
NA	NA	00[1]

[1]Indicates that Table ID: 041 (Country Subdivision) is included in the Variable Information data segment. Country subdivision field is not validated. Country subdivision value of the terminal is based on ISO3166-2 standard.This field should be included when using “00” as numeric state code in Data Element No. 102 (Terminal Identifier). Otherwise, it would result in transaction decline.

Appendix E. Valid Card Type Codes

The values for Element No. 14 (Card Type) can be used alone—as a unique data element—or preceded by a valid Transaction Type code (See Appendix G. Valid Transaction Type Codes.) to create a valid Prompt Code (Element No. 78).

This appendix contains all Card Type codes separated into the following tables describing their use:

- Valid Card Type Codes Used in the Table Load Response
- Valid Card Type Codes Used in Financial Transaction Prompt Codes
- Valid Card Type Codes Used in Special Transaction Prompt Codes

The following table contains Card Type codes, sent in the Card Type file in the Table Load response to the device, that identify card types the device accepts (001–086) and turn on specific features at the device (127–168):

Note: Code Nos. 001–086 are also used in 4-character Prompt Codes. Code Nos. 127–168 are not used in any Prompt Codes.

Valid Card Type Codes
Used in the Table Load Response

Code	Card Type	Card Type Description
001	BUYPASS fleet	Fleet
011	Debit checking	Debit and Follow-Up Signature Debit Single Message **Note:** Original Signature Debit Dual and/or Single Message transactions are always processed as Credit/Signature Debit Dual Message transaction and follow-up Signature Debit transactions are processed as Debit/Signature Debit Single Message transaction when Data Element 198 (Settlement Type) has a value of 'S'.
012	Debit saving	Debit
013	Debit ACH	Debit
020	Visa Fleet Credit	Fleet Credit and Signature Debit Dual and Single Message Original Transaction **Note:** In the case of a bill payment/recurring payment transaction, this Card Type code must be used.
040	Loyalty	Proprietary
041	Certegy (Formerly Telecredit)	Check
045	Generic check	Check
046	ECA/ TeleCheck® Service (ECA and Paper Warranty/ Verification)	Check

Valid Card Type Codes Used in the Table Load Response		
Code	**Card Type**	**Card Type Description**
050	Valero Energy Corporation	Proprietary
051	Shell	Proprietary
052	ExxonMobil Proprietary	Proprietary
053	ExxonMobil Fleet	Fleet
055	Valero UCC	Proprietary
056	Generic proprietary	Proprietary
057	Sohio/Gulf	Proprietary
058	Cenex	Proprietary
059	Wright Express	Fleet
060	Voyager℠	Fleet
061	Car Care One	Proprietary
064	Tesoro Petroleum Corporation	Proprietary
070	Valero Fleet	Fleet
071	Fleet One	Fleet
072	Tesoro UCC	Proprietary
073	EBT food stamps	EBT
074	EBT cash benefits	EBT
075	Citgo fleet	Proprietary
077	Unocal	Proprietary
078	Phone	Stored value
079	Stored value	Stored value
080	MasterCard® fleet	Fleet
081	Fuelman/Gascard®	Fleet
084	Sinclair®	Proprietary
085	Sunoco	Proprietary
086	eWIC	EBT
088	EBT	EBT childcare provider payment
090	Conoco	Proprietary
091	NGFC Card	NGFC (Next Generation Fleet Card)
092	WEX OTR Card	WEX OTR card
093	Money Code	WEX OTR Fleet (Money Code)
094	COMDATA Card	COMDATA card
095	Comdata Express Check	COMDATA Fleet (Check)
096	Comdata ComCheck Check	COMDATA Fleet (Check)
127	Address verification system	For manually keyed credit card sales transactions, configures the device to prompt for the cardholder's ZIP Code™/ZIP+4®.

Valid Card Type Codes Used in the Table Load Response		
Code	**Card Type**	**Card Type Description**
148	PAN truncation	For the customer's copy of a credit card sales receipt, truncates the account number to four digits and does not print the expiration date.
150	Debit Convenience Fee	Instructs the terminal to prompt for Debit Convenience Fee before sending a debit transaction
162	Address verification system	For manually keyed credit card sales transactions, configures the device to prompt for the cardholder's address.
163	Address verification system	For manually keyed credit card sale transactions, configures the device to prompt for the cardholder's address and ZIP Code™/ZIP+4®.
164	Auto close	Enables the Auto Close feature that closes the batch 30 minutes prior to the default cut time.
165	Gift card	Allows manually keyed gift card transactions.
166	Disable a function	Disables the Return/Refund function.
167	Disable a function	Disables the Void/Reversal function.
168	Disable a function	Disables the Preauthorization function except for proprietary cards.
171	BUYPASS TAP	Enables BUYPASS TAP function.
173	Store and Forward	Enables Store and Forward blocking for a designated range of time each day.
174	Card verification data	For manually keyed credit card sale transactions, configures the device to prompt for the card's verification data (CVC2. CVV2, CID) for inside sales only.

The following table contains Card Type codes sent by the device to the host as part of the 4-character Prompt Code used in financial transaction processing:

Note: These codes are preceded by a valid Transaction Type to create a valid Prompt Code. (See Appendix G. Valid Transaction Type Codes.)

Valid Card Type Codes Used in Financial Transaction Prompt Codes		
Code	**Card Type**	**Card Type Description**
001	BUYPASS fleet	Fleet
011	Debit checking	Debit
012	Debit saving	Debit
013	Debit ACH	Debit
020	Credit	Credit
040	Loyalty	Proprietary
041	Certegy (Formerly Telecredit)	Check
045	Generic check	Check
046	ECA/ TeleCheck® Service (ECA and Paper Warranty/ Verification)	Check
050	Valero Energy Corporation	Proprietary
051	Shell	Proprietary
052	ExxonMobil Proprietary	Proprietary
053	ExxonMobil Fleet	Fleet
055	Valero UCC	Proprietary
056	Generic proprietary	Proprietary
057	Sohio/Gulf	Proprietary
058	Cenex	Proprietary
059	Wright Express	Fleet
060	Voyager℠	Fleet
061	Car Care One	Proprietary
064	Tesoro Petroleum Corporation	Proprietary
070	Valero Fleet	Fleet
071	Fleet One	Fleet
072	Tesoro UCC	Proprietary
073	EBT food stamps	EBT
074	EBT cash benefits	EBT
075	Citgo fleet	Proprietary

Valid Card Type Codes Used in Financial Transaction Prompt Codes		
Code	**Card Type**	**Card Type Description**
077	Unocal	Proprietary
078	Phone	Stored value
079	Stored value	Stored value
080	MasterCard® fleet	Fleet
081	Fuelman/Gascard®	Fleet
083	Cardlock Fuel System, Inc. fleet	Fleet
084	Sinclair®	Proprietary
085	Sunoco	Proprietary
090	Conoco	Proprietary
Note: The Card Type code for Conoco has been identified as '89'. This code is internal for BUYPASS and used ONLY for BUYPASS backend processing. The available prompt code used to identify the Conoco Card Type is '90'. Merchants will use this code along with transaction type in Data Segment No. 100, Standard Message Data Segment to send the transaction to BUYPASS. For a list of valid entries, please see Appendix E. Valid Card Type Codes and Appendix G. Valid Transaction Type Codes.		

The following table contains a list of valid Card Type codes used to create valid 3-character Prompt Codes for special transactions. The device sends them to the host requesting data:

Note: A valid Prompt Code for a special transaction is composed of the 3-character Card Type data element only.

Valid Card Type Codes Used in Special Transaction Prompt Codes		
Code	**Card Type (Special Data)**	**Card Type Description**
900	Communications	Status transaction.
901	Custom receipt text	Retrieve custom text to be printed on all receipts
902	Dynamic card table	Retrieve dynamic card table data.
903	Site configuration data	Submit site configuration data.
904	Host discount data	Retrieve host discount data by product code for AFP transactions.
905	Fuel volume data	Submit fuel volume data.
981	Electronic mail	Retrieve nonproprietary electronic mail.
990	Totals	Single block, fixed-length response. Contains no print data.
995	Electronic mail	Submit electronic mail.
996	Electronic mail	Retrieve proprietary electronic mail.
997	Electronic mail	Reset electronic mail status flag.

Appendix F. Valid Payment Systems Product Codes

This appendix contains the following Product Code tables:

- Unused codes (Code No. 000)
- Motor fuel codes (Code Nos. 001–099)
- Automotive product/service codes (Code Nos. 100–149)
- Aviation fuel codes (Code Nos. 150–174)
- Aviation product/service codes (Code Nos. 175–224)
- Marine fuel codes (Code Nos. 225–249)
- Marine product/service codes (Code Nos. 250–299)
- Other fuel codes (Code Nos. 300–399)
- Merchandise codes (Code Nos. 400–599)
- Reserved (future use) codes (Code Nos. 600–799)
- Debit surcharge codes (Code Nos. 800–809)
- Reserved (proprietary use) codes (Code Nos. 810–889)
- FSA/HRA product codes (Code Nos. 890–894)
- Reserved (proprietary use) codes (Code Nos. 895–899)
- Negative codes (Code Nos. 900–949)
- Administrative codes (Code Nos. 950–999)

These codes are valid entries for Element No. 77, Product Code, which is part of the Product Data section in Data Segment No. 102, Product Code Data Segment.

Any product in these tables should be represented in messages to BUYPASS by the Product Code assigned to it in the following tables.

With regard to fuel purchases, a unique Product Code must be sent for each type of fuel purchase.

The following table contains unused codes:

Payment Systems Product Codes Unused Code Nos. (Code No. 000)	
Product Code	**Description**
000	Not used.

The following table contains motor fuel codes:

Note: A unique Product Code must be sent for each type of fuel purchase.

Payment Systems Product Codes Motor Fuel Codes (Code Nos. 001–099)	
Product Code	**Description**
001	Unleaded (88 octane or less)
002	Unleaded Plus (89–91 octane)
003	Super Unleaded (92 octane or higher)
004	Undefined Unleaded
005	Undefined Unleaded
006	Unleaded Methanol 5.7% Blend (88 octane or less)
007	Unleaded Plus Methanol 5.7% Blend (89–91 octane)
008	Super Unleaded Methanol 5.7% Blend (92 octane or higher)
009	Unleaded Methanol 7.7% Blend (88 octane or less)
010	Unleaded Plus Methanol 7.7% Blend (89–91 octane)
011	Unleaded Ethanol 5.7% Blend (88 octane or less)
012	Unleaded Plus Ethanol 5.7% Blend (89–91 octane)
013	Super Unleaded Ethanol 5.7% Blend (92 octane or higher)
014	Unleaded Ethanol 7.7% Blend (88 octane or less)
015	Unleaded Plus Ethanol 7.7% Blend (89–91 octane)
016	Methanol/Leaded
017	Ethanol/Leaded
018	Leaded
019	Diesel
020	Diesel—Premium
021	Undefined Diesel
022	Compressed natural gas
023	Liquid propane gas
024	Liquid natural gas (compressed)
025	M-85
026	E-85
027	Unleaded—Reformulated (88 octane or less)
028	Unleaded—Reformulated (89–91 octane)
029	Unleaded—Reformulated (92 octane or higher)
030	Undefined Unleaded—Reformulated
031	Undefined Unleaded—Reformulated
032	Diesel—Off-road/Farm Fuel (nontaxable)
033	Diesel—Refrigerator Fuel (nontaxable)
034	Undefined Diesel (nontaxable)

Payment Systems Product Codes Motor Fuel Codes (Code Nos. 001–099)	
Product Code	**Description**
035	Other fuel (nontaxable)
036	Racing fuel
037	Super Unleaded Methanol 7.7% Blend (92 octane or higher)
038	Unleaded Methanol 10% Blend (88 octane or less)
039	Unleaded Plus Methanol 10% Blend (89–91 octane)
040	Super Unleaded Methanol 10% Blend (92 octane or higher)
041	Upper Unleaded Ethanol 7.7% Blend (92 octane or higher)
042	Unleaded Ethanol 10% Blend (88 octane or less)
043	Unleaded Plus Ethanol 10% Blend (89–91 octane)
044	Super Unleaded Ethanol 10% Blend (92 octane or higher)
045	B2 Diesel Blend 2% Biodiesel
046	B5 Diesel Blend 5% Biodiesel
047	B10 Diesel Blend 10% Biodiesel
048	B11 Diesel Blend 11% Biodiesel
049	B15 Diesel Blend 15% Biodiesel
050	B20 Diesel Blend 20% Biodiesel
051	B100 Diesel Blend 100% Biodiesel
052	Ultra Low Sulfur #1
053	Ultra Low Sulfur #2
054	Ultra Low Sulfur Premium Diesel #2
055	Ultra Low Sulfur Biodiesel Blend 2%
056	Ultra Low Sulfur Biodiesel Blend 5%
057	Ultra Low Sulfur Biodiesel Blend 10%
058	Ultra Low Sulfur Biodiesel Blend 11%
059	Ultra Low Sulfur Biodiesel Blend 15%
060	Ultra Low Sulfur Biodiesel Blend 20%
061	Ultra Low Sulfur Biodiesel Blend 100%
062	DEF (Diesel Exhaust Fluid)
063	Premium Diesel #1
064	Unleaded Ethanol (15% Blend)
065	Unleaded Plus Ethanol (15% Blend)
066	Super Unleaded Ethanol (15% Blend)
067	Premium Diesel Blend <20% Biodiesel
068	Premium Diesel Blend >= 20% Biodiesel
069	B75 Diesel Blend 75% Biodiesel
070	B99 Diesel Blend 99% Biodiesel

Payment Systems Product Codes Motor Fuel Codes (Code Nos. 001–099)	
Product Code	**Description**
071-098	Undefined fuel
099	Miscellaneous fuel

The following table contains automotive product/service codes:

Payment Systems Product Codes Automotive Product/Service Codes (Code Nos. 100–149)	
Product Code	**Description**
100	General automotive merchandise
101	Motor oil
102	Car wash
103	Oil change
104	Oil filter
105	Work order
106	Antifreeze
107	Washer fluid
108	Brake fluid
109	Tires
110	Federal excise tax (tires)
111	Tire rotation
112	Batteries
113	Lube
114	Inspection
115	Labor
116	Towing
117	Road service
118	Auto accessories
119	Auto parts
120	Preventative maintenance
121	Air conditioning service
122	Engine service
123	Transmission service
124	Brake service
125	Exhaust service
126	Body work
127	Automotive glass
128	Synthetic oil
129	Lamps
130	Wipers
131	Hoses
132	Tire-related (wheel balance, value stem)
133	Repairs
134	Service packages

Payment Systems Product Codes Automotive Product/Service Codes (Code Nos. 100–149)	
Product Code	**Description**
135	Automotive parking
136	Truck tank cleaning
137	Other lubricants
138	Automotive fuel additives/treatment (injected)
139	Vehicle Rental
140	Air Filter
141	Vehicle Prep
142	Fuel System
143–148	Undefined part/service
149	Miscellaneous part/service

The following table contains aviation fuel codes:

Note: A unique Product Code must be sent for each type of fuel purchase.

Payment Systems Product Codes Aviation Fuel Codes (Code Nos. 150–174)	
Product Code	**Description**
150	Jet Fuel
151	Aviation Fuel 1
152	Aviation Fuel 2
153	Aviation Fuel 3
154	Aviation Fuel 4
155	Aviation Fuel 5
156–173	Undefined aviation fuel
174	Miscellaneous aviation fuel

The following table contains aviation product/service codes:

Payment Systems Product Codes Aviation Product/Service Codes (Code Nos. 175–224)	
Product Code	**Description**
175	Storage
176	Aircraft ground handling
177	Aircraft ground power unit
178	Aircraft labor
179	Aircraft work order
180	Aircraft maintenance
181	Aircraft service
182	Transportation
183	Deicing
184	Ramp fees
185	Catering
186	Hangar fee
187	Landing fee
188	Call out fee
189	Aircraft rental
190	Instruction fee
191	Flight plans/weather brief
192	Charter fee
193	Communication fee
194	Aircraft cleaning
195	Cargo handling
196	Aircraft accessories
197	Pilot Supplies
198	Aircraft Parking Fees
199	Aircraft Tiedown Fees
200	Aircraft Sanitation Fees
201	Aircraft Fuel Additive
202	AC Parts
203	Oxygen
204	De-fuel
205	Re-service
206	Static Dissipater Additive
207	Corrosion Inhibitor
208	Airport Fees
209	Overtime Fees

Payment Systems Product Codes Aviation Product/Service Codes (Code Nos. 175–224)	
Product Code	**Description**
210	IT/Bladder
211	Ground Equipment Service Fees
212	Secure Fees
213	Flow Fee
214–223	Undefined aviation
224	Miscellaneous aviation

The following table contains marine fuel codes:

Note: A unique Product Code must be sent for each type of fuel purchase.

Payment Systems Product Codes Marine Fuel Codes (Code Nos. 225–249)	
Product Code	**Description**
225	Marine Fuel 1
226	Marine Fuel 2
227	Marine Fuel 3
228	Marine Fuel 4
229	Marine Fuel 5
230	Marine—Other
231–248	Undefined marine fuel
249	Miscellaneous marine fuel

The following table contains marine product/service codes:

Payment Systems Product Codes Marine Product/Service Codes (Code Nos. 250–299)	
Product Code	**Description**
250	Marine services
251	Marine labor
252	Marine work order
253	Launch fee
254	Slip rental
255–298	Undefined marine service
299	Miscellaneous marine service

The following table contains other fuel codes:

Payment Systems Product Codes Other Fuel Codes (Code Nos. 300–399)	
Product Code	**Description**
300	Kerosene
301	White gas
302	Heating oil
303	Bottled propane (LPG)
304	Other fuel (nontaxable)
305	Kerosene -- Ultra Low Sulfur
306	Kerosene -- Low Sulfur (Non-taxable)
307	Kerosene -- Ultra Low Sulfur (Non-taxable)
308	EVC-1 – Level 1 charge = 110v 15 amp
309	EVC-2 – Level 2 charge = 240v 15-40 amp
310	EVC-3 – Level 3 charge = 480v 3 phase charge
311	Biodiesel Blend 2% Off-Road
312	Biodiesel Blend 5% Off-Road
313	Biodiesel Blend 10% Off-Road
314	Biodiesel Blend 11% Off-Road
315	Biodiesel Blend 15% Off-Road
316	Biodiesel Blend 20% Off-Road
317	Diesel #1 Off-Road
318	Diesel #2 Off-Road
319	Diesel #1 Premium Off-Road
320	Diesel #2 Premium Off-Road
321	Additive Dosage
322	Unleaded Ethanol Blends E16-E84
323	Low Octane Unleaded
324	Blended Diesel (#1 and #2)
325–398	Undefined other fuel
399	Miscellaneous other fuel

The following table contains merchandise codes:

Payment Systems Product Codes Merchandise Codes (Code Nos. 400–599)	
Product Code	**Description**
400	General merchandise
401	General ice
402–409	General undefined
410	General tobacco
411	Cigarettes
412	Tobacco—Other
413–419	Undefined tobacco
420	General packaged beverage
421	Packaged beverage (nonalcoholic)
422	Juice
423	Other packaged beverage
424–429	Undefined packaged beverage
430	General dispensed beverage
431	Hot dispensed beverage
432	Cold dispensed beverage
433	Frozen dispensed beverage
434	Other dispensed beverage
435–439	Undefined dispensed beverage
440	General snack
441	Salty snack
442	Alternative snack
443	Sweet snack—Packaged
444–449	Undefined snack
450	General candy
451–459	Undefined candy
460	General dairy
461	Fluid milk product
462	Packaged ice cream/novelty
463	Other dairy
464–469	Undefined dairy
470	General grocery
471	Groceries—Edible
472	Groceries—Nonedible
473	Groceries—Perishable
474	Bread—Packaged

Payment Systems Product Codes Merchandise Codes (Code Nos. 400–599)	
Product Code	**Description**
475	Frozen food
476–479	Undefined grocery
480	General alcohol
481	Beer—Alcoholic
482	Beer—Nonalcoholic
483	Wine
484	Liquor
485–489	Undefined alcohol
490	General deli
491	Packaged sandwich/deli product
492	Prepared food
493	Deli item
494–499	Undefined deli
500	General food service
501–509	Undefined food service
510	General lottery
511	Lottery—Instant
512	Lottery—Online
513	Lottery—Other
514–519	Undefined lottery
520	General money order
521	Money order—Vendor payment
522	Money order—Payroll check
523	Money order—Gift certificate
524	Money order—Refund check
525	Money order—Official check
526	Money order—Rebate check
527	Money order—Dividend check
528	Money order—Utility check
529	Undefined money order
530	General store service
531	Home delivery
532	Prepaid card—Purchase
533	Prepaid card—Activation/Recharge
534	Membership/Loyalty
535–539	Undefined store service

Payment Systems Product Codes Merchandise Codes (Code Nos. 400–599)	
Product Code	**Description**
540	General health and beauty care
541–549	Undefined health and beauty care
550	General publication
551–559	Undefined publication
560	PIN Activate Prepaid Card
561	PIN Return Prepaid Card
562	Enable Device/Handset Unlock
563	Disable Device/Handset Lock
564	3rd Party Prepaid Card Activate
565	3rd Party Prepaid Card Reload
566	Financial Prepaid Card Activate
567	Financial Prepaid Card Reload
568	Proprietary Prepaid Card Activate
569	Proprietary Prepaid Card Reload
570	General Purpose Activate
571	General Purpose Reload
572	Real Time Recharge
573	Wireless Real Time Recharge
574	Single Payee Bill Pay
575	Multiple Payee Bill Pay
576-590	Undefined Prepaid and Bill Pay
591–599	Undefined merchandise

The following table contains codes reserved for future use:

Payment Systems Product Codes Codes Reserved for Future Use (Code Nos. 600–799)	
Product Code	**Description**
600	DEF (Diesel Exhaust Fluid)
601	B99
602	B100
603	Additive
604	Kerosene
605	Propane
606-623	Undefined Packaged Fuels – Reserved for PCATS Future Use
624	Miscellaneous Packaged Fuels
625-649	Reserved for PCATS Future Use
650	Scales
651	Shower
652	Tire Repair
653	Lodging
654	Wash Out
655	Trailer Wash
656	RV Dump Fee
657	EV Charging Fee
658-699	Undefined Vehicle Product/Services – Reserved for PCATS Use
700-799	Reserved for PCATS Future Use

The following table contains debit surcharge codes:

Payment Systems Product Codes Debit Surcharge Codes (Code Nos. 800–809)	
Product Code	**Description**
800	Administrative—Debit surcharge **Note:** BUYPASS Product Code 800 is applicable for both credit (Visa/ MasterCard) and debit transactions.
801	Administrative—BUYPASS default tax
802	Administrative—BUYPASS default cash
803	Merchandise—BUYPASS default nonfuel/noncash/nontax
804	Merchandise—Other merchandise
805	Merchandise—Prepaid card
806	Merchandise—Split tender; use of prepaid card
807	Motor Fuels—Special
808	Motor Fuels—Extra
809	Negative Transaction—Tax exemption credit

The following table contains codes reserved for proprietary use:

Payment Systems Product Codes Codes Reserved for Proprietary Use (Code Nos. 810–889)	
Product Code	**Description**
Code Nos. 810–889 are reserved for proprietary use.	

The following table contains FSA/HRA Product codes:

Payment Systems Product Codes FSA/HRA Product Codes (Code Nos. 890–896)	
Product Code	**Description**
890	Clinical services
891	Dental services
892	Prescription/Rx only
893	Vision/Optical
894	Total qualified healthcare products (QHP)
895	Co-Payment
896	Transit

The following table contains codes reserved for proprietary use:

Payment Systems Product Codes Codes Reserved for Proprietary Use (Code Nos. 897–899)	
Product Code	**Description**
Code Nos. 897–899 are reserved for proprietary use.	

The following table contains negative codes:

Notes: If the transaction involves a fuel purchase, a unique Product Code must be sent for each type of fuel involved in the transaction.

Discounts and coupons are applied first to nonfuel purchase amounts.

Payment Systems Product Codes Negative Codes (Code Nos. 900–949)	
Product Code	**Description**
900	Discount 1
901	Discount 2
902	Discount 3
903	Discount 4
904	Discount 5
905	Coupon 1
906	Coupon 2
907	Coupon 3
908	Coupon 4
909	Coupon 5
910	Lottery pay out—Instant
911	Lottery pay out—Online
912	Lottery pay out—Other
913	Split tender
914	Tax Discount/Forgiven
915–948	Undefined negative
949	Miscellaneous negative administrative

The following table contains administrative codes:

Note: If a transaction involves a fuel purchase, a unique Product Code must be sent for each type of fuel purchase.

Payment Systems Product Codes Administrative Codes (Code Nos. 950–999)	
Product Code	**Description**
950	Tax 1
951	Tax 2
952	Tax 3
953	Tax 4
954	Tax 5
955	Cash back
956	Cash back fee
957	Fee 1
958	Fee 2
959	Fee 3
960	Fee 4
961	Fee 5
962	Miscellaneous Aviation Tax
963	GST/HST (Canadian)/VAT 1
964	PST/QST (Canadian) VAT 2
965	SWT Rate (Canadian)
966	Tax 6
967	Tax 7
968	Tax 8
969	Jet Federal Excise Tax
970	AvGas Federal Excise Tax
971–998	Undefined administrative
999	Miscellaneous administrative

Appendix G. Valid Transaction Type Codes

This appendix contains a list of valid Transaction Type codes.

Note: This note is specific to merchants processing MasterCard transactions:

Transactions routed to MasterCard will be defined as Pre Auth or Final as per MasterCard's request. The BUYPASS transactions submitted as Authorization/Completion/Auth Cancellation will be presented as Preauthorization* to MasterCard and transactions submitted as Sale/Sale Reversals/Merchandise Return/Void of Merchandise Return will be presented as Final Authorization** to MasterCard.

* Preauthorization is a request for an estimated amount, the amount of the original authorization may be adjusted if the final transaction amount is different than the original authorization amount requested, or the transaction might not be completed for reasons other than technical failure or lack of full issuer approval.

** Final authorization is a request for an amount greater than zero, where the final transaction amount is known, and the transaction is not expected to be canceled after the authorization request is approved in full.

MasterCard is revising the chargeback protection period for authorizations. Following are the revised chargeback protection period:

- 30 calendar days for MasterCard authorizations properly identified as preauthorization
- 7 calendar days for all other MasterCard authorizations
- Authorization holds must not exceed 30 or 7 calendar days durations as mentioned above

Note: For a list of valid Card Type codes, please see Appendix E. Valid Card Type Codes.

Prompt Codes are transmitted by a device at the time a transaction is initiated. A valid Prompt Code (Element No. 78) is composed of a Transaction Type code and a Card Type code.

Note: Any Prompt Code beginning with Transaction Type code *9* (See table below.) denotes a special transaction, not a financial transaction. Transaction Type code *9* can only be combined with Card Type codes *00* and *81–97*.

The Transaction Type code (one character) identifies the type of transaction being processed.

The following table contains a list of valid 1-character Transaction Type codes:

Valid Transaction Type Codes		
Code	**Transaction Type**	**Transaction Description**
0	POS Purchase/Capture	Point-of-sale purchase/preauthorized completion **Note:** This Transaction Type code is also used to complete an AFP prepay transaction initiated from an inside POS device.
3	POS Authorization Only	Point-of-sale authorization only (no draft capture)
4	Customer-activated Purchase/Capture	Customer-activated purchase completion
5	Customer-activated Authorization Only	Customer-activated authorization only (no draft capture)
6	Mail/Phone Purchase	Mail/Telephone order

Valid Transaction Type Codes		
Code	**Transaction Type**	**Transaction Description**
7	Merchandise return	Merchandise return/refund
8	Purchase reversal	Purchase reversal/void
9	Special Transactions	Special transactions such as totals, downloads, communications test, electronic mail, and special device prompts.
A	Account Verification	Account verification
B	Mail/Phone Authorization Only	Mail/Telephone authorization only
C	Mail/Phone Reversal	Mail/Telephone reversal/void
D	RBC Loyalty Lookup	Identify the loyalty number associated with an RBC card.
E	Balance Inquiry	Balance Inquiry—Credit, EBT cash benefits and stored value card
K	Activate	Activate a stored value card.
L	Deactivate	Deactivate a stored value card.
M	Balance Merge	Merge the balances of two stored value cards.
N	Replace Card	Replace a stored value card.
Q	Recharge Card	Recharge a stored value card.
S	Cancellation	Cancels a stored value card, credit Authorization Only transaction, and debit Authorization Only transaction.
T	Token Registration	TransArmor® token registration
U	Void of a Merchandise Return	Void of a merchandise return
V	Loyalty advice function	The following loyalty advice functions are supported: • Add account • Coupon redemption • Expiration Date update • Account inquiry • Points redemption • Reversal of coupon redeem • Reversal of points redeem • Sale update • Account update • Totals report, loyalty
Z	Time-out Reversal	Time-out reversal

Appendix H. Decline Codes Permitting Clerk/Customer Intervention

For some (Element No. 26), a device can be programmed to display a decline message prompting intervention by a clerk/customer.

This appendix contains the following tables:

- Decline Codes prompting data reentry
- Decline Codes identifying a timed out transaction
- Decline Codes identifying a declined Visa® recurring payment transaction
- Decline Code identifying a declined TransArmor Token Only transaction

Each table contains the following information about the Decline Code:

- Fast Code (Decline Code)
- Authorizer
- Decline Code description
- Device display

The following table contains a list of valid Decline Codes that can be used to prompt a clerk/customer to reenter indicated data:

Note: The device stores previously entered data. The clerk/customer needs to enter only the data indicated in the prompt.

Decline Codes Prompting Data Reentry			
Fast Code	**Authorizer**	**Device Display**	**Description**
!3	EBT	`INVALID PIN !3`	Invalid PIN; please retry.
!W	EBT	`INV VCHR DATA !W`	Invalid voucher/authorization number.
:H	GASC	`INVALID PIN :H`	Invalid person identifier number.
2D	HOST	`INVALID PIN 2D`	Invalid PIN.
3A	FLT	`REENTR EMPL # 3A`	Invalid employee number.
3F	FLT	`REENTR PIN NO 3F`	Bad PIN.
CB	TEXC	`INVALID PIN CB`	Invalid PIN.
FZ	DSCV	`INVALID PIN FZ`	Incorrect PIN.
I1	INAS	`INVLD ID I1`	Invalid ID number.
I2	INAS	`INVLD DRIVER I2`	Invalid driver number.
I3	INAS	`INVLD VEHICLE I3`	Invalid vehicle number.
KS	BASE	`INVALID PIN KS`	Incorrect PIN.
W2	GECC	`INV DRIVER W2`	Invalid driver ID.
W3	GECC	`INV VEHICLE W3`	Invalid vehicle ID.

The following table contains a list of Decline Codes identifying timed out transactions that should be resubmitted:

Decline Codes Identifying Timed-out Transactions			
Fast Code	**Authorizer**	**Device Display**	**Description**
4I	AR	PLEASE RETRY 4I	Transaction timed out.
6C	DIAL	TRY AGAIN 6C	Transaction timed out.

The following table contains a list of Decline Codes identifying a declined Visa® recurring payment transaction:

Decline Codes Identifying Declined Visa® Recurring Payment Transactions			
Fast Code	**Authorizer**	**Device Display**	**Description**
K=	Visa®	NOT APPROVED K=	The merchant must stop recurring payments.
K#	Visa®	NOT APPROVED K=	The merchant must not resubmit the same transaction.

The following table contains a Decline Code identifying a declined TransArmor transaction:

Decline Code Identifying Declined TransArmor Transaction			
Fast Code	**Authorizer**	**Device Display**	**Description**
4M	AR	PLEASE RETRY 4M	This Decline Code value can be used to indicate that BUYPASS has not been able to retrieve a token for the associated PAN for a TransArmor transaction.

Appendix I. Variable Information Data Layouts

This appendix contains data layout information for the following variable information used in conjunction with Element No. 111 (Variable Information Indicator).

- Universal Product Codes (UPC) information
- Program Identifier information
- Address verification service (AVS) information
- Card verification (CVV, CVV2, CVC2, CID, DTVV) information
- Point-of-service entry mode information
- Currency code information
- Standard Industrial Classification (SIC) information
- Balance information
- Partial Approval Indicator information
- Visa® Trusted Agent Program (TAP) information
- Tax Collector information
- Market-Specific Data Indicator information
- Special Payments Indicator information
- User data information
- Discover® Network Retrieval Reference Number information
- Visa payWave™ card information
- Suspected fraud reversal indicator information
- Card acceptor terminal identifier information
- Recharge Type Information
- Top Up Phone Number Information
- Settlement Data Acceptance Flag
- Originating Device Type Indicator
- Manual Check MICR Type
- Gift Card Number
- Order ID
- Customer Name
- Use Unique ID
- Universal Unique ID
- POS Condition Code
- Point-of-Service Terminal Capability Code
- POS Additional Data Information
- Re-Price Data Request Information
- Name Verification Service (NVS) Information
- Verified By Visa® Data
- MasterCard UCAF Data
- MCX Checkout Token
- Paydiant Tender ID
- Fraud Enhanced Data
- Final Amount Indicator
- Country Subdivision

- Cardholder Language
- Cardholder Postal Code (for AVS)
- Original Transaction Identifier
- Previous Transaction ID
- First Auth Amount
- Soft Descriptor – Merchant Name
- MasterCard Authentication Data
- Merchant Supplementary Data
- Additional Security Code
- Additional Card ID
- Additional TransArmor Data
- Remote Commerce Acceptor Identifier
- Merchant Tran ID
- Expérience Provider Value
- Stored Credentials Transacton Additonal Data
- Soft Descriptor – Merchant City
- Soft Descriptor – Merchant State
- Soft Descriptor – Merchant Country
- Real Time Account Updater Request Data
- Cardholder Additional Information
- Service Location Information
- Merchant Payment Gateway ID Data
- Digital Commerce Data
- Anticipated Amount
- EV- Additional Transaction Fee 1
- EV- Additional Transaction Fee 2
- Anticipated Completion Amount
- Enabler Verification Value
- Transaction Link Identifier
- Future Transaction Amount
- Discount Program Data
- Discount Line-Item Data
- Discount Amount
- Line-Item
- Visa Commercial Card Data
- Staged Wallet Id
- Visa Gateway Id
- Incremental Auth Data

The following table contains the following layout information for each type of variable information:

- Indicator name
- Data field attributes
- Data field description:

Variable Information Data Layouts		
UPC Information		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 001
Table Length	n	Variable
Table Data	n	Max. Len.: 35 **Note**: All Blackhawk transactions require Variable Information (Data Element No. 111) Table ID: 001, UPC Information. All Blackhawk PIN on Receipt requires Variable Information (Data Element No. 111) Table ID: 002, Program Identifier Information with a value of SP2. Safeway/Blackhawk requires UPC on all transaction types. If included in an InComm transaction, the value should represent an 11-12 digit UPC. The data in this field will be forwarded to InComm verbatim.
Program Identifier Information **Note**: Before implementing this information, contact your BUYPASS representative to discuss requirements.		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 002
Table Length	n3	Variable

Variable Information Data Layouts		
Program Identifier Information (continued)		
Table Data	an3	Max. Len.: 3 CHG = Chicago Gift CIT = ECA/ TeleCheck® Service CASH-IT ECA= ECA/ TeleCheck® Service(Electronic Check Authorizer) IDT = IDT Phone Card INC = InComm Stored Value Card Mall Program KWS = KwikSystems, LLC PPL = Visa® Prepaid PWC = Pennsylvania WIC SP2 = Safeway/Blackhawk Stored Value Card Mall Program (all transactions including PIN on Receipt and Top Up) SPK = Solspark Gift (EveryGift) SPP = Safeway/Blackhawk Stored Value Card Mall Program (all transactions excluding PIN on Receipt and Top Up) STB = Starbucks SVS = Stored Value Systems VAL = First Data™ Standard Gift Card WIC = Women, Infants, and Children WUG = Reserved for Western Union® Stored Value Card Mall Program

Variable Information Data Layouts

AVS Information

Field	Attributes	Description
Table ID	n	Fixed value: 003
Table Length	an	Variable
Table Data	an	Max. Len.: 29 The first 9 characters are always composed of one of the following: • ZIP Code™ (5 characters), followed by 4 spaces, for a total of 9 characters; or • ZIP+4® (9 characters) The remaining 00–20 characters contain street address information.

Card Verification Information

Field	Attributes	Description
Table ID	n	Fixed value: 004
Table Length	n	Variable
Table Data	n	Max. Len.: 4 American Express: 4 bytes Discover® Network: 3 bytes Visa®: 3 bytes MasterCard: 3 bytes **Note:** For Visa, this field is also used to send three-digit DTVV (Dynamic token verification value) cryptogram for Web browser based Auto fill transactions, alongside CVV2 data.

Note:
Web Browser Auto Filling of Tokens (Visa Only):

Cardholders participating in e-commerce transactions may use web browsers that offer auto-fill features, including the ability to auto fill cardholder names, shipping and billing addresses, and payment credentials to facilitate cardholder convenience and reduce the possibility of incorrectly entering account and payment information.

To enable increased security of sensitive cardholder information and authorization rates, certain card brands allow web browsers to replace PANs with e-commerce tokens, which provides cardholders the ability to auto fill token, token expiration date, and three-digit DTVV cryptogram on merchant card entry forms.

To help facilitate seamless payment experiences for cardholders, tokens provisioned by web

browsers will:

- Use a three-digit DTVV cryptogram which is collected by merchants that support CVV2 and sent in the same manner as a CVV2.
- Support transactions without a token cryptogram, for scenarios where the merchant does not collect and submit CVV2.
- When a DTVV is present in the authorization request Visa will validate it and decline the transaction if invalid.
- When a DTVV is not present Visa will send the transaction to the issuer to make the authorization decision without a cryptogram validation result.

Point-of-Service Entry Mode Information		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 005
Table Length	n	3
Table Data	n	Fixed Len.: 3 **Note**: The POS Entry Mode field must be populated for all transactions, which include EMV, Magstripe, and Key Entered transactions from all POS devices for any given solution. Please refer to Appendix J. Valid Point-of-Service Entry Mode Codes for valid entry information.

Variable Information Data Layouts

Currency Code Information		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 006
Table Length	n	Fixed
Table Data	n	Fixed Len.: 3 Identifies the currency of the country in which the transaction originated. [Default value: 840 (United States)]

SIC Information		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 007
Table Length	n	Variable: 4 or 5 or 6
Table Data	n	Max. Len.: 6 Identifies the industry type (SIC) for the business

		establishment involved in the transaction. In case of Discover Third Party Installments, this field represents the MCC/SIC code of third-party payment provider's underlying retailer

Balance Information		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 008
Table Length	n	Variable
Table Data	n	Max. Len.: 19 Identifies the account number for the account that is being reduced to $0.00 as the result of either a Balance Merge (Transaction Type code *M*) or a Replace (Transaction Type code *N*) transaction for a stored value gift card.

Variable Information Data Layouts

Partial Approval Indicator Information

Field	Attributes	Description
Table ID	n	Fixed value: 009
Table Length	an	Fixed Len.: 2
Table Data	an	Max. Len.: 1 Indicates whether or not a merchant supports credit/debit/fleet Partial Approvals. Valid values: 0 = Not supported 1 = Supported 5 = Supports receipt of card balance, but not partial approval

Visa® Trusted Agent Program (TAP) Information

Field	Attributes	Description
Table ID	n3	Fixed value: 010
Table Length	an3	Fixed value: 030 Identifies TAP code for Visa®. Positions 1–10: Visa® BID Value in hexadecimal representation (binary 40; Zero-filled, if not used and Visa TAP or First Data TPP/VAR ID data is included). Positions 11–22: Visa® TAP Value in hexadecimal representation (binary 48; Zero-filled, if not used and Visa BID or First Data TPP/VAR ID data is included). Positions 23–28: First Data Corporation TPP (third party processor)/VAR (value added reseller) ID (alphanumeric 6) (**Note**: Once this ID has been assigned, the vendor must send it to BUYPASS in Data Segment No. 111 [Variable Information Data Segment] of all transaction requests.) Positions 29–30: Reserved for future use (alphanumeric 2; space-filled)

Variable Information Data Layouts

Tax Collector Information

Field	Attributes	Description
Table ID	n3	Fixed value: 011
Table Length	a1	Fixed length of 1 character
Table Data	a	Max. Len.: 1 Indicates whether a transaction is taxable and if it is—whether the sales tax is included. Valid codes: Y = Transaction is taxable; sales tax is included N = Transaction is nontaxable; there is no sales tax Blank = Transaction is taxable; sales tax is not included

Market-Specific Data Indicator Information

Field	Attributes	Description
Table ID	n3	Fixed value: 012
Table Length	a1	Fixed length of 1 character
Table Data	a	Max. Len.: 1 Indicates whether a Market-Specific Data Indicator is present. Valid Codes: M = Medical (healthcare auto-substantiation service) B = Bill Payment E = eCommerce aggregate H = Hotel **Note**: Contact your BUYPASS representative for additional information about supported codes.

Special Payments Indicator Information

Field	Attributes	Description
Table ID	n3	Fixed value: 013
Table Length	a1	Fixed length of 1 character

Variable Information Data Layouts		
Table Data	a	Max. Len.: 1 Indicates whether special payment data is present. Valid Code: R (Recurring payment) I (Installment payment) **Note:** *R* is valid only if Card Type is "credit" (020) and the BIN is either a Visa® or MasterCard® or Discover® or Amex® cards.

User Data Information		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 014
Table Length	n2	Variable
Table Data	an	Max. Len.: 015 Identifies user data information in the request. This user data is not forwarded to the issuer. The data is echoed back in the response.

Discover® Network Retrieval Reference Number Information		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 015
Table Length	an	Fixed
Table Data	an	Len.: 015 Indicates that the Discover® Network Retrieval Reference Number from the associated financial response is included. **Note:** Required for all subsequent merchant initiated Discover Credential on File transactions

Variable Information Data Layouts		
Visa payWave™ Card Information		
Field	**Attributes**	**Description**
Table ID	n2	Fixed value: 016
Table Length	n3	Variable length up to 230
Card Sequence Number	an4	Card sequence number
Usage Tags		
Length of Tags	n3	Total length of all fields
Dataset ID	n2	Value: 01
Dataset Length	n3	Length of Tags minus 5
Amount Authorized Tag	an4	Value: 9F02
Amount Authorized Len	n2	Fixed length: 12
Amount Authorized	n12	Amount used to generate cryptogram
Application Cryptogram Tag	an4	Value: 9F26
Application Cryptogram Len	n2	Fixed length: 16
Application Cryptogram	an16	Hex cryptogram used for authentication
Application Transaction Counter Tag	an4	Value: 9F36
Application Transaction Counter Len	n2	Fixed length: 04
Application Transaction Counter	an4	Transaction count in Hex

Variable Information Data Layouts

Visa payWave™ Card Information *(continued)*

Field	Attributes	Description
Customer Exclusive Data Tag	an4	Value: 9F7C
Customer Exclusive Data Len	n2	Variable length of customer exclusive data
Customer Exclusive Data	an64	Customer Exclusive Data (Hex)
Form Factor Indicator Tag	an4	Value: 9F6E
Form Factor Indicator Len	n2	Fixed length: 08
Form Factor Indicator	n8	Contains indicators about device attributes
Issuer Application Data Tag	an4	Value: 9F10
Issuer Application Data Len	n2	Variable length of Issuer application Data
Issuer Application Data	an64	Issuer application data (hex)
Unpredictable Number Tag	an4	Value: 9F37
Unpredictable Number Len	n2	Fixed length: 08
Unpredictable Number	an8	Hex number used in generating cryptogram

Suspected Fraud Reversal Indicator Information

Field	Attributes	Description
Table ID	n3	Fixed value: 017
Table Length	n3	Fixed length of two digits
Table Data	n2	Fixed value: 34 Indicates suspected fraud in an eCommerce or card not present transaction. It is sent only in a purchase reversal transaction request. This indicator is required by MasterCard®.

Variable Information Data Layouts

Card Acceptor Terminal Identifier Information

Field	Attributes	Description
Table ID	n3	Fixed value: 018
Table Length	n3	Variable, maximum value of 004.
Table Data	n4	Required Indicates the Point-of-Sale (POS) entry point. If the Merchant's Terminal Identifier supports a single POS entry point, this field contains the Terminal Identifier's Device Number. If the Merchant's Terminal Identifier supports multiple POS entry points, the host/controller/device assigns a terminal number to each POS entry point and includes the terminal number in this field. This is a required field for the multi-controller TA-VE transactions.

Recharge Type Information

Field	Attributes	Description
Table ID	n3	Fixed value: 019
Table Length	n3	Fixed value: 002
Table Data	n2	Fixed length: 2 Required for PIN on Receipt and Top Up transactions. 00 = Default Value 01 = PIN on Receipt transaction 02 = Top Up transaction **Note:** All Blackhawk Top Up transactions and Pin on Receipt transactions require Variable Information (Data Element No. 111) Table ID: 002, Program Identifier Information with a value of SP2.

Variable Information Data Layouts

Top Up Phone Number Information

Field	Attributes	Description
Table ID	n3	Fixed value: 020
Table Length	n3	
Table Data	n20	Variable length of up to 20 digits. Required for Top Up transactions. Identifies a valid phone number for Top Up transactions. Example: ATL105+02+100+079+0+H313006519001+ Q079+9877890000004567=4912+++2600+02700+++083003+++1+ 111+070+019+002+02+020+010+4048902815+001+011+07675004 672+ 002+003+SP2+ **Note**: All Blackhawk Top Up transactions require Variable Information (Data Element No. 111) Table ID: 002, Program Identifier Information with a value of SP2.

Settlement Data Acceptance Flag

Field	Attributes	Description
Table ID	an3	Fixed value: 021
Table Length	n3	Value 001
Settlement Data Acceptance Flag	an1	This field is mandatory for requests in which the POS device will accept PINless debit in request and Settlement Data in response. Valid Values: 0 = Device does not accept Settlement Data in response 1 = Device accepts PINless POS Settlement Data only 2 = Device accepts Transaction Attributes Settlement Data and PINless POS Settlement Data X = Original transaction was processed as non-traditional Signature Debit (follow up transactions only) **Note:** Please check with your First Data Project Manager or Relationship Manager for availability of Signature Debit functionality.

Variable Information Data Layouts		
Originating Device Type Indicator Information		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 022
Table Length	n	Fixed Len.: 2
Table Data	n	Maximum Len. :2 Indicates whether originating device type data is present Valid Values: 00 = Card (Default) 01 = Mobile Network Operator (MNO) controlled removable secure element (SIM or UICC) personalized for use with a Mobile Phone or Smartphone 02 = Key fob 03 = Watch 04 = Mobile Tag 05 = Wristband 06 = Mobile Phone Case or Sleeve 07 = Mobile Phone or Smartphone with a fixed (non-removable) secure element controlled by the MNO 08 = Removable secure element not controlled by the MNO, for example, memory card personalized for use with a Mobile Phone or Smartphone 09 = Mobile Phone or Smartphone with a fixed (non-removable) secure element not controlled by the MNO 10 = MNO controlled removable secure element (SIM or UICC) personalized for use with a Tablet or E-Book 11 = Tablet or E-Book with a fixed (non-removable) secure element controlled by the MNO 12 = Removable secure element not controlled by the MNO, for example, (SD Card) personalized for use with a Tablet or E-Book 13 = Tablet or E-Book with fixed (non-removable) secure element not controlled by the MNO

Variable Information Data Layouts		
Originating Device Type Indicator Information(continued)		
Field	**Attributes**	**Description**
		14 = Mobile phone or smartphone with a payment application running in a host processor 15 = Tablet or E-Book with a payment application running in a host processor 16 = Mobile phone or smartphone with a payment application running in the TEE of a host processor 17 = Tablet or E-Book with a payment application running in the TEE of a host processor 18 = Watch with a payment application running in the TEE of a host processor 19 = Watch with a payment application running in a host processor 20 = Card 21 = Phone 22 = Tablet/e-reader 23 = Watch/Wristband 24 = Sticker 25 = PC 26 = Device Peripheral 27 = Tag 28 = Jewelry 29 = Fashion Accessory 30 = Garment 31 = Domestic Appliance 32 = Vehicle 33 = Media/Gaming Device 34 = VR Headset 35 = Card with contact chip and contactless interfaces; no magnetic stripe 36 = Card with contact chip interface only; no magnetic stripe, no contactless interface 37 = Card with contactless interface only; no magnetic stripe, no contact chip interface 38-99 = Reserved for future device types. Any value in this range may occur within devices and transaction data without prior notice
Manual Check MICR Type		

Field	Attributes	Description
Table ID	n3	Fixed value: 024
Table Length	n	Fixed Len.: 2
Table Data	n	MICR Type formats example “T$” T<aba>A<acct>C<checknum> - T123456780A88888888C1234" “18” <aba><acct> “12345678088888888” “09” raw micr (not touched) “t123456780t 8888888888o 0101" “19” <numerals bottom of check “12345678088888888009999”

Gift Card Number		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 025
Table Length	n3	Variable length of up to 19 bytes.
Table Data	an19	Indicates the Gift card number associated with the transaction.

Order ID		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 026
Table Length	n3	Variable length of up to 12 bytes.
Table Data	an12	Indicates the Order ID to identify multiple transactions that are part of a group (example: split tenders)

Customer Name		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 027
Table Length	n3	Variable length of up to 50 bytes
Table Data	an 50	Indicates the customer name

Use Unique ID		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 028
Table Length	n3	Fixed Length: 001
Table Data	an1	Identifies whether to return the Universal Unique Identifier in response. Valid values: 'Y' or 'N' or space The Universal Unique Identifier is returned in the response if a merchant sends the Use Unique Identifier with value 'Y'.
Universal Unique Identifier		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 029
Table Length	n3	Variable length of up to 41 bytes
Table Data	ans41	Universal Unique Identifier (UUID) – This is a unique identifier generated by First Data internal systems that uniquely identifies a transaction throughout the entire First Data infrastructure. **This table must be sent on subsequent transactions if, and only if, the original transaction received a UUID in Segment 112, Table 027. In that case, it must hold the UUID received in the original transaction.**

Variable Information Data Layouts		
POS Condition Code		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 030
Table Length	n3	Len.: 3 bytes
Table Data	an10	Identifies device class, presentation data, security condition, and terminal type. Fixed length of 10 digits. Required for all EMV transactions, all internet based transactions (Ex: Mobile Commerce, ecommerce and In App), and all Credential on File transactions. Optional for all other transactions. This field is sub divided as below: **Pos** **Data** 1–2 Terminal class 4–7 Presentation data 8 Security condition 9-10 Terminal Type **Note:** For Visa, Terminal/Merchant should send Visa CAVV Data when Terminal Type values ‘25’ or ‘26’ are used. **Note:** Please refer to Appendix Q., “Valid National Point-of-Service Condition Codes” for valid entry information.

Variable Information Data Layouts

Point-of-Service Terminal Capability Codes

Field	Attributes	Description
Table ID	n3	Fixed Value: 031
Table Length	n3	Fixed Len.: 2
Table Data	n2	Identifies the terminal capability code of the point-of-sale terminal. Fixed length of 2 digits. Terminal Capability values must be populated for all transactions, which include EMV, Magstripe, and Key Entered transactions from all POS devices for any given solution. Once certified, the respective terminal capability values do not change regardless of the POS Entry Mode. Valid codes/values: 00 = Unknown or unspecified 01 = Terminal not used (Example: eCommerce and In App) 02 = Magnetic stripe read capability (no contact or contactless chip) 03 = Bar code capability 04 = Optical character recognition (OCR) read capability 05 = Integrated circuit card (ICC) read capability 06 = Contactless chip read capability (no contact chip) 07 = Contactless magnetic stripe read capability (no contact or contactless chip capability) 09 = Terminal does not read card data 12 = Hybrid Integrated Circuit Card Reader (Contact Chip & Contactless Chip) Capabilities

POS Additional Information Indicator

Field	Attributes	Description
Table ID	n3	Fixed Value: 032
Table Length	n3	Len.: 3

Variable Information Data Layouts

POS Additional Information Indicator *(continued)*

Table Data	an100	Variable length up to 100 digits. Maximum Len. :100 Indicates whether POS Additional Data is present POS Additional Data contains following sub-table ID's.

Sub Table ID "01"

Purpose: Identifies the Terminal Type information.

Field	Attributes	Description
Sub-Table ID	an2	Fixed value: 01
Sub-Table Length	an3	Fixed Len.: 001
Sub-Table Data	an1	Terminal Type Valid Values: "M" = mPOS Example: 111+021+03200601001M+ **Note:** Applicable only to Amex, MasterCard, Maestro, and Discover. Mobile POS are always attended devices and not cardholder activated.

Sub Table ID "02"

Purpose: Identifies the Product group

Field	Attributes	Description
Sub-Table ID	an2	Fixed Value: 02
Sub-Table Length	an3	Fixed Len.: 004

Variable Information Data Layouts		
Sub-Table Data	an4	Specific to Amex Card Present transactions, where the Goods Sold Data identifies the product being purchased. If the transaction amount contains the purchase of a Gift Card, then the merchant must send a value of “1000” in this field to indicate that a Gift Card is purchased using an Amex Credit Card. Valid Values: “1000” = Gift Card is one of the products purchased Example: 111+029+032015020041000+

Variable Information Data Layouts

POS Additional Information Indicator *(continued)*

Sub Table ID "04"
Purpose: Identifies the Gateway that presents the transaction

Field	Attributes	Description
Sub-Table ID	an2	Fixed Value: 04
Sub-Table Length	an3	Fixed Len.: 001
Sub-Table Data	an1	Valid Values:"1" = FD uComm Gateway Example: 111+042+030010000000002203201204001101001M+

Sub Table ID "05"
Purpose: Identifies If Host Based Prompting is Supported for WEX.

Field	Attributes	Description
Sub-Table ID	an2	Fixed Value: 05
Sub-Table Length	an3	Fixed Len.: 001
Sub-Table Data	an1	Terminal Type Valid Values: "1" = Host Prompts supported "0" = Host Prompts not supported Example: 111+021+032006050011+

Sub Table ID "06"
Purpose: Identifies If NFC (Near Field Communication) Tokenization is Supported

Field	Attributes	Description
Sub-Table ID	an2	Fixed Value: 06
Sub-Table Length	an3	Fixed Len.: 001

Sub-Table Data	an1	Payment Tokenization Support. Valid Values: "1" = NFC Tokenization supported. "2" = NFC Tokenization not supported. Example: 111+021+032006060011+

Variable Information Data Layouts

POS Additional Information Indicator *(continued)*

Sub Table ID "07"

Purpose: Identifies wallet used for a transaction

Field	Attributes	Description
Sub-Table ID	an2	Fixed Value: 07
Sub-Table Length	an3	Fixed Len.: 002
Sub-Table Data	an2	Wallet Identifier Valid Values: "01" = Apple Pay "02" = Android Pay "03" = Samsung Pay "04" = Walmart Pay "05" = Walmart ScanN Go "06" = Sam's Club ScanN Go "07" = Click to Pay "08" = WEX (Wright Express) Wallet Example 1: 111+022+0320070700203+ Example 2: 111+022+0320070700205 - Walmart ScanNgo

Sub Table ID "08"

Purpose: Initial/Credential on File Indicator

Field	Attributes	Description
Sub-Table ID	an2	Fixed Value: 08
Sub-Table Length	an3	Fixed Len.: 001

Variable Information Data Layouts		
Sub-Table Data	a	Valid Values: "I" = Initial transaction for Customer COF establishment. "C" = Credential on File transaction. It is merchant's responsibility to send the indicator as Initial ("I") or Subsequent Credential on File ("C") from the terminal for all COF transactions (Authorization, Completion, Cancellation, Reversal etc.). Example: 111+021+03200608001C+ **Note:** For Visa/MasterCard/Discover/Amex: Value "C" is applicable for both Merchant Initiated COF transaction (MIT) and Customer Initiated COF transaction (CIT). For Customer Initiated COF transaction (CIT), the Transaction Initiation Indicator in Data Segment 111, Table ID 049 (Merchant Supplementary Data), Sub Table ID 02 should be 'C' and for Merchant Initiated COF transaction (MIT), the Transaction Initiation Indicator in Data Segment 111, Table ID 049 (Merchant Supplementary Data), Sub Table ID 02 should be 'M'. For Merchant initiated COF transactions (MIT) where transactions are based on the stand-in instructions with cardholder, Merchants should use the appropriate recurring/installment indicators and store the original transaction identifier to be used in the subsequent transactions. POS Entry Mode 10 should be used for all subsequent COF transactions (when the COF indicator is 'C').
Sub Table ID "09" Purpose: Identifies the Estimated Auth Indicator		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 09
Sub-Table Length	an3	Fixed Len.: 001
Sub-Table Data	an1	Estimated Auth indicator Valid Codes/Values: **Value** **Description** "1" Estimated Auth Indicator Refer to Appendix AE for details on Estimated Auth Indicator.

Sub Table ID "10" Purpose: Identifies wallet method used for a transaction		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 10
Sub-Table Length	an3	Fixed Len.: 001
Sub-Table Data	a	Wallet Method Indicator Valid Codes/Values: "S" = Staged Digital Wallet "P" = Pass-through Wallet Refer Appendix U for more details on Digital Wallet.
Sub Table ID "11" Reserved		
Sub Table ID "12" Purpose: Identifies the Alternate Routing Fallback Indicator for Debit Card Not Present (CNP) Ecommerce transactions.		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 12
Sub-Table Length	n3	Fixed Len.: 001

Sub-Table Data	an1	Alternate Routing Fallback Indicator is used by merchants to fall back on Credit rails when Debit routing is not available for Debit CNP Ecommerce transactions. If debit routing is not available, then the transaction will be rerouted to credit or declined based on the value chosen. ➢ This field is optional with value "Y" if the merchant opts for the ability to route the transaction to the credit association when the transaction cannot be routed to Debit network. ➢ If the transaction needs to be processed through debit only and should not be eligible for credit routing as fallback, then Merchant must send value "N" in this indicator. Valid Code/Value: "Y" – Alternate Routing supported (Transaction can be routed to credit when debit route is not available) "N" – Alternate Routing not supported (Transaction should NOT be routed to credit and should be declined when no debit routes available) Example 1: 111+022+03200612001Y+ Example 2: 111+022+03200612001N+ Note – When merchant is not sending the indicator, the value "Y" will be assumed by default.

Sub Table ID "13"

Purpose: Identifies Cryptocurrency purchase in a transaction

Field	Attributes	Description
Sub-Table ID	n2	Fixed Value: 13
Sub-Table Length	n3	Fixed Length: 001

Sub-Table Data	an1	Cryptocurrency purchase indicator **Value** **Description** "Y" Cryptocurrency purchase **Note** – In order to provide greater visibility into transactions that involve the purchase of cryptocurrency, Visa and MasterCard are mandating this attribute in all the related transactions. Only MCC '6051' is applicable to these transactions. **Value** **Description** "H" High risk security indicator **Note** – In order to provide greater visibility into transactions that involve high securities, MasterCard is mandating this attribute in all the related transactions. Only MCC '6211' is applicable to these transactions.
Sub Table ID "14" Purpose: Identifies Payment Account Reference (PAR) request indicator in a transaction		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 14
Sub-Table Length	n3	Fixed Length: 001
Sub-Table Data	an1	PAR (Payment Account Reference) data request indicator Valid Codes/Values: **Value** **Description** "Y" PAR Data request Merchants should include this indicator if they opt to receive PAR (Payment Account Reference) data from the authorizer.
Note: <Remaining bytes> reserved for future use.		
Re-Price Data Request Information		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 033
Table Length	n3	Variable length of up to 133 alphanumeric characters

Table Data	an15	Lists the different products available at the pump which can be re-priced. A max of 8 products can be sent. The format of the data should be as follows: Nbr of Products: N(2) Product Data: Repeated for number of products sent. Product Code: NAC Product Code of 3 bytes. Re-Price Type: 2 bytes. Valid values are as follows: **Note**: Additional values may be added. **Value** **Description** RP Retail Plus RM Retail Minus CM Cost Minus CP Cost Plus BO Best Of OD Other Discount RT Retail Price FP Fixed Price Unit Of Measure: 1 byte. Same as unit of measure sent in product data. Decimal Identifier Prc: 1 byte. Identifies the number of decimals in the unit price field. Unit Price: Price per unit up to 8 bytes. Delimiter: 1 byte ('\')
		Product Code Set: 3 bytes. Valid values are as follows: **Value** **Description** 001 NACS Example: 111+046+033+029+02001RTG33159\002RTG33259\001+

Name Verification Service (NVS) Information		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 034
Table Length	n3	Length 3 bytes

Table Data	an105	For Discover Card Type: Variable length up to 70 digits. Maximum Len. : 70 Cardholder full name contains the following information: Cardholder First Name (35 characters, left-aligned; space filled) Cardholder Last Name (35 characters, left-aligned; space filled) Example: ATL105+02+100+073+0+H313001682001+3020+6 XXXXXXXXXXXXXX7=4912++++02000+++000020+++1+ 111+087+034+070+FULL NAME For Visa Card Type: Middle Name is expected to follow First and Last Name. Variable length up to 105 digits. Maximum Len. : 105 Cardholder full name contains the following information: Cardholder First Name (35 characters, left-aligned; space filled) Cardholder Last Name (35 characters, left-aligned; space filled) Cardholder Middle Name (35 characters, left-aligned; space filled) Example: ATL105+02+100+073+0+H313001682001+3020+6 XXXXXXXXXXXXXX7=4912++++02000+++000020+++1+ 111+087+034+105+FULL NAME
Verified By Visa® Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 035
Table Length	n3	Length 3 bytes
Table Data	Binary 40	Variable length up to 40 characters. Maximum Len. : 40 Verified by Visa® Data contains the following information: XID (20 characters) CAVV (20 characters) Example: ATL105+02+100+073+0+H313001682001+3020+6011000993030877= 4912++++02000+++000020+++1+ 111+057+035+040+40 bytes of XID and CAVV in binary data format **Note:** When Element No. 111 (Variable Information Indicator) has a value of 035 (Verified By Visa Data), merchant should send the XID and CAVV in binary data format in the request message. For CAVV cryptogram usage details, refer to the Verified by Visa section in Appendix Y (3-D Secure).

Variable Information Data Layouts		
MasterCard UCAF Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 036
Table Length	n3	Length 3 bytes
Table Data	an	Variable length, either 3 or 35 characters. Maximum Len. : 35 MasterCard UCAF Data contains the following information: UCAF Security Level (3 characters). Valid values for UCAF Security Level are: Valid values for Positions 1–2: 21 = Channel encryption; cardholder certificate is not used. This is the preferred value. 24 = Digital secure remote payment with UCAF data transactions. Valid values for Position 3: 0 = UCAF data collection is not supported. 1 = UCAF data collection is supported; however, UCAF data was not populated. 2 = UCAF data collection is supported; and UCAF data is present. 3 = MARP static authentication. 4 = MasterCard Identity Check Insights. Merchant has chosen to share authentication data within authorization. 5 = Issuer risk based decisioning. 6 = Merchant risk based decisioning. 7 = Partial shipment or recurring payment. Liability will depend on the original UCAF values provided and match with the initial transaction. **Note:** As more transactions move from traditional web browsers to digital platforms and devices, the need to capture additional data for sophisticated intelligence is necessary. Enhancements in technology, artificial intelligence, and the EMV 3D Secure standardization have enabled MasterCard to improve their Decision Intelligence product suite. Identity Check Insights, formerly known as “Data only”, allows merchants to share data through the EMV 3DS network to generate authentication insights that could influence the issuer’s decisions to approve without authentication. There is no fraud liability shift to the issuer in an Identity Check Insight transaction message because the issuer is not receiving an authentication request.

		UCAF Data (32 characters) Example: ATL105+02+100+073+0+H313001682001+3020+6011000993030877=4912++++02000+++000020+++1+ 111+052+036+035+4328ABCD1234567890FEDC1234567890+ **Note:** Merchant has to send the UCAF data with subfield ID value as 43 and subfield length as 28 in the total 32 bytes of UCAF data, It is expected that subfield 43 with subfield length should be sent from the merchant in the request message. The UCAF data in this field should be used only for the Accountholder Authentication Value (MasterCard AAV). MasterCard has removed Identity Check values 'h' and 'j' from AAV. Merchant must not send these values going forward.For MasterCard Token and Digital Secure Remote Payment (DSRP) cryptogram, use the TAVV cryptogram field in the segment 123.

Variable Information Data Layouts

MCX Checkout Token

Field	Attributes	Description
Table ID	n3	Fixed value: 037
Table Length	n3	Fixed Value: 040
Table Data	an40	Checkout Token, left justified, space filled.

Paydiant Tender ID

Field	Attributes	Description
Table ID	n3	Fixed value: 038
Table Length	n3	Fixed Value: 012
Table Data	an12	Paydiant Tender ID, left justified, space filled.

Fraud Enhanced Data

Field	Attributes	Description
Table ID	n3	Fixed value: 039
Table Length	n3	Fixed length of 8 bytes.

Table Data	an8	Last registered user profile change date. It must be in DDMMCCYY format.

Final Amount Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 040
Table Length	n3	Fixed value: 001

Variable Information Data Layouts		
Table Data	an1	Indicates that the Settlement Amount may differ from the original Authorization Amount. Allowed on EMV initial transactions where the final Settlement Amount may be different from the original Authorization Amount. **Note:** This data element should NOT be sent in transactions where the final Settlement Amount will be the same as the original Authorization Amount. **Value** **Description** 1 Final transaction amount may change on EMV transactions only

Country Subdivision		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 041
Table Length	n3	Variable
Table Data	an3	Max. Len.: 3 Country subdivision value of the terminal based on ISO3166-2 standard. Should be included when using “00” as numeric state code in Data Element No. 102 (Terminal Identifier). For further details, refer to valid state codes for Non-US and Non-Canada merchants in Appendix D (Valid State Codes).

Cardholder Language		
Field	**Attributes**	**Description**

Table ID	n3	Fixed value: 042
Table Length	n3	Fixed Value: 003
Table Data	an3	Cardholder Language from the ISO 639-2 standard (e.g. – eng = English, fre = French).

Variable Information Data Layouts

Cardholder Postal Code (for AVS)

Field	Attributes	Description
Table ID	n3	Fixed value: 043
Table Length	n3	Variable
Table Data	an3	Max. Len.: 13 Cardholder's Postal Code used in conjunction with AVS for locations outside the US and Canada. Will be sent with international transactions to the appropriate authorizer formatted as they expect. AVS address information is to be included in the normal AVS data item.

Original Transaction Identifier

Field	Attributes	Description
Table ID	n3	Fixed value: 044
Table Length	n3	Fixed Value: 015
Transaction Identifier	an15	Original/Previous transaction identifier Note: Merchant should send the original/previous Visa/Amex Transaction Identifier for merchant-initiated Credential on File transactions. (Example: Recurring/installment transactions initiated by a merchant) The field is required for all merchant initiated Visa and Amex Credential On File transactions.

Examples for Visa Credential on File transactions:

Transaction	Request Table ID 032 and 044 (Variable Information Data)	Response Table ID 028 (Additional Information Data)
Consumer Agreement established for initial installment/recurring transaction. Indicator will be I – stands for Initial. PAN used 4005XXXXXXXX1114	111+021+03200608001I+ Transaction identifier will not be present in this request as it is an initial installment/recurring transaction	028015123456789012345
Installment1/Recurring1 for PAN 4005XXXXXXXX1114 Indicator will be C – stands for Credential on File.	111+043+03200608001C04401512345678901234 5+ (Transaction identifier retain from initial)	028015654321098765432
Installment2/Recurring2 for PAN 4005XXXXXXXX1114 Indicator will be C – stands for Credential on File.	111+043+03200608001C04401565432109876543 2+ (Transaction identifier retain from Installment1/Recurring1)	028015213456123456789

Previous Transaction ID

Field	Attributes	Description
Table ID	n3	Fixed value: 046
Table Length	n3	Variable
Table Data	an3	Max. Len.: 16 For certain back end authorizers, this can be used to contain information that the terminal must include in follow-up transactions.

First Auth Amount

Field	Attributes	Description

Table ID	N3	Fixed Value: 045
Table Length	N3	Length 3 bytes
Table Data	N..12	Variable length up to 12 digits with two assumed decimal places. Purpose: Indicates the first authorized amount in subsequent merchant initiated transactions. In a series of recurring/installment transactions for a Discover card, the merchant should capture the approved amount from the original response, and then present the same value as First Authorization Amount for all subsequent merchant initiated transactions. Ex: 111+026+045012000000004500

Soft Descriptor – Merchant Name

Field	Attributes	Description
Table ID	N3	Fixed Value: 047
Table Length	N3	Length 3 bytes
Table Data	an38	Variable length of up to 38 bytes of Alternate Merchant Name. Discover Third Party Installment transactions – In the Merchant Name field, Third Party Installment Payment Providers must include their Merchant name and the name of the underlying retailer in the format – Third party name*underlying merchant name e.g. “XYZ*A SMALL CO”

MasterCard Authentication Data

Field	Attributes	Description
Table ID	N3	Fixed Value: 048
Table Length	N3	Length 3 bytes Value : 001 or 037 Merchant should send the table length as ‘1’ when sending only Program Protocol, and table length should be ‘37’ when they support both Program Protocol and Directory Server Transaction ID.
Program Protocol	N1	This field is applicable to merchants sending e-Commerce transactions with 3D Secure version or EMV 3D Secure for authentication.

		Valid Code/Value: 1 = EMV 3-D Secure Version 2.1 2 = EMV 3-D Secure Version 2.2 3 = EMV 3-D Secure Version 2.3 4 = EMV 3-D Secure Version 2.4 5 = EMV 3-D Secure Version 2.5 6 = EMV 3-D Secure Version 2.6 7 = EMV 3-D Secure Version 2.7 8 = EMV 3-D Secure Version 2.8 9 = EMV 3-D Secure Version 2.9 Examples: 111+016+0480011+ 111+016+0480012+
Directory Server Transaction ID	ANS36	Directory Server Transaction ID is a universally unique transaction ID. It is applicable to merchants who participate in the MasterCard EMV 3D Secure program. If this field is present, the merchant must include the Program Protocol field. Example: 111+052+0480372f38e6948-5388-41a6-bca4-b49723c19437+
Merchant Supplementary Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value:049
Table Length	n3	Length: 3 bytes Variable length up to 500 bytes
Table Data	an500	Maximum Length: 500 Indicates whether the merchant supplementary data is present Merchant supplementary data contains the following sub-table IDs.
Note: Please use the below format for all Sub-Table IDs: Sub-Table ID - n2 - Valid values [01-99] Sub-Table Length - n1 - Valid values [1-9] Sub-Table Data - an[1-9] [Maximum length - 9 bytes]		
Sub -Table ID "01"		

Purpose: Identifies a deferred transaction.		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 01
Sub-Table Length	n1	Fixed Value: 1
Sub-Table Data	an1	Deferred Transaction indicator This field indicates that the transaction occurred while the merchant's connectivity was interrupted or unavailable, and the merchant approved the transaction, capturing the information to be resubmitted later. This indicator should be present in the transaction that is sent once the connectivity is restored. Valid Codes/Values: **Value** **Description** "Y" Deferred Transaction Example: 111+018+049004011Y+ **Note:** This indicator is currently supported for deferred transactions of the following types: • Visa Authorizations and Purchases • MasterCard Authorizations and Purchases • Blackhawk Activations
Sub -Table ID "02" Purpose: Identifies the entity initiating the transaction		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 02
Sub-Table Length	n1	Fixed Value: 1

Sub-Table Data	a1	Transaction Initiation Indicator This field indicates the entity initiating the transaction. Valid Codes/Values: **Value** **Description** 'M' Transaction initiated by Merchant (MIT) 'C' Transaction initiated by Cardholder (CIT) Example: 111+018+049004021M+ **Note: This field is required for all subsequent Credential on File transactions.**
Sub -Table ID "03" Purpose: Indicates available product/fleet information for WEX		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed value: 03
Sub-Table Length	n1	Fixed value: 1
Sub-Table Data	an1	Indicates whether or not a merchant supports receiving the available WEX products Valid values: 0 = Not supported 4= Supported
Sub -Table ID "04" Purpose: Identifies different business payment types		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 04
Sub-Table Length	N1	Fixed Value: 2
Sub-Table Data	An2	This field is used to indicate the business payment type associated with the transaction.

		Valid Codes/Values: Value Description "CB" Consumer Bill Payment Service. "IR" Purchase Repayment **Note:** This field is currently applicable to Visa and Mastercard transactions only. • The value 'CB' is applicable to VISA transactions only. • The value 'IR' is applicable to Mastercard transactions only. For Mastercard Purchase repayment transactions, it is merchant's responsibility to send Installment payment indicator and Merchant name as below • Installment payment indicator must be submitted in Data Segment 111, Table ID 013 (Special Payments Indicator Information) with valid value "I" (Installment payment). • Merchant name must be submitted in Data Segment 111, Table ID 047 [Soft Descriptor - Merchant name] in the format as follows: Installment Provider name followed by an asterisk (*) followed by the merchant's name, e.g., "XYZ*A SMALL CO" For Purcahse repayment details, refer to Appendix AB (Purchase repayment).

Sub -Table ID "05"

Purpose: Identifies product code of the highest priced item of consumer's transaction

Field	Attributes	Description
Sub-Table ID	n2	Fixed Value: 05
Sub-Table Length	n1	Variable Length up to 6
Sub-Table Data	n6	Maximum Length : 6 digits Identifies the Product code of the highest priced item of a consumer's transaction at the time of checkout at the POS. Product code will be

		considered for the transaction approval by TeleCheck Authorization system. **Note:** This field is applicable to Telecheck Debit (Connect Pay) transactions only.
Sub -Table ID "06" Purpose: Identifies the Mobile terminal classification code for Mobile POS transactions		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed value: 06
Sub-Table Length	n1	Fixed Length : 2
Sub-Table Data	An2	This field is specific to Amex, Discover, JCB, MasterCard, Maestro, and Diners mPOS transactions. It will be sent to Authorizer only if Segment 111 table ID 032 POS Additional Data Information Sub table ID '01' with value 'M' is also sent by the merchant. Valid Codes/Values: Value — Description "AC" — mPOS Accessory/dongle with contact and contactless interfaces, with or without PIN pad. "AS" — mPOS Accessory/dongle with contact and contactless interfaces combined with PIN on Glass support (SCRP, Software-based PIN on COTS). "CC" — Contactless Payment on COTS (CPoC) - Mobile device based contactless only mPOS without PIN support. "CS" — Contactless Payment on COTS (CPoC) - Mobile device based contactless only mPOS with PIN on Glass support. Note: **For Amex** This field is conditional. A merchant must send this field for all mPOS transactions. **For MasterCard, Maestro, Discover, JCB, and Dinners:** This field is optional for mPOS transactions. "CC" and "CS" are the only valid Codes/Values for Discover and Diners transactions. "AC","AS","CC", and "CS" are valid codes for MasterCard and Maestro.
Sub -Table ID "07"		

Purpose: Identifies Merchant Advice Code acceptance indicator from merchant		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 07
Sub-Table Length	N1	Fixed Value :1
Sub-Table Data	AN1	This field will be used to indicate that the Merchant Advice Code is accepted by the merchant. Valid Codes/Values: Value — Description "Y" — Merchant Advice Code acceptance indicator Merchants should include this indicator if they opt to receive Merchant Advice Code from the authorizer network.
Sub -Table ID "09" Purpose: Identifies InComm OTC Transaction Code		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 09
Sub-Table Length	n1	Fixed Value: 3
Sub-Table Data	n3	InComm OTC Transaction Code This field describes the InComm OTC transaction code. It describes which OTC transaction type is being used. Valid Codes/Values: **Value** — **Description** 310 — Benefit Eligibility & Balance Lookup 910 — Item Qualification 920 — Limited Spend 200 — Limited Spend Refund
Sub -Table ID "10" Purpose: Identifies Digital Authentication Framework (DAF) indicator from merchant		

Field	Attributes	Description
Sub-Table ID	n2	Fixed Value : 10
Sub-Table Length	N1	Fixed Value :1
Sub-Table Data	AN1	Digital Authentication Framework (DAF) Indicator Acceptance Flag. This field will be used to indicate that the DAF Indicator is accepted by the merchant. Fixed length of 1 byte Alphanumeric. Valid Codes/Values: Value Description "Y" DAF indicator acceptance flag. Example: 111+019+049004101Y+ Merchant receives the DAF Indicator only if it is sent in the Authorizer network's response and the Merchant sent the DAF Indicator acceptance flag. Note: This field is currently supported for Visa transactions only.
Sub -Table ID "11" Purpose: Identifies Agreement ID Acceptance Indicator from merchant		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value:11
Sub-Table Length	N1	Fixed Value:1
Sub-Table Data	AN1	Agreement ID Acceptance Indicator. This field will be used to indicate that Agreement ID is accepted by the merchant. Fixed length of 1 byte Alphanumeric. Valid Codes/Values:

		Value Description "Y" Agreement ID acceptance flag. Example: 111+019+049004111Y+ Merchant receives the Agreement ID only if it is sent in the Authorizer network's response and the Merchant sent the Agreement ID acceptance flag. Note: This field is currently supported for Visa transactions only.
Sub Table ID "13" Identifies Installment Payment Type		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 13
Sub-Table Length	n1	Fixed Value :1
Sub-Table Data	an1	This field is used to indicate the Installment Payment Type Valid Codes/Values: "S" - Merchant Installment Payment "T" - Third Party Installment Payment "V" - Issuer Installment Payment Note: This field is currently applicable to Discover card type.
Sub Table ID "16" Identifies Delayed Card Sale indicator from merchant		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 16
Sub-Table Length	n1	Fixed Value: 1
Sub-Table Data	an1	This field is used to indicate the Delayed Card Sale Indicator. Valid Codes/Values: "D" - Delayed Card Sale

		Note: This field is currently applicable to Discover card type only.
Sub Table ID "17" Identifies No Show Charges indicator from merchant		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 17
Sub-Table Length	n1	Fixed Value: 1
Sub-Table Data	an1	This field is used to indicate the No Show Charges Indicator. Valid Codes/Values: "N" - No Show Charges Note: This field is currently applicable to Discover card type only.
Sub Table ID "18" Identifies Purchase Restriction Method		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 18
Sub-Table Length	n1	Fixed Value: 1
Sub-Table Data	an1	This flag indicates to the authorizer whether the terminal can support receiving host based purchase restriction fields. Valid Codes/Values: "0" – Neither Chip nor Host Based "1" – Chip-Based Only "2" – Host-Based Only "3" – Both Chip and Host Based Note: • This field is applicable only to Visa Fleet EMV 2.0 and MasterCard Enhanced Fleet EMV transactions. • Purchase Restriction Method MUST be present for ALL MasterCard Enhanced EMV Fleet transactions (Product IDs MCF and MGF) Authorizations/Pre-Authorization and Completion transactions. • If the transaction is not MC Enhanced Fleet, then this indicator should not be sent to acquirers. • If this field is not sent, Buypass will default the value to Visa as No Restrictions supported.

		Availability Notice: This functionality for MasterCard Enhanced transactions is expected to be enabled in First Data production no earlier than June 2026. Until then, this field is intended for development and certification preparation only and should not be used for live production transactions.
Sub Table ID "20" Flag to indicate if a merchant can accept account type and account funding source information from Visa		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 20
Sub-Table Length	n1	Fixed Length: 1
Sub-Table Data	an1	This flag indicates if a merchant supports accepting account type information from Visa Authorizer. Valid Codes/Values: Value Description "Y" Merchant accepts account type information or account funding source information. **Note:** This field is currently supported for Visa transactions only.
Sub Table ID "21" Identifies Recurring or Installment transaction for card types other than Credit		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 21
Sub-Table Length	n1	Fixed Value: 1
Sub-Table Data	an1	This field is used to indicate a Recurring or Installment transaction for card types excluding Visa, MasterCard, Amex, and Discover. Valid Codes/Values: "R" Recurring Payment "I" Installment Payment
Sub Table ID "22" Identifies whether MasterCard and Maestro Transaction Link Identifier should be included in transaction response		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 22

Sub-Table Length	n1	Fixed Value: 1
Sub-Table Data	an1	Transaction Link Support Indicator for MasterCard and Maestro This field is used to indicate that the MasterCard and Maestro Transaction Link Identifier should be returned in the transaction response. Fixed length of 1 byte alphanumeric. Valid Codes/Values: **Value** **Description** "Y" Transaction Link Support Indicator Example: 111+019+049004221Y+ **Note:** This field is currently supported for MasterCard and Maestro transactions only.
Sub Table ID "23" Identifies Authentication Data Quality Program from merchant		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 23
Sub-Table Length	n1	Fixed Value: 1
Sub-Table Data	an1	Authentication Data Quality Program Acceptance Flag This field is used to indicate that the Authentication Data Quality Indicator is accepted by the merchant. Fixed length of 1 byte alphanumeric. Valid Codes/Values: **Value** **Description** "Y" Authentication Data Quality Acceptance Flag Example: 111+019+049004231Y+ A merchant receives the Authentication Data Quality Indicator only if it is sent in the Authorizer network's response and if the merchant has sent the Authentication Data Quality Acceptance Flag. **Note:** This field is currently supported for Visa transactions only.
Sub Table ID "24" Identifies Token Update First Use Acceptance Flag from merchant		

Field	Attributes	Description
Sub-Table ID	n2	Fixed Value: 24
Sub-Table Length	n1	Fixed Value: 1
Sub-Table Data	an1	Token Update First Use Acceptance Flag This field is used to indicate that the Token Update First Use Indicator is accepted by the merchant. Fixed length of 1 byte alphanumeric Valid Codes/Values: **Value** **Description** "Y" Token Update First Use Acceptance Flag Example: 111+019+049004241Y+ A merchant receives the Token Update First Use Indicator only if it is sent in the Authorizer network's response and the merchant has sent the Token Update First Use Acceptance Flag in the request. **Note:** This field is currently supported for Visa transactions only.
Sub -Table ID "25" Purpose: Flag to indicate if a merchant can accept Merchant Tran ID		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 25
Sub-Table Length	N1	Fixed Value:1
Sub-Table Data	AN1	Indicates that merchant can accept the Merchant Tran ID in response. Valid Code/Value: "Y" - Merchant accepts Merchant Tran ID
Sub -Table ID "26" **Transaction Link Action Support Indicator** Purpose: Flag to indicate that Transaction Link Action Indicators are desired in response		
Field	**Attributes**	**Description**
Sub-Table ID	N2	Fixed Value: 26

Sub-Table Length	N1	Fixed Length: 1
Sub-Table Data	AN1	Transaction Link Action Support Indicator for MasterCard/Maestro. This field indicates that the MasterCard/Maestro Transaction Link Action Indicator will be returned in the transaction response. Fixed length of 1 byte alphanumeric. Valid Codes/Values: Value Description "Y" – Transaction Link Action Indicator
Sub -Table ID "27" Purpose: Flag to indicate if a merchant can accept Applied Special Service Data from Visa		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 27
Sub-Table Length	N1	Fixed Value :1
Sub-Table Data	AN1	Indicates that merchant can accept the Applied Special Service Data from Visa in response. Valid Code/Value: "Y" - Merchant accepts Applied Special Service Data
Sub -Table ID "28" Purpose: The flag to indicate that Voyager Restriction Codes are desired in response		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 28
Sub-Table Length	N1	Fixed Value:1
Sub-Table Data	AN1	Indicates that merchant can accept Voyager Restriction Codes from Voyager in response. Valid Code/Value: "Y" - Merchant accepts Voyager Restriction Codes
Sub -Table ID "30" **Visa Category Code Support Indicator** Purpose: Flag to indicate that Visa Category Codes are desired in response		

Field	Attributes	Description
Sub-Table ID	N2	Fixed Value: 30
Sub-Table Length	N1	Fixed Length: 1
Sub-Table Data	AN1	Visa Category Code Support Indicator (for Visa). This field indicates that the Visa Category Code will be returned in the transaction response. Fixed length of 1 byte alphanumeric. Valid Codes/Values: Value Description "Y" - Visa Category Code
Note: <Remaining bytes> are reserved for future use.		
Additional Security Code		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 050
Table Length	n3	Length: 3 bytes
Security Code	an16	Variable length up to maximum length 16 Indicates whether an additional security code is sent from the terminal, and it has a variable length from 1-16 bytes. For SVS card transactions, this field must be 4 bytes.
Additional Card ID		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 051
Table Length	n3	Length: 3 bytes
Card ID	an16	Variable length up to maximum length 16 Indicates whether the merchant is sending an additional card ID, and it has a variable length from 1-16 bytes. For SVS card transactions, this field must be 4 bytes.

Additional TransArmor Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 052
Table Length	n3	Length: 3 bytes
Table Data	an100	Variable length of up to 100 digits Maximum length: 100 Contains additional data elements required by various TransArmor implementations. Field is in the TLV format. First 3 digits are length. It contains the following sub table IDs
Sub-Table ID "01" Purpose: KSN		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed value: 01
Sub-Table Length	n3	Length: 3 bytes
Sub-Table Data	an40	KSN information - variable length of up to 40 bytes This field is required for AES DUKPT, TDES, and Ingenico OnGuard encryption types.
Sub Table ID-"02" Purpose: Device Type		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 02
Sub-Table Length	n3	Length: 3 bytes
Sub-Table Data	an8	Device type information - variable length of up to 8 bytes This field is required for AES DUKPT encryption type. It is optional for TDES and Ingenico OnGuard encryption types.

Example:
111+051+052037010241234567812345678000000010200 3NCR+

Remote Commerce Acceptor Identifier

Field	Attributes	Description
Table ID	n3	Fixed Value: 053
Table Length	n3	Length: 3 bytes
Remote Commerce Acceptor Identifier	ans (b64 encoded)	Variable length up to a maximum of 028 characters This field contains the merchant identifier, such as the merchant business website URL or reverse domain name, as presented to the consumer during checkout. This field is applicable only to MasterCard transactions where the merchant is also sending the DSRP or Token Authentication Verification Value (TAVV) data field in segment 123.

Merchant Tran ID

Field	Attributes	Description
Table ID	n3	Fixed value: 054
Table Length	n3	Length 3 bytes
Table Data	ans95	Variable length up to 95 bytes. A value assigned by the merchant to identify the transaction. **Note:** For Moneris: This field is mandatory. It has a max length of 15 and maps to Moneris field Client Request ID 7b.

Experience Provider Value

Field	Attributes	Description
Table ID	n3	Fixed value: 055
Table Length	n3	Length 3 bytes
Table Data	An4	Variable length up to 4 bytes A value received from UCOM for each Experience Provider

		Valid Values: “WMPY” = Walmart + Experience Provider Value “AAPY” = Amazon Pay “GSPY” = Google Pay “WXPY” = Wex Pay Example: 111+019+055004WMPY+
Stored Credentials Transaction Additional Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value:056
Table Length	n3	Length: 3 bytes Variable length of up to 200
Table Data	ans200	Maximum Length: 200 Contains the additional data for the Merchant Initiated Transaction (MIT) and Cardholder Initiated Transaction (CIT). Stored Credentials Additional data contains the following sub-table IDs
Note: The format for all sub-table IDs are given below: Sub-Table ID - n2 - Valid values [01-24] Sub-Table Length – n2 - Valid values [01-35] Sub-Table Data - an[1-35] [Maximum length - 35 bytes] It is not necessary that all the sub-tables be populated by the merchant. The sub-tables required for a transaction can be populated.		
Sub Table ID “01” **Total payment count**		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 01

Sub-Table Length	n2	Fixed Value :02
Sub-Table Data	an2	Total payment count Indicates the number of Recurring Payments or Installments per the Cardholder Agreement with the Merchant. This is applicable only to Discover and Visa. **For Discover:** Valid Codes/Values: • 02 to 99 • UD - Not Defined • UC - Until Canceled. Note: The field is only used in the first payment transaction for these specific payment types and should be sent for Discover cards issued in India. **For Visa:** Valid Codes/Values: • 01 to 98 • 99 - Not Defined/ Until Canceled. Note: This field is required and should be sent only for recurring payments for Visa cards issued in India. If this field is provided in the initial transaction, the field must be provided in all subsequent transactions, and the value must be either same value as sent in the initial transaction or value used in the last “Modification” transaction. Refer to Variable Information Data segment 111 Table ID 056 Sub Table ID 12 (Recurring Payment Type) for more details.
Sub Table ID “02” Payment Type		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 02

Sub-Table Length	n2	Fixed Value :01
Sub-Table Data	an1	Payment Type This is applicable only to Discover and Visa. **For Discover:** Indicates whether a Recurring Payment or Installment is a fixed or variable amount. Valid Codes/Values: • F – Fixed • V – Variable Note: The field is only used in the first payment transaction for these specific payment types and should be sent for Discover cards issued in India. **For Visa:** Valid Codes/Values: • F – Fixed • V – Variable Note: This field is required and should be sent only for recurring payments for Visa cards issued in India. If this field is provided in the initial transaction, the field must be provided in all subsequent transaction and the value must be either same value as sent in the initial transaction or value used in the last “Modification” transaction. Refer to Variable Information Data segment 111 Table ID 056 Sub Table ID 12 (Recurring Payment Type) for more details.
Sub Table ID “03” MIT Amount		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 03
Sub-Table Length	n2	Fixed Value : 12
Sub-Table Data	n12	MIT Amount Indicates the amount of the Recurring Payment or Installment, and is in the same currency as Currency Code of Transaction Amount. This amount may be different from Transaction Amount.

		This tag has leading zeros and two decimal places ($50 as 000000005000) This is applicable only to Discover and Visa. **For Discover:** Note: This field is used in both initial and subsequent payments for these specific payment types and should be sent for Discover cards issued in India. **For Visa:** Note: This field is required and should be sent only for recurring payments for Visa cards issued in India. If this field is provided in the initial transaction, the field must be provided in all subsequent transaction and the value must be either same value as sent in the initial transaction or value used in the last "Modification" transaction. Refer to Variable Information Data segment 111 Table ID 056 Sub Table ID 12 (Recurring Payment Type) for more details.
Sub Table ID "04" Unique ID		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 04
Sub-Table Length	n2	Fixed Value : 14
Sub-Table Data	an14	Unique ID Merchant provided unique identifier for each Recurring or Installment Payment. This ID is used to reference Authorizations and Advice Messages. This is applicable only to Discover. Note: This field is used in both initial and subsequent payments for these specific payment types and should be sent for Discover cards issued in India.
Sub Table ID "05" Frequency of MIT		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 05
Sub-Table Length	n2	Fixed Value : 02

Sub-Table Data	n2	Frequency of MIT Indicates the frequency of a Recurring Payment or Installment. This is applicable only to Discover and Visa. **For Discover:** Valid Codes/Values: • 01 – Daily • 02 – Weekly • 03 – Biweekly • 04 – Monthly • 05 – Quarterly • 06 – Half-Yearly (Biannually) • 07 – Annually • 08 – Unscheduled (Type of MIT) Note: The field is only used in the first payment transaction for these specific payment types and should be sent for Discover cards issued in India. **For Visa:** • 01 – Daily • 02 – Weekly • 03 – Biweekly • 04 – Monthly • 05 – Quarterly • 06 – Half-Yearly (Biannually) • 07 – Annually • 08 – Unscheduled (Type of MIT) • 09 – Ten Days • 10 – Twice weekly • 11 – Every two months • 12 – Trimester Note: This field is required and should be sent only for recurring payments for Visa cards issued in India. If this field is provided in the initial transaction, the field must be provided in all subsequent transaction and the value must be either same value as sent in the initial transaction or value used in the last “Modification” transaction. Refer to Variable Information Data segment 111 Table ID 056 Sub Table ID 12 (Recurring Payment Type) for more details.
Sub Table ID “06” Validation Flag		

Field	Attributes	Description
Sub-Table ID	n2	Fixed Value : 06
Sub-Table Length	n2	Fixed Value : 01
Sub-Table Data	n1	Validation Flag Indicates whether validation is performed or not. Valid Codes/Values: • 1 – Validated • 0 – Not Validated This is applicable only to Discover and Visa. **For Discover:** Note: This field indicates if the card transaction is checked against the validation source or not. The field is used only in the subsequent payments for these specific payment types and should be sent for Discover cards issued in India. **For Visa:** Note: This field indicates if the recurring payment transaction has been validated offline during registration or not. This field is required and should be sent only for recurring payments for Visa cards issued in India. If this field is provided in the initial transaction, the field must be provided in all subsequent transaction and the value must be either same value as sent in the initial transaction or value used in the last "Modification" transaction. Refer to Variable Information Data segment 111 Table ID 056 Sub Table ID 12 (Recurring Payment Type) for more details.
Sub Table ID "07" Validation Reference		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 07
Sub-Table	n2	Fixed Value : 20

Length		
Sub-Table Data	ans20	Validation Reference A cryptogram generated by the validation source for each Cardholder Account, for an Issuer to validate a Card Transaction. This is applicable only to Discover. Note: The field is used only in the subsequent payments for these specific payment types and should be sent for Discover cards issued in India.
Sub Table ID "08" Sequence Indicator		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value : 08
Sub-Table Length	n2	Fixed Value : 02
Sub-Table Data	n2	Sequence Indicator Used to identify the Sequence of the transactions when multiple Installment payments are submitted. Sequence Indicator is populated in ascending order. This is applicable only to Discover.
Sub Table ID "09" MIT/CIT Category and Subcategory Data		
Field	**Attributes**	**Description**
Sub-Table ID	N2	Fixed Value: 09
Sub-Table Length	N2	Fixed Value: 04
Sub-Table Data	AN4	**This field will be used to indicate category and subcategory data for Cardholder initiated transaction (CIT) and Merchant initiated transaction (MIT).** <u>Cardholder initiated transaction (CIT)</u>

Scenario	**Valid Values**
Card holder-initiated Credential-on-file (adhoc)	C101
Card holder initiated Standing Order (variable amount and fixed frequency)	C102
Card holder-initiated Subscription (fixed amount and fixed frequency)	C103
Card holder-initiated Installment	C104

<table>
<tr><td></td><td></td><td>Merchant Initiated transaction (MIT)

<table><tr><th>Scenario</th><th>Valid Values</th></tr><tr><td>Merchant initiated Unscheduled Credential on File</td><td>M101</td></tr><tr><td>Merchant initiated Standing Order (variable amount and fixed frequency)</td><td>M102</td></tr><tr><td>Merchant initiated Subscription (fixed amount and fixed frequency)</td><td>M103</td></tr><tr><td>Merchant initiated Installment</td><td>M104</td></tr></table>
Note:
This field is currently applicable to MasterCard transactions only.</td></tr>
<tr><td colspan="3">Sub Table ID “12”
Recurring Payment Type</td></tr>
<tr><td>Field</td><td>Attributes</td><td>Description</td></tr>
<tr><td>Sub-Table ID</td><td>n2</td><td>Fixed Value : 12</td></tr>
<tr><td>Sub-Table Length</td><td>n2</td><td>Fixed Value : 01</td></tr>
<tr><td>Sub-Table Data</td><td>n1</td><td>Recurring Payment Type

Specifies the type of recurring payment.

This is applicable only to Visa.

Valid values are:
• 1 (Registration/first transaction)
• 2 (Subsequent transaction)
• 3 (Modification)
• 4 (Cancellation)

Note:
This field is required and should be sent only for recurring payments for Visa cards issued in India.

All subsequent transactions with Transaction amount equivalent to Amount less than or equal to 15,000 INR (India Rupee) should be sent as Non E-commerce transaction without the CAVV Cryptogram along with the value of “2” (Subsequent) and recurring transaction indicator (Variable Information Data segment 111, Table ID 13 – Special Payments Indicator Information as “R”).

All subsequent transactions with Transaction amount equivalent to Amount greater than 15,000 INR (India Rupee) should be sent as E-commerce transaction with Variable Information Data segment 111 Table ID 030 Position</td></tr>
</table>

		9-10 (Terminal Type) value as '25' (Cardholder was authenticated using an authentication methodology (Verified by Visa or 3DES)) with CAVV Cryptogram along with the value of "2" (Subsequent) and recurring transaction indicator (Variable Information Data segment 111, Table ID 13 – Special Payments Indicator Information as "R"). When there are modifications to one or more recurring payment parameters(Total payment count, Payment Type, MIT Amount, Frequency), the transaction should be sent as E-commerce transaction with Variable Information Data segment 111 Table ID 030 Position 9-10 (Terminal Type) value as '25' (Cardholder was authenticated using an authentication methodology (Verified by Visa or 3DES)) with CAVV Cryptogram along with the value of "3" (Modification) and recurring transaction indicator (Data Element 111, Table ID 13 –Special Payments Indicator Information as "R"). When the previously established recurring payment is being cancelled, the transaction should be sent as E-commerce Account verification transaction(zero dollar amount) with Variable Information Data segment 111 Table ID 030 Position 9-10 (Terminal Type) value as '25' (Cardholder was authenticated using an authentication methodology (Verified by Visa or 3DES)) as an with CAVV Cryptogram along with the value of "4" (Cancellation) and recurring transaction indicator (Data Element 111, Table ID 13 –Special Payments Indicator Information as "R").
Sub Table ID "13" **Registration Reference Number**		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 13
Sub-Table Length	n2	Fixed Value: 35
Sub-Table Data	an35	Registration reference number Contains the unique reference number for the recurring payment. This is applicable only to Visa. Note: This field is required and should be sent only for recurring payments for Visa cards issued in India If this field is provided in the initial transaction, the field must be provided in all subsequent transaction and the value must be the same value as sent in the initial transaction. Refer to Variable Information Data segment 111 Table ID 056 Sub Table ID 12 (Recurring Payment Type) for more details.
RBI recurring mandate for Visa transactions		

The Reserve Bank of India (RBI) mandates that when a cardholder first registers and transacts with a merchant, to subsequently have recurring payments debited from the card, additional factor authentication is required. For subsequent recurring transactions, additional factor authentication is not required. However, if a cardholder modifies or revokes a standing instruction, additional factor authentication is required.

Merchants are encouraged to subscribe to the Global BIN File for accurate identification of accounts issued from India.

All Visa recurring transactions for cards issued in India must contain new additional recurring data contained in Table ID 56 and 013 under Data Element No. 111.

Subsequent recurring transactions can only be performed for those cards, which have been successfully registered, and for which the first transaction was successfully authenticated and authorized.

The amount limit for Visa recurring transactions without additional factor authentication is equivalent to 15,000 INR (might change if the RBI mandate is revised in future). Transactions above the limit will require additional factor authentication and electronic commerce identification.

Soft Descriptor – Merchant City

Field	Attributes	Description
Table ID	N3	Fixed Value: 057
Table Length	N2	Length 2 bytes
Table Data	An20	Merchant city For Discover: Merchant City for Third Party Payment providers – this field will have the underlying retailer's city.

Soft Descriptor - Merchant State

Field	Attributes	Description
Table ID	n2	Fixed Value : 058
Table Length	n2	Fixed Value: 02
Table Data	An3	Merchant state For Discover: Merchant State for Third Party Payment providers – this field will have the underlying retailer's state

Soft Descriptor - Merchant Country

Field	Attributes	Description
Table ID	n2	Fixed Value: 059
Table Length	n2	Fixed Value: 03
Table Data	An3	Merchant country For Discover: Merchant Country for Third Party Payment providers – this field will have the underlying retailer's country. See Appendix AC for valid country codes.
Real Time Account Updater Request Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 060
Table Length	n3	Length: 3 bytes Variable length of up to 100
Table Data	ans100	Maximum Length: 100 Indicates the Real Time Account Updater Request data. Real Time Account Updater Request data contains the following sub-table IDs.
Note: This Real Time Account Updater Request Data table ID contains multiple sub-table IDs with specific indicators that First Data uses on requests to identify whether to perform First Data Auth Optimization processing and optionally on responses to send back account information updates from the Real Time Account Updater. It is important that the merchant should send only the applicable indicator in the request message, based on the type of request. The format for all sub-table IDs are given below: Sub-Table ID - n2 - Valid values [01-25] Sub-Table Length – n2 - Valid values [01-25] Sub-Table Data - an [1-25] [Maximum length - 25 bytes] Note: The sub-table IDs 01 and 02 are mutually exclusive. Merchant can send any one of them in the request, based on the type of request.		
Sub Table ID "01" Account Updater Request Indicator		
Field	**Attributes**	**Description**

Sub-Table ID	n2	Fixed Value: 01
Sub-Table Length	n2	Fixed Value: 01
Sub-Table Data	an1	This field is used to indicate whether the merchant can accept the most updated cardholder account information, such as new account number and/or new expiration date from First Data. The new account information updates will be only provided, if available. Valid Values: Y = Yes, send all account update information I = Info, send only indicators notifying merchant of what updates were provided. If the field is not provided, no information will be sent back to merchant even if First Data Auth Optimization is performed. Note: • This field is applicable to Visa and MasterCard. • Applicable only to Card Not Present transactions. • Applicable only to Auth, Sale, and Refund transactions • When this field is sent, the Auth Optimization Override Indicator field in table ID 060, sub-table 02 should not be sent.
Sub Table ID "02" Auth Optimization Override Indicator		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 02
Sub-Table Length	n2	Fixed Value: 01
Sub-Table Data	an1	This field is used to indicate whether the merchant wants to override the First Data Auth optimization services (including Real Time Account Updater) or not. Valid Values: O = Override, do not perform Auth Optimization on this authorization Note: • This field is applicable to Visa and MasterCard. • Applicable only to Card Not Present transactions. • Applicable only to Auth, Sale, and Refund transactions • When this field is sent, the Account Updater Request Indicator field in table ID 060, sub-table 01 should not be sent.
Customer Additional Information Data		
Field	**Attributes**	**Description**

Table ID	n3	Fixed Value: 062
Table Length	n3	Length: 3 bytes Variable length of up to 150
Table Data	ans150	Indicates whether Cardholder Additional Data is present. Cardholder Additional Data contains the following sub-table IDs.
Sub Table ID "01" Cardholder Phone number		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 01
Sub-Table Length	n3	Length: 3 bytes Max. length of up to 16 bytes.
Sub-Table Data	an16	This field contains the cardholder phone number for account verification transaction. Example: 111+030+062015010101234567890+ Note: Applicable only to Visa.
Sub Table ID "02" Cardholder Email Address		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 02
Sub-Table Length	n3	Length: 3 bytes Max. length of up to 99 bytes.
Sub-Table Data	an99	This field contains the cardholder email address for account verification transaction. Example: 111+035+06202002013card@holder.com+ Note: Applicable only to Visa.
Visa®/MC/AMEX Incremental Auth Lodging Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 063
Table Length	n3	Fixed Value: 011

Arrival Date	n4	Format: MMDD This field must be spaces if not used by merchant.
Duration	n2	Number of days of stay This field must be spaces if not used by merchant.
Terminal Transaction Number	n4	Used for tracking partial reversals and incomplete authorizations Mandatory for Lodging, Incremental Auth, and EV transactions This field is used only for Visa/MasterCard/Amex. The Terminal Transaction Number should be unique. When using the same card for Incremental Auth, the Terminal Transaction Number should be the same until the Incremental Auth series is complete. Please refer to the example below.
Preferred Customer Indicator	an1	P = Preferred customer

Examples of Visa Incremental Auth transaction:

Transaction	**Request**	**Response**
Original Auth/Estimated Auth transaction PAN used - 4005XXXXXXXX1114	Example: 063+011+0323050058P+ 063 – Table ID 011 – Table length 0323 – MMDD 05 – No. of days 0058 – Terminal transaction number P – Preferred customer indicator Non	Not applicable

	Travel & Entertainment Example: 063+011+<6 spaces>0058<1 space>+ 063 – Table ID 011 – Table length 4 spaces – Arrival date 2 spaces – Duration 0058 – Terminal transaction number 1 space in Preferred customer indicator. EV Example - 063+011+<6 spaces>0058(Terminal transaction number)<1 space>+	
Incremental Auth1 – PAN 4005XXXXXXXX1114 (Same merchant and card holder)	Visa ® Lodging Data- Table ID 063 Lodging Example: 063+011+0323050058P+	Not applicable

Stay extended	063 – Table ID 011 – Table length 0323 – MMDD 05 – No. of days 0058 – Terminal transaction number P – Preferred customer indicator Non-Travel & Entertainment Example: 063+011+<6 spaces>0058<1 space>+ 063 – Table ID 011 – Table length 4 spaces – Arrival date 2 spaces – Duration 0058 – Terminal transaction number 1 space in Preferred customer indicator. EV	

	Example - 063+011+<6 spaces>0058(Terminal transaction number)<1 space>+	
Completion - PAN 4005XXXXXXXXX1114 (Same merchant and card holder)	Visa Lodging Data- Table ID 063 Lodging Example: 063+011+0323050058P+ 063 – Table ID 011 – Table length 0323 – MMDD 05 – No. of days 0058 – Terminal transaction number P – Preferred customer indicator Non-Travel & Entertainment Example: 063+011+<6 spaces>0058<1 space>+	Not applicable

	063 – Table ID 011 – Table length 4 spaces – Arrival date 2 spaces – Duration 0058 – Terminal transaction number 1 space in Preferred customer indicator. EV Example - 063+011+<6 spaces>0058(Terminal transaction number)<1 space>+	
Service Location Information **Note:** This field is applicable to MasterCard and Maestro Debit transactions only.		
Field	**Attributes**	**Description**
Table ID	n2	Fixed Value: 064
Table Length	n3	Fixed Value: 03
Table Data	ans49	Variable length up to 49 bytes. Maximum Len: 49 This field will be used to receive Service Location Additional Information from terminal. Service Location Additional Data contains the following sub-table ID's. This field is applicable to card type 'MasterCard' when Payment Type is 'Credit'.

		It is also applicable to Maestro Debit transactions. If present in the original transaction, this field must be present in the subsequent transactions (e.g., Voids/Full Reversals, Completions). The value should be the same as submitted in the original transactions.

Note:
Service Location Information must be present for all transactions when it is different from the location identified in Acceptor Location.

Example:111+045+06403101010SLCITYNAME02003CNA03003630

Sub - Table ID "01"
Service Location City Name

Purpose: Identifies Service Location City Name which contains the name of the city where the cardholder received the services, if it is different from the location identified by the acceptor city name

Field	Attributes	Description
Sub-Table ID	n2	Fixed Value: 01
Sub-Table Length	n3	Variable Length up to 13
Sub-Table Data	ans..13	Maximum Length: 13 Service Location City Name must contain the name of the city, town, or village, as defined by country postal standards. It must be present for all transactions when different from the location identified in the Acceptor City Name. For the U.S. region, this field must match the U.S. Postal Service list of acceptable cities and spellings.

Sub - Table ID "02"
Service Location Country Subdivision Code

Purpose: Identifies Service Location Country Subdivision Code (State)

Field	Attributes	Description
Sub-Table ID	n2	Fixed Value: 02
Sub-Table Length	n3	Fixed 3 bytes
Sub-Table	ans3	Left justified filled with spaces. The Service Location Subdivision Code must be present for all transactions

Data		when different from the location identified in the Acceptor Subdivision Code. **Note:** For valid values, refer to Appendix D. Valid State Codes.
Sub - Table ID "03" **Service Location Country Code** Purpose: Identifies Service Location Country Code.		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 03
Sub-Table Length	n3	Fixed 3 bytes
Sub-Table Data	n3	Right justified filled with zeros in front. The Service Location Country Code must be present for all transactions when different from the location identified in the Acceptor Country Code. **Note:** For valid values, refer to Appendix AC. Valid Country Codes.
Sub - Table ID "04" **Service Location Postal Code** Purpose: Identifies Service Location Postal Code		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 04
Sub-Table Length	n3	Variable length up to 10 bytes
Sub-Table Data	an..10	Maximum Length: 10 The Service Location Postal Code must be present for all transactions when different from the Acceptor Postal Code and contain a valid code for countries that utilize postal codes. This information must be populated and formatted as defined by the country postal standards.
Merchant Payment Gateway ID Data **Note:** This field is applicable to MasterCard and Maestro as Merchant Payment Gateway.		
Field	**Attributes**	**Description**

Table ID	n3	Fixed Value: 065
Table Length	n3	Length: 3 bytes Fixed length of 11 bytes
Table Data	n11	For MasterCard and Maestro, Merchant Payment Gateway ID is provided by the MasterCard/Maestro network when a Merchant Payment Gateway (MPG) service provider registers with the network. If a payment gateway is involved, merchant must send the Merchant Payment Gateway ID to the acquirer in the transactions. If no Merchant Payment Gateway ID is found in the request, First Data will default the value to “00000999997” when sending to the network. If an external payment gateway is involved, the corresponding merchant gateway ID has to be sent. This field previously supported Visa's Enabler Verification Value, but with Visa's addition of support for the MPG data ID, we have introduced a new field for Enabler Verification Value (Sub Table: 64). It should no longer be populated here.
Digital Commerce Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 066
Table Length	n3	Length: 3 bytes Variable length of up to 200
Table Data	ans	Minimum Length: 006 Maximum Length: 200 Indicates the digital commerce solutions data from merchant. Digital commerce data contains the following sub-table IDs.
Note: The format for all sub-table IDs is given below: Sub-Table ID - n2 - Valid values Sub-Table Length - n3 - Valid values Sub-Table Data - Variable data		
Sub Table ID “01” Digital Commerce Domain Identifier		
Field	**Attributes**	**Description**

Sub-Table ID	n2	Fixed Value: 01
Sub-Table Length	n3	Fixed Length: 001
Sub-Table Data	n1	This field identifies the entity responsible for populating the digital commerce solutions indicators. Valid Value: 2 – Acquirer 0 – Multi-Domain **Note:** It is applicable to the MasterCard and Maestro card types only. If this field is sent in Auth, then it should also be sent in its completion. For reference: The value of "2" (Acquirer) is expected when the sub-table ID "02" contains the value "4" (Next Gen POI) or "5" (Pay by Link) or "6" (X QR). The value of "0" (Multi-Domain) is expected when the sub-table ID "02" contains the value "0" (Multi-Program).

Sub Table ID "02"

Digital Commerce Program Type

This field indicates whether the transaction originated using a digital commerce solution, such as NextGen POI. The digital commerce solution indicators will strengthen the security and performance of digital commerce transactions and help acquirers and issuers to easily identify them.

Field	Attributes	Description
Sub-Table ID	n2	Fixed Value: 02
Sub-Table Length	n3	Fixed Length: 001
Sub-Table Data	n1	This field is used to identify the type of digital commerce program. Valid Value: 0 - Multi-Program 3 - In-store Biometric 4 - Next Generation POI (Point of Interaction) 5 - Mastercard QR: Pay by Link 6 - Mastercard QR: X QR **Note:** It is applicable to the MasterCard and Maestro card types only. If this field is sent in Auth, then it should also be sent in its completion. For reference: This field is expected for the NextGen POI transactions or Multi-

		Program or In-Store biometric or Pay by Link or X QR transactions.
Sub Table ID "03" Digital Commerce Program Type Details		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 03
Sub-Table Length	n3	Length: 3 bytes Variable length up to 15
Sub-Table Data	ans..15	Maximum Length: 15 This field identifies the check-in entity type and entity name for the digital commerce transactions. **Note:** This field must be present for the NextGen POI transactions. It is applicable to MasterCard and Maestro card types only. If this field is sent in Auth, then it should also be sent in its completion.
Sub Table ID "04" Digital Commerce Check-in Details		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 04
Sub-Table Length	n3	Length: 3 bytes Variable length up to 40
Sub-Table Data	ans..40	Maximum Length: 40 This field identifies the method a check-in-entity applies to validate cardholder presence, such as reference number. **Note:** This field must be present for the NextGen POI transactions. It is applicable to MasterCard and Maestro card types only. If this field is sent in Auth, then it should also be sent in its completion.
Anticipated Amount		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 067

Table Length	n3	Length: 3 bytes Variable length of up to 12 bytes
Table Data	n12	Identifies the anticipated amount received from the merchant for verification and product eligibility transactions. It is a 12-character amount that is right justified and contains leading zeros. Amount includes implied decimal relative to currency code. **Example: 111+023+06700800500000+** **Note:** This field is applicable to Visa only.
EV- Additional Transaction Fee 1		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 068
Table Length	n3	Length: 3 bytes Variable length of up to 12 bytes
Table Data	..n12	Identifies Additional Transaction Fee 1 for EV transactions. Any additional fee associated with an EV charging that merchant charges is to be sent in this field. **Example: 111+023+06800800500000+**
EV- Additional Transaction Fee 2		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 069
Table Length	n3	Length: 3 bytes Variable length of up to 12 bytes
Table Data	..n12	Identifies Additional Transaction Fee 2 for EV transactions. Any additional fee associated with an EV charging that merchant charges is to be sent in this field. **Example: 111+023+06900800500000+**
Anticipated Completion Amount		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 070

Table Length	n3	Length: 3 bytes Variable length of up to 12 bytes
Table Data	..n12	In debit authorization requests, the terminal should send the anticipated amount of the completion transaction here. BUYPASS will use this information for choosing an authorizer. **Example: 111+019+0700044500+**
Enabler Verification Value		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 071
Table Length	n3	Length: 3 bytes Fixed value '005'
Table Data	An005	For Visa and Interlink, this field contains a 5-byte network-assigned value that is used by networks to identify enablers. A 'Visa Enabler Verification Value' (EVV) is a unique identifier assigned by Visa to a third-party payment processor, known as an 'Enabler'. It is a code used to verify the legitimacy of an enabler when processing Visa transactions. **Note:** Enablers are payment processing entities that are not acquiring members and are often the primary relationship owner with merchants and originators. Enablers own technical solutions through which the merchant or originator will access acceptance. Example: 111+020+0710053202+ **Note:** Applicable to Visa and Interlink only.
Transaction Link Identifier		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 072
Table Length	n3	Length: 3 bytes
Table Data	Ans.22	Variable length of up to 22 bytes. Send transaction link identifier for MasterCard/Maestro transactions. The value will be provided by MasterCard network in original transaction response (in Segment 112, table ID 038), and merchants should retain the

		value and send in request for subsequent transactions. For Moneris: This field maps to Moneris field Real Time Unique ID 9e. **Note:** Special characters are allowed.
Future Transaction Amount		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 073
Table Length	n3	Length: 3 bytes Fixed length of 12 bytes
Table Data	n12	This field contains a value that indicates the anticipated amount of a future completion. It will be used by MasterCard to improve service by preventing future declines for insufficient amount and other errors.
Discount Program Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 075
Table Length	n3	Variable Length: 53 bytes
Table Data	Ans..53	Identifies the Discount Program Data for Visa Merchant Discount Program. Merchants who are set up to use Visa Merchant Discount Program should send the discount related data to be passed to Visa. This field will contain following tags in TLV format – Tag 01 - Discount Method Length – 1 Values – 1 = Retail minus discount method 2 = Cost plus discount method 3 = Best of retail minus or cost plus Tag 02 - Discount Agent Length – 1 Values – M = Merchant V = Visa Tag 03 - Discount Plan ID Length – 4 Value – Varies

		Format - Contains a value to identify discount plan applied to a transaction Tag 04 - Discount Client ID Length – 15 Value – Varies Format - Contains value to identify fleet clients. Tag 05 – Net Fuel Price Length – 12 Value – Varies Format - Four Decimal Places implied
Discount Line-Item Data		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 076
Table Length	n3	Variable Length: 3 bytes
Table Data	Ans..256	This table ID will contain the line-item level data for each of the fuel product codes sent by merchant, if the merchant is participating in the Visa Merchant Discount Program. **Field - Number of Line Items** Length - 1 Description - Number of Line Items included in the additional Field Format – Numeric, Value Range 0-5 The following fields will repeat for number of line items provided in the above field. **Field - Product Code** Length - 3 Description – Fuel Product Code Format – Numeric, 3-digit code received from POS, with leading zeros **Field - Discount Rate-Percentage** Length - 4 Description – Contains the discount rate in percentage. Two decimal places are implied Format – Numeric, padded with leading zero **Field - Discount Rate-Unit Discount** Length - 8 Description – Contains unit discount amount in retail minus discount model Format – Numeric, padded with leading zero **Field – Flat Discount Rate** Length - 12 Description – Contains flat discount rate for the product code. Format – Numeric, padded with leading zero **Field – Total Discount Amount** Length - 12

		Description – Contains total discount amount for the line item. Format – Numeric, padded with leading zero **Field – Net Fuel Price** Length - 12 Description – Contains net line items, less taxes exempted or discounts. Format – Numeric, padded with leading zero Format - Four Decimal Places implied
Discount Amount		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 077
Table Length	n3	Length: 3 bytes Fixed length of 12 bytes
Table Data	n12	Identifies Total Discount Amount applied for fuel product code being sent.
Line Item		
Field	**Attributes**	**Description**
Table ID	N3	Fixed Value:078
Table Length	N3	Length: 3 bytes Variable length up to 999 • This table will now serve as the primary table for all future Visa fleet line-item details for Fuel/Electric Vehicle transactions, excluding previously processed. • This table will also be used for Non Fuel products for MasterCard Enhanced Fleet Transactions. **Note:** This table ID will contain the Line-Item level data for each of the fuel product codes sent by Merchant. All applicable taxes must already be included in the Transaction Amount in the Standard Segment. Any product codes sent here must also be sent in Product Code Data Segment 102.
Table Data	ans…	Maximum Length: 999
Note: Tag: 78 – Identifier for this structure. Length: • 3 bytes, variable (up to 999).		

- Represents total blob size, including:
  - ◊ Length of line items array
  - ◊ All sub-tables

Line Items

- **ProductCodeTag:** A tag that identifies the Fuel Product Code.
- **Length:** A 3 bit field indicating how many sub tables belong to this product code.
- Each line item sub-table uses **TLV (Tag-Length-Value)**.

Example:

```
[78][TotalLength]
   [ProductCodeTag][Length]
      [Sub-Table ID][Sub-Table Length][Sub-Table Data]
      [Sub-Table ID][Sub-Table Length][Sub-Table Data]
      [Sub-Table ID][Sub-Table Length][Sub-Table Data]
   [ProductCodeTag][Length]
      [Sub-Table ID][Sub-Table Length][Sub-Table Data]

[78][084]                          -- Outer TLV: Fleet Line Item Group
   [001][40]                        -- Product Code: 001 (Unleaded), 40 length
      [01][04][FLSV]                 -- Service Type: Full Service
      [05][12][000000001500]              -- Local Tax Amount: $15.00
      [06][04][0025]                -- Local Tax Rate: 2.5%
      [09][04][0500]                -- National Tax Rate: 5.00%
   [019][32]                       -- Product Code: 019 (Diesel), 32 length
      [01][04][SLSV]                 -- Service Type: Self Service
      [08][12][000000000800]              -- National Tax Amount: $8.00
      [09][04][0500]                -- National Tax Rate: 5.00
```

Product Line Item Group

Product Code	n3	**Numeric, 3-digit code received from POS, with leading zeros**
Product Length	n3	**Numeric 3-digit length.**

Sub Table ID "01"
Service Type

Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 01
Sub-Table Length	n2	Fixed Value: 04
Sub-Table Data	a4	Contains these values to indicate the type of service provided at the acceptor location: FLSV = Full service

		HSDI = High-speed dispense SLSV = Self-service
Sub Table ID "02" Product Category/Description		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 02
Sub-Table Length	n2	Fixed Value: 35
Sub-Table Data	a…	Contains product category or detailed description.This tag is optional for fuel transactions.
Sub Table ID "03" Taxable Indicator		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 03
Sub-Table Length	n2	Fixed Value: 01
Sub-Table Data	n1	Contains a value to indicate whether the item is taxable: Valid Codes/Values: 0 = False 1 = True
Sub Table ID "04" Local Tax Included Indicator		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 04
Sub-Table Length	n2	Fixed Value: 01
Sub-Table Data	n1	Contains a value to indicate whether the unit cost includes local tax Valid Codes/Values: 1 = True
Sub Table ID "05" Local Tax Amount		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 05

Sub-Table Length	n2	Fixed Value: 12
Sub-Table Data	n12	Contains the amount of local tax when tag 04 contains a value of 1 (True). Two decimal places are implied.
Sub Table ID "06" Local Tax Rate		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 06
Sub-Table Length	n2	Fixed Value: 04
Sub-Table Data	n4	Contains the VAT/Local tax rate (in percentage) for fuel purchased. Two decimal places are implied.
Sub Table ID "10" Other Tax Included Indicator		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 10
Sub-Table Length	n2	Fixed Value: 01
Sub-Table Data	n1	Contains a value to indicate whether the unit cost includes national tax: Valid Codes/Values: 1 = True
Sub Table ID "11" Other Tax Amount		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 11
Sub-Table Length	n2	Fixed Value: 12
Sub-Table Data	n12	Contains the amount of other tax when tag 10 contains a value of 1 (True). Two decimal places are implied.
Sub Table ID "12" Other Tax Rate		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 12

Sub-Table Length	n2	Fixed Value: 04
Sub-Table Data	n4	Contains other tax rate (in percentage) for fuel purchased. Two decimal places are implied.
Sub Table ID "14" Extended Unit of Measure		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 14
Sub-Table Length	n2	Fixed Value: 04
Sub-Table Data	an4	Contains one of the 36 abbreviations of possible unit of measures for non-fuel product items set by MasterCard Enhanced Fleet Transactions. Ex. ACRE = Acre CMET = Centimeter Note : See Extended Unit of Measure Appendix for possible values. **Availability Notice**: This functionality is expected to be enabled in First Data production no earlier than June 2026. Until then, this field is intended for development and certification preparation only and should not be used for live production transactions.
Visa Commercial Card Data		
Field	**Attributes**	**Description**
Table ID	N3	Fixed Value:079
Table Length	N3	Length: 3 bytes Variable length up to 999 This table will now serve as the primary table for all future Visa commercial card aggregate data.
Table Data	ans…	Maximum Length: 999
Sub Table ID "01" Local Tax Included Indicator		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 01
Sub-Table Length	n2	Fixed Value: 01
Sub-Table Data	n1	Contains a value to indicate whether the unit cost includes local tax

		Valid Codes/Values: 1 = True
Sub Table ID "02" Local Tax Amount		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 02
Sub-Table Length	n2	Fixed Value: 12
Sub-Table Data	n12	Contains the amount of local tax when tag 01 contains a value of 1 (True). Two decimal places are implied.
Sub Table ID "03" Local Tax Rate		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 03
Sub-Table Length	n2	Fixed Value: 04
Sub-Table Data	n4	Contains the VAT/Local tax rate (in percentage) for fuel purchased. Two decimal places are implied.
Sub Table ID "07" Other Tax Included Indicator		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 07
Sub-Table Length	n2	Fixed Value: 01
Sub-Table Data	n1	Contains a value to indicate whether the unit cost includes national tax: Valid Codes/Values: 1 = True
Sub Table ID "08" Other Tax Amount		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 08
Sub-Table Length	n2	Fixed Value: 12

Sub-Table Data	n12	Contains the amount of other tax when tag 07 contains a value of 1 (True). Two decimal places are implied.
Sub Table ID "09" Other Tax Rate		
Field	**Attributes**	**Description**
Sub-Table ID	an2	Fixed Value: 09
Sub-Table Length	n2	Fixed Value: 04
Sub-Table Data	n4	Contains other tax rate (in percentage) for fuel purchased. Two decimal places are implied.
Staged Wallet ID		
Field	**Attributes**	**Description**
Table ID	N3	Fixed Value: 080
Table Length	N2	Fixed Length: 11
Table Data	an11	Contains the staged digital wallet ID. Identifies the **digital wallet provider** that issued the token used in the transaction.
Visa Gateway ID		
Field	**Attributes**	**Description**
Table ID	N3	Fixed Value: 081
Table Length	n2	Fixed Value: 11
Table Data	an11	Indicates whether the transaction was initiated by the merchant or by a third-party gateway provider. This field is mandatory for transactions submitted by a gateway service provider (e.g., CyberSource) acting on behalf of a merchant. It helps Visa identify the source of the transaction for routing and compliance purposes.
Incremental Auth Data		
Field	**Attributes**	**Description**
Table ID	N2	Fixed Value: 82
Table Length	N3	Fixed Length : 004

Terminal Transaction Number	N4	Used for tracking partial reversals and incomplete authorizations. The Terminal Transaction Number should be unique for each Original Auth (aka Estimated Auth). This field should be sent with the same value for all subsequent transactions tied to this original transaction (Incremental Auths, Partial or Full Reversals, IA Completions). Therefore, when using the same card for Incremental Auth, the Terminal Transaction Number should be the same until the Incremental Auth series is completed. **Discover:** Mandatory for Original/Estimated Auth and all the subsequent transactions. **Visa, MasterCard, Amex:** This is optional, though highly encouraged to be populated along with Visa/MC/Amex Incremental Auth Lodging Data in Variable Information Segment 111 table ID 063.
Note: <Remaining bytes> are reserved for future use.		

Appendix J. Valid Point-of-Service Entry Mode Codes

This appendix lists valid entries for Variable Information (Element No. 113).

A valid 3-digit Point-of-Service Entry Mode code has the following components:

- 2-digit PAN Entry Mode code in position nos. 1–2
- 1-digit Terminal Capability Mode code in position no. 3

The following table contains valid PAN Entry Mode codes:

Note: The POS Entry Mode values listed below must persist throughout the lifecycle of the transactions, except as noted below the Valid PAN Entry Mode Codes table.

Valid PAN Entry Mode Codes	
Pos. Nos. 1–2	**Description**
00	Unspecified
01	Manual/keyed
02	Magnetic stripe read with partial track data[1] sent to the host
04	Bar Code Scan Optical character recognition (OCR) read for EMV
05	Integrated circuit card (ICC) read – CVV data reliable (used for EMV chip read)
07	Contactless chip read (used for contactless chip)
10	Credential on file[2]
79	Contact and Contactless chip fallback to manual
80	Chip card capable – unaltered track data read (used for contact EMV fall back to swiped)
82	Contactless mobile commerce device[3]
86	Dual interface contactless chip to contact chip
90	Magnetic stripe read with full track data[1] sent to the host
91	Full contactless magnetic stripe read with track data[1] sent to the host
95	Integrated circuit card (ICC) read – CVV data unreliable

[1]Due to credit authorizer (American Express, Diners Club®, Discover® Network, JCB®, MasterCard®, MasterCard® Fleet, Visa®, and Visa® Fleet) and debit authorizer regulatory requirements regarding storage of full track data, previously authorized credit and debit transactions (Voids and Completions) can only pass the account number and expiration date to BUYPASS. These transactions will use the same PAN Entry Mode code as their corresponding original Purchase or Authorization transaction, with one exception: They will use a PAN Entry Mode code *02* when referencing an original Purchase or Authorization transaction in which a PAN Entry Mode code *90* was used.

[2]Entry Mode 10 is applicable to Visa Credit, MasterCard Credit, Discover Credit, and Amex transactions where merchant or cardholder is initiating a transaction based on the credentials already stored on the file. The value is also applicable to the subsequent Debit Completions and Debit Reversals when the original transaction was initiated as a Credit COF transaction. The value is NOT applicable to the first transaction where credentials are being verified or stored. For Discover, POS Entry Mode '10' must NOT be present in the following scenarios:

- QR Code/Bar Code/Payment Code transactions
- Card Sales with card account information stored by an Issuer or provisioned in a digital wallet

[3]For Discover, Entry Mode 82 must NOT be assigned to card transactions for which the cardholder uses their stored card account information. Discover Credential on File transactions must use the POS Entry Mode value of '10'.

The following table contains valid Terminal PIN Entry Capability Mode codes:

Valid Terminal PIN Entry Capability Mode Codes[1]	
Position 3	**Description**
0	Unspecified
1	PIN entry capability (including both EMV Contact and Contactless)
2	No PIN entry capability (including both EMV Contact and Contactless)
3	mPOS Software-based PIN Entry Capability
6	PIN pad inoperative
8	Contactless Magnetic Stripe Read Capability

[1]If more than one valid Terminal Capability Mode code exists for a POS device, the following hierarchy logic should be followed:

If a Contactless MSR is attached to the POS device and it has no chip capability, then all transactions will use Terminal PIN Entry Capability Mode code 8; but

For EMV Contact and Contactless the valid PIN Entry Capability Mode code is 1 and not 8. The PIN Entry Capability Mode code 8 is reserved only for Contactless MSR; or

If no Contactless MSR is attached, and the POS device has the PIN debit feature enabled, then all transactions will use Terminal PIN Entry Capability Mode code 1; or

If no Contactless MSR is attached, and the POS device has the PIN debit feature disabled, then all transactions will use Terminal PIN Entry Capability Mode code 2.

PIN entry capability mode code 3 is applicable only to MasterCard Mobile POS transactions.

mPOS transactions can be classified based on the presence of PIN pad (Hardware-based) or without PIN pad (Software-based).

mPOS Devices Phone/Tablet + Dongle with PIN Pad should have PIN Entry Capability Mode code as 1.

mPOS Devices Phone/Tablet + Dongle without PIN Pad should have PIN Entry Capability Mode code as 3.

Appendix K. Additional Information Data Layouts

This appendix contains data layout information for the following additional information used in conjunction with Element No. 116 (Additional Information Indicator) and Element No. 118 (Additional Information):

- Balance information for gift cards, EBT cards, phone cards, or credit/debit cards (Code *001*)
- Reserved code information (Code *002*)
- Address verification service (AVS) or Name Verification Service(NVS) or both information (Code *003*)
- Card verification value (CVV, CVV2, CVC2, CID, DTVV) information (Code *004*)
- ECA/ TeleCheck® Trace ID information (Code *005*)
- ECA/ TeleCheck® Denial Record Number information (Code *006*)
- ECA/ TeleCheck® Return Check information (Code *007*)
- Loyalty information – Version 1 (Code *008*)
- Visa® product result information (Code *009*)
- Loyalty information – Version 2 (Code *010*)
- User data information (Code *011*)
- Discover® Network Retrieval Reference Number (Code *012*)
- Expiration Date for TransArmor - VeriFone edition processing (Code *013*)
- BUYPASS Card Type Code (Code 016)
- PIN on Receipt Information (Code 017)
- PINless Debit Information (Code 018)
- Visa Spend Qualified Indicator Information (Code 019)
- Host Prompts Information (Code 020)
- Re-Price Data Response Information (Code 021)
- CAVV Result (Code 022)
- MCX Reference Number (Code 023)
- Carwash Indicator (Code 024)
- Language Indicator (Code 025)
- DST Response (Code 026)
- Universal Unique Identifier (Code 027)
- Transaction Identifier (Code 028)
- PAR (Payment Account Reference) Data Information (Code 029)
- Merchant Advice Code (Code 030)
- DAF Indicator (Code 031)
- Agreement ID (Code 032)
- Host-based Purchase Restriction (Code 034)
- Cardholder Additional Information Result Code (Code 035)
- Account Type (Code 036)
- Account Funding Source (Code 037)
- Transaction Link Identifier (Code 038)

- Transaction Link Action Indicator (Code 039)
- Fraud Score (Code 040)
- Fraud Score Reason Code (Code 041)
- Authentication Data Quality Indicator (Code 042)
- Token Update First Use Indicator (Code 043)
- Visa Category Code (Code 047)

- The following table contains the following layout information for each type of additional information:
- Indicator name
- Data field attributes
- Data field description

Additional Information Data Layouts

Balance Information		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 001
Table Length	n	Fixed
Maximum Length	n	Len.: 6 Identifies the gift card, EBT card, phone card, or credit/debit card balance information being transmitted in a Financial Transaction response by BUYPASS.

Reserved Indicator Information		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 002
Table Length	N/A	N/A
Maximum Length	N/A	N/A

Additional Information Data Layouts				
AVS or NVS or Both Information				
Field	**Attributes**	**Description**		
Table ID	n3	Fixed value: 003		
Table Length	n3	Fixed		
Maximum Length	an10	Variable length up to 10 digits. Maximum Len. : 10 Identifies the AVS or NVS or both response from the authorizer. AVS: Value: AVScr where *c* is: + or – + = AVS positive response from card issuer. - = AVS negative response from card issuer. where *r* is: the actual AVS response from the issuer Example: 112050003005AVS+A00100600123401201512345678000000 NVS: Value : NVScr where *c* is + or – + = NVS positive response from card issuer - = NVS negative response from card issuer where *r* is: the actual response from the issuer. Example: 112050003005NVS+M00100600123401201512345678000000 BOTH It contains both AVS and NVS response as below, 112055003010AVS+ANVS+M001006001234012015123456780000000 The following table summarizes AVS and NVS Response Codes by Issuer.		

Additional Information Data Layouts								
Valid AVS Response Codes and Their Corresponding Issuer Descriptions								
Code	**Amex®**	**Discover®**	**MasterCard®**	**Visa®**	**STAR®**	**Pulse®**	**Accel®**	**Nyce®**
A	Address only matches	Address & 5-digit ZIP match	Address matches, ZIP does not	Address matches, ZIP does not	Address matches, Postal Code does not	Address matches, zip does not match	Address matches, but ZIP code does not match	Address matches, ZIP code does not match
E	Not applicable	Not applicable	Not applicable	Not applicable	Err, ineligible transaction, or content error	Error; ineligible transaction	Not applicable	Not applicable
G	Not applicable	Address information not verified for international transactions	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Additional Information Data Layouts

Valid AVS Response Codes and Their Corresponding Issuer Descriptions *(continued)*

Code	Amex®	Discover®	MasterCard®	Visa®	STAR®	Pulse®	Accel®	Nyce®
N	Neither ZIP nor address match	Neither ZIP nor address match	Neither ZIP nor address match	Neither ZIP nor address match	Address and Postal Code do not match	Neither ZIP nor address match	Neither the address or the ZIP code match	Neither ZIP nor address matches
R	System unavailable, retry	Not applicable	System unavailable, retry	System unavailable, retry	System unavailable/Timed out	Retry; system unavailable or timed out	Not applicable	Retry, system unable to process
S	AVS not supported	AVS not supported	AVS not supported	Not applicable	Issuer does not support AVS	AVS not supported	AVS not supported	Address verification not supported
T	Not applicable	9-digit matches, address does not	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
U	Information not available	System unavailable, retry	Information not available	Verification on Data or Token data not sent	Address information unavailable	No data from issuer	Not applicable	No data from issuer
W	Not applicable	Information not available	For U.S., 9-digit ZIP matches, address does not. For non-US, ZIP matches, address does not	Not applicable	9-digit Postal Code matches, address does not	Nine-digit zip matches, address does not match	9-digit ZIP code matches, but address does not match	9-digit ZIP code matches, address does not match

Additional Information Data Layouts

Valid AVS Response Codes and Their Corresponding Issuer Descriptions *(continued)*

Code	Amex®	Discover®	MasterCard®	Visa®	STAR®	Pulse®	Accel®	Nyce®
X	Not applicable	Address & 9-digit ZIP match	For U.S., all digits match. For non-U.S., ZIP and address match	Not applicable	Address and 9-digit Postal Code match	Address and nine-digit zip match	Address and 9-digit ZIP code match	9-digit ZIP code and address match
Y	Address & ZIP match	Address only matches	Address & 5-digit ZIP match	Address & 5- or 9-digit ZIP match	Address and 5-digit Postal Code match	Address and five-digit zip match	Address and 5-digit ZIP code match	5-digit ZIP code matches
Z	ZIP only matches	5-digit ZIP matches, address does not	5-digit ZIP matches, address does not	Either 5- or 9-digit ZIP match, address does not	5-digit Postal Code matches, address does not	Five-digit zip matches, address does not match	5-digit ZIP code matches, but address does not match	5-digit ZIP code matches, address does not match
Zero	NA	NA	NA	NA	NA	AVS not performed yet or not requested	Not applicable	Stand-in is performed when there is no response from an issuer

Valid NVS Response Codes and Their Corresponding Issuer Descriptions

Code	Discover®
M	First Name and Last Name match
F	First Name Matches, Last Name does not match
L	First Name does not match, Last Name matches
N	Nothing matches
W	No data from Issuer/Authorization system

Additional Information Data Layouts

Valid AVS Response Codes
and
Their Corresponding Issuer Descriptions
(continued)

Code	Amex®	Discover®	MasterCard®	Visa®	STAR®	Pulse®	Accel®	Nyce®
U	Retry, system unable to process							
Code	**Visa®**							
A	First Name, Middle Name and Last Name match							
R	Partial Match							
N	Nothing matches							
O	Name match not performed							
S	Name match not supprted							

Additional Information Data Layouts

CVV Information

Field	Attributes	Description
Table ID	n	Fixed value: 004
Table Length	n	Fixed
Maximum Length	an	Len.: 1 Identifies the CVV response from the authorizer. Valid values: **American Express**: **Code Description** Y Card identifier (CID) match N CID nonmatch U CID not checked **Discover® Network**: **Code Description** M CID match N CID nonmatch P Not processed S CID should be on the card, but the merchant has indicated that it is not present. U Issuer is not certified. **MasterCard**: **Code Description** M Valid or matched CVC2 N Invalid CVC2 value P CVC2 not processed U Issuer unregistered for CVC2 **Visa**: **Code Description** M CVV2 and dCVV2 Match Indicates that Visa or the issuer was able to verify the CVV2 and dCVV2 value provided by the merchant and it matched. N CVV2 and dCVV2 No Match Indicates that Visa or the issuer was able to verify the CVV2 and dCVV2 value provided by the merchant and it did not

Additional Information Data Layouts		
		match. P Not Performed Indicates that Visa or the issuer participates in the CVV2 and dCVV2 service but did not perform CVV2 and dCVV2 verification due to system settings, because the transaction was declined, or because stand-in processing (STIP) has responded to an issuer-unavailable response. S System Error (Retry) Indicates that Visa or the issuer could not perform CVV2 and dCVV2 verification because the merchant did not provide the required information (CVV2 value, dCVV2 value, expiry date) in the request. U Unable to Verify Indicates that the issuer is not participating in CVV2 and dCVV2 service or has not provided Visa with encryption keys needed to perform verification. Note: The same codes apply to validation result code for Web browser based Auto fill transactions with DTVV, alongside CVV2 result codes. **STAR®:** **Code Description** M Match N No match P Not processed S Merchant/ Cardholder states expected CVV2 is not on card U No keys, not certified, or both **ACCEL®:** **Code Description** Blank Validation not attempted N CVV2/CVC2 Failed validation Y CVV2/CVC2 Passed validation **PULSE®:** **Code Description**

Additional Information Data Layouts

		P CVV2/CVC2/CID not verified M CVV2/CVC2/CID verified and matched N CVV2/CVC2/CID verified and not Matched S CVV2/CVC2/CID not on card U CVV2/CVC2/CID no keys **NYCE®:** **Code Description** P CVV2/CVC2/CID not processed M CVV2/CVC2/CID matched N CVV2/CVC2/CID did not match S CVV2/CVC2/CID not on card U CVV2/CVC2/CID no keys, not certified, or both

ECA/ TeleCheck® Trace ID

Field	Attributes	Description
Table ID	n	Fixed value: 005
Table Length	n	Fixed
Maximum Length	an	Len.: 22 Identifies the ECA/ TeleCheck® Trace ID from ECA/ TeleCheck® service. This indicates an approval from ECA/ TeleCheck® service.

ECA/ TeleCheck® Denial Record Number

Field	Attributes	Description
Table ID	n	Fixed value: 006
Table Length	n	Fixed
Maximum Length	an	Len.: 7 Identifies the Denial Record Number from ECA/ TeleCheck® service.

ECA/ TeleCheck® Return Check Data

Field	Attributes	Description
Table ID	n	Fixed value: 007
Table Length	n	Variable

Additional Information Data Layouts		
Maximum Length	an	Len.: 106 Identifies the Return Check Fee and Return Fee Data from ECA/ TeleCheck® service. **Len.** **Description** 6 Return Check Fee 100 Return Fee Data

Additional Information Data Layouts

Loyalty Information – Version 1

Field	Attributes	Description
Table ID	n	Fixed value: 008
Table Length	n	Fixed
Maximum Length	an	Len.: 229 Identifies the additional information for loyalty cards
	3n	Loyalty Processing Status **Code** **Description** 000 Success 010 Failed 100 Account not found 110 Account add failed 120 Duplicate card 130 Account already exists 140 Loyalty card is expired 150 Invalid loyalty request 160 Invalid merchant 200 Loyalty processing failed 210 Insufficient points in account 300 Transaction not found 310 Transaction already reversed 320 Transaction found, but data does not match 330 Reversal failed 340 Credit service failed 900 System failure
	6n	New points (points to be accumulated)
	6n	Total points (program point balance)
	7n	Adjusted sale amount (net result after discount applied)
	7n	Discounted sale amount (discount amount applied)
	200an	Loyalty print data (receipt printing information)

Additional Information Data Layouts

Visa® Product Result Information

Field	Attributes	Description
Table ID	n	Fixed value: 009
Table Length	n	Fixed
Maximum Length	an	Len.: 2 Identifies the Visa® card-level product result returned on all Visa® transactions.

Loyalty Information – Version 2

Field	Attributes	Description
Table ID	n	Fixed value: 010
Table Length	n	Fixed
Maximum Length	an	Len.: 60 Identifies the additional information for loyalty cards
	3n	Loyalty Processing Status **Code** **Description** 000 Success 010 Failed 100 Account not found 110 Account add failed 120 Duplicate card 130 Account already exists 140 Loyalty card is expired 150 Invalid loyalty request 160 Invalid merchant 200 Loyalty processing failed 210 Insufficient points in account 300 Transaction not found 310 Transaction already reversed 320 Transaction found, but data does not match 330 Reversal failed 340 Credit service failed 900 System failure

Additional Information Data Layouts

Loyalty Information – Version 2 (*cont.*)		
Field	**Attributes**	**Description**
Maximum Length (*cont.*)	6n	New points (points to be accumulated)
	6n	Total points (program point balance)
	7n	Adjusted sale amount (net result after discount applied)
	7n	Discounted sale amount (discount amount applied)
	19n	Unit of Work

User Data Information		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 011
Table Length	n2	Fixed (left justified, space-filled)
Maximum Length	an	Len.: 15 Identifies the user data received in the request and echoed back in the response. This data is not forwarded to the issuer.

Discover® Network Retrieval Reference Number		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 012
Table Length	an	Fixed
Maximum Length	an	Len.: 015 Identifies the Discover® Network Retrieval Reference Number.

Additional Information Data Layouts

Card Expiration Date

Field	Attributes	Description
Table ID	n3	Fixed Value: 013
Table Length		Fixed length, field not present
Maximum Length	n	Len.: 4 digits Identifies the expiration date in a TA-VE processing transaction. If no value has been stored at the device, the default value of *1249* is sent. **Note:** Depending on the context of the transaction, this value can be either MMYY or YYMM.

Additional Information Data Layouts

BUYPASS Card Type Code

Field	Attributes	Description
Table ID	n3	Fixed Value: 016
Table Length		Fixed length, field not present
Maximum Length	n1	Len.: 3 Indicates the BUYPASS Card Type Code in the transaction.

PIN on Receipt Information

Field	Attributes	Description
Table ID	n3	Fixed Value: 017
Table Length	n3	.
Maximum Length	an30	Variable length of up to 30 alphanumeric characters. Identifies the receipt data to be printed on a transaction receipt for PIN on Receipt transactions. **Note:** All Blackhawk PIN on Receipt requires Variable Information (Data Element No. 111) Table ID: 002, Program Identifier Information with a value of SP2.

PINless Debit Information

Field	Attributes	Description
Additional Information Indicator	n3	Fixed Value: 018
Table Length	n3	Fixed len: 003
Card Accepted As	an3	This field is mandatory for requests in which the POS device has sent Segment 111, Table 21 (Settlement Data Acceptance Flag). **Value** **Description** 020 Transaction was processed as a credit card (or Signature Debit eligible transaction that went to a Credit (CC) Authorizer) 011 Transaction was processed as a debit card (or Signature Debit eligible transaction that went to a Debit (DB)

Additional Information Data Layouts		
		Authorizer as Single Message) 025 Transaction was processed via Signature Debit Dual message Authorizer This field is Conditional in response to the Card Not Present Pinless eComm/Mobile Commerce transactions. When a transaction is presented as credit and processed as debit, the value in the response will be 011, which indicates that the transaction went to a Debit Authorizer and was approved. This will act as an indicator for the merchant to send subsequent transactions (completions, reversals) as Debit, since the original transaction was processed as Debit.

Visa Spend Qualified Indicator Information		
Field	**Attributes**	**Description**
Table ID	n2	Fixed Value: 019
Table Length	n3	Fixed Len.: 001
Spend Qualified Indicator	an1	Identifies spend qualified indicator information from Visa. Visa uses Spend Qualified Indicator to establish annual point-of-sale (POS) spending requirements.

Host Prompts Information		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 020
Table Length	n3	Variable length of up to 15 alphanumeric characters
Table Data	an15	Identifies the additional prompts that need to be requested and sent on the Completion message along with the original prompts.

Additional Information Data Layouts

Host Prompts Information (*cont.*)

Field	Attributes	Description
		Terminal Type. The data returned can have max of 5 prompts with each prompt being represented by 3 bytes. Example: 112024020012DRVDLSZIPDAT Valid 3 byte tags that can be sent back to POS are: **Value** **Description** USE User ID VID Vehicle ID DRV Driver ID ODO Odometer LIC Driver License Number DLS Driver License State/Province Abbreviation DLN Driver License Naming PON Work Order/P.O. Number INV Invoice Number TRP Trip Number UNT Unit Number TLH Trailer hours/Refer hours DOB Date Of Birth ZIP ZIP/Postal Code DAT Data END Entered Data (Alphanumeric) JOB Job Number MID Maintenance ID DPT Department VIN VIN TRA Tractor Number HUB Hubometer TLR Trailer Number CBA Cashback Amount VHT Vehicle Tag

Additional Information Data Layouts

Re-Price Data Response Information

Field	Attributes	Description
Table ID	n3	Fixed Value: 021
Table Length	n3	Variable length of up to 133 alphanumeric characters
Table Data	an15	Lists the different products available at the pump which can be re-priced. A max of 8 products can be sent. The format of the data should be as follows: Nbr of Products: N(2) Product Data: Repeated for number of products sent. Product Code: NAC Product Code of 3 bytes. Re-Price Type: 2 bytes. Valid values are as follows: **Note**: Additional values may be added. **Value** **Description** RP Retail Plus RM Retail Minus CM Cost Minus CP Cost Plus BO Best Of OD Other Discount RT Retail Price FP Fixed Price Unit Of Measure: 1 byte. Same as unit of measure sent in product data. Decimal Identifier Prc: 1 byte. Identifies the number of decimals in the unit price field. Unit Price: Price per unit up to 8 bytes. Delimiter: 1 byte ('\') Product Code Set: 3 bytes. Valid values are as follows: **Value** **Description** 001 NACS Example: 112041021029020010DG33159\002RTG33259\001

Additional Information Data Layouts

CAVV Result

Field	Attributes	Description
Table ID	n3	Fixed value: 022
Table Length	n3	Length 3 bytes
CAVV Result	n	Fixed Length: 1 byte Specifies the CAVV response received from the issuer. Example:- 112057003005AVS+A00100600123401201512345678000000022001Y Valid CAVV Result values are as follows: **Code** — **Description** Blank — CAVV not verified. 0 — CAVV not verified due to erroneous data submitted or no CAVV passed when expected. 1 — CAVV failed verification–cardholder authentication. 2 — CAVV passed verification–cardholder authentication. 3 — CAVV passed verification–attempted authentication 4 — CAVV failed verification–attempted authentication 5 — Reserved for future use. 6 — CAVV not verified. Issuer not participating in CAVV verification. 7 — CAVV failed verification—attempt. 8 — CAVV passed verification—attempt. 9 — CAVV failed verification—attempt. A — CAVV passed verification—attempt. B — CAVV passed verification—information only, no liability shift. C — CAVV was not verified—attempted authentication. D — CAVV was not verified—cardholder authentication **Note:** Discover supports 0,1, and 2 values Visa supports all of the values mentioned above

MCX Reference Number

Field	Attributes	Description
Table ID	n3	Fixed value: 023

Additional Information Data Layouts		
Table Length	n3	Fixed Value: 012
Table Data	an12	Reference number for printing on customer receipt. Taken from MCX ISO DE37.
Carwash Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 024
Table Length	n3	Fixed Value: 001
Table Data	an1	Single character indicator that specifies user's carwash preference.
Language Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 025
Table Length	n3	Fixed Value: 001
Table Data	an1	Single character indicator that specifies user's language preference.
DST Response		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 026
Table Length	n3	Fixed Value: 016
Table Data	an16	DST Challenge Response consisting of 10 bytes of DST Challenge value and 6 bytes of DST Challenge response value.
Universal Unique Identifier		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 027
Table	n3	Variable length of up to 41 alphanumeric special characters

Additional Information Data Layouts		
Length		
Table Data	an41	Universal Unique Identifier – This is a unique identifier generated by First Data internal systems that uniquely identifies a transaction throughout the entire First Data infrastructure. Returned in the response message only when merchant requested this data by sending Use Unique Identifier as 'Y' in the request message
Transaction Identifier		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 028
Table Length	n3	Fixed Value: 015
Table Data	an15	Transaction Identifier **Note:** Authorizer will send this data in response message. Merchant should store this data for Visa® and Amex merchant-initiated Credential on File transactions. (Example: Recurring/installment transactions initiated using stored Visa/Amex credentials)This value will be sent back in response only for transaction requests coming with the Credential on File Indicator as "I" or "C", otherwise this field will not beincluded in the response.
Table ID 029 – Reserved for Internal Use		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 029
Table Length	n3	Fixed Length: 029
Table Data	An29	Identifies the Payment Account Reference (PAR) data received from the authorizer Fixed length of 29 characters Merchants receive PAR Data from the authorizer only if they have requested for it in the POS Additional Data table ID '32' sub table ID '14' with value 'Y'. Source: Authorizer

Additional Information Data Layouts

Merchant Advice Code

Field	Attributes	Description
Table ID	n3	Fixed Value: 030
Table Length	N3	Fixed Length: 002
Table Data	AN2	Indicates Merchant Advice Code received from the authorizer network. Fixed length of 2 bytes Alphanumeric. Valid values: 01 = New account information available 02 = Cannot approve at this time, try again later 03 = Do not try again 04 = Token requirements not fulfilled for this token type 21 = Payment Cancellation 22 = Merchant does not qualify for product code 24 = Retry after 1 hour 25 = Retry after 24 hours 26 = Retry after 2 days 27 = Retry after 4 days 28 = Retry after 6 days 29 = Retry after 8 days 30 = Retry after 10 days 40 = Consumer non-reloadable prepaid card 41 = Consumer single-use virtual card number 43 = Consumer multi-use virtual card number Note: Currently Merchant Advice Code is supported only for MasterCard network. Merchant should NOT retry on getting the below Merchant Advice Codes • 03 [Do not try again] • 21 [Payment Cancellation] • 22 [Merchant does not qualify for product code] Merchants will receive Merchant Advice Code only if they send

Additional Information Data Layouts		
		Merchant Advice Code acceptance indicator in Merchant supplementary Data table ID “049’ sub table ID ‘07” with value ‘Y’. ‘Merchant does not qualify for product code' will be returned by authorizer on declines when a merchant is not registered for the Purchase repayment program. MasterCard may add new merchant advice codes at any point of time. Merchant should be ready to accept any new Merchant advice code values. Source: Authorizer
DAF Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 031
Table Length	N3	Fixed Length: 002
Table Data	AN2	Indicates DAF indicator received from the authorizer network. Fixed length of 2 bytes Alphanumeric. Valid values: 01 = DAF (Digital Authentication Framework) Indicator 02 = Visa Secure 03 = Visa Data Only 04 = Visa Payment Passkey with Visa secure 05 = Visa Payment Passkey with VTS 06 = IDX 3rd Party 07 = VTS Data Only Note: Merchant receives DAF Indicator only if it is sent in the Authorizer network’s response and the Merchant sent the DAF Indicator acceptance flag in Merchant Supplementary Data sub-table ID “10” with value “Y””.

Additional Information Data Layouts		
		Source: Authorizer Note: Currently DAF Indicator is supported only for Visa network.
Agreement ID		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 032
Table Length	N3	Fixed Length: 004
Table Data	ANS4	Indicates Agreement ID received from the authorizer network. Fixed length of 4 bytes (ANS) and it is left-aligned; space filled. Note: Merchant receives Agreement ID only if it is sent in the Authorizer network's response and the Merchant sent the Agreement ID acceptance flag in Merchant supplementary Data table ID "049' sub table ID '11" with value 'Y'. Source: Authorizer Note: Currently Agreement ID is supported only for VISA network.
Host-based Purchase Restriction		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 034
Table Length	n3	Length: 3 bytes
Table Data	ans32	Variable length of up to 32 bytes. This field contains Host-based Purchase Restrictions received from the network. The Merchant receives Host-based Purchase Restrictions only if the Merchant has sent the Purchase Restriction Method value "2" or "3" in Merchant supplementary Data sub table ID '18", and if a value is returned in the network response. Note: Host-based Purchase Restrictions are applicable only to both Visa Fleet EMV 2.0 and MasterCard Enhanced Fleet EMV transactions

Additional Information Data Layouts		
		. Note: For a full list of values supported in this field, please refer to the Conexxus PSPC reference document.
Cardholder Additional Information Result Code		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 035
Table Length	n3	Length: 3 bytes
Table Data	ans50	Maximum Len.: 50 bytes. Indicates whether Cardholder Additional Information result code is present. Cardholder Additional Information result code contains the following sub-table IDs.
Sub Table ID "01" Purpose: Identifies the Cardholder Account Name result codes		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed Value: 01
Sub-Table Length	n2	Fixed Value: 03
Sub-Table Data	Ans3	This field contains the Cardholder name result codes received from the Visa network if the merchant sends Table ID 34 (NVS Information) in Variable information segment 111 for a verification transaction. It contains 3 bytes of data. Below are the valid values for all the 3 bytes of data. First byte refers to the Account nume match decision for First Name. Second byte refers to the account name match decision for Last name. Third byte refers to the account name match decision for Middle name. The valid values are: M - Match P - Partial Match N - No Match SPACES – Not applicable or name match decision not received from Visa network.

Additional Information Data Layouts

		Ex.1: MNP - It means First name full match, Last name no match, and Middle name partial match. Ex 2: M N - It means First name full match, spaces (no account match decision for Last name), and Middle name no match. **Note:** Currently applicable to Visa only.
Sub Table ID "02" Purpose: Identifies the Cardholder Phone result code		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed value: 02
Sub-Table Length	n2	Fixed value: 01
Sub-Table Data	n1	This field contains the Cardholder Phone result code received from the Visa network. Valid Values: 1 – Verified 2 – Failed 3 – Not performed Note: Applicable only to Visa.
Sub Table ID "03" Purpose: Identifies the Cardholder Email result code		
Field	**Attributes**	**Description**
Sub-Table ID	n2	Fixed value: 03
Sub-Table Length	n2	Fixed value: 01
Sub-Table Data	n1	This field contains the Cardholder Email address result code received from the Visa network. Valid Values: 1 – Verified 2 – Failed 3 – Not performed

Additional Information Data Layouts		
		Note: Applicable only to Visa.
Account Type		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 036
Table Length	n3	Fixed Value: 002
Table Data	n2	This field identifies the 'from' account type received from network. Valid Values: **00 Not applicable or not specified** **20 Checking account** **30 Credit card account** **35 Deferred debit account** **36 Charge account** **60 Prepaid account** **Note:** Applicable to Visa card type only. A merchant receives Account Type only if it is sent in the Authorizer network's response and if the merchant has sent the Merchant Supplementary Data sub-table ID "20" with the value "Y".
Account Funding Source		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 037
Table Length	n3	Fixed Value: 001

Additional Information Data Layouts		
Table Data	an1	This field contains the product account funding source value from Visa. Valid Values: **C (Credit)** **D (Debit)** **P (Prepaid)** **Note:** A merchant receives Account Funding Source only if it is sent in the Authorizer network's response and if the merchant has sent the Visa Account Flag in the Merchant Supplementary Data sub-table ID "20" with the value "Y".
Transaction Link Identifier		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 038
Table Length	n	Fixed
Maximum Length	ans	Len: 22 This field contains the Transaction Link Identifier. It will only be included if sub-table 22 of Merchant Supplementary Data contains the value "Y". For Moneris: This field maps to Moneris field Real Time Unique ID 9e. **Note:** Special characters are allowed.
Transaction Link Action Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 039
Table Length	n3	Fixed Length: 001
Table Data	an1	This one-byte field indicates the Transaction Link Action Indicator from MasterCard/Maestro. Values are: 1 – The customer did not provide the TLID. Mastercard/Maestro

Additional Information Data Layouts

		responds with the corresponding TLID in the segment 112, table ID 38. 2 - The TLID provided by a customer is unmatched. Mastercard/Maestro responds with the corresponding TLID in the segment 112, table ID 38. 3 - Mastercard/Maestro was unable to match this lifecycle message to any original message. Mastercard responds with the corresponding TLID in the segment 112, table ID 38. This field can be returned in a response message only when a merchant sends the Transaction Link Action Support Indicator in the Variable segment 111, table ID 049, sub table ID 26.
Fraud Score		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 040
Table Length	n	Fixed
Maximum Length	n	Len: 3 This field contains the fraud score returned by MasterCard. It is only included if the field Future Transaction Amount was included in the transaction request.
Fraud Score Reason Code		
Field	**Attributes**	**Description**
Table ID	n	Fixed value: 041
Table Length	n	Fixed
Maximum Length	n	Len: 2 This field contains the reason code explaining the fraud score in table ID 040. It will only be included if the field Future Transaction Amount has been included in the transaction request.
Authentication Data Quality Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 042
Table Length	n3	Fixed Length: 001

Additional Information Data Layouts		
Table Data	an1	This field indicates whether a transaction meets the Visa Secure Authentication Data Quality requirements and is received from Authorizer. Fixed length of 1 byte alphanumeric. Valid values: 1 = Does not meet Visa Secure authentication data quality requirements 2 = Meets Visa Secure authentication data quality requirements **Note:** A merchant receives Authentication Data Quality Indicator only if it is sent in the Authorizer network's response and if the merchant has sent the Data Quality Indicator Acceptance Flag with the value "Y" in Merchant Supplementary Data sub-table ID "23". Source: Authorizer **Note:** Authentication Data Quality Indicator is currently supported for the Visa network only.
Token Update First Use Indicator		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 043
Table Length	n3	Fixed Length: 001
Table Data	an1	This field indicates that it is the first time a network token is being used since the underlying PAN or PAN expiration date was updated in the Network Digital Credential Updater system (e.g. VDCU or Visa Digital Credential Updater). When it is not the first time a token is being used since a PAN update, the field will be absent. Valid values: Y = First use since update **Note:** A merchant receives the Token Update First Use Indicator only if it is sent in the Authorizer network's response and if the merchant has sent the Token Update First Use Acceptance Flag with the value "Y" in Merchant Supplementary Data sub-table ID "24".

Additional Information Data Layouts		
		Source: Authorizer **Note:** This field is applicable to Visa e-commerce/card-on-file and e-commerce enabler token types.
Merchant Tran ID		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 044
Table Length	n3	Fixed
Table Data	an..95	Variable length up to 95 bytes. A value assigned by the merchant to identify the transaction. BUYPASS will echo this field in responses only if Merchant Supplementary Data subfield 25 is set to "Y". For Moneris: This field has a max length of 15, and maps to Moneris field Client Request ID 7b.
Applied Special Service		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 045
Table Length	n3	Fixed Length: 001
Table Data	n	Identify if the merchant has used the eligible authentication or data sharing method and provided the required data elements. It will only be included if subfield 27 of Merchant Supplementary Data is populated with the value “Y”.
Voyager Restriction Code		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 046
Table Length	n3	Fixed Length: 001
Table Data	N1	Contains product restriction from Voyager in an unspecified format. It will only be included if subfield 28 of Merchant Supplementary Data is populated with the value “Y”.
Visa Category Code		

Additional Information Data Layouts		
Field	**Attributes**	**Description**
Table ID	n3	Fixed Value: 047
Table Length	n3	Fixed Length: 001
Table Data	n1	This one-byte field indicates the category of a Visa 'declined' response code. Values are: **1** – Issuer never approves **2** – Issuer cannot approve at this time **3** – Data quality/revalidate payment information This field can be returned in a response message only when the card type is 'Visa', and the merchant sends the Visa Category Code Support Indicator in the Variable segment 111, table ID 049, sub table ID 30.

Appendix L. Valid Currency Codes

This appendix contains a list of currencies and their corresponding valid Currency Codes for Element No. 20 (Currency Code). These codes identify the currency used in a transaction.

Information in the "Visa®, MC, or Both" column indicates whether the Currency Code is used for Visa®, MasterCard, or both card types.

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
Afghani	Afghanistan	971	Both
Algerian Dinar	Algeria	012	Both
Angola Kwanza	Angola	973	Both
Argentine Peso	Argentina	032	Both
Armenian Dram	Armenia	051	Both
Australian Dollar	Australia Christmas Island Cocos (Keeling) Islands Heard and McDonald Islands Kiribati Nauru Norfolk Island Tuvalu	036	Both
Azerbaijanian Manat	Azerbaijan	944	Both
Bahamian Dollar	Bahamas	044	Both
Belarusian Ruble	Belarus	933	Both
Bahraini Dinar	Bahrain	048	Both
Baht	Thailand	764	Both
Balboa	Panama	590	Both
Barbados Dollar	Barbados	052	Both
Belize Dollar	Belize	084	Both
Bermudian Dollar	Bermuda	060	Both
Bhutanese Ngultrum	Bhutan	064	Both
Bolivar Soberano	Venezuela	928	Both
Boliviano	Bolivia	068	Both
Brazilian Real	Brazil	986	Both
Brunei Dollar	Brunei Darussalam	096	Both

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
Burundi Franc	Burundi	108	Both
Canadian Dollar	Canada	124	Both
Cape Verde Escudo	Cape Verde	132	Both
Cayman Islands Dollar	Cayman Islands	136	Both
Cedi	Ghana	936	Both
CFP Franc	French Polynesia New Caledonia Wallis and Futuna	953	Both
CFA Franc BCEAO[2]	Benin Burkina Faso Cote D'Ivoire (Ivory Coast) Guinea-Bissau Mali Niger Senegal Togo	952	Both
CFA Franc BEAC[3]	Cameroon Central African Republic Chad Congo Equatorial Guinea Gabon	950	Both
Chilean Peso	Chile	152	Both
Chinese People's Bank Dollar	China	158	MC
Chinese Yuan Renminbi	China	156	Both
Colombian Peso	Columbia	170	Both
Comoro Franc	Comoros	174	Both
Congolese Franc	Democratic Republic of the Congo	976	Both
Convertible Mark	Bosnia and Herzegovina	977	Both
Cordoba Oro	Nicaragua	558	Both
Costa Rican Colon	Costa Rica	188	Both

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
Dalasi	Gambia	270	Both
Danish Krone	Denmark Faroe Islands Greenland	208	Both
Denar	Macedonia	807	Both
Djibouti Franc	Djibouti	262	Both
Dobra	Sao Tome and Principe	930	Both
Dominican Peso	Dominican Republic	214	Both
Dong	Vietnam	704	Both
East Caribbean Dollar	Anguilla Antigua and Barbuda Dominica Grenada Montserrat St. Kitts-Nevis St. Lucia St. Vincent and the Grenadines	951	Both
Egyptian Pound	Egypt	818	Both
El Salvador Colon	El Salvador	222	MC
Eritrean Nakfa	Eritrea	232	Visa®
Ethiopian Birr	Ethiopia	230	Both
Ethiopian Birr	Eritrea	230	MC
Euro	SAINT Barthélemy SAINT MARTIN, French part	978	MC
Euro	France, Metropolitan	978	Visa®

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
Euro	Aland Islands Andorra Austria Belgium Bulgaria Croatia* Cyprus Estonia European Union Countries Finland France French Guiana French Southern Territories Germany Greece Guadeloupe Holy See (Vatican City State) Ireland Italy Kosovo, United Nations Mission in Kosovo (UNMIK) Latvia Lithuania Luxembourg Malta Martinique Mayotte Monaco Montenegro Netherlands Portugal Reunion St. Pierre and Miquelon San Marino Slovakia Slovenia Spain	978	Both

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
Falkland Islands Pound	Falkland Islands (Malvinas)	238	Both
Fiji Dollar	Fiji	242	Both
Forint	Hungary	348	Both
Gibraltar Pound	Gibraltar	292	Both
Gourde	Haiti	332	Both
Guarani	Paraguay	600	Both
Guilder	Aruba	533	Both
Guinea Franc	Guinea	324	Both
Guyana Dollar	Guyana	328	Both
Hong Kong Dollar	Hong Kong	344	Both
Hryvnia	Ukraine	980	Both
Iceland Krona	Iceland	352	Both
Indian Rupee	Bhutan	356	MC
Indian Rupee	India	356	Both
Iraqi Dinar	Iraq	368	Both
Jamaican Dollar	Jamaica	388	Both
Jordanian Dinar	Jordan	400	Both
Kenyan Shilling	Kenya	404	Both
Kina	Papua New Guinea	598	Both
Kip	Lao People's Democratic Republic	418	Both
Koruna	Czech Republic	203	Both
Kuwaiti Dinar	Kuwait	414	Both
Lari	Georgia	981	Both
Lebanese Pound	Lebanon	422	Both
Lek	Albania	008	Both
Lempira	Honduras	340	Both
Leone	Sierra Leone	925	Both
Liberian Dollar	Liberia	430	Both
Libyan Dinar	Libyan Arab Jamahiriya	434	Visa®

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
Lilangeni	Swaziland	748	Both
Loti	Lesotho	426	Both
Malagasy Ariary	Madagascar	969	Both
Malaysian Ringgit	Malaysia	458	Both
Malawi Kwacha	Malawi	454	Both
Manat	Turkmenistan	934	Both
Mauritius Rupee	Mauritius	480	Both
Mexican Peso	Mexico	484	Both
Moldovan Lau	Moldova, Republic of	498	Both
Moroccan Dirham	Morocco Western Sahara	504	Both
Mozambique Metical	Mozambique	943	Both
Myanmar Kyat	Myanmar	104	Both
Naira	Nigeria	566	Both
Namibian Dollar	Namibia	516	Both
Nepalese Rupee	Nepal	524	Both
Caribbean Guilder	Curacao Sint Maarten (Dutch Part)	532	Both
New Israeli Shekel	Israel	376	Both
New Taiwan Dollar	The Republic of China (Taiwan)	901	Both
New Zealand Dollar	Cook Islands New Zealand Niue Pitcairn Tokelau	554	Both
Norwegian Krone	Bouvet Island Norway Svalbard and Jan Mayen	578	Both
Norwegian Krone	Antarctica	578	MC
Nuevo Sol	Peru	604	Both
Ouguiya	Mauritania	929	Both

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
Pa'anga	Tonga	776	Both
Pakistan Rupee	Pakistan	586	Both
Pataca	Macao	446	Both
Peso Uruguayo	Uruguay	858	Both
Philippine Peso	Philippines	608	Both
Pound Sterling	Guernsey Isle of Man Jersey United Kingdom	826	Both
Pound Sterling	South Georgia and South Sandwich Islands	826	Both
Pula	Botswana	072	Both
Qatari Rial	Qatar	634	Both
Quetzal	Guatemala	320	Both
Rand	Lesotho Namibia	710	Visa®
Rand	South Africa	710	Both
Rial Omani	Oman	512	Both
Riel	Cambodia	116	Both
Romanian Leu	Romania	946	Both
Rufiyaa	Maldives	462	Both
Rupiah	Indonesia	360	Both
Russian Ruble	Russian Federation	643	Both
Rwanda Franc	Rwanda	646	Both
Saudi Riyal	Saudi Arabia	682	Both
Serbian Dinar	Serbia	891	MC
Serbian Dinar	Serbia, Republic of	941	Both
Seychelles Rupee	Seychelles	690	Both
Singapore Dollar	Singapore	702	Both
Som	Kyrgyzstan	417	Both
Solomon Islands Dollar	Solomon Islands	090	Both

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
Somali Shilling	Somalia	706	Both
South Sudanese Pound	Republic of South Sudan	728	Both
Spanish Peseta (ESA)	Brazil	996	MC
Sri Lanka Rupee	Sri Lanka	144	Both
St. Helena Pound	St. Helena	654	Both
Sudan Pound	Sudan	938	Both
Suriname Dollar	Suriname	968	Both
Swedish Krona	Sweden	752	Both
Swiss Franc	Liechtenstein Switzerland	756	Both
Tajikistan Somoni	Tajikistan	972	Both
Taka	Bangladesh	050	Both
Tala	Samoa	882	Both
Tanzanian Shilling	Tanzania, United Republic of	834	Both
Tenge	Kazakhstan	398	Both
Trinidad and Tobago Dollar	Trinidad and Tobago	780	Both
Tugrik	Mongolia	496	Both
Tunisian Dinar	Tunisia	788	Both
Turkish Lira	Turkey	949	Both
UAE Dirham	United Arab Emirates	784	Both
Uganda Shilling	Uganda	800	Both
U.S. Dollar	Libyan Arab Jamahiriya Palestine Panama	840	MC

Valid Currency Codes			
Currency	**Country**	**Currency Code**	**Visa, MC, or Both**
U.S. Dollar	American Samoa Bonaire, Sint Eustatius and Saba British Indian Ocean Territory Ecuador El Salvador Guam Marshall Islands Micronesia Northern Mariana Islands Palau Puerto Rico Timor-Leste Turks and Caicos Islands United States U.S. Minor Outlying Islands Virgin Islands, British Virgin Islands, U.S.	840	Both
Uzbekistan Sum	Uzbekistan	860	Both
Vatu	Vanuatu	548	Both
Won	Korea, Republic of	410	Both
Yemeni Rial	Yemen	886	Both
Yen	Japan	392	Both
Zambian Kwacha	Zambia	967	Both
Zimbabwe Gold	Zimbabwe	924	Both
Zloty	Poland	985	Both

[2] CFA Franc BCEAO; Responsible authority:Banque Centrale des Etats de l'Afrique de l'Quest

[3] CFA Franc BEAC; Responsible authority:Banque des Etats de l'Afrique Centrale

**Effective 1st January 2026, Euro is the currency for Bulgaria.

*Effective 1st January 2023, Euro is the currency for Croatia.

Appendix M. Element Format Examples for EBT Program Data

This appendix contains the following two format examples for Element No. 164 (EBT Program Data):

- One when used in a request that contains only one set of subelements
- One when used in a request that contains two sets of subelements

The following Example describes the element's format when used in a Request containing only one set of subelements—A HIP purchase/return amount:

Example. 024502098508 40C000000001234

Valid Codes/Values for the Element when Used in A Request Containing a HIP Purchase/Return Amount	
Data	**Description**
Subelement 1: Total Length	
024	Total Length of all data in the entire message Format: n3; right-aligned; zero-filled The value "024" indicates that 24 characters follow.
Subelement 2: Program Data	
50	New Label: TAG (Indicates the type of HIP data to follow.) Format: n2 The value "50" indicates a HIP purchase/return amount follows.
20	New Label: LEN (Indicates the length of the Detail data to follow. Format: n2 The value "020" indicates that 20 characters follow.

Valid Codes/Values for the Element when Used in A Request Containing a HIP Purchase/Return Amount	
Data	**Description**
Subelement 2: Program Data (*continued*)	
98	New Label: ACCOUNT TYPE Format: n2 Valid value: 98 for HIP Account Type
50	New Label: AMOUNT TYPE Format: n2 Valid value: 40 for HIP Amount Type
840	New Label: CURRENCY CODE Format: n3 Valid value: 840 for US Dollars
C	New Label: AMOUNT DESCRIPTOR Valid values: 0, C, or D (0 or C = credit amount; and D = debit amount)
000000001234	New Label: DETAIL HIP purchase/return amount with two assumed decimal places; right-aligned, padded with zeroes. The value "000000001234" indicates that the HIP purchase/return amount is $12.34.

The following Example describes the element's format when used in a Request containing two sets of subelements—A HIP purchase/return amount and Internet shipping information:

Example.
065502098508 40C000000001234IT37AAAAAAAAAAAAAAAAAAAAAAAAAAZZZZZZZZZ

Valid Codes/Values for the Element when Used in a Request Containing a HIP Purchase/Return Amount and Internet Shipping Information	
Data	**Description**
Subelement 1: Total Length	
065	Total Length of all data in the entire message Format: n3; right-aligned; zero-filled The value "065" indicates that 65 characters follow.
Subelement 2: Program Data	
50	New Label: TAG (Indicates the type of HIP data to follow.) Format: n2 Description: The value "50" indicates a HIP purchase/return amount follows.
20	New Label: LEN (Indicates the length of the Detail data to follow. Format: n2 The value "020" indicates that 20 characters follow.
98	New Label: Account Type Format: n2 Valid value: 98 for HIP Account Type

Valid Codes/Values for the Element when Used in a Request Containing a HIP Purchase/Return Amount and Internet Shipping Information	
Data	**Description**
Subelement 2: Program Data (*continued*)	
50	New Label: Amount Type Format: n2 Valid value: 40 for HIP Amount Type
840	New Label: Currency Code Format: n3 Valid value: 840 for US Dollars
C	New Label: Amount Descriptor Valid values: 0, C, or D (0 or C = credit amount; and D = debit amount)
000000001234	New Label: DETAIL HIP purchase/return amount with two assumed decimal places; right-aligned, padded with zeroes. The value "000000001234" indicates that the HIP purchase/return amount is $12.34.
Subelement 3: Program Data (Second Program Data Subelement)	
IT	New Label: TAG (Indicates the TYPE OF hip data to follow. Format: an2 The value "IT" indicates a HIP Internet purchase shipping address/zip code follows.
37	New Label: LEN (Indicates the length of the DETAIL data to follow.) Format: n2 The value "37" indicates that 37 characters of the shipping address follow.
AAAAAAAAAAAAAAAAAAAAAAAAAAAA ZZZZZZZZZ	New Label: DETAIL HIP Internet purchase shipping address/zip code. The value "AAAAAAAAAAAAAAAAAAAAAAAAAAAA" indicates the 28-character shipping address—beginning with the street address. The value "ZZZZZZZZZ" indicates the 9-digit zip code.

Appendix N. First Data™ Premium Gift Card Magnetic Stripe Data Layout

This appendix contains the following table that describes the magnetic stripe data layout for Track II for the First Data™ Premium Gift Card:

Magnetic Stripe Data Layout Track II			
Field No.	**Len.**	**Name**	**Description**
1	1	Start Sentinel	This character is not displayed as part of the Track data and is cut from the packet.
2	16	Account Number	A Visa standard account number with the exception that the number is required to be 16 digits and uses the First Data™ Premium Gift Card ISO assigned BIN.
3	1	Separator	Separator (=)
4	4	Expiration Date	Format: YYMM The Track expiration date.
5	3	Service Code	
6	1	PVKI	Not currently used by the First Data™ Premium Gift Card system.
7	4	PVV	
8	8	Discretionary Data	See Note below this table.
9	1	End Sentinel	This character is not included in the packet that is transmitted to the host application.
10	1	LRC	Longitudinal Redundancy Check

Note: The card's serial number is created from the first four bytes of the Discretionary Data and the nine bytes that immediately follow the 6-byte BIN in the Account Number field of the Track data. The Account Number field is formatted as follows:

Layout Field No. 2 (Account Number)		
Size	**Name**	**Description**
6	BIN	
9	Partial Account Number (*B*)	
1	Visa Check Digit	modulo

Appendix O. Payment Token Terminologies

This appendix contains a list of general terminologies related to payment tokenization transactions.

Payment Tokenization Terminologies and Descriptions	
Term	**Description**
Cryptogram Token Data	In App transactions coming from merchants will be sent with security data that contains In App token cryptogram. Merchant can send 28 bytes or 56 bytes of Cryptogram Data in base64 format. If merchant sends 56 bytes of token data, then BUYPASS considers it as token data block A and token data block B. If merchant sends only 28 bytes of token data, then BUYPASS considers it as token data block A.
Payment Token	A surrogate value for a PAN is generated by and maintained in the token vault. This value is consistent with BUYPASS Platform (BUYPASS).message requirements, and is a 13 to 19-digit numeric value that must pass basic validation rules of an account number, including the LUHN check. Payment tokens are generated within a BIN range that has been designated as a token BIN range and flagged accordingly in all appropriate BIN tables. Payment tokens must not have the same value as a cardholder PAN.
Payment Token Expiration Date	The expiration date of the token is populated in the PAN Expiration Date field during transaction processing to ensure interoperability and minimize the impact of tokenization implementation.
Payment Token Requestor	Enrolled entity requesting payment token PANs for initiating payments. Example: Merchants, Vendors, Aggregators, Acquirers, etc.
Payment Token Service Provider	The operator of the token vault which is the licensed BIN provider; Example: Card brand networks, Issuers etc.
POS Entry Mode	A value that is used to indicate the token presentment mode through which the token is presented for payment.

Appendix P. TransArmor VeriFone Edition: Responses to Administrative Command

Note: For detailed information on TransArmor, refer the BUYPASS®_Platform_ATL105_Specification_Updates_for_TransArmor_Processing document, available on the Direct Platform Specifications Portal.

Appendix Q. Valid National Point-of-Service Condition Codes

This appendix lists valid entries for Variable Information (Element No. 113), Table ID 030

The National Point-of-Service Condition Codes data element contains the following elements:

- Terminal Class (Position Nos. 1–3)
- Presentation Type (Position Nos. 4–7)
- Security Condition (Position No. 8)
- Terminal Type (Position Nos. 9–10)

This appendix contains the following information about valid entries for these components:

- Valid Terminal Class (Position Nos. 1–3) Entries
- Valid Terminal Type (Position Nos. 9–10) Entries
- ISO Codes for Position Nos. 1–3 and 9–10
- Valid Presentation Type (Position Nos. 4–7) Entries
- ISO Codes for Position Nos. 4–7)
- Valid Security Condition (Position No. 8) Entries

The following table contains information about valid entries for Terminal Class (Position Nos. 1–3):

National Point-of-Service Condition Code **Terminal Class (Position Nos. 1–3)** **Valid Entries** (Within the National Point-of-Service Condition Code, Position Nos. 1–3 identify the Terminal Class, and Position Nos. 9–10 identify the Terminal Type.)	
Position No. 1	
Value	**Description**
0	Attended
1	Unattended
2–7	Reserved for national use
8–9	Reserved for private use
Position No. 2	
Value	**Description**
0	Customer-operated
1	Card acceptor-operated
2	Administrative
3–7	Reserved for national use
8–9	Reserved for private use
Position No. 3	
Value	**Description**
0	On premise
1	Off premise
2–7	Reserved for national use
8–9	Reserved for private use

The following table contains information about valid entries for Terminal Type (Position Nos. 9–10):

National Point-of-Service Condition Code Terminal Type (Position Nos. 9–10) Valid Entries (Within the National Point-of-Service Condition Code, Position Nos. 9–10 identify the Terminal Type.)	
Value	**Description**
00	Administrative terminal
01	Point-of-service terminal
02	ATM
03	Home terminal
04	ECR
05	Dial terminal
06	Traveler's check machine
07	Fuel machine
08	Scrip machine
09	Coupon machine
10	Ticket machine
11	Point-of-banking terminal
19	Call Center Operator
20	Voice response unit (VRU).
21	Channel encryption; cardholder certificate is not used
24	Digital secure remote payment with UCAF data transactions
25	Internet (nonrecurring) (secure using Verified by Visa® or 3DES)
26	Internet (secure; 3DES attempted)
27	Internet (secure using Secure Socket Layer [SSL])
28	Internet (nonsecure)
29	Internet recurring payment transaction after initial transaction was processed with a mobile device-issued payment token (MasterCard Only)

The following table shows a few examples of translations from the internal Terminal Type to the ISO codes. For all possible combinations of the ISO codes, please refer to National Point-of-Service Condition CodeTerminal Class (Position Nos. 1–3) and National Point-of-Service Condition CodeTerminal Type (Position Nos. 9–10).The values in Position Nos. 1 and 2 are variable from an ISO acquirer.

Codes for Position Nos. 1–3 and 9–10		
Position Nos. 1–3	**Position Nos. 9–10**	**Terminal Class/Type Description**
020	00	Attended on premise administrative terminal
021	00	Attended off premise administrative terminal
100	01	Unattended on premise customer-operated point-of-service terminal
001	01	Attended off premise customer-operated point-of-service terminal
100	02	Unattended on premise customer-operated ATM
101	02	Unattended off premise customer-operated ATM
000	03	Attended on premise customer-operated home terminal
001	03	Attended off premise customer-operated home terminal
000	04	Attended on premise customer-operated ECR
010	04	Attended on premise Card acceptor-operated ECR
001	04	Attended off premise customer-operated ECR
000	05	Attended on premise customer-operated dial terminal
001	05	Attended off premise customer-operated dial terminal
100	06	Unattended on premise customer-operated traveler's check machine
101	06	Unattended off premise customer-operated traveler's check machine
100	07	Unattended on premise customer-operated fuel machine
101	07	Unattended off premise customer-operated fuel machine
100	08	Unattended on premise customer-operated scrip machine
101	08	Unattended off premise customer-operated scrip machine
100	09	Unattended on premise customer-operated coupon machine
101	09	Unattended off premise customer-operated coupon machine
000	10	Attended on premise customer-operated ticket machine
001	10	Attended off premise customer-operated ticket machine
000	11	Attended on premise customer-operated point-of-banking terminal
001	11	Attended off premise customer-operated point-of-banking terminal
101	20	Unattended off premise customer-operated voice response unit (VRU)
101	25	Unattended off premise Customer-operated internet (nonrecurring). The cardholder was authenticated using an authentication methodology (Verified by Visa or 3DES).
101	26	Unattended off premise customer-operated internet (secure; 3DES attempted).The merchant is capable of and attempted to authenticate the cardholder using 3DES but the cardholder did not participate.
101	27	Unattended off premise customer-operated internet [secure using Secure Sockets Layer (SSL)].The cardholder’s payment card was

		protected with a form of internet security, such as SSL, but authentication was not performed.
101	28	Unattended off premise customer-operated internet (nonsecure). A non-secure transaction in which the cardholder's payment card data was transmitted with no security method.

The following table contains information about valid entries for Presentation Type (Position Nos. 4–7):

Presentation Type	
Position No. 4	
Value	**Description**
0	Customer present
1	Customer not present
2	Mail and/or telephone order (customer not present)
3–7	Reserved for national use
8	Preauthorized purchase
9	Reserved for private use
Position No. 5	
Value	**Description**
0	Card present
1	Card not present
2–7	Reserved for national use
8	Preauthorized purchase
9	Reserved for private use
Position No. 6	
Value	**Description**
0	Device does not have card retention capability
1	Device has card retention capability
2–7	Reserved for national use
8–9	Reserved for private use
Position No. 7	
Value	**Description**
0	Original presentment
1	First re-presentment
2	Second re-presentment
3	Third re-presentment
4	Previously authorized request
5	Resubmit
6–7	Reserved for national use
8–9	Reserved for private use

The following table shows the ISO values for Position Nos. 4–7:

Codes for Position Nos. 4–7	
Position Nos. 4–7	**Presentation Type Description**
0000	Customer and card present
0100	Customer present; card not present
1000	Customer not present; card present
1100	Customer not present; card not present
2100	Mail and/or telephone order
8800	Preauthorized purchase

The following table contains information about the valid entries for Security Condition (Position No. 8):

Security Condition	
Value	**Description**
0	No security concern
1	Suspected fraud
2	Identification verified
3	Electronic commerce transaction with digital signature
4–7	Reserved for national use.
8–9	Reserved for private use.

Appendix R. EMV Chip Data Example

This appendix contains an EMV chip data packet example.

Field	Attribute	Description
Length	N3	Example: 0nnn
Tag/length/value data	ANSB …999	EMV chip tag fields included as applicable

Note: The data in this example is BCK Packed/Hex Nibbles in format _nn\

Note: The below table does not contain complete list of tags. For complete list of EMV tags, please contact your BUYPASS representative.

EMV Chip Data Example	
EMV Tag Fields	**Description**
01\06	EMV/Chip Data length (BCD packed)
9F\06\07\A0\00\00\00\98\08\40	Application Identifier (AID)
9F\02\06\00\00\00\00\09\00\	Amount, authorized (numeric)
9F\26\08\56\BF\29\94\72\BD\B0\C7	Application cryptogram
82\02\5C\00\	Application interchange profile
9F\36\02\45\67\	Application Transaction Counter (ATC)
9F\37\04\12\13\54\15\	Unpredictable number
95\05\40\80\00\80\00\	Terminal verification results
9C\01\00\	Transaction type
9F\10\11\06\01\1A\03\90\00\00\11\22\33\44\55\66\77\88\ 99\AA	Issuer application data
84\07\A0\00\00\00\04\10\10\	Dedicated File (DF) name
9F\27\01\80\	Cryptogram information data
9F\34\03\5E\03\00\	CVM results
5F\2A\02\09\78\	Transaction currency code
9F\1A\02\08\40	Terminal country code
9A\03\06\08\02\	Transaction date
9F\33\03\E0\B8\C8\	Terminal capabilities

Appendix S. Valid CA Public Key File Record Layout

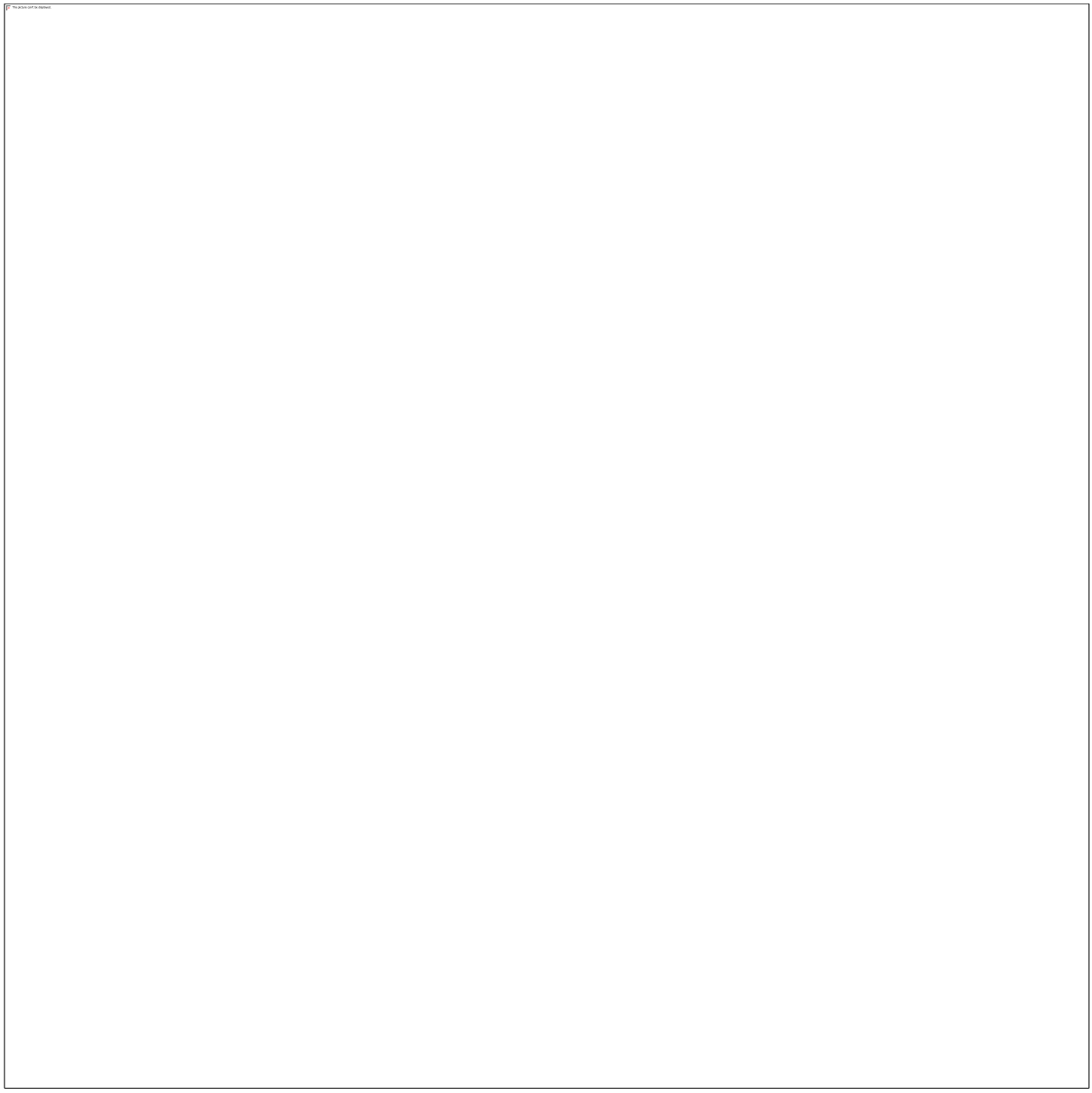
This appendix contains the record layout information of the CA Public Key File used in conjunction with element CA Public Key File Data.

The file contains a single header record followed by multiple Public Key records. All records are ASCII, use commas as Field Separators, and are terminated with a CRLF (Hex 0D0A).

The following table illustrates the record layout of the CA Public Key file:

CA Public Key File Record Layout				
Header Record **Note:** All fields are comma-separated and the record is terminated with a CRLF.				
Field Name	**Type**	**Max. Len.**	**Value**	**Usage**
File Name	AN	8	CA_KEYS	Identifies the type of file. Fixed value: CA_KEYS
File Version	N	4		Specifies the file version for tracking purposes (i.e. 0100).
Key Records This section is repeated for each Public Key sent in the Key Load. **Note:** All fields are comma-separated and the record is terminated with a CRLF.				
Field Name	**Type**	**Max. Len.**	**Value**	**Usage**
Expiry Data	N	8		Specifies when the key expires. Format: MMDDYYYY
Certification Authority Hash Algorithm Indicator	N	2	01	Identifies the hash algorithm used to produce the Hash Result in the digital signature scheme. Value: 01 **Note:** At the present time only a value of *01* (SHA-1) is supported.
Certification Authority Hash Public Key Algorithm Indicator	N	2	01	Identifies the digital signature algorithm to be used with the Certification Authority Public Key. Value: 01 **Note:** At the present time only a value of *01* (RSA) is supported.
Registered Application Provider Identifier (RID)	AN	10		Identifies the payment system to which the Certification Authority Public Key is associated (eg. A000000003).

CA Public Key File Record Layout				
Key Records This section is repeated for each Public Key sent in the Key Load. **Note:** All fields are comma-separated and the record is terminated with a CRLF.				
Field Name	**Type**	**Max. Len.**	**Value**	**Usage**
Index	Hex	2		Identifies the Certification Authority Public Key in conjunction with the RID.
Modulus	Hex	Var.		Value of the modulus part of the Certification Authority Public Key.
Exponent	N	2 or 6	03 or 2^{16+1}	Value of the exponent part of the Certification authority Public Key. Value: 3 or 2^{16+1}
Certification Authority Public Key Check Sum	Hex	40		A check value calculated on the concatenation of all parts of the Certification Authority Public Key (RID, Certification Authority Public Key Index, Certification Authority Public Key Modulus, Certification Authority Public Key Exponent) using SHA-1.



Appendix T. EMV Additional Information Data Layouts

This appendix contains data layout information for the following EMV additional information used in conjunction with EMV Additional Information Indicator and EMV Additional Information:

- EMV Table Data (Code *001*)
- Card Authentication Results Code (CARC)

The following table contains the following layout information for each type of EMV additional information:

- Indicator name
- Data field attributes
- Data field description

EMV Additional Information Data Layouts

EMV Table Data

Field	Attributes	Description
Table ID	n3	Fixed value: 001
Table Length	n	Fixed
Maximum Length	n6	Len.: 6 Identifies the EMV table data which is used to notify the chip when the MasterCard X-Code system was unable to go online. This is to ensure proper processing by a chip card personalized for full grade processing. **Request** **Valid Codes/Values:** **Code** — **Description** EMVYES — This code indicates that the device is able to receive EMV Table Data. It is only sent in an EMV Transaction request. EMVNOT — This code indicates that the device is unable to receive EMV Table Data. It is only sent in an EMV Transaction request. **Response** If this field is included in an authorization response, the device will echo it back on any subsequent advice or batch upload request.

EMV Additional Information Data Layouts		
Card Authentication Results Code (CARC)		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 002
Table Length	n	Fixed
Maximum Length	n6	Len.: 1 Identifies the CARC value returned by Visa in Bit No. 44.8 of the response.

Appendix U. Visa Digital Wallet

A Visa Digital Wallet can be of two categories:

- Pass-through Digital Wallet
- Staged Digital Wallet

Pass-through Digital Wallet: A digital wallet in which the payment device/method loaded into the wallet is actually used to conduct the overall transaction but the wallet form factor (e.g., mobile device) substitutes the original form factor (e.g., plastic card).

A Pass-through Digital Wallet provides functionality to a cardholder that facilitates the following:

- Allows usage at multiple merchants.
- Stores a credential (account or token) supplied by the cardholder to the Digital Wallet Operator.
- Completes a transaction by transferring the stored credential to the merchant without interrupting the flow of funds.

A pass-through Digital Wallet Operator (DWO) must ensure the cardholder consents in writing to the use of his or her account in the wallet according to the new rules, and must not perform Visa payment services for another DWO.

Staged Digital Wallet: A Staged Digital Wallet couples two transactions, Purchase and Funding, each of which individually relies on and retains the character of the original, underlying legacy transaction methods. This wallet provides functionality to a cardholder that facilitates the following:

- Allows usage at multiple retailers.
- Uses a Visa account number supplied by the cardholder to the Staged Digital Wallet Operator (SDWO).
- Uses a separate account or accounts assigned by the SDWO to the cardholder.
- Completes a transaction (via an SDWO) in more than one stage, in any order, as follows:
  - ◊ Purchase Transaction: Pays the retailer using the account assigned by the SDWO.
  - ◊ Funding Transaction: Uses the Visa account number provided by the cardholder to fund or reimburse the Staged Digital Wallet.

Appendix V. Moneris Data Layout

This appendix contains data layout information for the following data used in conjunction with Data Segment No. 135 (Moneris Data Segment (Request)) and Data Segment No. 136 (Moneris Data Segment (Response)).

The following table provides layout information of the Moneris Data Layout Request:

Moneris Data Layout Request				
SPDH Header				
Field	**Attributes**	**Description**		
Table ID	n3	Fixed value: 001		
Table Length	n3	Fixed Value: 048		
Table Data	an48	**Name**	**Length**	**Format**
		Device Type	2	A
		Transmission Number	2	N
		Moneris Terminal ID	8	A
		Filler	8	A
		Clerk ID	6	A
		Local Date/Time	12	A
		Message Type	1	A
		Message Subtype	1	A
		Transaction Code	2	N
		Processing Flags	3	N
		Base24 Response Code	3	N
Moneris Terminal ID				
Table ID	n3	Fixed value: 002		
Table Length	n3	Fixed Value: 008		
Table Data	an8	Moneris Terminal ID (as provided by Moneris)		
Moneris Merchant ID				
Table ID	n3	Fixed value: 003		
Table Length	n3	Fixed Value: 013		
Table Data	an13	Moneris Merchant ID (as provided by Moneris)		

<table>
<tr><th colspan="7">Moneris Data Layout Request</th></tr>
<tr><th colspan="7">Moneris Language Indicator</th></tr>
<tr><td>Field</td><td>Attributes</td><td colspan="5">Description</td></tr>
<tr><td>Table ID</td><td>n3</td><td colspan="5">Fixed value: 004</td></tr>
<tr><td>Table Length</td><td>n3</td><td colspan="5">Fixed Value: 001</td></tr>
<tr><td rowspan="7">Table Data</td><td rowspan="7">an13</td><td colspan="5">Moneris Language Indicator from the following matrix:</td></tr>
<tr><td colspan="5">Moneris Language Matrix</td></tr>
<tr><td>Value</td><td>Terminal</td><td>PED</td><td>Merchant Receipt</td><td>Cardholder Receipt</td></tr>
<tr><td>3</td><td>English</td><td>English</td><td>English</td><td>English</td></tr>
<tr><td>4</td><td>French</td><td>French</td><td>French</td><td>French</td></tr>
<tr><td>5</td><td>English</td><td>French</td><td>English</td><td>French</td></tr>
<tr><td>6</td><td>French</td><td>English</td><td>French</td><td>English</td></tr>
<tr><th colspan="7">Auth Amount for Completion</th></tr>
<tr><td>Table ID</td><td>n3</td><td colspan="5">Fixed value: 005</td></tr>
<tr><td>Table Length</td><td>n3</td><td colspan="5">Variable. Max: 7</td></tr>
<tr><td>Table Data</td><td>an3</td><td colspan="5">Amount from auth transaction sent on completion.</td></tr>
<tr><th colspan="7">Sequence Number</th></tr>
<tr><td>Table ID</td><td>n3</td><td colspan="5">Fixed value: 006</td></tr>
<tr><td>Table Length</td><td>n3</td><td colspan="5">Fixed Value: 010</td></tr>
<tr><td>Table Data</td><td>an10</td><td colspan="5">Sequence number of transaction to be sent to Moneris.</td></tr>
<tr><th colspan="7">MAC</th></tr>
<tr><td>Table ID</td><td>n3</td><td colspan="5">Fixed value: 007</td></tr>
<tr><td>Table Length</td><td>n3</td><td colspan="5">Fixed Value: 016</td></tr>
<tr><td>Table Data</td><td>an16</td><td colspan="5">MAC for this transaction via algorithm supplied by Moneris.</td></tr>
<tr><th colspan="7">Application Account Type</th></tr>
<tr><td>Table ID</td><td>n3</td><td colspan="5">Fixed value: 008</td></tr>
<tr><td>Table Length</td><td>n3</td><td colspan="5">Fixed Value: 001</td></tr>
<tr><td>Table Data</td><td>an1</td><td colspan="5">Debit account type (for MAC calculation)</td></tr>
</table>

Moneris Data Layout Request		
Terminal Capabilities and Contactless Processing		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 009
Table Length	n3	Fixed Value: 002
Table Data	an2	Terminal Capabilities and Contactless Processing Flags as required by Moneris.

The following table provides layout information of the Moneris Data Layout Request:

Moneris Data Layout Response				
SPDH Header				
Field	**Attributes**	**Description**		
Table ID	n3	Fixed value: 001		
Table Length	n3	Fixed Value: 048		
Table Data	an48	**Name**	**Length**	**Format**
		Device Type	2	A
		Transmission Number	2	N
		Moneris Terminal ID	8	A
		Filler	8	A
		Clerk ID	6	A
		Local Date/Time	12	A
		Message Type	1	A
		Message Subtype	1	A
		Transaction Code	2	N
		Processing Flags	3	N
		Base24 Response Code	3	N
Moneris Terminal ID				
Table ID	n3	Fixed value: 002		
Table Length	n3	Fixed Value: 008		
Table Data	an8	Moneris Terminal ID (as provided by Moneris)		

Moneris Data Layout Response		
Moneris Merchant ID		
Table ID	n3	Fixed value: 003
Table Length	n3	Fixed Value: 013
Table Data	an13	Moneris Merchant ID (as provided by Moneris)
Sequence Number		
Table ID	n3	Fixed value: 004
Table Length	n3	Fixed Value: 010
Table Data	an10	Sequence number returned.
MAC		
Table ID	n3	Fixed value: 005
Table Length	n3	Fixed Value: 016
Table Data	an16	MAC for this transaction via algorithm supplied by Moneris.
Encryption Key		
Table ID	n3	Fixed value: 006
Table Length	n3	Variable Max: 51
Table Data	an	3 character indicator identifying which keys are included followed by up to three 16 byte keys. Key indicators are: Character 1: “Y” if MAC Encryption key is present Character 2: “Y” if Data Encryption Key is present Character 3: “Y” if PIN Encryption Key is present Characters 4-end: Relevant Keys in the above order.

Appendix W. Download Data Layout

This appendix contains data layout information for the following data used in conjunction with Data Segment No. DL7 (Supplemental Terminal Data Segment)

The following table provides layout information of the Supplemental Terminal Data Segment:

Download Data Layouts		
Site Language		
Field	**Attributes**	**Description**
Table ID	n3	Fixed value: 001
Table Length	n3	Fixed Value: 003
Table Data	an1	ISO 639-2 language code associated with this terminal.
Postal Code		
Table ID	n3	Fixed value: 002
Table Length	n3	Fixed Value: 013
Table Data	an1	Identifies the 13 character international postal code associated with the transmitting device.

Appendix X. Reserved

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Appendix Y. 3-D Secure

Introduction: "3-D Secure" is a general term for an advanced authentication solution implemented to reduce eCommerce fraud by verifying a cardholder's identity in real time. Each of the major card brands has a 3-D Secure offering: Visa "Verified by Visa", MasterCard "SecureCode", American Express "Safekey". This appendix provides detailed information on these services.

How it Works: The 3-D Secure protocol operates in three domains:

- Acquirer Domain – The Acquirer accepts eCommerce transaction data from the merchant and passes it to Visa/MasterCard/Amex/Discover (as applicable).
- Issuer Domain – The Issuer is responsible for determining whether authentication is available for the card account presented in a purchase.
- Interoperability Domain – Infrastructure created by the payment networks and software providers to enable the 3-D Secure protocol between the Acquirer Domain and the Issuer Domain

The 3-D Secure process involves the following steps:

1. The cardholder enters card information on the merchant's site (that uses 3-D Secure) to purchase goods.
2. Merchant Plug-in (MPI) contacts the Directory Server (DS) to initiate an authentication request.
3. The DS routes the authentication request to the appropriate Issuer Access Control Server (ACS).
4. The DS sends the Issuer ACS response to the MPI.
5. The cardholder is directed to the Issuer's 3-D Secure browser page. The Issuer authenticates cardholder's identity in a variety of ways, such as asking to enter password which the cardholder has previously set up with the Issuer, or requesting to enter an authentication code/OTP that has been sent to the cardholder's registered email/mobile number
6. The Issuer approves or declines the Authorization Request and returns the Authorization Response (containing cryptographic value along with ECI/Terminal Type (Position Nos. 9–10) values 25, 26, 27, 28 or 29 (as applicable) and other data) to the merchant. Refer to the below grid for the cryptographic values returned for each card brands:

Card Brand	Cryptographic Value
Visa	CAVV
MasterCard	AAV
Amex	AEVV

7. The cardholder is redirected to the merchant's site and the purchase is completed.

Note: Contact your 3-D Secure representative for further details.

VERIFIED BY VISA:

Verified by Visa (VbV) is an online service that verifies a cardholder's ownership of an account, in real time, during an online payment transaction. It gives Visa card Issuers the ability to confirm a cardholder's identity using a variety of authentication methods, including passwords, chip cards and digital certificates, providing merchants with real time authentication results during the checkout process. Verified by Visa is based on the 3-D Secure Protocol, which uses Secure Sockets Layer (SSL) encryption to collect and protect payment card information transmitted via the Internet.

Merchants participating in Visa token service may (optionaly) support the TAVV in combination with 3-D Secure (3DS) CAVV for payment token-based transactions with 3-D Secure.

Effective October 2018, Visa will support two separate cryptograms in the authorization message:

1) The TAVV Cryptogram for payment token validation

2) The 3DS CAVV cryptogram for cardholder authentication

Note: Please contact your First Data representative for the availability of this functionality.

Refer to the below table for usage of both these cryptograms:

Note: The usages provided below are not exhaustive and all encompassing.

Scenario	CAVV	TAVV
Verified-by-Visa only	Cardholder Authentication Verification Value (CAVV) as defined in Table ID: 035 (Verified By Visa® Data) of Appendix I (Variable Information Data Layouts).	Not applicable
Digital Wallet (Payment Token) only	Cardholder Authentication Verification Value (CAVV) – Revised Format (Data Element 195)	Not applicable
Verified-by-Visa and Digital Wallet	Cardholder Authentication Verification Value (CAVV) – Revised Format (Data Element 195)	Digital Wallet Payment Token Cryptogram (TAVV) (Data Element 237)

AMERICAN EXPRESS SAFEKEY

American Express SafeKey (Amex SafeKey) is an online service that verifies a cardholder's ownership of an account, in real time, during an online payment transaction. Amex SafeKey gives Amex card Issuers the ability to confirm a cardholder's identity using a variety of authentication methods, including passwords, chip cards and digital certificates, providing merchants with real time authentication results during the checkout process. Amex SafeKey is based on the 3-D Secure Protocol, which uses Secure Sockets Layer (SSL) encryption to collect and protect payment card information transmitted via the Internet.

SafeKey transactions coming from merchants will be sent with indicator "SK" followed by 56 bytes of SafeKey data in base64 format. When the first 28 bytes of value is not applicable, it should be space filled.

Note: American Express Safekey is not available for OptBlue merchants.

MASTERCARD SECURECODE

MasterCard SecureCode is a solution designed to authenticate cardholders when paying online. SecureCode offers a mechanism for securing the Internet channel by strongly authenticating the cardholder at the point of interaction by providing a unique transaction-specific token that provides evidence that the cardholder originated the transaction. SecureCode uses MasterCard's Universal Cardholder Authentication Field (UCAF) infrastructure to communicate the authentication information among the cardholder, Issuer, merchant and Acquirer.

MasterCard SecureCode supports the 3-D Secure Protocol. MasterCard SecureCode requires merchants to install 3-D Secure v1.0.2 (or better) compliant Merchant Server Plug-In software.

MASTERCARD DIGITAL SECURE REMOTE PAYMENT

MasterCard has introduced a new data element for Digital Payment containing both cardholder authentication and token authentication in a more comprehensive manner to validate token transactions in electronic commerce environment.

This feature enables merchants to take advantage of the security validation when both DSRP cryptogram and SecureCode/Identity Check AAVare present in same message.

Merchants can support both Digital Secure Remote Payment (DSRP) cryptogram and SecureCode/Identity Check 3DS Accountholder Authentication Value (AAV) in the same request message. However, the UCAF Security level code (position '1 -2') should be '21' (Channel Encrypted).

Merchants should use the UCAF data field for the MasterCard AAV cryptogram (Data segment 111 table ID 36) and should use the TAVV field (Segment 123) for all Token and DSRP cryptograms.

Scenario	UCAF	DSRP
AAV Cryptogram only	MasterCard UCAF Data (Data segment 111 table ID 36)	
Token or DSRP Cryptogram only		Digital Wallet Payment Token Cryptogram (TAVV) (Data segment 123 Element 237)
AAV Cryptogram and DSRP Cryptogram	MasterCard UCAF Data (Data segment 111 table ID 36)	Digital Wallet Payment Token Cryptogram (TAVV) (Data segment 123 Element 237)

Appendix Z. Stored Credential

For definitions and use cases, refer to section 10.13.2.4.1 Stored Credentials. The use cases apply to Visa, MasterCard, Discover, and Amex.

When storing credentials for the first time:

- If no amount is due when credentials are stored for the first time, a $0.00 Account Verification transaction can be sent along with Value 'I' (Initial) in Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08
- If an amount is due when credentials are stored for cardholder-initiated stored credential transactions, Value 'I' (Initial) in Data Element 111, Table ID 32 – POS Additional Data Information Sub table ID 08
- If an amount is due when credentials are stored for recurring or installment, the transaction must be identified as recurring ('R') or installment ('I')
- The original POS Entry mode (for example: 02, 90 etc. respectively as defined in Appendix J) will be sent
- For Unscheduled Merchant Initiated or Cardholder initiated Credential-on-File transactions, no recurring or installment indicators are required
- Visa/Amex Transaction ID (Data Element 112, Table ID 28) and Discover Network Reference number (Data Element 112, Table ID 12) from response should be stored and used for allsubsequent merchant-initiated COF transactions
- Authorization amount should be stored for Discover transactions as it is later required for all subsequent merchant initiated Discover transactions
- For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to Merchant Initiated transaction (MIT) or Cardholder initiated transaction (CIT) for initial transaction while storing credential on file. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details

When initiating a recurring transaction using a stored credential MIT (Merchant Initiated Transaction) without active participation of the cardholder:

- The transaction must be identified as 'R' (Recurring)
- For all card types, Data Element 111, Table ID 32 – POS Additional Data Information, Sub table ID 08 should have value 'C' for Subsequent Credential on File
- For all card types, POS Entry Mode value of '10' must be submitted in Data Element 111, Table ID 05 - Point-of-Service Entry Mode
- For Merchant initiated COF transactions (MIT) where transactions are based on the stand-in instructions with cardholder, Merchants should use the appropriate recurring indicators and store the original transaction identifier to be used in the subsequent transactions.
- Include the Merchant Initiated (MIT) indicator using Data Element 111, Table ID 49, Sub Table ID 02, Value 'M'.
- CVV2 Data must not be sent
- Visa/Amex Transaction ID (Data Element 111, Table ID 44) and Discover Network Reference Number (Data Element 111, Table ID 15) should be included in this subsequent request

- First Authorization Amount (Data Element 111, Table ID 45) should be included for Discover MIT transactions
- Stored Credential Additional Data (Data Element 111, Table ID 56, Sub Table ID 01 to 08) should be included for Discover MIT transactions if the card is issued in India and amount is above $52
- For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to corresponding recurring MIT transactions (Standing order (Variable amount and fixed frequency)) or Subscription (fixed amount and fixed frequency). Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.
- For Visa, merchant needs to send Stored Credential Additional Data for Recurring payments for cards issued in India. Refer Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Additional Data), Sub Table ID 01 to 13 for more details.

When initiating an installment transaction using a stored credential MIT (Merchant Initiated Transaction) without active participation of the cardholder:

- The transaction must be identified as 'I' (Installment)
- For all card types, Data Element 111, Table ID 32 – POS Additional Data Information, Sub table ID 08 should have value 'C' for Subsequent Credential on File
- For all card types, POS Entry Mode value of '10' must be submitted in Data Element 111, Table ID 05 - Point-of-Service Entry Mode

- For Merchant initiated COF transactions (MIT) where transactions are based on the stand-in instructions with cardholder, Merchants should use the appropriate installment indicators and store the original transaction identifier to be used in the subsequent transactions
- Include the Merchant initiated (MIT) indicator using Data Element 111, Table ID 49, Sub Table ID 02, Value 'M'
- CVV2 Data must not be sent
- Visa/Amex Transaction ID (Data Element 111, Table ID 44) and Discover Network Reference Number (Data Element 111, Table ID 15) should be included in this subsequent request
- First Authorization Amount (Data Element 111, Table ID 45) should be included for Discover MIT transactions
- Stored Credential Additional Data (Data Element 111, Table ID 56, Sub Table ID 01 and 08) should be included for Discover MITinstallment transactions,
- Installment Payment Type (Data Element 111, Table ID 49, Sub table ID 13) should be included for Discover MITinstallment transactions,
- For Discover Soft Descriptor – Marchant name (Data Element 111, Table ID 047) Should contain the third party payment provider and the underlying retailer name in the format - 'Third Party*Marchant name'. e.g. "XYZ*A SMALL CO".
- For Discover Soft Descriptor – Merchant location (Data Element 111, Table ID 057, 058 an 059). For Third party payment providers the address of the underlying retailer should be included.
- For Discover - SIC information (Data Element 111, Table ID 007) should be the SIC/MCC code of the third party payment provider's underlying retailer/merchant

- For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to MIT Installment transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.

When initiating an unscheduled transaction using a stored credential MIT (Merchant Initiated Transaction) without active participation of the cardholder:

- The transaction must not be identified as Installment or Recurring
- Include the Merchant Initiated (MIT) indicator using Data Element 111, Table ID 49, Sub Table ID 02, Value 'M'
- For all card types, Data Element 111, Table ID 32 – POS Additional Data Information, Sub table ID 08 should have value 'C' for Subsequent Credential on File
  For all card types, POS Entry Mode value of '10' must be submitted in Data Element 111, Table ID 05 - Point-of-Service Entry Mode.
- Visa/Amex Transaction ID (Data Element 111, Table ID 44) and Discover Network Reference Number (Data Element 111, Table ID 15) should be included in this subsequent request
- First Authorization Amount (Data Element 111, Table ID 45) should be included for Discover MIT transactions
- Stored Credential Additional Data (Data Element 111, Table ID 56, Sub Table ID 01 to 08) should be included for Discover MIT transactions if the card is issued in India and amount is above $52.
- For Mastercard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable for MIT Unscheduled credential on file transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.
- For Discover card type Delayed Card Sale transactions, Variable information Data Element 111, Table ID 16 Delayed Card Sale Indicator must be 'D'.
- For Discover card type No Show Charge transactions, Variable information Data Element 111, Table ID 17 No Show Charge Indicator must be 'N'.

When initiating a recurring transaction using a stored credential CIT (Cardholder-Initiated Transaction)

- The transaction must be identified as 'R' (Recurring)
- CVV2 Data must not be sent
- Data Element 111, Table ID 32 – POS Additional Data Information, Sub table ID 08 should have Value 'I' (Initial)
- Include the Customer Initiated (CIT) indicator using Data Element 111, Table ID 49, Sub Table ID 02, Value 'C'.
- Data Element 111, Table ID 05 - Point-of-Service Entry Mode, original POS Entry Mode value (for example: 02, 90 etc. respectively as defined in Appendix J) must be submitted for initial transaction.
- For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to CIT Standing order (Variable amount and fixed frequency) and Subscription (fixed amount and fixed frequency) transactions. Refer to

Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.

- For Visa, merchant needs to send Stored Credential Additional Data for Recurring payments for cards issued in India. Refer Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Additional Data), Sub Table ID 01 to 13 for more details.

When initiating an installment transaction using a stored credential CIT (Cardholder-Initiated Transaction)

- The transaction must be identified as 'I' (Installment)
- CVV2 Data must not be sent
- Data Element 111, Table ID 32 – POS Additional Data Information, Sub table ID 08 should have Value 'I' (Initial)
- Include the Cardholder Initiated (CIT) indicator using Data Element 111, Table ID 49, Sub Table ID 02, Value 'C'.
- Data Element 111, Table ID 05 - Point-of-Service Entry Mode, original POS Entry Mode value (for example: 02, 90 etc. respectively as defined in Appendix J) must be submitted for initial transaction.
- For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to CIT Installment transactions. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.

When initiating a transaction using a stored credential CIT (Cardholder-Initiated Transaction)

- POS Entry Mode value of '10' must be submitted in Data Element 111, Table ID 05 - Point-of-Service Entry Mode

- For all card types, Data Element 111, Table ID 32 – POS Additional Data Information, Sub table ID 08 should have value 'C' for Subsequent Credential on File

- Include the Cardholder Initiated (CIT) indicator using Data Element 111, Table ID 49, Sub Table ID 02, Value 'C'
- CVV2 Data is optional
- For MasterCard card type, Merchant needs to send MIT/CIT category and subcategory data with valid value as applicable to CIT unscheduled credential on file transaction. Refer to Variable Information Indicator (Data Element 111), Table ID 56 (Stored Credentials Transaction Additional Data), Sub Table ID 09 for more details.

Appendix AA. TransArmor Processing Considerations

TransArmor is a processing option provided by FirstData that provides more secure transaction processing using encryption and tokenization.

When initializing a terminal for TransArmor Processing, a TransArmor Key Load should be performed to obtain the Key ID and Key to be used for encryption purposes. In addition, a RegiStart transaction should be performed to register the device with TransArmor. These functions are described in the ATL105 Specifications Update for TransArmor Processing document.

The available encryption types for TransArmor processing are:

- RSA encryption
- TA-VE Cipher Hidden Encryption
- TDES/DUKPT
- Ingenico On Guard FPE
- AES DUKPT

The following are specific considerations required in transaction processing in order to use TransArmor.

Data Element 2 – Account Number

For TransArmor processing, the Account Number field is replaced by the TAP header and EDATA. The general format for this data is:

TAPnnnnnnxyyyyyyyyyyyzzzz[EDATA]

Where,

TAP =	3-byte TransArmor Indicator
nnnnnn =	6-byte Sequence Number of the transaction that matches the value in Element No. 86 (Sequence Number)
x =	1-byte [EDATA] Identifier that describes the content of [EDATA]
yyyyyyyyyyy =	11-byte Key ID
zzzz =	4-byte Token Type ID. **Valid Values**: XXXX = Single use or MultiPay Token; 4-byte alphanumeric value assigned by First Data
[EDATA] =	contents are dependent on the information in the TAP header and whether the transaction is an initial transaction or a subsequent. Maximum length of encrypted data is 400 bytes.

Note: On all subsequent transactions, the unencrypted expiration date should be sent in Data Element 12, Card Discretionary Block Data, as it would for non-TransArmor transactions.

Note: In all cases, the TransArmor token to be used in the subsequent transaction is contained in the Account Number field in the initial transaction response.

Note: In the case of the Initial Transaction Request for the PKI Encryption and Tokenization method, 2 zeros followed by the first 6 bytes of the Merchant Number assigned by BUYPASS

should be prepended to the Account Number before encrypting. Please see the description for Element No. 102 (Terminal Identifier) for additional information about the Merchant Number.

The following table lays out the expected data in each field by transaction type and [EDATA] identifier.

EDATA identifier	Description	Transaction Type	Key ID	EDATA
1	PKI, TDES/DUKPT, Ingenico OnGuard FPE, and AES DUKPT Encryption and Tokenization	Initial	For PKI, as received in TransArmor Key and KeyID Load Response. Key ID will be sent in the Additional TransArmor Data. For TDES/DUKPT, OnGuard FPE, and AES DUKPT, 00000000000. Key ID will be sent in the Additional TransArmor Data.	Encrypted Track1 Data
2		Initial		Encrypted Track2 Data
3		Initial		Encrypted manually-keyed PAN. Encrypted data for a manually entered PAN should be followed by <fs> and MMYY unencrypted expiration date.
4	Token Only	Initial		Unencrypted Data (Track1, Track2, or manually-keyed PAN)
6	TransArmor Verifone Edition (TA-VE) Encryption and Tokenization. PIN Pad is identified via Buypass terminal number	Initial	5-byte Merchant Domain, followed by semicolon (;), followed by 5-byte Merchant Brand. Provided by First Data.	Up to 16 character alphanumeric processor code provided by Buypass at boarding that identifies this location to TransArmor, followed by a colon (":"), followed by Encrypted Track Data. Encrypted Data for a manually entered PAN should be followed by <fs> and MMYY unencrypted expiration as for unencrypted manually entered PANs.
7	TA-VE, but PIN pad is identified via pump/lane number (Data Element 79)	Initial		
M	Multi-Controller TA-VE, but PIN pad is identified via the Card	Initial		

	Acceptor Terminal ID (Segment 111, Table ID 18)			
0	Token	Subsequent	0 filled, with or without ";" as per the initial transaction.	Token received in initial transaction or stored on-file.

Data Element 12 – Card Discretionary Data

For TransArmor processing, the expiration date must be stored for subsequent transactions associated with a token.

Data Element 26 – Decline Code

For a TransArmor Token Only transaction that has been declined because BUYPASS cannot retrieve the necessary token for the associated PAN, the decline code 4M will be returned to the POS.

Data Element 30 – Download Indicator

A value of 8 in this field indicates that the terminal needs to update its Key ID and Key because the current Key ID is due to expire or has expired.

Data Element 48 – Load Type

A value of K in this field indicates that the load being requested is a TransArmor Key and Key ID request.

Data Element 83 – Response Code

The following values are specific to TransArmor Key Load processing:

K Approved—TransArmor Load

L Rejected for one of the following reasons—

1) Merchant is not TransArmor enabled

2) Signing Key is invalid or outdated

Data Element 102 – Terminal Identifier

For TransArmor PKI Encryption and Tokenization Load Request, the Device Type should be "++" regardless of the actual device type of the terminal.

Segment 111, Table ID 52 – Additional TransArmor Data

In some TransArmor implementations, a longer Key ID and/or a Special device type can be specified using this Data Element.

Transaction Examples

In the following examples, the encrypted block is for illustrative purposes only and will not necessarily translate into valid data. In all examples, "." represents the field separator character.

For VSP/PKI encryption, a sample Account Number field may look like this. All 3 varieties of VSP/PKI Encryption (Track1, Track2, and Manually-keyed) are formatted the same with the exception of the EDATA identifier.

Authorization request:

```
ATL105.3.100.424.1.6348500151004.3020.TAP028320166257982250LKLLZXAoaMX
2gVzxdjP6IK8hoJ20gjdFKwyVCq8C4+Or+c/3Ah3Cy2B8tGO2FhefKsxdGO9vVVoyqP79k
ZT3Yw8dPO05+mr6Zk3vEvMh4U/4VhBcnrztcz6CS2q47Q+qdj1j34jVY1jqjr58Xg1VKy3
hZl+km5YxC/OrDDhtg3ttl5pTI6GZMJHmcGFJikaLV/Twb+Mdvv3GUjz5NFymzQThGpEBw
W0eUpWcr6Jxh6jpyEKQhNIMmP2vkvznuovQZUw8qMssLLZaFjOHcHoMT9p3+9/zlXaRZeS
tDJ2xY0IWcayWm7Rhviez99pkpHjZ56t6ZXk5ckTufot+ZM6cY2xxxx=x...00001200..
.028320...1.102.077.S02421O000000006\200000100\00000600\441O000000006\
200000100\00000600.111.071.0470230000DG008D7F113437F0ACA00500390103001
0000000000103100205.
```

Where,

- TAP is the TransArmor indicator
- 028320 is the sequence number
- 1 is the EDATA identifier (encrypted Track1)
- 66257982250 is the key id
- LKLL is the token type
- The rest is the EDATA block

Authorization reply:

```
280505201542028320000001200523702754680671372001 0
0200050611APPROVED  523702               112014009002VI
```

Where,

- 7546806713720010 is the returned token

Completion request:

```
ATL105.3.100.126.1.6348500151004.0020.TAP028320066257982250LKLL7546806
713720010.4912....00001200...028320.523702.0505201542.1.102.077.S02421
O000000006\200000100\00000600\441O000000006\200000100\00000600.111.071
.0470230000DG108D7F11343BDE4E2005003901030010000000000103100205.
```

Where,

- TAP is the TransArmor indicator
- 028320 is the sequence number
- 0 is the EDATA identifier (Token)
- 66257982250 is the key id
- LKLL is the token type
- 7546806713720010 is the token

Completion reply:

```
080505201542028320000001200523702754680671372001 0          14
050601APPROVED  523702
```

Where,

- 7546806713720010 is, once again, the Token

For TAVE encryption, a sample Account Number field may look like this. All 3 varieties of TAVE encryption (Buypass terminal based, pump/lane number base, and card acceptor term ID based) are formatted the same with the exception of the EDATA identifier.

Authorization request:

```
ATL105.3.100.142.0.KA48500055001.3020.TAP0283216BYDEV;BYDEV00010384462
0923:B4818521131690165^VISATESTCARD^49122016503225734....00001000...02
8321...1.102.044.S01490O0000000010\200000100\00001000.111.071.047023000
866008D7F1158010E476005003901030010000000000103100205.
```

Where,

- TAP is the TransArmor indicator
- 028321 is the sequence number
- 6 is the EDATA identifier (TAVE)
- BYDEV;BYDEV is the key id
- 0001 is the token type
- 03844620923 is the processor code
- : separates the processor code from the encrypted portion of the EDATA
- The rest is the encrypted EDATA

Authorization reply:

```
220505201458028321000001000868604053338263743016 5
0200050611APPROVED  868604               1120210090022B0131249
```

Where,

- 0533382637430165 is the returned Token

Cancellation request:

```
ATL105.3.100.116.0.KA48500055001.S020.TAP0283210BYDEV;BYDEV00010533382
637430165.4912....00001000...028321.868604..1.102.044.S01490O0000000010
\200000100\00001000.111.071.047023000866108D7F11580490C330050039010300
10000000000103100205.
```

Where,

- TAP is the TransArmor indicator
- 028321 is the sequence number

- 0 is the EDATA identifier (TAVE)
- BYDEV;BYDEV is the key id
- 0001 is the token type
- 0533382637430165 is the Token

Cancellation reply:

```
020505201458028321000001000868604053338263743016 5          14
050601CANCELLED 86860
```

Where,

- 0533382637430165 is the Token

Appendix AB. Purchase Repayment

Purchase repayments are type of installment payments submitted by third party installment provider or payment facilitators. Third party installment providers referred here are the merchants that are different from the retailer selling goods or services.

Use case for Purchase repayments includes post purchase payments, Buy now Pay later services provided by third party service provider.

Appendix AC. Valid Country Codes

This appendix contains a list of valid codes for Table ID 59.

These codes identify the country in which the acquiring institution resides.

Country Codes			
COUNTRY	**Code**	**Country**	**Code**
Afghanistan	004	British Virgin Islands	092
Albania	008	Brunei Darussalam	096
Algeria	012	Bulgaria	100
American Samoa	840	Burkina Faso	854
Andorra	020	Burundi	108
Angola	024	Cambodia	116
Anguilla	660	Cameroon	120
Antarctica (no universal currency)		Canada	124
Antigua and Barbuda	028	Cape Verde	132
Argentina	032	Cayman Islands	136
Armenia	051	Central African Republic	140
Aruba	533	Chad	148
Australia	036	Chile	152
Austria	040	China	156
Azerbaijan	031	Christmas Island	162
Bahamas	044	Cocos (Keeling) Island	166
Bahrain	048	Columbia	170
Bangladesh	050	Comoros	174
Barbados	052	Congo	178
Belarus	112	Cook Islands	184
Belgium	056	Costa Rica	188
Belize	084	Croatia	191
Benin	204	Cyprus	196
Bermuda	060	Czech Republic	203
Bhutan	064	Democratic Republic of Congo	180
Bolivia	068	Denmark	208
Bosnia – Herzegovina	070	Djibouti	262
Botswana	072	Dominica	212
Bouvet Island	074	Dominican Republic	214
Brazil	076	East Timor	626
British Indian Ocean Territory	086	Haiti	332

Country Codes			
ACQUIRING INSTITUTION'S COUNTRY	**Code**	**Acquiring Institution's Country**	**Code**
Ecuador	218	Heard Island and McDonald Island	334
Egypt	818	Holy See (See Vatican City State.)	N/A
El Salvador	222	Honduras	340
Equatorial Guinea	226	Hong Kong	344
Eritrea	232	Hungary	348
Estonia	233	Iceland	352
Ethiopia	231	India	356
European Economic and Monetary Union	N/A	Indonesia	360
Faeroe Islands	234	Iraq	368
Falkland Islands (Malvinas)	238	Ireland	372
Fiji	242	Israel	376
Finland	246	Italy	380
France	250	Ivory Coast	384
France, Metropolitan	249	Jamaica	388
French Guiana	254	Japan	392
French Polynesia	258	Jordan	400
French Southern Territory	260	Kazakhstan	398
Gabon	266	Kenya	404
Gambia	270	Kiribati	296
Georgia	268	Korea, Republic of	410
Germany	276	Kuwait	414
Ghana	288	Kyrgyzstan	417
Gibraltar	292	Lao People's Democratic Republic	418
Greece	300	Latvia	428
Greenland	304	Lebanon	422
Grenada	308	Lesotho	426
Guadeloupe	312	Liberia	430
Guam	316	Libyan Arab Jamahiriya	434
Guatemala	320	Liechtenstein	438
Guinea	324	Norfolk Island	574
Guinea-Bissau	624	Northern Mariana Islands	580
Guyana	328	Norway	578

Country Codes			
COUNTRY	**Code**	**Country**	**Code**
Lithuania	440	Oman	512
Luxembourg	442	Pakistan	586
Macau	446	Palau	585
Macedonia	807	Palestinian Territory, Occupied	275
Madagascar	450	Panama	591
Malawi	454	Papua New Guinea	598
Malaysia	458	Paraguay	600
Maldives	462	Peru	604
Mali	466	Philippines	608
Malta	470	Pitcairn Island	612
Marshall Islands	584	Poland	616
Martinique	474	Portugal	620
Mauritania	929	Puerto Rico	630
Mauritius	480	Qatar	634
Mayotte	175	Reunion	638
Mexico	484	Romania	642
Micronesia	583	Russian Federation	643
Moldova	498	Rwanda	646
Monaco	492	St. Helena	654
Mongolia	496	St. Kitts-Nevis	659
Montenegro	499	St. Lucia	662
Montserrat	500	St. Pierre and Miquelon	666
Morocco	504	St. Vincent and the Grenadines	670
Mozambique	508	Samoa	882
Myanmar	104	San Marino	674
Namibia	516	Sao Tome and Principe	930
Nauru	520	Saudi Arabia	682
Nepal	524	Senegal	686
Netherlands	528	Serbia, Republic of	688
New Caledonia	540	Seychelles	690
New Zealand	554	Sierra Leone	694
Nicaragua	558	Singapore	702
Niger	562	Slovakia	703
Nigeria	566	United States	840
Niue	570	U.S. Minor Outlying Islands	581
		United States Virgin Islands	850

Valid Acquiring Institution Country Codes			
ACQUIRING INSTITUTION'S COUNTRY	**Code**	**Acquiring Institution's Country**	**Code**
Slovenia	705	Uruguay	858
Solomon Islands	090	Uzbekistan	860
Somalia	706	Vanuatu	548
South Africa	710	Vatican City State (Holy See)	336
South Georgia and South Sandwich Islands	239	Venezuela	862
Spain	724	Vietnam	704
Sri Lanka	144	Virgin Islands: See British Virgin Islands and United States Virgin Islands.	
Sudan	729	Wallis and Futuna Islands	876
Suriname	740	Western Sahara	732
Svalbard and Jan Mayen Islands	744	Yemen	887
Swaziland	748	Zambia	894
Sweden	752	Zimbabwe	716
Switzerland	756	Saint Martin	663
Taiwan	158	Sint Maarten	534
Tajikistan	762	South Sudan	728
Tanzania, United Republic of	834	Syria	760
Thailand	764	Jersey	832
Togo	768	Isle of Man	833
Tokelau	772	Bonaire, Sint Eustatius and Saba	535
Tonga	776	Aland Islands	248
Trinidad and Tobago	780	Cuba	192
Tunisia	788	Iran	364
Turkey	792	Korea-north	408
Turkmenistan	795	Curaçao	531
Turks and Caicos Islands	796	Northern Mariana Islands	580
Tuvalu	798	Saint Barthelemy	652
Uganda	800		
Ukraine	804		
United Arab Emirates	784		
United Kingdom	826		
United Nations Interim Administration Mission in Kosovo (UNMIK)	900		

Appendix AD. Real Time Account Updater

Definition:

The Real Time Account Updater enables the secure electronic exchange of account information updates between participating issuers and acquirers for merchants who securely store the Cardholder account information for future recurring / subsequent transactions (for example, credentials on file).

Transaction processing rules:

The transaction must be from a participating acquirer and must be a card not present transaction as identified by any of the following criteria:

- Credential on file transaction
- Instalment payment
- Recurring payment

The merchant must register and certify with First Data to participate in the Real Time Account Updater process.

Note: For further details and requirements, please contact your Account Representative.

Participating merchant should send the Account Updater Request Indicator field (Variable Information segment 111, Table ID 060, Sub table ID 01) as trigger in request messages for all Instalment, Recurring, and Credential on File transactions. First Data will determine whether the transaction qualifies for Real Time Account Updater before requesting account information update from the association.

Note: Merchants using Real Time Account Updater should refer to Account Number - Data Element 2 and Card Discretionary Block Data - Data Element 12 for additional information.

Appendix AE. Visa Estimated and Incremental Authorization Transactions

* MCC 4121, 5411 - Estimated and Incremental authorizations are allowed for Card Not Present transactions only.

* MCC 5552 – Estimated and Incremental authorizations for EV transactions.

For permitted merchants, an estimated or initial authorization must be included in an estimated or initial authorization request, and must be used when the final amount is unknown. In addition, merchants and acquirers must ensure that each incremental authorization requests:

- Only follows an estimated / initial authorization.
- Contains an incremental authorization indicator - BUYPASS Host identifies the incremental authorization transaction by the approval code returned (Authorization Identification Response). Once identified as incremental authorization transaction, the Host will send out the incremental indicator on behalf of the merchant to Visa.
- Contains the same Sequence Number as the estimated or initial transaction request.
- Contains the same Transaction Identifier in Segment as the estimated or initial transaction request.
- Contains the same Terminal transaction number (Table ID 63 in Segment 111) as the estimated or initial transaction request.

Example: Customer A checks in a hotel for 3 days. The merchant uses Customer A's credit or signature debit card (common AID - routed to Visa if processed as dual message) and sends the estimated auth indicator as '1' in the Acquirer Trace Data – Sub Table ID 09 in the initial authorization request. At the end of Day 3, Customer A extends his stay at the same hotel and also avails a few other services, such as laundry service. The subsequent transactions must not carry the estimated authorization indicator in Sub Table ID 09. BUYPASS Host will identify the transactions as incremental based on the approval code that the merchant sends in Authorization Identification Response in the subsequent transactions. Merchants must send the final amount in the completion transaction message.

For cruise lines, lodging merchants, and Car/ Vehicle Rental merchants, an approval response to the estimated authorization and any subsequent incremental authorization will expire 14 days after the initial estimated authorization. For other rental merchants, such as trailer parks and campgrounds, an approval response to the estimated authorization and any subsequent incremental authorization will expire seven days after the initial estimated authorization. Merchants with stays or rentals that exceed this period will be subject to chargebacks for no authorizations unless they complete the first transaction and then initiate the second transaction on or before day 14 or day seven, as applicable by MCCs.

Appendix AF. Estimated and Incremental Authorization Transactions

The Incremental Auth product is a means for a merchant to open an authorization, and modify it over time, before eventually submitting the final amount in the form of completion. This functionality has typically been used in the lodging and car rental industries, where the Initial Authorization may occur days before the transaction is completed, but is being expanded to support additional industries.

The Incremental Auth (IA) product supports four transaction types:

- Original Auth (aka Estimated Auth)
- Incremental Auth
- Partial Reversal (aka IA Reversal)
- Incremental Auth Completion (aka IA Completion)

This functionality is mandatory for Discover and is not limited by MCC.

For Visa, Mastercard, and American Express, the use of this functionality is strongly recommended in addition to populating the Estimated and Incremental Authorization fields for the applicable lodging and EV transactions.

REQUIREMENTS:

- All IA transaction types should **contain Incremental Auth Data (Variable Information segment 111 Table 82). This is mandatory for Discover.**
    - Incremental Auth Data contains the **Transaction ID** which must be unique (per terminal) to any open Incremental Auth purchase.
- Incremental Auths, IA Reversals, and IA Completions are all subsequent transactions, and must be preceded by an Original Auth.
- Subsequent IA transactions must contain the **Approval Code (DE 39)** that was received on the approved Original Auth.
- Once an IA Completion is received, the IA is closed, and no more subsequent transactions will be approved for that Terminal Transaction ID.